Birational Geometry Seminar

Chapters 1–5 and Appendices A–D

Revised reading notes

Original numbering and notation retained.
References are retained at the relevant results and proofs.
Chapter 6 is outside the scope of this volume.

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Chapter 1
Ample and Nef Line Bundles

In this chapter all schemes will be considered over $\mathbb{C}$ unless another ground field is explicitly specified. Schemes are of finite type over the ground field unless stated otherwise. Some results hold over more general fields; characteristic-zero assumptions are retained where they are needed.

1.1 Cartier Divisors, Line Bundles and Linear Series

1.1A Cartier Divisors and Line Bundles

> Cartier divisors are important for studying maps from a given variety to a projective space.
>
> Robin Hartshorne

Definition 1.1.1 (Cartier divisor)**.** A Cartier divisor on X is a global section of the quotient sheaf $\mathcal{K}_X^*/\mathcal{O}_X^*$, where $\mathcal{K}_X$ is the sheaf of total quotient rings:

$$\mathrm{Div}(X) := \Gamma(X, \mathcal{K}_X^*/\mathcal{O}_X^*).$$

Concretely, $D \in \mathrm{Div}(X)$ is represented by a collection $\{(U_i, f_i)\}$, where the U_i cover X, $f_i \in \Gamma(U_i, \mathcal{K}_X^*)$, and on $U_{ij} = U_i \cap U_j$,

$$f_i = g_{ij} f_j \qquad \text{for some } g_{ij} \in \Gamma(U_{ij}, \mathcal{O}_X^*).$$

The function f_i is called a local equation for D at any point $x \in U_i$. Two such collections represent the same Cartier divisor if, on a common refinement V_k, their local equations satisfy

$$f_k = h_k f_k' \qquad \text{for some } h_k \in \Gamma(V_k, \mathcal{O}_X^*).$$

Although the local equations are multiplied, we use additive notation:

$$D = \{(U_i, f_i)\}, \quad D' = \{(U_i, f_i')\}, \quad D + D' = \{(U_i, f_i f_i')\}.$$

Here a common refinement has been chosen. The support of D is the set of points where a local equation is not a unit in $\mathcal{O}_{X,x}$. The divisor is effective if every f_i is regular, hence a nonzerodivisor, on U_i. We write $D \geq 0$, and $D \geq D'$ if $D - D' \geq 0$.

Remark 1.1.2. For a general scheme, $\mathcal{K}_X$ is the sheaf associated to

$$U \longmapsto S(U)^{-1}\Gamma(U, \mathcal{O}_X),$$

where $S(U)$ consists of sections whose germs are nonzerodivisors at every point of U. If X is locally Noetherian, then

$$\Gamma(U, \mathcal{K}_X) = A_{\mathrm{tot}} \quad (U = \operatorname{Spec} A \text{ affine}), \qquad \mathcal{K}_{X,x} = (\mathcal{O}_{X,x})_{\mathrm{tot}}.$$

In particular, on an integral scheme $\mathcal{K}_X$ is the sheaf of rational functions.

Definition 1.1.3 (Cycles)**.** Let X be a variety, or a scheme, pure of dimension n. A k-cycle on X is a finite $\mathbb{Z}$-linear combination of integral closed subvarieties of dimension k. The group of these cycles is denoted by $Z_k(X)$. A Weil divisor is an $(n-1)$-cycle; we write $\mathrm{WDiv}(X) = Z_{n-1}(X)$.

Remark 1.1.4. There is a cycle map

$$\mathrm{Div}(X) \longrightarrow \mathrm{WDiv}(X), \qquad D \longmapsto [D] = \sum_V \mathrm{ord}_V(D)[V].$$

Here V runs through the integral codimension-one subvarieties. If X is normal, then $\mathcal{O}_{X,\eta_V}$ is a discrete valuation ring and $\mathrm{ord}_V(D) = v_V(f_i)$ whenever $\eta_V \in U_i$. The compatibility of the local equations makes this independent of i. For a general pure-dimensional Noetherian scheme, the order is defined using lengths: write $f_i = a/b$ in $A = \mathcal{O}_{X,\eta_V}$ with a, b nonzerodivisors, and set

$$\mathrm{ord}_V(D) = \mathrm{length}_A(A/aA) - \mathrm{length}_A(A/bA).$$

One should not call this a discrete valuation without the normality assumption. If X is normal, the cycle map is injective; if X is nonsingular, or more generally locally factorial, it is an isomorphism. Unless otherwise specified, "divisor" will mean Cartier divisor.

Definition 1.1.5. A principal Cartier divisor is $\mathrm{div}(f)$ for $f \in \Gamma(X, \mathcal{K}_X^*)$. Two Cartier divisors are linearly equivalent, written $D_1 \sim_{\mathrm{lin}} D_2$, if

$$D_1 - D_2 = \mathrm{div}(f) \qquad \text{for some } f \in \Gamma(X, \mathcal{K}_X^*).$$

Remark 1.1.6 (Pullback of Cartier divisors)**.** Let D be a Cartier divisor on X and let $f : Y \to X$ be a morphism. One would like to pull back its local equations:

$$D = \{(U_i, f_i)\} \longmapsto f^*D = \{(f^{-1}U_i, f_i \circ f)\}.$$

A sufficient condition, for locally Noetherian schemes, is

$$f(\eta) \notin \mathrm{Supp}(D) \qquad \text{for every } \eta \in \mathrm{Ass}(Y).$$

Equivalently, no associated subvariety of Y is mapped into $\mathrm{Supp}(D)$. An associated subvariety may be embedded; it need not be an irreducible component. If Y is reduced, it is enough to check its irreducible components.

For an effective D, this is precisely the condition that the pulled-back local equations be nonzerodivisors. If an associated point maps into D, the corresponding local equation becomes a zerodivisor, so it does not define an effective Cartier divisor on Y. The problem is not merely whether the function vanishes identically on an open set.

Now we discuss the general construction, following [GW20, 11.16].

Inverse Image of Cartier Divisors

We can always pull back line bundles, but we cannot always pull back a specified Cartier divisor. In practice, one often chooses a linearly equivalent representative with suitable support. This is a form of the "moving lemma" philosophy; such a choice still requires justification.

Key Question

Let $f : X \to Y$ be a morphism of schemes. When does $\mathcal{O}_Y \to f_*\mathcal{O}_X$ extend to a homomorphism $\mathcal{K}_Y \to f_*\mathcal{K}_X$? Such an extension induces

$$\mathcal{K}_Y^*/\mathcal{O}_Y^* \longrightarrow f_*(\mathcal{K}_X^*/\mathcal{O}_X^*)$$

and hence a homomorphism

$$f^* : \mathrm{Div}(Y) \longrightarrow \mathrm{Div}(X).$$

Definition 1.1.7 (Pullback)**.** Suppose $\mathcal{O}_Y \to f_*\mathcal{O}_X$ extends to $\mathcal{K}_Y \to f_*\mathcal{K}_X$. For a Cartier divisor D on Y, its image under the induced map above is called the pullback f^*D.

Proposition 1.1.8. *The pullback has the following properties:*

(a) $f^*(D_1 + D_2) = f^*D_1 + f^*D_2$;

(b) $f^*\mathcal{O}_Y(D) \simeq \mathcal{O}_X(f^*D)$.

The pullback of line bundles gives $f^* : \mathrm{Pic}(Y) \to \mathrm{Pic}(X)$ *for every morphism* f*, even when not all Cartier divisors can be pulled back.*

Proof. We give a sketch. For (a), use $(f_1 f_2)\circ f = (f_1 \circ f)(f_2 \circ f)$. For (b), both line bundles are locally generated by the inverse of a pulled-back local equation. □

Theorem 1.1.9 (Pullback Condition)**.** *Let* $f : X \to Y$ *be a morphism of schemes. Assume one of the following:*

(a) f *is flat;*

(b) X *is reduced and locally Noetherian, and each irreducible component of* X *dominates an irreducible component of* Y*.*

Then $f^\flat : \mathcal{O}_Y \to f_*\mathcal{O}_X$ *extends to* $\mathcal{K}_Y \to f_*\mathcal{K}_X$*.*

Proof. Recall the presheaf in Remark 1.1.2. It is enough to show that a section which is a nonzerodivisor at every point remains so after pullback.

Case (a): flatness preserves the injectivity of multiplication by a nonzerodivisor.

Case (b): on a reduced locally Noetherian scheme, the zerodivisors are the union of the minimal primes. A generic point of an irreducible component of X maps to the generic point of an irreducible component of Y. A regular section on Y is a unit at such a generic point: its local ring has dimension zero, and a nonzerodivisor in a zero-dimensional local ring is a unit. Its pullback is therefore nonzero at every generic point of X, hence is a nonzerodivisor on X. □

Remark 1.1.10 (Divisor of zeros)**.** Let $s \in \Gamma(X, \mathcal{L})$ be a global section of a line bundle. Assume its local equations are nonzerodivisors. For locally Noetherian X, this means that s does not vanish at any associated point. If X is reduced, it is enough that s not vanish identically on any irreducible component. Then $(s)_0 = \operatorname{div}(s)$ is an effective Cartier divisor.

Line Bundles

A Cartier divisor $D = \{(U_i, f_i)\}$ determines the line bundle $\mathcal{O}_X(D) \subseteq \mathcal{K}_X$, with

$$\mathcal{O}_X(D)|_{U_i} = \mathcal{O}_{U_i} f_i^{-1}.$$

Thus, for an open subset $U \subseteq U_i$,

$$\Gamma(U, \mathcal{O}_X(D)) = \{h \in \Gamma(U, \mathcal{K}_X) \mid h f_i \in \Gamma(U, \mathcal{O}_X)\}.$$

The zero section is included. If X is integral, a nonzero rational function h belongs to this space precisely when $\operatorname{div}(h) + D|_U \geq 0$. If $f_i = g_{ij} f_j$, the local frames $e_i = f_i^{-1}$ satisfy $e_j = g_{ij} e_i$ on U_{ij}. This specifies the transition convention.

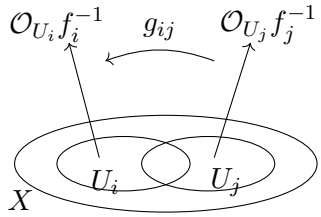


Figure 1.1: Line bundle and local frames

Moreover, the exact sequence of sheaves of abelian groups

$$1 \longrightarrow \mathcal{O}_X^* \longrightarrow \mathcal{K}_X^* \longrightarrow \mathcal{K}_X^*/\mathcal{O}_X^* \longrightarrow 1$$

induces the exact sequence

$$\Gamma(X, \mathcal{K}_X^*) \longrightarrow \operatorname{Div}(X) \longrightarrow H^1(X, \mathcal{O}_X^*) \simeq \operatorname{Pic}(X).$$

Consequently, $\mathcal{O}_X(D_1) \simeq \mathcal{O}_X(D_2)$ if and only if $D_1 \sim_{\text{lin}} D_2$. For a Weil divisor D on a normal variety, $\mathcal{O}_X(D)$ denotes the divisorial sheaf; it need not be a line bundle unless D is Cartier.

Example 1.1.11 (Divisor of a section of a line bundle)**.** Let X be an integral variety and $\mathcal{L}$ a line bundle. For a nonzero section $s \in H^0(X, \mathcal{L})$, choose local trivializations $\varphi_i : \mathcal{L}|_{U_i} \to \mathcal{O}_{U_i}$ and define

$$\operatorname{div}(s) = (s)_0 := \{(U_i, \varphi_i(s|_{U_i}))\}.$$

This is an effective Cartier divisor. If $\mathcal{L} = \mathcal{O}_X(D)$, the local trivialization

$$\mathcal{O}_X(D)|_{U_i} = \mathcal{O}_{U_i} f_i^{-1} \longrightarrow \mathcal{O}_{U_i}$$

is multiplication by f_i, so

$$\operatorname{div}(s) = \{(U_i, s|_{U_i} f_i)\}.$$

If D is effective, there is a canonical section $s_D = 1 \in H^0(X, \mathcal{O}_X(D))$, viewed as a rational function. Thus

$$\operatorname{div}(s_D) = \{(U_i, 1 \cdot f_i)\} = D,$$

which recovers the original Cartier divisor.

Now we recall some standard facts in algebraic geometry.

Canonical Bundle and Cartier Divisor

Let X be a smooth projective variety over $\mathbb{C}$ of dimension n. The canonical line bundle is

$$\omega_X := \bigwedge^n \Omega_X^1 = \Omega_X^n.$$

A corresponding Cartier divisor K_X, with $\mathcal{O}_X(K_X) \simeq \omega_X$, is called a canonical divisor. It is determined up to linear equivalence. The symbol K_X here is different from the sheaf $\mathcal{K}_X$ of rational functions.

Kodaira Vanishing Theorem

Let L be an ample Cartier divisor on a smooth projective variety X over a field of characteristic zero. Then

$$H^i(X, \mathcal{O}_X(K_X + L)) = 0 \qquad \text{for every } i > 0.$$

We will prove this in Chapter 4, using the covering trick.

1.1B Linear Series

Let X be an integral variety, $\mathcal{L}$ a line bundle on X, and $V \subseteq H^0(X, \mathcal{L})$ a nonzero finite-dimensional subspace. We set

$$|V| := \mathbb{P}_{\text{sub}}(V),$$

the projective space of one-dimensional subspaces of V. When X is complete, its points correspond to the divisors of nonzero sections of V, up to multiplication by a nonzero scalar. This is the linear series of Cartier divisors associated to V. For a complete variety, $H^0(X, \mathcal{L})$ is finite-dimensional [Har77, III, Remark 8.8.1], so $V = H^0(X, \mathcal{L})$ gives the complete linear series. For a Cartier divisor D, we write $|D|$ for the complete linear series of $\mathcal{O}_X(D)$.

Morphism to a Projective Space

Evaluation of sections in V gives an $\mathcal{O}_X$-linear morphism

$$\mathrm{eval}_V : V \otimes_{\mathbb{C}} \mathcal{O}_X \longrightarrow \mathcal{L}, \qquad \sum_i v_i \otimes f_i \longmapsto \sum_i f_i v_i|_U$$

on an open set $U \subseteq X$. Here the target is $\mathcal{L}(U)$. At a point x, the value lies in the one-dimensional $\kappa(x)$-vector space $\mathcal{L}(x) := \mathcal{L}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x)$, identified with $\kappa(x)$ only after choosing a local trivialization.

Definition 1.1.12 (Base ideal)**.** The base ideal of $|V|$ is

$$\mathfrak{b}(|V|) = \mathfrak{b}(X, |V|) \subseteq \mathcal{O}_X,$$

the image of the map $V \otimes_{\mathbb{C}} \mathcal{L}^\vee \to \mathcal{O}_X$ induced by eval_V and $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}^\vee \simeq \mathcal{O}_X$. Here $\mathcal{L}^\vee = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$, and locally $v \otimes \lambda \mapsto \lambda(v)$.

Definition 1.1.13 (Scheme-theoretic base locus)**.** The base scheme of $|V|$ is the closed subscheme defined by $\mathfrak{b}(|V|)$. Its underlying closed set is[1]

$$\mathrm{Bs}(|V|) := \mathrm{Supp}(\mathcal{O}_X/\mathfrak{b}(|V|)).$$

For a complete linear series, we also write $\mathfrak{b}(|\mathcal{L}|)$ or $\mathfrak{b}(|D|)$.

Remark 1.1.14.

$$\mathrm{Bs}(|V|) = \{x \in X \mid s(x) = 0 \text{ in } \mathcal{L}(x) \text{ for every } s \in V\}.$$

Example 1.1.15. Suppose X is a complete variety and D a Cartier divisor. For integers $m, \ell \geq 1$, there is an inclusion

$$\mathfrak{b}(|\ell D|)\, \mathfrak{b}(|mD|) \subseteq \mathfrak{b}(|(\ell + m)D|).$$

Indeed, multiplication of sections gives

$$\mu : H^0(X, \mathcal{O}_X(\ell D)) \otimes H^0(X, \mathcal{O}_X(mD)) \longrightarrow H^0(X, \mathcal{O}_X((\ell + m)D)), \quad s \otimes t \mapsto st.$$

Tensoring this map with $\mathcal{O}_X(-(\ell + m)D)$ and then evaluating gives the product of the two base ideals, which is contained in the image of the full evaluation map for $|(\ell + m)D|$.

[1] For finite-dimensionality of the complete linear series on a projective scheme, see [Har77, III, 5.2].

Definition 1.1.16 (Free linear series)**.** The linear series $|V|$ is free, or base-point-free, if $\mathrm{Bs}(|V|) = \varnothing$, equivalently $\mathfrak{b}(|V|) = \mathcal{O}_X$. A Cartier divisor or line bundle is free if its complete linear series is free. We also say the line bundle is generated by its global sections, or globally generated.

Proposition 1.1.17. *The linear series $|V|$ is free if and only if, for every point $x \in X$, there is a section $s \in V$ such that $s(x) \neq 0$.*

Proof. The condition $\mathfrak{b}(|V|) = \mathcal{O}_X$ is equivalent to the surjectivity of $V \otimes \mathcal{L}_x^\vee \to \mathcal{O}_{X,x}$ for every x. After trivializing $\mathcal{L}_x$, this says that the local coefficients of the sections generate the unit ideal. In a local ring this holds if and only if at least one coefficient is a unit, equivalently

$$s_x \notin \mathfrak{m}_x \mathcal{L}_x \quad \Longleftrightarrow \quad s(x) \neq 0 \text{ in } \mathcal{L}(x)$$

for some $s \in V$. The module $\mathfrak{m}_x \mathcal{L}_x$ is identified with $\mathfrak{m}_x$ only after a trivialization has been chosen. □

Assume now $\dim V \geq 2$, and set $B = \mathrm{Bs}(|V|)$. The linear series determines a morphism

$$\varphi = \varphi_{|V|} : X - B \longrightarrow \mathbb{P}(V).$$

Here, consistently with the quotient convention, $\mathbb{P}(V)$ parametrizes one-dimensional quotients of V; thus $\mathbb{P}(V) = \mathbb{P}_{\mathrm{sub}}(V^\vee)$. It is different from the projective space $|V| = \mathbb{P}_{\mathrm{sub}}(V)$. For a closed point $x \in X - B$, the corresponding quotient is

$$V \longrightarrow \mathcal{L}(x), \qquad s \longmapsto s(x),$$

whose kernel is the hyperplane of sections vanishing at x. At a base point this map is zero and does not define a quotient line.

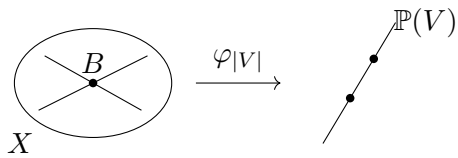


Figure 1.2: The formula for the morphism is undefined on the base locus

For a basis $s_0, \ldots, s_r$ of V, the homogeneous-coordinate expression is

$$\varphi : x \longmapsto [s_0(x) : \cdots : s_r(x)].$$

Since X is integral and $V \neq 0$, the open set $X - B$ is nonempty and dense. Thus $\varphi_{|V|}$ is a rational map $X \dashrightarrow \mathbb{P}(V)$. If $|V|$ is free, it is a morphism on all of X.

When $B = \varnothing$, the construction can be reversed [Har77, II, Remark 7.8.1]. Suppose $\varphi : X \to \mathbb{P} = \mathbb{P}(V)$ is a morphism whose image is not contained in a

hyperplane, with $V = H^0(\mathbb{P}, \mathcal{O}_{\mathbb{P}}(1))$. If $\varphi^* s = 0$ for some $s \in V$, the image lies in the hyperplane defined by s, so $s = 0$. Hence

$$\varphi^* : V \hookrightarrow H^0(X, \varphi^* \mathcal{O}_{\mathbb{P}}(1))$$

is injective. Its image generates $\varphi^* \mathcal{O}_{\mathbb{P}}(1)$, and the associated morphism is precisely φ. In particular, if H is a hyperplane which does not contain the image, this line bundle is $\mathcal{O}_X(\varphi^* H)$.

Example 1.1.18 (Projection)**.** Suppose $W \subseteq V$ has dimension at least two. Then[2]

$$\mathfrak{b}(|W|) \subseteq \mathfrak{b}(|V|), \qquad \mathrm{Bs}(|V|) \subseteq \mathrm{Bs}(|W|).$$

On $X - \mathrm{Bs}(|W|)$ one has

$$\varphi_{|W|} = \pi \circ \varphi_{|V|}, \qquad \pi : \mathbb{P}(V) - \mathbb{P}(V/W) \longrightarrow \mathbb{P}(W),$$

where π is linear projection with centre $\mathbb{P}(V/W)$.

If $|W|$ is free and X is complete, then $|V|$ is free as well. The image $Z = \varphi_{|V|}(X)$ is a closed projective subvariety of $\mathbb{P}(V)$ and avoids the centre. The restricted projection $\pi|_Z : Z \to \mathbb{P}(W)$ is proper: its source is proper and its target is separated. It is also affine, because the linear projection above is an affine morphism and Z is closed in its domain. Therefore $\pi|_Z$ is finite. Thus the two morphisms differ by a finite projection of $\varphi_{|V|}(X)$. Completeness makes the image closed; it does not assert that $\varphi_{|V|} : X \to \mathbb{P}(V)$ is a closed immersion.

1.1C Intersection Numbers via the Cohomological Approach

We follow [Deb01, Section 1.2]. In this subsection schemes are of finite type over a field k. When discussing varieties over an algebraically closed field, we use Hartshorne's convention.

For a projective subvariety $X \subseteq \mathbb{P}_k^N$ of dimension n, the function $m \mapsto \chi(X, \mathcal{O}_X(m))$ is a polynomial of degree n. Indeed, for $m \gg 0$, Serre vanishing gives $\chi(X, \mathcal{O}_X(m)) = h^0(X, \mathcal{O}_X(m))$, whose value is the Hilbert polynomial [Har77, I, 7.5; III, 5.2]. Its leading term is

$$P_X(m) = \frac{\deg X}{n!} m^n + O(m^{n-1}).$$

If H is a hyperplane section, $\deg X$ will be denoted by (H^n). To define $(D_1 \cdots D_n)$ for Cartier divisors on a proper scheme of dimension n, we use Euler characteristics, following [Kle66]; see also [KM98, 1.36], [Deb01, 1.5], [Laz04a, 1.1.24], [Iit82, Chapter 8] and [Ful98, 18.3.6].

[2]For closed images of proper morphisms, properness of the restricted projection, and proper affine morphisms being finite, respectively, see [Har77, II, Exercise 4.4], [Har77, II,ex4.8], [Har77, II, Exercise 4.6].

Theorem 1.1.19 (Asymptotic Riemann–Roch)**.** *Let $D_1, \dots, D_r$ be Cartier divisors on a proper scheme X and $\mathcal{F}$ a coherent sheaf on X. The function*[3]

$$(m_1, \dots, m_r) \longmapsto \chi(X, \mathcal{F}(m_1 D_1 + \cdots + m_r D_r))$$

agrees on $\mathbb{Z}^r$ with a polynomial with rational coefficients of total degree at most $\dim \operatorname{Supp} \mathcal{F}$*. In particular, for integral X of dimension $n \geq 1$,*

$$\chi(X, mD) := \chi(X, \mathcal{O}_X(mD)) = \frac{(D^n)}{n!} m^n + O(m^{n-1}),$$
$$\chi(X, \mathcal{F}(mD)) = \operatorname{rank}(\mathcal{F}) \frac{(D^n)}{n!} m^n + O(m^{n-1}).$$

Here (D^n) is the self-intersection number defined below, and $\operatorname{rank}(\mathcal{F}) = \dim_{k(X)} \mathcal{F}_\eta$ *is the length of the stalk at the generic point. In dimension zero the Euler characteristic is constant.*

Proof. We use induction on $\dim \operatorname{Supp} \mathcal{F}$. A coherent sheaf on a Noetherian scheme admits a finite filtration by coherent subsheaves whose successive quotients are torsion-free coherent sheaves on integral closed subschemes. Euler characteristic is additive in exact sequences, so it is enough to consider integral X and torsion-free $\mathcal{F}$.

Choose a nonempty open set U on which $\mathcal{F}$ is free of rank q. Torsion-freeness gives an embedding

$$\mathcal{F} \hookrightarrow \mathcal{K}_X^q,$$

which may be chosen to identify $\mathcal{F}|_U$ with $\mathcal{O}_U^q$. Set $\mathcal{G} = \mathcal{F} \cap \mathcal{O}_X^q$ inside $\mathcal{K}_X^q$. There are exact sequences

$$0 \to \mathcal{G} \to \mathcal{F} \to \mathcal{G}_1 \to 0, \qquad 0 \to \mathcal{G} \to \mathcal{O}_X^q \to \mathcal{G}_2 \to 0.$$

The coherent sheaves $\mathcal{G}_1$ and $\mathcal{G}_2$ vanish on U, so their supports have smaller dimension. Tensoring both sequences with $\mathcal{O}_X(\sum m_i D_i)$ and using induction reduces the polynomial assertion to $\mathcal{F} = \mathcal{O}_X$. The same calculation also shows that the top-degree part for $\mathcal{F}$ is q times the top-degree part for $\mathcal{O}_X$.

Step 1: First treat a single divisor D, with X integral. Set

$$I_1 = \mathcal{O}_X(-D) \cap \mathcal{O}_X, \qquad I_2 = \mathcal{O}_X(D) \cap \mathcal{O}_X$$

inside $\mathcal{K}_X$, and let Y_j be the subscheme defined by I_j. Both Y_j have smaller dimension than X. Since tensoring with a line bundle preserves intersections,

$$I_1(D) = \mathcal{O}_X \cap \mathcal{O}_X(D) = I_2.$$

[3]For the Hilbert polynomial, degree, and the finite-difference argument used here, see [Har77, III, 5.2(b)], [Har77, p. 52], [Har77, I, 7.3(b)].

Thus there are exact sequences

$$0 \to I_1(mD) \to \mathcal{O}_X(mD) \to \mathcal{O}_{Y_1}(mD) \to 0,$$
$$0 \to I_2((m-1)D) \to \mathcal{O}_X((m-1)D) \to \mathcal{O}_{Y_2}((m-1)D) \to 0.$$

Using $I_1(mD) = I_2((m-1)D)$ gives

$$\chi(X, mD) - \chi(X, (m-1)D) = \chi(Y_1, mD) - \chi(Y_2, (m-1)D).$$

By induction the right side is a rational polynomial of degree at most $\dim X - 1$. A function on $\mathbb{Z}$ whose first difference is such a polynomial agrees, for every integer, with a rational polynomial of degree at most $\dim X$: take a polynomial antidifference, and note that the difference between the two functions is constant on $\mathbb{Z}$. This also handles the zero polynomial. The dimension-zero case is immediate, since tensoring by a line bundle does not change lengths.

Step 2: For several divisors we use the following elementary lemma.

Lemma 1.1.20. *Let $d \geq 0$, and let $f : \mathbb{Z}^r \to \mathbb{Z}$ be a function which is a polynomial of degree at most d in each variable separately when the others are fixed. Then f agrees on $\mathbb{Z}^r$ with a polynomial in r variables with rational coefficients.*[4]

Proof. Use induction on r. The assertion for $r = 1$ is the hypothesis. For $r > 1$, write

$$f(m_1, \ldots, m_r) = \sum_{j=0}^{d} f_j(m_1, \ldots, m_{r-1}) m_r^j,$$

where the coefficient functions take values in $\mathbb{Q}$, not necessarily in $\mathbb{Z}$. Choose distinct integers $c_0, \ldots, c_d$. By induction,

$$\sum_{j=0}^{d} f_j(m_1, \ldots, m_{r-1}) c_i^j = P_i(m_1, \ldots, m_{r-1}) \qquad (0 \leq i \leq d)$$

for rational polynomials P_i. In matrix form,

$$\begin{pmatrix} 1 & c_0 & \cdots & c_0^d \\ 1 & c_1 & \cdots & c_1^d \\ \vdots & \vdots & & \vdots \\ 1 & c_d & \cdots & c_d^d \end{pmatrix} \begin{pmatrix} f_0 \\ f_1 \\ \vdots \\ f_d \end{pmatrix} = \begin{pmatrix} P_0 \\ P_1 \\ \vdots \\ P_d \end{pmatrix}.$$

The Vandermonde matrix is invertible over $\mathbb{Q}$, so every f_j is a rational linear combination of the P_i, hence a polynomial. □

[4] For the one-variable finite-difference criterion, see [Har77, I, 7.3(a)].

To apply the lemma, fix all but one of the m_i. The one-variable argument applies to the line bundle obtained by the fixed twists, so the degree in that variable is at most $\dim X$. Therefore

$$\chi(X, m_1D_1 + \cdots + m_rD_r) = P(m_1, \ldots, m_r)$$

for a rational polynomial P. If $P \neq 0$ has total degree d, choose integers $n_1, \ldots, n_r$ at which its top homogeneous part is nonzero. Then $P(n_1T, \ldots, n_rT)$ has degree d. Setting $D' = n_1D_1 + \cdots + n_rD_r$, Step 1 shows that this degree is at most $\dim X$. This proves the total-degree bound. The top-coefficient formulas now follow from Definition 1.1.21 and Remark 1.1.22, together with the reduction for a sheaf of rank q above. □

Definition 1.1.21. Let $D_1, \ldots, D_r$ be Cartier divisors on a proper scheme X, where $r \geq \dim X$. The intersection number $(D_1 \cdots D_r)$ is the coefficient of $m_1 \cdots m_r$ in the polynomial

$$\chi(X, m_1D_1 + \cdots + m_rD_r).$$

Proposition 1.1.25 shows that it is an integer. By Theorem 1.1.19, it vanishes when $r > \dim X$.

Remark 1.1.22. If $D_1 = \cdots = D_n = D$ and $\dim X = n$, write $\chi(X, mD) = a_n m^n + O(m^{n-1})$. The coefficient of $m_1 \cdots m_n$ in[5]

$$a_n(m_1 + \cdots + m_n)^n + \text{terms of total degree at most } n-1$$

is $n!a_n$. Hence $(D^n) = n!a_n$, giving the asymptotic Riemann–Roch formula. If X is projective and H is a hyperplane section, Serre vanishing for $\mathcal{O}_X(mH)$ with $m \gg 0$ gives

$$(H^n) = \deg X.$$

This is the degree of the embedded variety X, not the degree of a Cartier divisor viewed as a line bundle on a higher-dimensional variety.

Example 1.1.23. Let X be a smooth projective variety of dimension n, and suppose effective Cartier divisors $D_1, \ldots, D_n$ meet properly in finitely many points. Then

$$(D_1 \cdots D_n) = \sum_{x \in D_1 \cap \cdots \cap D_n} \operatorname{length}_{\mathcal{O}_{X,x}} \frac{\mathcal{O}_{X,x}}{(f_1, \ldots, f_n)} [\kappa(x) : k].$$

Over an algebraically closed field, this is the number of intersection points counted with multiplicity. For a transverse intersection all multiplicities are one. The same length formula holds whenever the local equations form a regular sequence, for example for a proper intersection on a Cohen–Macaulay pure-dimensional scheme. On a general scheme, merely counting the points or the length of their scheme-theoretic intersection is not sufficient without such a hypothesis.

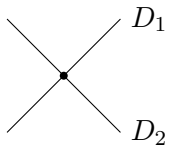


Figure 1.3: Transverse intersection

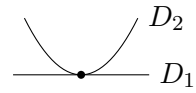


Figure 1.4: Nontransverse intersection

If $Y \subseteq X$ is a proper closed subscheme of dimension at most s, we set

$$(D_1 \cdots D_s \cdot Y) := (\mathcal{O}_X(D_1)|_Y \cdots \mathcal{O}_X(D_s)|_Y).$$

Here the restrictions are restrictions of line bundles; a Cartier divisor itself need not pull back to Y. The polynomial definition applies directly to line bundles. More generally, if $\dim \operatorname{Supp} \mathcal{F} \leq r$, define $(D_1 \cdots D_r \cdot \mathcal{F})$ as the coefficient of $m_1 \cdots m_r$ in $\chi(X, \mathcal{F}(\sum m_i D_i))$.

Example 1.1.24. For a smooth projective integral curve C over an algebraically closed field, Riemann–Roch gives[6]

$$\chi(C, mD) = m \deg(D) + 1 - g(C).$$

Thus $(D \cdot C)$ is the degree of the restricted line bundle. If $C \subseteq X$ is an integral, possibly singular, proper curve and $\tau : \widehat{C} \to C$ is its normalization, then

$$(D \cdot C) = \deg(\tau^*(\mathcal{O}_X(D)|_C)).$$

This agrees with the preceding polynomial definition: in $0 \to \mathcal{O}_C \to \tau_* \mathcal{O}_{\widehat{C}} \to Q \to 0$, the cokernel has zero-dimensional support, and twisting does not change its Euler characteristic. Hence the linear coefficients of the two Euler polynomials agree. Over a perfect field $\widehat{C}$ is smooth; over an arbitrary field it is regular, and degree is still defined by Euler characteristic. For a nonreduced or reducible curve, one must also retain the multiplicities of its one-dimensional components.

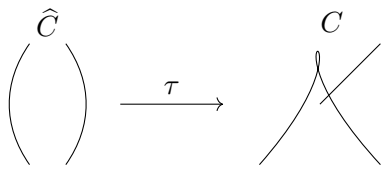


Figure 1.5: Normalization separates the branches

Proposition 1.1.25. *Let $D_1, \ldots, D_n$ be Cartier divisors on a proper scheme X of dimension n.*

[5]Integrality of the intersection number follows from Corollary 1.1.26.

[6]For Riemann–Roch on a smooth projective curve, see [Har77, IV, 1.3].

(a) $(D_1, \dots, D_n) \mapsto (D_1 \cdots D_n)$ *is multilinear, symmetric and integer-valued.*

(b) If D_n *is effective and* Y *is its associated closed subscheme, then* $(D_1 \cdots D_n) = (D_1 \cdots D_{n-1} \cdot Y)$.

Proof. For (a), symmetry follows from the definition. To prove integrality, use the following finite-difference identity. For a polynomial $P(T_1, \dots, T_r)$ of total degree at most r, the coefficient of $T_1 \cdots T_r$ is

$$\sum_{I \subseteq \{1,\dots,r\}} (-1)^{|I|} P(-m_I),$$

where m_I is the indicator vector of I. Indeed, the alternating sum kills every monomial missing a variable, and among monomials of total degree at most r only $T_1 \cdots T_r$ contains all the variables. Consequently,

$$(D_1 \cdots D_r) = \sum_{I \subseteq \{1,\dots,r\}} (-1)^{|I|} \chi\left(X, -\sum_{i \in I} D_i\right) \in \mathbb{Z}.$$

For multilinearity, the same alternating sum with $n+1$ divisors vanishes, since the Euler polynomial has degree at most n. Take the divisors $D_1, D_1', D_2, \dots, D_n$. Put $S_I = \sum_{i \in I} D_i$ and $\varepsilon_I = (-1)^{|I|}$ for $I \subseteq \{2, \dots, n\}$. Expanding according to the first two divisors gives

$$\begin{aligned} 0 = \sum_I \varepsilon_I \big(&\chi(X, -S_I) - \chi(X, -D_1 - S_I) - \chi(X, -D_1' - S_I) \\ &+ \chi(X, -D_1 - D_1' - S_I)\big). \end{aligned}$$

On the other hand,

$$\begin{aligned} ((D_1 + D_1') \cdot D_2 \cdots D_n) &= \sum_I \varepsilon_I \big(\chi(X, -S_I) - \chi(X, -D_1 - D_1' - S_I)\big), \\ (D_1 \cdot D_2 \cdots D_n) &= \sum_I \varepsilon_I \big(\chi(X, -S_I) - \chi(X, -D_1 - S_I)\big), \\ (D_1' \cdot D_2 \cdots D_n) &= \sum_I \varepsilon_I \big(\chi(X, -S_I) - \chi(X, -D_1' - S_I)\big). \end{aligned}$$

Subtracting the first expression from the sum of the last two gives zero by the previous identity. This proves additivity in the first variable, and symmetry gives multilinearity.

For (b), fix the index n in the alternating-sum formula. The exact sequence

$$0 \longrightarrow \mathcal{O}_X(-D_n) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_Y \longrightarrow 0$$

remains exact after twisting by $\mathcal{O}_X(-\sum_{i\in I} D_i)$. Thus

$$\begin{aligned}(D_1 \cdots D_n) &= \sum_{I\subseteq\{1,\dots,n-1\}} (-1)^{|I|}\left[\chi\left(X, -\sum_{i\in I} D_i\right) - \chi\left(X, -D_n - \sum_{i\in I} D_i\right)\right] \\ &= \sum_{I\subseteq\{1,\dots,n-1\}} (-1)^{|I|}\chi\left(Y, -\sum_{i\in I} D_i\right) = (D_1 \cdots D_{n-1} \cdot Y).\end{aligned}$$

□

Projection Formula

Let $\pi : X \to Y$ be a proper morphism between varieties and let $C \subseteq X$ be an integral proper curve. Define the pushforward cycle by

$$\pi_*[C] = \begin{cases} 0, & \pi(C) \text{ is a point,} \\ d[\pi(C)], & \pi(C) \text{ is a curve,} \end{cases}$$

where $d = [k(C) : k(\pi(C))]$ is the degree, including any inseparable degree.

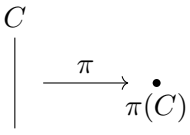


Figure 1.6: A contracted curve

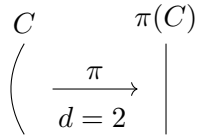


Figure 1.7: A map of degree two

Corollary 1.1.26 (Projection formula for curves)**.** *If D is a Cartier divisor on Y, then*

$$(\pi^*\mathcal{O}_Y(D) \cdot C) = (D \cdot \pi_*[C]).$$

We also write the left side as $(\pi^*D \cdot C)$, interpreting the pullback as a line bundle when the Cartier divisor itself cannot be pulled back.

Proof. If $\pi(C)$ is a point, the restricted pullback line bundle is trivial, so both sides are zero. Otherwise, pass to normalizations of the curves. The degree of the induced finite map remains d, and pullback multiplies line-bundle degree by d. Therefore

$$(\pi^*D \cdot C) = d\deg(\mathcal{O}_Y(D)|_{\pi(C)}) = (D \cdot d[\pi(C)]).$$

Here the degree on a singular curve is interpreted as in Example 1.1.24. □

Example 1.1.27 (Expansion of Har77, II, 6.9)**.** A finite dominant morphism $\pi : Y \to X$ between integral varieties induces a finite extension $\pi^* : k(X) \hookrightarrow k(Y)$. Its degree is $\deg \pi = [k(Y) : k(X)]$. The same definition applies to a dominant generically finite morphism.

The next result comes from [Kle66, p. 299, 6]; see also the proof in Appendix D.6.4 of the notes. For an integral n-dimensional variety Y, the cycle $\pi_*[Y]$ is zero if $\dim \pi(Y) < n$; otherwise it is $[k(Y) : k(\pi(Y))][\pi(Y)]$. Intersection with a cycle is extended linearly.

Proposition 1.1.28. *Let $\pi : Y \to X$ be a morphism between proper integral varieties and let $D_1, \ldots, D_r$ be Cartier divisors on X, where $r \geq \dim Y$. Then*

$$(\pi^* D_1 \cdots \pi^* D_r \cdot Y) = (D_1 \cdots D_r \cdot \pi_*[Y]).$$

The pullbacks on the left are interpreted as line bundles. If π is dominant and generically finite, so that $\dim Y = \dim X$, this becomes

$$(\pi^* D_1 \cdots \pi^* D_r) = \deg(\pi)(D_1 \cdots D_r).$$

In particular, for a birational morphism the intersection numbers agree.

Remark 1.1.29. In Lazarsfeld's notation, the projection formula for a dominant generically finite morphism between n-dimensional varieties reads

$$\int_Y \pi^* D_1 \cdots \pi^* D_n = \deg(\pi) \int_X D_1 \cdots D_n.$$

For line bundles one writes $\int_X c_1(L_1) \cdots c_1(L_n)$. On a general scheme, the first Chern class is an operator on Chow groups,

$$c_1(L) \cap - : A_k(X) \longrightarrow A_{k-1}(X).$$

It acts on cycle classes modulo rational equivalence; it is not a canonical operation producing a particular representative cycle. Applying n such operators to $[X]$ and taking the degree of the resulting zero-cycle gives the intersection number. See [EH16, 1.30] or [Ful98]. On a smooth projective complex variety, $c_1(L) \in H^2(X, \mathbb{Z})$ also gives the topological description by cup product and integration. Poincaré duality applies to the underlying compact complex manifold; see [GH78].

Numerical Equivalence

Let X be a proper scheme.

Definition 1.1.30. Two Cartier divisors D_1, D_2 are numerically equivalent, written $D_1 \equiv_{\text{num}} D_2$, if

$$(D_1 \cdot C) = (D_2 \cdot C) \qquad \text{for every integral curve } C \subseteq X.$$

A divisor is numerically trivial if $D \equiv_{\text{num}} 0$. Such divisors form a subgroup $\text{Num}(X) \subseteq \text{Div}(X)$.

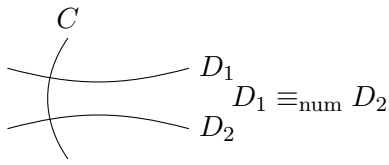


Figure 1.8: Equal intersection numbers with every curve

Definition 1.1.31 (Numerical divisor classes and the Picard number)**.** The group of numerical classes of Cartier divisors is[7]

$$N^1(X) := \operatorname{Div}(X)/\operatorname{Num}(X).$$

It is a free abelian group of finite rank, and its rank $\rho(X) := \operatorname{rank} N^1(X)$ is the Picard number. The usual Néron–Severi group $\operatorname{NS}(X)$ is instead the group of line bundles modulo algebraic equivalence. We keep $N^1(X)$ for numerical classes and do not identify these two groups without qualification. For a smooth projective complex variety, $N^1(X) \simeq \operatorname{NS}(X)/\operatorname{NS}(X)_{\text{tors}}$.

Remark 1.1.32. Equivalently,

$$\rho(X) = \dim_{\mathbb{R}} N^1(X)_{\mathbb{R}}, \qquad N^1(X)_{\mathbb{R}} := N^1(X) \otimes_{\mathbb{Z}} \mathbb{R}.$$

We will discuss this real vector space in the section on $\mathbb{R}$-Cartier divisors.

1.2 The Classical Theory

Question

Given a Cartier divisor D on a projective variety X, what should it mean for D to be positive?

Idea

Ask that $\mathcal{O}_X(D)$ be the hyperplane line bundle for a suitable projective embedding. In this case D is called very ample. If D is effective, it is then a hyperplane section for the embedding defined by the complete linear series. For a specified noneffective representative, it is its linear equivalence class that contains the hyperplane sections.

Good Idea

More conveniently, require some positive multiple of D to be very ample. In this case we say D is ample.

[7]For the Néron–Severi finiteness theorem and its relation to numerical equivalence, see [Har77, V, Exercise 1.7], [GH78, p. 461].

Definition 1.2.1 (Ampleness and very ampleness on a complete scheme)**.** Let X be a complete scheme and L a line bundle on X.

(i) L is very ample if there is a closed immersion $i : X \hookrightarrow \mathbb{P}^N$ with $L \simeq i^*\mathcal{O}_{\mathbb{P}^N}(1)$.

(ii) L is ample if $L^{\otimes m}$ is very ample for some integer $m > 0$.

A Cartier divisor D is ample or very ample if $\mathcal{O}_X(D)$ is so.

Example 1.2.2 (Ample line bundles on a curve)**.** For an integral complete curve X and a line bundle L on X,

$$L \text{ is ample} \quad \Longleftrightarrow \quad \deg L > 0.$$

For a nonsingular curve this is [Har77, IV, 3.3]; the criterion also holds for an integral singular curve, using degree as in Example 1.1.24.

Example 1.2.3 (Intersection products)**.** If $D_1, \ldots, D_n$ are ample Cartier divisors on an n-dimensional projective variety X, where $n \geq 1$, then $(D_1 \cdots D_n) > 0$. By taking a common positive multiple and using multilinearity, it is enough to treat very ample divisors. Choose general effective representatives successively, avoiding the associated subvarieties of the preceding intersections. They then meet properly, and their intersection through D_{n-1} has a nonzero effective one-cycle

$$[D_1 \cap \cdots \cap D_{n-1}] = \sum_C a_C[C], \qquad a_C > 0.$$

Consequently,

$$(D_1 \cdots D_n) = \sum_C a_C \deg(\mathcal{O}_X(D_n)|_C) > 0.$$

We use here the fact, proved below, that an ample line bundle restricts to an ample line bundle on a closed subvariety. For $n = 1$ the assertion is exactly the curve criterion. Arbitrary effective representatives need not meet properly.

1.2A Cohomological Properties

As in [Har77], we already know the following in the projective case. We now state it for a complete scheme, not assumed projective at the outset.

Theorem 1.2.4. *Let L be a line bundle on a complete scheme X. The following are equivalent:*[8]

(i) L is ample.

(ii) For every coherent sheaf $\mathcal{F}$ there is an integer $m_1 = m_1(\mathcal{F}, L) > 0$ such that

$$H^i(X, \mathcal{F} \otimes L^{\otimes m}) = 0 \qquad (i > 0,\ m \geq m_1).$$

[8]For the cohomological and global-generation criteria for ampleness, see [SP25, Tag 0B5T], [SP25, Tag 01Q3].

(iii) For every coherent sheaf $\mathcal{F}$ there is an integer $m_2 = m_2(\mathcal{F}, L) > 0$ such that $\mathcal{F} \otimes L^{\otimes m}$ is globally generated for every $m \geq m_2$.

(iv) There is an integer $m_3 = m_3(L) > 0$ such that $L^{\otimes m}$ is very ample for every $m \geq m_3$.

See also [SP25, Tags 0B5T and 01Q3] for the cohomological and generation criteria.

Example 1.2.5 (Sums of Cartier divisors)**.** Let D, E be Cartier divisors on a projective scheme X, with D ample. Choose m_1 so that $m_1 D$ is very ample, and m_2 so that $qD + E$ is free for every $q \geq m_2$. If $m \geq m_1 + m_2$, then

$$mD + E = m_1 D + ((m - m_1)D + E)$$

is the sum of a very ample divisor and a globally generated divisor, hence is very ample [Har77, II, Exercise 7.5(d)]. In particular it is ample. The bound depends on both D and E.

Example 1.2.6 (Ample line bundles on a product)**.** If L and M are ample line bundles on projective schemes X and Y, then $\mathrm{pr}_1^* L \otimes \mathrm{pr}_2^* M$ is ample on $X \times Y$.[9]

Proof. Choose $N > 0$ such that $L^{\otimes N}$ and $M^{\otimes N}$ are very ample, giving embeddings $i_1 : X \hookrightarrow \mathbb{P}^r$ and $i_2 : Y \hookrightarrow \mathbb{P}^s$. Their product followed by the Segre embedding is

$$i : X \times Y \hookrightarrow \mathbb{P}^r \times \mathbb{P}^s \hookrightarrow \mathbb{P}^{rs+r+s}.$$

By the formula for the Segre hyperplane line bundle,

$$(\mathrm{pr}_1^* L \otimes \mathrm{pr}_2^* M)^{\otimes N} = \mathrm{pr}_1^* L^{\otimes N} \otimes \mathrm{pr}_2^* M^{\otimes N} \simeq i^* \mathcal{O}_{\mathbb{P}^{rs+r+s}}(1).$$

Thus a positive power is very ample, which proves the assertion. □

Proposition 1.2.7 (Finite Pullbacks, I)**.** *Let $f : Y \to X$ be a finite morphism of complete schemes, and let L be an ample line bundle on X. Then f^*L is ample on Y. In particular, if $Y \subseteq X$ is a closed subscheme, then $L|_Y$ is ample.*

Remark 1.2.8. This is the finite-morphism version of the familiar fact that an ample line bundle restricts to an ample line bundle on a closed subscheme. The assertion can be checked by the cohomological criterion above.[10]

[9] For the twisting sheaves and the ampleness criterion used in the Segre argument, see [Har77, II, 5.12 and Exercise 5.11], [Har77, II, 7.6].

[10] For finite morphisms, coherence, the projection formula, and comparison of cohomology under an affine morphism, see [Har77, II, Exercise 5.7(a)+(d)], [Har77, II, Exercise 5.1(d)], [Har77, II, Exercise 5.5(c)], [Har77, III, Exercise 8.2].

Proof of Proposition 1.2.7. Let $\mathcal{F}$ be coherent on Y. Since f is finite, $f_*\mathcal{F}$ is coherent and higher direct images of quasi-coherent sheaves vanish. The projection formula therefore gives

$$\begin{aligned} H^i(Y, \mathcal{F} \otimes f^*L^{\otimes m}) &\simeq H^i(X, f_*(\mathcal{F} \otimes f^*L^{\otimes m})) \\ &\simeq H^i(X, f_*\mathcal{F} \otimes L^{\otimes m}) = 0 \qquad (i > 0,\ m \gg 0). \end{aligned}$$

Theorem 1.2.4 proves ampleness. A closed immersion is finite, which gives the restriction assertion. □

Corollary 1.2.9 (Globally generated line bundles)**.** *Let X be a complete scheme, let L be globally generated, and let*[11]

$$\varphi = \varphi_{|L|} : X \longrightarrow \mathbb{P}H^0(X, L)$$

be the resulting morphism. The following are equivalent:

(i) L is ample;

(ii) φ is finite;

(iii) $(L \cdot C) = \deg(L|_C) > 0$ for every integral curve $C \subseteq X$.

Proof. If φ is finite, then $L \simeq \varphi^*\mathcal{O}(1)$ is ample by Proposition 1.2.7. An ample L has positive degree on every integral curve by restriction and the curve criterion.

Conversely, φ is proper because X is complete and projective space is separated. If it is not finite, properness and the equivalence "proper and quasi-finite $\Longleftrightarrow$ finite" give a positive-dimensional fibre. A positive-dimensional proper variety contains an integral curve C, on which

$$L|_C \simeq (\varphi|_C)^*\mathcal{O}(1) \simeq \mathcal{O}_C.$$

Thus $(L \cdot C) = 0$, contradicting (iii). This proves all the equivalences. The same argument works for any generating linear series defining φ with $L \simeq \varphi^*\mathcal{O}(1)$. □

Proposition 1.2.10. *Let X be a complete scheme and L a line bundle on X.*[12]

(i) L is ample on X if and only if L_{red} is ample on X_{red}.

(ii) L is ample on X if and only if it is ample on each irreducible component with its reduced structure.

[11] For proper quasi-finite morphisms being finite, see [Har77, III, Exercise 11.2].

[12] For cohomology under a closed immersion, see [Har77, III, 2.10].

Proof. For (i), one direction follows from the finite closed immersion $X_{\mathrm{red}} \hookrightarrow X$. For the converse, let $\mathcal{N}$ be the nilradical, with $\mathcal{N}^r = 0$, and fix a coherent sheaf $\mathcal{F}$. The finite filtration

$$\mathcal{F} \supseteq \mathcal{N}\mathcal{F} \supseteq \cdots \supseteq \mathcal{N}^r\mathcal{F} = 0$$

has quotients

$$Q_i = \mathcal{N}^i\mathcal{F}/\mathcal{N}^{i+1}\mathcal{F} = \mathcal{N}^i\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_{X_{\mathrm{red}}},$$

which are coherent on X_{red}. Hence

$$H^j(X, Q_i \otimes L^{\otimes m}) = H^j(X_{\mathrm{red}}, Q_i \otimes L_{\mathrm{red}}^{\otimes m}) = 0 \qquad (j > 0,\ m \gg 0).$$

Choose a single bound for the finitely many Q_i. Twist $0 \to \mathcal{N}^{i+1}\mathcal{F} \to \mathcal{N}^i\mathcal{F} \to Q_i \to 0$ by $L^{\otimes m}$. Decreasing induction on i then gives vanishing for $\mathcal{F} \otimes L^{\otimes m}$, and Theorem 1.2.4 applies.

For (ii), by (i) assume X is reduced, and write $X = X_1 \cup \cdots \cup X_r$. Restriction gives the forward implication. For the converse, let I be the ideal of X_1 and J the ideal of $X' = X_2 \cup \cdots \cup X_r$, both with their reduced structures. Since X is reduced, $I \cap J = 0$, hence $IJ = 0$. In the sequence

$$0 \longrightarrow I\mathcal{F} \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}/I\mathcal{F} \longrightarrow 0,$$

the first sheaf is annihilated by J and is therefore a coherent sheaf on X', whereas the last is a coherent sheaf on X_1. Induction on the number of components gives the desired cohomology vanishing for both end terms after twisting by $L^{\otimes m}$. It then gives vanishing for the middle term, proving ampleness. The first sheaf need not vanish at points of $X_1 \cap X'$; its support is contained in X'. □

The next result can be found in [EGA III, 4.7.1] or [KM98, 1.41].

Theorem 1.2.11 (Ampleness in families)**.** *Let $f : X \to T$ be a proper morphism of Noetherian schemes and let L be a line bundle on X. For $t \in T$, write*[13]

$$X_t = X \times_T \operatorname{Spec} \kappa(t), \qquad L_t = L|_{X_t}.$$

If L_0 is ample on X_0 for a point $0 \in T$, then there is an open neighbourhood U of 0 such that L_t is ample on X_t for every $t \in U$. No flatness assumption on f is needed.

[13]For the higher direct images, projection formula, morphism to a projective bundle, and openness of the finite locus used in this argument, see [Har77, III, 8.5], [Har77, III, 2.10 and 5.3], [Har77, III, Exercise 8.3], [Har77, II, 7.12], [GW20, 12.89], [SP25, Tag 02OH].

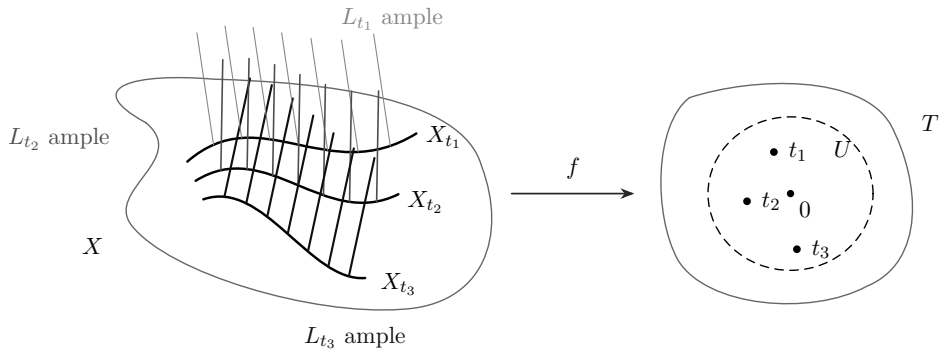


Figure 1.9: Ampleness holds on fibres over an open neighbourhood

Proof. We keep the three steps of the argument. First work over the local base $\operatorname{Spec} A = \operatorname{Spec} \mathcal{O}_{T,0}$, where $\mathfrak{m}$ is the maximal ideal and $A/\mathfrak{m} = \kappa(0)$. In Step 3 we extend the finitely many sections used to an actual open neighbourhood in the original T. This permits 0 to be any point, not necessarily a closed point of T.

Step 1: We claim that for every coherent sheaf $\mathcal{F}$ there is an integer $m(\mathcal{F})$ such that, over this local base,

$$R^i f_*(\mathcal{F} \otimes L^{\otimes m}) = 0 \qquad (i \geq 1,\ m \geq m(\mathcal{F})). \tag{1.1}$$

Use decreasing induction on the cohomological index, with the assertion at each index made for all coherent sheaves. It starts above a bound for the cohomological dimension of f. Choose generators $u_1, \ldots, u_p$ of $\mathfrak{m}$. Pulling back $A^p \xrightarrow{u} A \to \kappa(0) \to 0$ and tensoring with $\mathcal{F}$ gives

$$\mathcal{F}^p \xrightarrow{u} \mathcal{F} \longrightarrow \mathcal{F}_0 \longrightarrow 0, \qquad \mathcal{F}_0 = \mathcal{F} \otimes \mathcal{O}_{X_0}.$$

Let K be the kernel of the first map and I its image. Thus

$$0 \to K \to \mathcal{F}^p \to I \to 0, \qquad 0 \to I \to \mathcal{F} \to \mathcal{F}_0 \to 0.$$

Suppose the assertion is known at index $i \geq 2$. For $m \gg 0$, the induction hypothesis gives $R^i f_*(K \otimes L^{\otimes m}) = 0$, and ampleness of L_0 gives

$$H^{i-1}(X_0, \mathcal{F}_0 \otimes L_0^{\otimes m}) = 0.$$

The two long exact sequences yield a surjection

$$M_m^{\oplus p} \xrightarrow{(u_1,\ldots,u_p)} M_m, \qquad M_m = H^{i-1}(X, \mathcal{F} \otimes L^{\otimes m}).$$

This is a finite A-module by properness. The image is $\mathfrak{m}M_m$, so $M_m = \mathfrak{m}M_m$, and Nakayama's lemma gives $M_m = 0$. Since T is affine, this is the required higher-direct-image vanishing at index $i-1$. We only use the displayed map given by the u_j; no assertion that cohomology commutes with tensoring by $A/\mathfrak{m}$ is needed.

Step 2: We next show that, for $m \gg 0$, the evaluation morphism

$$\rho_m : f^* f_* L^{\otimes m} \longrightarrow L^{\otimes m}$$

is surjective. Apply Step 1 to the ideal sheaf $\mathcal{I}_{X_0/X}$. The sequence $0 \to \mathcal{I}_{X_0/X} \to \mathcal{O}_X \to \mathcal{O}_{X_0} \to 0$ then gives a surjection

$$H^0(X, L^{\otimes m}) \longrightarrow H^0(X_0, L_0^{\otimes m}) \qquad (m \gg 0). \tag{1.2}$$

All sufficiently large powers of L_0 are globally generated. Composing this restriction with evaluation on X_0, over the field $\kappa(0)$, shows that ρ_m is surjective along X_0, by Nakayama's lemma. Its cokernel is coherent. The image under the proper map f of the support of this cokernel is closed and misses the closed point of $\operatorname{Spec} A$. Every nonempty closed subset of the spectrum of a local ring contains its closed point, so this support is empty. Thus ρ_m is surjective on the local-base family.

Step 3: Fix one sufficiently large integer m. On the affine local base, the coherent module $f_* L^{\otimes m}$ is generated by finitely many sections $s_0, \ldots, s_r$. Composing their pullback with ρ_m gives a surjection $\mathcal{O}_X^{r+1} \to L^{\otimes m}$.

Return to an affine neighbourhood of 0 in the original T. Sections of a quasi-coherent sheaf on the quasi-compact separated scheme X commute with localization on this base. The finitely many s_i therefore extend after shrinking that neighbourhood once. The failure of these extended sections to generate has closed support, whose proper image misses 0; remove this image. The same sections now give a morphism over the resulting neighbourhood U,

$$\varphi : X_U \longrightarrow \mathbb{P}^r_U, \qquad L^{\otimes m}|_{X_U} \simeq \varphi^* \mathcal{O}_{\mathbb{P}^r_U}(1).$$

The integer m remains fixed throughout these shrinkings.

On X_0, the morphism φ_0 is finite. Indeed, a positive-dimensional fibre would contain a curve C, and the pullback formula would give $m \deg(L_0|_C) = 0$, contrary to ampleness of L_0. The locus in X_U where φ is not quasi-finite is closed and disjoint from X_0. Its image under the proper map f is closed; remove that image from U. Then φ is quasi-finite and proper, hence finite. For each $t \in U$, Proposition 1.2.7 now shows that $L_t^{\otimes m} \simeq \varphi_t^* \mathcal{O}_{\mathbb{P}^r_{\kappa(t)}}(1)$ is ample, and therefore L_t is ample. □

Example 1.2.12 (Asymptotic Riemann–Roch in the ample case)**.** Let D be ample on an integral projective variety X of dimension $n \geq 1$. Then

$$h^0(X, \mathcal{O}_X(mD)) = \frac{(D^n)}{n!} m^n + O(m^{n-1}).$$

More generally, for a coherent sheaf $\mathcal{F}$,

$$h^0(X, \mathcal{F} \otimes \mathcal{O}_X(mD)) = \operatorname{rank}(\mathcal{F}) \frac{(D^n)}{n!} m^n + O(m^{n-1}).$$

This follows immediately from Theorem 1.1.19 and Serre vanishing. The zero-dimensional case is constant.

Example 1.2.13 (Upper bounds on h^0)**.** If E is any Cartier divisor on an integral projective variety X of dimension n, then there is a constant $C > 0$ such that[14]

$$h^0(X, \mathcal{O}_X(mE)) \leq Cm^n \qquad (m \gg 0).$$

Proof. Fix an ample divisor D. For a fixed sufficiently large a, $\mathcal{O}_X(aD - E)$ has a nonzero global section s, so $aD - E$ is linearly equivalent to an effective divisor. Since X is integral, multiplication by s^m gives an injection

$$\mathcal{O}_X(mE) \hookrightarrow \mathcal{O}_X(maD), \qquad t \longmapsto s^m t.$$

Taking global sections and using the preceding example gives

$$h^0(X, mE) \leq h^0(X, maD) = \frac{(aD)^n}{n!} m^n + O(m^{n-1}) \leq Cm^n.$$

For $n = 0$ the bound is immediate from finite length. □

Example 1.2.14 (Resolutions of a sheaf)**.** Let D be ample on a projective variety X. Any coherent sheaf $\mathcal{F}$ admits a resolution[15]

$$\cdots \longrightarrow \mathcal{O}_X(-p_1 D)^{\oplus a_1} \longrightarrow \mathcal{O}_X(-p_0 D)^{\oplus a_0} \longrightarrow \mathcal{F} \longrightarrow 0$$

for suitable integers $0 \ll p_0 \ll p_1 \ll \cdots$ and finite ranks a_i. The resolution need not terminate.

Proof. Choose $p_0 \gg 0$ such that $\mathcal{F}(p_0 D)$ is globally generated, and choose finitely many generating sections. After twisting back, this gives a surjection $\mathcal{O}_X(-p_0 D)^{\oplus a_0} \to \mathcal{F}$. Its kernel is coherent, so the same construction applies to the kernel with a sufficiently large p_1. Repeat this process. □

1.2B Numerical Properties

The ampleness of Cartier divisors can be characterized numerically.

Theorem 1.2.15 (Nakai–Moishezon–Kleiman criterion)**.** *Let L be a line bundle on a projective scheme X. Then L is ample if and only if*

$$\int_V c_1(L)^{\dim V} = (L^{\dim V} \cdot V) > 0$$

for every positive-dimensional integral closed subvariety $V \subseteq X$. In particular, a Cartier divisor D is ample if and only if $(D^{\dim V} \cdot V) > 0$ for all such V.

[14] For regular sections and effective Cartier divisors, see [SP25, Tag 01X0].

[15] Compare the construction of resolutions by twists in [Har77, II, 5.18].

Before giving a proof, we derive two important consequences. First, ampleness depends only on the numerical equivalence class.

Corollary 1.2.16 (Numerical nature of ampleness)**.** *If D_1, D_2 are numerically equivalent Cartier divisors on a projective scheme X, then D_1 is ample if and only if D_2 is ample.*

We need a lemma to prove this corollary.

Lemma 1.2.17. *Let X be a complete scheme, and suppose $D_i \equiv_{\text{num}} D_i'$ for Cartier divisors $D_1, \ldots, D_k, D_1', \ldots, D_k'$. Then*

$$(D_1 \cdots D_k \cdot V) = (D_1' \cdots D_k' \cdot V)$$

for every integral closed subvariety $V \subseteq X$ of dimension k.

Proof. For $E \equiv_{\text{num}} 0$, the class $c_1(\mathcal{O}_X(D_2)) \cdots c_1(\mathcal{O}_X(D_k)) \cap [V]$ in $A_1(X)$ is represented by a one-cycle γ. Thus

$$(E \cdot D_2 \cdots D_k \cdot V) = (E \cdot \gamma) = 0.$$

Apply this to $E = D_1 - D_1'$, then replace the other divisors one at a time. Multilinearity proves the lemma. We use Chow groups and Chern class operators here; X need not have a Chow ring as a smooth variety does. □

Proof of Corollary 1.2.16. The lemma gives $(D_1^{\dim V} \cdot V) = (D_2^{\dim V} \cdot V)$ for every V. Theorem 1.2.15 therefore applies to the two divisors simultaneously. □

Definition 1.2.18 (Ample classes)**.** A numerical class $\delta \in N^1(X)$ is ample if it is the class of an ample Cartier divisor.

Example 1.2.19 (Varieties with Picard number one)**.** If X is a projective variety with $\rho(X) = 1$, then any nonzero effective Cartier divisor D is ample. Indeed, fix an ample Cartier divisor H. The positivity $(D \cdot H^{n-1}) > 0$ shows that $[D]$ and $[H]$ lie on the same half-line in $N^1(X)_{\mathbb{R}}$. Since these are integral classes in a rank-one group, there are positive integers a, b with $aD \equiv_{\text{num}} bH$. By Corollary 1.2.16, aD is ample, and so is D.

We next discuss the converse to finite pullback. In the implication below, the assumption that f^*L is ample already forces f to be finite. Thus finiteness need not be imposed separately.

Corollary 1.2.20 (Finite Pullbacks, II)**.** *Let $f : Y \to X$ be a surjective morphism of projective schemes, and let L be a line bundle on X. Then*[16]

$$f^*L \text{ ample on } Y \quad \Longrightarrow \quad L \text{ ample on } X.$$

[16] For the finite multisection argument, positivity of degree, and representation of line bundles by Cartier divisors, see [GW20, 12.80], [KM98, 1.35(3)], [Har77, II, 6.15]. For the decomposition into a difference of very ample divisors used in the proof, see also Lemma 4.1.13.

Proof. A positive-dimensional fibre would contain an integral curve C, on which f^*L is trivial. This contradicts ampleness. Hence f is quasi-finite, and since it is proper it is finite.

Let $V \subseteq X$ be a positive-dimensional integral subvariety. There is an irreducible component W of $f^{-1}(V)$ which dominates V, and we give W its reduced structure. The restricted map $f|_W : W \to V$ is finite and surjective, so $\dim W = \dim V$ and

$$0 < \int_W c_1(f^*L)^{\dim V} = \deg(W \to V) \int_V c_1(L)^{\dim V}.$$

Theorem 1.2.15 proves the result. No hyperplane cuts are needed here. For a general projective surjective map, hyperplane cuts can produce a subvariety generically finite over V; they do not in general guarantee finiteness over every point of V. □

Now we return to the proof of the numerical criterion.

Proof of Theorem 1.2.15. The forward implication follows by choosing $m > 0$ such that $L^{\otimes m}$ is very ample. For the resulting embedding and every V,

$$m^{\dim V} \int_V c_1(L)^{\dim V} = \deg(V \text{ in this embedding}) > 0.$$

For the converse, use Proposition 1.2.10 to reduce to an integral projective variety X, and write $L \simeq \mathcal{O}_X(D)$. We induct on $n = \dim X$. In dimension zero every line bundle is ample. In dimension one the assertion is the curve criterion. Assume the result in dimensions less than n.

Step 1: We claim that $H^0(X, \mathcal{O}_X(mD)) \neq 0$ for $m \gg 0$. By asymptotic Riemann–Roch,

$$\chi(X, mD) = \frac{(D^n)}{n!} m^n + O(m^{n-1}), \qquad (D^n) > 0.$$

Fix a very ample H, and then fix an integer $a \gg 0$ such that $aH + D$ and aH are both very ample. Choose effective representatives $A \sim_{\text{lin}} aH + D$ and $B \sim_{\text{lin}} aH$, so $D \sim_{\text{lin}} A - B$. These divisors are fixed while m tends to infinity. Their defining sections give exact sequences

$$0 \to \mathcal{O}_X(mD - B) \xrightarrow{\cdot A} \mathcal{O}_X((m+1)D) \to \mathcal{O}_A((m+1)D) \to 0,$$
$$0 \to \mathcal{O}_X(mD - B) \xrightarrow{\cdot B} \mathcal{O}_X(mD) \to \mathcal{O}_B(mD) \to 0.$$

All integral subvarieties of A and B satisfy the required positivity. They have smaller dimension, so induction makes $\mathcal{O}_A(D)$ and $\mathcal{O}_B(D)$ ample. The higher cohomology of the right-hand terms vanishes for $m \gg 0$. Thus, for $i \geq 2$,

$$H^i(X, mD) \simeq H^i(X, mD - B) \simeq H^i(X, (m+1)D).$$

The dimensions $h^i(X, mD)$ are eventually constant for $i \geq 2$. Hence

$$\chi(X, mD) = h^0(X, mD) - h^1(X, mD) + c$$

for some constant c, possibly zero. Since $\chi(X, mD) \to +\infty$, we have $h^0(X, mD) \to +\infty$. This proves the claim. Choose a nonzero section of one such positive multiple and replace D by its effective zero divisor. This is a replacement by a linearly equivalent representative of a positive multiple, and does not change the ampleness question.

Step 2: We show that $\mathcal{O}_X(mD)$ is globally generated for $m \gg 0$. Outside $\operatorname{Supp} D$, the canonical section s_D^m does not vanish. It remains to lift enough sections from the subscheme D. Use

$$0 \to \mathcal{O}_X((m-1)D) \to \mathcal{O}_X(mD) \to \mathcal{O}_D(mD) \to 0.$$

Induction makes $\mathcal{O}_D(D)$ ample. Thus for $m \gg 0$, $\mathcal{O}_D(mD)$ is globally generated and $H^1(D, \mathcal{O}_D(mD)) = 0$. The long exact sequence gives a surjection

$$H^1(X, (m-1)D) \longrightarrow H^1(X, mD).$$

The nonnegative integers $h^1(X, mD)$ are eventually nonincreasing, hence eventually constant. The displayed surjections are then isomorphisms, so exactness implies that

$$H^0(X, mD) \longrightarrow H^0(D, \mathcal{O}_D(mD))$$

is surjective. Therefore there are no base points on D, and the canonical section takes care of the complement. This proves global generation. We did not assume $H^1(X, mD) = 0$.

Step 3: For every integral curve $C \subseteq X$, the hypothesis gives $(mD \cdot C) > 0$. Apply Corollary 1.2.9 to the globally generated line bundle $\mathcal{O}_X(mD)$: it is ample, and hence $\mathcal{O}_X(D)$ is ample. □

Remark 1.2.21 (Nakai's criterion on proper schemes)**.** The statement remains true for a complete scheme not assumed projective. Projectivity was used above to write D as the difference of two very ample divisors. The following argument replaces that step by the two ideal sheaves already used in the proof of Theorem 1.1.19. See [Deb01, Section 1.2] and [KM98, 1.37].

Theorem 1.2.22 (Nakai–Moishezon criterion)**.** *A Cartier divisor D on a proper scheme X is ample if and only if*[17]

$$(D^{\dim Y} \cdot Y) > 0$$

for every positive-dimensional integral closed subscheme $Y \subseteq X$. One may include zero-dimensional integral subschemes as well, since their degrees over the ground field are automatically positive.

[17] For the correspondence between regular sections and effective Cartier divisors, see [SP25, Tag 01X0].

Proof. If D is ample, a positive multiple defines a projective embedding and

$$m^{\dim Y}(D^{\dim Y} \cdot Y) = \deg(Y \text{ in this embedding}) > 0.$$

Conversely, by Proposition 1.2.10 it is enough to treat integral X. Induct on $n = \dim X$; the zero-dimensional case is immediate. Choose the usual inclusion $\mathcal{O}_X(D) \subseteq \mathcal{K}_X$ and set

$$I_1 = \mathcal{O}_X(-D) \cap \mathcal{O}_X, \qquad I_2 = \mathcal{O}_X(D) \cap \mathcal{O}_X.$$

The subschemes Y_j defined by I_j have smaller dimension, and $I_1(D) = I_2$. By induction, $\mathcal{O}_X(D)|_{Y_j}$ is ample. In the exact sequences

$$0 \to I_1(mD) \to \mathcal{O}_X(mD) \to \mathcal{O}_{Y_1}(mD) \to 0,$$
$$0 \to I_2((m-1)D) \to \mathcal{O}_X((m-1)D) \to \mathcal{O}_{Y_2}((m-1)D) \to 0,$$

the right-hand terms have vanishing higher cohomology for $m \gg 0$. Consequently, for $i \geq 2$,

$$h^i(X, mD) = h^i(X, I_1(mD)) = h^i(X, I_2((m-1)D)) = h^i(X, (m-1)D).$$

These dimensions are eventually constant. Since $(D^n) > 0$ and $n > 0$, asymptotic Riemann–Roch shows that $\chi(X, mD) \to +\infty$. It follows that $h^0(X, mD) - h^1(X, mD)$, and therefore $h^0(X, mD)$, tends to infinity. A nonzero section makes some positive multiple of D linearly equivalent to an effective divisor. Replace D by this divisor.

By induction $\mathcal{O}_D(D)$ is ample. The sequence

$$0 \to \mathcal{O}_X((m-1)D) \to \mathcal{O}_X(mD) \to \mathcal{O}_D(mD) \to 0$$

gives, for $m \gg 0$, surjections $H^1(X, (m-1)D) \to H^1(X, mD)$. Their dimensions eventually stabilize, so $H^0(X, mD) \to H^0(D, \mathcal{O}_D(mD))$ is surjective. The right-hand line bundle is globally generated, while s_D^m is nonvanishing outside $\operatorname{Supp} D$. Thus $\mathcal{O}_X(mD)$ is globally generated on all of X.

It defines a proper morphism $\varphi : X \to \mathbb{P}^r$. A positive-dimensional fibre would contain a curve C and give $\mathcal{O}_X(mD)|_C \simeq \mathcal{O}_C$, contrary to $(D \cdot C) > 0$. Thus φ is finite, and $\mathcal{O}_X(mD) \simeq \varphi^*\mathcal{O}(1)$ is ample by Proposition 1.2.7. Therefore D is ample. □

Example 1.2.23 (Further characterization of ampleness)**.** Let X be a projective scheme and D a Cartier divisor on X. Then D is ample if and only if for every positive-dimensional integral subvariety $V \subseteq X$, there are an integer $m = m(V) > 0$ and a nonzero section $s_V \in H^0(V, \mathcal{O}_V(mD))$ which vanishes at some point of V. This characterization is due to Kleiman [Kle66].[18]

[18] For the effective divisor associated to a nonzero section on an integral scheme, see [SP25, Tag 01X0].

Proof. If D is ample, a sufficiently large multiple is very ample on V, and a hyperplane through a point of V but not containing V supplies the required section.

For the converse, Proposition 1.2.10 reduces to integral X. In dimension zero every line bundle is ample. On an integral curve, the zero divisor of s_V is nonempty and effective, so $m \deg(D|_V) > 0$; the curve criterion applies. Now induct on dimension. Use $V = X$ and replace D by the nonzero effective divisor of s_X, which is linearly equivalent to a positive multiple of the original divisor. The condition is preserved by this replacement, since powers of a section remain nonzero and still vanish at the same point.

Every integral subvariety of the subscheme D satisfies the hypothesis, so induction and Proposition 1.2.10 make $\mathcal{O}_D(D)$ ample. The same exact-sequence and stabilization argument as in Theorem 1.2.15, Step 2, makes $\mathcal{O}_X(mD)$ globally generated for $m \gg 0$. The curve case already gives $(D \cdot C) > 0$ for every integral curve $C \subseteq X$. Corollary 1.2.9 proves ampleness. □

Example 1.2.24 (Growth of cohomology)**.** Let X be a projective scheme of dimension n and D a Cartier divisor. For every coherent sheaf $\mathcal{F}$ and every $i \geq 0$,

$$h^i(X, \mathcal{F}(mD)) = O(m^n) \qquad (m \to +\infty).$$

Proof. Write $D \sim_{\text{lin}} A - B$ with A, B very ample effective divisors chosen so that neither contains an associated subvariety of $\mathcal{F}$. Such choices exist over the infinite ground field. The corresponding multiplication maps are injective, giving

$$0 \to \mathcal{F}(mD - B) \xrightarrow{\cdot A} \mathcal{F}((m+1)D) \to \mathcal{F}|_A((m+1)D) \to 0,$$
$$0 \to \mathcal{F}(mD - B) \xrightarrow{\cdot B} \mathcal{F}(mD) \to \mathcal{F}|_B(mD) \to 0.$$

Induct on dimension. For $n = 0$, higher cohomology vanishes and h^0 is bounded by the fixed length of $\mathcal{F}$. For $n > 0$, both restricted sheaves have support of dimension at most $n - 1$. Put $b_m = h^i(X, \mathcal{F}(mD - B))$. The two long exact sequences give

$$\begin{aligned} -h^{i-1}(A, \mathcal{F}|_A((m+1)D)) &\leq h^i(X, \mathcal{F}((m+1)D)) - b_m \\ &\leq h^i(A, \mathcal{F}|_A((m+1)D)), \\ -h^{i-1}(B, \mathcal{F}|_B(mD)) &\leq h^i(X, \mathcal{F}(mD)) - b_m \\ &\leq h^i(B, \mathcal{F}|_B(mD)), \end{aligned}$$

where $h^{-1} = 0$ when $i = 0$. Induction bounds the outer terms by $O(m^{n-1})$. Subtracting the two middle expressions therefore gives

$$\left| h^i(X, \mathcal{F}((m+1)D)) - h^i(X, \mathcal{F}(mD)) \right| = O(m^{n-1}).$$

Summing this estimate from a fixed starting integer to m gives $h^i(X, \mathcal{F}(mD)) = O(m^n)$. □

1.3 $\mathbb{Q}$-Cartier Divisors and $\mathbb{R}$-Cartier Divisors

Goal

Make a small perturbation of a Cartier divisor class and see what happens.

1.3A Definition of $\mathbb{Q}$-Cartier Divisors

A $\mathbb{Q}$-Cartier divisor is a rational linear combination of integral Cartier divisors.

Definition 1.3.1 ($\mathbb{Q}$-Cartier divisors)**.** For a variety or scheme X, set

$$\mathrm{Div}_{\mathbb{Q}}(X) := \mathrm{Div}(X) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

A $\mathbb{Q}$-Cartier divisor is an element of this vector space, represented by a finite sum $D = \sum_i c_i A_i$, with $c_i \in \mathbb{Q}$ and $A_i \in \mathrm{Div}(X)$. Clearing denominators gives $D = A/r$ for an integer $r > 0$ and an integral Cartier divisor A. The map $\mathrm{Div}(X) \to \mathrm{Div}_{\mathbb{Q}}(X)$ kills torsion: for $c \in \mathbb{Q} \setminus \{0\}$, the element cA is zero if and only if A is torsion in $\mathrm{Div}(X)$. The divisor D is effective if it admits an expression $D = \sum_i c_i A_i$ with $c_i \geq 0$ and all A_i effective integral Cartier divisors.

Definition 1.3.2 (Supports)**.** A closed subset $E \subseteq X$ supports $D \in \mathrm{Div}_{\mathbb{Q}}(X)$ if there is an expression $D = \sum_i c_i A_i$ such that

$$\bigcup_i \mathrm{Supp}\, A_i \subseteq E.$$

Such an expression, and such a supporting set, need not be unique. On a normal variety, the associated Weil $\mathbb{Q}$-divisor has a canonical support, obtained from its nonzero coefficients along prime divisors. On a general scheme we will specify a supporting set when it is needed.

The usual operations extend to $\mathbb{Q}$-Cartier divisors.

Definition 1.3.3 (Equivalences and operations on $\mathbb{Q}$-Cartier divisors)**.** Assume X is complete.

(i) For a closed subscheme $V \subseteq X$ of dimension k, the intersection product

$$\mathrm{Div}_{\mathbb{Q}}(X)^k \longrightarrow \mathbb{Q}, \qquad (D_1, \ldots, D_k) \longmapsto (D_1 \cdots D_k \cdot V)$$

is defined by extension of scalars from the integral intersection product.

(ii) Divisors D_1, D_2 are numerically equivalent, written $D_1 \equiv_{\mathrm{num}} D_2$ or $D_1 \equiv_{\mathrm{num},\mathbb{Q}} D_2$, if their intersection numbers with every integral curve agree. The vector space of numerical classes is denoted by $N^1(X)_{\mathbb{Q}}$.

(iii) Write $D_1 \sim_{\mathbb{Q}} D_2$ if a positive integral multiple of $D_1 - D_2$ is the image of a principal Cartier divisor in $\mathrm{Div}_{\mathbb{Q}}(X)$. Equivalently, after a common positive multiple, integral representatives can be chosen which are linearly equivalent in the usual sense. This is $\mathbb{Q}$-linear equivalence; we reserve $\sim_{\mathrm{lin}}$ for ordinary linear equivalence of integral Cartier divisors when a distinction matters.

(iv) If $f : Y \to X$ is a morphism and no associated subvariety of Y maps into a chosen supporting set for D, the local equations can be pulled back and extended $\mathbb{Q}$-linearly to define f^*D. This is the condition of Remark 1.1.6, applied to the chosen representatives.

(v) For an arbitrary morphism of complete varieties, or of projective schemes, pullback of line bundles induces a functorial map

$$f^* : N^1(X)_{\mathbb{Q}} \longrightarrow N^1(Y)_{\mathbb{Q}}.$$

It agrees with divisor pullback whenever the latter is defined, because the projection formula shows that numerical equivalence is preserved. Here line bundles on the source have Cartier representatives. For arbitrary proper schemes without this property, the same functorial statement is formulated using numerical classes of line bundles; one should not infer it solely by extending an everywhere-defined Cartier-divisor pullback, which may not exist.

Remark 1.3.4. The operations are computed from $D = \sum_i c_i A_i$ by linearity. In particular,

$$D \equiv_{\mathrm{num}} 0 \quad \Longleftrightarrow \quad \sum_i c_i (A_i \cdot C) = 0 \text{ for every curve } C.$$

There is an isomorphism

$$N^1(X)_{\mathbb{Q}} \simeq N^1(X) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

Indeed, write a numerically trivial rational divisor as A/r with $r > 0$. Then A is numerically trivial. Thus its kernel is exactly the rational span of numerically trivial integral Cartier divisors.

Remark 1.3.5. Two integral Cartier divisors in distinct ordinary linear equivalence classes can become $\mathbb{Q}$-linearly equivalent. Their difference then has torsion class in $\mathrm{Pic}(X)$. Numerical equivalence of integral divisors does not change after extension to $\mathbb{Q}$.

Continue to assume X is complete, as required in Nakai's criterion.

Definition 1.3.6 (Ampleness for $\mathbb{Q}$-Cartier divisors)**.** A divisor $D \in \mathrm{Div}_{\mathbb{Q}}(X)$ is ample if the following equivalent conditions hold:

(i) $D = \sum_i c_i A_i$, with $c_i \in \mathbb{Q}_{>0}$ and A_i ample integral Cartier divisors;

(ii) some positive integral multiple rD has an ample integral representative;

(iii) $(D^{\dim V} \cdot V) > 0$ for every positive-dimensional integral closed subvariety $V \subseteq X$.

The equivalence follows by clearing denominators and applying Nakai's criterion. Thus ampleness is preserved under numerical equivalence, and we can speak of ample classes in $N^1(X)_{\mathbb{Q}}$.

We now prove that ampleness is open under small perturbations.

Proposition 1.3.7. *Let X be a projective variety, let H be an ample $\mathbb{Q}$-Cartier divisor, and let $E_1, \dots, E_r$ be $\mathbb{Q}$-Cartier divisors. Then*

$$H + \varepsilon_1 E_1 + \cdots + \varepsilon_r E_r$$

is ample for all sufficiently small rational ε_i, of either sign. In particular this holds for a single divisor E.

Proof. Clear a common denominator to assume that $H, E_1, \dots, E_r$ are integral. By Example 1.2.5, choose a fixed $m > 0$ such that all $mH \pm E_i$ are ample. If $m \sum_i |\varepsilon_i| < 1$, then

$$H + \sum_i \varepsilon_i E_i = \left(1 - m \sum_i |\varepsilon_i|\right) H + \sum_{\varepsilon_i \neq 0} m|\varepsilon_i| \left(H + \frac{\operatorname{sgn}(\varepsilon_i)}{m} E_i\right).$$

Every coefficient displayed is positive, and $H \pm E_i/m = (mH \pm E_i)/m$ is ample. The expression is therefore a positive rational combination of ample divisors. For instance, it is enough to take $|\varepsilon_i| < 1/(mr)$ when $r > 0$. □

Remark 1.3.8 (Weil $\mathbb{Q}$-divisors)**.** For an integral variety X, set

$$\mathrm{WDiv}_{\mathbb{Q}}(X) := \mathrm{WDiv}(X) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

Its elements are finite rational combinations of prime divisors. The cycle map extends to

$$[\,] : \mathrm{Div}_{\mathbb{Q}}(X) \longrightarrow \mathrm{WDiv}_{\mathbb{Q}}(X), \qquad D \longmapsto \sum_V \mathrm{ord}_V(D)[V],$$

with the orders extended rationally from Remark 1.1.4. If X is normal, this is injective, and the orders are computed by the codimension-one valuations. In this case we identify $\mathrm{Div}_{\mathbb{Q}}(X)$ with its image in $\mathrm{WDiv}_{\mathbb{Q}}(X)$. A Weil $\mathbb{Q}$-divisor is $\mathbb{Q}$-Cartier if it lies in this image, equivalently if a positive multiple is an integral Cartier divisor. The operations above then apply to it.

Example 1.3.9 ($\mathbb{Q}$-Cartier need not be Cartier)**.** Consider the quadric cone[19]

$$X = \{x_0x_2 - x_1^2 = 0\} \subseteq \mathbb{P}^3, \qquad O = [0:0:0:1].$$

This surface is singular at its vertex O, but it is normal: it is a hypersurface, hence Cohen–Macaulay, and is regular in codimension one. Let $E = \{x_0 = x_1 = 0\} \subseteq X$ be a ruling line through O. It is a Weil divisor but is not Cartier at O, since its ideal (x_0, x_1) in the local ring at the vertex needs two generators. However, the tangent hyperplane $x_0 = 0$ cuts out the Cartier divisor $A = 2E$ as a Weil divisor. Hence E is $\mathbb{Q}$-Cartier, and

$$E^2 = \left(\frac{1}{2}A \cdot \frac{1}{2}A\right) = \frac{1}{4}A^2 = \frac{1}{4}\deg X = \frac{1}{2}.$$

Thus the example illustrates the distinction between Cartier and $\mathbb{Q}$-Cartier on a normal singular surface, not a failure of normality.

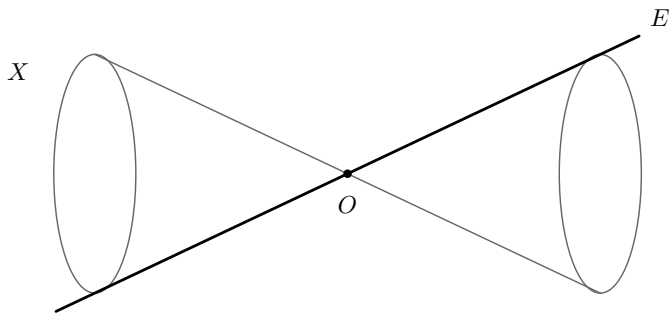


Figure 1.10: The normal quadric cone in $\mathbb{P}^3$

1.3B $\mathbb{R}$-Cartier Divisors and Their Ampleness

Definition 1.3.10 ($\mathbb{R}$-Cartier divisors)**.** An $\mathbb{R}$-Cartier divisor on a variety or scheme X is an element of

$$\mathrm{Div}_{\mathbb{R}}(X) := \mathrm{Div}(X) \otimes_{\mathbb{Z}} \mathbb{R},$$

represented by a finite sum $D = \sum_i c_i A_i$, with $c_i \in \mathbb{R}$ and $A_i \in \mathrm{Div}(X)$. Write $D_1 \sim_{\mathbb{R}} D_2$ if $D_1 - D_2$ is a finite real linear combination of principal divisors. If X is complete, intersection products extend to real-valued products and depend only on numerical equivalence. In particular,

$$D \equiv_{\mathrm{num}} 0 \quad \Longleftrightarrow \quad \sum_i c_i (A_i \cdot C) = 0 \text{ for every integral curve } C \subseteq X.$$

Effectivity means that such an expression can be chosen with all $c_i \geq 0$ and all A_i effective. Supports and pullbacks are handled as above, with the same qualifications about Cartier representatives.

[19]For this non-Cartier divisor whose double is Cartier, see [Har77, II, Example 6.11.3].

Example 1.3.11. There is an isomorphism

$$N^1(X)_{\mathbb{R}} \simeq N^1(X) \otimes_{\mathbb{Z}} \mathbb{R}.$$

The key fact is linear algebra: a numerically trivial $\mathbb{R}$-Cartier divisor is a real combination of numerically trivial integral Cartier divisors. To see it, write $B = \sum_{i=1}^{s} b_i B_i$. Numerical triviality is the system

$$\sum_i b_i (B_i \cdot C) = 0 \qquad \text{for all integral curves } C.$$

The integer row vectors $((B_1 \cdot C), \ldots, (B_s \cdot C))$ span a finite-dimensional rational vector space, so finitely many equations give the same system. Its real kernel has a rational basis; clearing denominators gives integral kernel vectors, each corresponding to a numerically trivial integral divisor. This proves the asserted fact and the isomorphism. We keep the subscript $\mathbb{R}$ when it helps distinguish the vector space from the integral group $N^1(X)$.

Definition 1.3.12 (Ampleness for $\mathbb{R}$-Cartier divisors)**.** Assume X is complete. An $\mathbb{R}$-Cartier divisor is ample if it can be written as a finite positive real combination

$$D = \sum_i c_i A_i, \qquad c_i > 0,$$

where the A_i are ample integral Cartier divisors. In particular, a positive real combination of ample real divisors is ample.

Remark 1.3.13 (Nakai inequalities for $\mathbb{R}$-Cartier divisors)**.** If D is ample, multilinearity and positivity of mixed ample intersections give

$$(D^{\dim V} \cdot V) > 0$$

for every positive-dimensional integral subvariety $V \subseteq X$. The converse is less immediate because denominators cannot be cleared. For projective varieties it is the theorem of Campana and Peternell [CP90; Laz04a, Theorem 2.3.18]. The converse also holds for complete schemes; a reference covering that generality is Fujino–Miyamoto [FM21, Theorem 1.3]. We will not use this real Nakai criterion to prove the elementary properties below.

Proposition 1.3.14 (Ample classes for $\mathbb{R}$-Cartier divisors)**.** *The ampleness of an $\mathbb{R}$-Cartier divisor depends only on its numerical equivalence class.*

Proof. It is enough to prove that $D + B$ is ample if D is ample and $B \equiv_{\text{num}} 0$. By Example 1.3.11, B is a real combination of numerically trivial integral Cartier divisors. We first treat an ample integral divisor A and a numerically trivial integral divisor B_0. For every rational r, the divisor $A + rB_0$ is ample by the rational Nakai criterion. For arbitrary real r, choose rationals $r_1 < r < r_2$ and $0 < t < 1$ with $r = tr_1 + (1-t)r_2$. Then

$$A + rB_0 = t(A + r_1 B_0) + (1-t)(A + r_2 B_0)$$

is a positive real combination of ample rational divisors, hence is ample.

For the general case, write $D = \sum_j c_j A_j$ and $B = \sum_{i=1}^{s} b_i B_i$. If $s > 0$, divide the first ample summand into s parts:

$$D + B = \sum_{i=1}^{s} \frac{c_1}{s}\left(A_1 + \frac{s b_i}{c_1} B_i\right) + \sum_{j \geq 2} c_j A_j.$$

Every term is ample by the special case. If $B = 0$, nothing is required. □

We next show that ampleness is open in the usual topology of $N^1(X)_{\mathbb{R}}$.

Example 1.3.15 (Openness of ampleness for $\mathbb{R}$-Cartier divisors)**.** Let X be projective, let H be an ample $\mathbb{R}$-Cartier divisor, and let $E_1, \ldots, E_r$ be $\mathbb{R}$-Cartier divisors. Then $H + \sum_i \varepsilon_i E_i$ is ample for all sufficiently small real ε_i, of either sign.

Proof. For integral H, E_i, the exact positive-combination formula in the proof of Proposition 1.3.7 works equally well for real ε_i: the coefficients are now real and the summands remain ample rational divisors. Thus that proof already gives a real neighbourhood, not just rational points.

In general, express the finitely many E_i using finitely many integral Cartier divisors. Their new coefficients are real linear combinations of the ε_i, so it is enough to treat integral E_i. Write $H = \sum_j c_j A_j$ with $c_j > 0$ and A_j ample and integral, and choose a rational $0 < c < c_1$. Then

$$H + \sum_i \varepsilon_i E_i = c\left(A_1 + \sum_i \frac{\varepsilon_i}{c} E_i\right) + (c_1 - c) A_1 + \sum_{j \geq 2} c_j A_j.$$

For sufficiently small real ε_i, the first parenthesis is ample by the case just proved. The other nonzero summands are ample as well. □

Corollary 1.3.16. *If X is projective, $N^1(X)_{\mathbb{R}}$ is spanned by classes of ample integral Cartier divisors.*

Proof. Choose integral Cartier divisors $D_1, \ldots, D_\rho$ whose classes form a basis, and fix an ample Cartier divisor H. For a fixed sufficiently large m, all $mH + D_i$ and mH are ample. Since

$$[D_i] = [mH + D_i] - [mH],$$

ample integral classes span the whole space. One can consequently select a basis from this finite spanning set. The openness statement concerns the real ample cone; the integral ample classes themselves form a discrete subset, not an open subset of $N^1(X)_{\mathbb{R}}$. □

1.4 Nef Line Bundles and Cartier Divisors

We have seen that if X is a projective variety, a class $\delta \in N^1(X)_{\mathbb{Q}}$ is ample if and only if it satisfies the Nakai inequalities

$$(\delta^{\dim V} \cdot V) > 0$$

for all irreducible subvarieties $V \subseteq X$ of positive dimension. So we want to make this inequality weaker:

$$(\delta^{\dim V} \cdot V) \geq 0.$$

This is related to nefness and Kleiman's theorem: for these non-negative inequalities, it suffices to test curves.

1.4A Definitions and Formal Properties

Definition 1.4.1 (Nef line bundles and Cartier divisors)**.** Let X be a complete variety or scheme. A line bundle L on X is numerically effective, or nef, if

$$\int_C c_1(L) \geq 0$$

for every irreducible curve $C \subseteq X$. Similarly, an integral, $\mathbb{Q}$- or $\mathbb{R}$-Cartier divisor D is nef if $(D \cdot C) \geq 0$ for every such curve.

Since intersection with a curve depends only on numerical equivalence, this defines nef classes in $N^1(X)$, $N^1(X)_{\mathbb{Q}}$ and $N^1(X)_{\mathbb{R}}$. Any ample class is nef.

Remark 1.4.2 (Chow's lemma)**.** In dealing with nefness, we often use Chow's lemma to reduce statements about complete varieties or schemes to the projective case. There is a projective variety or scheme X' and a proper surjective morphism

$$f : X' \longrightarrow X$$

which is an isomorphism over a dense open subset of X. On a general proper scheme the existence of an ample line bundle is not automatic. On the projective model one has ample line bundles available; nefness and intersection numbers are then compared by pullback and the projection formula.

We record the formal properties of nefness, as in [Kle66].

Example 1.4.3 (Formal properties of nefness)**.** Let X be complete and L a line bundle on X.

(i) If $f : Y \to X$ is proper and L is nef, then f^*L is nef. In particular, restrictions to closed subschemes remain nef.

(ii) If, moreover, f is surjective and f^*L is nef, then L is nef.

(iii) L is nef if and only if L_{red} is nef on X_{red}.

(iv) L is nef if and only if its restriction to each irreducible component of X is nef.

For (i), apply the projection formula to each curve in Y, including curves contracted by f. For (ii), given a curve $C \subseteq X$, choose a curve $C' \subseteq Y$ mapping onto it with positive degree d. Such a curve can be obtained on a projective model of a component of $f^{-1}(C)$ by general hyperplane sections and then pushed down to Y. The projection formula gives $(f^*L \cdot C') = d(L \cdot C)$. The last two statements follow because the reduced irreducible curves are unchanged.

Example 1.4.4. Let X be complete and L a globally generated line bundle on X. Then L is nef.

Proof. Properness implies that $H^0(X, L)$ is finite-dimensional over $\mathbb{C}$ ([Har77, III, Remark 8.8.1]). Global generation gives a morphism

$$\varphi_{|L|} : X \longrightarrow \mathbb{P}H^0(X, L), \qquad L \simeq \varphi_{|L|}^*\mathcal{O}(1).$$

By the projection formula,

$$(L \cdot C) = (\mathcal{O}(1) \cdot (\varphi_{|L|})_*[C]) \geq 0.$$

The pushforward is either zero or an effective one-cycle. □

Example 1.4.5 (Cartier divisor with nef normal bundle)**.** Let X be complete and $D \subseteq X$ an effective Cartier divisor. If the normal bundle $N_{D/X} = \mathcal{O}_D(D)$ is nef, then D is nef.

Proof. For a curve $C \subseteq D$,

$$(D \cdot C) = \deg \mathcal{O}_C(D) = \int_C c_1(\mathcal{O}_D(D)) \geq 0.$$

For a curve not contained in D, the effective divisor restricts to an effective divisor on the normalization of C, so again $(D \cdot C) \geq 0$. □

In particular, if X is a smooth projective surface and $C \subseteq X$ is an irreducible curve with $C^2 \geq 0$, then C is nef.

1.4B Kleiman's Theorem

We discuss the fundamental result due to [Kle66].

Theorem 1.4.6 (Kleiman's theorem)**.** *Let X be a complete variety or scheme. If D is a nef $\mathbb{R}$-Cartier divisor, then*

$$(D^k \cdot V) \geq 0$$

for every irreducible subvariety $V \subseteq X$ of dimension k. Similarly, $\int_V c_1(L)^{\dim V} \geq 0$ for every nef line bundle L.

Before giving a proof, we discuss some applications.

Corollary 1.4.7. *Let X be projective, D a nef $\mathbb{R}$-Cartier divisor and H an ample $\mathbb{R}$-Cartier divisor. Then $D + \varepsilon H$ is ample for every $\varepsilon > 0$. Conversely, if D, H are $\mathbb{R}$-Cartier divisors and $D + \varepsilon H$ is ample for all sufficiently small $\varepsilon > 0$, then D is nef.*

Proof. "$\Leftarrow$": For every curve C,

$$(D \cdot C) + \varepsilon(H \cdot C) > 0.$$

Letting $\varepsilon \to 0$ gives $(D \cdot C) \geq 0$.

"$\Rightarrow$": Replacing εH by H, it suffices to prove that $D + H$ is ample. We first treat the case in which $D + H$ is numerically equivalent to a rational Cartier divisor, and then make a small perturbation.

Rational case

Fix an irreducible $V \subseteq X$ of positive dimension k. Expand

$$((D + H)^k \cdot V) = \sum_{s=0}^{k} \binom{k}{s} (H^s D^{k-s} \cdot V).$$

Since H is a positive real combination of integral ample divisors, $H^s \cdot [V]$ has an effective real cycle representative: take sufficiently high very ample multiples and general successive hyperplane sections, and then divide by their degrees. Kleiman's theorem applied to its components gives $(H^s D^{k-s} \cdot V) \geq 0$. The last term $(H^k \cdot V)$ is strictly positive by Example 1.2.3 and linearity. Thus $D + H$ satisfies the Nakai inequalities. Its numerical class is rational, so Nakai's criterion and Proposition 1.3.14 imply ampleness.

General case

Choose ample integral Cartier divisors $H_1, \ldots, H_r$ whose classes form a basis of $N^1(X)_{\mathbb{R}}$. By Example 1.3.15,

$$H(\varepsilon_1, \ldots, \varepsilon_r) := H - \varepsilon_1 H_1 - \cdots - \varepsilon_r H_r$$

remains ample for all sufficiently small $\varepsilon_i \geq 0$. For $0 < \varepsilon_i \ll 1$, the classes of $D + H(\varepsilon_1, \ldots, \varepsilon_r)$ fill an open subset, and rational classes are dense. We can therefore choose such ε_i so that $D' = D + H(\varepsilon_1, \ldots, \varepsilon_r)$ has rational numerical class. The previous case makes D' ample. Hence

$$D + H = D' + \varepsilon_1 H_1 + \cdots + \varepsilon_r H_r$$

is ample as a positive sum of ample real divisors. □

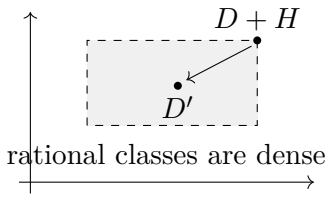


Figure 1.11: Give a small perturbation.

Corollary 1.4.8 (Test ampleness by intersecting with curves)**.** *Let X be projective, H an ample $\mathbb{R}$-Cartier divisor and D an $\mathbb{R}$-Cartier divisor. Then D is ample if and only if there is an $\varepsilon > 0$ such that*

$$\frac{D \cdot C}{H \cdot C} \geq \varepsilon$$

for every irreducible curve $C \subseteq X$.

Proof. "$\Leftarrow$": Since $H \cdot C > 0$, the inequality says that $D - \varepsilon H$ is nef. Corollary 1.4.7 makes $D = (D - \varepsilon H) + \varepsilon H$ ample.

"$\Rightarrow$": If D is ample, openness of ampleness (Example 1.3.15) makes $D - \varepsilon H$ ample for $0 < \varepsilon \ll 1$. Intersecting with C gives the required inequality. □

Example 1.4.9 (Comparison of ample degrees on curves)**.** If H_1, H_2 are ample Cartier divisors on a projective variety X, there are $M, m \in \mathbb{Q}_{>0}$ such that

$$m(H_1 \cdot C) \leq (H_2 \cdot C) \leq M(H_1 \cdot C)$$

for every irreducible curve $C \subseteq X$.

Proof. By openness of ampleness, choose rational $M \gg 1$ and $0 < m \ll 1$ so that $MH_1 - H_2$ and $H_2 - mH_1$ are ample. Intersecting with C gives both bounds. □

Theorem 1.4.10 (Seshadri's criterion)**.** *Let X be a projective variety and D a Cartier divisor. Then D is ample if and only if there is an $\varepsilon > 0$ such that*

$$\frac{D \cdot C}{\operatorname{mult}_x C} \geq \varepsilon$$

for every closed point $x \in X$ and every irreducible curve C through x.

Proof. "$\Rightarrow$": If an effective Cartier divisor E_x passes through x and does not contain C, its local intersection number satisfies

$$(E_x \cdot C) \geq i(E_x, C; x) \geq \operatorname{mult}_x C.$$

Choose $m > 0$ so that mD is very ample. For every pair (x, C), a hyperplane through x not containing C gives such an $E_x \sim_{\text{lin}} mD$. Thus $(D \cdot C) \geq \frac{1}{m} \operatorname{mult}_x C$, with the same m for all pairs.

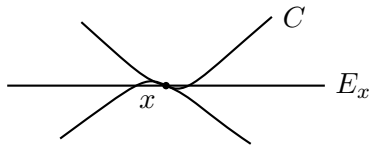


Figure 1.12: Intersection number bounded below by multiplicity.

"$\Leftarrow$": Use induction on $n = \dim X$. Dimension zero is immediate; for $n = 1$, the hypothesis gives positive degree. The same inequality holds on every proper irreducible subvariety $V \subset X$, so induction and Nakai's criterion give $(D^{\dim V} \cdot V) > 0$. It remains to prove $D^n > 0$. Assume $n \geq 2$, fix a smooth point $x \in X$, and consider

$$\mu : X' = \mathrm{Bl}_x(X) \longrightarrow X, \qquad E = \mu^{-1}(x) \simeq \mathbb{P}^{n-1}.$$

Here $\mathcal{O}_E(E) = \mathcal{O}_{\mathbb{P}^{n-1}}(-1)$.

Step 1: We claim that $\mu^*D - \varepsilon E$ is nef on X'. If so, Kleiman's theorem gives

$$\begin{aligned} 0 &\leq ((\mu^*D - \varepsilon E)^n)_{X'} \\ &= \sum_{s=0}^{n} \binom{n}{s} (-\varepsilon)^s ((\mu^*D)^{n-s} E^s)_{X'} \\ &= (D^n)_X - \varepsilon^n. \end{aligned}$$

Indeed, $\mu^*D|_E$ is trivial, so the mixed terms vanish for $1 \leq s \leq n-1$, whereas

$$(E^n)_{X'} = \int_E c_1(\mathcal{O}_E(-1))^{n-1} = (-1)^{n-1}.$$

The subscript indicates where the intersection number is computed. Consequently $D^n \geq \varepsilon^n > 0$.

Step 2: To check nefness, take an irreducible curve $C' \subseteq X'$. If C' is not contained in E, it is the strict transform of $C = \mu(C')$, and

$$(E \cdot C') = \mathrm{mult}_x C, \qquad (\mu^*D \cdot C') = (D \cdot C).$$

The multiplicity is taken to be zero if $x \notin C$. Thus

$$((\mu^*D - \varepsilon E) \cdot C') = (D \cdot C) - \varepsilon \,\mathrm{mult}_x C \geq 0.$$

If $x \notin C$, use $D \cdot C > 0$, obtained by applying the hypothesis at any closed point of C. If $C' \subseteq E$, then

$$(\mu^*D \cdot C') = 0, \qquad -\varepsilon(E \cdot C') = \varepsilon \deg \mathcal{O}_E(1)|_{C'} > 0.$$

This proves the claim. □

Remark 1.4.11 ($(C' \cdot E) = \operatorname{mult}_x C$)**.** In the preceding blow-up, the strict transform C' is the blow-up of C at the ideal $\mathfrak{m}_x \mathcal{O}_C$. The exceptional divisor on this curve is $E_C = E|_{C'}$, with

$$\mathfrak{m}_x \mathcal{O}_{C'} = \mathcal{O}_{C'}(-E_C).$$

The multiplicity formula in [Ful98, p. 79], applied to the one-dimensional scheme C at x, gives $\deg E_C = \operatorname{mult}_x C$. Therefore $(E \cdot C') = \deg E_C$. In particular, one must apply the formula to the curve here; the top self-intersection of the exceptional divisor in the ambient variety is a different intersection number.

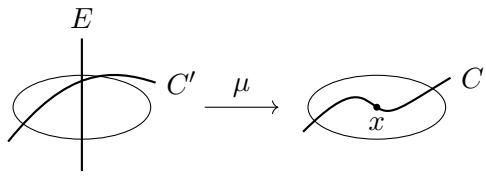


Figure 1.13: Strict transform and exceptional divisor.

Proposition 1.4.12 (Nefness in families)**.** *Let $f : X \to T$ be a surjective proper morphism of varieties, and L a line bundle on X. Put $X_t = f^{-1}(t)$ and $L_t = L|_{X_t}$. If L_0 is nef for a given $0 \in T$, there is a countable union $B \subseteq T$ of proper closed subvarieties, not containing 0, such that L_t is nef for every $t \in T \setminus B$.*

Proof. By relative Chow's lemma there is a proper surjective modification $\mu : X' \to X$ such that $f \circ \mu : X' \to T$ is projective. Nefness of L_t is equivalent to nefness of $\mu_t^* L_t$ by Example 1.4.3. Thus we may assume f is projective. Choose a relatively ample line bundle A on X; every A_t is ample. There is no need to represent L by a divisor avoiding every fibre. By Corollary 1.4.7,

$$L_t \text{ is nef} \quad \Longleftrightarrow \quad L_t^{\otimes m} \otimes A_t \text{ is ample for every } m \geq 1.$$

For each m, ampleness in families (Theorem 1.2.11) gives an open neighbourhood U_m of 0 on which $L_t^{\otimes m} \otimes A_t$ is ample. Take $B = \bigcup_{m \geq 1} (T \setminus U_m)$, decomposing each closed subset into its finitely many irreducible components if necessary. Every such component is proper and avoids 0. For $t \notin B$, all the displayed powers are ample, and L_t is nef. □

Now we turn to the proof of Kleiman's theorem.

Proof of Theorem 1.4.6. It suffices to work on each reduced irreducible subvariety and prove its top intersection non-negative. By Chow's lemma and the projection formula, we may replace this subvariety by a projective birational model. Thus assume X is projective and integral, of dimension n. We use induction on n: for $n = 1$ this is the definition of nefness, and the zero-dimensional assertion

is immediate. Assume the assertion in smaller dimensions; the issue is to show $D^n \geq 0$.

Rational case

Suppose first that D is a $\mathbb{Q}$-Cartier divisor. Fix an ample integral Cartier divisor H and consider

$$P(t) = (D + tH)^n \qquad (t \in \mathbb{R}).$$

We must prove $P(0) \geq 0$. Suppose, to the contrary, that $P(0) < 0$.

Step 1: For $k < n$,

$$(D^k \cdot H^{n-k}) \geq 0.$$

Indeed, general members of a sufficiently high very ample multiple of H intersect properly to give an effective k-cycle representing $m^{n-k}H^{n-k} \cdot [X]$. Apply induction to its components and divide by m^{n-k}. Therefore every non-constant coefficient of $P(t)$ is non-negative, and the coefficient of t^n is $H^n > 0$. It follows that P is strictly increasing for $t > 0$, with a unique positive root $t_0 > 0$.

Step 2: For every rational $t > t_0$, the divisor $D + tH$ is ample. By Nakai's criterion, we check

$$((D + tH)^k \cdot V) > 0$$

for each positive-dimensional irreducible $V \subseteq X$. For $V = X$, this is $P(t) > 0$. For $V \subsetneq X$, expand

$$((D + tH)^k \cdot V) = \sum_{s=0}^{k} \binom{k}{s} t^{k-s} (D^s \cdot H^{k-s} \cdot V).$$

Induction, applied directly when $s = k$ and to general hyperplane sections when $s < k$, makes every term non-negative. The term $t^k(H^k \cdot V)$ is positive. This proves the claim.

Step 3: Write $P(t) = Q(t) + R(t)$, where

$$Q(t) = D \cdot (D + tH)^{n-1}, \qquad R(t) = tH \cdot (D + tH)^{n-1}.$$

For rational $t > t_0$, the ample rational divisor $D + tH$ has an effective rational complete-intersection one-cycle. Its intersection with the nef D is non-negative, so $Q(t) \geq 0$. By continuity, $Q(t_0) \geq 0$. On the other hand,

$$R(t) = \sum_{s=0}^{n-1} \binom{n-1}{s} t^{n-s} (D^s \cdot H^{n-s}).$$

All coefficients are non-negative and the coefficient H^n of t^n is positive. Since $t_0 > 0$, we get $R(t_0) > 0$, contradicting $P(t_0) = Q(t_0) + R(t_0) = 0$.

General case

For a nef real divisor D, choose ample integral divisors $H_1, \dots, H_r$ whose classes form a basis of $N^1(X)_{\mathbb{R}}$. The divisors

$$D(\varepsilon_1, \dots, \varepsilon_r) = D + \varepsilon_1 H_1 + \cdots + \varepsilon_r H_r$$

are nef for $\varepsilon_i > 0$, simply because their degrees on curves are non-negative. Rational numerical classes are dense in these open sets, so we may take such classes with every $\varepsilon_i \to 0$. The rational case gives $D(\varepsilon_1, \dots, \varepsilon_r)^n \geq 0$. Continuity of intersection products yields $D^n \geq 0$. The same proof applies to line bundles, using their first Chern classes on the projective models. □

Example 1.4.13 (Intersection products of nef classes)**.** Let X be a complete variety or scheme of dimension n, and let $\delta_1, \dots, \delta_n$ be nef classes. Then

$$\int_X \delta_1 \cdots \delta_n \geq 0.$$

Proof. Reduce to the projective case by taking the top-dimensional components, with their multiplicities, and applying Chow's lemma and the projection formula. Choose ample classes H_i. Corollary 1.4.7 makes $\delta_i + \varepsilon_i H_i$ ample for $\varepsilon_i > 0$, and Example 1.2.3 gives

$$\int_X (\delta_1 + \varepsilon_1 H_1) \cdots (\delta_n + \varepsilon_n H_n) > 0.$$

Let every $\varepsilon_i \to 0$. □

Remark 1.4.14 (Mumford's counterexample)**.** Let X be projective and L a line bundle. If $\int_C c_1(L) > 0$ for every irreducible curve C, does it follow that $\int_V c_1(L)^{\dim V} > 0$ for every positive-dimensional irreducible V? The answer is no. This is the famous counterexample due to Mumford, which we will discuss later in the notes.

Definition 1.4.15 (Minimal surface)**.** A smooth projective surface X is minimal if it contains no (-1)-curves, that is, no curve $E \simeq \mathbb{P}^1$ with $E^2 = -1$.

Example 1.4.16 (Minimal surfaces)**.** Let X be a smooth projective surface with $\kappa(X) \geq 0$, so that $|mK_X| \neq \varnothing$ for some $m > 0$. Then X is minimal if and only if K_X is nef.

Proof. "⇒": Suppose K_X is not nef and take an irreducible curve C with $K_X \cdot C < 0$. Fix $D \in |mK_X|$ and write $D = \sum_i a_i C_i$, with $a_i > 0$. Since

$$(D \cdot C) = m(K_X \cdot C) < 0,$$

C must be a component, say $C = C_1$. Distinct curves have non-negative intersection, so

$$a_1 C^2 \leq D \cdot C < 0.$$

Adjunction now gives

$$-2 \leq 2p_a(C) - 2 = (K_X + C) \cdot C.$$

Both $K_X \cdot C$ and C^2 are negative integers. Hence they are both -1 and $p_a(C) = 0$, which implies that C is smooth and rational. Thus C is a (-1)-curve, contradicting minimality.

"$\Leftarrow$": For any (-1)-curve E, adjunction gives $K_X \cdot E = -1$. Such a curve cannot exist if K_X is nef. This is also the criterion in [Mat02, 1-1-4]. □

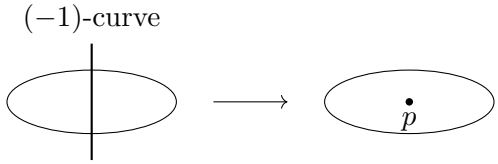


Figure 1.14: Contraction of a (-1)-curve.

Remark 1.4.17. Why do we choose the special divisor K_X here? Its role is made precise by adjunction: $K_X \cdot C + C^2 = 2p_a(C) - 2$. Together with the effective pluricanonical divisor, this forces a curve of negative canonical degree to be a (-1)-curve. Riemann–Roch is related to this formula, but the same conclusion does not hold for an arbitrary divisor in place of K_X.

1.4C Cones

Kleiman's work [Kle66] allows us to translate positivity questions into linear algebra in a finite-dimensional real vector space V. A cone $K \subseteq V$ is a subset stable under multiplication by positive scalars; it is convex if it is also stable under addition. We do not require every cone to contain 0, since the ample cone generally does not contain 0. The nef cone and the cones of curves below do contain it.

Definition 1.4.18 (Ample and nef cones)**.** The ample cone $\mathrm{Amp}(X) \subseteq N^1(X)_{\mathbb{R}}$ consists of all ample $\mathbb{R}$-Cartier divisor classes. The nef cone $\mathrm{Nef}(X) \subseteq N^1(X)_{\mathbb{R}}$ consists of all nef classes. Both are convex cones.

Concretely, $\mathrm{Amp}(X)$ consists of the finite sums $\sum_i a_i \delta_i$, where $a_i > 0$ and the δ_i are ample integral (or rational) classes.

Remark 1.4.19. As soon as $\rho(X) = \dim N^1(X)_{\mathbb{R}} \geq 3$, these cones can have quite complicated structure. For example, they need not be polyhedral.

Remark 1.4.20 (Visualization)**.** It is sometimes convenient to represent a cone by its intersection with a hyperplane not passing through the origin.

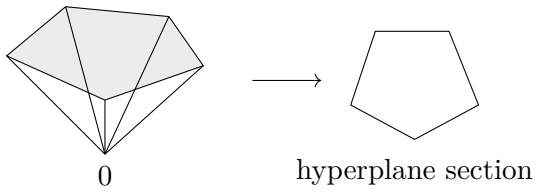


Figure 1.15: Representing cones.

We use the usual Euclidean topology on $N^1(X)_{\mathbb{R}}$ when discussing closures and interiors.

Theorem 1.4.21 (Kleiman)**.** *Let X be a projective variety or scheme. Then*

(i) $\mathrm{Nef}(X) = \overline{\mathrm{Amp}(X)}$*;*

(ii) $\mathrm{Amp}(X) = \mathrm{int}(\mathrm{Nef}(X))$.

Proof. The nef cone is closed, since it is the intersection of the closed half-spaces defined by non-negative degree on each curve. The ample cone is open by Example 1.3.15 and lies in the nef cone. Hence

$$\overline{\mathrm{Amp}(X)} \subseteq \mathrm{Nef}(X), \qquad \mathrm{Amp}(X) \subseteq \mathrm{int}(\mathrm{Nef}(X)).$$

For the converse inclusions, fix an ample Cartier divisor H. If D is nef, Corollary 1.4.7 makes $D + \varepsilon H$ ample for every $\varepsilon > 0$. Letting $\varepsilon \to 0$ puts D in $\overline{\mathrm{Amp}(X)}$. If $[D] \in \mathrm{int}(\mathrm{Nef}(X))$, then $D - \varepsilon H$ remains nef for $0 < \varepsilon \ll 1$, and

$$D = (D - \varepsilon H) + \varepsilon H$$

is ample by the same corollary. □

Remark 1.4.22 (Non-projective complete varieties)**.** Kleiman [Kle66, pp. 325–326] proves (ii) also for a complete variety whose Zariski topology is generated by complements of effective Cartier divisors, for example a smooth or $\mathbb{Q}$-factorial complete variety. For such an X, the nef cone has nonempty interior precisely when X is projective. The projective hypothesis cannot simply be omitted from all the assertions above without qualification.

Intersection with divisors gives a dual description in terms of curves.

Definition 1.4.23 (Numerical equivalence classes of curves)**.** Let X be complete. The space $Z_1(X)_{\mathbb{R}}$ consists of finite real linear combinations $\gamma = \sum_i a_i[C_i]$ of irreducible curves. Two such cycles are numerically equivalent if

$$(D \cdot \gamma_1) = (D \cdot \gamma_2) \quad \text{for every } D \in \mathrm{Div}_{\mathbb{R}}(X).$$

The quotient is denoted by $N_1(X)_{\mathbb{R}}$. Intersection induces a perfect pairing of finite-dimensional real vector spaces

$$N^1(X)_{\mathbb{R}} \times N_1(X)_{\mathbb{R}} \longrightarrow \mathbb{R}, \qquad (\delta, \gamma) \longmapsto (\delta \cdot \gamma).$$

Numerical equivalence for integral and rational one-cycles is defined similarly. We give $N_1(X)_{\mathbb{R}}$ its Euclidean topology.

Indeed, the curve functionals span the dual of $N^1(X)_{\mathbb{R}}$: otherwise a nonzero numerical divisor class would pair to zero with every curve. This also explains finite-dimensionality and perfectness of the pairing.

Definition 1.4.24 (Cone of curves)**.** The cone of curves is

$$\mathrm{NE}(X) = \left\{ \sum_i a_i [C_i] : a_i \geq 0,\ C_i \subseteq X \text{ irreducible curves} \right\} \subseteq N_1(X)_{\mathbb{R}},$$

where the sums are finite. Its closure $\overline{\mathrm{NE}}(X)$ is the closed cone of curves, also called the Kleiman–Mori cone.

Remark 1.4.25. The cone $\mathrm{NE}(X)$ need not be closed.

For a cone $K \subseteq V$, its dual is

$$K^* = \{\varphi \in V^* : \varphi(v) \geq 0 \text{ for every } v \in K\}.$$

We use the following elementary convexity fact ([BV04, p. 53]).

Lemma 1.4.26. *If K is a closed convex cone containing 0 in a finite-dimensional real vector space, then $K^{**} = K$ under $V^{**} = V$.*

Proposition 1.4.27. *The closed cone of curves is dual to the nef cone:*[20]

$$\overline{\mathrm{NE}}(X) = \{\gamma \in N_1(X)_{\mathbb{R}} : (\delta \cdot \gamma) \geq 0 \text{ for all } \delta \in \mathrm{Nef}(X)\}.$$

Proof. Take $V = N_1(X)_{\mathbb{R}}$ and $K = \overline{\mathrm{NE}}(X)$. By definition and continuity, $K^* = \mathrm{Nef}(X)$. The preceding lemma gives $K = \mathrm{Nef}(X)^*$. □

For a nonzero numerical class $[D] \in N^1(X)_{\mathbb{R}}$, write

$$\varphi_D : N_1(X)_{\mathbb{R}} \longrightarrow \mathbb{R}, \qquad \gamma \longmapsto (D \cdot \gamma),$$

and set

$$D^{\perp} = \{\gamma : D \cdot \gamma = 0\}, \qquad D_{>0} = \{\gamma : D \cdot \gamma > 0\}.$$

These are a hyperplane and an open half-space, respectively. The notations $D_{\geq 0}, D_{\leq 0}, D_{<0}$ are analogous.

[20] The numerical curve space is the one introduced in Definition 1.4.23.

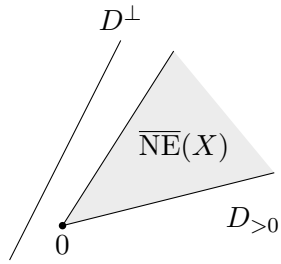


Figure 1.16: Test for ampleness via the closed cone of curves.

Theorem 1.4.28 (Kleiman's criterion: ampleness via cones)**.** *Let X be projective and D an $\mathbb{R}$-Cartier divisor. Then D is ample if and only if*

$$\overline{\mathrm{NE}}(X) \setminus \{0\} \subseteq D_{>0}.$$

Moreover, for an ample Cartier divisor H and an integer k, the set

$$W_k = \{\gamma \in \overline{\mathrm{NE}}(X) : (H \cdot \gamma) \leq k\}$$

is compact. In particular, there are only finitely many numerical classes of effective integral one-cycles Z with $(H \cdot Z) \leq k$.

Proof. For the first assertion, we follow [Deb01].

"$\Rightarrow$": If D is ample, its degree is non-negative on $\overline{\mathrm{NE}}(X)$ by continuity. Suppose $D \cdot \gamma = 0$ for a nonzero $\gamma \in \overline{\mathrm{NE}}(X)$. By the perfect pairing there is a real divisor E with $E \cdot \gamma < 0$. Then $(D + tE) \cdot \gamma < 0$ for every $t > 0$, so $D + tE$ is not nef. This contradicts openness of ampleness for small t.

"$\Leftarrow$": Choose a norm on $N_1(X)_{\mathbb{R}}$. The set

$$S = \{\gamma \in \overline{\mathrm{NE}}(X) : \|\gamma\| = 1\}$$

is compact, since it is a closed subset of the unit sphere. If X has positive dimension, S is nonempty. Positivity gives a lower bound $D \cdot \gamma \geq a > 0$ on S. For a fixed ample H, choose an upper bound $H \cdot \gamma \leq b$ with $b > 0$. Thus

$$(D - \tfrac{a}{b}H) \cdot \gamma \geq 0 \quad (\gamma \in S),$$

and by scaling the same holds on $\overline{\mathrm{NE}}(X)$. Hence $D - \frac{a}{b}H$ is nef, and $D = (D - \frac{a}{b}H) + \frac{a}{b}H$ is ample by Corollary 1.4.7. In dimension zero the criterion is vacuous and every line bundle is ample.

For compactness, we keep the argument of [KM98, 1.19]. The set W_k is closed, and is empty if $k < 0$. If it were unbounded for $k \geq 0$, take $\gamma_i \in W_k$ with $\|\gamma_i\| \to \infty$. After a subsequence,

$$\frac{\gamma_i}{\|\gamma_i\|} \longrightarrow \gamma \in \overline{\mathrm{NE}}(X), \qquad \|\gamma\| = 1.$$

But

$$0 \leq H \cdot \gamma = \lim_i \frac{H \cdot \gamma_i}{\|\gamma_i\|} = 0,$$

contradicting ampleness of H. Finally choose integral Cartier divisors $D_1, \ldots, D_\rho$ whose classes form a real basis of $N^1(X)_{\mathbb{R}}$. The linear isomorphism

$$\gamma \longmapsto (D_1 \cdot \gamma, \ldots, D_\rho \cdot \gamma) \quad \text{from } N_1(X)_{\mathbb{R}} \text{ to } \mathbb{R}^\rho$$

sends integral one-cycle classes into $\mathbb{Z}^\rho$. They therefore form a discrete subset, which has finite intersection with W_k. □

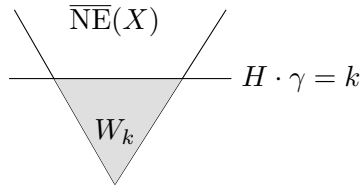


Figure 1.17: The part of $\overline{\mathrm{NE}}(X)$ with H-degree at most k is compact.

Example 1.4.29. For a projective X, the cone $\overline{\mathrm{NE}}(X)$ contains no line. Indeed, if $\gamma, -\gamma \in \overline{\mathrm{NE}}(X)$ and $\gamma \neq 0$, an ample divisor would have positive intersection with both, which is impossible. More generally a line contained in this closed cone would, by rescaling and taking limits in both directions, give such γ and $-\gamma$.

Definition 1.4.30 (Extremal faces)**.** Let K be a convex cone containing 0. An extremal face is a convex subcone $F \subseteq K$ such that

$$v, w \in K, \quad v + w \in F \quad \Longrightarrow \quad v, w \in F.$$

A ray $\mathbb{R}_{\geq 0} v$ with $v \neq 0$ which is an extremal face is called an extremal ray, and is denoted by r or R.

Example 1.4.31 (Curves on a surface)**.** On a smooth projective surface X, one-cycles and Cartier divisors are identified ([Har77, II, Proposition 6.11]). Numerical equivalence gives $N^1(X)_{\mathbb{R}} = N_1(X)_{\mathbb{R}}$, so all these cones live in the same space.

(i) $\mathrm{Nef}(X) \subseteq \overline{\mathrm{NE}}(X)$, with equality if and only if $C^2 \geq 0$ for every irreducible curve $C \subseteq X$.

(ii) If C is irreducible and $C^2 \leq 0$, then

$$\overline{\mathrm{NE}}(X) = \overline{\mathrm{NE}}(X)_{C \geq 0} + \mathbb{R}_{\geq 0}[C], \qquad \overline{\mathrm{NE}}(X)_{C \geq 0} := \overline{\mathrm{NE}}(X) \cap C_{\geq 0}.$$

(iii) Such a $[C]$ lies on the boundary of $\overline{\mathrm{NE}}(X)$. If $C^2 < 0$, it spans an extremal ray.

Proof. For (i), an ample integral divisor has an effective positive multiple, which on a surface is a one-cycle. Thus $\mathrm{Amp}(X) \subseteq \overline{\mathrm{NE}}(X)$, and taking closures gives $\mathrm{Nef}(X) \subseteq \overline{\mathrm{NE}}(X)$. If every $C^2 \geq 0$, then every curve is nef, since distinct curves have non-negative intersection. Closedness of the nef cone gives the reverse inclusion. Conversely, equality makes every curve class nef, hence $C^2 \geq 0$.

For (ii), first take an effective real cycle and separate its C-component: $\gamma_j = a_j[C]+\beta_j$, with $a_j \geq 0$ and β_j effective without C as a component. Then $C\cdot\beta_j \geq 0$. For a convergent sequence $\gamma_j \to \gamma \in \overline{\mathrm{NE}}(X)$, fix ample H. The inequalities

$$0 \leq a_j(H \cdot C) \leq H \cdot \gamma_j, \qquad 0 \leq H \cdot \beta_j \leq H \cdot \gamma_j$$

make a_j bounded and β_j contained in a compact slice of $\overline{\mathrm{NE}}(X)$ by Theorem 1.4.28. Passing to a subsequence gives $\gamma = a[C]+\beta$ with $a \geq 0$ and $\beta \in \overline{\mathrm{NE}}(X)_{C\geq 0}$.

For (iii), if $C^2 = 0$, then C is nef and the nonzero functional $\gamma \mapsto C\cdot\gamma$ vanishes at $[C]$. Its zero hyperplane supports $\overline{\mathrm{NE}}(X)$, so $[C]$ is a boundary point. If $C^2 < 0$ and $u + v = [C]$, apply (ii) to write $u = a[C] + u_0$, $v = b[C] + v_0$, where $a, b \geq 0$ and $u_0, v_0 \in \overline{\mathrm{NE}}(X)_{C\geq 0}$. Then

$$(1 - a - b)[C] = u_0 + v_0.$$

Intersection with C shows $1 - a - b \leq 0$; intersection with an ample H shows $1 - a - b \geq 0$. Hence $u_0 + v_0 = 0$, and positivity of H implies $u_0 = v_0 = 0$. This proves extremality. To see the boundary assertion directly, the divisor

$$H + \frac{H \cdot C}{-C^2}C$$

is nonzero and nef, and has zero intersection with C. It therefore defines a supporting hyperplane through $[C]$. □

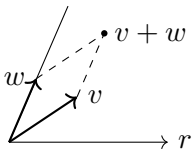


Figure 1.18: $v + w$ does not lie on the extremal ray r.

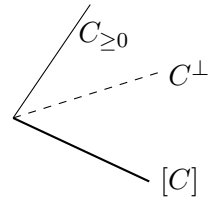


Figure 1.19: A negative curve spans an extremal ray.

1.4D Applications of Cones on Examples

To begin, we recall the simplest blow-down, following [Har77] and [Mat02]. For this result we allow any algebraically closed ground field k.

Theorem 1.4.32 (Castelnuovo's contraction theorem)**.** *Let X be a smooth projective surface over an algebraically closed field k, and let $E \subseteq X$ be a (-1)-curve. There is a proper birational morphism*[21]

$$\mu : X \longrightarrow X_0$$

onto a smooth projective surface, such that $\mu(E) = p$ and μ is the blow-up of X_0 at p.

Proof. We want a globally generated line bundle whose morphism contracts precisely E. In particular, it should have degree zero on E and positive degree on every other curve. Choose a very ample divisor A and put $d = A \cdot E > 0$, reserving k for the ground field. After replacing A by a sufficiently high multiple, assume $H^1(X, \mathcal{O}_X(A)) = 0$ by Serre vanishing, and redefine $d = A \cdot E$. Set

$$L = A + dE, \qquad L_n = A + nE \quad (0 \leq n \leq d+1).$$

Since $E^2 = -1$, we have $L \cdot E = 0$. For every curve $C \neq E$, $L \cdot C = A \cdot C + d(E \cdot C) > 0$.

Step 1: We claim that $H^1(X, \mathcal{O}_X(L_n)) = 0$ for $0 \leq n \leq d+1$. The case $n = 0$ is our choice of A. For $n \geq 1$, use

$$0 \longrightarrow \mathcal{O}_X(L_{n-1}) \longrightarrow \mathcal{O}_X(L_n) \longrightarrow \mathcal{O}_E(L_n) \longrightarrow 0.$$

Since $E \simeq \mathbb{P}^1$ and $\deg \mathcal{O}_E(L_n) = d - n$, we have $\mathcal{O}_E(L_n) \simeq \mathcal{O}_{\mathbb{P}^1}(d-n)$. For $n \leq d+1$ its first cohomology vanishes. The long exact sequence and induction prove the claim.

Step 2: We claim that $|L_n|$ is base point free for $0 \leq n \leq d$. The subsystem $|A| + nE$ shows that all possible base points lie on E. For $n \geq 1$, Step 1 makes

$$H^0(X, \mathcal{O}_X(L_n)) \longrightarrow H^0(E, \mathcal{O}_E(d-n))$$

surjective. Since $d - n \geq 0$, the bundle $\mathcal{O}_E(d-n)$ is globally generated. At each point of E, choose a section nonzero at that point and lift it. Thus there are no base points on E either. The case $n = 0$ is immediate from very ampleness of A.

Step 3: Consider

$$\varphi = \varphi_{|A+dE|} : X \longrightarrow X_1 \subseteq \mathbb{P}^N.$$

The equality $L \cdot E = 0$ implies that $\varphi(E) = p_1$ is a point: otherwise the projection formula would give positive degree on E. Away from E, the sections from $|A| + dE$ recover the embedding defined by $|A|$, since the canonical section of dE is invertible there. The projection onto these coordinates therefore gives an inverse to φ on

[21] For cohomology on the exceptional projective line and factorization of birational morphisms of smooth surfaces, see [Har77, III, 5.1], [Bea78, II, 11].

its image away from p_1. These coordinates all vanish on E and never vanish simultaneously away from E. Hence

$$X \setminus E \simeq X_1 \setminus \{p_1\}, \qquad \varphi^{-1}(p_1) = E \quad \text{set-theoretically.}$$

Step 4: Normalize X_1:

$$\nu : X_0 \longrightarrow X_1.$$

Since X is normal and φ is dominant, it factors as $\varphi = \nu \circ f$ with $f : X \to X_0$ proper and birational. The surface X_0 is projective, since ν is finite and X_1 is projective. The morphism f is an isomorphism away from E and maps E to a single point p over p_1. Indeed, $\nu^{-1}(p_1)$ is finite, f is surjective, and E is connected. By normality of X_0,

$$f_*\mathcal{O}_X = \mathcal{O}_{X_0}$$

([Har77, III, Corollary 11.4]).

Step 5: We check smoothness at p using formal functions. Put $R = \mathcal{O}_{X_0,p}$, $\mathfrak{m} = \mathfrak{m}_p$ and $I = \mathcal{O}_X(-E)$ near $f^{-1}(p)$. At this stage we know only

$$\sqrt{\mathfrak{m}\mathcal{O}_X} = I, \qquad \mathfrak{m}\mathcal{O}_X \subseteq I.$$

The two ideals have the same radical on a noetherian neighbourhood, so their powers are cofinal: for some $r > 0$, $I^r \subseteq \mathfrak{m}\mathcal{O}_X \subseteq I$. The theorem on formal functions [Har77, III, Theorem 11.1] consequently gives

$$\widehat{R} \simeq \varprojlim_n B_n, \qquad B_n := H^0(E_n, \mathcal{O}_{E_n}), \quad E_n = V(I^n).$$

We have not yet asserted $\mathfrak{m}\mathcal{O}_X = I$.

Because E is Cartier with self-intersection -1,

$$I/I^2 \simeq \mathcal{O}_{\mathbb{P}^1}(1), \qquad I^n/I^{n+1} \simeq \mathcal{O}_{\mathbb{P}^1}(n).$$

The exact sequences of infinitesimal neighbourhoods give

$$0 \longrightarrow H^0(\mathbb{P}^1, \mathcal{O}(n)) \longrightarrow B_{n+1} \longrightarrow B_n \longrightarrow 0 \qquad (n \geq 1),$$

since $H^1(\mathbb{P}^1, \mathcal{O}(n)) = 0$. Also $B_1 = k$. Choose compatible elements x, y in the inverse limit whose images in $\ker(B_2 \to B_1) = H^0(\mathcal{O}(1))$ form a basis. Such compatible lifts exist because every transition map is surjective.

Now keep track of multiplication. The natural maps

$$\operatorname{Sym}^n H^0(\mathcal{O}(1)) \longrightarrow H^0(\mathcal{O}(n))$$

are isomorphisms; the monomials $x^n, x^{n-1}y, \ldots, y^n$ give their usual bases. Induction in the preceding exact sequences therefore gives compatible *ring* isomorphisms

$$k[x,y]/(x,y)^n \simeq B_n.$$

For $n = 1$ this is $k \simeq B_1$. In the induction step the map is an isomorphism both on the quotient B_n and on the displayed degree-n kernel, hence on B_{n+1}. Taking inverse limits yields

$$\widehat{\mathcal{O}}_{X_0,p} \simeq k[[x, y]].$$

Thus $\mathcal{O}_{X_0,p}$ is a regular local ring of dimension two, and X_0 is smooth at p.

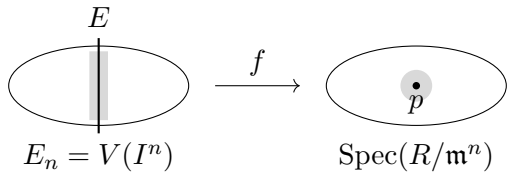


Figure 1.20: The two systems of infinitesimal neighbourhoods have cofinal ideals.

Step 6: We identify f with the blow-up at p. Choose regular parameters $u, v \in \mathfrak{m}$ whose images in $\ker(B_2 \to B_1)$ form the basis used above. Their pullbacks generate $I/I^2 \simeq \mathcal{O}_E(1)$ at every point of E, since its two standard sections generate it. Nakayama's lemma applied to $I/(\mathfrak{m}\mathcal{O}_X)$ gives

$$\mathfrak{m}\mathcal{O}_X = I = \mathcal{O}_X(-E)$$

near E; away from E both ideals are the unit ideal. Thus the universal property of the blow-up gives a factorization

$$X \xrightarrow{u_0} \mathrm{Bl}_p(X_0) \longrightarrow X_0.$$

On E, the morphism u_0 is defined by the complete system $|\mathcal{O}_{\mathbb{P}^1}(1)|$, hence maps E isomorphically onto the exceptional curve of $\mathrm{Bl}_p(X_0)$. No other curve can be contracted by u_0, because f contracts only E. Thus u_0 is proper and quasi-finite, hence finite. It is birational with normal target, so it is an isomorphism. Set $\mu = f$. □

Remark 1.4.33. The statement in [KM98] is given for smooth proper surfaces. Over an algebraically closed field this is consistent with the formulation above: Zariski's projectivity theorem says that a smooth complete algebraic surface is projective. Here the objects are schemes of finite type over the ground field, as throughout the chapter.

Corollary 1.4.34. *For every smooth projective surface, there is a finite sequence of blow-down morphisms*

$$X = X_0 \longrightarrow X_1 \longrightarrow \cdots \longrightarrow X_m$$

such that $X_i \simeq \mathrm{Bl}_{p_i}(X_{i+1})$, every X_i is smooth projective, and X_m is minimal.

Proof. If X_i contains a (-1)-curve, contract it by Castelnuovo's theorem. For a blow-up of a smooth point,

$$N^1(X_i)_{\mathbb{R}} \simeq N^1(X_{i+1})_{\mathbb{R}} \oplus \mathbb{R}[E_i],$$

so $\rho(X_{i+1}) = \rho(X_i) - 1$. Since these are non-negative integers, the process terminates. □

Definition 1.4.35 (Relative cone of curves)**.** For a morphism $\pi : X \to Y$ of projective varieties, the relative cone $\mathrm{NE}(\pi) \subseteq \mathrm{NE}(X)$ is generated by the classes of curves contracted by π. Here $\mathrm{NE}(\pi)$ is not defined to include its closure.

For an ample divisor H on Y, a curve C is contracted exactly when $\pi_*[C] = 0$, equivalently $(\pi^*H \cdot C) = 0$. Since every summand of an effective real cycle has non-negative π^*H-degree,

$$\mathrm{NE}(\pi) = \mathrm{NE}(X) \cap \ker \pi_* = \mathrm{NE}(X) \cap (\pi^*H)^{\perp}.$$

Consequently $\mathrm{NE}(\pi)$ is closed *in* $\mathrm{NE}(X)$; this sentence does not assert that it is closed in $N_1(X)_{\mathbb{R}}$.

We need the following rigidity statements; compare [Deb01, 1.15] and [KM98, 1.6]. We return to the chapter's ground field $\mathbb{C}$.

Lemma 1.4.36 (Rigidity lemma I)**.** *Let X, Y, Y' be varieties, let $\pi : X \to Y$ be proper and surjective, and let $\pi' : X \to Y'$ be a morphism. Assume that the natural map $\mathcal{O}_Y \to \pi_*\mathcal{O}_X$ is an isomorphism; in particular, every fibre of π is connected.*[22]

(a) If π' contracts the fibre over y_0, there is an open neighbourhood Y_0 of y_0 and a factorization

$$\pi'|_{\pi^{-1}(Y_0)} : \pi^{-1}(Y_0) \xrightarrow{\pi} Y_0 \longrightarrow Y'.$$

(b) If π' contracts every fibre, then $\pi' = h \circ \pi$ for a unique morphism $h : Y \to Y'$.

Proof. Properness of π' is unnecessary. Consider

$$g = (\pi, \pi') : X \longrightarrow Y \times Y'.$$

It is proper, since it factors as the closed graph embedding $X \to X \times Y'$ followed by the proper map $\pi \times \mathrm{id}$. Give its closed image Z the reduced structure, and write

$$X \xrightarrow{g} Z \xrightarrow{p} Y, \qquad p' : Z \longrightarrow Y', \qquad \pi = p \circ g, \quad \pi' = p' \circ g.$$

The projection p is proper. Indeed, after every base change, g remains surjective; the image under p of a closed subset is the image under the proper π of its closed

[22] For properness, the open finite locus, and the injectivity corresponding to dominance, see [Har77, II, Exercise 4.8], [SP25, Tag 02OH], [Har77, II, Exercise 2.18(b)].

inverse image under g. Thus p is universally closed, and it is also separated and of finite type.

For (a), $p^{-1}(y_0) = g(\pi^{-1}(y_0))$ is a single point. A proper morphism with finite fibre is finite over some neighbourhood Y_0 of y_0 (Stacks, Tag 02OH). Set $X_0 = \pi^{-1}(Y_0)$ and $Z_0 = p^{-1}(Y_0)$. Dominance between varieties gives injections

$$\mathcal{O}_{Y_0} \subseteq (p|_{Z_0})_* \mathcal{O}_{Z_0} \subseteq (\pi|_{X_0})_* \mathcal{O}_{X_0} = \mathcal{O}_{Y_0}.$$

Hence $(p|_{Z_0})_* \mathcal{O}_{Z_0} = \mathcal{O}_{Y_0}$. Since $p|_{Z_0}$ is finite, it is an isomorphism. The desired factor is $p'|_{Z_0} \circ (p|_{Z_0})^{-1}$.

For (b), every fibre of p is finite, so p is finite everywhere and the same argument gives $Z \simeq Y$. Uniqueness follows from surjectivity of π (and the reduced structures). □

Lemma 1.4.37 (Rigidity lemma II)**.** *Let Y be an irreducible variety and $f : Y \to Z$ a proper surjective morphism whose fibres are connected and all have dimension n. If a morphism $g : Y \to X$ contracts one fibre $f^{-1}(z_0)$, then it contracts every fibre of f.*[23]

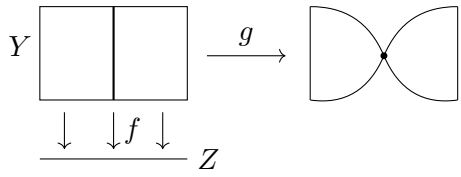
Proof. As above, $(f, g) : Y \to Z \times X$ is proper. Let W be its reduced closed image, and factor

$$f : Y \xrightarrow{h} W \xrightarrow{p} Z.$$

Both maps are proper and surjective. Properness of p follows by the universally closed argument in the preceding proof; surjectivity alone would not allow us to lift a prescribed K-point without extending K. We have $p^{-1}(z) = h(f^{-1}(z))$ and $p^{-1}(z_0)$ is a point.

Upper semicontinuity of fibre dimension gives a nonempty open $U \subseteq Z$ such that every fibre of p over U is finite. Since $f^{-1}(z)$ is connected, this finite image is one point. Thus the fibres of h over $p^{-1}(U)$ have dimension n. The irreducibility of W and upper semicontinuity for the proper h show that every fibre of h has dimension at least n.

For $w \in p^{-1}(z)$, we have $h^{-1}(w) \subseteq f^{-1}(z)$ and $\dim f^{-1}(z) = n$. Hence $h^{-1}(w)$ has dimension n and contains an n-dimensional irreducible component of $f^{-1}(z)$. Distinct w give disjoint fibres, and there are only finitely many such components. Therefore $p^{-1}(z)$ has only finitely many points. It is connected as the image of the connected $f^{-1}(z)$, so it is a single point. This says exactly that g contracts $f^{-1}(z)$. □



[23] For properness and upper semicontinuity of fibre dimension, see [Har77, II, 4.8(e)], [Har77, II, Exercise 3.22(d)].

Only one fibre contracted: impossible under Lemma 1.4.37.

Proposition 1.4.38. *Let X, Y, Y' be projective varieties and $\pi : X \to Y$ a morphism.*

(a) The subcone $\mathrm{NE}(\pi)$ is an extremal face of $\mathrm{NE}(X)$.

(b) Assume $\pi_*\mathcal{O}_X = \mathcal{O}_Y$, and let $\pi' : X \to Y'$ be a morphism. If $\mathrm{NE}(\pi) \subseteq \mathrm{NE}(\pi')$, there is a unique morphism $h : Y \to Y'$ with $\pi' = h \circ \pi$. Consequently, among morphisms satisfying $\pi_*\mathcal{O}_X = \mathcal{O}_Y$, the morphism π is determined up to isomorphism by $\mathrm{NE}(\pi)$.

Proof. For (a), take $a = \sum_i a_i[C_i]$ and $a' = \sum_j a'_j[C'_j]$ in $\mathrm{NE}(X)$, with all coefficients positive after deleting zero terms. If $a + a' \in \mathrm{NE}(\pi)$, pushing forward gives

$$\sum_i a_i \pi_*[C_i] + \sum_j a'_j \pi_*[C'_j] = 0 \quad \text{in } N_1(Y)_{\mathbb{R}}.$$

Intersect with an ample divisor H on Y. Each term is non-negative, and is positive unless the corresponding curve is contracted. Hence all these curves are contracted, so $a, a' \in \mathrm{NE}(\pi)$. The point is that no nonzero effective real one-cycle on a projective variety is numerically trivial.

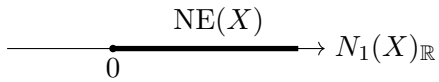
For (b), each curve contracted by π is contracted by π'. On each projective irreducible component of a fibre of π, a morphism with positive-dimensional image would map some curve onto a curve (by taking general sections over a curve in the image). Thus π' maps every such component to a point. Connectedness of the fibre makes all these points equal. Rigidity lemma I(b) gives the unique factor h. If two morphisms satisfying the stated pushforward condition have the same relative cone, apply the factorization in both directions. Surjectivity makes the two factors inverse isomorphisms. □

Remark 1.4.39. An ample divisor H on the projective target Y has positive degree on every irreducible curve. If $C \subseteq X$ is not contracted, then

$$\pi_*[C] = \deg(C \to \pi(C))[\pi(C)]$$

has positive H-degree. Thus $\pi_*[C] = 0$ numerically forces C to be contracted. This does not mean that distinct curves on Y cannot be numerically equivalent.

Example 1.4.40 (Picard number one)**.** If X is projective and $\dim N_1(X)_{\mathbb{R}} = 1$, for example $X = \mathbb{P}^n$ with $n \geq 1$, then $\mathrm{NE}(X) = \overline{\mathrm{NE}}(X)$ is a single ray. Its only extremal faces are $\{0\}$ and $\mathrm{NE}(X)$ itself. They correspond to the identity $X \to X$ and the map to a point, respectively.[24]



[24] For the divisor class group of projective space, see [Har77, II, 6.4].

Example 1.4.41 (Product of projective spaces)**.** Let $X = \mathbb{P} \times \mathbb{P}'$ be the product of two positive-dimensional projective spaces. Then $N_1(X)_{\mathbb{R}}$ has dimension two, spanned by the class l of a line in $\mathbb{P} \times \{p'\}$ and the class l' of a line in $\{p\} \times \mathbb{P}'$. They are dual to the two pulled-back hyperplane classes. The cone is

$$\mathrm{NE}(X) = \overline{\mathrm{NE}}(X) = \mathbb{R}_{\geq 0} l + \mathbb{R}_{\geq 0} l'.$$

Its two extremal rays are generated by l and l'. They correspond, respectively, to the projections $X \to \mathbb{P}'$ and $X \to \mathbb{P}$. Indeed, the projection to $\mathbb{P}'$ contracts the lines lying in the first factor, and the projection to $\mathbb{P}$ contracts those in the second.

1.4E Fujita's Vanishing Theorem

Fujita's theorem makes Serre-type vanishing uniform with respect to twists by nef Cartier divisors.

Theorem 1.4.42 (Fujita's vanishing theorem)**.** *Let X be a complex projective scheme and H an ample integral Cartier divisor. For every coherent sheaf $\mathcal{F}$ there is an integer $m(\mathcal{F}, H)$ such that*

$$H^i(X, \mathcal{F} \otimes \mathcal{O}_X(mH + D)) = 0$$

for every $i > 0$, every $m \geq m(\mathcal{F}, H)$ and every nef integral Cartier divisor D.

The essential point is that $m(\mathcal{F}, H)$ is independent of D. Applying Serre vanishing separately for each D would not give this uniformity. The proof below uses characteristic-zero vanishing for a resolution; that vanishing theorem is an input independent of Fujita's theorem.

Proof. Replacing X by the closed subscheme defined by the annihilator of $\mathcal{F}$, we may assume $\operatorname{Supp} \mathcal{F} = X$ set-theoretically. We use induction on dimension, allowing every coherent sheaf in the induction hypothesis.

Step 1: In dimension zero, all higher cohomology vanishes. Henceforth assume the assertion known in smaller dimensions.

Step 2: Reduce to the case in which X is reduced. Let N be the nilradical of $\mathcal{O}_X$, with $N^r = 0$. Use the filtration

$$\mathcal{F} \supseteq N\mathcal{F} \supseteq N^2\mathcal{F} \supseteq \cdots \supseteq N^r\mathcal{F} = 0.$$

Every quotient $Q_i = N^i\mathcal{F}/N^{i+1}\mathcal{F}$ is annihilated by N, hence is a coherent $\mathcal{O}_{X_{\mathrm{red}}}$-module. If the theorem holds on reduced schemes, it applies to each Q_i, since $H|_{X_{\mathrm{red}}}$ is ample and $D|_{X_{\mathrm{red}}}$ is nef. Take the maximum of the finitely many bounds. The sequences

$$0 \to N^{i+1}\mathcal{F} \to N^i\mathcal{F} \to Q_i \to 0,$$

twisted by $\mathcal{O}_X(mH + D)$, prove vanishing for $N^i\mathcal{F}$ by decreasing induction on i, and finally for $\mathcal{F}$.

Step 3: Reduce from reduced X to integral X. Write $X = X_1 \cup \cdots \cup X_s$ with its reduced irreducible components. Let I be the ideal of X_1 and J the ideal of $X_2 \cup \cdots \cup X_s$, with reduced structure. Then $IJ = 0$, because $I \cap J = 0$ on the reduced union. In

$$0 \longrightarrow I\mathcal{F} \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}/I\mathcal{F} \longrightarrow 0,$$

the first term is annihilated by J and is a sheaf on the other components; the quotient is a sheaf on X_1. Induction on the number of components, followed by the long exact cohomology sequence, proves the assertion using the maximum of the two bounds.

Step 4: We may assume H very ample. Choose $\ell > 0$ such that ℓH is very ample. If the assertion holds with ℓH, apply it to $\mathcal{F}(rH)$ for $0 \leq r < \ell$. Choose a common non-negative bound q_0 for these finitely many sheaves. Write $m = q\ell + r$; when $m \geq \ell q_0$, we have $q \geq q_0$, and

$$\mathcal{F}(mH + D) = \mathcal{F}(rH) \otimes \mathcal{O}_X(q\ell H + D)$$

has the required vanishing.

Step 5: It suffices to prove the assertion for $\mathcal{F} = \mathcal{O}_X$ on integral X. We induct on $\operatorname{rank}\mathcal{F}$. Rank zero means that $\mathcal{F}$ is supported in smaller dimension, so dimension induction applies. If the rank is positive, take $a \gg 0$ so that $\mathcal{F}(aH)$ is globally generated, and choose a global section s which is *nonzero at the generic point* η. It defines

$$\alpha : \mathcal{O}_X \longrightarrow \mathcal{F}(aH)$$

which is injective at η. Its kernel is an ideal subsheaf of the torsion-free $\mathcal{O}_X$, so it is zero. Explicitly, on an affine open $\operatorname{Spec} A$ the kernel is an ideal $I \subseteq A$ with $I \otimes_A K(A) = 0$; the injection $I \hookrightarrow K(A)$ forces $I = 0$. Complete this to

$$0 \longrightarrow \mathcal{O}_X \xrightarrow{\alpha} \mathcal{F}(aH) \longrightarrow \mathcal{G} \longrightarrow 0, \qquad \operatorname{rank}\mathcal{G} = \operatorname{rank}\mathcal{F} - 1.$$

After twisting by $\mathcal{O}_X((m-a)H + D)$, the long exact sequence gives the claim for $\mathcal{F}$ from the claims for $\mathcal{O}_X$ and $\mathcal{G}$. It is enough to take

$$m(\mathcal{F}, H) \geq a + \max\{m(\mathcal{O}_X, H), m(\mathcal{G}, H)\}.$$

Step 6: Let $f : Y \to X$ be a projective resolution of singularities. The coherent sheaf $f_*\omega_Y$ is torsion-free of rank one. For $b \gg 0$, its dual twisted by bH is globally generated. Choose a section nonzero at the generic point to obtain an injection

$$f_*\omega_Y \longrightarrow \mathcal{O}_X(bH).$$

The kernel is zero because it is generically zero and is contained in a torsion-free sheaf. The cokernel $\mathcal{Q}$ has smaller-dimensional support, giving

$$0 \longrightarrow f_*\omega_Y \longrightarrow \mathcal{O}_X(bH) \longrightarrow \mathcal{Q} \longrightarrow 0.$$

For $m > 0$ and D nef, $mH + D$ is ample by Corollary 1.4.7. Kollár's vanishing theorem [Laz04a, Theorem 4.3.8] gives

$$H^i(X, f_*\omega_Y \otimes \mathcal{O}_X(mH + D)) = 0 \qquad (i > 0).$$

The dimension induction hypothesis applies to $\mathcal{Q}$, uniformly in D. Thus, for $m \geq \max\{1, m(\mathcal{Q}, H)\}$,

$$H^i(X, \mathcal{O}_X((b + m)H + D)) = 0 \qquad (i > 0).$$

This proves the case $\mathcal{F} = \mathcal{O}_X$ and completes all the reductions. □

Remark 1.4.43. Fujita vanishing also holds over algebraically closed fields of positive characteristic; see [Fuj11]. The resolution-and-vanishing proof above is a characteristic-zero proof, so a different argument is needed there.

Theorem 1.4.44 (Higher cohomology of nef Cartier divisors, II)**.** *Let X be a projective variety or scheme of dimension n, and let D be a nef Cartier divisor. For every coherent sheaf $\mathcal{F}$ and every $0 \leq i \leq n$,*

$$h^i(X, \mathcal{F}(mD)) = O(m^{n-i}) \qquad (m \to +\infty).$$

For $i > n$ the cohomology is zero.

Proof. The case $i = 0$ follows from Example 1.2.24, without using nefness. For higher cohomology, induct on n, with dimension zero immediate. By Fujita vanishing, choose a fixed very ample divisor class H such that

$$H^i(X, \mathcal{F}(mD + H)) = 0 \qquad (i > 0,\ m \geq 0).$$

Here the variable m multiplies the nef divisor; it is the fixed ample multiple used to choose H that is made sufficiently large. Choose a general member of $|H|$ avoiding every associated subvariety of $\mathcal{F}$ and of $\mathcal{O}_X$, and denote this effective Cartier divisor again by H. Multiplication by its section is injective on $\mathcal{F}$, so

$$0 \longrightarrow \mathcal{F}(mD) \longrightarrow \mathcal{F}(mD + H) \longrightarrow \mathcal{F}(mD + H) \otimes \mathcal{O}_H \longrightarrow 0.$$

For $i \geq 1$, the long exact sequence gives

$$\begin{aligned} h^i(X, \mathcal{F}(mD)) &\leq h^{i-1}(H, (\mathcal{F}(H) \otimes \mathcal{O}_H)(mD)) \\ &= O(m^{(n-1)-(i-1)}) = O(m^{n-i}) \end{aligned}$$

by induction, since $D|_H$ is nef. The cohomology groups are isomorphic for $i \geq 2$; for $i = 1$ one has only a surjection from the group on H. No lower bound on the nef exponent m is needed for the chosen vanishing. □

Remark 1.4.45. The associated-prime condition is exactly what ensures injectivity. On an affine open, write $\mathcal{F} = \widetilde{M}$ and let h be a local equation for H. After trivializing the line-bundle factors, the map $\mathcal{F}(mD) \to \mathcal{F}(mD + H)$ is multiplication by h on M. It is injective if and only if h is not a zero divisor on M, equivalently

$$h \notin \bigcup_{\mathfrak{p} \in \operatorname{Ass} M} \mathfrak{p}.$$

If h lies in an associated prime, there is some nonzero element of M annihilated by h; this need not hold for every element. This is why we choose the general divisor to avoid the associated subvarieties.

Corollary 1.4.46 (Asymptotic Riemann–Roch for nef Cartier divisors)**.** *Let X be a projective scheme of dimension n and D a nef Cartier divisor. Then*

$$h^0(X, \mathcal{O}_X(mD)) = \frac{D^n \cdot [X]_n}{n!} m^n + O(m^{n-1}),$$

where $[X]_n$ is the top-dimensional cycle, including its multiplicities. For integral X, the usual notation is

$$h^0(X, \mathcal{F}(mD)) = \operatorname{rank}(\mathcal{F}) \frac{D^n}{n!} m^n + O(m^{n-1}).$$

For an arbitrary projective scheme and coherent $\mathcal{F}$, the general form is

$$h^0(X, \mathcal{F}(mD)) = \frac{D^n \cdot Z_n(\mathcal{F})}{n!} m^n + O(m^{n-1}),$$

with

$$Z_n(\mathcal{F}) = \sum_{\dim V = n} \operatorname{length}_{\mathcal{O}_{X,\eta_V}} (\mathcal{F}_{\eta_V})[V].$$

In particular, if $\mathcal{F}$ is locally free of constant rank r, then $Z_n(\mathcal{F}) = r[X]_n$, recovering $rD^n/n!$ with the usual scheme-theoretic intersection convention.

Proof. Asymptotic Riemann–Roch gives the stated leading coefficient for $\chi(X, \mathcal{F}(mD))$. Theorem 1.4.44 makes every higher-cohomology contribution $O(m^{n-1})$, so the same leading coefficient holds for h^0. In dimension zero the formula is the exact length formula. □

1.5 Ampleness in the Relative Setting

Ampleness can be a relative concept, in the spirit of Grothendieck's relative point of view. Let $f : X \to T$ be a proper morphism of schemes and $\mathcal{F}$ a coherent sheaf on X. Our finite-type convention makes T noetherian and quasi-compact. The sheaf $f_*\mathcal{F}$ is coherent, and we can form

$$\pi : \mathbb{P}(f_*\mathcal{F}) := \operatorname{Proj}_T \operatorname{Sym}(f_*\mathcal{F}) \longrightarrow T.$$

Its fibre over t is the projective space of one-dimensional quotients of $(f_*\mathcal{F}) \otimes \kappa(t)$. This vector space need not equal $H^0(X_t, \mathcal{F}_t)$ unless the relevant cohomology-and-base-change statement holds. If $f_*\mathcal{F}$ is free of rank $n+1$ on $U \subseteq T$, then $\pi^{-1}(U) \simeq \mathbb{P}^n_U$. The natural evaluation map

$$f^* f_* \mathcal{F} \longrightarrow \mathcal{F}$$

is the relative analogue of global generation. For $T = \operatorname{Spec} \mathbb{C}$, it is $H^0(X, \mathcal{F}) \otimes_{\mathbb{C}} \mathcal{O}_X \to \mathcal{F}$, as in Example 1.4.4.

1.5A Relative Ampleness

Definition 1.5.1 (Ampleness for a map)**.** Let $f : X \to T$ be proper and L a line bundle on X.

(i) L is f-free if $\rho : f^* f_* L \to L$ is surjective. It is f-very ample if, in addition, the resulting morphism
$$j : X \longrightarrow \mathbb{P}(f_* L)$$
is a closed immersion over T.

(ii) L is f-ample if $L^{\otimes m}$ is f-very ample for some $m > 0$.

An integral Cartier divisor is relatively ample or very ample when its associated line bundle is. Relative ampleness for $\mathbb{Q}$-Cartier divisors is defined by a positive multiple clearing denominators.

Remark 1.5.2 (Relative ampleness is local on the base)**.** Both properties are local on T. For very ampleness, one can equivalently use an open cover $\{U_\alpha\}$ and closed immersions[25]

$$j_\alpha : X_\alpha = f^{-1}(U_\alpha) \hookrightarrow \mathbb{P}^{n_\alpha}_{U_\alpha}, \qquad L|_{X_\alpha} \simeq j_\alpha^* \mathcal{O}(1).$$

See [GW20, 13.56 and 13.63]. For ampleness, the exponents can be made uniform on a finite subcover, using a common positive multiple. If $U \subseteq T$ is affine, L is f-ample over U exactly when $L|_{f^{-1}(U)}$ is ample in the absolute sense.

A proper morphism with a relatively ample line bundle is projective in the convention of embedding into $\mathbb{P}(\mathcal{E})$ for a coherent sheaf $\mathcal{E}$ on T. This is the convention used here. An embedding into one fixed $\mathbb{P}^N_T$ is a stronger global formulation over a general base; locally on T it is available. Compare Stacks, Tags 01VV and 01W7.

Example 1.5.3. For a vector bundle $\mathcal{E}$ on T, the Serre line bundle $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$ is relatively very ample, hence ample, for $\pi : \mathbb{P}(\mathcal{E}) \to T$.

The cohomological properties are as follows.

[25] For relative ampleness and projective morphisms, see [SP25, Tag 01VV and 01W7].

Theorem 1.5.4. *Let $f : X \to T$ be proper and L a line bundle. The following are equivalent:*

(i) L is f-ample.

(ii) For every coherent $\mathcal{F}$, there is $m_1(\mathcal{F}) > 0$ such that

$$R^i f_*(\mathcal{F} \otimes L^{\otimes m}) = 0 \quad (i > 0,\ m \geq m_1(\mathcal{F})).$$

(iii) For every coherent $\mathcal{F}$, there is $m_2(\mathcal{F}) > 0$ such that

$$f^* f_*(\mathcal{F} \otimes L^{\otimes m}) \longrightarrow \mathcal{F} \otimes L^{\otimes m}$$

is surjective for every $m \geq m_2(\mathcal{F})$.

(iv) There is $m_3 > 0$ such that $L^{\otimes m}$ is f-very ample for all $m \geq m_3$.

These are the relative Serre criteria; they can be checked on a finite affine cover of the noetherian base.

Example 1.5.5. Suppose $Y \xrightarrow{\mu} X \xrightarrow{f} T$, where μ is finite, and put $g = f \circ \mu$. If L is f-ample, then $\mu^* L$ is g-ample.

Proof. Take an affine cover $\{U_i\}$ of T. Then $L|_{f^{-1}(U_i)}$ is ample, and the restriction $g^{-1}(U_i) \to f^{-1}(U_i)$ of μ is finite. Finite pullback preserves ampleness, without a surjectivity assumption. Thus $\mu^* L$ is relatively ample on every U_i, hence on T. □

Theorem 1.5.6 (Fibrewise ampleness)**.** *Let $f : X \to T$ be proper and L a line bundle. For a closed point $t \in T$, put $X_t = f^{-1}(t)$ and $L_t = L|_{X_t}$. Then L is f-ample if and only if L_t is ample for every closed point $t \in T$. Equivalently, one may test every geometric fibre.*

Proof. If L is f-ample, restrict a relative embedding to a fibre. A positive power of L_t is very ample, so L_t is ample. Conversely, the last step in the proof of ampleness in families (Theorem 1.2.11) gives, near each closed t, an open neighbourhood U_t, an integer $m_t > 0$ and a finite morphism

$$\varphi_t : X_{U_t} \longrightarrow \mathbb{P}^{r_t} \times U_t$$

with

$$L^{\otimes m_t}|_{X_{U_t}} \simeq \varphi_t^* \mathcal{O}_{\mathbb{P}^{r_t} \times U_t}(1).$$

The preceding example makes this power relatively ample over U_t, and hence so is L. These open sets cover T: their complement, if nonempty, is a closed subset of a finite-type scheme over a field and contains a closed point. Relative ampleness is local on the base, so the claim follows. The formulation with geometric fibres follows by base change and descent of ampleness under field extension. □

Corollary 1.5.7 (Relative Nakai–Moishezon criterion)**.** *For a proper $f : X \to T$, a $\mathbb{Q}$-Cartier divisor D is f-ample if and only if*

$$(D^{\dim V} \cdot V) > 0$$

for every positive-dimensional irreducible closed subvariety $V \subseteq X$ which maps to a closed point of T.

The intersection is computed on that proper fibre, using the associated line bundles. This avoids requiring the Cartier divisor itself to restrict as a Cartier divisor when its support contains V.

Proposition 1.5.8. *Let $f : X \to T$ be a morphism of projective schemes, L a line bundle on X, and A an ample line bundle on T. Then L is f-ample if and only if $L \otimes f^*A^{\otimes m}$ is ample for all $m \gg 0$. For the reverse implication, one such m already suffices.*

Proof. "$\Rightarrow$": Choose $r > 0$ such that $L^{\otimes r}$ is f-free. For $p \gg 0$, the coherent sheaf $f_*L^{\otimes r} \otimes A^{\otimes p}$ is globally generated. Pull back its generators and use the evaluation surjection to obtain an everywhere-defined morphism

$$\varphi : X \longrightarrow \mathbb{P} \times T, \quad \mathbb{P} = \mathbb{P}H^0(T, f_*L^{\otimes r} \otimes A^{\otimes p}),$$

with

$$L^{\otimes r} \otimes f^*A^{\otimes p} \simeq \varphi^* \operatorname{pr}_1^* \mathcal{O}_{\mathbb{P}}(1).$$

This morphism is finite: it is proper, and on every fibre of f the pulled-back $\mathcal{O}(1)$ is the ample $L_t^{\otimes r}$, so a positive-dimensional fibre of φ would contain a curve of degree zero, a contradiction. For every integer $q \geq 1$,

$$L^{\otimes r} \otimes f^*A^{\otimes (p+q)} \simeq \varphi^*(\mathcal{O}_{\mathbb{P}}(1) \boxtimes A^{\otimes q})$$

is ample by finite pullback. In particular, for all $m \gg 0$, take $q = rm - p \geq 1$. Then

$$(L \otimes f^*A^{\otimes m})^{\otimes r} = L^{\otimes r} \otimes f^*A^{\otimes rm}$$

is ample, proving the assertion for the original L.

"$\Leftarrow$": Restriction to X_t gives

$$(L \otimes f^*A^{\otimes m})|_{X_t} \simeq L_t,$$

up to tensoring by the one-dimensional vector space $A(t)^{\otimes m}$. Thus each L_t is ample, and Theorem 1.5.6 applies. □

1.5B Intersections in the Relative Setting

Corollary 1.5.7 characterizes relative ampleness by intersections on fibres. This motivates relative numerical equivalence. Let $f : X \to T$ be proper. As in Section 1.3, Cartier notation agrees with line-bundle notation on varieties and on schemes where line bundles have Cartier representatives. For arbitrary schemes without this property, the numerical groups and real positivity statements below are understood using numerical classes of line bundles. This gives the same formulas wherever Cartier representatives are available.

Definition 1.5.9 (Relative subvariety)**.** A closed irreducible subvariety $Z \subseteq X$ is relative over T if $f(Z)$ is a closed point of T.

Such a Z is proper over the ground field, so its intersection numbers with line bundles on X are defined even when X itself is not proper absolutely.

Definition 1.5.10 (Relative numerical equivalence)**.** Two Cartier divisors D_1, D_2 are relatively numerically equivalent, written $D_1 \equiv_T D_2$ or $D_1 \equiv_f D_2$, if

$$(D_1 \cdot C) = (D_2 \cdot C) \quad \text{for every relative curve } C \subseteq X.$$

The subgroup of relatively numerically trivial divisors is $\operatorname{Num}(X/T) = \{D : D \equiv_T 0\}$. The analogous definition for line bundles uses their degrees on relative curves.

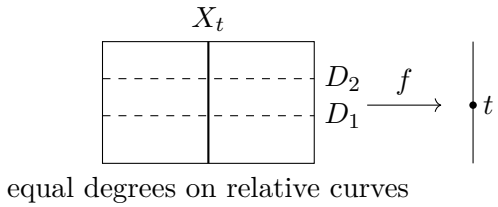


Figure 1.21: Relative numerical equivalence.

Definition 1.5.11 (Relative numerical divisor group)**.** In Cartier notation, the relative numerical divisor group is

$$N^1(X/T) = \operatorname{Div}(X)/\operatorname{Num}(X/T).$$

It is also called the relative Néron–Severi group with the numerical convention adopted in these notes. On arbitrary schemes, use $\operatorname{Pic}(X)$ modulo line bundles of degree zero on all relative curves, as specified above. Set $\rho(X/T) = \operatorname{rank} N^1(X/T)$.

If X is complete, absolute numerical triviality implies relative numerical triviality, so there is a surjection

$$N^1(X) \longrightarrow N^1(X/T), \qquad N^1(X/T) \simeq \frac{\operatorname{Div}(X)/\operatorname{Num}(X)}{\operatorname{Num}(X/T)/\operatorname{Num}(X)}$$

in Cartier notation. This proves finite generation in this case. For non-proper X, one cannot appeal to the absolute group as previously defined only for complete schemes.

For completeness, over $\mathbb{C}$ one can see the general finiteness as follows. By Chow's lemma and resolution, take a proper surjective morphism $\mu : Y \to X$ with Y a finite disjoint union of smooth quasi-projective varieties. Choose smooth projective compactifications $\overline{Y}$. Every line bundle on Y extends to $\overline{Y}$ by taking the closure of a Cartier divisor on the smooth compactification. Thus the finite numerical groups of $\overline{Y}$ surject onto the numerical groups on Y tested against curves relative to T. Pullback from X is injective on these numerical groups: a relative curve on X has a curve above it mapping onto it with positive degree, and the projection formula detects its degree. Hence $N^1(X/T)$ is finitely generated and torsion-free. In particular, $\rho(X/T)$ is finite.

Definition 1.5.12 (Real coefficients)**.** For real Cartier divisors, write $D_1 \equiv_T D_2$ if their degrees on every relative curve agree, with intersections extended real-linearly. The resulting vector space is

$$N^1(X/T)_{\mathbb{R}} = N^1(X/T) \otimes_{\mathbb{Z}} \mathbb{R}.$$

The corresponding definition applies to real linear combinations of line bundles under the general scheme convention.

The identification follows as in Example 1.3.11: the relative curve conditions are homogeneous linear equations with integral coefficients.

Definition 1.5.13 (Numerical classes of relative curves)**.** Let $f : X \to T$ be proper. The space $Z_1(X/T)_{\mathbb{R}}$ consists of finite real sums $\gamma = \sum_i a_i[C_i]$ of relative irreducible curves. Two such cycles are numerically equivalent if

$$(D \cdot \gamma_1) = (D \cdot \gamma_2)$$

for every Cartier divisor D on X (or every line bundle, with the general scheme convention). Denote the quotient by $N_1(X/T)_{\mathbb{R}}$. There is a perfect pairing

$$N^1(X/T)_{\mathbb{R}} \times N_1(X/T)_{\mathbb{R}} \longrightarrow \mathbb{R}.$$

In particular both spaces are finite-dimensional, and we give them their Euclidean topology. Integral and rational cycle classes are defined similarly.

For a proper morphism $g : Y \to X$ over T and a relative integral subvariety $Z \subseteq Y$, proper pushforward is

$$g_*[Z] = \begin{cases} 0, & \dim g(Z) < \dim Z, \\ d[g(Z)], & \dim g(Z) = \dim Z, \end{cases} \qquad d = [k(Z) : k(g(Z))].$$

The target cycle lies on X, and the projection formula holds.

Proposition 1.5.14. *Let $f : X \to T$ and $g : Y \to X$ be proper.*

(1) If $L \equiv_T 0$, then $g^*L \equiv_T 0$. Thus pullback induces a group homomorphism

$$g^* : N^1(X/T) \longrightarrow N^1(Y/T),$$

and an $\mathbb{R}$-linear map after tensoring with $\mathbb{R}$.

(2) If g is surjective, this map is injective.

Proof. For (1), take a relative curve $C \subseteq Y$. If g contracts it, $(g^*L \cdot C) = 0$. Otherwise

$$(g^*L \cdot C) = (L \cdot g_*[C]) = d(L \cdot g(C)) = 0.$$

For (2), every relative curve $C \subseteq X$ is dominated by a relative curve $C' \subseteq Y$ of positive degree d, as in Example 1.4.3(ii). Thus

$$(L \cdot C) = \frac{1}{d}(g^*L \cdot C'),$$

which proves the assertion. For $T = \operatorname{Spec} \mathbb{C}$ this is the absolute statement. □

Definition 1.5.15. A real Cartier divisor is f-ample if it is a finite positive real combination of f-ample integral Cartier divisors. The rational definition is analogous and agrees with clearing denominators. For real line-bundle classes, make the same definition in $\operatorname{Pic}(X) \otimes \mathbb{R}$; this is the interpretation in general scheme statements. When T is a point, these are the absolute definitions.

Proposition 1.5.16 (Openness of relative ampleness)**.** *Let $f : X \to T$ be a projective morphism, and let H be an f-ample real divisor (or real line-bundle class). Given finitely many real divisors $E_1, \ldots, E_r$, the class*

$$H + \varepsilon_1 E_1 + \cdots + \varepsilon_r E_r$$

is f-ample for all sufficiently small real ε_i.

Proof. **Step 1:** First assume $H, E_1, \ldots, E_r$ integral. Relative Serre ampleness, checked on a finite affine cover of T, makes all the $2r$ divisors $mH \pm E_i$ relatively ample for one sufficiently large m. This is the relative form of Example 1.2.5. When X, T are projective, we can also use the original absolute argument: choose ample A on T and $n \gg 0$ so that $H + nf^*A$ is ample by Proposition 1.5.8. Then

$$m(H + nf^*A) \pm E_i = (mH \pm E_i) + mnf^*A$$

is ample for $m \gg 0$, and restriction to fibres makes $mH \pm E_i$ relatively ample. For any real ε_i with $m \sum_i |\varepsilon_i| < 1$, use the exact decomposition

$$H + \sum_i \varepsilon_i E_i = \left(1 - m \sum_i |\varepsilon_i|\right) H + \sum_{\varepsilon_i \neq 0} m|\varepsilon_i| \left(H + \frac{\operatorname{sgn}(\varepsilon_i)}{m} E_i\right).$$

Every divisor in parentheses is relatively ample, so the positive combination is relatively ample. This argument handles real, not just rational, perturbation coefficients. Clearing denominators handles rational H, E_i as well.

Step 2: Write $H = \sum_j a_j A_j$, with $a_j > 0$ and integral relatively ample A_j. Expand the finitely many real E_i in finitely many integral divisors; small original coefficients give small coefficients in this larger finite list. We may therefore assume the E_i integral. Fix a rational $0 < c < a_1$ and write

$$H + \sum_i \varepsilon_i E_i = c\left(A_1 + \sum_i \frac{\varepsilon_i}{c} E_i\right) + (a_1 - c)A_1 + \sum_{j \geq 2} a_j A_j.$$

For sufficiently small ε_i, Step 1 makes the first parenthesized divisor relatively ample; the remaining terms are already positive combinations of relatively ample divisors. This proves the claim. □

1.5C Nefness in the Relative Setting

We now take the relative analogue of nefness.

Definition 1.5.17 (Relative nefness)**.** For a proper morphism $f : X \to T$, a line bundle L is f-nef if its restrictions to the geometric fibres are nef. Equivalently, in our finite-type setting one can test the inequalities

$$\int_C c_1(L) \geq 0$$

on every relative curve $C \subseteq X$. The definition extends linearly to real Cartier divisors or real line-bundle classes.

Remark 1.5.18. These definitions and the relative ampleness and nefness results also hold over algebraically closed ground fields of arbitrary characteristic. For general numerical finiteness in that setting, use the relative Néron–Severi finiteness theorem; the resolution argument given above was a proof over $\mathbb{C}$. This remark does not change the ground field of any of our already defined varieties.

Theorem 1.5.19 (Relative Kleiman theorem)**.** *Let $f : X \to T$ be proper. If D is an f-nef real Cartier divisor, then*

$$(D^k \cdot V) \geq 0$$

for every relative irreducible subvariety V of dimension k. For every f-nef line bundle L, one likewise has

$$\int_V c_1(L)^{\dim V} \geq 0.$$

Proof. If $f(V) = t$, compute the intersection on the complete fibre X_t:

$$(D^k \cdot V) = (D_t^k \cdot V) \geq 0$$

by the absolute Kleiman theorem. Here D_t means restriction of the associated real line-bundle class, so it is defined even when the Cartier support contains the fibre. □

Corollary 1.5.20. *Let $f : X \to T$ be projective, D an f-nef real divisor and H an f-ample real divisor. Then $D + \varepsilon H$ is f-ample for every $\varepsilon > 0$. Conversely, if D, H are real divisors and $D + \varepsilon H$ is f-ample for all sufficiently small $\varepsilon > 0$, then D is f-nef.*

Proof. "⇐": For every relative curve C,

$$(D \cdot C) + \varepsilon(H \cdot C) > 0.$$

Let $\varepsilon \to 0$ to get $D \cdot C \geq 0$.

"⇒": On each projective fibre, the absolute Corollary 1.4.7 makes $D_t + \varepsilon H_t$ ample. Consequently

$$((D + \varepsilon H)^{\dim V} \cdot V) > 0$$

for every positive-dimensional relative subvariety V. For rational classes the conclusion follows from Corollary 1.5.7. For real classes we use the real version of the relative Nakai–Moishezon criterion [FM21, Theorem 1.5], which has the same intersection condition for real line bundles on proper morphisms. Thus $D + \varepsilon H$ is f-ample. □

The same relative numerical criterion shows that f-ampleness of real classes depends only on relative numerical equivalence. Other relative statements below will be checked with their own hypotheses; one should not infer every absolute assertion simply by substituting relative curves.

1.5D Kleiman's Criterion in the Relative Setting

Definition 1.5.21 (Relative ample and nef cones)**.** For a proper $f : X \to T$, the relative ample cone $\mathrm{Amp}(X/T) \subseteq N^1(X/T)_{\mathbb{R}}$ is the cone of f-ample real classes, and the relative nef cone $\mathrm{Nef}(X/T)$ is the cone of f-nef classes.

The ample cone consists of finite positive combinations of relatively ample integral (or rational) classes. As in Section 1.4, we do not add 0 to it by convention; in relative dimension zero the zero class itself can be relatively ample.

Theorem 1.5.22. *Let $f : X \to T$ be projective. Then*

(1) $\mathrm{Nef}(X/T) = \overline{\mathrm{Amp}(X/T)}$*;*

(2) $\mathrm{Amp}(X/T) = \mathrm{int}(\mathrm{Nef}(X/T))$.

Proof. The relative nef cone is closed, since its defining inequalities are closed half-spaces. Proposition 1.5.16 gives openness of relative ampleness. More explicitly, apply it to a finite set of integral classes spanning $N^1(X/T)_{\mathbb{R}}$. Thus

$$\overline{\mathrm{Amp}(X/T)} \subseteq \mathrm{Nef}(X/T), \qquad \mathrm{Amp}(X/T) \subseteq \mathrm{int}(\mathrm{Nef}(X/T)).$$

Fix an f-ample integral class H. If D is f-nef, then $D + \varepsilon H$ is f-ample for all $\varepsilon > 0$ by Corollary 1.5.20. Taking the limit gives the first reverse inclusion. If $[D] \in \mathrm{int}(\mathrm{Nef}(X/T))$, then $D - \varepsilon H$ is f-nef for $0 < \varepsilon \ll 1$, and

$$D = (D - \varepsilon H) + \varepsilon H$$

is f-ample by the same corollary. This is the relative version of the proof of Theorem 1.4.21. □

Definition 1.5.23 (Relative cone of curves)**.** The cone of relative curves is

$$\mathrm{NE}(X/T) = \left\{ \sum_i a_i [C_i] : a_i \geq 0,\ C_i \text{ relative irreducible curves} \right\} \subseteq N_1(X/T)_{\mathbb{R}}.$$

Its closure $\overline{\mathrm{NE}}(X/T)$ is the closed cone of relative curves.

Remark 1.5.24. The cone $\mathrm{NE}(X/T)$ need not be closed.

Proposition 1.5.25. *The closed relative cone of curves is dual to the relative nef cone:*

$$\overline{\mathrm{NE}}(X/T) = \{\gamma \in N_1(X/T)_{\mathbb{R}} : (\delta \cdot \gamma) \geq 0 \text{ for every } \delta \in \mathrm{Nef}(X/T)\}.$$

Proof. By definition and continuity, $\overline{\mathrm{NE}}(X/T)^* = \mathrm{Nef}(X/T)$. Apply Lemma 1.4.26 to the closed convex cone $\overline{\mathrm{NE}}(X/T)$. □

Theorem 1.5.26 (Relative Kleiman criterion)**.** *Let $f : X \to T$ be projective and D a real divisor. Then D is f-ample if and only if*

$$\overline{\mathrm{NE}}(X/T) \setminus \{0\} \subseteq D_{>0}.$$

For any f-ample integral divisor H and any integer k, the set

$$W_k = \{\gamma \in \overline{\mathrm{NE}}(X/T) : (H \cdot \gamma) \leq k\}$$

is compact. In particular it contains only finitely many numerical classes of effective relative integral one-cycles of H-degree at most k.

Proof. "⇒": If D is f-ample, it is non-negative on the closed cone by continuity. If $D \cdot \gamma = 0$ for some nonzero $\gamma \in \overline{\mathrm{NE}}(X/T)$, the perfect pairing gives a real divisor E with $E \cdot \gamma < 0$. Hence $(D + tE) \cdot \gamma < 0$ for every $t > 0$. This contradicts Proposition 1.5.16, which makes $D + tE$ relatively ample for sufficiently small $t > 0$.

"⇐": Choose a norm and put

$$S = \{\gamma \in \overline{\mathrm{NE}}(X/T) : \|\gamma\| = 1\}.$$

This is a compact subset of the unit sphere. When it is nonempty, choose bounds $D \cdot \gamma \geq a > 0$ and $H \cdot \gamma \leq b$, $b > 0$, for a fixed relatively ample integral H. Then

$$(D - \tfrac{a}{b}H) \cdot \gamma = D \cdot \gamma - \tfrac{a}{b}(H \cdot \gamma) \geq 0 \quad (\gamma \in S).$$

By scaling, $D - \frac{a}{b}H$ is f-nef, and Corollary 1.5.20 proves $D = (D - \frac{a}{b}H) + \frac{a}{b}H$ relatively ample. If S is empty, there are no relative curves: any such curve would have positive H-degree and give a point of S. Thus all fibres are zero-dimensional and f is finite. Every line bundle is then relatively ample, so the assertion holds also in this case.

For compactness, W_k is closed and is empty if $k < 0$. If it were unbounded for $k \geq 0$, choose $\gamma_i \in W_k$ with $\|\gamma_i\| \to \infty$ and pass to a subsequence such that

$$\frac{\gamma_i}{\|\gamma_i\|} \longrightarrow \gamma \in \overline{\mathrm{NE}}(X/T), \qquad \|\gamma\| = 1.$$

Then

$$H \cdot \gamma = \lim_i \frac{H \cdot \gamma_i}{\|\gamma_i\|} = 0,$$

contrary to relative ampleness. Finally take an integral basis over $\mathbb{R}$ of relative numerical divisor classes. Pairing with it embeds integral relative curve classes into $\mathbb{Z}^{\rho(X/T)}$, proving their discreteness and therefore the asserted finiteness. □

We finish by comparing relative classes with absolute classes when the source and target are projective.

Proposition 1.5.27. *Let $f : X \to Y$ be a morphism of projective varieties. Then:*[26]

(1) Proper pushforward induces an $\mathbb{R}$-linear map $f_* : N_1(X)_{\mathbb{R}} \to N_1(Y)_{\mathbb{R}}$.

(2) $f_* \mathrm{NE}(X) \subseteq \mathrm{NE}(Y)$ and $f_* \overline{\mathrm{NE}}(X) \subseteq \overline{\mathrm{NE}}(Y)$.

(3) There is a natural identification of $N_1(X/Y)_{\mathbb{R}}$ with the subspace of $N_1(X)_{\mathbb{R}}$ spanned by the contracted curves, and

$$N_1(X/Y)_{\mathbb{R}} \subseteq \ker f_*.$$

Equality need not hold.

[26] For the separating hyperplane theorem, see [BV04, 2.5.1].

(4) Under this identification,

$$\overline{\mathrm{NE}}(X/Y) = \overline{\mathrm{NE}}(X) \cap N_1(X/Y)_{\mathbb{R}}.$$

Proof. For (1), well-definedness follows from the projection formula: if $\gamma \equiv 0$ on X, then for every divisor D on Y,

$$D \cdot f_*\gamma = f^*D \cdot \gamma = 0.$$

For (2), the pushforward of an effective cycle is effective or zero. The second inclusion follows by continuity and passage to limits.

For (3), numerical equivalence of relative cycles tests exactly the same divisors on X as absolute numerical equivalence does. This gives the injection into $N_1(X)_{\mathbb{R}}$. Every contracted curve has zero pushforward, so the inclusion in $\ker f_*$ follows. For a concrete example of strict inclusion, take an elliptic curve E and $f : E \times E \to E$ the second projection. All its fibres have the same numerical class, so $N_1(X/Y)_{\mathbb{R}} = \mathbb{R}[E \times \{0\}]$ has dimension one. The classes

$$F_1 = E \times \{0\}, \quad F_2 = \{0\} \times E, \quad \Delta$$

are linearly independent: their intersection matrix has zero diagonal and all off-diagonal entries equal to one, hence determinant two. But $f_*F_2 = f_*\Delta = [E]$, so $[\Delta] - [F_2] \in \ker f_*$ and is not a multiple of $[F_1]$. Compare [KM98, 1.46].

For (4), the inclusion from left to right is immediate. Suppose a class ξ belongs to the right-hand side but not to $K = \overline{\mathrm{NE}}(X/Y)$. Separation gives a linear functional ℓ on $N_1(X/Y)_{\mathbb{R}}$ with $\ell \geq 0$ on K and $\ell(\xi) < 0$. Add a sufficiently small positive multiple of a relatively ample functional. The resulting functional is strictly positive on $K \setminus \{0\}$ and still negative at ξ. These conditions persist under a sufficiently small perturbation: use the compact unit slice of K for uniform strict positivity. Thus it may be chosen rational, and represented by a $\mathbb{Q}$-Cartier divisor H on X.

The relative Kleiman criterion makes H f-ample. By Proposition 1.5.8, a sufficiently ample integral divisor M on Y makes $H + f^*M$ ample on X, after clearing denominators if needed. Since $\xi \neq 0$ lies in $\overline{\mathrm{NE}}(X)$, absolute Kleiman gives

$$0 < (H + f^*M) \cdot \xi = H \cdot \xi + M \cdot f_*\xi = H \cdot \xi < 0,$$

a contradiction. This proves (4). □

Chapter 2
Rational Curves & Bend and Break

2.1 Parametrizing Morphisms

2.1A Rational Curves on the Exceptional Locus

The more rational curves X contains, the more complicated the birational geometry of X.

János Kollár

Rational curves play an important role in classifying algebraic varieties. For example, over an algebraically closed field of characteristic zero, a smooth projective surface has a minimal surface X_m with no (-1)-curves, as in Corollary 1.4.34. There are two cases:

- either X is not covered by rational curves, in which case its minimal surface is unique and K_{X_m} is nef;
- or X is covered by rational curves, in which case minimal surfaces need not be unique, and their canonical divisors are not nef.

The characteristic-zero assumption in this introductory classification is relevant; it is not a statement about all surfaces in positive characteristic.

[Bea78, II, 14(b)] gives a simple example for the second case. Start with $\mathbb{P}^2$ blown up at two distinct points to get $\widehat{Q}$, and then blow down the strict transform of the line joining them. This gives the smooth quadric surface $\mathbb{P}^1 \times \mathbb{P}^1$. Thus $\mathbb{P}^2$ and $\mathbb{P}^1 \times \mathbb{P}^1$ are two distinct birational minimal surfaces.

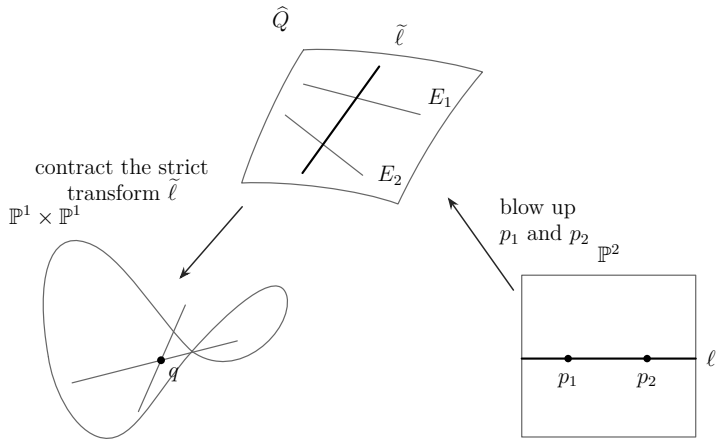


Figure 2.1: Two distinct minimal surfaces.

From now on, until the characteristic-zero convention in Section 2.3A, varieties are over an arbitrary algebraically closed field k. We follow [Deb01, 1.10]. A rational curve means an integral projective curve whose normalization is $\mathbb{P}^1_k$.

Definition 2.1.1. Let $\pi : Y \to X$ be a birational morphism of varieties. The exceptional locus $\mathrm{Ex}(\pi)$ is the locus of points of Y where π is not a local isomorphism. We denote it by E.[27]

The set E is closed, since the local-isomorphism locus is open. The rational inverse π^{-1} has its domain of definition in X, not in Y.

Assume now that X is normal and π is proper. Then π is surjective and has connected fibers, by Zariski's connectedness theorem [GW23, 24.58]. A zero-dimensional fiber is a single point; properness and the quasi-finite locus reduce a neighborhood to a finite birational morphism, which is an isomorphism because X is normal. Thus a fiber meets E exactly when it is positive-dimensional. Connectedness then implies that the entire fiber lies in E: a point where π is a local isomorphism would be isolated in that fiber. Consequently,

$$E = \pi^{-1}(\pi(E)), \qquad \pi(E) = \{x \in X \mid \dim \pi^{-1}(x) \geq 1\}.$$

The latter is closed in X, by upper semicontinuity of fiber dimension for a proper morphism. If it had a codimension-one component W, then

$$\dim \pi^{-1}(W) \geq \dim W + 1 = \dim Y.$$

This would force the proper closed subset $\pi^{-1}(W)$ to equal the irreducible variety Y, a contradiction. Hence $\operatorname{codim}_X \pi(E) \geq 2$. The largest open set on which the inverse is defined is $X \setminus \pi(E)$.

Remark 2.1.2 (Proper and birational is surjective)**.** Since π is proper, $\pi(Y)$ is closed. Since it is birational, this image contains a dense open subset of X. Thus $\pi(Y) = X$.[28]

Proposition 2.1.3 (Van der Waerden)**.** *Let $\pi : Y \to X$ be a birational morphism of varieties. If X is normal and $\mathbb{Q}$-factorial, then every irreducible component of* $\mathrm{Ex}(\pi)$ *has codimension one in Y.*[29]

Proof. Let y be a closed point of E and $x = \pi(y)$. Identify the function fields using π^*. The local inclusion $\mathcal{O}_{X,x} \subset \mathcal{O}_{Y,y}$ is strict; otherwise the morphism would be a local isomorphism at y. As both residue fields are k, we can choose $t \in \mathfrak{m}_y \setminus \mathcal{O}_{X,x}$ by subtracting its residue from an element outside $\mathcal{O}_{X,x}$.

[27] For fibre dimension and the dimension inequality used in describing the exceptional locus, see [Har77, II, Exercise 3.22(d)], [GW20, 14.112], [Sha13, p. 75], [Har77, I, Exercise 1.10(d)].

[28] For the closed-image property of a proper morphism, see [Har77, II, Exercise 4.4].

[29] For the identification of a morphism of line bundles with a section of their tensor quotient, see [Har77, II, Exercise 5.1(b)].

Write $\operatorname{div}_X(t) = D' - D''$ with effective Weil divisors having no common components. By $\mathbb{Q}$-factoriality, some mD' and mD'' are Cartier. Their local equations can be chosen so that

$$t^m = u/v, \qquad u, v \in \mathcal{O}_{X,x}.$$

Here $v \in \mathfrak{m}_x$, for otherwise $t^m \in \mathcal{O}_{X,x}$ and normality would imply $t \in \mathcal{O}_{X,x}$. Also $u = t^m v \in \mathfrak{m}_y \cap \mathcal{O}_{X,x} = \mathfrak{m}_x$. The subscheme $Z = (u = v = 0)$ has codimension two near x, since the two divisors have no common components. On a neighborhood of y where t is regular, its inverse image is defined by

$$t^m v = v = 0,$$

hence by the single nonzero equation $v = 0$. Every irreducible component through y of this local zero locus has codimension one. Such a component is exceptional: outside E, a local isomorphism cannot map a divisor into the codimension-two set Z. Thus every closed point of E lies on a codimension-one component of E. This proves the assertion. □

Ramification divisor of a morphism between smooth varieties

Assume X and Y are smooth of the same dimension n. The map

$$\pi^*\Omega_X \longrightarrow \Omega_Y$$

induces $\pi^*\omega_X \to \omega_Y$, hence a section of

$$\pi^*\omega_X^{-1} \otimes \omega_Y \cong \mathcal{O}_Y(K_Y - \pi^*K_X).$$

If π is dominant and generically separable, this section is nonzero. Its zero divisor is the effective ramification divisor $\operatorname{Ram}(\pi)$, locally cut out by the Jacobian determinant. Thus, choosing compatible canonical divisors,

$$K_Y = \pi^*K_X + \operatorname{Ram}(\pi).$$

For a birational morphism, separability is automatic and $\operatorname{Supp}\operatorname{Ram}(\pi) = \operatorname{Ex}(\pi)$. The multiplicities in $\operatorname{Ram}(\pi)$ need not be one.

Proposition 2.1.4 (Abhyankar's lemma)**.** *Let X and Y be varieties with X smooth, and let $\pi : Y \to X$ be a proper birational morphism which is not an isomorphism. Through a general point of each component of $\operatorname{Ex}(\pi)$, there is a rational curve contracted by π.*[30]

Proof. Replace Y by its normalization. This is finite over Y, and a nonconstant image of a rational curve is rational by Lüroth's theorem. Thus it suffices to work on the normalization and descend the curves. By Proposition 2.1.3, the

[30] For Abhyankar's lemma and the blow-up constructions used here, see [Deb01, 1.43], [KM98, 1.3], [Har77, II, 7.14], [Har77, II, 7.16], [Har77, II, 8.24].

exceptional components are divisors. Fix one such component E with generic point η_E; Y is smooth near η_E.

We construct successive blow-ups of the centers of E, working on neighborhoods of η_E. Let $X_0 = X$. As long as the center Z_{i-1} of E on X_{i-1} has codimension at least two, take an open neighborhood U_{i-1} of its generic point on which the center is smooth, and blow it up:

$$\varepsilon_i : X_i = \mathrm{Bl}_{Z_{i-1}\cap U_{i-1}} U_{i-1} \longrightarrow U_{i-1}.$$

The ideal of the center pulls back to a nonzero principal ideal in the DVR $\mathcal{O}_{Y,\eta_E}$. After shrinking a neighborhood V_i of η_E in Y, it is invertible there. The universal property of the blow-up gives a birational morphism $\pi_i : V_i \to X_i$ and the diagram

$$\begin{array}{ccccc} V_i & \xrightarrow{\pi_i} & X_i & \xrightarrow{\varepsilon_i} & U_{i-1} \\ & & & & \subseteq X_{i-1}. \end{array}$$

This is the precise reason that the lift exists; smoothness of E alone would not make the pulled-back ideal invertible everywhere.

Let E_i be the exceptional divisor of ε_i and

$$c_i = \mathrm{codim}_{U_{i-1}}(Z_{i-1} \cap U_{i-1}) - 1 > 0.$$

The blow-up formula gives

$$K_{X_i} = \varepsilon_i^* K_{U_{i-1}} + c_i E_i.$$

Moreover, $\pi_i^* E_i$ has positive integral multiplicity along E. Since the remaining ramification divisor on the smooth neighborhood V_i is effective, repeated pullback gives, at η_E,

$$K_Y - \pi^* K_X \ \geq \ (c_1 + \cdots + c_i)E.$$

More explicitly, each summand is c_j times the pullback of E_j, whose E-multiplicity is at least one. The fixed integer

$$a = \mathrm{ord}_E(K_Y - \pi^* K_X)$$

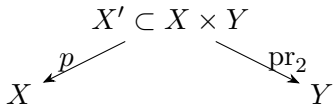
therefore satisfies $a \geq c_1 + \cdots + c_i \geq i$. The process must terminate.

At the last step, the center of E is a divisor on the smooth X_i, and it is the exceptional divisor E_i of the last blow-up. A birational morphism to a normal variety is an isomorphism at the generic point of a divisor that it dominates. Thus E is birational to E_i, which is a projective-space bundle over the smooth center of ε_i.

Through a general point of E_i choose a line in its projective-space fiber. The birational inverse to E gives a rational map from that line to Y, defined near the chosen point. Its composite with π is constant. Since $Y \to X$ is proper, the valuative criterion extends this map over the whole $\mathbb{P}^1$. Its image is a nonconstant rational curve through the chosen general point of E, contracted by π. Here we use the original proper morphism; shrinking Y did not make the resulting open neighborhood proper. $\square$

Corollary 2.1.5. *Let X be smooth and Y proper, with no rational curves. Every rational map $f : X \dashrightarrow Y$ is defined everywhere.*[31]

Proof. Let $X' \subset X \times Y$ be the closure of the graph. The first projection $p : X' \to X$ is proper and birational. If it is not an isomorphism, Proposition 2.1.4 supplies a rational curve contracted by p. The second projection also contracts it because Y contains no rational curves. But a curve in $X \times Y$ cannot be contracted by both projections. Thus $\mathrm{Ex}(p)$ is empty, p is an isomorphism, and f is a morphism.



An exceptional rational curve would map
to a point under both projections.

Figure 2.2: The two projections cannot both contract a curve in the graph.

□

Remark 2.1.6. The valuation-theoretic form of the termination argument is discussed in [KM98, 2.45] and the appendix of [Kol96]. In the smooth-target situation above, the fixed ramification coefficient at η_E supplies the required termination bound. Thus the proof must include this step; it is not enough simply to repeat the blow-ups indefinitely.[32]

Proposition 2.1.7 (Negativity lemma in its simplest form)**.** *Let X and Y be smooth projective varieties and $\pi : Y \to X$ a birational morphism which is not an isomorphism. There is a rational curve C on Y, contracted by π, such that $(K_Y \cdot C) < 0$.*[33]

Proof. Put $E = \mathrm{Ex}(\pi)$ and $B = \pi(E)$, so $\operatorname{codim}_X B \geq 2$. Choose a closed point $x \in B$. Using sufficiently high very ample multiples, take $\dim X - 2$ general hypersurface sections through x to cut out a smooth projective surface S such that $S \cap B$ is finite. Smoothness at the fixed point is justified by Lemma 2.1.8; proper intersection with B follows by general choice. When $\dim X = 2$, take $S = X$.

On S, choose two smooth irreducible members C_0, C of the same sufficiently ample linear system, with $C_0 \cap B = \{x\}$ and $C \cap B = \varnothing$. Hence

$$(K_X \cdot C_0) = (K_X \cdot C).$$

[31] This application of Abhyankar's lemma is also given in [KM98, 1.5].

[32] For the version of Abhyankar's lemma under discussion, see [KM98, 1.3].

[33] For connected hyperplane sections and the surface blow-up facts used in the reduction, see [Har77, III, 7.9], [Bea78, II.11], [Bea78, Lem II.2].

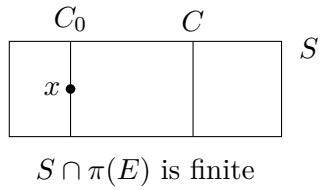


Figure 2.3: Construction of the two smooth curves C_0 and C.

Let $C' = \pi^{-1}(C)$ and let C_0' be the strict transform of C_0. Since C' avoids E,

$$(K_Y \cdot C') = (\pi^* K_X + \mathrm{Ram}(\pi)) \cdot C' = (K_X \cdot C).$$

The curve C_0' meets E: its proper map to C_0 is surjective, and the fiber above x lies in E. It is not contained in E, so

$$\begin{aligned}(K_Y \cdot C_0') &> (\pi^* K_X \cdot C_0') = (K_X \cdot C_0) \\ &= (K_X \cdot C) = (K_Y \cdot C').\end{aligned}$$

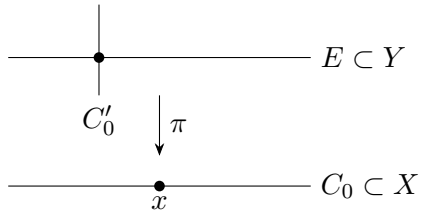


Figure 2.4: The strict transform C_0' meets the exceptional locus.

Resolve the rational map $\pi^{-1}|_S : S \dashrightarrow Y$ by point blow-ups:

$$g : \widetilde{S} \xrightarrow{\varepsilon} S \dashrightarrow Y, \qquad \pi \circ g = \iota_S \circ \varepsilon.$$

All centers lie over the finite set $S \cap B$. This resolves the map from the surface; it does not assert a factorization of the original higher-dimensional morphism π.

$$\begin{array}{ccc} \widetilde{S} & \xrightarrow{g} & Y \\ \varepsilon \downarrow & & \downarrow \pi \\ S & \hookrightarrow & X. \end{array}$$

Figure 2.5: Blowing up points, including possibly infinitely near points, to resolve the map.

Write $C'' = \varepsilon^* C$ and let C_0'' be the strict transform of C_0. Then $g_* C'' = C'$ and $g_* C_0'' = C_0'$, while

$$\varepsilon^* C_0 = C_0'' + \sum_i m_i E_i, \qquad m_i \geq 0.$$

Here the E_i are exceptional prime curves on the final surface. They are smooth rational curves, but need not all still have self-intersection -1; the m_i are their total-transform multiplicities. Since $C \sim_{\text{lin}} C_0$ on S, pushing the relation forward gives an equality modulo rational equivalence of 1-cycles on Y:

$$C' \sim_{\text{rat}} C'_0 + \sum_i m_i g_* E_i.$$

In particular,

$$(K_Y \cdot C') = (K_Y \cdot C'_0) + \sum_i m_i (K_Y \cdot g_* E_i).$$

The strict inequality above implies that some $(K_Y \cdot g_* E_i) < 0$. Its image is therefore a curve, rational by Lüroth's theorem, and it is contracted by π because $\varepsilon(E_i)$ is a point. The mapping degree in $g_* E_i$ is positive, so $K_Y \cdot g(E_i) < 0$ as required.

$$E_i \cong \mathbb{P}^1 \xrightarrow{g} g(E_i) \subset E \subset Y \xrightarrow{\pi} \{x_i\} \subset X.$$

Figure 2.6: A noncontracted image $g(E_i)$ with negative canonical degree.

□

When taking a first look at this argument, I thought the fixed-point Bertini step was trivial, but my advisor had doubts about it. We therefore give a complete proof here. The following formulation records exactly the sufficiently ample version used above.

Lemma 2.1.8. *Let X be a smooth projective variety of dimension at least two, $x \in X$ a closed point, and L a very ample divisor. For $m \gg 0$, a general member of $|mL|$ passing through x is smooth and irreducible. Thus it is a hyperplane section for the embedding given by $|mL|$.*[34]

Proof. Let $\pi : \widetilde{X} = \mathrm{Bl}_x X \to X$, with $E \cong \mathbb{P}^{n-1}$. For $m \gg 0$, $m\pi^* L - E$ is very ample. By Bertini, a general

$$H' \in |m\pi^* L - E|$$

is smooth and irreducible, and does not contain E. Its image H is smooth away from x. The pullback line bundle $\pi^* \mathcal{O}_X(mL)|_E$ is trivial, since E maps to a point, whereas $\mathcal{O}_{\widetilde{X}}(-E)|_E \cong \mathcal{O}_E(1)$. Therefore

$$\mathcal{O}_{\widetilde{X}}(H')|_E \cong \mathcal{O}_E(1).$$

[34] For the very ample system on the blow-up and Bertini's theorem, see [Har77, II, Exercise 7.14(b)], [Har77, III, 10.9].

Locally at x, the equation f of H has a nonzero linear initial form F, because its total transform has multiplicity exactly one along E. Thus

$$f = F + \text{terms of order at least two},$$

so H is smooth at x. The identity $\pi_*\mathcal{O}_{\widetilde{X}}(-E) = \mathfrak{m}_x$ identifies this linear system with the members of $|mL|$ through x, proving the general-member assertion. □

2.1B Parametrizing Rational Curves

Let k be algebraically closed, not necessarily of characteristic zero. For a projective variety $X \subseteq \mathbb{P}^N_k$, we aim to construct a scheme $\mathrm{Mor}(\mathbb{P}^1_k, X)$ parametrizing morphisms $\mathbb{P}^1_k \to X$. This gives a precise meaning to deforming a morphism.

To begin with, take $X = \mathbb{P}^N_k$, with $N \geq 1$. A morphism $f : \mathbb{P}^1_k \to \mathbb{P}^N_k$ is given by homogeneous polynomials $F_0, \ldots, F_N$ of the same degree $d \geq 0$:

$$f([u : v]) = [F_0(u, v) : \cdots : F_N(u, v)],$$

with no common zero in $\mathbb{P}^1_k$. Two such tuples of the same degree define the same morphism precisely when they differ by one common nonzero scalar. The scalar cannot be chosen independently for each coordinate.

Each F_i has $d + 1$ coefficients, so the tuples up to simultaneous scaling form the projective space

$$\mathbb{P}\big(k^{(d+1)(N+1)}\big) \cong \mathbb{P}^{(d+1)(N+1)-1}_k.$$

Inside it, let $\mathrm{Mor}_d(\mathbb{P}^1_k, \mathbb{P}^N_k)$ consist of tuples with no common zero. The degree here is $\deg f^*\mathcal{O}_{\mathbb{P}^N}(1)$; it includes the mapping degree onto the image.

Lemma 2.1.9. *The subset* $\mathrm{Mor}_d(\mathbb{P}^1_k, \mathbb{P}^N_k) \subset \mathbb{P}^{(d+1)(N+1)-1}_k$ *is Zariski open, hence a quasi-projective variety.*

Proof. If $d = 0$, every nonzero tuple of constants defines a constant morphism, and this subset is all of $\mathbb{P}^N_k$. Suppose $d > 0$, and put $I = (F_0, \ldots, F_N) \subset k[U, V]$. The absence of a projective common zero is equivalent, by the Nullstellensatz, to

$$\sqrt{I} = (U, V), \qquad \text{equivalently} \qquad (U, V)^m \subset I \text{ for some } m \geq d.$$

Equivalently, for some $m \geq d$, the linear map

$$\begin{aligned} \varphi_m : (k[U, V]_{m-d})^{N+1} &\longrightarrow k[U, V]_m, \\ (G_0, \ldots, G_N) &\longmapsto \sum_i G_i F_i \end{aligned}$$

is surjective. Its matrix entries are linear in the coefficients of the F_i, and surjectivity means that at least one $(m + 1)$-minor is nonzero. This is open for each m. Taking the union of these open sets proves the assertion. Equivalently, the bad locus is the intersection, over all m, of the closed rank-deficiency loci. □

Thus all morphisms are parametrized by the locally noetherian disjoint union

$$\mathrm{Mor}(\mathbb{P}^1_k, \mathbb{P}^N_k) = \coprod_{d\geq 0} \mathrm{Mor}_d(\mathbb{P}^1_k, \mathbb{P}^N_k).$$

Remark 2.1.10. The degree-d morphisms fit into a universal evaluation morphism

$$\begin{aligned} f^{\mathrm{univ}} : \mathbb{P}^1_k \times \mathrm{Mor}_d(\mathbb{P}^1_k, \mathbb{P}^N_k) &\longrightarrow \mathbb{P}^N_k, \\ ([u:v], [F_0 : \cdots : F_N]) &\longmapsto [F_0(u,v) : \cdots : F_N(u,v)]. \end{aligned}$$

If $X \subseteq \mathbb{P}^N_k$ is a closed subscheme defined by homogeneous equations $G_1, \ldots, G_a$, the condition that the image lies in X is

$$G_j(F_0(U,V), \ldots, F_N(U,V)) = 0 \quad (1 \leq j \leq a).$$

Each coefficient of these polynomials must vanish. These equations define the closed subscheme $\mathrm{Mor}_d(\mathbb{P}^1_k, X)$ of the open scheme above, and

$$\mathrm{Mor}(\mathbb{P}^1_k, X) = \coprod_{d\geq 0} \mathrm{Mor}_d(\mathbb{P}^1_k, X).$$

These are quasi-projective schemes; they need not be reduced or irreducible, even when X is a variety.

2.1C Parametrizing Morphisms

We admit the following theorem generalizing the preceding construction.

Theorem 2.1.11 (Grothendieck's Hilbert scheme theorem)**.** *Let X be a quasi-projective k-scheme and Y a projective k-scheme. There is a scheme $\mathrm{Mor}(Y, X)$ locally of finite type over k, parametrizing morphisms $Y \to X$ in the following sense:*

(a) There is a universal evaluation morphism

$$f^{\mathrm{univ}} : Y \times_k \mathrm{Mor}(Y, X) \longrightarrow X.$$

(b) For every k-scheme T and morphism $f : Y \times_k T \to X$, there is a unique morphism $\varphi : T \to \mathrm{Mor}(Y, X)$ such that

$$f = f^{\mathrm{univ}} \circ (\mathrm{id}_Y \times \varphi).$$

The universal property is expressed by the commutative diagram

$$\begin{array}{ccc} Y \times T & & \\ {\scriptstyle \mathrm{id}_Y \times \varphi}\downarrow & \searrow^{f} & \\ Y \times \mathrm{Mor}(Y,X) & \xrightarrow[f^{\mathrm{univ}}]{} & X \end{array}$$

The theorem applies in particular to varieties, and also to the finite and nonreduced schemes used below. The space is constructed by parametrizing graphs in $Y \times X$.

Fix very ample line bundles A on Y and L on X, and use the product polarization $A \boxtimes L$. The Hilbert polynomial to fix is the polynomial of the graph:

$$P_f(m) = \chi(Y, A^{\otimes m} \otimes f^* L^{\otimes m}).$$

The locus $\mathrm{Mor}_P(Y, X)$ with this polynomial is quasi-projective and of finite type, and

$$\mathrm{Mor}(Y, X) = \coprod_P \mathrm{Mor}_P(Y, X).$$

There are only countably many such polynomials. For a curve Y, fixing this polynomial is equivalent to fixing $\deg f^* L$, since A and Y are fixed. In higher dimension, one must retain the $A^{\otimes m}$ factor in this boundedness statement.

Remark 2.1.12. Projectivity of the source is essential here. For example, infinitesimal deformations of the zero map $\mathbb{A}^1_k \to \mathbb{A}^N_k$ form $k[t]^{\oplus N}$, which is infinite-dimensional. Thus this functor cannot be represented by a scheme locally of finite type over k.

In moduli language, the contravariant functor

$$\mathcal{M}\wr\nabla(Y, X) : \mathrm{Sch}_k^{\mathrm{op}} \to \mathrm{Sets}, \qquad T \longmapsto \mathrm{Hom}_k(Y \times_k T, X)$$

is represented by $\mathrm{Mor}(Y, X)$. It is naturally isomorphic to the functor

$$T \longmapsto \mathrm{Hom}_k(T, \mathrm{Mor}(Y, X)).$$

Given $\varphi : T \to \mathrm{Mor}(Y, X)$, its evaluation is

$$\mathrm{ev} : Y \times T \to X, \qquad (y, t) \longmapsto \varphi(t)(y).$$

This is a fine moduli problem for morphisms with the source and target fixed; we do not quotient by automorphisms of the source.

Example 2.1.13. If $Y = \mathrm{Spec}\, k$, then $\mathrm{Mor}(Y, X) \cong X$ as schemes. Its k-points are $\mathrm{Hom}_k(\mathrm{Spec}\, k, X) = X(k)$, rather than the scheme X itself. The universal evaluation is the canonical isomorphism $\mathrm{Spec}\, k \times X \to X$. More generally, if Y is a reduced projective zero-dimensional k-scheme, then Y is a finite disjoint union of k-points and

$$\mathrm{Mor}(Y, X) \cong X^{|Y|}.$$

Example 2.1.14 (Dual numbers)**.** For $Y = \mathrm{Spec}\, k[\epsilon]/(\epsilon^2)$, a k-morphism $Y \to X$ consists of a point $x \in X(k)$ and a tangent vector in $T_{X,x}$. If X is smooth, $\mathrm{Mor}(Y, X)$ is the total space of the tangent bundle T_X; see [GW20, (6.4)].

Example 2.1.15. We use the universal property to determine $T_{\mathrm{Mor}(Y,X),[f]}$. Its elements are morphisms

$$\varphi : \operatorname{Spec} k[\epsilon]/(\epsilon^2) \to \mathrm{Mor}(Y, X)$$

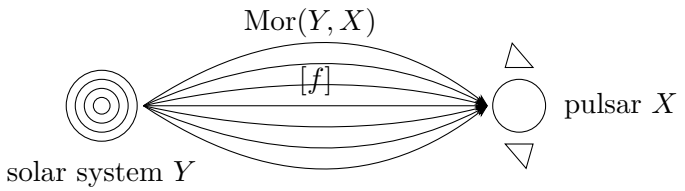
whose restriction to $\operatorname{Spec} k$ is $[f]$. They correspond to extensions

$$f_\epsilon : Y \times \operatorname{Spec} k[\epsilon]/(\epsilon^2) \to X, \qquad f_\epsilon|_{Y\times \operatorname{Spec} k} = f.$$

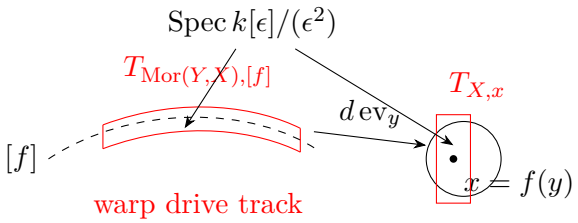
These are first-order deformations of the *morphism* f, with Y fixed. The diagram is

$$\begin{array}{ccc} Y \times \operatorname{Spec} k[\epsilon]/(\epsilon^2) & \xrightarrow{f_\epsilon} & X \\ \mathrm{id} \times \varphi \downarrow & & \| \\ Y \times \mathrm{Mor}(Y, X) & \xrightarrow{f^{\mathrm{univ}}} & X. \end{array}$$

We can view this as a family of tracks from a fixed source to the target. The "solar system / pulsar" picture illustrates the parameter space of maps:



A tangent vector at the parameter point $[f]$ is an infinitesimal change of the track. At a chosen $y \in Y$, it induces the infinitesimal motion of the landing point $x = f(y)$:



Proposition 2.1.16 (Tangent space of $\mathrm{Mor}(Y, X)$)**.** *For Y projective, X quasi-projective, and a morphism $f : Y \to X$, there is a canonical isomorphism*

$$T_{\mathrm{Mor}(Y,X),[f]} \cong H^0(Y, \mathcal{H}om(f^*\Omega_X, \mathcal{O}_Y)).$$

Proof. First compute locally on affine schemes $\operatorname{Spec} B$ and $\operatorname{Spec} A$. This is a calculation of deformations of ring maps; it does not require a representing scheme for morphisms between arbitrary affine schemes. If $f^\# : A \to B$ is the original map, a first-order lift has the form

$$f_\epsilon^\#(a) = f^\#(a) + \epsilon g(a).$$

It is a k-algebra homomorphism precisely when g is k-linear and

$$\begin{aligned} f_\epsilon^\#(aa') &= f^\#(aa') + \epsilon g(aa') \\ &= f^\#(a)f^\#(a') + \epsilon(f^\#(a)g(a') + f^\#(a')g(a)), \end{aligned}$$

that is, $g(aa') = f^\#(a)g(a') + f^\#(a')g(a)$. By the universal property of differentials,

$$\mathrm{Der}_k(A, B) \cong \mathrm{Hom}_A(\Omega_{A/k}, B) \cong \mathrm{Hom}_B(\Omega_{A/k} \otimes_A B, B).$$

Here B is an A-module via $f^\#$.

In general, take affine opens $V_i \subset Y$, $U_i \subset X$ with $f(V_i) \subset U_i$. The underlying topological spaces do not change under the dual-number thickening, so a lift restricts to $V_i \times \operatorname{Spec} k[\epsilon]/(\epsilon^2) \to U_i$. The derivations just computed must agree on overlaps. They therefore glue to a global section of $\mathcal{H}om(f^*\Omega_X, \mathcal{O}_Y)$, and every such section gives a lift. This proves the assertion. □

Corollary 2.1.17 (A very important result)**.** *If X is smooth along $f(Y)$, then*[35]

$$T_{\mathrm{Mor}(Y,X),[f]} \cong H^0(Y, f^*T_X).$$

Proof. The sheaf Ω_X is locally free on a neighborhood of $f(Y)$, hence

$$\mathcal{H}om(f^*\Omega_X, \mathcal{O}_Y) \cong (f^*\Omega_X)^\vee \cong f^*T_X.$$

□

Example 2.1.18. For smooth projective X, $\mathrm{Aut}(X)$ is an open subscheme of $\mathrm{Mor}(X, X)$. At an automorphism f, its tangent space is $H^0(X, f^*T_X) \cong H^0(X, T_X)$. If $H^0(X, T_X) = 0$, the automorphism scheme is zero-dimensional and reduced, hence discrete over k. It need not be finite: local finite type does not imply finite type for the entire automorphism scheme.

We now discuss the local structure of the parameter space.

Lemma 2.1.19. *Let $(R, \mathfrak{m})$ be a noetherian local k-algebra, essentially of finite type, with residue field $R/\mathfrak{m} = k$. Let $I \subset \mathfrak{m}$ satisfy $\mathfrak{m}I = 0$. Let Y be projective and $f : Y \to X$ a morphism, with X smooth along $f(Y)$. For an extension*[36]

$$f_{R/I} : Y \times \operatorname{Spec}(R/I) \to X,$$

there is an obstruction class in $H^1(Y, f^*T_X) \otimes_k I$ whose vanishing is equivalent to the existence of a lift $f_R : Y \times \operatorname{Spec} R \to X$.

[35] For duals of locally free sheaves and the Hom identification used here, see [Har77, II, Exercise 5.1(b)].

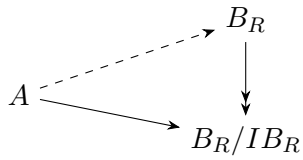
[36] The infinitesimal lifting property used here is Proposition B.4.8 (with the difference of lifts described in Lemma B.4.7).

Proof. Since $\mathfrak{m}I = 0$, we have $I^2 = 0$, and I is a finite-dimensional k-vector space. The source Y_R is flat over R, so its ideal for $Y_{R/I} \subset Y_R$ is $\mathcal{O}_Y \otimes_k I$, supported on the closed fiber.

The given map has image in the smooth locus of X. Indeed, the inverse image of its complement is closed in the proper R/I-scheme $Y_{R/I}$; its image in $\operatorname{Spec}(R/I)$ is closed and misses the closed point, so it is empty. Cover Y_R by affine opens whose reductions map into smooth affine opens of X. On each such open, formal smoothness supplies a lift. In ring notation, this is the lifting across the square-zero surjection

$$B_R \twoheadrightarrow B_R/IB_R, \qquad IB_R \cong (B_R/\mathfrak{m}B_R) \otimes_k I.$$

The affine lifting diagram is



where the diagonal map is the sought lift. For a smooth affine target $\operatorname{Spec} A$, two lifts differ by a derivation. On the closed fiber, writing $B = B_R/\mathfrak{m}B_R$, the space of differences is

$$\begin{aligned}\operatorname{Hom}_A(\Omega_{A/k}, B \otimes_k I) &\cong \operatorname{Hom}_B(\Omega_{A/k} \otimes_A B, B \otimes_k I)\\ &\cong \operatorname{Hom}_B(\Omega_{A/k} \otimes_A B, B) \otimes_k I.\end{aligned}$$

This is the local section space of $f^*T_X \otimes_k I$.

On overlaps, the differences of local lifts form a Čech 1-cocycle. The cocycle condition follows by adding the differences of three lifts. Since the source is separated, affine intersections compute the cohomology of this quasi-coherent sheaf. The resulting class lies in $H^1(Y, f^*T_X) \otimes_k I$. If it vanishes, subtracting a 0-cochain from the local lifts makes them agree, giving a global lift. Conversely, a global lift makes this cocycle a coboundary.

When R is Artinian the usual product affine cover of Y suffices. The affine-cover argument on Y_R also handles the essentially finite type local rings used in Theorem 2.1.23; their spectra are not necessarily nilpotent thickenings of a single point. □

Remark 2.1.20. Choose a k-basis of I, of dimension s. Then

$$\begin{aligned}\operatorname{Hom}_B(\Omega_A \otimes_A B, B \otimes_k I) &\cong \operatorname{Hom}_B(\Omega_A \otimes_A B, B^{\oplus s})\\ &\cong \operatorname{Hom}_B(\Omega_A \otimes_A B, B)^{\oplus s}\\ &\cong \operatorname{Hom}_B(\Omega_A \otimes_A B, B) \otimes_k I.\end{aligned}$$

One must choose a basis, not just an arbitrary list of ideal generators and identify its length with $\dim_k I$.

Remark 2.1.21 (An important case)**.** Take $R = k[\epsilon]/(\epsilon^m)$, $I = (\epsilon^{m-1})$, $\mathfrak{m} = (\epsilon)$, with $m \geq 2$. The obstruction to extending a deformation from $Y \times \operatorname{Spec} k[\epsilon]/(\epsilon^{m-1})$ to $Y \times \operatorname{Spec} k[\epsilon]/(\epsilon^m)$ lies in $H^1(Y, f^*T_X)$, using the basis ϵ^{m-1} of I.

Lemma 2.1.22. *Let $(R, \mathfrak{m})$ be a noetherian local ring and $I \subset \mathfrak{m}^2$ an ideal. If the projection $\pi : R \to R/I$ admits a ring section $s : R/I \to R$, then $I = 0$.*

Proof. For $a, b \in \mathfrak{m}$, write $s(\pi(a)) = a + a'$, $s(\pi(b)) = b + b'$ with $a', b' \in I$. Then

$$s(\pi(ab)) = (a + a')(b + b') \in ab + \mathfrak{m}I.$$

Every element of $\mathfrak{m}^2$ is a finite sum of such products, so for $x \in I \subset \mathfrak{m}^2$,

$$0 = s(\pi(x)) \in x + \mathfrak{m}I.$$

Thus $I = \mathfrak{m}I$, and Nakayama's lemma gives $I = 0$. □

Theorem 2.1.23 (Local structure of $\operatorname{Mor}(Y, X)$)**.** *Let X, Y be projective varieties and $f : Y \to X$ a morphism with X smooth along $f(Y)$. Near $[f]$, the scheme $\operatorname{Mor}(Y, X)$ can be defined by at most $h^1(Y, f^*T_X)$ equations in a smooth variety of dimension $h^0(Y, f^*T_X)$.*

Proof. Because $\operatorname{Mor}(Y, X)$ is locally of finite type, an affine neighborhood of $[f]$ is cut out by finitely many polynomials $P_1, \ldots, P_m$ in $\mathbb{A}^n$. The rank of their Jacobian at $[f]$ is exactly

$$r = n - \dim T_{\operatorname{Mor}(Y,X),[f]} = n - h^0(Y, f^*T_X).$$

Choose r of the equations whose differentials are independent. They cut out a variety V, smooth near $[f]$, of dimension $h^0(Y, f^*T_X)$. After shrinking, $\operatorname{Mor}(Y, X)$ is a closed subscheme of V and their tangent spaces at $[f]$ coincide.

Set $R = \mathcal{O}_{V,[f]}$, with maximal ideal $\mathfrak{m}$, and let I define the local ring of $\operatorname{Mor}(Y, X)$. Equality of tangent spaces gives $I \subset \mathfrak{m}^2$. By Nakayama's lemma, it suffices to show

$$\dim_k I/\mathfrak{m}I \leq h^1(Y, f^*T_X).$$

The natural map $\operatorname{Spec} R/I \to \operatorname{Mor}(Y, X)$ gives $f_{R/I}$. Lemma 2.1.19, applied to the local ring $R/\mathfrak{m}I$ and its small ideal $I/\mathfrak{m}I$, gives an obstruction

$$o = \sum_{i=1}^{h^1} a_i \otimes \bar{b}_i \in H^1(Y, f^*T_X) \otimes_k I/\mathfrak{m}I,$$

where the a_i form a basis and $b_i \in I$. Put

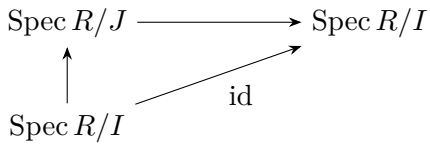
$$J = \mathfrak{m}I + (b_1, \ldots, b_{h^1}).$$

The image of o in $H^1(Y, f^*T_X) \otimes_k I/J$ is zero, so the map lifts to $Y \times \operatorname{Spec} R/J \to X$.

By the universal property, this gives $\operatorname{Spec} R/J \to \operatorname{Mor}(Y, X)$. Since R/J is local and its closed point maps to $[f]$, this morphism factors through the local scheme $\operatorname{Spec} \mathcal{O}_{\operatorname{Mor}(Y,X),[f]} = \operatorname{Spec} R/I$. This is the universal property of localization; $\operatorname{Spec} R/I \to \operatorname{Mor}(Y, X)$ need not be an open immersion. Its restriction to $\operatorname{Spec} R/I$ is the original map, so we obtain a section of

$$R/J \twoheadrightarrow R/I.$$

Equivalently, the diagram on local schemes is



The kernel I/J lies in the square of the maximal ideal of R/J. Lemma 2.1.22 therefore gives $I = J$. Thus the images of the b_i generate $I/\mathfrak{m}I$, as required. Shrinking the neighborhood realizes these local generators as at most h^1 defining equations. □

Corollary 2.1.24. *Every irreducible component of* $\operatorname{Mor}(Y, X)$ *through* $[f]$ *has dimension at least*

$$h^0(Y, f^*T_X) - h^1(Y, f^*T_X).$$

This follows from the dimension bound for a scheme cut out by a given number of equations in a smooth variety. It is the classical conclusion of [Kol96, I, 2.16], over an arbitrary algebraically closed field.

2.1D Morphisms with Fixed Points

Fix a closed subscheme $B \subseteq Y$ and a morphism $g : B \to X$. Consider restriction

$$\rho : \operatorname{Mor}(Y, X) \to \operatorname{Mor}(B, X), \qquad f \longmapsto f|_B,$$

and denote its scheme-theoretic fiber over $[g]$ by $\operatorname{Mor}(Y, X; g)$. Here Theorem 2.1.11 is used for the projective scheme B, which need not be reduced.

If $f|_B = g$ and X is smooth along $f(Y)$, the tangent map is restriction

$$d\rho : H^0(Y, f^*T_X) \longrightarrow H^0(B, g^*T_X).$$

Tensoring $0 \to \mathcal{I}_B \to \mathcal{O}_Y \to \mathcal{O}_B \to 0$ with the locally free f^*T_X and taking cohomology gives

$$\begin{aligned} 0 \to H^0(Y, f^*T_X \otimes \mathcal{I}_B) \to H^0(Y, f^*T_X) &\xrightarrow{d\rho} H^0(B, g^*T_X) \\ &\to H^1(Y, f^*T_X \otimes \mathcal{I}_B) \to \cdots . \end{aligned}$$

Thus $T_{\mathrm{Mor}(Y,X;g),[f]} = H^0(Y, f^*T_X \otimes \mathcal{I}_B)$.

Theorem 2.1.25 (Local structure with fixed points)**.** *With X, Y, B, g, f as above, $\mathrm{Mor}(Y, X; g)$ is locally defined near $[f]$ by at most $h^1(Y, f^*T_X \otimes \mathcal{I}_B)$ equations in a smooth variety of dimension $h^0(Y, f^*T_X \otimes \mathcal{I}_B)$.*[37]

The obstruction proof is the same as above, with differences of lifts required to vanish along B. Local lifts agreeing with the prescribed map on B exist by formal smoothness: combine the maps on $Y_{R/I}$ and B_R, which agree on $B_{R/I}$, and lift across their square-zero union in Y_R. The sheaf of differences is $f^*T_X \otimes \mathcal{I}_B \otimes_k I$. The cocycle argument and Theorem 2.1.23 then apply.

2.1E Morphisms from a Curve

Let C be a smooth projective curve and $f : C \to X$ a morphism, with X smooth along $f(C)$. By Corollary 2.1.24,

$$\dim_{[f]} \mathrm{Mor}(C, X) \geq h^0(C, f^*T_X) - h^1(C, f^*T_X) = \chi(C, f^*T_X).$$

Writing $n = \dim X$, the Riemann–Roch calculation is

$$\mathrm{ch}(f^*T_X) = n + c_1(f^*T_X), \qquad \mathrm{td}(T_C) = 1 + \tfrac{1}{2}c_1(T_C).$$

Taking the degree-one part of the product,

$$\begin{aligned}
\chi(C, f^*T_X) &= \deg\big(\mathrm{ch}(f^*T_X)\,\mathrm{td}(T_C)\big)_1 \\
&= \deg c_1(f^*T_X) + \frac{n}{2}\deg c_1(T_C) \\
&= \deg f^*\omega_X^{-1} + \frac{n}{2}\deg(-K_C) \\
&= -(K_X \cdot f_*C) + n(1 - g(C)).
\end{aligned}$$

Here $c_1(f^*T_X) = c_1(\det f^*T_X)$ and f_*C is the pushforward cycle, including the mapping degree. Thus

$$\boxed{\dim_{[f]} \mathrm{Mor}(C, X) \geq -(K_X \cdot f_*C) + (1 - g(C)) \dim X.}$$

For a finite subscheme $B \subset C$, the same exact sequence gives

$$\chi(C, f^*T_X \otimes \mathcal{I}_B) = \chi(C, f^*T_X) - \chi(B, f^*T_X|_B).$$

Since B is zero-dimensional,

$$\chi(B, f^*T_X|_B) = h^0(B, f^*T_X|_B) = \mathrm{length}(B) \dim X.$$

[37] For the Chern-class identity and the two deformation-dimension formulas, see [Har77, Appendix A, C.5], [KM98, p. 12, line −7], [KM98, p. 13, line 3].

By Theorem 2.1.25,

$$\begin{aligned}\dim_{[f]} \operatorname{Mor}(C, X; f|_B) &\geq \chi(C, f^*T_X \otimes \mathcal{I}_B) \\ &= -(K_X \cdot f_*C) + (1 - g(C) - \operatorname{length}(B)) \dim X.\end{aligned}$$

In particular, for a single point $c_0 \in C$,

$$\dim_{[f]} \operatorname{Mor}(C, X; f|_{\{c_0\}}) \geq -(K_X \cdot f_*C) - g(C) \dim X.$$

This is the dimension estimate used in [KM98, pp. 12–13] and in the bend-and-break arguments below.

2.2 "Bend and Break" Lemmas

Olivier Debarre writes in [Deb01, Chapter 3]:

> The whole story began in 1979, with Mori's spectacular proof of a conjecture of Hartshorne characterizing projective spaces as the only smooth projective varieties with ample tangent bundle. The techniques that Mori introduced to solve this conjecture have turned out to have more far-reaching applications than Hartshorne's conjecture itself.

Recall that a 1-cycle is a finite formal sum $\sum_i n_i[C_i]$ of integral curves, with $n_i \in \mathbb{Z}$. It is rational if its components are rational curves. For a proper morphism, pushforward includes both multiplicity and mapping degree:

$$f_*\Big(\sum_i m_i[C_i]\Big) = \sum_{\dim f(C_i)=1} m_i \deg(C_i/f(C_i))[f(C_i)].$$

Contracted components contribute zero. In particular, a nonreduced fiber must be taken with its scheme-theoretic multiplicities.

The original bend-and-break lemma says that a curve moving nontrivially as a morphism while keeping one point image fixed degenerates to an effective cycle with a rational component through that point. In these proofs we resolve rational maps from smooth surfaces by point blow-ups; this supplies rational exceptional curves directly, without requiring the higher-dimensional Abhyankar lemma.

Lemma 2.2.1 (Bend and Break I)**.** *Let X be projective, C a smooth projective curve, $c_0 \in C$, and $f : C \to X$ a nonconstant morphism. Assume there are a smooth connected nonproper pointed curve $t_0 \in T$ and a morphism $F : C \times T \to X$ such that*

(1) $F|_{C\times\{t_0\}} = f$;

(2) $F(\{c_0\} \times T) = \{f(c_0)\}$;

(3) $F|_{C\times\{t\}} \neq f$ *for general* $t \in T$.

Then there are a possibly constant morphism $f' : C \to X$ *and a nonzero effective rational* 1*-cycle* $Z = \sum_i a_i[Z_i]$ *such that*

$$f_*C \sim_{\text{alg}} f'_*C + Z, \qquad f(c_0) \in \operatorname{Supp} Z.$$

In particular, X *contains a rational curve through* $f(c_0)$.

Remark 2.2.2. If X is smooth along $f(C)$, Section 2.1E shows that it suffices to have[38]

$$-(K_X \cdot f_*C) - g(C) \dim X \geq 1.$$

Indeed a positive-dimensional component through $[f]$ contains a nonconstant curve through that point; normalization and restriction to an open subset give the required smooth parameter curve.

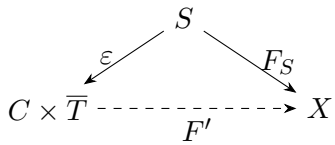
Proof of Lemma 2.2.1. Let $\overline{T}$ be the smooth compactification and $F' : C \times \overline{T} \dashrightarrow X$ the induced rational map. We claim it is undefined at some point of $\{c_0\} \times \overline{T}$.

Suppose it were defined along that whole curve. Its indeterminacy locus is closed, and projection to C is proper, so there is an open neighborhood U of c_0 such that F' is a morphism on $U \times \overline{T}$. The fiber $\{c_0\} \times \overline{T}$ is contracted. By the rigidity lemma, after shrinking U if necessary, F' is constant on each fiber of $U \times \overline{T} \to U$. Hence

$$F'(c,t) = F'(c,t_0) = f(c) \qquad (c \in U).$$

For each $t \in T$, the two morphisms $C \to X$ agree on the dense open U, hence everywhere because X is separated. This contradicts (3).

Choose an indeterminacy point $a = (c_0, t)$ and resolve by point blow-ups:



Take the resolution to be an isomorphism above the regular locus of F'. The total transform of the special fiber is

$$\varepsilon^*(C \times \{t\}) = C' + \sum_i m_i E_i,$$

[38] For extension from a smooth curve and resolution of indeterminacy on a smooth surface, see [Har77, I, 6.8], [Bea78, II, 7].

where $C' \cong C$ is its strict transform, the E_i are rational exceptional curves, and $m_i > 0$. Put $f' = F_S|_{C'}$ and

$$Z = (F_S)_*\Big(\sum_i m_i E_i\Big).$$

The coefficients in Z include the degrees of the nonconstant maps $E_i \to F_S(E_i)$. By Lüroth's theorem, each nonconstant image is rational.

The morphism $p : S \to \overline{T}$ is flat: over a smooth curve, the integral surface has torsion-free, hence flat, local modules. Its fibers, with multiplicities, are algebraically equivalent. Pushing forward and comparing with t_0 gives

$$f_*C \sim_{\text{alg}} f'_*C + Z.$$

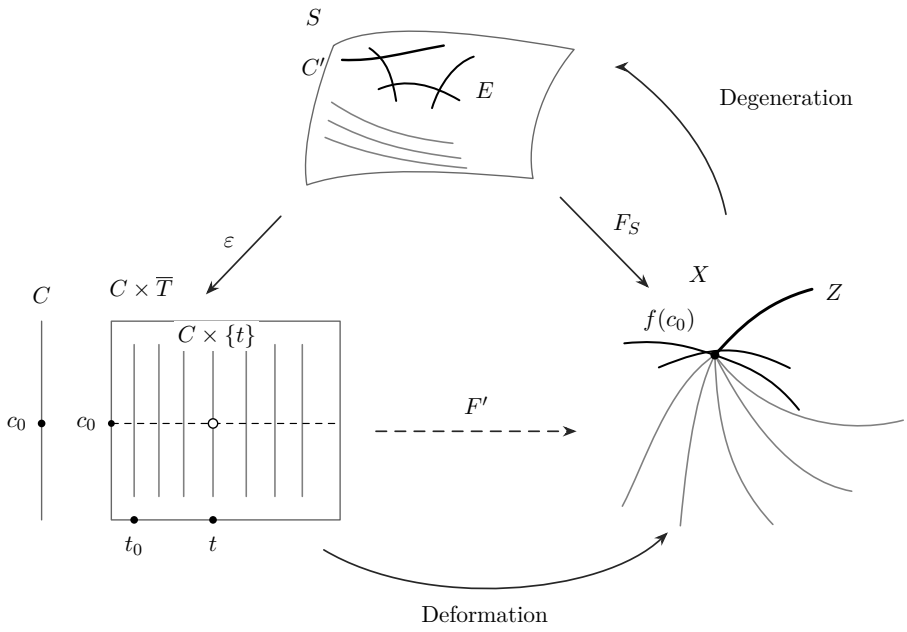


Figure 2.7: Bend and Break I: a nonconstant exceptional component supplies a rational curve through the fixed point.

It remains to prove both $Z \neq 0$ and the marked-point assertion. The connected exceptional tree $\varepsilon^{-1}(a)$ cannot be entirely contracted by F_S. If it were, the graph of F' would have a single point over a; its projection to the normal surface $C \times \overline{T}$ would be finite and birational near a, hence an isomorphism. This would extend F' across a, a contradiction.

The strict transform of $\{c_0\} \times \overline{T}$ meets this tree and maps constantly to $f(c_0)$. Follow a path in the tree from that intersection to the first component with nonconstant image. All preceding contracted components map to the same point $f(c_0)$. Thus this first nonconstant image contains $f(c_0)$ and occurs in Z with positive coefficient. This proves the two assertions.

□

Remark 2.2.3 (Thanks to YuHong Zhou)**.** Under (2) and (3), the curve T cannot be proper. If it were, rigidity applied to $C \times T \to C$ would give[39]

$$F(c,t) = F(c,t_0) = f(c),$$

contradicting (3). Thus the phrase "possibly nonproper" in [KM98] is harmless: the other hypotheses already exclude the proper case. Writing "nonproper" makes this consequence explicit; it does not contradict their statement.

Lemma 2.2.4 (Bend and Break II)**.** *Let X be projective and $f : \mathbb{P}^1 \to X$ a nonconstant morphism. Assume there are a smooth connected pointed curve $t_0 \in T$ and a morphism $F : \mathbb{P}^1 \times T \to X$ such that*

(1) $F|_{\mathbb{P}^1 \times \{t_0\}} = f$;

(2) $F(\{0\} \times T) = \{f(0)\}$ and $F(\{\infty\} \times T) = \{f(\infty)\}$;

(3) the image of F has dimension two.

Then $f_*\mathbb{P}^1$ is algebraically equivalent to a nonintegral effective rational 1-cycle Z whose support contains $f(0)$ and $f(\infty)$. Thus Z is reducible, or is a multiple rational curve $a[\Gamma]$ with $a > 1$.

Remark 2.2.5. The condition[40]

$$\dim_{[f]} \operatorname{Mor}(\mathbb{P}^1, X; f|_{\{0,\infty\}}) \geq 2$$

is sufficient for the *conclusion* of the lemma. To see this, let $\Gamma = f(\mathbb{P}^1)_{\mathrm{red}}$ and $d = [k(\mathbb{P}^1) : k(\Gamma)]$. The nonconstant morphism to its image is finite. If $d > 1$, then $f_*\mathbb{P}^1 = d[\Gamma]$ already has the required form.

If $d = 1$, f is the normalization of its image. Morphisms with this same image and the same degree differ by an automorphism of its normalization. Fixing the two point images leaves only finitely many choices for their preimages on the normalization, and for each such choice the remaining automorphism group is one-dimensional, isomorphic to $\mathbb{G}_m$. Hence the fixed-image locus through $[f]$ has dimension at most one. A component of dimension at least two therefore contains a parameter curve whose images actually move, so condition (3) holds for that family.

For multiple covers, variations with the same image need not come from $\mathbb{G}_m$; hence the dimension condition alone does not imply (3) for an arbitrary f. The degree d here is the function-field degree, including any inseparable part; connected fibers are not an additional case outside this definition.

[39] For projectivity of a smooth proper curve, see [Har77, II, 4.10.2].

[40] For the Hodge index theorem and the divisor classes on a ruled surface, see [Har77, V, 1.9], [Har77, V, 2.3].

If X is smooth along $f(\mathbb{P}^1)$, Section 2.1E shows that

$$-(K_X \cdot f_*\mathbb{P}^1) - \dim X \geq 2$$

is sufficient for the same conclusion.

Proof of Lemma 2.2.4. Compactify T to $\overline{T}$. Take a $\mathbb{P}^1$-bundle $\pi : P \to \overline{T}$ containing $\mathbb{P}^1 \times T$ with its given projection, and resolve the induced rational map $F' : P \dashrightarrow X$ by a sequence of point blow-ups

$$S = S_r \longrightarrow S_{r-1} \longrightarrow \cdots \longrightarrow S_1 \longrightarrow P, \qquad F_S : S \to X.$$

Among all such bundle models and resolutions, choose one with the number r minimal. We may take each resolution to be an isomorphism over the regular locus of the rational map. Put $p = \pi \circ \varepsilon : S \to \overline{T}$.

Each fiber $p^{-1}(t)$ is a connected tree of rational curves, with its total-transform multiplicities. As in the preceding lemma, the cycles

$$Z_t = (F_S)_*[p^{-1}(t)]$$

are effective, rational, and algebraically equivalent to $f_*\mathbb{P}^1$. The closures of the two marked sections lift to sections of p and map to $f(0)$ and $f(\infty)$. Thus these points belong to the image of each fiber. They also belong to $\operatorname{Supp} Z_t$: the fiber has some nonconstant component because its ample degree equals the positive degree of $f_*\mathbb{P}^1$, and a connected chain of contracted components attaches to such a component with the same image point.

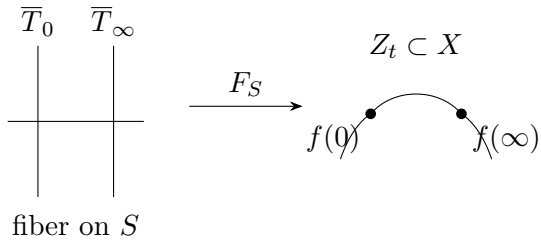


Figure 2.8: The support of the fiber cycle contains both marked images.

Assume, seeking a contradiction, that every Z_t is integral. There is then a unique component R of each fiber that is not contracted by F_S, and

$$1 = \operatorname{mult}_R(p^{-1}(t)) \deg(R/F_S(R)).$$

Both positive integers are therefore one.

The case $r = 0$

Now $F' : P \to X$ is a morphism. For an ample divisor H on X, put $L = F'^*H$. The image has dimension two, so $L^2 > 0$. Both distinct sections $\overline{T}_0, \overline{T}_\infty$ are contracted, hence

$$L \cdot \overline{T}_0 = L \cdot \overline{T}_\infty = 0.$$

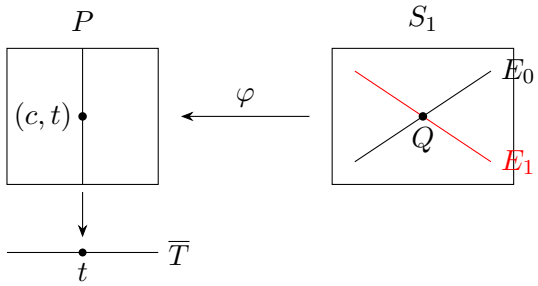
Hodge index gives $\overline{T}_0^2 < 0$ and $\overline{T}_\infty^2 < 0$. Their numerical classes are independent: proportionality, intersected with a fiber, would force them to be equal, contrary to their negative self-intersection and $\overline{T}_0 \cdot \overline{T}_\infty \geq 0$. Together with L they would give three independent classes in $N^1(P)_\mathbb{R}$. This contradicts $\rho(P) = 2$ for a ruled surface [Har77, V, 2.3].

The case $r > 0$

Consider the first blow-up $\varphi : S_1 \to P$, at a point on the fiber over t. Its fiber has two components E_0, E_1 meeting transversally at Q, with E_0 the strict transform of the original fiber and E_1 exceptional. Both are (-1)-curves, since

$$E_0^2 = (\varphi^*[\pi^{-1}(t)] - E_1)^2 = 0 - 1 = -1.$$

Write $\varepsilon_1 : S \to S_1$ for the remaining blow-ups.



By minimality, some component of $\varepsilon_1^{-1}(E_1)$ is not contracted by F_S. Otherwise this connected inverse image would map to a point, and descent through the normal surface would extend F' over the first center, making that blow-up unnecessary. Thus the unique noncontracted component R lies in $\varepsilon_1^{-1}(E_1)$.

Every exceptional component above the crossing Q has fiber multiplicity at least two. The first blow-up of Q has multiplicity $1+1 = 2$; a later blow-up either preserves a component's multiplicity at a smooth point or adds multiplicities at an intersection. Therefore R, whose multiplicity is one, cannot lie over Q. It follows that all of $\varepsilon_1^{-1}(E_0)$ is contracted by F_S, and, by connectedness, to a single point.

The same graph-descent argument shows that the induced rational map $S_1 \dashrightarrow X$ is regular along E_0. A minimal resolution has no further blow-up centers there. We may now contract E_0 on S_1. This is the elementary transformation of a ruled surface: it gives another $\mathbb{P}^1$-bundle $P' \to \overline{T}$, identical to P over T. Since E_0 is contracted by the map to X, that map descends near its contraction point. The remaining blow-ups, disjoint from E_0, give a resolution over P' using $r - 1$ blow-ups. This contradicts the choice of r.

Thus some Z_t is nonintegral, proving the lemma. □

Remark 2.2.6. The two fixed domain points can be any distinct $q = [q_1 : q_2]$ and $q' = [q_1' : q_2']$. The matrix

$$\begin{pmatrix} q_1' & q_1 \\ q_2' & q_2 \end{pmatrix} \in \mathrm{PGL}_2(k)$$

sends $0 = [0 : 1]$ to q and $\infty = [1 : 0]$ to q'. Reparametrization gives the same conclusion: the support of the resulting cycle contains $f(q)$ and $f(q')$. These points need not lie on different components, and the argument does not require simultaneous indeterminacy on both marked sections.

If all ε-exceptional components of a particular fiber are contracted, the push-forward of that fiber comes from its strict transform $R_0 \cong \mathbb{P}^1$, which has fiber multiplicity one. If its cycle is nonintegral, then

$$(F_S)_*[p^{-1}(t)] = d[\Gamma], \qquad d = \deg(F_S|_{R_0} : R_0 \to \Gamma) > 1.$$

This is a sufficient mechanism for a multiple curve, but not the only one: several noncontracted components may have the same image and their coefficients add. The inequality concerns the map on this special component; it does not imply that the original f has degree greater than one onto its own image.

For a stronger bend-and-break statement with more fixed points, see [Deb01, 3.5].

2.3 Application: Finding Rational Curves

2.3A Rational Curves on Fano Varieties

We are ready to formulate and prove Mori's result on rational curves on Fano varieties. Here a Fano variety is smooth, projective, of positive dimension, with ample anticanonical divisor.

For this section and Section 2.4 the original ground field k is algebraically closed of characteristic zero. The symbols X, C denote the original varieties over k. Models over finitely generated rings will be written $\mathcal{X}, \mathcal{C}$, and their geometric reductions in positive characteristic will be denoted X_p, C_p or X_P, C_P. This convention is maintained when applying Frobenius.

Example 2.3.1 (The simplest Fano varieties)**.** For a smooth projective hypersurface $X_d \subset \mathbb{P}^{n+1}$ of degree d, with $n \geq 1$, adjunction gives

$$K_{X_d} = (K_{\mathbb{P}^{n+1}} + dH)|_{X_d} = (d - n - 2)H|_{X_d}.$$

Thus X_d is Fano exactly when $d < n + 2$.

Example 2.3.2 (Constructing Fano varieties)**.** Let Y be Fano and $D_1, \ldots, D_r$ nef divisors such that[41]

[41] For smoothness and the exact sequence of relative differentials, see [SP25, Tag 01V8], [SP25, Tag 02K4]. The ampleness of the sum of an ample and a nef class is Corollary 1.4.7.

$$M = -K_Y - D_1 - \cdots - D_r$$

is ample. Then

$$X = \mathbb{P}_Y\Big(\bigoplus_{i=1}^{r} \mathcal{O}_Y(D_i)\Big)$$

is Fano. We use the quotient convention for the projective bundle and write D for the class of $\mathcal{O}_X(1)$.

For $\pi : X \to Y$, the relative Euler sequence [Har77, III, Exercise 8.4(b)] is

$$0 \to \Omega_{X/Y} \to \pi^*\Big(\bigoplus_i \mathcal{O}_Y(D_i)\Big) \otimes \mathcal{O}_X(-1) \to \mathcal{O}_X \to 0.$$

Taking determinants gives

$$\det \Omega_{X/Y} \cong \pi^*\mathcal{O}_Y\Big(\sum_i D_i\Big) \otimes \mathcal{O}_X(-r).$$

The map π is smooth because locally it is the projection of a projective-space bundle; on its standard affine charts it is a polynomial extension. Thus

$$0 \to \pi^*\Omega_Y \to \Omega_X \to \Omega_{X/Y} \to 0$$

is exact, and

$$-K_X \sim_{\text{lin}} rD + \pi^*M.$$

Each $M + D_i$ is ample. The vector bundle $\bigoplus_i \mathcal{O}_Y(M + D_i)$ is ample [Laz04b, 6.1.13]. Tensoring a bundle by a line bundle leaves its projectivization unchanged, with

$$\varphi : X' = \mathbb{P}_Y\Big(\bigoplus_i \mathcal{O}_Y(M + D_i)\Big) \xrightarrow{\sim} X,$$
$$\mathcal{O}_{X'}(1) \cong \varphi^*(\mathcal{O}_X(1) \otimes \pi^*\mathcal{O}_Y(M)).$$

Hence $D + \pi^*M$ is ample. Moreover D is nef, since the tautological quotient of the pullback of a direct sum of nef line bundles has nonnegative degree on every curve. More explicitly, on the normalization of a curve in X, some summand maps nontrivially to that quotient line bundle, so its degree is at least the degree of that summand, which is nonnegative. Therefore

$$rD + \pi^*M = (D + \pi^*M) + (r-1)D$$

is ample, and X is Fano.

Lemma 2.3.3. *Let R be a finitely generated $\mathbb{Z}$-algebra.*

(1) For every maximal ideal $\mathfrak{m} \subset R$, the residue field $R/\mathfrak{m}$ is finite.

(2) Closed points are Zariski dense in $\operatorname{Spec} R$.

Proof. (1) Put $L = R/\mathfrak{m}$. If L had characteristic zero, it would be a finitely generated $\mathbb{Q}$-algebra and a field, hence a finite extension of $\mathbb{Q}$ by Zariski's lemma. Let $x_1, \ldots, x_s$ generate it as a $\mathbb{Z}$-algebra. Choose $N > 0$ clearing the denominators in monic algebraic equations for these elements over $\mathbb{Q}$. Then each x_i is integral over $\mathbb{Z}[1/N]$, so $L = \mathbb{Z}[1/N][x_1, \ldots, x_s]$ is integral over $\mathbb{Z}[1/N]$. A subring over which a field is integral is itself a field, but $\mathbb{Z}[1/N]$ is not a field. This is a contradiction.

Thus $\operatorname{char} L = p > 0$. The field L is a finitely generated algebra over $\mathbb{F}_p$, hence is a finite extension by Zariski's lemma, so it is finite.

(2) Take a nonempty basic open $D(a) = \operatorname{Spec} R_a$. Nonemptiness, rather than merely $a \neq 0$, is needed if R has nilpotents. Choose a maximal ideal $\mathfrak{n}$ of R_a. Its residue field is finite by (1). If $\mathfrak{m}$ is the inverse image in R, then $R/\mathfrak{m}$ embeds in that finite field and is a finite domain, hence a field. Thus $\mathfrak{m}$ is a closed point in $D(a)$. The same argument applied to any quotient of R shows that closed points are dense in every closed subscheme as well. □

Lemma 2.3.4. *A homogeneous system of equations with integer coefficients which has a nontrivial solution over* $\overline{\mathbb{F}}_p$ *for infinitely many primes* p *has a nontrivial solution over every algebraically closed field of characteristic zero.*

More generally, let R *be a finitely generated integral* $\mathbb{Z}$*-algebra of characteristic zero, and take homogeneous equations with coefficients in* R*. If their projective zero scheme has nonempty geometric fibers over a Zariski dense subset of* $\operatorname{Spec} R$*, then it has a point over every algebraically closed field containing* $\operatorname{Frac} R$*, with the given embedding of* R*.*

Proof. The equations define a closed subscheme $Z \subset \mathbb{P}^N_R$. Since $Z \to \operatorname{Spec} R$ is proper, its image is closed. It contains the given dense subset, so it is all of $\operatorname{Spec} R$. In particular, the generic fiber Z_η is nonempty.

Let $K = \operatorname{Frac} R$. A closed point of the finite type K-scheme Z_η has residue field finite over K. This field embeds over K into any algebraically closed extension of K, giving the required point after base change. For the original integer-coefficient statement, take $R = \mathbb{Z}$: infinitely many closed points are dense, and $K = \mathbb{Q}$ embeds into every characteristic-zero field. For a general R, the condition that the chosen algebraically closed field contain $\operatorname{Frac} R$ cannot be omitted; for example, $\mathbb{Q}(t)$ cannot embed into $\overline{\mathbb{Q}}$. □

Remark 2.3.5 (The functor-of-points view)**.** For an R-algebra field K,

$$\mathbb{P}^N_R(K) = \operatorname{Hom}_R(\operatorname{Spec} K, \mathbb{P}^N_R) \cong (K^{N+1} \setminus \{0\})/K^*.$$

Thus a projective point is a nonzero tuple up to simultaneous nonzero scaling. On an affine chart, the corresponding description is

$$\operatorname{Hom}_{R\text{-alg}}(R[T_1, \ldots, T_N]/(f_1, \ldots, f_r), A) \cong \{(a_1, \ldots, a_N) \in A^N \mid f_j(a) = 0 \text{ for all } j\}.$$

For $Z = V_+(F_1, \dots, F_r)$ this gives

$$Z(K) = \{[x_0 : \cdots : x_N] \in \mathbb{P}^N(K) \mid F_j(x) = 0 \text{ for all } j\}.$$

If K is algebraically closed, nonemptiness of the finite type fiber Z_K is equivalent to the existence of a K-point, by the Nullstellensatz. The algebraic-closedness condition matters.

The same strategy also applies to open conditions such as the absence of a common zero in the coordinates of a morphism. For those parameter spaces, properness need not hold. Instead one uses Chevalley's theorem: the image of a finite type morphism is constructible. A dense constructible subset of an integral noetherian base contains a nonempty open subset, hence its generic point. This is the form used in Step 8 of Theorem 2.3.6.

Theorem 2.3.6 ([Mor82])**.** *For every point x of a Fano variety X, there is a rational curve D through x such that*[42]

$$0 < -(D \cdot K_X) \leq \dim X + 1.$$

Proof. This is done in several steps. Write $n = \dim X$.

Step 1: To find this rational curve, we intend to use the bend-and-break technique. Thus we have to find a morphism $f : C \longrightarrow X$ which we will be able to deform, then we bend these deformations together over a fixed point, and break them into some rational curves.

Pick an integral projective curve $C' \subseteq X$ passing through x, and let $f : C \longrightarrow C' \hookrightarrow X$ be its normalization. Then C is a smooth projective curve. Pick $c_0 \in C$ such that $f(c_0) = x$, which is the target point we are bending.

Step 2: According to the formula in Section 2.1E,

$$\dim_{[f]} \operatorname{Mor}(C, X; f|_{\{c_0\}}) \geq -(K_X \cdot f_*C) - g(C) \cdot n.$$

Thus whenever the quantity $-(K_X \cdot f_*C) - g(C)n$ is positive, there must be a non-trivial one-dimensional family of deformations of the map $f : C \longrightarrow X$ keeping the image of c_0 fixed. So using Bend and Break I, we obtain a rational curve in X through x.

Step 3: Now we show how to get the positive quantity

$$-(K_X \cdot f_*C) - g(C) \cdot n > 0.$$

To do this, we need to get $-(K_X \cdot f_*C)$ to be big enough. We consider three cases:

[42]For Frobenius, generic flatness, and constancy of Euler characteristics in flat proper families, see [Har77, IV, 2.4.3], [FGA05, 5.12], [GW20, 10.84], [FGA05, 5.10(2)].

(1) $g(C) = 0$.

Then automatically $-(K_X \cdot f_*C) > 0$, since $-K_X$ is ample. Actually, by Lüroth's theorem, the image of C is already a rational curve through x.

(2) $g(C) = 1$.

For the same reason we have $-(K_X \cdot f_*C) > 0$, but as we want it to be big enough, we apply some finite endomorphism of C fixing the point c_0.

In particular, take c_0 as the origin of the elliptic curve C. Via the injective ring homomorphism [Har77, IV, 4.10]

$$\mathbb{Z} \longrightarrow \mathrm{End}(C, c_0),$$

we can pick the multiplication-by-r endomorphism $[r]_C$ and compose it with f. Then

$$-(K_X \cdot (f \circ [r]_C)_*C) - n = -r^2(K_X \cdot f_*C) - n,$$

where the equality follows from $\deg [r]_C = r^2$ [Har77, IV, 4.17]. This is positive for $r \gg 0$, and $(f \circ [r]_C)(c_0) = x$. Thus a multiple of the original cycle admits deformations with the image of one point fixed.

$$(C, c_0) \xrightarrow[\deg = r^2]{[r]_C} (C, c_0) \xrightarrow{f} (X, x).$$

Figure 2.9. The endomorphism $[r]_C$ raises the mapping degree while preserving the genus and the marked point.

(3) $g(C) \geq 2$.

By the Hurwitz formula for a separable endomorphism of degree d, we have

$$2g(C) - 2 = d \cdot (2g(C) - 2) + \deg R_{\mathrm{ram}}.$$

If $d > 1$, the right hand side is strictly larger than the left, since $\deg R_{\mathrm{ram}} \geq 0$. So a curve of genus at least 2 has no separable endomorphism of degree greater than 1. In characteristic zero, every finite endomorphism is separable.

Thus we can try to consider another smooth projective curve $h : C'' \longrightarrow C$ and deform $f \circ h$. But the problem here is that even though

$$((f \circ h)_*C'' \cdot K_X) = \deg h \cdot (f_*C \cdot K_X),$$

the genus also changes. For example, if C'' is a connected r-sheeted unbranched cover of C, then by the Hurwitz formula again,

$$2g(C'') - 2 = r \cdot (2g(C) - 2).$$

Hence

$$\begin{aligned} &-(K_X \cdot (f \circ h)_*C'') - g(C'') \cdot n \\ &\quad = -r\big((K_X \cdot f_*C) + g(C) \cdot n\big) + (r-1)n. \end{aligned}$$

This does not necessarily become positive even by making r large enough.

Step 4: Thus we are only in trouble in the case $g(C) > 1$, because in characteristic zero C does not admit an endomorphism of high degree. To overcome this, we will first reduce the data to positive characteristic. In the situation of Step 5, write C_p for the smooth projective reduction of C over $\overline{\mathbb{F}}_p$, defined over $\mathbb{F}_p$. Raising its homogeneous coordinates to the p-th power gives an $\overline{\mathbb{F}}_p$-endomorphism

$$F_p : C_p \longrightarrow C_p, \qquad \deg F_p = p.$$

Its iterates $F_p^m : C_p \longrightarrow C_p$ have degree p^m, while the domain remains the same curve C_p and hence has the same genus. Thus Frobenius will be applied to the reduction C_p, not to the original characteristic-zero curve C.

Next we construct these reductions. Step 6 treats the general case, where the residue field may be $\mathbb{F}_q$ and we use the q-power Frobenius on C_P.

Step 5: First suppose that X, C, the morphism f, and the point c_0 are all defined over $\mathbb{Q}$. After clearing denominators, take equations with integral coefficients

$$\begin{aligned} h_1(x_0, \ldots, x_N), \ldots, h_a(x_0, \ldots, x_N) \quad &\text{for } X, \\ c_1(y_0, \ldots, y_M), \ldots, c_b(y_0, \ldots, y_M) \quad &\text{for } C. \end{aligned}$$

We also spread out the graph of f and the marked point c_0. After inverting finitely many primes, these data give models

$$\mathcal{X} \longrightarrow U, \qquad \mathcal{C} \longrightarrow U, \qquad f_U : \mathcal{C} \longrightarrow \mathcal{X}, \qquad U = \operatorname{Spec} \mathbb{Z}[1/N_0],$$

together with a section $c_{0,U} : U \longrightarrow \mathcal{C}$, meaning that its composite with $\mathcal{C} \to U$ is id_U. It selects a marked point on each fibre. Set $x_U = f_U \circ c_{0,U}$. We may shrink U so that $\mathcal{X} \longrightarrow U$ and $\mathcal{C} \longrightarrow U$ are smooth and projective with geometrically integral fibres, of relative dimensions n and 1 respectively.

The sheaves

$$\mathcal{O}_{\mathcal{C}}, \qquad f_U^* \omega_{\mathcal{X}/U}, \qquad f_U^* T_{\mathcal{X}/U}$$

are vector bundles on $\mathcal{C}$ and are flat over U. Since $\mathcal{C} \longrightarrow U$ is proper, their fibrewise Euler characteristics are locally constant. For instance, writing $f_p : C_p \longrightarrow X_p$ for the geometric fibre over (p),

$$(K_{X_p} \cdot (f_p)_* C_p) = \deg f_p^* \omega_{X_p} = \chi(f_p^* \omega_{X_p}) - \chi(\mathcal{O}_{C_p}).$$

As U is connected, the quantities

$$(K_{X_p} \cdot (f_p)_* C_p), \qquad g(C_p), \qquad \chi(f_p^* T_{X_p})$$

are constant, and agree with their values on the original curve.

Moreover, ω_X^{-1} extends to $\omega_{\mathcal{X}/U}^{-1}$. By openness of ampleness on the fibres of a proper morphism, we may shrink U again so that $-K_{X_p}$ is ample for every geometric fibre. Hence

$$A := -(K_{X_p} \cdot (f_p)_* C_p) = -(K_X \cdot f_* C) > 0.$$

The geometric fibres C_p and X_p are defined over $\mathbb{F}_p$. Raising the homogeneous coordinates of C_p to the p-th power gives the Frobenius endomorphism F_p, a finite morphism of degree p. Since the section $c_{0,U}$ was included in the data, $c_{0,p}$ is $\mathbb{F}_p$-rational and $F_p^m(c_{0,p}) = c_{0,p}$. Therefore the base-pointed deformation space of the composite morphism

$$C_p \xrightarrow{F_p^m} C_p \xrightarrow{f_p} X_p$$

has dimension bounded below by

$$\begin{aligned}\dim_{[f_p \circ F_p^m]} \operatorname{Mor}(C_p, X_p; (f_p \circ F_p^m)|_{\{c_{0,p}\}}) \\ \geq -p^m(K_{X_p} \cdot (f_p)_* C_p) - g(C_p) \cdot n \\ = p^m A - g(C)n.\end{aligned}$$

This is positive for $m \gg 0$. Then as in Step 2, we produce a rational curve A_p on X_p through x_p for all primes represented in U.

Step 6: Suppose now that we are in the general case. The coefficients of the h_j defining X, the c_j defining C, the equations defining the graph $\Gamma(f)$, and the coordinates of c_0 need not be rational. These finitely many coefficients lie in a finitely generated $\mathbb{Z}$-subalgebra $R \subseteq k$. After adjoining the necessary coefficients and localizing R, we obtain models and sections as in Step 5 over a non-empty affine open subset $U \subseteq \operatorname{Spec} R$. We may again arrange smoothness, projectivity, geometric integrality, constancy of the above numerical quantities, and ampleness of the anticanonical bundles on the geometric fibres.

Now let $P \in U$ be a closed point. By Lemma 2.3.3, its residue field $\kappa(P)$ is finite. Set $q = |\kappa(P)|$, and denote the geometric fibres by C_P and X_P. The q-power Frobenius endomorphism $F_q : C_P \longrightarrow C_P$ fixes $c_{0,P}$, and its m-th iterate has degree q^m. Thus

$$\begin{aligned}\dim_{[f_P \circ F_q^m]} \operatorname{Mor}(C_P, X_P; (f_P \circ F_q^m)|_{\{c_{0,P}\}}) \\ \geq -q^m(K_{X_P} \cdot (f_P)_* C_P) - g(C_P) \cdot n > 0\end{aligned}$$

for $m \gg 0$. This gives a rational curve A_P through x_P for every closed point $P \in U$.

Step 7: In this step, we wish to obtain the required inequality, so we wish to replace A_P with another rational curve B_P through x_P such that

$$0 < -(K_{X_P} \cdot B_P) \leq n + 1.$$

To do this, notice that if $-(K_{X_P} \cdot A_P) > n+1$, then we can take the normalization morphism

$$\varphi : \mathbb{P}^1_{\kappa(P)} \longrightarrow A_P \hookrightarrow X_P.$$

In particular, $\varphi_*\mathbb{P}^1 = A_P$ as 1-cycles. Choose $q_0 \in \mathbb{P}^1$ with $\varphi(q_0) = x_P$, and choose $q_1 \neq q_0$. According to Section 2.1E, with these two point images fixed, we have

$$\begin{aligned}\dim_{[\varphi]} \operatorname{Mor}(\mathbb{P}^1, X_P; \varphi|_{\{q_0,q_1\}}) &\geq -(K_{X_P} \cdot \varphi_*\mathbb{P}^1) - n \\ &= -(K_{X_P} \cdot A_P) - n > 1.\end{aligned}$$

So this deformation space has dimension at least two.

But notice that φ is birational onto its image. Parametrizations with the same image differ by an automorphism of its normalization. Keeping the two point images fixed allows only finitely many choices of their preimages on the normalization; for each such choice, the remaining automorphisms form a one-dimensional family. Hence the fixed-image locus near $[\varphi]$ has dimension at most one, and we can choose a one-dimensional family in which the image of $\mathbb{P}^1$ actually moves.

By Bend and Break II, we obtain

$$A_P = \varphi_*\mathbb{P}^1 \sim_{\text{alg}} \sum_i a_i C_i, \qquad a_i \in \mathbb{Z}_{>0}, \qquad x_P \in \bigcup_i C_i,$$

where the C_i are integral rational curves and the effective cycle on the right is non-integral: it is reducible or a multiple rational curve. Since $-K_{X_P}$ is ample, each component satisfies

$$0 < -(K_{X_P} \cdot C_i) < -(K_{X_P} \cdot A_P).$$

This strict inequality also holds when the right hand side is a single multiple curve aC with $a > 1$.

Choose a component passing through x_P, take its normalization, and continue this process of deformation and decomposition. As long as the anticanonical degree is greater than $n + 1$, we find a rational curve through x_P of strictly smaller positive integral degree. This process must terminate in finitely many steps, giving our desired rational curve B_P with

$$x_P \in B_P, \qquad 0 < -(K_{X_P} \cdot B_P) \leq n + 1.$$

Step 8: In this step, we must conclude the existence of a rational curve on the original Fano variety X, in particular in characteristic zero, from the existence of bounded degree rational curves B_P for every closed point $P \in U$.

Choose a positive integer ℓ such that $-\ell K_X$ is very ample, and fix the corresponding embedding $X \hookrightarrow \mathbb{P}^N_k$. By enlarging R and shrinking U if necessary, spread out this fixed embedding to

$$\mathcal{X} \hookrightarrow \mathbb{P}^N_U, \qquad \mathcal{O}_{\mathcal{X}}(1) \simeq \omega^{-\ell}_{\mathcal{X}/U}.$$

All the preceding arguments apply to these models. Since a closed immersion remains a closed immersion after base change, this gives embeddings of all the geometric fibres into a fixed projective space. The degree of B_P under this embedding is

$$-\ell(K_{X_P} \cdot B_P) \leq \ell(n+1).$$

Set $D_0 = \ell(n+1)$. For $1 \leq e \leq D_0$, consider the relative space of morphisms of degree e, measured by this embedding, with the image of $0 = [0:1] \in \mathbb{P}^1$ fixed at x_U. Their disjoint union is

$$\mathcal{M} = \coprod_{e=1}^{D_0} \operatorname{Mor}_e(\mathbb{P}^1_U, \mathcal{X}; 0 \mapsto x_U).$$

This is a scheme of finite type over U. On each geometric fibre, Mor_e parametrizes tuples of homogeneous polynomials of degree e, up to simultaneous non-zero scaling, with no common zero, satisfying the equations of the target and the prescribed point condition. Thus these are actual morphisms, and $e \geq 1$ excludes constant maps. We allow all degrees between 1 and D_0 instead of multiplying coordinates by a common factor.

For every closed point $P \in U$, the normalization of B_P, after reparametrizing so that 0 maps to x_P, gives a geometric point of $\mathcal{M}_P$. By Chevalley's theorem, the image of $\mathcal{M} \longrightarrow U$ is constructible. Since U is an integral scheme of finite type over $\mathbb{Z}$, its closed points are dense. This constructible image contains every closed point, so it is dense. A dense constructible subset of an integral noetherian scheme contains a non-empty open subset, hence contains the generic point η of U.

Therefore $\mathcal{M}_\eta$ is non-empty. Write $K = \operatorname{Frac} R \subseteq k$. A closed point of $\mathcal{M}_\eta$ has residue field finite over K, and this field embeds into the original algebraically closed field k over K. After base change, we obtain a non-constant morphism

$$g : \mathbb{P}^1_k \longrightarrow X, \qquad g(0) = x, \qquad 1 \leq \deg g^*\mathcal{O}_X(1) \leq D_0.$$

Here we return to the original field k containing R; no embedding of $\operatorname{Frac} R$ into an arbitrary algebraically closed field is required.

Step 9: Let $D = g(\mathbb{P}^1)_{\mathrm{red}}$. By Lüroth's theorem, D is a rational curve, and $x \in D$. If δ is the degree of g onto its image, then $g_*\mathbb{P}^1 = \delta D$, so

$$\ell\delta\big(-(K_X \cdot D)\big) = \deg g^*\mathcal{O}_X(1) \leq \ell(n+1).$$

Hence $-(K_X \cdot D) \leq n+1$. This number is strictly positive since $-K_X$ is ample. Thus D satisfies the required inequality. □

2.3B Finding Rational Curves when K_X is not Nef

Now let us weaken our hypothesis on the smooth projective variety X. We no longer require $-K_X$ to be ample; we only assume that K_X is not nef, which means

that there is an integral curve $C' \subseteq X$ such that

$$-(K_X \cdot C') > 0.$$

This imposes no nefness assumption on $-K_X$. We also fix an ample Cartier divisor H on X.

Theorem 2.3.7 ([Mor82])**.** *Let X be a smooth projective variety and H an ample Cartier divisor on X. Assume that there is an integral curve $C' \subseteq X$ such that $-(C' \cdot K_X) > 0$. Then there is a rational curve $E \subseteq X$ such that*[43]

$$0 < -(E \cdot K_X) \leq \dim X + 1, \qquad \frac{-(E \cdot K_X)}{E \cdot H} \geq \frac{-(C' \cdot K_X)}{C' \cdot H}.$$

Proof. For finding rational curves, we can use a similar technique as before, passing to fields of positive characteristic to ensure that the dimension of the deformation space is large enough. Write $n = \dim X$.

Step 1: Let $f : C \longrightarrow C' \hookrightarrow X$ be the normalization of C'. It is birational onto its image and $f_*C = C'$ as 1-cycles. Choose $c_0 \in C$. If

$$\dim_{[f]} \operatorname{Mor}(C, X; f|_{\{c_0\}}) \geq -(K_X \cdot f_*C) - g(C) \cdot n > 0,$$

then f can deform with the image of c_0 fixed. As before, using Bend and Break I, we obtain

$$f_*C \sim_{\text{alg}} f'_*C + Z,$$

where $f' : C \longrightarrow X$ may be constant and Z is a non-zero effective rational 1-cycle. We keep the pushforward cycles here, since the remaining morphism need not be birational onto its image.

Step 2 (Repeating bend and break): In order to achieve the above lower bound on the deformation space, we pass to positive characteristic as in Steps 5–6 of Theorem 2.3.6. Spread out X, C, f, c_0, and also H, over a non-empty affine integral scheme U of finite type over $\mathbb{Z}$. After shrinking U, take smooth projective models $\mathcal{X}, \mathcal{C}$, a morphism $f_U : \mathcal{C} \longrightarrow \mathcal{X}$ extending f, and a relatively ample line bundle $\mathcal{H}$ extending $\mathcal{O}_X(H)$.

For a closed point $P \in U$, use the same notation $f_P : C_P \to X_P$ for the geometric fibres, and write H_P for the corresponding divisor class. Set

$$A = -(K_{X_P} \cdot (f_P)_*C_P), \quad B = H_P \cdot (f_P)_*C_P, \quad G = g(C_P)n.$$

These quantities are independent of P, with $A, B > 0$. In particular,

$$M := \frac{-(C' \cdot K_X)}{C' \cdot H} = \frac{-(K_{X_P} \cdot (f_P)_*C_P)}{H_P \cdot (f_P)_*C_P} = \frac{A}{B} > 0.$$

[43] For preservation of intersection numbers under algebraic equivalence, see [Ful98, 10.3.2].

Here constancy follows from the same Euler characteristic argument as in Theorem 2.3.6, applied also to $f_U^*\mathcal{H}$.

Put $q = |\kappa(P)|$, and compose f_P with the m-th iterate of the q-power Frobenius $F_q : C_P \longrightarrow C_P$. Writing $Q = q^m$, we get

$$\begin{aligned}\dim_{[f_P \circ F_q^m]} \operatorname{Mor}(C_P, X_P; (f_P \circ F_q^m)|_{\{c_{0,P}\}}) \\ \geq -Q(K_{X_P} \cdot (f_P)_* C_P) - G = QA - G,\end{aligned}$$

which is positive for $m \gg 0$. The ratio M does not change after this composition, since both intersection numbers are multiplied by Q.

Set $f_{P,m}^0 = f_P \circ F_q^m$ and

$$Y_j = (f_{P,m}^j)_* C_P, \qquad Y_0 = Q(f_P)_* C_P.$$

Whenever $-(K_{X_P} \cdot Y_{j-1}) > G$, we can deform the remaining morphism with the image of $c_{0,P}$ fixed and apply Bend and Break I again, without composing with Frobenius again. This gives

$$Y_{j-1} \sim_{\text{alg}} Y_j + W_j,$$

where W_j is a non-zero effective rational cycle. The image of the marked point is fixed for the deformation at each individual stage; we do not need it to remain the same between stages.

Recall that algebraic equivalence implies numerical equivalence [Ful98, 10.3]. Since H_P is ample, at each step

$$H_P \cdot Y_j = H_P \cdot Y_{j-1} - H_P \cdot W_j < H_P \cdot Y_{j-1}.$$

These are non-negative integers, so the process must stop after finitely many steps, with

$$Q(f_P)_* C_P \sim_{\text{alg}} Y_s + Z_s, \qquad Z_s = \sum_{j=1}^{s} W_j, \qquad -(K_{X_P} \cdot Y_s) \leq G.$$

This is our stopping condition; it does not assert that the remaining morphism has no deformations. The cycle Y_s may be zero.

Let

$$\begin{aligned} a &= -(K_{X_P} \cdot Y_s), \quad & b &= -(K_{X_P} \cdot Z_s), \\ c &= H_P \cdot Y_s, & d &= H_P \cdot Z_s. \end{aligned}$$

Then

$$a \leq G, \qquad c \geq 0, \qquad a + b = QA, \qquad c + d = QB.$$

For large m, it follows that $b \geq QA - G > 0$ and $0 < d \leq QB$. We only need the upper bound on a; no lower bound on a is assumed.

Step 3: We claim the following. For any $\varepsilon > 0$, if $m \gg 0$, then there exists an integral component E_P of the rational cycle Z_s such that

$$\frac{-(E_P \cdot K_{X_P})}{E_P \cdot H_P} > M - \varepsilon.$$

To see this, we use the following lemma.

Lemma 2.3.8. *If $c, d > 0$ and $a, b \in \mathbb{R}$, then*

$$\frac{a+b}{c+d} \leq \max\left\{\frac{a}{c}, \frac{b}{d}\right\}.$$

Indeed, $(a+b)/(c+d)$ is the weighted average of a/c and b/d, with positive weights c and d. The same argument applies to any finite number of terms.

In our situation, for Q sufficiently large, we have

$$\frac{b}{d} \geq \frac{QA - G}{QB} = M - \frac{G}{QB} > M - \varepsilon.$$

This estimate also applies when $c = 0$; there is no need to form a/c. Write

$$Z_s = \sum_i r_i E_{P,i}, \qquad r_i \in \mathbb{Z}_{>0},$$

and set $b_i = -(K_{X_P} \cdot E_{P,i})$ and $d_i = H_P \cdot E_{P,i} > 0$. By the weighted-average argument,

$$\max_i \left\{\frac{b_i}{d_i}\right\} \geq \frac{\sum_i r_i b_i}{\sum_i r_i d_i} = \frac{b}{d} > M - \varepsilon.$$

Thus the required component E_P exists.

Step 4: From now on, take $0 < \varepsilon < M/2$. Suppose that

$$-(E_P \cdot K_{X_P}) > n + 1$$

for the rational curve E_P obtained above. Take its normalization $\varphi : \mathbb{P}^1 \longrightarrow E_P \hookrightarrow X_P$ and fix the images of two distinct points $q_0, q_1 \in \mathbb{P}^1$. Then

$$\dim_{[\varphi]} \mathrm{Mor}(\mathbb{P}^1, X_P; \varphi|_{\{q_0,q_1\}}) \geq -(E_P \cdot K_{X_P}) - n > 1.$$

As in Step 7 of Theorem 2.3.6, the fixed-image locus has dimension at most one. Hence we can move the image curve and apply Bend and Break II to obtain a non-integral effective rational cycle

$$E_P \sim_{\mathrm{alg}} \sum_i a_i E'_{P,i}.$$

Using Lemma 2.3.8 with the multiplicities a_i, at least one component E'_P satisfies

$$\frac{-(E'_P \cdot K_{X_P})}{E'_P \cdot H_P} > M - \varepsilon. \tag{2.1}$$

Since $M-\varepsilon>0$, this component has positive anticanonical degree. Also,

$$0<H_P\cdot E'_P<H_P\cdot E_P,$$

because H_P is ample and the cycle is non-integral. This includes the case of a multiple rational curve.

If $-(E'_P\cdot K_{X_P})>n+1$, take its normalization and repeat. This process cannot continue infinitely, since at each step the positive integer $H_P\cdot E_P$ decreases. Eventually we arrive at a rational curve, again called E_P, such that

$$0<-(E_P\cdot K_{X_P})\le n+1,\qquad \frac{-(E_P\cdot K_{X_P})}{E_P\cdot H_P}>M-\varepsilon.$$

No nefness of $-K_{X_P}$ is needed here: positivity on the selected component follows from its ratio, and termination follows from H_P.

These inequalities imply

$$1\le E_P\cdot H_P<\frac{n+1}{M-\varepsilon}<\frac{2(n+1)}{M}.$$

Therefore, for all $0<\varepsilon<M/2$ and all closed points P, the possible ratios belong to the same finite set

$$\mathcal{S}=\left\{\frac{a}{b}:a,b\in\mathbb{Z}_{>0},\quad a\le n+1,\quad b<\frac{2(n+1)}{M}\right\}.$$

If $\mathcal{S}$ contains numbers below M, choose $\varepsilon>0$ smaller than $M/2$ and smaller than the minimum of the positive numbers $M-s$ for $s\in\mathcal{S}$, $s<M$. If it contains no such number, any $0<\varepsilon<M/2$ will do. The ratio $>M-\varepsilon$ must then be at least M. Consequently, for every closed point $P\in U$, we obtain a rational curve E_P with

$$0<-(E_P\cdot K_{X_P})\le n+1,\qquad \frac{-(E_P\cdot K_{X_P})}{E_P\cdot H_P}\ge M.$$

This makes precise the argument of letting $\varepsilon\to 0$: we use finitely many numerical possibilities, without taking a limit of the curves themselves.

Furthermore, $H_P\cdot E_P\le (n+1)/M$, uniformly in P. We can now reason as in Step 8 of Theorem 2.3.6, keeping both intersection numbers in the parameter space.

Choose $\ell>0$ such that ℓH is very ample, and spread out the corresponding fixed embedding, after enlarging R and shrinking U if necessary, with

$$\mathcal{O}_{\mathcal{X}}(1)\simeq\mathcal{H}^{\otimes\ell}.$$

The preceding argument applies to every closed fibre of the resulting model. For each of the finitely many pairs of integers

$$1\le a\le n+1,\qquad b>0,\qquad \frac{a}{b}\ge M,$$

let $\mathcal{N}_{a,b}$ be the locus in $\mathrm{Mor}_{\ell b}(\mathbb{P}^1_U, \mathcal{X})$ where

$$\deg g^*\omega^{-1}_{\mathcal{X}/U} = a, \qquad \deg g^*\mathcal{H} = b.$$

The degrees of these pullback line bundles are locally constant in families of morphisms, so these loci are open and closed in their respective fixed-degree morphism spaces. Thus

$$\mathcal{N} = \coprod_{(a,b)} \mathcal{N}_{a,b}$$

is of finite type over U.

The normalization of each E_P gives a geometric point of $\mathcal{N}_P$. Chevalley's theorem therefore implies, as before, that $\mathcal{N}$ has a non-empty generic fibre. Base changing to the original algebraically closed field k, we obtain a non-constant morphism $g : \mathbb{P}^1_k \longrightarrow X$ with

$$a = \deg g^*\omega_X^{-1}, \qquad b = \deg g^*\mathcal{O}_X(H), \qquad 1 \leq a \leq n+1, \quad \frac{a}{b} \geq M.$$

Let $E = g(\mathbb{P}^1)_{\mathrm{red}}$, which is rational by Lüroth's theorem, and let δ be the degree of g onto E. Then

$$-(K_X \cdot E) = \frac{a}{\delta}, \qquad H \cdot E = \frac{b}{\delta}.$$

Hence

$$0 < -(K_X \cdot E) \leq n+1, \qquad \frac{-(K_X \cdot E)}{H \cdot E} = \frac{a}{b} \geq M,$$

which is the required conclusion. □

From this theorem we see the strength of the bend-and-break technique: Frobenius makes the deformation space large in positive characteristic, and bounded degree allows us to recover a rational curve over the original field.

2.4 Mori's Cone Theorem

Recall that in Chapter 1 we introduced some concepts like cones, extremal rays and the nef cone. For a projective variety X, the cone of effective curves is a subcone of the $\mathbb{R}$-vector space $N_1(X)_{\mathbb{R}}$. Related to the perfect pairing

$$N^1(X)_{\mathbb{R}} \times N_1(X)_{\mathbb{R}} \longrightarrow \mathbb{R},$$

we have

$$\mathrm{NE}_{\mathbb{Q}}(X) = \left\{ \sum_i a_i[C_i] \;\middle|\; C_i \subseteq X \text{ an irreducible curve, } a_i \in \mathbb{Q}_{\geq 0} \right\},$$

$$\mathrm{NE}(X) = \left\{ \sum_i a_i[C_i] \;\middle|\; C_i \subseteq X \text{ an irreducible curve, } a_i \in \mathbb{R}_{\geq 0} \right\},$$

$$\overline{\mathrm{NE}}(X) = \text{the closure of } \mathrm{NE}(X) \text{ in } N_1(X)_{\mathbb{R}}.$$

Here every sum is finite. The nef cone lies in $N^1(X)_{\mathbb{R}}$ and is the dual cone of $\overline{\mathrm{NE}}(X)$. Recall also that for a Cartier divisor D,

$$D_{\geq 0} := \{\gamma \in N_1(X)_{\mathbb{R}} \mid (D \cdot \gamma) \geq 0\}, \qquad D^{\perp} := \{\gamma \in N_1(X)_{\mathbb{R}} \mid (D \cdot \gamma) = 0\}.$$

We use the notation

$$\overline{\mathrm{NE}}(X)_{D \geq 0} := \overline{\mathrm{NE}}(X) \cap D_{\geq 0},$$

and similarly for > 0, ≤ 0 and < 0, also for $\mathbb{R}$-Cartier divisors. Now we are ready to state the first result of Mori for smooth varieties of arbitrary dimension.

2.4A Cone Theorem in the Nonsingular Case

Theorem 2.4.1 (Cone Theorem)**.** *Let X be a nonsingular projective variety such that K_X fails to be nef.*[44]

(1) There are at most countably many rational curves $C_i \subseteq X$ such that $0 < -(C_i \cdot K_X) \leq \dim X + 1$, and

$$\overline{\mathrm{NE}}(X) = \overline{\mathrm{NE}}(X)_{K_X \geq 0} + \sum_i \mathbb{R}_{\geq 0}[C_i].$$

(2) For any $\varepsilon > 0$ and any ample Cartier divisor H, we have

$$\overline{\mathrm{NE}}(X) = \overline{\mathrm{NE}}(X)_{(K_X + \varepsilon H) \geq 0} + \sum_{i \in I(H,\varepsilon)} \mathbb{R}_{\geq 0}[C_i],$$

where $I(H, \varepsilon) = \{i \mid (C_i \cdot (K_X + \varepsilon H)) < 0\}$ is finite.

Proof. From Grothendieck's Theorem 2.1.11 we know that the scheme $\mathrm{Mor}(\mathbb{P}^1_k, X)$ is the disjoint union of countably many schemes of finite type over k, hence has at most countably many irreducible components. Pick a component B, with its reduced structure, and consider the flat projective family

$$\pi : \mathbb{P}^1_k \times B \longrightarrow B$$

and the universal evaluation map

$$f^{\mathrm{univ}} : \mathbb{P}^1_k \times \mathrm{Mor}(\mathbb{P}^1_k, X) \longrightarrow X.$$

For any Cartier divisor D on X, the pullback of $\mathcal{O}_X(D)$ is a line bundle on $\mathbb{P}^1_k \times B$ and is flat over B. Thus the function

$$b \longmapsto \chi(\mathbb{P}^1_b, f_b^* \mathcal{O}_X(D))$$

[44] For separation of convex sets and the closedness assertion in the cone theorem, see [BV04, 2.5.1], [Kol96, p. 161].

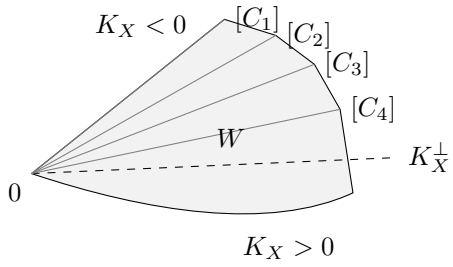
is locally constant. Since B is irreducible, it is connected, so this function is constant. By Riemann–Roch on $\mathbb{P}^1$, the intersection number $((f_b)_*[\mathbb{P}^1] \cdot D)$ is constant. Hence the *pushforward* 1*-cycles* parametrized by B are numerically equivalent. Every rational curve has a normalization map $\mathbb{P}^1 \to X$ whose pushforward is its curve cycle, so the numerical classes of rational curves also form an at most countable set.

In each numerical class γ represented by a rational curve with

$$0 < -(\gamma \cdot K_X) \leq \dim X + 1,$$

we pick one such curve C_i. This gives an at most countable index set I, and we define

$$W := \overline{\left(\overline{\mathrm{NE}}(X)_{K_X \geq 0} + \sum_{i \in I} \mathbb{R}_{\geq 0}[C_i] \right)}.$$



Step 1: We aim to prove $W = \overline{\mathrm{NE}}(X)$. Assume the contrary, so the natural inclusion $W \subseteq \overline{\mathrm{NE}}(X)$ is strict. By separation of convex cones, followed by a small ample perturbation and rational approximation, we may choose a Cartier divisor D whose associated linear function $(D \cdot -)$ is positive on $W - \{0\}$ but negative somewhere in $\overline{\mathrm{NE}}(X)$. Indeed, strict positivity can be checked on a compact slice cut out by an ample divisor, so it survives a sufficiently small perturbation; we then clear the denominators.

Let H be an ample Cartier divisor, and let $\mu > 0$ be the largest real number such that $H + \mu D$ is nef. This number is finite because D is negative somewhere in $\overline{\mathrm{NE}}(X)$. The class $H + \mu D$ lies on the boundary of the nef cone. Thus, by Kleiman's criterion, there is a nonzero $z \in \overline{\mathrm{NE}}(X)$ with

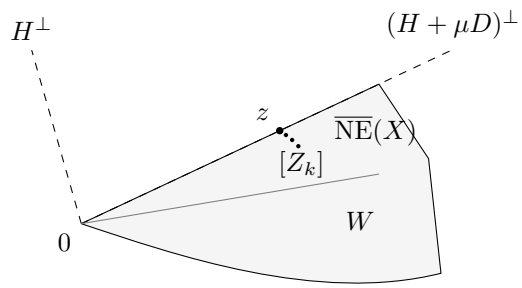
$$((H + \mu D) \cdot z) = 0.$$

Then $(D \cdot z) < 0$ because $(H \cdot z) > 0$, so $z \notin W$. Moreover $(K_X \cdot z) < 0$, since $\overline{\mathrm{NE}}(X)_{K_X \geq 0} \subseteq W$.

Choose effective 1-cycles with rational coefficients $Z_k = \sum_j a_{kj}[Z_{kj}]$, where $a_{kj} > 0$, such that

$$Z_k \longrightarrow z \qquad \text{as } k \longrightarrow \infty.$$

The situation is illustrated schematically below.



Fix a rational number $0 < \mu' < \mu$. The $\mathbb{Q}$-Cartier divisor

$$L := H + \mu' D = \left(1 - \frac{\mu'}{\mu}\right) H + \frac{\mu'}{\mu}(H + \mu D)$$

is ample, being the sum of an ample and a nef $\mathbb{R}$-divisor. For k sufficiently large, $-(Z_k \cdot K_X) > 0$. By Lemma 2.3.8,

$$\max_j \frac{-(Z_{kj} \cdot K_X)}{(Z_{kj} \cdot L)} \;\geq\; \frac{-(Z_k \cdot K_X)}{(Z_k \cdot L)} > 0.$$

Rearrange the indices so that the maximum is achieved for Z_{k0}. Apply Theorem 2.3.7 to Z_{k0} and L: formally, apply it to a positive Cartier multiple of L, then cancel that multiple in the ratio. We obtain a rational curve E_k such that

$$0 < -(E_k \cdot K_X) \leq \dim X + 1 \tag{2.2}$$

and

$$\frac{-(E_k \cdot K_X)}{(E_k \cdot L)} \geq \frac{-(Z_{k0} \cdot K_X)}{(Z_{k0} \cdot L)} \geq \frac{-(Z_k \cdot K_X)}{(Z_k \cdot L)}.$$

By (2.2), E_k is numerically equivalent to one of the curves C_i. Thus $(E_k \cdot D) > 0$, by the choice of D, and consequently

$$\frac{-(E_k \cdot K_X)}{(E_k \cdot H)} \geq \frac{-(E_k \cdot K_X)}{(E_k \cdot L)} \geq \frac{-(Z_k \cdot K_X)}{(Z_k \cdot L)}.$$

On the other hand, there is a positive constant $c \gg 0$ such that $K_X + cH$ is ample. Hence, for every such curve,

$$\frac{-(E_k \cdot K_X)}{(E_k \cdot H)} < c.$$

But $-(z \cdot K_X) > 0$ and $((H + \mu D) \cdot z) = 0$. We can first choose a rational $\mu' < \mu$ sufficiently close to μ that

$$\frac{-(z \cdot K_X)}{(z \cdot (H + \mu' D))} > c.$$

Keep this μ' fixed. Since $Z_k \to z$, the same strict inequality holds with Z_k in place of z for k sufficiently large, a contradiction. This concludes Step 1.

Step 2: We prove part (2) of the theorem. Fix H and $\varepsilon > 0$. If $(C_i \cdot (K_X + \varepsilon H)) < 0$, then

$$(C_i \cdot H) < \frac{-(C_i \cdot K_X)}{\varepsilon} \leq \frac{\dim X + 1}{\varepsilon}.$$

By the second statement of Theorem 1.4.28, there are only finitely many such numerical classes. Since we chose one C_i per numerical class, there are only finitely many such C_i. Set

$$V_{H,\varepsilon} := \overline{\mathrm{NE}}(X)_{(K_X+\varepsilon H)\geq 0} + \sum_{i \in I(H,\varepsilon)} \mathbb{R}_{\geq 0}[C_i].$$

This cone is closed. Indeed, in a convergent sequence $u_m + \sum_i a_{mi}[C_i]$ in this cone, with $a_{mi} \geq 0$, intersection with H bounds $H \cdot u_m$ and each a_{mi}. The set $\{u \in \overline{\mathrm{NE}}(X) \mid H \cdot u \leq M\}$ is compact for every M. We can therefore pass to a subsequence on which all these terms converge, and the limit remains in $V_{H,\varepsilon}$.

Clearly $V_{H,\varepsilon} \subseteq \overline{\mathrm{NE}}(X)$. It also contains $\overline{\mathrm{NE}}(X)_{K_X \geq 0}$ and every ray $\mathbb{R}_{\geq 0}[C_i]$, and hence, being closed, it contains W. By Step 1, $W = \overline{\mathrm{NE}}(X)$, which proves part (2).

Step 3: We prove part (1), including the removal of the closure in the definition of W. Recall the elementary finite-dimensional convexity fact that a closed convex cone with a compact base is generated by its extremal rays, using finite nonnegative sums. To see its applicability here, intersect $\overline{\mathrm{NE}}(X)$ with $H \cdot z = 1$. This is a compact convex set; the finite-dimensional Minkowski theorem and Carathéodory's theorem express every point as a finite convex combination of extreme points. These correspond to extremal rays of $\overline{\mathrm{NE}}(X)$.

Let $R = \mathbb{R}_{\geq 0} r$ be a K_X-negative extremal ray. Choose $\varepsilon > 0$ small enough that $(K_X + \varepsilon H) \cdot r < 0$. By Step 2,

$$r = u + \sum_{i \in I(H,\varepsilon)} a_i [C_i], \qquad u \in \overline{\mathrm{NE}}(X)_{(K_X+\varepsilon H)\geq 0}, \quad a_i \geq 0.$$

Extremality implies that every nonzero summand belongs to R. Since $R - \{0\}$ is $(K_X + \varepsilon H)$-negative, we must have $u = 0$. Thus some C_i generates R. Every remaining extremal ray is contained in $\overline{\mathrm{NE}}(X)_{K_X \geq 0}$. The convexity fact above now gives

$$\overline{\mathrm{NE}}(X) = \overline{\mathrm{NE}}(X)_{K_X \geq 0} + \sum_i \mathbb{R}_{\geq 0}[C_i],$$

as required. In particular, each K_X-negative extremal ray is generated by a rational curve of anticanonical degree at most $\dim X + 1$. □

Example 2.4.2 (Fano varieties)**.** Let X be a Fano variety. Then $\overline{\mathrm{NE}}(X) \subseteq N_1(X)_{\mathbb{R}}$ is a rational polyhedral cone, spanned by the classes of finitely many

rational curves. Apply part (2) of the cone theorem with $H = -K_X$ and $\varepsilon = \frac{1}{2}$. Since $K_X + \varepsilon H = \frac{1}{2}K_X$ is negative on $\overline{\mathrm{NE}}(X) - \{0\}$, the nonnegative part is $\{0\}$.

Now we introduce a key step: suitable extremal rays correspond to morphisms of a variety. This connects the cone of curves with more traditional geometric properties of the variety.

Definition 2.4.3. Let X be a normal projective variety and $F \subseteq \overline{\mathrm{NE}}(X)$ an extremal face. A surjective morphism $\mathrm{cont}_F : X \to Z$ to a normal projective variety is called the *contraction of* F if the following hold:

(1) $\mathrm{cont}_F(C)$ is a point for an irreducible curve $C \subseteq X$ if and only if $[C] \in F$, and F is the closed cone generated by these curve classes;

(2) $(\mathrm{cont}_F)_*\mathcal{O}_X = \mathcal{O}_Z$; in particular, cont_F has connected fibres.

Remark 2.4.4. By Proposition 1.4.38, the contraction of F is unique up to isomorphism if it exists. Moreover, a morphism $g : X \to Z$ between normal projective varieties with $g_*\mathcal{O}_X = \mathcal{O}_Z$ is determined by the curves it contracts. Its relative cone $\mathrm{NE}(g)$, as before, is generated by their classes; its closure is the corresponding extremal face of $\overline{\mathrm{NE}}(X)$. Thus g is the contraction of this face. Now we give a lemma which will be used in Chapter 5.

Lemma 2.4.5. *Let X be a normal projective variety and let $r \subseteq \overline{\mathrm{NE}}(X)$ be an extremal ray. Assume that X is $\mathbb{Q}$-factorial and that the contraction of r between normal projective varieties exists,*

$$f = \mathrm{cont}_r : X \longrightarrow Z.$$

Then one of the following holds:

- *f is birational and $\mathrm{Ex}(f)$ is an irreducible divisor (we say that f is a* **divisorial contraction***);*
- *f is birational and $\mathrm{Ex}(f)$ has codimension at least two (we say that f is a* **small contraction***);*
- $\dim Z < \dim X$ *(we say that f is* **of fibre type***).*

Proof. If $\dim Z = \dim X$, the equality $f_*\mathcal{O}_X = \mathcal{O}_Z$ implies that f is birational. Thus the only assertion to prove is that $\mathrm{Ex}(f)$ is irreducible when it contains a divisor.

Assume that $\mathrm{Ex}(f)$ contains a prime divisor D. Since Z is normal, $f(D)$ has codimension at least two in Z. After replacing a projective embedding of Z by a sufficiently high Veronese embedding, choose a general integral hyperplane section H containing $f(D)$. Equivalently, choose a general member of a sufficiently positive linear system with base locus $f(D)$.

Let $H' \subseteq X$ be the strict transform of H. The proper morphism $H' \to H$ is surjective, so H' meets $f^{-1}(z)$ for every $z \in f(D)$. On the other hand, H' does not contain D. Thus for general $z \in f(D)$, the fibre $f^{-1}(z)$ meets H' but is not contained in it.

This fibre is connected. Since X is projective, we can choose an irreducible curve $C \subseteq f^{-1}(z)$ which meets H' but is not contained in it: a component of the fibre outside H' meets H', and hyperplane sections through an intersection point give such a curve. By $\mathbb{Q}$-factoriality, H' is $\mathbb{Q}$-Cartier, so $(H' \cdot C) > 0$. Since C is contracted by f, we have $r = \mathbb{R}_{\geq 0}[C]$.

The Cartier divisor H has pullback

$$f^*H = H' + \sum_i a_i E_i,$$

where the E_i are the exceptional prime divisors that occur in the pullback, and $a_i > 0$. Locally these coefficients are the orders of vanishing of a local equation of H pulled back to X. Each E_i is $\mathbb{Q}$-Cartier. Moreover, $(f^*H \cdot C) = (H \cdot f_*C) = 0$, hence

$$-(H' \cdot C) = \left(\sum_i a_i E_i \cdot C \right) < 0.$$

It follows that there is a divisor $E_i = E' \subseteq \mathrm{Ex}(f)$ with $(E' \cdot C) < 0$.

For a proper birational morphism to a normal variety, every point of $\mathrm{Ex}(f)$ lies on a positive-dimensional fibre, by Zariski's main theorem. The fibres are connected, so through any $x \in \mathrm{Ex}(f)$ there is an irreducible contracted curve C'. Since f contracts the ray r,

$$[C'] = \lambda [C] \qquad \text{for some } \lambda \in \mathbb{R}_{>0}.$$

Therefore $(E' \cdot C') = \lambda (E' \cdot C) < 0$. An effective $\mathbb{Q}$-Cartier divisor has nonnegative intersection with a curve not contained in its support, so $C' \subseteq E'$. Hence $x \in E'$. We conclude that $\mathrm{Ex}(f) = E'$, as wanted. □

In general, an extremal face need not admit a contraction to a projective variety, as the following example shows.

Example 2.4.6. For this example, work over $k = \mathbb{C}$. Let $g : X \to \mathbb{P}^2$ be the blow-up at twelve distinct very general points $p_1, \ldots, p_{12}$ on a smooth plane cubic D. Let $C \subseteq X$ be the strict transform of D. By the canonical divisor formula for a blow-up [Har77, II, Exercise 8.5(b)],

$$\begin{aligned}(K_X \cdot C) &= \left(g^* K_{\mathbb{P}^2} + \sum_{i=1}^{12} E_i \right) \cdot C \\ &= \deg \mathcal{O}_D(-3) + \sum_{i=1}^{12} (E_i \cdot C) \\ &= -3 \times 3 + 12 = 3.\end{aligned}$$

Since $g^*D = C + \sum_{i=1}^{12} E_i$, we also have

$$C^2 = \left(g^*D - \sum_{i=1}^{12} E_i\right)^2 = 9 - 12 = -3.$$

Thus, by Example 1.4.31, $[C]$ spans an extremal ray r of $\overline{\mathrm{NE}}(X)$. We claim that r has no contraction to a projective variety.

Indeed, suppose that such a contraction $\mathrm{cont}_r : X \to Z$ exists. Since $r \neq \overline{\mathrm{NE}}(X)$, the variety Z is not a point. Pulling back an ample line bundle on Z, we obtain a nontrivial line bundle on X

$$L = g^*\mathcal{O}_{\mathbb{P}^2}(d) \otimes \mathcal{O}_X\left(\sum_{i=1}^{12} a_i E_i\right)$$

whose restriction to C is trivial. Identifying C with D, this gives

$$\mathcal{O}_D(d) \otimes \mathcal{O}_D\left(\sum_{i=1}^{12} a_i p_i\right) \simeq \mathcal{O}_D.$$

Its degree is $3d + \sum_i a_i$, so a relation is possible only if $3d + \sum_i a_i = 0$. For every nonzero integer tuple satisfying this degree condition, at least one a_i is nonzero. Varying that p_i, multiplication by a_i on the elliptic curve is nonconstant. Thus the displayed relation holds on a proper closed subset of D^{12}. There are only countably many integer tuples. Choosing the points very generally avoids all these subsets, so the only possible relation has $d = a_1 = \cdots = a_{12} = 0$. This contradicts the choice of L. Hence cont_r does not exist, as claimed.

2.4B Application: Minimal Models of Surfaces

For the rest of this section, all surfaces are smooth and projective over k. On a smooth surface, curves are Cartier divisors, and intersection identifies $N^1(X)_{\mathbb{R}}$ with $N_1(X)_{\mathbb{R}}$. We use this identification when writing z^2 for $z \in N_1(X)_{\mathbb{R}}$. To begin with, we need two lemmas.

Lemma 2.4.7. *If D is a Cartier divisor on a smooth projective surface X with $D^2 > 0$, then either $|nD| \neq \varnothing$ for all $n \gg 0$, or $|-nD| \neq \varnothing$ for all $n \gg 0$.*

Proof. By Riemann–Roch on a surface and Serre duality, dropping the nonnegative superabundance terms h^1, we get

$$h^0(nD) + h^0(K_X - nD) \geq \frac{n^2}{2}D^2 - \frac{n}{2}(D \cdot K_X) + \chi(\mathcal{O}_X),$$
$$h^0(-nD) + h^0(K_X + nD) \geq \frac{n^2}{2}D^2 + \frac{n}{2}(D \cdot K_X) + \chi(\mathcal{O}_X).$$

The right-hand sides grow quadratically. Set $B = h^0(2K_X)$. If both $H^0(K_X - nD)$ and $H^0(K_X + nD)$ are nonzero, multiplication by a fixed nonzero section of either line bundle injects the other space into $H^0(2K_X)$, since X is integral. Thus both dimensions are at most B. If one space is zero, the corresponding inequality already gives sections of nD or $-nD$.

For n sufficiently large that both right-hand sides exceed B, it follows that $h^0(nD) > 0$ or $h^0(-nD) > 0$. These signs cannot vary with n: for an ample divisor H, a nonzero effective divisor has positive intersection with H, whereas $(nD \cdot H)$ and $(-nD \cdot H)$ have opposite signs. This also shows $(D \cdot H) \neq 0$. Hence one fixed sign works for all $n \gg 0$. □

Corollary 2.4.8. *Let X be a smooth projective surface with an ample Cartier divisor H. The set*[45]

$$Q := \{z \in N_1(X)_{\mathbb{R}} \mid z^2 > 0\}$$

has two connected components,

$$Q^+ := \{z \in Q \mid (z \cdot H) > 0\}, \qquad Q^- := \{z \in Q \mid (z \cdot H) < 0\}.$$

Furthermore, $Q^+ \subseteq \overline{\mathrm{NE}}(X)$.

Proof. We have the intersection form

$$(-,-) : N_1(X)_{\mathbb{R}} \times N_1(X)_{\mathbb{R}} \longrightarrow \mathbb{R}, \qquad (z_1, z_2) \longmapsto (z_1 \cdot z_2).$$

Write $\rho = \rho(X)$. By the Hodge index theorem,

$$N_1(X)_{\mathbb{R}} = \mathbb{R}[H] \oplus H^{\perp}, \qquad H^{\perp} = \{z \in N_1(X)_{\mathbb{R}} \mid (z \cdot H) = 0\},$$

where the intersection form is positive definite on the one-dimensional space $\mathbb{R}[H]$ and negative definite on the $(\rho - 1)$-dimensional space $H^{\perp}$.

More precisely, for a basis $\{D_i\}_{i=1}^{\rho}$, let $M = (D_i \cdot D_j)_{i,j}$. Writing $z_1 = \sum_i x_i D_i$ and $z_2 = \sum_i y_i D_i$, we have

$$(z_1 \cdot z_2) = x^T M y.$$

The real symmetric matrix M can be orthogonally diagonalized. After also rescaling the basis vectors, Sylvester's law of inertia gives a basis $\{e_i\}_{i=1}^{\rho}$, orthogonal for the intersection form, in which the matrix is

$$\begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & -1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & -1 \end{pmatrix}.$$

[45] For the inertia of a real symmetric bilinear form, see [Lan00, p. 577 Cor4.2].

We may take $e_1 = [H]/\sqrt{H^2}$ and choose the remaining basis vectors in $H^\perp$, so

$$[H] = (\sqrt{H^2}, 0, \ldots, 0).$$

For $z = \sum_i x_i e_i \in Q$, we have

$$x_1^2 - \sum_{i\geq 2} x_i^2 > 0, \qquad \text{or equivalently} \qquad |x_1| > \sqrt{\sum_{i\geq 2} x_i^2}.$$

This gives the two open convex, hence connected, components

$$Q^+ = \left\{ x_1 > \sqrt{\sum_{i\geq 2} x_i^2} \right\}, \qquad Q^- = \left\{ x_1 < -\sqrt{\sum_{i\geq 2} x_i^2} \right\}.$$

Since $(z \cdot H) = x_1\sqrt{H^2}$, these are exactly the components in the statement.

For the second assertion, first take a rational class $[D] \in Q^+$. After clearing denominators, we may assume that D is Cartier. By Lemma 2.4.7, either nD or $-nD$ is linearly equivalent to an effective divisor for $n \gg 0$. An effective nonzero divisor has positive intersection with H, so $(D \cdot H) > 0$ excludes the negative sign. Hence $[D] = \frac{1}{n}[nD] \in \mathrm{NE}(X)$. Rational classes are dense in the open set Q^+, and $\overline{\mathrm{NE}}(X)$ is closed. Thus $Q^+ \subseteq \overline{\mathrm{NE}}(X)$. In particular, every point of Q^+ is an interior point of $\overline{\mathrm{NE}}(X)$. □

Now we give the main result in this section (even in this chapter!), classifying the possible contractions on surfaces when K_X fails to be nef.

Theorem 2.4.9 (MMP for Surfaces)**.** *Let X be a smooth projective surface and $R \subseteq \overline{\mathrm{NE}}(X)$ a K_X-negative extremal ray. Then the contraction morphism $\mathrm{cont}_R : X \to Z$ exists and is of one of the following types:*[46]

(1) Z is a smooth projective surface and X is obtained from Z by blowing up a closed point; $\rho(Z) = \rho(X) - 1$.

(2) Z is a smooth projective curve and X is a geometrically ruled surface over Z, that is, a $\mathbb{P}^1$-bundle; $\rho(X) = 2$.

(3) Z is a point, $\rho(X) = 1$ and $-K_X$ is ample. (In fact, $X \simeq \mathbb{P}^2$, but this requires an additional argument.)

Proof. **Step 1:** By the cone theorem, Theorem 2.4.1 and its proof, we may choose an irreducible rational curve $C \subseteq X$ with $R = \mathbb{R}_{\geq 0}[C]$ and

$$0 < -(K_X \cdot C) \leq \dim X + 1 = 3.$$

[46] For arithmetic genus zero, the final surface classification, Serre duality, and ruled surfaces, see [Har77, IV, Exercise 1.8(b)], [Mat02, 1-4-4], [Har77, III, 7.7], [Har77, V, 2.8].

Hence $(K_X \cdot C) \in \{-1, -2, -3\}$. Recall that the arithmetic genus is defined by

$$p_a(C) := 1 - \chi(C, \mathcal{O}_C) = h^1(C, \mathcal{O}_C),$$

since C is integral and proper. As a consequence of Riemann–Roch for surfaces, we have the adjunction formula [Har77, V, Exercise 1.3(a)]:

$$2p_a(C) - 2 = (C \cdot (C + K_X)) = C^2 + (K_X \cdot C).$$

Since $p_a(C) \geq 0$ (notice that C may be singular!), this implies $C^2 \geq -2 - (K_X \cdot C) \geq -1$. Equality $C^2 = -1$ holds if and only if $(K_X \cdot C) = -1$ and $p_a(C) = 0$. In that case $C \simeq \mathbb{P}^1$, so C is a (-1)-curve. Castelnuovo's contraction theorem contracts exactly C to a smooth point on a smooth projective surface Z. This is the contraction of R, and $\rho(Z) = \rho(X) - 1$, giving case (1).

Step 2: It remains to deal with $C^2 > 0$ and $C^2 = 0$. Suppose first that $C^2 > 0$. By Corollary 2.4.8, $[C] \in Q^+$ is an interior point of $\overline{\mathrm{NE}}(X)$. Since $[C]$ also spans an extremal ray, this forces $\rho(X) = 1$ and $\overline{\mathrm{NE}}(X) = R$. Thus $X \to \operatorname{Spec} k$ is the contraction of R. By assumption $(K_X \cdot C) < 0$, so $-K_X$ is positive on $\overline{\mathrm{NE}}(X) - \{0\}$ and is ample by Kleiman's criterion, Theorem 1.4.28. For the further conclusion $X \simeq \mathbb{P}^2$, see the classification of smooth del Pezzo surfaces, or [Mat02, I-4-4].

Step 3: Now consider $C^2 = 0$. Adjunction gives $2p_a(C) - 2 = (K_X \cdot C) < 0$, so $p_a(C) = 0$, $(K_X \cdot C) = -2$ and $C \simeq \mathbb{P}^1$.

We prove that a multiple of C gives the desired contraction. For a fixed ample divisor H, $(K_X - mC) \cdot H < 0$ for $m \gg 0$, since $C \cdot H > 0$. Thus, by Serre duality,

$$H^2(X, \mathcal{O}_X(mC)) \simeq H^0(X, \mathcal{O}_X(K_X - mC))^\vee = 0 \qquad (m \gg 0).$$

Riemann–Roch on the surface now gives

$$h^0(X, \mathcal{O}_X(mC)) \geq \chi(X, \mathcal{O}_X(mC)) = -\frac{m}{2}(K_X \cdot C) + \chi(\mathcal{O}_X) = m + \chi(\mathcal{O}_X),$$

which grows linearly. In particular, for m sufficiently large, $h^0(X, \mathcal{O}_X(mC)) \geq 2$.

Write the complete linear system as $|mC| = F + |M|$, where F is its fixed part. Since mC itself is a member of $|mC|$, we have $F \leq mC$ as divisors. Because C is irreducible, this implies $F = kC$ for an integer $0 \leq k < m$. Removing the fixed part gives the complete linear system

$$|M| = |(m - k)C|.$$

Set $m' = m - k > 0$. Then $|m'C|$ has no fixed component and has dimension at least one.

Next we verify that $|m'C|$ is base point free. Since it has no fixed component, we may choose $D \in |m'C|$ which does not contain C. Then

$$(D \cdot C) = m'C^2 = 0,$$

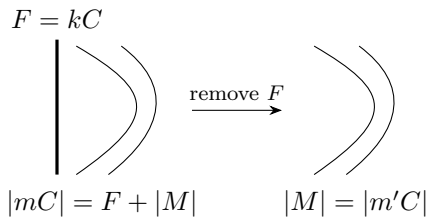


Figure 2.10: Removing the fixed component from $|mC|$ (schematic).

so D is disjoint from C. But $m'C$ is another member of the same system. These two members have disjoint supports, and hence $|m'C|$ has no base points.

Let $\phi : X \to Y \subseteq \mathbb{P}^N_k$ be the corresponding morphism. Its image has dimension one: it is nonconstant, and a two-dimensional image would imply $(m'C)^2 > 0$. Take the Stein factorization

$$\phi : X \xrightarrow{f} Z \xrightarrow{h} Y \subseteq \mathbb{P}^N_k, \qquad Z = \operatorname{Spec}_Y(\phi_*\mathcal{O}_X).$$

Here $f_*\mathcal{O}_X = \mathcal{O}_Z$, h is finite, and Z is a normal projective curve, hence smooth. We have not yet assumed that $f = \operatorname{cont}_R$.

Put $A = h^*\mathcal{O}_Y(1)$, so $f^*A \simeq \mathcal{O}_X(m'C)$. By the projection formula and $f_*\mathcal{O}_X = \mathcal{O}_Z$, the section with zero divisor $m'C$ is the pullback of a section of A. Its zero divisor on Z is supported at one point z_0, since its pullback is supported on the irreducible curve C. Hence the scheme-theoretic fibre over z_0 is aC for some integer $a > 0$.

The morphism f is flat, since it is a dominant morphism from a smooth surface to a smooth curve. All its fibres are numerically equivalent as divisors. A general fibre G is smooth and connected by generic smoothness in characteristic zero and the Stein factorization. Thus $G \equiv aC$, and adjunction gives

$$2g(G) - 2 = G^2 + (K_X \cdot G) = a(K_X \cdot C) = -2a.$$

Since $g(G) \geq 0$ and $a > 0$, we get $a = 1$ and $g(G) = 0$. Therefore every fibre has numerical class $[C]$.

Let $\sum_i a_i C_i$ be any fibre, with $a_i \in \mathbb{Z}_{>0}$ and C_i its distinct irreducible components. We now have

$$\sum_i a_i [C_i] = [C] \in R.$$

Since R is extremal, every $[C_i]$ lies in R. Consequently $C_i^2 = 0$ and $(K_X \cdot C_i) < 0$. Adjunction gives

$$2p_a(C_i) - 2 = (K_X \cdot C_i) < 0,$$

so $p_a(C_i) = 0$, $C_i \simeq \mathbb{P}^1$ and $(K_X \cdot C_i) = -2$. It follows that

$$-2 = (K_X \cdot C) = \left(K_X \cdot \sum_i a_i C_i\right) = -2 \sum_i a_i.$$

Thus $\sum_i a_i = 1$, and every fibre is irreducible, reduced, and isomorphic to $\mathbb{P}^1$.

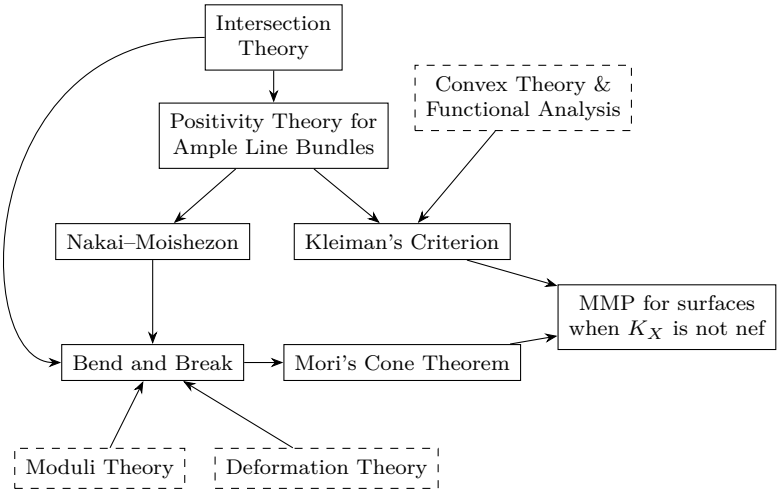
Moreover, an irreducible curve E is contracted by f if and only if $[E] \in R$. One direction follows from the description of the fibres. For the other, if $[E] \in R$, then $(f^*A \cdot E) = m'(C \cdot E) = 0$; since A is ample on Z, E must be contracted. Hence $f = \mathrm{cont}_R$.

Finally, Tsen's theorem gives a $k(Z)$-rational point on the genus-zero generic fibre. This yields a section over an open subset of Z, which extends over all of the smooth curve Z by properness. Thus X is a geometrically ruled surface over Z. By [Har77, V, Proposition 2.2], it is the projectivization of a rank-two locally free sheaf on Z:

$$X \simeq \mathbb{P}_Z(\mathcal{E}).$$

In particular, $\rho(X) = \rho(Z) + 1 = 2$, as also follows from [Har77, II, Exercise 7.9(a)]. This is case (2). □

Summary:



Dashed boxes: only basic tools needed.
Solid boxes: main results to understand.

Figure 2.11: Knowledge framework.

Chapter 3
Linear Series

In this chapter all schemes will be considered over $\mathbb{C}$ unless another ground field is explicitly specified. Schemes are of finite type over the ground field unless stated otherwise. Some results hold over more general fields; characteristic-zero assumptions are retained where they are needed.

3.1 Asymptotic Theory

3.1A Basic Definitions

"This definition may look a little bit scary."
Caucher Birkar on the asymptotic definition of Kodaira dimension

The asymptotic behavior of the linear series $|L^{\otimes m}|$ as $m \to \infty$, due to Iitaka and Ueno, is that up to birational equivalence they define an essentially unique rational mapping $\varphi : X \dashrightarrow Y$, and L is close to trivial (in a suitable sense) along the fibres of φ. Throughout this section X always means a projective variety.

Definition 3.1.1 (Semigroup and exponent of a line bundle)**.** Let L be a line bundle on a projective variety X. The semigroup of L consists of those non-negative powers of L that have a non-zero section:

$$N(L) = N(X, L) = \{m \geq 0 \mid H^0(X, L^{\otimes m}) \neq 0\}.$$

Clearly $N(L) = \{0\}$ if $H^0(X, L^{\otimes m}) = 0$ for all $m > 0$. Assuming that $N(L) \neq \{0\}$, let $e = e(L)$ be the greatest common divisor of $\{m \in N(L) \mid m > 0\}$, called the *exponent* of L. Every element of $N(L)$ is a multiple of e, and conversely every sufficiently large multiple of e belongs to $N(L)$. The semigroup $N(X, D)$ and exponent $e(D)$ of a Cartier divisor D are defined via $L = \mathcal{O}_X(D)$.

For completeness, the last assertion is an elementary fact about semigroups. After dividing by e, choose finitely many elements with greatest common divisor 1, and choose a positive element a. Their non-negative sums represent every residue class modulo a: the subsemigroup they generate in the finite group $\mathbb{Z}/a\mathbb{Z}$ is a subgroup, and is the whole group. Choose one representative b_r in the semigroup for each residue r. Every sufficiently large integer congruent to r is $b_r + ta$ with $t \geq 0$. Here "sufficiently divisible" means divisible by a fixed positive integer; when a lower bound is also needed, we say "sufficiently large and divisible."

Example 3.1.2. Let T be a projective variety of dimension $d \geq 1$ carrying a non-trivial torsion line bundle L_η of order e in $\mathrm{Pic}(T)$, so that $L_\eta^{\otimes e} \cong \mathcal{O}_T$. Let Y

be any projective variety of dimension k, and B a very ample line bundle on Y. Set[47]

$$X = Y \times T, \qquad L = \mathrm{pr}_1^* B \otimes \mathrm{pr}_2^* L_\eta.$$

Then $e(L) = e$ and $N(L) = e\mathbb{Z}_{\geq 0}$, because the Künneth formula gives

$$H^0(X, L^{\otimes m}) \cong H^0(Y, B^{\otimes m}) \otimes H^0(T, L_\eta^{\otimes m}).$$

Here we use the fact that a non-trivial torsion line bundle $\mathcal{L}$ on a projective variety has no non-zero section. Indeed, if $s \neq 0$ were a section and $\mathcal{L}^{\otimes a} \cong \mathcal{O}_T$, then s^a would be a non-zero constant. Hence s would be nowhere vanishing and would trivialize $\mathcal{L}$, a contradiction.

For $m > 0$ in $N(L)$, consider the rational map

$$\varphi_m = \varphi_{|L^{\otimes m}|} : X \dashrightarrow \mathbb{P}H^0(X, L^{\otimes m})$$

associated to the complete linear series, and denote by $Y_m = \overline{\varphi_m(X)}$ the closure of its image. Here projective space uses the quotient convention, as in Chapter 1. Our purpose is to understand the asymptotic behavior of these rational maps as $m \to \infty$.

Definition 3.1.3 (Iitaka dimension)**.** Assume that X is normal. The Iitaka dimension of L is

$$\kappa(L) = \kappa(X, L) = \max_{\substack{m \in N(L) \\ m > 0}} \dim \overline{\varphi_m(X)},$$

provided that $N(L)$ is non-trivial; otherwise set $\kappa(L) = -\infty$. If X is non-normal, pass to its normalization $\nu : X' \to X$ and set

$$\kappa(X, L) = \kappa(X', \nu^* L).$$

The definition for Cartier divisors is similar.

Thus either $\kappa(X, L) = -\infty$, or $0 \leq \kappa(X, L) \leq \dim X$. With this convention, on a non-normal variety $\kappa(X, L)$ need not measure the growth of $H^0(X, L^{\otimes m})$ itself; see Example 3.1.7.

Example 3.1.4. For the pair (X, L) in Example 3.1.2 and a positive multiple m of e,

$$H^0(X, L^{\otimes m}) \cong H^0(Y, B^{\otimes m}) \otimes \mathbb{C}.$$

The image $\varphi_m(X)$ is the image of Y under the embedding defined by $B^{\otimes m}$. Applying the same observation after normalization if necessary gives $\kappa(X, L) = \dim Y = k$.

[47] For the fact about regular functions on proper integral schemes used in this example, see [SP25, Tag 0BED].

Example 3.1.5 (Kodaira dimension)**.** Let X be a smooth projective variety and K_X a canonical divisor. Then $\kappa(X) = \kappa(X, K_X)$ is the Kodaira dimension of X. For a singular variety, its Kodaira dimension is defined using any smooth projective model; this is independent of the model by birational invariance of the pluricanonical spaces.

Example 3.1.6 (Iitaka dimension of restrictions)**.** Iitaka dimension can be unpredictable under restriction to subvarieties. Let $X = \mathrm{Bl}_p \mathbb{P}^2$, with exceptional curve E, and let H be the pullback of a line. Both $\mathcal{O}_X(H)$ and $\mathcal{O}_X(H+E)$ have Iitaka dimension 2, but their restrictions to E have Iitaka dimensions 0 and $-\infty$, respectively. Indeed, by the projection formula $(H \cdot E) = 0$, and

$$\mathcal{O}_X(H)|_E \cong \mathcal{O}_{\mathbb{P}^1}, \qquad \mathcal{O}_X(H+E)|_E \cong \mathcal{O}_{\mathbb{P}^1}(-1).$$

Thus the Iitaka dimension drops upon restriction. On the other hand, on $X = \mathbb{P}^1 \times \mathbb{P}^1$ take $L = \mathrm{pr}_1^* \mathcal{O}_{\mathbb{P}^1}(-1) \otimes \mathrm{pr}_2^* \mathcal{O}_{\mathbb{P}^1}(1)$. For $m > 0$,

$$H^0(X, L^{\otimes m}) \cong H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(-m)) \otimes H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(m)) = 0.$$

Hence $\kappa(X, L) = -\infty$, whereas for $Y = \{\text{point}\} \times \mathbb{P}^1$ we have $\kappa(Y, L|_Y) = 1$.

Example 3.1.7 (Non-normal varieties)**.** Linear series can behave differently from their pullbacks to the normalization. Let X be a nodal plane cubic. By [Har77, II, Exercise 6.7], $\mathrm{Pic}^0(X) \cong \mathbb{G}_m$. Choose a non-torsion line bundle L, corresponding to $\lambda \in \mathbb{C}^*$ which is not a root of unity. The normalization $\nu : \mathbb{P}^1 \to X$ identifies two points p, q, and $\nu^* L \cong \mathcal{O}_{\mathbb{P}^1}$. The descent data identify the fibres at p, q by multiplication by λ (after choosing trivializations). Thus the pullback of a section of $L^{\otimes m}$ is a constant c satisfying

$$s'(p) = \lambda^m s'(q), \qquad \text{that is,} \qquad c = \lambda^m c.$$

For every $m > 0$ this forces $c = 0$. Consequently

$$H^0(X, L^{\otimes m}) = 0 \quad (m > 0), \qquad H^0(\mathbb{P}^1, (\nu^* L)^{\otimes m}) = \mathbb{C}.$$

Our normalization convention gives $\kappa(X, L) = 0$, although $N(X, L) = \{0\}$. This explains the normality hypothesis when Iitaka dimension is described by the original spaces of sections.

Example 3.1.8. Assume that X is normal and $\kappa(X, L) = \kappa \geq 0$. Then $\dim \varphi_m(X) = \kappa$ for all sufficiently large $m \in N(X, L)$.

Proof. Choose $k > 0$ such that $\dim \varphi_k(X) = \kappa$. If $m \in N(L)$ is sufficiently large, then $m - k$ is a sufficiently large multiple of $e(L)$, and hence belongs to $N(L)$. Multiplication by a non-zero section of $L^{\otimes (m-k)}$ gives an injection

$$H^0(X, L^{\otimes k}) \hookrightarrow H^0(X, L^{\otimes m}).$$

The associated linear projection $\nu_{m-k} : \mathbb{P}H^0(X, L^{\otimes m}) \dashrightarrow \mathbb{P}H^0(X, L^{\otimes k})$ gives $\varphi_k = \nu_{m-k} \circ \varphi_m$. Therefore $\dim Y_m \geq \dim Y_k = \kappa$, and the reverse inequality is the definition. □

Now we consider morphisms for which pullback preserves spaces of sections.

Lemma 3.1.9 (Coincidence for pullbacks)**.** *Let $f : X \to Y$ be a projective surjective morphism of projective varieties satisfying $f_*\mathcal{O}_X = \mathcal{O}_Y$, and let L be a line bundle on Y. Then*

$$H^0(X, f^*(L^{\otimes m})) \cong H^0(Y, L^{\otimes m}) \qquad (m \geq 0).$$

In particular, $\kappa(Y, L) = \kappa(X, f^*L)$, with the normalization convention of Definition 3.1.3.

Proof. The projection formula gives

$$f_*(f^*L^{\otimes m}) = f_*(\mathcal{O}_X \otimes f^*L^{\otimes m}) \cong f_*\mathcal{O}_X \otimes L^{\otimes m} \cong L^{\otimes m}.$$

Taking global sections proves the first assertion. In the normal case the complete linear-series maps have the same images, since f is surjective, which proves the assertion about Iitaka dimension in that case. For general X, Y, normalize both and take the Stein factorization of the induced morphism $X^\nu \to Y^\nu$. Its fibre-space part preserves Iitaka dimension by the normal case just proved, and its finite part preserves Iitaka dimension by finite-pullback invariance for normal projective varieties (see [Pop, Exercise 1.2.5]). This gives the stated identity with the normalization convention. □

Definition 3.1.10 (Algebraic fibre space)**.** An algebraic fibre space is a surjective projective morphism $f : X \to Y$ of varieties such that $f_*\mathcal{O}_X = \mathcal{O}_Y$.[48]

For an arbitrary projective surjective morphism $g : V \to W$, Stein factorization [Har77, III, 11.5] expresses g as

$$V \xrightarrow{g'} W' \xrightarrow{h} W, \qquad W' = \operatorname{Spec}_W(g_*\mathcal{O}_V),$$

where g' is an algebraic fibre space and h is finite. Thus g is a fibre space if and only if its finite Stein factor is trivial. The theorem on formal functions shows that the fibres of a fibre space are connected. Conversely, if Y is normal and $f : X \to Y$ has connected fibres, write $f = h \circ g$ for its Stein factorization. For a closed point $y \in Y$,

$$f^{-1}(y) = \coprod_{y' \in h^{-1}(y)} g^{-1}(y').$$

Every set on the right is non-empty, so $h^{-1}(y)$ consists of one point. In characteristic zero a finite dominant map is generically separable, and on a dense open subset its fibres have $[\mathbb{C}(Y') : \mathbb{C}(Y)]$ points. Thus this degree is 1. The finite birational map h is an isomorphism because Y is normal. If $f : X \to Y$ is a fibre space, X is normal, and $\mu : X' \to X$ is a projective birational morphism, then $\mu_*\mathcal{O}_{X'} = \mathcal{O}_X$ and $(f \circ \mu)_*\mathcal{O}_{X'} = \mathcal{O}_Y$. Hence the composition is again a fibre space.

[48] For the pushforward of the structure sheaf under a proper birational morphism to a normal variety, see [Har77, III, 11.4].

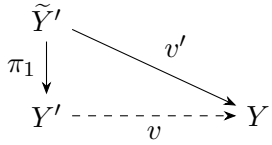
Example 3.1.11 (Fibre spaces and function fields)**.** Let $f : X \to Y$ be a projective surjective morphism of normal projective varieties. Then f is an algebraic fibre space if and only if $\mathbb{C}(Y)$ is algebraically closed in $\mathbb{C}(X)$.

Proof. "⇒": Suppose not. There is a finite extension

$$\mathbb{C}(Y) \subsetneq K \subseteq \mathbb{C}(X), \qquad [K : \mathbb{C}(Y)] \geq 2.$$

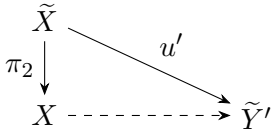
Choose a normal projective model Y' of K. The field inclusions give rational maps $X \dashrightarrow Y' \dashrightarrow Y$ factoring f, with the second map generically finite of degree at least 2. We eliminate indeterminacies in two steps, as in [Har77, II, 7.17.3].

Step 1: Embed Y in projective space and resolve the rational map $v : Y' \dashrightarrow Y$:



Normalize the modification if necessary. We now have a rational map $X \dashrightarrow \widetilde{Y}'$.

Step 2: Embed $\widetilde{Y}'$ in projective space and resolve this map:



Thus $f \circ \pi_2 = v' \circ u'$. Since X is normal, $\pi_{2*}\mathcal{O}_{\widetilde{X}} = \mathcal{O}_X$, so $\widetilde{X} \to Y$ remains an algebraic fibre space. On a dense open subset of Y, the map v' is finite étale of degree $[K : \mathbb{C}(Y)] \geq 2$. Its closed fibres have at least two points, and their inverse images under the proper surjective map u' disconnect the corresponding fibres of $\widetilde{X} \to Y$, a contradiction. Here it is the closed fibres on this open set (or the *geometric* generic fibre) that have several points; $\operatorname{Spec} K$ itself has only one.

"⇐": Run the Stein factorization $X \to Y' \to Y$. The extension $\mathbb{C}(Y')/\mathbb{C}(Y)$ is finite and lies in $\mathbb{C}(X)$, hence is trivial. Since Y is normal, $Y' \to Y$ is an isomorphism. □

The function-field criterion also makes sense in the birational category: a dominant rational map of normal varieties is a rational fibre space when the induced inclusion of function fields is algebraically closed.

Example 3.1.12 (Injectivity of Picard groups)**.** Let X, Y be projective varieties and $f : X \to Y$ an algebraic fibre space. Then $f^* : \operatorname{Pic}(Y) \to \operatorname{Pic}(X)$ is injective.

Proof. Let $B = \mathcal{O}_Y(D)$ satisfy $f^*B \cong \mathcal{O}_X$. By Lemma 3.1.9,

$$H^0(Y, B) = H^0(X, f^*B) \neq 0, \qquad H^0(Y, B^\vee) = H^0(X, f^*B^\vee) \neq 0.$$

There are effective Cartier divisors $E_1 \sim_{\text{lin}} D$ and $E_2 \sim_{\text{lin}} -D$. The effective divisor $E_1 + E_2$ is linearly equivalent to zero. Its defining section, viewed as a section of $\mathcal{O}_Y$, is a non-zero constant, so $E_1 + E_2 = 0$. Hence $E_1 = E_2 = 0$ and $D \sim_{\text{lin}} 0$. □

Example 3.1.13 (Normality of fibre spaces)**.** If $f : X \to Y$ is an algebraic fibre space and X is normal, then Y is normal.[49]

Proof. The universal property of the normalization $\nu : Y' \to Y$ gives $f = \nu \circ g$ for a morphism $g : X \to Y'$. Since these maps are dominant,

$$\mathcal{O}_Y = (\nu \circ g)_*\mathcal{O}_X = \nu_* g_* \mathcal{O}_X \supseteq \nu_* \mathcal{O}_{Y'} \supseteq \mathcal{O}_Y.$$

Thus $\nu_*\mathcal{O}_{Y'} = \mathcal{O}_Y$, and, since ν is finite,

$$Y' = \operatorname{Spec}_Y(\nu_*\mathcal{O}_{Y'}) \cong \operatorname{Spec}_Y \mathcal{O}_Y = Y.$$

□

Example 3.1.14 (Exceptional twists of birational pullbacks)**.** Let $\mu : X' \to X$ be a proper birational morphism of varieties, with X' normal. Let L be a line bundle on X, and E a μ-exceptional effective Cartier divisor on X': every component of E maps to a subset of codimension at least 2 in X. Then the natural inclusion

$$H^0(X', \mu^*L) \longrightarrow H^0(X', \mu^*L \otimes \mathcal{O}_{X'}(E))$$

is an isomorphism.

Proof. Factor μ through the normalization $\nu : X^\nu \to X$, writing $\mu = \nu \circ \mu'$. For $L^\nu = \nu^*L$ we have $\mu^*L = (\mu')^*L^\nu$. Since ν is finite, the exceptional divisors for μ and μ' coincide. Thus we may assume X normal. Now $\mu_*\mathcal{O}_{X'} = \mathcal{O}_X$, and the projection formula gives

$$H^0(X', \mu^*L) \cong H^0(X, L).$$

Set $Z = \mu(\operatorname{Supp} E)$, $U = X \setminus Z$, and $U' = \mu^{-1}(U)$. A section of $\mu^*L(E)$ restricts to a section of μ^*L on U', hence to a section of L on U. Choose open sets V_i trivializing L. By normality and $\operatorname{codim}_X Z \geq 2$, the restriction map

$$H^0(V_i, \mathcal{O}_{V_i}) \longrightarrow H^0(V_i \cap U, \mathcal{O}_{V_i})$$

is an isomorphism. These extensions glue uniquely to a section of L on X. Its pullback agrees with the original section on the dense open U' and hence everywhere. This proves surjectivity of the natural inclusion. □

[49] For recovering an affine morphism as a relative spectrum, see [Har77, II, Exercise 5.17(d)].

Definition 3.1.15 (Section ring associated to a line bundle)**.** For a line bundle L on a projective variety X, its graded ring or section ring is the graded $\mathbb{C}$-algebra

$$R(L) = R(X, L) = \bigoplus_{m \geq 0} H^0(X, L^{\otimes m}).$$

The section ring $R(D) = R(X, D)$ of a Cartier divisor is defined similarly.

Example 3.1.16. For $X = \mathbb{P}^n$ and $L = \mathcal{O}_{\mathbb{P}^n}(1)$, we have $R(L) = \mathbb{C}[T_0, \ldots, T_n]$, the homogeneous coordinate ring of $\mathbb{P}^n$.

The section ring is finitely generated in this example, but need not be so in general. This property greatly influences the geometry of L.

Definition 3.1.17 (Finitely generated line bundles and Cartier divisors)**.** A line bundle L on a projective variety is finitely generated if $R(X, L)$ is a finitely generated $\mathbb{C}$-algebra. A Cartier divisor D is finitely generated if $\mathcal{O}_X(D)$ is so.

Passing to additive notation, let D be a Cartier divisor on a projective variety X. We consider the set-theoretic base locus

$$\mathrm{Bs}(|mD|) = \bigcap_{D' \in |mD|} \mathrm{Supp}(D'),$$

with $\mathrm{Bs}(|mD|) = X$ if $|mD| = \varnothing$. As in Chapter 1, this is the base scheme equipped with its reduced structure.

Definition 3.1.18 (Stable base locus)**.** The stable base locus of D is the algebraic set[50]

$$\mathbf{B}(D) = \bigcap_{m \geq 1} \mathrm{Bs}(|mD|).$$

The intersection is set-theoretic, rather than an intersection of schemes.

Proposition 3.1.19. *The stable base locus is the unique minimal element of* $\{\mathrm{Bs}(|mD|)\}_{m \geq 1}$*. In particular, there exists* $m_0 > 0$ *such that*

$$\mathbf{B}(D) = \mathrm{Bs}(|km_0D|) \qquad \textit{for all } k \geq 1.$$

Proof. For $m, l \geq 1$, multiplication of sections gives $\mathfrak{b}(|mD|)^l \subseteq \mathfrak{b}(|lmD|)$, hence $\mathrm{Bs}(|lmD|) \subseteq \mathrm{Bs}(|mD|)$. Since X is noetherian, the family of closed sets has a minimal element, say $\mathrm{Bs}(|m_0D|)$. For every m,

$$\mathrm{Bs}(|mm_0D|) \subseteq \mathrm{Bs}(|m_0D|) \cap \mathrm{Bs}(|mD|).$$

Minimality makes the first inclusion into $\mathrm{Bs}(|m_0D|)$ an equality. Thus $\mathrm{Bs}(|m_0D|)$ lies in every member of the family, and equals $\mathbf{B}(D)$. The same argument gives $\mathrm{Bs}(|km_0D|) = \mathbf{B}(D)$ for every $k \geq 1$. □

[50] For the inclusion of base ideals under multiplication of sections, see Example 1.1.15.

Remark 3.1.20 (Realizing the stable base locus)**.** It is not always possible to take $m_0 = 1$. If $N(D) \neq \{0\}$ and $e(D) = e > 1$, then $\mathbf{B}(D) \neq X$, whereas $\mathrm{Bs}(|mD|) = X$ whenever $e \nmid m$. Thus equality $\mathbf{B}(D) = \mathrm{Bs}(|mD|)$ is possible only when $e \mid m$. Even when $e(D) = 1$, equality can fail for arbitrarily large m.

Example 3.1.21 (Multiples)**.** For every Cartier divisor D and every integer $p \geq 1$, $\mathbf{B}(pD) = \mathbf{B}(D)$.

Proof. By Proposition 3.1.19 choose $m_0, m_0' > 0$ such that

$$\mathbf{B}(D) = \mathrm{Bs}(|km_0D|), \qquad \mathbf{B}(pD) = \mathrm{Bs}(|k'm_0'pD|)$$

for positive k, k'. Taking $k = m_0'p$ and $k' = m_0$ identifies the two sets. □

Remark 3.1.22 ($\mathbb{Q}$-Cartier divisors)**.** For a $\mathbb{Q}$-Cartier divisor D, choose $p > 0$ such that pD is an integral Cartier divisor and define $\mathbf{B}(D) = \mathbf{B}(pD)$. The preceding example shows that this is independent of p.

Example 3.1.23 (Scheme structures on the base locus)**.** Although the underlying sets can stabilize, the base schemes need not. Let $X = \mathrm{Bl}_p \mathbb{P}^2$, with exceptional divisor E and pullback of a line H, and set $D = H + E$. We have $|mD| = |mH| + mE$. Indeed, the exact sequence

$$0 \longrightarrow \mathcal{O}_X(mD - E) \xrightarrow{\cdot s_E} \mathcal{O}_X(mD) \longrightarrow \mathcal{O}_E(mD|_E) \longrightarrow 0$$

and $(mD \cdot E) = -m < 0$ show that multiplication by s_E induces an isomorphism on global sections. Consequently $|mD| = |mD - E| + E$. Repeating the argument until all m copies of E have been removed gives $|mD| = |mH| + mE$, with $|mH|$ free. The evaluation map defining the base ideal therefore gives

$$\mathfrak{b}(|mD|) = \mathcal{O}_X(-mE)\, \mathfrak{b}(|mH|) = \mathcal{O}_X(-mE).$$

The base scheme is the thickening mE defined by $\mathcal{I}_E^m$, so these schemes are distinct even though their underlying set is always E.

3.1B Semiample Line Bundles

We are investigating the asymptotic behavior of the mappings φ_k determined by $|L^{\otimes k}|$. This subsection focuses on the important case where some power of L is free. On a complete scheme the spaces of global sections are finite-dimensional, so the corresponding projective spaces are defined.

Definition 3.1.24. A line bundle L on a complete scheme is *semi-ample* if $L^{\otimes m}$ is globally generated, that is, $|L^{\otimes m}|$ is free, for some $m > 0$. A Cartier divisor D is semi-ample if $\mathcal{O}_X(D)$ is so.

Fix a semi-ample line bundle L on a projective variety, and write

$$M(X, L) = \{m \in \mathbb{Z}_{\geq 0} \mid L^{\otimes m} \text{ is free}\} \subseteq N(X, L).$$

Let $f = f(L)$ be the greatest common divisor of its positive elements. By the semigroup observation following Definition 3.1.1, $L^{\otimes kf}$ is free for all $k \gg 0$. For $m > 0$ in $M(X, L)$, write

$$\varphi_m : X \longrightarrow Y_m \subseteq \mathbb{P}H^0(X, L^{\otimes m})$$

for the map of the complete linear series, viewed as a morphism onto its image.

Theorem 3.1.25 (Semi-ample fibrations)**.** *Let X be a normal projective variety and L a semi-ample line bundle. There is an algebraic fibre space $\varphi : X \to Y$ such that, for every sufficiently large $m \in M(X, L)$, $Y_m \cong Y$ and $\varphi_m = \varphi$ up to this isomorphism. Furthermore, there is an ample line bundle A on Y such that $\varphi^* A \cong L^{\otimes f}$, where $f = f(L)$.*[51]

Thus the morphisms stabilize to a fibre space structure. The bundle $L^{\otimes f}$ is trivial on each fibre, since

$$L^{\otimes f}|_{X_y} \cong (\varphi|_{X_y})^*(A \otimes k(y)) \cong \mathcal{O}_{X_y}.$$

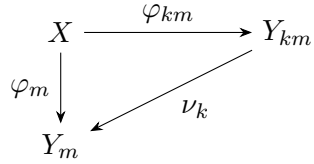
By Lemma 3.1.9 and Example 3.1.13, Y is normal and

$$H^0(X, L^{\otimes kf}) \cong H^0(Y, A^{\otimes k}) \qquad (k \geq 0).$$

Before proving the theorem, consider the multiplication map

$$\mu_k : \operatorname{Sym}^k H^0(X, L^{\otimes m}) \longrightarrow H^0(X, L^{\otimes km}).$$

Its image W_k determines a free subseries giving the kth Veronese re-embedding of Y_m. The linear projection associated to $W_k \subseteq H^0(X, L^{\otimes km})$, followed by the inverse of that Veronese embedding on its image, gives a finite morphism $\nu_k : Y_{km} \to Y_m$ such that



Finiteness follows because the projection is defined everywhere on Y_{km} and pulls back the ample hyperplane bundle on its image to the ample hyperplane bundle of Y_{km}, so it contracts no positive-dimensional fibre. The very ample bundle $A_m = \mathcal{O}_{\mathbb{P}}(1)|_{Y_m}$ satisfies $\varphi_m^* A_m \cong L^{\otimes m}$. Also

$$H^0(Y_m, A_m) \cong H^0(X, L^{\otimes m}).$$

[51] For the comparison of the morphisms by finite linear projection, see Example 1.1.18; injectivity of pullback along an algebraic fibre space is Example 3.1.12.

Indeed, restriction from the ambient projective space and pullback to X are injective, and their composition is the identity on the space of sections defining this *complete* linear series. This equality does not yet require φ_m to be a fibre space.

Lemma 3.1.26. *In the situation of Theorem 3.1.25, fix $m > 0$ in $M(X, L)$. For all sufficiently large integers k, the factorization*

$$X \xrightarrow{\varphi_{km}} Y_{km} \xrightarrow{\nu_k} Y_m$$

is the Stein factorization of φ_m. In particular, Y_{km} and φ_{km} are independent of k up to isomorphism for $k \gg 0$.

Proof. Let $X \xrightarrow{\psi} V \xrightarrow{\mu} Y_m$ be the Stein factorization. The variety V is normal, and μ is finite. Hence $B = \mu^* A_m$ is ample, and $B^{\otimes k}$ is very ample for $k \gg 0$. Since

$$\psi^* B^{\otimes k} \cong L^{\otimes km}, \qquad H^0(X, L^{\otimes km}) \cong H^0(V, B^{\otimes k}),$$

we can choose a basis $t_0, \dots, t_N$ of the latter space and write

$$\begin{aligned} \varphi_{km}(x) &= [\psi^* t_0(x) : \cdots : \psi^* t_N(x)] \\ &= [t_0(\psi(x)) : \cdots : t_N(\psi(x))] = \varphi_{|B^{\otimes k}|}(\psi(x)). \end{aligned}$$

The last map embeds V in projective space. Thus $Y_{km} \cong V$ and φ_{km} is ψ up to this isomorphism. □

Proof of Theorem 3.1.25. Replace L by $L^{\otimes f}$ to assume $f = 1$, so every sufficiently large power of L is free. Choose relatively prime large integers a, b. By Lemma 3.1.26 there are integers K_a, K_b such that the maps φ_{ka}, respectively φ_{kb}, stabilize for $k \geq K_a$, respectively $k \geq K_b$. Choose distinct primes $l_a \geq K_a$, $l_b \geq K_b$ not dividing ab, and set $p = l_a a$, $q = l_b b$. Then $\gcd(p, q) = 1$, and

$$\varphi_{kp} = \varphi_p, \qquad \varphi_{kq} = \varphi_q \qquad (k \geq 1)$$

up to isomorphism. In particular, comparison at the common multiple pq gives $Y_p \cong Y_{pq} \cong Y_q$. Denote the resulting fibre space by $\varphi : X \to Y$. The variety Y carries very ample bundles A_p, A_q with $\varphi^* A_p = L^{\otimes p}$ and $\varphi^* A_q = L^{\otimes q}$. Choose integers r, s with $rp + sq = 1$, and set

$$A = A_p^{\otimes r} \otimes A_q^{\otimes s}.$$

Then $\varphi^* A = L$. The injectivity of $\varphi^* : \operatorname{Pic}(Y) \to \operatorname{Pic}(X)$ gives $A^{\otimes p} \cong A_p$ and $A^{\otimes q} \cong A_q$. In particular, A is ample. It remains to identify all sufficiently large exponents. For $c, d \geq 1$, the product map

$$\operatorname{Sym}^c H^0(Y, A^{\otimes p}) \otimes \operatorname{Sym}^d H^0(Y, A^{\otimes q}) \longrightarrow H^0(Y, A^{\otimes (cp+dq)})$$

gives a subseries embedding Y by the corresponding Segre–Veronese map. Moreover, Lemma 3.1.9 identifies this complete space of sections with $H^0(X, L^{\otimes (cp+dq)})$.

Thus φ_{cp+dq} is φ followed by the embedding associated to the complete series on Y. Every sufficiently large integer is of the form $cp + dq$ with $c, d \geq 1$. Equivalently, one can use directly that $A^{\otimes m}$ is very ample for $m \gg 0$. Restoring the original L gives $\varphi^* A = L^{\otimes f}$ and the required conclusion for all sufficiently large $m \in M(X, L)$. □

Semi-ample fibrations lead to finite generation properties.

Example 3.1.27 (Surjectivity of multiplication maps)**.** Let X be a normal projective variety and L a globally generated line bundle. There exists $m_0 = m_0(L)$ such that[52]

$$H^0(X, L^{\otimes a}) \otimes H^0(X, L^{\otimes b}) \longrightarrow H^0(X, L^{\otimes (a+b)})$$

is surjective for $a, b \geq m_0$. More generally, for any coherent sheaf $\mathcal{F}$,

$$H^0(X, \mathcal{F} \otimes L^{\otimes a}) \otimes H^0(X, L^{\otimes b}) \longrightarrow H^0(X, \mathcal{F} \otimes L^{\otimes (a+b)})$$

is surjective for $a, b \gg 0$.

Proof. Take the Stein factorization of the map associated to $|L|$:

$$X \xrightarrow{\varphi} Y \xrightarrow{\nu} Y_1 \subseteq \mathbb{P}H^0(X, L).$$

The bundle $A = \nu^* \mathcal{O}_{Y_1}(1)$ is both ample and globally generated, and $\varphi^* A = L$. The projection formula gives

$$\varphi_*(\mathcal{F} \otimes L^{\otimes m}) \cong \varphi_* \mathcal{F} \otimes A^{\otimes m}.$$

Thus it suffices to prove surjectivity of

$$H^0(Y, \mathcal{G} \otimes A^{\otimes a}) \otimes H^0(Y, A^{\otimes b}) \longrightarrow H^0(Y, \mathcal{G} \otimes A^{\otimes (a+b)}), \qquad \mathcal{G} = \varphi_* \mathcal{F}.$$

The sheaf $\mathcal{G}$ is coherent. For some m_0 it is m_0-regular with respect to the ample globally generated bundle A:

$$H^i(Y, \mathcal{G} \otimes A^{\otimes (m_0 - i)}) = 0 \qquad (i > 0).$$

Indeed, for $i > \dim Y$ this is Grothendieck vanishing; for the remaining finitely many i, choose $m_0 \geq n_0 + \dim Y$ using Serre vanishing. Castelnuovo–Mumford regularity [Laz04a, Theorem 1.8.5] now gives the surjectivity of multiplication by $H^0(Y, A)$ for $a \geq m_0$. Iterating, the product with $H^0(Y, A)^{\otimes b}$ is onto for every $b \geq 0$. It factors through the displayed multiplication map with $H^0(Y, A^{\otimes b})$, which is therefore also onto. Taking $\mathcal{F} = \mathcal{O}_X$ gives the first assertion. □

[52] For the vanishing input in the regularity argument, see [Har77, III, 5.2–5.3].

Theorem 3.1.28 (Finite generation of semi-ample line bundles)**.** *Let L be a semi-ample line bundle on a normal projective variety X. Then $R(X, L)$ is a finitely generated $\mathbb{C}$-algebra.*

Proof. Choose $k > 0$ such that $L^{\otimes k}$ is free. By Example 3.1.27,

$$H^0(X, L^{\otimes ka}) \otimes H^0(X, L^{\otimes kb}) \longrightarrow H^0(X, L^{\otimes k(a+b)})$$

is onto for $a, b \geq m_0$. Thus the Veronese subalgebra

$$R(X, L)^{(k)} = \bigoplus_{m \geq 0} H^0(X, L^{\otimes km})$$

is generated by its pieces of degrees at most $2m_0 - 1$: every higher piece is a product of the piece of degree m_0 with one of lower degree. For $0 \leq i < k$, apply the sheaf version of Example 3.1.27 to the *free* bundle $L^{\otimes k}$ and the coherent sheaf $L^{\otimes i}$. It shows that

$$\bigoplus_{m \geq 0} H^0(X, L^{\otimes i} \otimes (L^{\otimes k})^{\otimes m})$$

is a finitely generated module over $R(X, L)^{(k)}$ (fix a sufficiently large degree in the multiplication map and induct on m). Their finite direct sum is $R(X, L)$, so this ring is finitely generated. □

Zariski proved a striking result about finite base loci in [Zar62, II, 6.2], and Fujita generalized it in [Fuj80].

Theorem 3.1.29 (Zariski–Fujita)**.** *Let L be a line bundle on a projective variety X such that* $\mathrm{Bs}(|L|)$ *is a finite set. Then L is semi-ample: $L^{\otimes m}$ is free for some $m > 0$.*

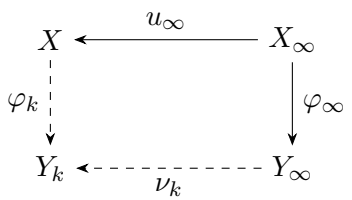
3.1C Iitaka Fibration

We now give the asymptotic analysis of the rational mappings

$$\varphi_k = \varphi_{|L^{\otimes k}|} : X \dashrightarrow Y_k \subseteq \mathbb{P}H^0(X, L^{\otimes k})$$

in the general setting, always viewing φ_k as a rational map to Y_k.

Theorem 3.1.30 (Iitaka fibrations)**.** *Let X be a normal projective variety, and L a line bundle such that $\kappa(X, L) > 0$. For all sufficiently large $k \in N(X, L)$, the rational maps $\varphi_k : X \dashrightarrow Y_k$ are birationally equivalent to a fixed algebraic fibre space $\varphi_\infty : X_\infty \to Y_\infty$ of normal projective varieties. More precisely, there is a commutative diagram*



where u_∞ is a birational morphism and ν_k is birational. One has $\dim Y_\infty = \kappa(X,L)$. *Setting $L_\infty = u_\infty^* L$, a very general fibre F of φ_∞ satisfies $\kappa(F, L_\infty|_F) = 0$.*

Definition 3.1.31 (Iitaka fibration associated to a line bundle)**.** The morphism $\varphi_\infty : X_\infty \to Y_\infty$ in the theorem is an Iitaka fibration associated to L. It is unique up to birational equivalence, whereas a semi-ample fibration is unique up to isomorphism. For a Cartier divisor D use the line bundle $\mathcal{O}_X(D)$. When $\kappa(X,L) = 0$, the Iitaka fibration is the map to a point.

Definition 3.1.32 (Iitaka fibration of a variety)**.** For a variety X of non-negative Kodaira dimension, its Iitaka fibration is the Iitaka fibration associated to the canonical bundle on a smooth projective model. When $\kappa(X) = -\infty$, this construction does not produce an Iitaka fibration.[53]

Proof of Theorem 3.1.30. Fix $m > 0$ in $N(X,L)$ with $\dim Y_m = \kappa(X,L)$.

Step 1: We first prove that, for $k \gg 0$, all maps φ_{km} are birationally equivalent to a fixed algebraic fibre space $\psi_{(m)} : X_{(m)} \to Y_{(m)}$ of normal varieties. Blow up the base ideal $\mathfrak{b} = \mathfrak{b}(|L^{\otimes m}|)$:

$$\varepsilon : \widetilde{X} = \operatorname{Proj}_X \Big(\bigoplus_{d \geq 0} \mathfrak{b}^d \Big) \longrightarrow X.$$

The inverse-image ideal is invertible, $\varepsilon^{-1}\mathfrak{b} \cdot \mathcal{O}_{\widetilde{X}} = \mathcal{O}_{\widetilde{X}}(-E)$, where E is an effective Cartier divisor. It need not be ε-exceptional: the original series can already have divisorial fixed components. Normalize, writing $\nu : X_{(m)} \to \widetilde{X}$ and $u_m = \varepsilon \circ \nu$. The evaluation map gives a surjection

$$H^0(X, L^{\otimes m}) \otimes \mathcal{O}_X \longrightarrow \mathfrak{b} \otimes L^{\otimes m}.$$

Pulling back and then taking the image of $u_m^*\mathfrak{b} \to \mathcal{O}_{X_{(m)}}$ gives a surjection onto $(u_m^{-1}\mathfrak{b} \cdot \mathcal{O}_{X_{(m)}}) \otimes u_m^* L^{\otimes m}$. Here

$$\begin{aligned} u_m^{-1}\mathfrak{b} \cdot \mathcal{O}_{X_{(m)}} &= \nu^{-1}(\varepsilon^{-1}\mathfrak{b} \cdot \mathcal{O}_{\widetilde{X}}) \cdot \mathcal{O}_{X_{(m)}} \\ &= \nu^* \mathcal{O}_{\widetilde{X}}(-E) = \mathcal{O}_{X_{(m)}}(-F_m), \qquad F_m = \nu^* E. \end{aligned}$$

The pullback of a local equation of E remains non-zero on the integral variety $X_{(m)}$, because ν is dominant. Thus it defines the indicated effective Cartier divisor; no flatness of ν is being assumed. Set

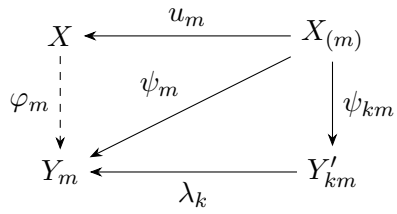
$$M_m = \mathcal{O}_{X_{(m)}}(-F_m) \otimes u_m^* L^{\otimes m}.$$

[53] For blow-ups, twisting the corresponding graded algebras, products of projective spaces, and the birational pushforward of the structure sheaf, see [Har77, II, 7.12.2], [Har77, II, Exercise 5.11], [Har77, III, 11.4]. The function-field characterization of algebraic fibre spaces used repeatedly here is Example 3.1.11.

This line bundle is globally generated. Since X is normal, $H^0(X_{(m)}, u_m^* L^{\otimes m}) = H^0(X, L^{\otimes m})$, and all these sections vanish along F_m. Consequently

$$u_m^* |L^{\otimes m}| = |M_m| + F_m, \qquad H^0(X_{(m)}, M_m) = H^0(X, L^{\otimes m}).$$

The morphism $\psi_m : X_{(m)} \to Y_m$ defined by $|M_m|$ resolves φ_m. For the morphism ψ_{km} defined by $|M_m^{\otimes k}|$, we obtain the diagram



where λ_k is finite. By semi-ample fibrations, ψ_{km} stabilizes for $k \gg 0$ to a fibre space $\psi_{(m)} : X_{(m)} \to Y_{(m)}$. On the other hand, multiplication by the section of kF_m embeds $H^0(X_{(m)}, M_m^{\otimes k})$ in $H^0(X, L^{\otimes km})$. Thus, as rational maps,

$$\psi_{(m)} = \mu_k \circ \varphi_{km} \circ u_m, \qquad \mu_k : Y_{km} \dashrightarrow Y_{(m)}.$$

Since λ_k is finite, $\dim Y_{(m)} = \dim Y_m = \kappa(X, L)$, and hence

$$\dim Y_{(m)} \le \dim Y_{km} \le \kappa(X, L) = \dim Y_{(m)}.$$

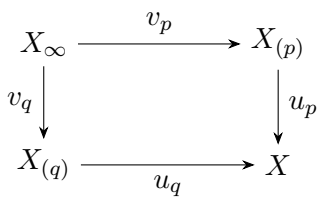
Therefore μ_k is generically finite. The inclusion

$$\mathbb{C}(Y_{(m)}) \subseteq \mathbb{C}(Y_{km}) \subseteq \mathbb{C}(X)$$

and the algebraic closedness of $\mathbb{C}(Y_{(m)})$ in $\mathbb{C}(X)$ force its degree to be 1. Thus μ_k is birational, proving Step 1.

Step 2: We show that these fibre spaces are birationally isomorphic and choose a common model. Replacing L by $L^{\otimes e}$, with $e = e(L)$, we may assume $e = 1$. Positive powers do not change Iitaka dimension: products of sections give Veronese subseries with the same image dimension. We restore the original line bundle at the end of the proof. Choose relatively prime integers $p, q > 1$, both sufficiently large that $\dim Y_p = \dim Y_q = \kappa(X, L)$. Choose $m \gg 0$ so that the powers $M_p^{\otimes p^{m-1}}$ and $M_q^{\otimes q^{m-1}}$ give the stabilized fibre spaces $\psi_{(p)}$ and $\psi_{(q)}$ of Step 1.

To construct a common resolution, form $Z = X_{(p)} \times_X X_{(q)}$. On a dense open $U \subseteq X$ over which u_p, u_q are isomorphisms, $Z_U \cong U$. Take the reduced closure of Z_U in Z and normalize it. This gives a normal projective variety X_∞ and birational morphisms v_p, v_q with $u_p \circ v_p = u_q \circ v_q =: u_\infty$:



Consider the globally generated line bundle

$$M_{p,q} = v_p^* M_p^{\otimes p^{m-1}} \otimes v_q^* M_q^{\otimes q^{m-1}}.$$

By construction, these two factors are pulled back from very ample bundles $\mathcal{O}_{Y_{(p)}}(1)$ and $\mathcal{O}_{Y_{(q)}}(1)$ for the chosen stabilized embeddings. Their external tensor product is very ample on $Y_{(p)} \times Y_{(q)}$, and its pullback by

$$\pi = (\psi_{(p)} \circ v_p, \psi_{(q)} \circ v_q) : X_\infty \longrightarrow Y_{(p)} \times Y_{(q)}$$

is $M_{p,q}$. Let Y_∞ be the normalization of the image of the *complete* series $|M_{p,q}|$, and let $\varphi_\infty : X_\infty \to Y_\infty$ be the induced morphism. The map from the complete-series image to $\pi(X_\infty)$ is finite: it pulls back the ample hyperplane bundle to the hyperplane bundle. Thus Y_∞ maps finitely onto $\pi(X_\infty)$, which is a closed subvariety of the product. We obtain

$$\begin{array}{ccccc}
X_{(p)} & \xleftarrow{v_p} & X_\infty & \xrightarrow{v_q} & X_{(q)} \\
\downarrow{\scriptstyle \psi_{(p)}} & & \downarrow{\scriptstyle \varphi_\infty} & & \downarrow{\scriptstyle \psi_{(q)}} \\
Y_{(p)} & \xleftarrow[w_p]{} & Y_\infty & \xrightarrow[w_q]{} & Y_{(q)}
\end{array}$$

Both w_p, w_q are surjective, so $\dim Y_\infty \geq \kappa(X, L)$. On the other hand,

$$M_{p,q} \hookrightarrow u_\infty^* L^{\otimes (p^m + q^m)},$$

and its complete series is a subseries of the latter complete series. Hence $\dim Y_\infty \leq \dim Y_{p^m+q^m} \leq \kappa(X, L)$. Thus equality holds and w_p, w_q are generically finite. The maps $\psi_{(p)} \circ v_p$ and $\psi_{(q)} \circ v_q$ are fibre spaces, because v_p, v_q are birational morphisms to normal varieties. The function-field criterion now makes w_p, w_q birational. It also shows that φ_∞ is a fibre space: $\mathbb{C}(Y_\infty) = \mathbb{C}(Y_{(p)})$ is algebraically closed in $\mathbb{C}(X_\infty)$. By construction, Y_∞ carries an ample globally generated bundle $A_{p,q}$ with $\varphi_\infty^* A_{p,q} = M_{p,q}$.

For $c, d \geq 1$, multiply by a fixed non-zero section of $v_p^* M_p^{\otimes (c-1)p^{m-1}} \otimes v_q^* M_q^{\otimes (d-1)q^{m-1}}$ to obtain

$$\begin{aligned} H^0(X_\infty, M_{p,q}) &\hookrightarrow H^0(X_\infty, v_p^* M_p^{\otimes cp^{m-1}} \otimes v_q^* M_q^{\otimes dq^{m-1}}) \\ &\hookrightarrow H^0(X_\infty, u_\infty^* L^{\otimes (cp^m + dq^m)}). \end{aligned}$$

It follows that φ_∞ factors rationally through $\varphi_{cp^m+dq^m} \circ u_\infty$, with a generically finite map $\mu_{cp^m+dq^m} : Y_{cp^m+dq^m} \dashrightarrow Y_\infty$. As before, this map is birational because φ_∞ is a fibre space. Since $\gcd(p^m, q^m) = 1$, every sufficiently large integer has the form $cp^m + dq^m$ with $c, d \geq 1$. Taking $\nu_k = \mu_k^{-1}$ gives the diagram in the theorem.

Step 3: It remains to show that a very general fibre F has $\kappa(F, L_\infty|_F) = 0$, where $L_\infty = u_\infty^* L$. Choose a very general closed point $y \in Y_\infty$ and set $F = \varphi_\infty^{-1}(y)$. We impose the following countably many open conditions. The fibre F is normal and integral; each ν_j with $j \gg 0$ is defined at y; F is not contained in the base locus of any non-empty $|L_\infty^{\otimes j}|$; and cohomology and base change holds at y for every sheaf $L_\infty^{\otimes k}$, $k \geq 1$. The first condition holds on a dense open set: the generic fibre of a normal variety is normal, and a characteristic-zero fibre space has geometrically integral generic fibre. Generic flatness and cohomology and base change give the other required open sets. Over $\mathbb{C}$, their countable intersection contains very general points.

The map $\varphi_j \circ u_\infty$ is constant on the domain where it is defined in F. Thus all restricted sections are multiples of a single non-zero section s_F:

$$s_i|_F = c_i s_F.$$

This section may have zeros. Consequently

$$\rho_j : H^0(X_\infty, L_\infty^{\otimes j}) \longrightarrow H^0(F, L_\infty^{\otimes j}|_F)$$

has rank one for all sufficiently large j (we are still in the case $e(L) = 1$). In particular, some positive power of $L_\infty|_F$ has a non-zero section, giving $\kappa(F, L_\infty|_F) \geq 0$.

Choose a very ample line bundle B on Y_∞. We claim that there is $m_0 > 0$ and a non-zero section

$$\sigma \in H^0(X_\infty, L_\infty^{\otimes m_0} \otimes \varphi_\infty^* B^\vee).$$

Choose F also so that $\sigma|_F \neq 0$; this is one more dense open condition. For fixed $k \geq 1$ and $r > 0$, multiplication by σ^r and restriction give the commutative square

$$\begin{array}{ccc} H^0(X_\infty, L_\infty^k \otimes \varphi_\infty^* B^r) & \xrightarrow{\cdot\sigma^r} & H^0(X_\infty, L_\infty^{k+rm_0}) \\ \downarrow{\scriptstyle \beta_{k,r}} & & \downarrow{\scriptstyle \rho_{k+rm_0}} \\ H^0(F, (L_\infty^k \otimes \varphi_\infty^* B^r)|_F) & \xrightarrow[\cdot\sigma^r|_F]{} & H^0(F, L_\infty^{k+rm_0}|_F) \end{array}$$

The lower horizontal map is injective because F is integral and $\sigma|_F$ is non-zero. Let $\mathcal{G}_k = \varphi_{\infty *}(L_\infty^{\otimes k})$. The projection formula and the base-change condition identify $\beta_{k,r}$ with evaluation

$$\begin{aligned} H^0(X_\infty, L_\infty^{\otimes k} \otimes \varphi_\infty^* B^{\otimes r}) &= H^0(Y_\infty, \mathcal{G}_k \otimes B^{\otimes r}) \\ &\longrightarrow (\mathcal{G}_k \otimes B^{\otimes r}) \otimes k(y) \\ &\cong H^0(F, L_\infty^{\otimes k}|_F) \otimes B^{\otimes r}|_y. \end{aligned}$$

For fixed k and $r \gg 0$, $\mathcal{G}_k \otimes B^{\otimes r}$ is globally generated. Its evaluation map is

$$H^0(Y_\infty, \mathcal{G}_k \otimes B^{\otimes r}) \otimes \mathcal{O}_{Y_\infty} \twoheadrightarrow \mathcal{G}_k \otimes B^{\otimes r}.$$

Tensoring its stalk at y with $k(y) = \mathbb{C}$ proves that $\beta_{k,r}$ is surjective. The right vertical map has rank one for $r \gg 0$, so the square and the lower injection imply $\operatorname{rank} \beta_{k,r} \leq 1$. Hence

$$h^0(F, L_\infty^{\otimes k}|_F) \leq 1 \qquad \text{for every } k \geq 1.$$

Together with the non-vanishing of some positive power this gives the desired Iitaka dimension zero.

It remains to prove the claim. The ample bundle $A = A_{p,q}$ satisfies $\varphi_\infty^* A = M_{p,q}$. For some $m_1 \gg 0$, $H^0(Y_\infty, A^{\otimes m_1} \otimes B^\vee) \neq 0$. Pull back a non-zero section and use

$$M_{p,q}^{\otimes m_1} \hookrightarrow L_\infty^{\otimes (p^m + q^m) m_1}$$

to obtain σ, with $m_0 = (p^m + q^m) m_1$. Finally restore the original L: the maps for $L^{\otimes e}$ are exactly the maps for the original exponents divisible by e, which include all sufficiently large elements of $N(L)$. On the fibre, taking a positive power does not change Iitaka dimension. This completes the proof. □

Remark 3.1.33. At the beginning of the proof we removed the base locus of a linear system, a useful technique; see [Kaw24, 1.2.5]. We give the general version below.

Example 3.1.34. Let $|V|$ be a non-empty linear series of Cartier divisors on a normal variety X. There is a projective birational morphism $f : Y \to X$ from a normal variety such that[54]

$$f^*|V| = |V_1| + F,$$

where F is an effective fixed Cartier divisor and $|V_1|$ is free.

Proof. Let $0 \neq V \subseteq H^0(X, \mathcal{O}_X(D))$ be the finite-dimensional space of sections. Write the image of evaluation as

$$\operatorname{im}(V \otimes \mathcal{O}_X \to \mathcal{O}_X(D)) = \mathfrak{b} \otimes \mathcal{O}_X(D).$$

Let $\varepsilon : \widetilde{X} = \operatorname{Bl}_{\mathfrak{b}} X \to X$ be the blow-up, $\nu : Y \to \widetilde{X}$ its normalization, and $f = \varepsilon \circ \nu$. Then

$$f^{-1}\mathfrak{b} \cdot \mathcal{O}_Y = \nu^{-1}(\varepsilon^{-1}\mathfrak{b} \cdot \mathcal{O}_{\widetilde{X}}) \cdot \mathcal{O}_Y = \nu^* \mathcal{O}_{\widetilde{X}}(-E) = \mathcal{O}_Y(-F).$$

Here F is effective Cartier, as in Step 1 above. To see projectivity explicitly, if $\dim V = n + 1$, the evaluation surjection induces a surjection of graded algebras

$$\operatorname{Sym}(\mathcal{O}_X^{\oplus (n+1)}) \twoheadrightarrow \bigoplus_{d \geq 0} \mathfrak{b}^d \otimes \mathcal{O}_X(dD).$$

[54] For twisting a graded algebra without changing its Proj and for projectivity, see [Har77, II, Exercise 7.11(b)], [Har77, II, 7.10(b)].

Twisting each degree by $\mathcal{O}_X(dD)$ does not change relative Proj, so

$$\mathrm{Bl}_{\mathfrak{b}} X = \mathrm{Proj}_X\Big(\bigoplus_{d\geq 0} \mathfrak{b}^d\Big) \cong \mathrm{Proj}_X\Big(\bigoplus_{d\geq 0} \mathfrak{b}^d \otimes \mathcal{O}_X(dD)\Big) \hookrightarrow \mathbb{P}^n_X.$$

Thus ε, and hence f, is projective. The pullback evaluation map now gives

$$V \otimes \mathcal{O}_Y \twoheadrightarrow \mathcal{O}_Y(f^*D - F),$$

so the corresponding residual series $|V_1|$ is free and F is the fixed part of $f^*|V|$. □

Corollary 3.1.35. *Let L be a line bundle on a normal projective variety X, and suppose $\kappa = \kappa(X, L) \geq 0$. There are constants $a, A > 0$ such that*

$$am^\kappa \leq h^0(X, L^{\otimes m}) \leq Am^\kappa$$

for all sufficiently large $m \in N(X, L)$.

Proof. If $\kappa = 0$, every non-empty positive complete linear series has projective dimension zero, so $h^0(X, L^{\otimes m}) = 1$ for positive $m \in N(L)$, and the assertion follows. Assume $\kappa > 0$. Spaces of sections are unchanged by a birational modification of a normal variety. We may therefore use the Iitaka fibration $\varphi : X \to Y$ and, by resolving X, also assume X nonsingular. Step 3 of the theorem gives an ample line bundle B on Y and $m_0 > 0$ with $H^0(X, L^{\otimes m_0} \otimes \varphi^* B^\vee) \neq 0$. Multiplying by powers of this section gives

$$H^0(Y, B^{\otimes l}) = H^0(X, \varphi^* B^{\otimes l}) \hookrightarrow H^0(X, L^{\otimes l m_0}).$$

Asymptotic Riemann–Roch and Serre vanishing give

$$h^0(Y, B^{\otimes l}) = \frac{l^\kappa}{\kappa!} \int_Y c_1(B)^\kappa + O(l^{\kappa-1}).$$

The leading coefficient is positive, proving the lower bound along multiples of m_0. To cover all sufficiently large $m \in N(L)$, let $e = e(L)$; necessarily $e \mid m_0$. Choose a sufficiently large multiple n of e such that $n + r \in N(L)$ for $r \in \{0, e, \ldots, m_0 - e\}$. Divide

$$m - n = l m_0 + r, \qquad 0 \leq r < m_0.$$

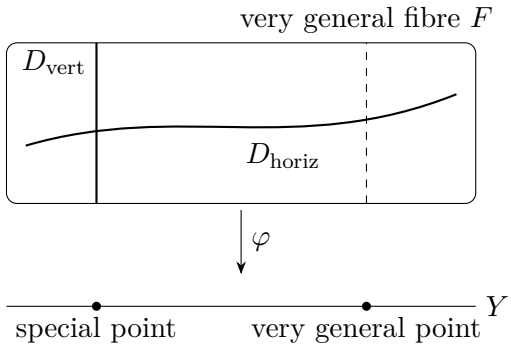
Then $e \mid r$, and multiplying by a non-zero section of $L^{\otimes(n+r)}$ gives $h^0(X, L^{\otimes m}) \geq h^0(X, L^{\otimes l m_0}) \geq C l^\kappa$. Since $l = (m - n - r)/m_0$ and there are finitely many r, this is at least am^κ for $m \gg 0$.

For the upper bound, first suppose $H^0(X, L) \neq 0$, and write $L = \mathcal{O}_X(D)$ with D effective. On the nonsingular X, decompose

$$D = D_{\mathrm{vert}} + D_{\mathrm{horiz}},$$

where each component of D_{vert} maps to a proper subvariety of Y and each component of D_{horiz} dominates Y. These are Cartier divisors since X is nonsingular. We assert that, under the usual realization by rational functions,

$$H^0(X, \mathcal{O}_X(mD_{\mathrm{vert}})) = H^0(X, \mathcal{O}_X(mD)) \qquad (m \geq 1). \tag{3.1}$$



Granting the assertion, choose an ample Cartier divisor H on Y with $D_{\mathrm{vert}} \le \varphi^* H$. Then

$$\begin{aligned} h^0(X, \mathcal{O}_X(mD)) &= h^0(X, \mathcal{O}_X(mD_{\mathrm{vert}})) \\ &\le h^0(X, \varphi^*\mathcal{O}_Y(mH)) = h^0(Y, \mathcal{O}_Y(mH)) \\ &= \frac{(H^\kappa)}{\kappa!} m^\kappa + O(m^{\kappa-1}) \le A m^\kappa \end{aligned}$$

for $m \gg 0$. To prove (3.1), one inclusion is immediate from effectivity. For the other, every section s of $\mathcal{O}_X(mD)$ must vanish along mD_{horiz}. Indeed, its zero divisor is $(s)_0 = \mathrm{div}(s) + mD$, and the required inequality is

$$\mathrm{div}(s) + mD_{\mathrm{vert}} = (s)_0 - mD_{\mathrm{horiz}} \ge 0.$$

If a section failed this inequality at a horizontal prime divisor, its restriction to a very general fibre would have a different zero divisor from the restriction of the canonical section of mD. The fibre avoids D_{vert}, meets this horizontal divisor, and can be chosen to avoid the other exceptional intersection conditions. These two restricted sections would be linearly independent. This contradicts $h^0(F, \mathcal{O}_F(mD)) \le 1$, proved in Step 3 of the theorem.

For the construction of H, write $D_{\mathrm{vert}} = \sum_i a_i \Gamma_i$ and $Z_i = \varphi(\Gamma_i) \subsetneq Y$. Choose a very ample divisor A_Y on Y. For $m_i \gg 0$, $\mathcal{I}_{Z_i} \otimes \mathcal{O}_Y(m_i A_Y)$ is globally generated and has a non-zero section s_i. Put $H_i = (s_i)_0$, so that $Z_i \subseteq \mathrm{Supp}\, H_i$ and

$$\varphi^* H_i = b_i \Gamma_i + E_i, \qquad b_i \ge 1, \quad E_i \ge 0.$$

Choose positive integers c_i with $c_i b_i \ge a_i$. Then $H = \sum_i c_i H_i$ is ample and

$$\varphi^* H - D_{\mathrm{vert}} = \sum_i (c_i b_i - a_i)\Gamma_i + \sum_i c_i E_i \ge 0.$$

If $D_{\mathrm{vert}} = 0$, simply choose any effective ample H.

Finally remove the assumption $H^0(X, L) \ne 0$. Choose $p > 0$ in $N(L)$ and apply the upper-bound argument to $L^{\otimes p}$, whose Iitaka fibration and dimension are the same. For each residue $r \in \{0, e, \ldots, p-e\}$ choose $b_r \in N(L)$ with $b_r \equiv -r$

(mod p). Such integers exist by the semigroup property. If $m \in N(L)$ and $m \equiv r$ (mod p), multiplication by a non-zero section of $L^{\otimes b_r}$ gives

$$h^0(X, L^{\otimes m}) \leq h^0(X, L^{\otimes (m+b_r)}) \leq C \left(\frac{m+b_r}{p}\right)^{\kappa} \leq A' m^{\kappa}$$

for $m \gg 0$. There are only finitely many b_r, so a uniform A' works. □

Remark 3.1.36. For a normal projective variety and $\kappa = \kappa(X, L) \geq 0$, it is now known that[55]

$$\lim_{\substack{m \to \infty \\ m \in N(X,L)}} \frac{h^0(X, L^{\otimes m})}{m^{\kappa}}$$

exists and is positive and finite. This follows from the limit theorem for graded linear series of Okounkov, Lazarsfeld–Mustaţă, and Kaveh–Khovanskii; see [Cut12, Theorem 3.1]. Indeed, that theorem gives the limit along multiples of the index $e(L)$, and these are exactly the sufficiently large elements of $N(L)$. Without this restriction, zeros in the other residue classes can prevent a limit when $e(L) > 1$. In the big case the index is 1, as we will see in Corollary 3.2.10; the full sequence therefore has a limit.

Remark 3.1.37. For a normal projective variety, the preceding corollary gives an asymptotic definition of $\kappa(X, L)$: if every positive power has no non-zero section, set $\kappa(X, L) = -\infty$; otherwise it is the largest integer $d \geq 0$ such that

$$\limsup_{m \to \infty} \frac{h^0(X, L^{\otimes m})}{m^d} > 0.$$

One can restrict to $m \in N(L)$ without changing this limsup. This is the asymptotic definition used in the lecture referred to at the beginning of the chapter. For non-normal X, apply it to $\nu^* L$ on the normalization to retain Definition 3.1.3.

Proof. Write κ_{class} and κ_{def} for the two definitions. *Case 1.* If $h^0(X, L^{\otimes m}) = 0$ for every $m > 0$, no non-negative integer satisfies the inequality, and both definitions give $-\infty$. *Case 2.* Otherwise, Corollary 3.1.35 gives

$$\frac{h^0(X, L^{\otimes m})}{m^{\kappa_{\text{class}}}} \geq a > 0$$

for all sufficiently large $m \in N(L)$, so $\kappa_{\text{def}} \geq \kappa_{\text{class}}$. If $d > \kappa_{\text{class}}$, the upper bound gives

$$0 \leq \frac{h^0(X, L^{\otimes m})}{m^d} \leq A m^{\kappa_{\text{class}} - d} \longrightarrow 0.$$

Thus no such d can occur, proving equality of the definitions. □

[55] For the existence of the limit in the big case, see [Laz04b, 11.4.7].

3.2 Big Line Bundles and Cartier Divisors

3.2A Basic Properties of Big Cartier Divisors

Definition 3.2.1 (Big)**.** A line bundle L on a projective variety X is big if $\kappa(X, L) = \dim X$. A Cartier divisor D is big if $\mathcal{O}_X(D)$ is so.

Example 3.2.2. The pullback of an ample line bundle under a dominant generically finite morphism $f : X \to Y$ of projective varieties is big.

Proof. We use "generically finite" for a dominant map with finite extension of function fields. Passing to the normalization of X, we may assume X normal. Put $\dim X = \dim Y = n$ and let A be ample on Y. Dominance gives $\mathcal{O}_Y \hookrightarrow f_*\mathcal{O}_X$, hence

$$A^{\otimes m} \hookrightarrow A^{\otimes m} \otimes f_*\mathcal{O}_X \cong f_*(f^*A^{\otimes m}).$$

Consequently

$$\begin{aligned} h^0(X, f^*A^{\otimes m}) &\geq h^0(Y, A^{\otimes m}) \\ &= \frac{m^n}{n!}\int_Y c_1(A)^n + O(m^{n-1}) \geq Cm^n \end{aligned}$$

for some $C > 0$ and $m \gg 0$. The growth characterization of Iitaka dimension in Remark 3.1.37 proves bigness. The zero-dimensional case is immediate. □

Example 3.2.3. If X is normal, L is big if and only if, for some $m > 0$, the rational map $\varphi_m : X \dashrightarrow Y_m$ is birational onto its image.

Proof. One direction is immediate from the definition. Conversely, for a big line bundle the Iitaka fibration has $\dim X_\infty = \dim Y_\infty$. The field extension $\mathbb{C}(Y_\infty) \subseteq \mathbb{C}(X_\infty)$ is finite and the first field is algebraically closed in the second, so they are equal. The Iitaka fibration, and hence φ_m for sufficiently large $m \in N(L)$, is birational. If $\dim X = 0$, both sides are automatic. □

Although the definition uses normalization, in the big case we can measure growth on the original variety. The following criterion is often used as the definition; see [KM98, 2.59].

Lemma 3.2.4. *Let X be a projective variety of dimension n, not necessarily normal. A line bundle L is big if and only if there is $C > 0$ such that*[56]

$$h^0(X, L^{\otimes m}) \geq Cm^n$$

for all sufficiently large and divisible m.

[56] For the cohomological error term on a lower-dimensional support, see Example 1.2.24.

Proof. Let $\nu : X' \to X$ be the normalization and $L' = \nu^* L$. The exact sequence

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \nu_* \mathcal{O}_{X'} \longrightarrow \eta \longrightarrow 0$$

has η supported in dimension at most $n-1$. By the projection formula,

$$\begin{aligned} h^0(X, L^{\otimes m}) &\leq h^0(X', L'^{\otimes m}) \\ &\leq h^0(X, L^{\otimes m}) + h^0(X, \eta \otimes L^{\otimes m}). \end{aligned}$$

The last term is $O(m^{n-1})$ by the cohomology growth bound of Chapter 1. Thus the two spaces have the same leading growth in degree n. If L is big, apply Corollary 3.1.35 on X' along sufficiently large multiples of $e(L')$ and absorb the lower-order error to obtain the stated bound on X. Conversely, such a lower bound on X also holds on X', forcing $\kappa(X', L') = n$ by Corollary 3.1.35 and the general $O(m^n)$ bound. For $n = 0$ the normalization is an isomorphism and the assertion is immediate. □

Example 3.2.5 (Blow-up of projective space)**.** Let $S \subseteq \mathbb{P}^n$ be a finite reduced set of s points, with ideal sheaf $\mathcal{I}$, and let $\mu : X = \mathrm{Bl}_S \mathbb{P}^n \to \mathbb{P}^n$. Write $E = E_S$ for the divisor defined by the inverse-image ideal and $H = H_S$ for the pullback of a hyperplane. For positive integers d, r put $D = dH - rE$. Then

$$\begin{aligned} \mu_* \mathcal{O}_X(mD) &= \mu_* (\mathcal{O}_X(mdH) \otimes \mathcal{I}_E^{mr}) \\ &= \mathcal{O}_{\mathbb{P}^n}(md) \otimes \mathcal{I}^{mr}. \end{aligned}$$

Here $\mathcal{I}^{mr}$ is the ordinary ideal power, rather than its tensor power; for the blow-up of distinct smooth points, $\mu_* \mathcal{O}_X(-mrE) = \mathcal{I}^{mr}$. Thus $H^0(X, \mathcal{O}_X(mD))$ is the space of degree md hypersurfaces vanishing to order at least mr at each point of S. We have

$$h^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(md)) = \binom{n+md}{n} = \frac{d^n}{n!} m^n + O(m^{n-1}).$$

Vanishing to order mr at one smooth point imposes at most

$$\binom{mr-1+n}{n} = \frac{r^n}{n!} m^n + O(m^{n-1})$$

linear conditions, obtained by setting all Taylor coefficients of degree less than mr to zero. No independence of these conditions at different points is needed for the inequality

$$h^0(X, \mathcal{O}_X(mD)) \geq \frac{d^n - sr^n}{n!} m^n + O(m^{n-1}).$$

Therefore D is big whenever $d^n - sr^n > 0$.

The first essential result is due to Kodaira.

Proposition 3.2.6 (Kodaira's lemma)**.** *Let D be a big Cartier divisor on a projective variety X, and F any effective Cartier divisor. Then*[57]

$$H^0(X, \mathcal{O}_X(mD - F)) \neq 0$$

for all sufficiently large and divisible m.

Proof. Let $\dim X = n$. From

$$0 \longrightarrow \mathcal{O}_X(mD - F) \longrightarrow \mathcal{O}_X(mD) \longrightarrow \mathcal{O}_F(mD) \longrightarrow 0$$

we compare $h^0(X, mD) \geq Cm^n$ with $h^0(F, \mathcal{O}_F(mD)) = O(m^{n-1})$. For sufficiently large and divisible m the first is larger, so the kernel on sections is non-zero. If $F = 0$ the assertion is immediate. □

Kodaira's lemma gives the following characterization.

Corollary 3.2.7 (Characterization of big Cartier divisors)**.** *For a Cartier divisor D on a projective variety X, the following are equivalent:*

(i) D is big.

(ii) For any ample integral Cartier divisor A, there are $m > 0$ and an effective Cartier divisor N with $mD \sim_{\text{lin}} A + N$.

(iii) The same statement holds for some ample Cartier divisor A.

(iv) There are an ample Cartier divisor A, an integer $m > 0$ and an effective Cartier divisor N with $mD \equiv_{\text{num}} A + N$.

Proof. "(i)⇒(ii)": Choose $r \gg 0$ such that $rA \sim_{\text{lin}} H_r$ and $(r+1)A \sim_{\text{lin}} H_{r+1}$ are effective; Serre's global generation theorem gives this choice. Apply Kodaira's lemma with $F = H_{r+1}$ to get

$$mD \sim_{\text{lin}} H_{r+1} + N' \sim_{\text{lin}} A + (H_r + N'), \qquad N' \geq 0.$$

Take $N = H_r + N'$. The implications (ii)⇒(iii)⇒(iv) are immediate. For (iv)⇒(i), the Cartier divisor $mD - N$ is numerically equivalent to A, hence ample by Nakai–Moishezon. Choose $t > 0$ such that $H = t(mD - N)$ is very ample. Then $tmD \sim_{\text{lin}} H + tN$. Multiplication by the section of ktN embeds the sections of kH in those of $ktmD$. The growth criterion proves D big. □

Corollary 3.2.8 (Numerical nature of bigness)**.** *Bigness of a Cartier divisor depends only on its numerical equivalence class.*

[57] The growth bound on the fixed divisor is Example 1.2.24.

Example 3.2.9 (Cohomological characterization of big Cartier divisors)**.** A Cartier divisor D on a projective variety X is big if and only if, for every coherent sheaf $\mathcal{F}$, there is an integer $m = m(\mathcal{F}) > 0$ such that $\mathcal{F} \otimes \mathcal{O}_X(mD)$ is generated by its global sections on some non-empty open subset of X.

Proof. "$\Rightarrow$": Choose an ample Cartier divisor A such that $\mathcal{F}(A)$ is globally generated. Write $mD \sim_{\text{lin}} A+N$ with $N \geq 0$ by Corollary 3.2.7. Multiplying generators of $\mathcal{F}(A)$ by the canonical section of N generates $\mathcal{F}(mD)$ on $X \setminus \operatorname{Supp} N$. "$\Leftarrow$": Take $\mathcal{F} = \mathcal{O}_X(-A)$ for a very ample A. Generation on a non-empty open set gives a non-zero section of $\mathcal{O}_X(mD - A)$. Thus $mD \sim_{\text{lin}} A + N$ with $N \geq 0$, and D is big by Corollary 3.2.7. □

Concerning the exponent of a big divisor, we obtain:

Corollary 3.2.10 (Exponent of big Cartier divisors)**.** *If D is a big Cartier divisor on a projective variety, then $e(D) = 1$. Consequently all sufficiently large multiples of D are effective up to linear equivalence.*

Proof. Choose a very ample H such that $H - D \sim_{\text{lin}} H_1$ is effective. Corollary 3.2.7 and its proof give $m \geq 2$ and $N \geq 0$ with $mD \sim_{\text{lin}} H + N$. Then

$$(m-1)D \sim_{\text{lin}} (H - D) + N \sim_{\text{lin}} H_1 + N$$

is effective as well. The two consecutive integers $m, m-1$ belong to $N(D)$, so $e(D) = 1$. The final assertion is the semigroup fact of Definition 3.1.1. □

Combining this with normalization in Lemma 3.2.4 shows that its growth bound, and the non-vanishing in Kodaira's lemma, in fact hold for every sufficiently large integer m when D is big.

Corollary 3.2.11 (Restrictions of big Cartier divisors)**.** *Let L be a big line bundle on a projective variety X. There is a proper closed subset $V \subsetneq X$ such that, for every subvariety $Y \subseteq X$ not contained in V, $L|_Y$ is big. In particular, restriction to a general member of a very ample linear series is big (componentwise if that member has dimension zero).*

Proof. Write $L = \mathcal{O}_X(D)$ and $mD \sim_{\text{lin}} H+N$ with H very ample and N effective. If $Y \not\subseteq \operatorname{Supp} N$, restriction gives the sum of a very ample divisor and an effective Cartier divisor on Y. Apply Corollary 3.2.7 again, and take $V = \operatorname{Supp} N$. □

Example 3.2.12 (Restriction fails to preserve bigness)**.** On $X = \operatorname{Bl}_p \mathbb{P}^2$, let E be exceptional and H the pullback of a line. The bundle $\mathcal{O}_X(H + E)$ is big, but $\mathcal{O}_X(H + E)|_E \cong \mathcal{O}_{\mathbb{P}^1}(-1)$ is not big.

An integral Cartier divisor is big if and only if some, or every, positive multiple is big. This leads to the following definition.

Definition 3.2.13 (Big $\mathbb{Q}$-Cartier divisors)**.** A $\mathbb{Q}$-Cartier divisor D is big if mD is integral Cartier and big for some positive integer m.

By passing to integral multiples, bigness is a numerical property for $\mathbb{Q}$-Cartier divisors. We now give Siu's numerical criterion.

Theorem 3.2.14 (Numerical criterion for bigness)**.** *Let X be a projective variety of dimension $n \geq 1$, and D, E nef $\mathbb{Q}$-Cartier divisors. If*[58]

$$(D^n) > n(D^{n-1} \cdot E),$$

then $D - E$ is big.

Proof. Fix an ample Cartier divisor H. The polynomial

$$f(t) = ((D + tH)^n) - n((D + tH)^{n-1} \cdot (E + tH))$$

is positive at 0, hence remains positive for small $t > 0$. Replace D, E by $D + \varepsilon H, E + \varepsilon H$ with small positive rational ε. Both become ample, and their difference is *unchanged*. Multiplying both by a common positive integer, we may assume they are integral and very ample. Choose general divisors $E_1, E_2, \ldots \in |E|$ successively so that the local equation of E_j is a non-zero-divisor modulo the sum of the previous divisors. Equivalently, avoid the finitely many associated points of the previous Cartier thickening. Then intersection of their ideal sheaves is their product, and the sections vanishing on all E_i, $1 \leq i \leq m$, are exactly those vanishing on $\sum_{i=1}^m E_i \sim mE$. Thus there is an exact sequence at the first two terms

$$0 \longrightarrow H^0(X, m(D-E)) \longrightarrow H^0(X, mD) \longrightarrow \bigoplus_{i=1}^m H^0(E_i, \mathcal{O}_{E_i}(mD)).$$

For $m \gg 0$, the dimensions on the right are independent of i. Indeed, the sequence for $E_i \sim E$ and Serre vanishing for $\mathcal{O}_X(mD - E)$ express them as $h^0(X, mD) - h^0(X, mD - E)$, uniformly for every E_i. Riemann–Roch gives

$$\begin{aligned} h^0(X, m(D-E)) &\geq h^0(X, mD) - \sum_{i=1}^m h^0(E_i, \mathcal{O}_{E_i}(mD)) \\ &\geq \frac{(D^n) - n(D^{n-1} \cdot E)}{n!} m^n + O(m^{n-1}). \end{aligned}$$

Here the restriction Hilbert polynomial has leading term $(D^{n-1} \cdot E)m^{n-1}/(n-1)!$; for $n = 1$ it is the constant length (E). The leading coefficient is positive, proving bigness. □

Theorem 3.2.15 (Bigness of nef Cartier divisors)**.** *A nef Cartier divisor D on a projective variety of dimension n is big if and only if $(D^n) > 0$.*

[58] For adding a small ample class to a nef class, see Corollary 1.4.7.

Proof. "⇐": For $n \geq 1$, apply Theorem 3.2.14 with $E = 0$. The case $n = 0$ is immediate. "⇒": Write $mD \sim_{\text{lin}} H + N$, with H very ample and $N \geq 0$. Since D is nef, $(D^{n-1} \cdot N) \geq 0$, hence

$$m(D^n) = (D^{n-1} \cdot (H + N)) \geq (D^{n-1} \cdot H).$$

We proceed by induction. For a curve, the same decomposition gives $m \deg D = \deg H + \deg N > 0$, proving the base case directly. For $n > 1$, choose a general integral member of $|H|$. By Corollary 3.2.11, $D|_H$ is big, and it is also nef. The induction hypothesis gives $(D^{n-1} \cdot H) > 0$, so $(D^n) > 0$. □

Remark 3.2.16 (Alternative approach to Theorem 3.2.15)**.** The asymptotic result for nef divisors deduced from Fujita vanishing in Corollary 1.4.46 gives[59]

$$h^0(X, mD) = \frac{(D^n)}{n!} m^n + O(m^{n-1}).$$

Thus Lemma 3.2.4 gives the same characterization directly.

Example 3.2.17 (Big line bundles with negative numerics)**.** Without nefness, intersection numbers of big line bundles can be negative. Let $X = \text{Bl}_p \mathbb{P}^2$, with exceptional divisor E and a very ample divisor A. Then $D_l = A + lE$ is big for every integer $l \geq 0$, but

$$(D_l^2) = (A^2) + 2l(A \cdot E) - l^2 \longrightarrow -\infty.$$

Example 3.2.18 (Alternative characterization of big and nef Cartier divisors)**.** A Cartier divisor D on a projective variety is nef and big if and only if there is an effective Cartier divisor N such that $D - \frac{1}{k}N$ is ample for all $k \gg 0$.

Proof. "⇐": Bigness follows from the ample-plus-effective decomposition. For every curve C, $(D \cdot C)$ is the limit of the positive numbers $((D - \frac{1}{k}N) \cdot C)$, so D is nef. "⇒": Write $mD \equiv_{\text{num}} A + N$, with A ample and $N \geq 0$. For $k > m$,

$$kD - N \equiv_{\text{num}} (k - m)D + A$$

is the sum of a nef and an ample divisor, hence ample. □

3.2B Pseudo-effective and Big Cones

We now rephrase Kodaira's lemma in the language of numerical geometry. As before, X is a projective variety.

Definition 3.2.19 (Big $\mathbb{R}$-Cartier divisors)**.** An $\mathbb{R}$-Cartier divisor $D \in \text{Div}_{\mathbb{R}}(X)$ is big if it can be written $D = \sum_i a_i D_i$, where each D_i is a big integral Cartier divisor and $a_i > 0$ is real.

[59] The Fujita vanishing input is Theorem 1.4.42.

The definition is stable under addition and positive scalar multiplication.

Proposition 3.2.20 (Formal properties of big $\mathbb{R}$-Cartier divisors)**.** *Let D, D' be $\mathbb{R}$-Cartier divisors on X.*

(1) If $D \equiv_{\text{num}} D'$, then D is big if and only if D' is big.

(2) D is big if and only if $D \equiv_{\text{num}} A+N$, where A is ample and N is an effective $\mathbb{R}$-Cartier divisor.

Proof. (1) For rational divisors, clear denominators and use Corollary 3.2.8. For real divisors one more argument is needed. As in Proposition 1.3.14, numerical relations in the span of finitely many integral Cartier divisors are defined over $\mathbb{Q}$. Thus a numerically trivial real Cartier divisor can be written as $P = \sum_j b_j P_j$ with P_j numerically trivial integral Cartier divisors. If B is big and integral, choose rational numbers on either side of each b_j. Expressing $(b_j)_j$ as a convex combination of the rational vertices of this box expresses $B + P$ as a convex combination of rational divisors numerically equivalent to B. They are big, so $B + P$ is big. If $D = \sum_i a_i D_i$ is big, apply the same argument to $D_1 + P/a_1$ and leave the other summands unchanged. This proves invariance under adding P, and interchanging D, D' proves (1).

(2) If D is big, apply Corollary 3.2.7 to each integral summand and take the same positive real combination of the ample and effective parts. Conversely, an ample real divisor is a positive combination of ample, hence big, integral divisors. To add an effective real divisor it suffices, one summand at a time, to treat $B+sN$, with B big integral, N effective integral and $s > 0$ real. For rational s this is Corollary 3.2.7. In general choose $0 < s_1 < s < s_2$ rational and $t \in (0, 1)$ with $s = ts_1 + (1 - t)s_2$. Then

$$B + sN = t(B + s_1 N) + (1 - t)(B + s_2 N)$$

is a positive real combination of big rational divisors. Finally use (1) to pass from $A + N$ to the numerically equivalent divisor D. □

Example 3.2.21 (Big and nef $\mathbb{R}$-Cartier divisors)**.** If D is a nef and big real Cartier divisor, there is an effective real Cartier divisor N such that $D - \frac{1}{k}N$ is ample for every sufficiently large positive integer k.

Proof. Write $D \equiv_{\text{num}} A + N$ by Proposition 3.2.20. For $k > 1$,

$$D - \frac{1}{k}N \equiv_{\text{num}} \left(1 - \frac{1}{k}\right) D + \frac{1}{k}A$$

is the sum of a nef real divisor and an ample real divisor, hence ample. □

Corollary 3.2.22 (Openness of big Cartier divisors)**.** *If D is a big real Cartier divisor and $E_1, \ldots, E_t$ are arbitrary real Cartier divisors, then $D+\varepsilon_1 E_1+\cdots+\varepsilon_t E_t$ remains big for all sufficiently small real $|\varepsilon_i|$.*

Proof. Write $D \equiv_{\text{num}} A+N$ with A ample and N effective. The divisor $A+\sum_i \varepsilon_i E_i$ remains ample by openness of ampleness. Apply Proposition 3.2.20(2). □

Numerical invariance allows us to speak of big real classes in $N^1(X)_{\mathbb{R}}$.

Definition 3.2.23 (Big and pseudo-effective cones)**.** The big cone $\text{Big}(X) \subseteq N^1(X)_{\mathbb{R}}$ is the convex cone of all big real Cartier divisor classes. The pseudo-effective cone $\overline{\text{Eff}}(X) \subseteq N^1(X)_{\mathbb{R}}$ is the closure of the convex cone generated by effective real Cartier divisor classes. A real Cartier divisor is pseudo-effective if its class lies in $\overline{\text{Eff}}(X)$.

Theorem 3.2.24. *The big cone is the interior of the pseudo-effective cone, and the pseudo-effective cone is the closure of the big cone:*

$$\text{Big}(X) = \text{int}(\overline{\text{Eff}}(X)), \qquad \overline{\text{Eff}}(X) = \overline{\text{Big}(X)}.$$

Proof. The big cone is open, and every big class is the sum of an ample class and an effective class. Ample classes are positive real combinations of classes with effective positive multiples, so $\text{Big}(X) \subseteq \overline{\text{Eff}}(X)$. Taking closures gives one inclusion. Conversely, if $\eta \in \overline{\text{Eff}}(X)$, choose effective *real* classes $\eta_k \to \eta$ and fix an ample class α. Then

$$\eta = \lim_{k\to\infty} (\eta_k + k^{-1}\alpha),$$

and each class on the right is big. This proves equality of closures. Finally, a non-empty open convex set in a finite-dimensional real vector space equals the interior of its closure. For example, a separating hyperplane at any point outside the open convex set shows that the point cannot be interior to its closure. Apply this to $\text{Big}(X)$. □

Remark 3.2.25 (Pseudo-effective Cartier divisors on a surface)**.** On a smooth projective surface, divisors and curves are identified via the intersection pairing. Under this identification $\overline{\text{Eff}}(X) = \overline{\text{NE}}(X)$ and[60]

$$\overline{\text{Eff}}(X) = \text{Nef}(X)^*.$$

Thus D is pseudo-effective if and only if $(D \cdot H) \geq 0$ for every nef divisor H. This says that D lies in the *dual of the nef cone*; it does not assert that D itself is nef.

3.2C Volume of a Big Cartier Divisor

There is an interesting invariant measuring the asymptotic growth of $|mD|$ as $m \to \infty$.

[60] On a surface the duality with the nef cone is Proposition 1.4.27; duality does not identify the pseudo-effective cone with the nef cone.

Definition 3.2.26 (Volume of a line bundle)**.** For a line bundle L on an n-dimensional projective variety X, define

$$\operatorname{vol}(L) = \operatorname{vol}_X(L) = \limsup_{m\to\infty} \frac{h^0(X, L^{\otimes m})}{m^n/n!}.$$

The volume $\operatorname{vol}(D) = \operatorname{vol}_X(D)$ of a Cartier divisor is defined via $\mathcal{O}_X(D)$.

The cohomology growth bound makes volume finite, and $\operatorname{vol}(L) > 0$ if and only if L is big. If L is nef, Fujita vanishing and asymptotic Riemann–Roch give

$$h^0(X, L^{\otimes m}) = \frac{\int_X c_1(L)^n}{n!} m^n + O(m^{n-1}), \qquad \operatorname{vol}(L) = \int_X c_1(L)^n.$$

For a zero-dimensional variety all these bundles are trivial and the volume is 1. The proofs below involving hyperplane divisors are given for $n \geq 1$; the zero-dimensional assertions are immediate.

Example 3.2.27 (Volume on a blow-up of projective space)**.** Blow up $\mathbb{P}^n$ at s distinct points and put $D = dH - rE$, as in Example 3.2.5. The computation there gives

$$h^0(X, mD) \geq \frac{d^n - sr^n}{n!} m^n + O(m^{n-1}), \qquad \operatorname{vol}(D) \geq d^n - sr^n.$$

Proposition 3.2.28 (Properties of the volume)**.** *Let D be a big Cartier divisor on an n-dimensional projective variety.*

(i) For every positive integer a, $\operatorname{vol}(aD) = a^n \operatorname{vol}(D)$.

(ii) Given any Cartier divisor N and $\varepsilon > 0$, there is an integer $p_0 = p_0(D, N, \varepsilon)$ such that

$$\frac{|\operatorname{vol}(pD - N) - \operatorname{vol}(pD)|}{p^n} < \varepsilon \qquad (p > p_0).$$

We begin with two lemmas.

Lemma 3.2.29. *Let L be a line bundle and A a very ample divisor on X, with $n \geq 1$. For very general $E, E' \in |A|$, one has*[61]

$$h^0(E, L^{\otimes m}|_E) = h^0(E', L^{\otimes m}|_{E'}) \quad \text{for all } m \geq 1,$$

and hence $\operatorname{vol}_E(L|_E) = \operatorname{vol}_{E'}(L|_{E'})$. For $n = 1$, the volume on the zero-dimensional divisor means its length; the same assertion holds.

[61] For the local flatness criterion and upper semicontinuity, see [SP25, Tag 00MK], [Har77, III, 12.8].

Proof. Let $V = H^0(X, \mathcal{O}_X(A))$ and let $P = |A|$ be the space of lines in V. With our quotient convention this is $P = \mathbb{P}(V^\vee)$. Choose a basis $s_0, \dots, s_N$ of V and dual homogeneous coordinates $y_0, \dots, y_N$ on P. On $X \times P$ the line bundle

$$\mathcal{H} = \mathrm{pr}_1^*\, \mathcal{O}_X(A) \otimes \mathrm{pr}_2^*\, \mathcal{O}_P(1)$$

has the universal section $s = \sum_i s_i \boxtimes y_i$. Its zero divisor $\mathcal{D} \subseteq X \times P$ has fibre over $t = [c_0 : \cdots : c_N]$ equal to the divisor of $\sum_i c_i s_i$, namely $E_t \in |A|$. We have the two projections

$$X \xleftarrow{\;p_1\;} \mathcal{D} \subseteq X \times P \xrightarrow{\;p_2\;} P = |A|$$

The exact sequence

$$0 \longrightarrow \mathcal{H}^{-1} \xrightarrow{\cdot s} \mathcal{O}_{X\times P} \longrightarrow \mathcal{O}_{\mathcal{D}} \longrightarrow 0$$

shows that $\mathcal{D} \to P$ is flat. Indeed, the first two sheaves are flat over P, and on every fibre multiplication by the corresponding non-zero section is injective, since X is geometrically integral. The local criterion for flatness then makes the quotient flat over P. For each $m \geq 1$, the sheaf $(\mathrm{pr}_1^* L^{\otimes m})|_{\mathcal{D}}$ is also flat over P. Properness and upper semicontinuity give a dense open $U_m \subseteq P$ on which $h^0(E_t, L^{\otimes m}|_{E_t})$ is constant. Choose E, E' in the intersection of all U_m. For $n \geq 2$, also impose Bertini's open condition that they be integral; for $n = 1$ they are finite divisors and their section dimensions equal their lengths. The equality of all section dimensions proves the equality of volumes. Notice that bigness of L was not needed in this proof. □

Lemma 3.2.30. *For any Cartier divisor D and any fixed integer $a > 0$,*

$$\limsup_{m\to\infty} \frac{h^0(X, mD)}{m^n/n!} = \limsup_{k\to\infty} \frac{h^0(X, akD)}{(ak)^n/n!}.$$

Proof. If D is not big, its volume is zero, and both sides vanish. Assume D big, so every sufficiently large integer belongs to $N(D)$. For $r \geq 0$ put

$$v_r = \limsup_{k\to\infty} \frac{h^0(X, (ak+r)D)}{(ak+r)^n/n!}.$$

For any fixed r_0, division with a shifted remainder set partitions the sequence into the a subsequences with $r \in \{r_0 + 1, \dots, r_0 + a\}$. Hence $\mathrm{vol}(D) = \max\{v_{r_0+1}, \dots, v_{r_0+a}\}$. Choose r_0 so large that $h^0(X, rD) \neq 0$ for $r \geq r_0$, and then choose q with $qa - (r_0 + a) \geq r_0$. For each of these remainders choose

$$D_r \in |rD|, \qquad D'_r \in |(qa - r)D|.$$

Multiplication gives

$$\mathcal{O}_X(kaD) \hookrightarrow \mathcal{O}_X((ka + r)D) \hookrightarrow \mathcal{O}_X((k + q)aD),$$

and therefore

$$h^0(X, kaD) \le h^0(X, (ka + r)D) \le h^0(X, (k + q)aD).$$

After division by $(ka)^n/n!$ and taking limsup, both outer expressions have limsup v_0: the fixed shifts change the denominators by factors tending to 1. Thus $v_0 \le v_r \le v_0$, proving the assertion. □

Proof of Proposition 3.2.28. By Lemma 3.2.30,

$$\operatorname{vol}(aD) = \limsup_k \frac{h^0(X, akD)}{k^n/n!} = a^n \limsup_k \frac{h^0(X, akD)}{(ak)^n/n!} = a^n \operatorname{vol}(D).$$

The same argument proves (i) for every Cartier divisor, including non-big ones. For (ii), write $N \sim_{\text{lin}} A - B$ with A, B very ample effective divisors. Since D is big, choose $r > 0$ and $B_1 \ge 0$ with $rD - B \sim_{\text{lin}} B_1$. Then

$$pD - N \sim_{\text{lin}} (p + r)D - (A + B_1).$$

Since $(p + r)^n/p^n \to 1$, it suffices to establish $\operatorname{vol}(qD - F)/q^n \to \operatorname{vol}(D)$ for the effective divisor $F = A + B_1$. Thus we may assume N effective. For a second effective divisor N', multiplication of sections gives

$$\operatorname{vol}(pD - (N + N')) \le \operatorname{vol}(pD - N) \le \operatorname{vol}(pD).$$

Choose N' so that $N + N'$ is very ample. It is enough to prove (ii) for this larger divisor, so we reduce to N very ample.

Choose distinct very general divisors $E_1, E_2, \ldots \in |N|$, with the proper-intersection condition used in the proof of Theorem 3.2.14. By Lemma 3.2.29 they have the same section dimensions in all powers of $\mathcal{O}_X(D)$ as a fixed very general $E \in |N|$. For each $m, p > 0$, the exact restriction sequence on sections gives

$$\begin{aligned} h^0(X, m(pD - N)) &\ge h^0(X, mpD) - \sum_{i=1}^{m} h^0(E_i, \mathcal{O}_{E_i}(mpD)) \\ &= h^0(X, mpD) - mh^0(E, \mathcal{O}_E(mpD)). \end{aligned}$$

The divisors E_i remain distinct in this sequence; only their section *dimensions* are replaced by the same number. Divide by $m^n/n!$. Take a subsequence on which the first term on the right approaches $\operatorname{vol}_X(pD)$, and bound the limsup of the subtracted term by $n \operatorname{vol}_E(pD|_E)$. This gives

$$\operatorname{vol}_X(pD - N) \ge \operatorname{vol}_X(pD) - n \operatorname{vol}_E(pD|_E).$$

Homogeneity in dimension $n - 1$ gives $\operatorname{vol}_E(pD|_E) = p^{n-1} \operatorname{vol}_E(D|_E)$, including the length interpretation when $n = 1$. Consequently

$$\operatorname{vol}(D) \ge \frac{\operatorname{vol}(pD - N)}{p^n} \ge \operatorname{vol}(D) - \frac{n}{p} \operatorname{vol}_E(D|_E),$$

and (ii) follows by letting $p \to \infty$. □

Remark 3.2.31. The second statement hints at the continuity properties of volume.

Remark 3.2.32 (Volume of a $\mathbb{Q}$-Cartier divisor)**.** For a $\mathbb{Q}$-Cartier divisor D, choose $a > 0$ such that aD is integral Cartier and define $\mathrm{vol}(D) = a^{-n}\,\mathrm{vol}(aD)$. Homogeneity shows independence of a. Equivalently, take the defining limsup over the integers m for which mD is integral Cartier.

Example 3.2.33 (Volume of a difference of nef Cartier divisors, I)**.** For nef $\mathbb{Q}$-Cartier divisors D, E,

$$\mathrm{vol}(D - E) \geq (D^n) - n(D^{n-1} \cdot E).$$

Indeed, when D, E are ample this is the section estimate in the proof of Theorem 3.2.14, followed by clearing denominators and homogeneity. In the nef case apply that estimate to $D+\varepsilon H, E+\varepsilon H$ with $\varepsilon > 0$ rational. Their difference is $D-E$, and letting $\varepsilon \to 0$ on the intersection-polynomial side proves the result. No continuity theorem for volume is needed at this stage.

Using Fujita vanishing, we next deduce numerical invariance of volume.

Proposition 3.2.34 (Numerical nature of volume)**.** *If D, D' are numerically equivalent $\mathbb{Q}$-Cartier divisors on X, then $\mathrm{vol}(D) = \mathrm{vol}(D')$. Thus $\mathrm{vol}(\xi)$ is defined for $\xi \in N^1(X)_{\mathbb{Q}}$.*

Lemma 3.2.35. *There is a fixed Cartier divisor N such that*

$$H^0(X, \mathcal{O}_X(N + P)) \neq 0$$

for every numerically trivial Cartier divisor P. In particular, $N \pm mP_0$ is linearly equivalent to an effective divisor for every numerically trivial P_0 and every $m \in \mathbb{Z}$.

Proof. Fix a very ample divisor B. By Fujita vanishing there is a large multiple F of B such that $H^i(X, \mathcal{O}_X(F+A)) = 0$ for $i > 0$ and every nef Cartier divisor A. For $1 \leq i \leq n$, use $A = P+(n-i)B$. Castelnuovo–Mumford regularity shows that $F + P + nB$ is globally generated. Thus $N = F + nB$ works for all P. Applying this to $\pm mP_0$ proves the last assertion. □

Proof of Proposition 3.2.34. Clear denominators to assume the divisors integral. By numerical invariance of bigness, either both volumes are zero or both divisors are big. In the latter case write $D' = D + P$, with P numerically trivial. Fix the divisor N of Lemma 3.2.35, independent of p. A non-zero section of $\mathcal{O}_X(N - pP)$ gives, for every $m > 0$, an inclusion

$$\mathcal{O}_X\big(m(p(D + P) - N)\big) \hookrightarrow \mathcal{O}_X(mpD).$$

It follows that $\mathrm{vol}(p(D+P)-N) \leq p^n\,\mathrm{vol}(D)$. By Proposition 3.2.28(ii), division by p^n and $p \to \infty$ give $\mathrm{vol}(D + P) \leq \mathrm{vol}(D)$. Replace D by $D + P$ and P by $-P$ for the opposite inequality. □

Proposition 3.2.36 (Birational invariance of volume, I)**.** *Let $\nu : X' \to X$ be a birational morphism of projective varieties. For an integral or $\mathbb{Q}$-Cartier divisor D, put $D' = \nu^* D$. Then $\mathrm{vol}_{X'}(D') = \mathrm{vol}_X(D)$.*[62]

Proof. It suffices to treat integral D. Projection formula gives

$$H^0(X', \mathcal{O}_{X'}(mD')) = H^0(X, \nu_* \mathcal{O}_{X'} \otimes \mathcal{O}_X(mD)).$$

The quotient η in

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \nu_* \mathcal{O}_{X'} \longrightarrow \eta \longrightarrow 0$$

is supported in dimension at most $n-1$. Therefore

$$\begin{aligned} h^0(X, \mathcal{O}_X(mD)) &\leq h^0(X', \mathcal{O}_{X'}(mD')) \\ &\leq h^0(X, \mathcal{O}_X(mD)) + h^0(X, \eta \otimes \mathcal{O}_X(mD)) \\ &= h^0(X, \mathcal{O}_X(mD)) + O(m^{n-1}). \end{aligned}$$

Dividing by $m^n/n!$ and taking limsup proves the assertion. □

Theorem 3.2.37 (Continuity of volume)**.** *Let X be a projective variety of dimension $n \geq 1$, and fix a norm $\| \ \|$ on $N^1(X)_{\mathbb{R}}$. There is $C > 0$ such that*

$$|\operatorname{vol}(\xi) - \operatorname{vol}(\xi')| \leq C\big(\max\{\|\xi\|, \|\xi'\|\}\big)^{n-1} \|\xi - \xi'\|$$

for all $\xi, \xi' \in N^1(X)_{\mathbb{Q}}$. The case $\xi = \xi' = 0$ is interpreted as the trivial equality.

Proof. Both sides are homogeneous of degree n, so multiply both classes by a common large integer. Fix very ample Cartier divisors $A_1, \ldots, A_r$ whose classes form a basis of $N^1(X)_{\mathbb{Q}}$. By numerical invariance it suffices to treat

$$D = \sum_i a_i A_i, \qquad D' = \sum_i a_i' A_i, \qquad a_i, a_i' \in \mathbb{Z}.$$

All norms on a finite-dimensional space are equivalent, so use $\|D\| = \max_i |a_i|$. The following argument applies to arbitrary D, D'; no bigness assumption is required.

First suppose $a_i - a_i' = b_i \geq 0$, and write $D' = D - B$ with $B = \sum_i b_i A_i \geq 0$. Choose very general $E_j \in |A_j|$ meeting all the chosen A_i properly, and put $D_j = D|_{E_j}$. We claim

$$\operatorname{vol}(D - B) \geq \operatorname{vol}(D) - n \sum_{j=1}^{r} b_j \operatorname{vol}_{E_j}(D_j).$$

[62] For the growth estimate on the cokernel supported in lower dimension, see Example 1.2.24.

For the case $B = b_1 A_1$, choose mb_1 distinct very general divisors $F_\alpha \in |A_1|$ as in the proof of Proposition 3.2.28. Then

$$0 \longrightarrow \mathcal{O}_X\Big(mD - \sum_{\alpha=1}^{mb_1} F_\alpha\Big) \longrightarrow \mathcal{O}_X(mD) \longrightarrow \bigoplus_{\alpha=1}^{mb_1} \mathcal{O}_{F_\alpha}(mD)$$

is exact at the first two terms. Lemma 3.2.29 identifies every section dimension on the right with $h^0(E_1, mD_1)$. Thus

$$h^0(X, m(D - b_1 A_1)) \geq h^0(X, mD) - mb_1 h^0(E_1, mD_1).$$

Divide by $m^n/n!$, use a subsequence realizing $\mathrm{vol}(D)$, and bound the other term by its limsup. This proves

$$\mathrm{vol}(D - b_1 A_1) \geq \mathrm{vol}(D) - nb_1 \,\mathrm{vol}_{E_1}(D_1).$$

This argument uses the definition of volume; it does not assert an asymptotic Riemann–Roch equality for the possibly non-nef D_1.

For several A_j, induct on the number of non-zero b_j. If $\widetilde{D} = D - \sum_{j<r} b_j A_j$, the one-term case gives

$$\begin{aligned}\mathrm{vol}(\widetilde{D} - b_r A_r) &\geq \mathrm{vol}(\widetilde{D}) - nb_r \,\mathrm{vol}_{E_r}(\widetilde{D}|_{E_r}) \\ &\geq \mathrm{vol}(D) - n\sum_{j<r} b_j \,\mathrm{vol}_{E_j}(D_j) - nb_r \,\mathrm{vol}_{E_r}(D_r).\end{aligned}$$

Here restriction of $\sum_{j<r} b_j A_j$ to E_r is effective, so multiplication of sections gives $\mathrm{vol}_{E_r}(\widetilde{D}|_{E_r}) \leq \mathrm{vol}_{E_r}(D_r)$. We may impose the finitely many required very general conditions simultaneously.

Now put $D^+ = \sum_i |a_i| A_i$. The divisor $D^+ - D$ restricts effectively to E_j, so $\mathrm{vol}_{E_j}(D_j) \leq \mathrm{vol}_{E_j}(D^+|_{E_j})$. The latter restriction is nef. Hence

$$\begin{aligned}\mathrm{vol}_{E_j}(D^+|_{E_j}) &= ((D^+)^{n-1} \cdot A_j)_X \\ &= \Big((\sum_i |a_i| A_i)^{n-1} \cdot A_j\Big)_X \leq C_j \|D\|^{n-1},\end{aligned}$$

where C_j depends only on the fixed very ample basis. Since $\sum_j b_j \leq r\|B\|$, choose $C = 2nr \max_j C_j$ to obtain

$$0 \leq \mathrm{vol}(D) - \mathrm{vol}(D - B) \leq \frac{C}{2}\|D\|^{n-1}\|B\|.$$

For the general case write $D' = D + E - F$, where E, F are non-negative integral combinations of the A_i with disjoint sets of non-zero coefficients. Apply the preceding inequality to (D, F) and (D', E):

$$\begin{aligned}0 \leq \mathrm{vol}(D) - \mathrm{vol}(D - F) &\leq \frac{C}{2}\|D\|^{n-1}\|F\|, \\ 0 \leq \mathrm{vol}(D') - \mathrm{vol}(D' - E) &\leq \frac{C}{2}\|D'\|^{n-1}\|E\|.\end{aligned}$$

Since $D - F = D' - E$ and $\max\{\|E\|, \|F\|\} = \|E - F\| = \|D' - D\|$, the triangle inequality gives

$$|\operatorname{vol}(D) - \operatorname{vol}(D')| \leq C \max\{\|D\|, \|D'\|\}^{n-1} \|D - D'\|.$$

Homogeneity and equivalence of norms finish the proof. □

Corollary 3.2.38 (Volume of real classes)**.** *The function* $\xi \mapsto \operatorname{vol}(\xi)$ *on* $N^1(X)_{\mathbb{Q}}$ *extends uniquely to a continuous function* $\operatorname{vol} : N^1(X)_{\mathbb{R}} \to \mathbb{R}_{\geq 0}$.

Proof. For a rational sequence $\xi_i \to \xi$, the preceding estimate makes $\operatorname{vol}(\xi_i)$ a Cauchy sequence, since the norms $\|\xi_i\|$ are bounded. Applying the same estimate to two approximating sequences proves that the limit is independent of the sequence. The estimate also proves continuity, and density of rational classes proves uniqueness. □

Example 3.2.39 (Volume on the blow-up of projective space)**.** Let $X = \operatorname{Bl}_P \mathbb{P}^n$, with $n \geq 2$, and let $e, h \in N^1(X)_{\mathbb{R}} \cong \mathbb{R}^2$ be the classes of the exceptional divisor E and the pullback H of a hyperplane. Write a class as $xh - ye$. The nef cone is generated by h and $h - e$, so its region in these coordinates is $x \geq y \geq 0$. In the

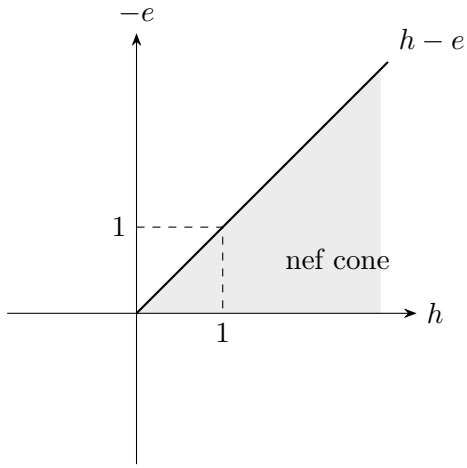


Figure 3.1: Plane of $N^1(X)_{\mathbb{R}}$, with coordinates $xh - ye$.

nef region, volume is the top self-intersection:

$$\operatorname{vol}(xh - ye) = x^n - y^n \qquad (x \geq y \geq 0).$$

Indeed, $(H^n) = 1$, $(E^n) = (-1)^{n-1}$, and $(H^{n-j} \cdot E^j) = 0$ for $1 \leq j < n$ by the projection formula. On the other hand, for integers $k, l \geq 0$, lE is the fixed part of $|kH + lE|$. If $\pi : X \to \mathbb{P}^n$ is the blow-up, then

$$\begin{aligned} H^0(X, \mathcal{O}_X(kH + lE)) &= H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(k) \otimes \pi_* \mathcal{O}_X(lE)) \\ &= H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(k)), \end{aligned}$$

because $\pi_*\mathcal{O}_X(lE) = \mathcal{O}_{\mathbb{P}^n}$ by Hartogs across the point P. This is where $n \geq 2$ is needed. Hence in the region $x \geq 0$, $y \leq 0$ spanned by h, e, volume is x^n. Outside these two regions it is zero: for rational classes this follows from the description by polynomials, which have non-negative degree and cannot vanish at P to order greater than their degree. Extend by continuity. Altogether,

$$\mathrm{vol}(xh - ye) = \begin{cases} x^n - y^n, & x \geq y \geq 0, \\ x^n, & x \geq 0,\ y \leq 0, \\ 0, & \text{otherwise.} \end{cases}$$

The formulas agree on the overlapping boundary $y = 0$.

By rational approximation and continuity, we obtain the following results.

Example 3.2.40 (Volume of a difference of nef Cartier divisors, II)**.** For real nef classes ξ, η on an n-dimensional projective variety,

$$\mathrm{vol}(\xi - \eta) \geq (\xi^n) - n(\xi^{n-1} \cdot \eta).$$

To approximate within nef classes, first add a small positive ample class to both, then approximate the resulting ample classes by rational ample classes. Apply Example 3.2.33 and use continuity of volume and intersection products.

Example 3.2.41 (Volume increases in effective directions)**.** If $\xi \in N^1(X)_{\mathbb{R}}$ is big and $e \in N^1(X)_{\mathbb{R}}$ is effective, then $\mathrm{vol}(\xi) \leq \mathrm{vol}(\xi + e)$. For rational divisors, this follows by multiplying sections by the defining section of a positive integral multiple of the effective divisor. Approximate its non-negative coefficients and those of ξ by rational numbers, then use continuity.

Example 3.2.42 (Birational invariance of volume, III)**.** If $\nu : X' \to X$ is a birational morphism of projective varieties, then $\mathrm{vol}_X(\xi) = \mathrm{vol}_{X'}(\nu^*\xi)$ for every $\xi \in N^1(X)_{\mathbb{R}}$. This follows from Proposition 3.2.36 by rational approximation and continuity of volume and of the linear pullback map.

3.3 Bigness in Relative Version

In this section we consider bigness for a morphism. By way of preparation, let $f : X \to S$ be a proper surjective morphism from a normal variety to a projective variety, and assume its geometric generic fibre is irreducible. Write $\eta \in S$ for the generic point, $K = k(\eta)$ and $\overline{K}$ for an algebraic closure. Proper varieties can be treated by pulling line bundles back to projective birational models; the growth argument justifying this extension is given in Lemma 3.3.11 below.

Definition 3.3.1 (Relative Iitaka dimension)**.** The relative Iitaka dimension of a line bundle L on X is

$$\kappa(X/S, L) = \kappa(X_{\bar{\eta}}, L|_{X_{\bar{\eta}}}), \qquad X_{\bar{\eta}} = X \times_S \operatorname{Spec} \overline{K}.$$

We keep $\kappa(X, L)$ for the absolute Iitaka dimension.

Remark 3.3.2. If we drop irreducibility of the geometric generic fibre, it is still a normal reduced scheme in our characteristic-zero setting, but may have several disjoint components. Indeed, X_η is normal: on affine charts its rings are localizations of the normal rings of X. The extension $\overline{K}/K$ is separable, so normality is preserved by this field extension [SP25, Tag 0C3M]. For $\bar{x}$ lying over $x \in X_\eta$, the local description is[63]

$$\mathcal{O}_{X_{\bar{\eta}},\bar{x}} \cong (\mathcal{O}_{X_\eta,x} \otimes_K \overline{K})_{\mathfrak{q}_{\bar{x}}}.$$

Since a noetherian normal scheme is a finite disjoint union of normal integral schemes [SP25, Tag 033M], $X_{\bar{\eta}}$ is a disjoint union of normal varieties. Under our irreducibility assumption it is itself a normal variety.

Definition 3.3.3 (Relative bigness)**.** A line bundle L on a normal variety X is big relative to f, or f-big, if $\kappa(X/S, L) = \dim X_{\bar{\eta}}$, that is, if its restriction to the geometric generic fibre is big. A Cartier divisor D is f-big if $\mathcal{O}_X(D)$ is so.

Remark 3.3.4 (Equivalent definition of relative bigness)**.** In [KMM87] relative bigness is defined using the generic fibre X_η. This is equivalent to the geometric generic-fibre definition. Flat base change [Har77, III, 9.3] gives

$$H^0(X_\eta, L^{\otimes m}|_{X_\eta}) \otimes_K \overline{K} \cong H^0(X_{\bar{\eta}}, L^{\otimes m}|_{X_{\bar{\eta}}}).$$

Thus the dimensions of these spaces over their respective ground fields are the same, and so are the Iitaka dimensions and the bigness conditions. We use either description as convenient. The same definitions extend to non-normal varieties by normalization, as in the absolute case.

Lemma 3.3.5. *Let $f : X \to S$ be a projective surjective morphism of varieties, with $\dim X_\eta = n$; normality is not required. A line bundle L is f-big if and only if there is $C > 0$ such that*

$$\operatorname{rank} f_* L^{\otimes m} \geq C m^n \qquad \textit{for every sufficiently large } m.$$

Proof. At the generic point,

$$\operatorname{rank} f_* L^{\otimes m} = \dim_K (f_* L^{\otimes m})_\eta = h^0(X_\eta, L^{\otimes m}|_{X_\eta}).$$

Apply the growth criterion of Lemma 3.2.4, with the all-large-exponents consequence of Corollary 3.2.10. The same proof works over K, or after extension to $\overline{K}$ by the preceding remark. This proves both directions. Surjectivity matters: otherwise the direct image has generic rank zero on S. □

Proposition 3.3.6 (Relative Kodaira's lemma)**.** *Let $f : X \to S$ be as in Lemma 3.3.5, D an f-big Cartier divisor and F an effective Cartier divisor on X. Then*

$$f_* \mathcal{O}_X(mD - F) \neq 0 \qquad (m \gg 0).$$

[63] For geometric reducedness and geometric normality over a perfect field, see [SP25, Tag 030U], [SP25, Tag 0C30].

Proof. The generic fibre of this sheaf is $H^0(X_\eta, \mathcal{O}_X(mD - F)|_{X_\eta})$, which is non-zero for $m \gg 0$ by Kodaira's lemma on the generic fibre. The restriction of F is effective, possibly zero. Thus the direct image is non-zero. □

Corollary 3.3.7 (Characterization of relatively big Cartier divisors)**.** *Let $f : X \to S$ be a projective surjective morphism of varieties, with S projective, and let D be Cartier on X. The following are equivalent:*[64]

(i) D is f-big.

(ii) For every f-ample Cartier divisor A, there are $m > 0$ and an effective Cartier divisor N on X such that $mD \sim_S A + N$.

(iii) The same statement holds for some f-ample Cartier divisor A.

(iv) There are an f-ample Cartier divisor A, $m > 0$, and an effective Cartier divisor N with $mD \equiv_S A + N$.

Here $\sim_S$ and $\equiv_S$ have the relative meanings of Appendix A. In (iii) one may also arrange $\sim_{\mathrm{lin}}$ by absorbing a pullback from S into A, which preserves f-ampleness.

Proof. "(i)⇒(ii)": Fix A f-ample. For $r \gg 0$, twisting rA and $(r+1)A$ by suitable pullbacks of sufficiently ample divisors on S makes them globally generated. Choose effective representatives

$$rA \sim_S H_r, \qquad (r+1)A \sim_S H_{r+1}.$$

Relative Kodaira's lemma gives a non-zero coherent sheaf $f_*\mathcal{O}_X(mD - H_{r+1})$ for $m \gg 0$. Twist it by a sufficiently ample $\mathcal{O}_S(B)$ to obtain a non-zero global section. By projection formula,

$$\begin{aligned} 0 \neq H^0(S, f_*\mathcal{O}_X(mD - H_{r+1}) \otimes \mathcal{O}_S(B)) \\ = H^0(X, \mathcal{O}_X(mD - H_{r+1} + f^*B)). \end{aligned}$$

Thus $mD + f^*B \sim_{\mathrm{lin}} H_{r+1} + N'$ with $N' \geq 0$, and, relatively,

$$mD \sim_S A + (H_r + N').$$

Take $N = H_r + N'$. The implications (ii)⇒(iii)⇒(iv) are immediate. For (iv)⇒(i), $mD - N$ is relatively numerically equivalent to an f-ample divisor, and hence is f-ample by the fibrewise Nakai–Moishezon criterion. Its restriction to the generic fibre is ample; adding the effective $N|_{X_\eta}$ shows that $D|_{X_\eta}$ is big. □

Corollary 3.3.8. *Under these hypotheses, bigness of a Cartier divisor relative to f depends only on its relative numerical equivalence class.*

[64] Relative linear equivalence is discussed in Appendix A.

Remark 3.3.9. Since the rank is computed on the generic fibre, one may likewise define relative volume by

$$\mathrm{vol}_{X/S}(L) = \limsup_m \frac{\mathrm{rank}\, f_* L^{\otimes m}}{m^n/n!} = \mathrm{vol}_{X_\eta}(L|_{X_\eta}), \qquad n = \dim X_\eta.$$

Real relative bigness and the corresponding generic-fibre cones can be treated similarly. These constructions must be read on the generic fibre, or through restriction to it; they should not be identified without explanation with all of $N^1(X/S)$, which can contain non-zero vertical classes restricting to zero on the generic fibre.

Remark 3.3.10. The proof of Lemma 3.2.4 uses the $O(m^n)$ cohomology estimate for projective schemes, where a Cartier divisor can be written as a difference of very ample divisors. To extend the growth criterion to proper varieties, it suffices to prove the following proper version, as needed for [KM98, 3.22]. The definitions via a projective birational model then agree with the intrinsic growth criterion. On the other hand, a proper morphism with an f-ample line bundle is projective in the convention of Remark 1.5.2; thus the relative ample decomposition in Corollary 3.3.7 naturally retains its projective hypothesis.[65]

Lemma 3.3.11 (Growth of cohomology for proper schemes)**.** *Let X be a proper scheme of dimension n, D a Cartier divisor and $\mathcal{F}$ a coherent sheaf on X. For every $i \geq 0$,*[66]

$$h^i(X, \mathcal{F} \otimes \mathcal{O}_X(mD)) = O(m^n) \qquad (m \to \infty).$$

Proof. We induct on dimension. In dimension zero, X is finite over the ground field, higher cohomology vanishes, and tensoring by a line bundle preserves length, so the assertion holds. A sheaf supported in smaller dimension can be viewed as a sheaf on the closed subscheme defined by its annihilator, to which the induction hypothesis applies.

Step 1: By Chow's lemma choose a projective scheme X' of dimension n and a proper morphism $f : X' \to X$ which is an isomorphism over an open set U meeting every irreducible component of X. One may first choose a dense quasi-projective such U and apply the version of Chow's lemma preserving it. For the natural map $\mathcal{F} \to f_* f^* \mathcal{F}$, set

$$\mathcal{K} = \ker(\mathcal{F} \to f_* f^* \mathcal{F}), \quad \mathcal{I} = \mathrm{im}(\mathcal{F} \to f_* f^* \mathcal{F}), \quad \mathcal{Q} = \mathrm{coker}(\mathcal{F} \to f_* f^* \mathcal{F}).$$

[65] The projective cohomology estimate used here is Example 1.2.24.

[66] The spectral sequence used here is Theorem D.6.3; its eventual stabilization follows from Theorem D.5.2 and Lemma D.3.4. The projective estimate on the modification is Example 1.2.24.

Both $\mathcal{K}$ and $\mathcal{Q}$ are supported outside U, in dimension at most $n-1$. The two exact sequences

$$0 \to \mathcal{K} \to \mathcal{F} \to \mathcal{I} \to 0, \qquad 0 \to \mathcal{I} \to f_* f^* \mathcal{F} \to \mathcal{Q} \to 0$$

remain exact after tensoring by $\mathcal{O}_X(mD)$. Their long exact cohomology sequences and induction give

$$\begin{aligned} |h^i(X, \mathcal{F}(mD)) - h^i(X, \mathcal{I}(mD))| &\le h^i(X, \mathcal{K}(mD)) + h^{i+1}(X, \mathcal{K}(mD)), \\ |h^i(X, (f_* f^* \mathcal{F})(mD)) - h^i(X, \mathcal{I}(mD))| &\le h^{i-1}(X, \mathcal{Q}(mD)) + h^i(X, \mathcal{Q}(mD)). \end{aligned}$$

Here $h^{-1} = 0$. Thus

$$h^i(X, \mathcal{F}(mD)) = h^i(X, (f_* f^* \mathcal{F})(mD)) + O(m^{n-1}).$$

It suffices to bound the right-hand cohomology group.

Step 2: Put $\mathcal{G}_m = f^*\mathcal{F} \otimes f^*\mathcal{O}_X(mD)$. The projection formula identifies $f_*\mathcal{G}_m = (f_* f^* \mathcal{F}) \otimes \mathcal{O}_X(mD)$. Consider the Leray spectral sequence (Appendix D):

$$E_2^{p,q}(m) = H^p(X, R^q f_* \mathcal{G}_m) \implies H^{p+q}(X', \mathcal{G}_m).$$

For each total degree i it induces a decreasing filtration

$$0 = \Phi^{i+1} H^i \subseteq \Phi^i H^i \subseteq \cdots \subseteq \Phi^0 H^i = H^i(X', \mathcal{G}_m),$$

with exact sequences

$$0 \longrightarrow \Phi^{p+1} H^i \longrightarrow \Phi^p H^i \longrightarrow E_\infty^{p,i-p}(m) \longrightarrow 0.$$

Taking dimensions and telescoping gives

$$\begin{aligned} h^i(X', \mathcal{G}_m) &= \sum_{p=0}^{i} (\dim \Phi^p H^i - \dim \Phi^{p+1} H^i) \\ &= \sum_{p+q=i} \dim E_\infty^{p,q}(m). \end{aligned}$$

For $q > 0$, the fixed coherent sheaf $R^q f_*(f^*\mathcal{F})$ is supported outside U. The higher-direct-image projection formula gives

$$R^q f_* \mathcal{G}_m \cong R^q f_*(f^*\mathcal{F}) \otimes \mathcal{O}_X(mD).$$

Induction therefore yields $\dim E_2^{p,q}(m) = O(m^{n-1})$ for $q > 0$. Every later term is a subquotient of $E_2^{p,q}$, so

$$h^i(X', \mathcal{G}_m) = \dim E_\infty^{i,0}(m) + O(m^{n-1}). \tag{3.2}$$

In the bottom row there is no outgoing differential for $r \geq 2$, since its target has second index $1 - r < 0$. Hence

$$E_{r+1}^{i,0}(m) = E_r^{i,0}(m) / \operatorname{im}(d_r : E_r^{i-r,r-1}(m) \to E_r^{i,0}(m)).$$

Each image has dimension at most $\dim E_2^{i-r,r-1}(m) = O(m^{n-1})$. There are only finitely many possible incoming differentials, uniformly in m: once $r > i$, the first index of the source is negative. Therefore

$$\dim E_\infty^{i,0}(m) = \dim E_2^{i,0}(m) + O(m^{n-1}).$$

Combining this with (3.2) gives

$$h^i(X, f_*\mathcal{G}_m) = h^i(X', \mathcal{G}_m) + O(m^{n-1}).$$

Since X' is projective, the projective cohomology growth estimate applies to the fixed sheaf $f^*\mathcal{F}$ and the fixed divisor f^*D, so $h^i(X', \mathcal{G}_m) = O(m^n)$. Step 1 completes the proof. □

The same quotient-sheaf argument as in Proposition 3.2.36 now proves birational invariance of volume, and hence of bigness, for proper varieties. For real Cartier divisors on a proper variety, retain the definition by positive real combinations of big integral Cartier divisors. This is equivalent to bigness after pullback to any projective birational model. Indeed, the integral and rational statements follow from the growth comparison just proved. In the finite-dimensional real span of the finitely many Cartier divisors involved, the inverse image of the big cone on the projective model is open and convex. A real point in it is a convex combination of nearby rational points still in it. Their integral multiples are big on the original variety by the integral statement. This proves the real statement and also its independence of the chosen model.

Example 3.3.12. Let X be a proper variety and N a nef and big Cartier divisor. If T is a nef real Cartier divisor, then $N + T$ is nef and big.[67]

Proof. By Chow's lemma choose a projective variety X' and a proper birational morphism $f : X' \to X$. The preceding discussion makes f^*N big, and it is nef by pullback of nefness. By Example 3.2.18, for a fixed effective Cartier divisor E and $k \gg 0$ we can write

$$f^*N = A_k + \frac{1}{k}E,$$

where A_k is an ample *rational* Cartier divisor. Since f^*T is nef, $f^*T + A_k$ is ample. Thus

$$f^*(N + T) = (f^*T + A_k) + \frac{1}{k}E$$

is big, and it is also nef. Birational invariance of real bigness gives bigness of $N + T$ on X; its nefness also follows directly from that of N and T. □

[67] For nef pullbacks, nef plus ample, and the ample-plus-effective characterization of big classes, see Example 1.4.3, Corollary 1.4.7, and Example 3.2.21, respectively.

There is a bigness analogue of Proposition 1.5.8.

Proposition 3.3.13. *Let $f : X \to S$ be a projective surjective morphism of varieties onto a proper variety S. Let D be Cartier on X and M an absolutely big Cartier divisor on S. Then D is f-big if and only if $D + mf^*M$ is absolutely big for all $m \gg 0$.*

Proof. First assume S projective, so X is also projective. "⇒": By Corollary 3.3.7(iii), absorbing a base pullback into the relatively ample part, write

$$rD \sim_{\text{lin}} A + N, \qquad r > 0, \quad A \text{ } f\text{-ample}, \quad N \geq 0.$$

Since M is big on S, there are $q > 0$, an ample divisor H and an effective divisor E with $qM \sim_{\text{lin}} H + E$. Then

$$\begin{aligned} rq(D + mf^*M) &\sim_{\text{lin}} qA + qN + rmf^*(H + E) \\ &= \underbrace{qA + rmf^*H}_{P_m} + \underbrace{qN + rmf^*E}_{G_m}. \end{aligned}$$

The divisor P_m is ample for $m \gg 0$ by Proposition 1.5.8, while $G_m \geq 0$ because f is surjective. Kodaira's characterization therefore shows that $D + mf^*M$ is big. "⇐": If $D + mf^*M$ is big for even one integer m, there are $t > 0$, an ample Cartier divisor A and $E \geq 0$ with

$$t(D + mf^*M) \sim_{\text{lin}} A + E.$$

Thus $tD \sim_S A + E$, where A is also f-ample. Corollary 3.3.7 gives relative bigness of D.

For a merely proper base S, choose a projective birational modification $\sigma : S' \to S$ by Chow's lemma, and let X' be the reduced main component of $X \times_S S'$ dominating X:

$$\begin{array}{ccc} X' & \xrightarrow{\tau} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow[\sigma]{} & S \end{array}$$

The variety X' is projective, since f' is projective and S' is projective. The maps σ, τ are birational, and the generic fibre is unchanged. Thus D is f-big if and only if τ^*D is f'-big, and σ^*M is big by proper birational invariance. Apply the projective-base case to (f', τ^*D, σ^*M), and use

$$\tau^*(D + mf^*M) = \tau^*D + m(f')^*\sigma^*M.$$

Proper birational invariance of bigness transfers the conclusion back to X, in both directions. This proves the stated proper-base version. □

3.4 Examples

3.4A Zariski's Construction

3.4B Cutkosky's Construction

3.4C Okounkov's Construction

Chapter 4
Vanishing Theorems

In this chapter, we are still working with varieties over the complex numbers, and unless otherwise stated, all varieties are assumed to be non-singular. Schemes are of finite type over $\mathbb{C}$.

4.1 Preliminaries

If the cohomology of a sheaf F comes from topological cohomology, then there is a Kodaira-type vanishing theorem.

János Kollár

4.1A Simple Normal Crossings and Resolution of Singularities

Let X be a non-singular variety of dimension n.

Definition 4.1.1. An effective divisor $D = \sum D_i$ on X has *simple normal crossings* if D is reduced, each component D_i is smooth, and D is defined in a neighborhood of any point by an equation in local analytic coordinates of the type

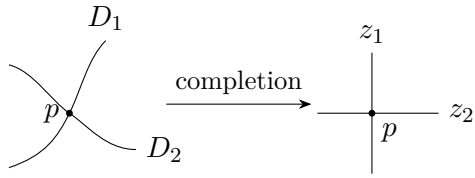
$$z_1 \cdots z_k = 0,$$

for some $k \leq n$. A $\mathbb{Q}$-divisor $E = \sum a_i D_i$ has *simple normal crossing support* if its underlying reduced support $\sum_{a_i \neq 0} D_i$ has simple normal crossings.

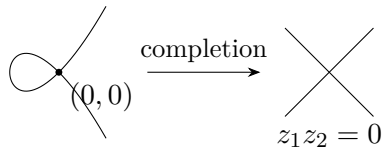
Remark 4.1.2. In the above case we say that D is a simple normal crossing divisor, or an snc divisor. Local analytic coordinates are coordinates in the analytic local ring $\mathbb{C}\{z_1, \ldots, z_n\}$. We can also describe this condition using formal coordinates in the completion

$$\widehat{\mathcal{O}}_{X,p} \cong \mathbb{C}[[z_1, \ldots, z_n]],$$

where the isomorphism follows from the Cohen structure theorem; see [Eis95, 10.16]. The completion is a formal, rather than an analytic, local ring. For smooth components, the snc condition is equivalently that their local equations extend to a regular system of parameters. The number $k \leq n$ is determined by the components of D containing p: if $k = 0$, then $p \notin \operatorname{Supp} D$; if $k \geq 1$, exactly k components contain p.



Remark 4.1.3. In our definition we emphasize that each component must be smooth; this is why we call it "simple" normal crossing. Some references on complex manifolds allow locally smooth branches without requiring the global irreducible components to be smooth, and call this normal crossing. For example, in the affine plane $\mathbb{A}^2$ with coordinates x, y, the nodal curve $y^2 = x^2(x+1)$ is irreducible but has a node at the origin. Therefore it is a normal crossing divisor on the complex manifold, but not a simple normal crossing divisor.



In other words, intersections among the components of an snc divisor should occur "as transversely as possible": locally they look like distinct coordinate hyperplanes.

Example 4.1.4 (Hyperplane arrangements)**.** Let $D = \sum H_i$ be the reduced divisor on $\mathbb{P}^n$ whose components are a collection of hyperplanes $H_i \subseteq \mathbb{P}^n$, corresponding to a finite set of points $S \subseteq (\mathbb{P}^n)^\vee$. Then D is snc if and only if the points of S are in linearly general position, i.e. for $r < n$, no $r+2$ points lie on an r-dimensional linear subspace of $(\mathbb{P}^n)^\vee$.

Proof. By the rank–nullity theorem, the hyperplanes of $\mathbb{P}^n$ correspond to the points of its dual space.

"$\Rightarrow$", by contraposition: suppose S is not in linearly general position, so there are $r+2$ points lying on an r-dimensional linear subspace, with $r < n$. Their linear forms

$$L_1, \ldots, L_{r+2} \in \mathrm{Hom}(\mathbb{C}^{n+1}, \mathbb{C})$$

span a subspace of dimension at most $r+1$, and hence satisfy

$$a_1 L_1 + \cdots + a_{r+2} L_{r+2} = 0$$

with the a_i not all zero. The codimension of $Y = \bigcap_{i=1}^{r+2} H_i$ equals the dimension of their span, which is at most $r+1 \leq n$. Thus Y is non-empty. At a point $p \in Y$, dehomogenizing by a linear form nonzero at p gives dependent local equations. They cannot be part of a regular system of parameters in $\mathcal{O}_{\mathbb{P}^n,p}$, or of formal coordinates in $\widehat{\mathcal{O}}_{\mathbb{P}^n,p}$. Thus D is not snc.

"$\Leftarrow$", again by contraposition: suppose D is not snc at p. The equations $L_1, \ldots, L_k$ of the hyperplanes containing p are linearly dependent. Write $d = \dim \mathrm{Span}\{L_1, \ldots, L_k\} < k$. Since their common kernel contains a representative of p, we have $d \leq n$. The corresponding points lie in a projective space of dimension $r = d-1 < n$, and $k \geq d+1 = r+2$. Thus S is not in linearly general position. □

The normal crossing condition is very important because it singles out a class of divisors whose singularities we can understand completely.

Theorem 4.1.5 (Resolution of singularities)**.** *Let (X, D) be a pair where X is a complex algebraic variety, possibly singular, and $D \subseteq X$ is an effective Cartier divisor.*

(i) There is a projective birational morphism $\mu : X' \longrightarrow X$, with X' non-singular and with divisorial exceptional locus $\mathrm{Exc}(\mu)$*, such that the reduced divisor supported on*

$$\mathrm{Supp}(\mu^* D) \cup \mathrm{Exc}(\mu)$$

is snc.

(ii) We can construct X' via a sequence of blow-ups along smooth centers in the successive bad loci of the pair. In particular, μ can be chosen to be an isomorphism over

$$X \setminus (\mathrm{Sing}(X) \cup \mathrm{Sing}(D)).$$

We call X' a log resolution of D, or of the pair (X, D). Log resolutions also exist over arbitrary fields of characteristic zero by Hironaka's theorem. Notice that $\mu^* D$ itself may have multiplicities; the assertion is about its support together with the exceptional locus.

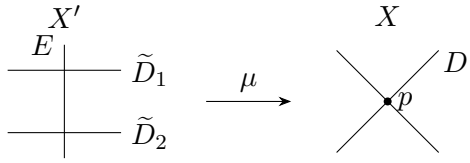


Figure 4.1: A schematic log resolution. The reduced total transform, including the exceptional divisors, is snc.

Corollary 4.1.6. *Assume X is smooth, and that $\mu : X' \longrightarrow X$ is a log resolution of (X, D) constructed by a sequence of blow-ups along smooth centers.*[68]

(i) For the canonical divisors of X and X', we have

$$\mu_* \mathcal{O}_{X'}(K_{X'}) = \mathcal{O}_X(K_X), \qquad R^i \mu_* \mathcal{O}_{X'}(K_{X'}) = 0 \quad (i > 0).$$

(ii) If X is projective with an ample divisor H, then for suitable integers $p \gg 0$ and $b_j > 0$,

$$\mu^*(pH) - \sum_j b_j E_j$$

[68] For blow-ups, cohomology of projective space, formal functions, faithful flatness of completion, and relative ampleness, see [Har77, II, 8.24], [Har77, III, 5.1], [Har77, III, 11.1], [Eis95, 7.2], [Har77, II, 7.10(b)]. The composition of higher direct images is governed by Theorem D.6.3, and degeneration on one axis by Proposition D.5.11.

is ample on X', where the sum is over the μ-exceptional prime divisors.

Proof. We factor the resolution as

$$X' = X_m \xrightarrow{\pi_m} X_{m-1} \longrightarrow \cdots \longrightarrow X_1 \xrightarrow{\pi_1} X_0 = X,$$

where π_k blows up a smooth center Z_{k-1} of codimension $c_k \geq 2$. Blow-ups of Cartier centers can be omitted. Treating connected components of the centers separately, we may take each exceptional divisor E_k to be prime. Write $\mu_k = \pi_1 \circ \cdots \circ \pi_k$.

(i) First consider a single blow-up. By [Har77, II, Exercise 8.5],

$$K_{X_1} = \pi_1^* K_X + (c_1 - 1)E_1.$$

The projection formula gives

$$R^i\pi_{1*}\mathcal{O}_{X_1}(K_{X_1}) \cong R^i\pi_{1*}\mathcal{O}_{X_1}((c_1-1)E_1) \otimes \mathcal{O}_X(K_X).$$

Thus put $\mathcal{F} = \mathcal{O}_{X_1}((c_1-1)E_1)$.

- For $i = 0$, algebraic Hartogs and $\operatorname{codim}_X Z_0 \geq 2$ give $\pi_{1*}\mathcal{F} = \mathcal{O}_X$.
- For $i > 0$, each fiber over Z_0 is $\mathbb{P}^{c_1-1}$ and

  $$\mathcal{F}|_{\pi_1^{-1}(x)} \cong \mathcal{O}_{\mathbb{P}^{c_1-1}}(-(c_1-1)).$$

  Its higher cohomology vanishes. However, π_1 is not flat, so we still need an argument with infinitesimal neighborhoods.

Work over an affine open $U \subseteq X$, and let I be the ideal of $Z_0 \cap U$. Since $I\mathcal{O}_{\pi_1^{-1}(U)} = \mathcal{O}(-E_1)$, the pullback of the thickened center $Z_k = V(I^k)$ is $F_k = kE_1$. Denote $\mathcal{F}_k = \mathcal{F}|_{F_k}$. Thus we have the Cartesian diagram

$$\begin{array}{ccc} F_k & \longrightarrow & \pi_1^{-1}(U) \\ \downarrow & & \downarrow \pi_1 \\ Z_k & \longrightarrow & U. \end{array}$$

These are thickenings of the exceptional divisor over the center. Because E_1 is Cartier, the actual short exact sequence is

$$0 \longrightarrow \mathcal{O}_{E_1}(-kE_1) \longrightarrow \mathcal{O}_{F_{k+1}} \longrightarrow \mathcal{O}_{F_k} \longrightarrow 0.$$

Writing $\mathcal{O}_{E_1}(1) = \mathcal{O}_{X_1}(-E_1)|_{E_1}$ and twisting by $\mathcal{F}$ gives

$$0 \longrightarrow \mathcal{O}_{E_1}(k - c_1 + 1) \longrightarrow \mathcal{F}_{k+1} \longrightarrow \mathcal{F}_k \longrightarrow 0.$$

The map $E_1 \to Z_0$ is a projective space bundle of relative dimension $c_1 - 1$. The projective space cohomology calculation gives

$$R^i(E_1 \to Z_0)_*\mathcal{O}_{E_1}(a) = 0 \quad \text{for } i > 0,\ a \geq -(c_1 - 1).$$

Since $Z_0 \cap U$ is affine, this also gives the corresponding vanishing of higher cohomology on $E_1 \cap \pi_1^{-1}(U)$. Starting with $\mathcal{F}_1 = \mathcal{O}_{E_1}(-(c_1 - 1))$, the long exact sequence therefore proves, by induction, that

$$H^i(F_k, \mathcal{F}_k) = 0 \qquad (i > 0,\ k \geq 1).$$

The theorem of formal functions, applied to the ideal I, now gives

$$H^0(\widehat{U, R^i\pi_{1*}\mathcal{F}})^{I} \cong \varprojlim_k H^i(F_k, \mathcal{F}_k) = 0.$$

The sheaf $R^i\pi_{1*}\mathcal{F}$ is coherent and supported on Z_0, because π_1 is an isomorphism outside Z_0. Its vanishing completion implies its vanishing: locally at a point of Z_0 this follows from Nakayama's lemma, or from faithful flatness of completion. Hence $R^i\pi_{1*}\mathcal{F} = 0$ for $i > 0$.

Now suppose the assertion holds for k blow-ups. The Grothendieck spectral sequence for $\mu_{k+1} = \mu_k \circ \pi_{k+1}$ is

$$E_2^{p,q} = R^p\mu_{k*}(R^q\pi_{k+1*}\mathcal{O}_{X_{k+1}}(K_{X_{k+1}})) \Longrightarrow R^{p+q}\mu_{k+1*}\mathcal{O}_{X_{k+1}}(K_{X_{k+1}}).$$

The single blow-up case shows that it collapses to the p-axis, so

$$R^n\mu_{k+1*}\mathcal{O}_{X_{k+1}}(K_{X_{k+1}}) \cong R^n\mu_{k*}\mathcal{O}_{X_k}(K_{X_k}).$$

The right side is $\mathcal{O}_X(K_X)$ when $n = 0$ and is zero when $n > 0$.

(ii) We again use induction. For a single blow-up, $\mathcal{O}_{X_1}(-E_1) = \mathcal{O}_{X_1}(1)$ is π_1-*very ample*. Tensoring it with the pullback of a sufficiently high power of an ample line bundle on X makes it very ample on X_1. Thus

$$A_1 = \pi_1^*(p_1 H) - E_1$$

is very ample for some $p_1 > 0$. Suppose that on X_k we have

$$A_k = \mu_k^*(p_k H) - \sum_{i=1}^{k} c_i^{(k)} E_i^{(k)}, \qquad c_i^{(k)} > 0,$$

where $E_i^{(k)}$ is the strict transform of E_i. For a suitable $p'_{k+1} > 0$,

$$A_{k+1} = \pi_{k+1}^*(p'_{k+1} A_k) - E_{k+1}$$

is very ample. Since

$$\pi_{k+1}^* E_i^{(k)} = E_i^{(k+1)} + m_{k+1,i} E_{k+1}, \qquad m_{k+1,i} \geq 0,$$

we obtain

$$\begin{aligned} A_{k+1} &= \mu_{k+1}^*(p'_{k+1}p_kH) - \sum_{i=1}^{k} p'_{k+1}c_i^{(k)}E_i^{(k+1)} \\ &\quad - \left(1 + \sum_{i=1}^{k} p'_{k+1}c_i^{(k)}m_{k+1,i}\right) E_{k+1} \\ &= \mu_{k+1}^*(p_{k+1}H) - \sum_{i=1}^{k+1} c_i^{(k+1)}E_i^{(k+1)}. \end{aligned}$$

All these coefficients are positive, as required. □

Remark 4.1.7 (Tangent description of simple normal crossings)**.** The coordinate definition says that the smooth components intersect transversally everywhere. We can also describe this by a regular system of parameters, i.e. minimal generators of $\mathfrak{m}_p$ whose classes form a basis of the Zariski cotangent space $\mathfrak{m}_p/\mathfrak{m}_p^2$.

At a closed point p, if the components through p have local equations $f_1, \ldots, f_k$, the condition is that these extend to parameters $f_1, \ldots, f_n$ of $\mathcal{O}_{X,p}$. Then

$$D: \ f_1 \cdots f_k = 0, \qquad \mathfrak{m}_p/\mathfrak{m}_p^2 = \mathrm{Span}_{\mathbb{C}}\{\bar{f}_1, \ldots, \bar{f}_n\}.$$

For $R = \mathcal{O}_{X,p}$, the natural isomorphism $\mathrm{gr}_{\mathfrak{m}} R \cong \mathrm{gr}_{\widehat{\mathfrak{m}}} \widehat{R}$ identifies the initial classes. More precisely, the chosen parameters give the isomorphism

$$\mathbb{C}[[z_1, \ldots, z_n]] \xrightarrow{\sim} \widehat{R}, \qquad z_i \longmapsto f_i.$$

Analytically, their independent differentials give local coordinates by the inverse function theorem. Thus these descriptions agree.

4.1B Covering Trick

We introduce an exquisite method to extract "roots" of divisors or line bundles.

Cyclic coverings.

We begin with the local description of the m-fold cyclic covering branched along a given divisor. Suppose X is affine, with $H^0(X, \mathcal{O}_X) = \mathbb{C}[X]$, and $s \in \mathbb{C}[X]$ is nonzero. We wish to construct a scheme Y on which the m-th root of s makes sense. In $X \times \mathbb{A}^1$, with coordinate t on the second factor, set

$$Y = V(t^m - s), \qquad \mathbb{C}[Y] = \mathbb{C}[X][t]/(t^m - s).$$

Projection gives a finite flat morphism $\pi : Y \to X$ of degree m, branched over $D = \mathrm{div}(s)$ when $m > 1$. The function $s' = t|_Y$ satisfies

$$(s')^m = \pi^* s.$$

Thus we have extracted the desired root. The scheme Y need not be irreducible. This local construction can be globalized as follows.

Proposition 4.1.8 (Cyclic coverings)**.** *Let X be a variety, and L a line bundle on X. Suppose a positive integer $m \geq 1$ and a nonzero section $s \in H^0(X, L^{\otimes m})$ are given, defining an effective Cartier divisor $D = \operatorname{div}(s)$. Then there is a finite flat covering*[69]

$$\pi : Y \longrightarrow X$$

of degree m, where Y is a scheme, such that $L' = \pi^* L$ carries a section

$$s' \in H^0(Y, L'), \qquad (s')^m = \pi^* s.$$

The Cartier divisor $D' = \operatorname{div}(s')$ maps isomorphically to D. Moreover, if X and D are non-singular, then so are Y and D'.

As above, we can regard s' as the m-th root of s.

Proof. Locally we trivialize L on affine open sets and carry out the preceding construction. To see directly that these local constructions glue, consider the geometric line bundle

$$\mathbf{L} = \mathbf{V}(L^\vee) = \operatorname{Spec}_X \operatorname{Sym}(L^\vee), \qquad p : \mathbf{L} \longrightarrow X.$$

Since $p_* \mathcal{O}_{\mathbf{L}} = \operatorname{Sym}(L^\vee)$, a section of $p^* L$ corresponds to a module homomorphism

$$\begin{aligned} H^0(\mathbf{L}, p^* L) &= \operatorname{Hom}_{\mathcal{O}_{\mathbf{L}}}(\mathcal{O}_{\mathbf{L}}, p^* L) \\ &\cong \operatorname{Hom}_{p_* \mathcal{O}_{\mathbf{L}}}(p_* \mathcal{O}_{\mathbf{L}}, p_* p^* L) \\ &\cong \operatorname{Hom}_{\operatorname{Sym}(L^\vee)}(\operatorname{Sym}(L^\vee), L \otimes \operatorname{Sym}(L^\vee)). \end{aligned}$$

Here we use the equivalence for an affine morphism and the projection formula. In rank one, symmetric powers coincide with tensor powers, and

$$\begin{aligned} L \otimes \operatorname{Sym}(L^\vee) &= \bigoplus_{n \geq 0} L \otimes (L^\vee)^{\otimes n} \\ &\cong L \oplus \mathcal{O}_X \oplus L^\vee \oplus (L^\vee)^{\otimes 2} \oplus \cdots . \end{aligned}$$

The inclusion of $\operatorname{Sym}(L^\vee)$ into the second and subsequent summands, with this degree shift, is a $\operatorname{Sym}(L^\vee)$-module map. It defines the tautological section

$$T \in H^0(\mathbf{L}, p^* L).$$

Take $Y \subseteq \mathbf{L}$ to be the zero scheme of

$$T^m - p^* s \in H^0(\mathbf{L}, p^* L^{\otimes m}),$$

and put $s' = T|_Y$. For $\pi = p|_Y$, we immediately have $(s')^m = \pi^* s$.

[69] For relative Spec and the correspondence of modules under an affine morphism, see [Har77, II, Exercise 5.17(c)], [Har77, II, Exercise 5.17(e)].

Let $U = \operatorname{Spec} A$ trivialize L, with $H^0(U, L) = Ae$, and let t be the dual basis. Write $s|_U = fe^{\otimes m}$. Then

$$\mathbf{L}|_U = \operatorname{Spec} A[t] = U \times \mathbb{A}^1, \qquad T = t\, p^* e,$$

so Y has equation $t^m - f = 0$ over U. The divisors D and D' have equations $f = 0$ and $t = 0$, respectively, and

$$(A[t]/(t^m - f))/(t) \cong A/(f).$$

Also, t is a nonzerodivisor in $A[t]/(t^m - f)$: if $tb = 0$, then $fb = t^m b = 0$, and f is a nonzerodivisor on this free A-module. Thus D' is Cartier and maps isomorphically to D. Finally,

$$A[t]/(t^m - f) = A \oplus At \oplus \cdots \oplus At^{m-1}$$

is free of rank m, proving finiteness, flatness and surjectivity.

For smoothness, if $m = 1$ the cover is the graph of f. If $m > 1$, the differential of $F = t^m - f$ is

$$dF = mt^{m-1}\, dt - df.$$

At a point away from D', the coefficient of dt is nonzero. At a point of D', smoothness of D gives $df \neq 0$. Since the dt direction is independent of the directions on X, dF never vanishes on Y. Thus Y is non-singular, and so is $D' \cong D$. □

Remark 4.1.9 (Geometrical grasp)**.** Restrict to $U = \operatorname{Spec} A$, where $H^0(U, L) = Ae$. Then $\mathbf{L}|_U = \operatorname{Spec} A[t]$, where t is dual to e, and by the definition of pullback,

$$H^0(\mathbf{L}|_U, p^* L) = Ae \otimes_A A[t].$$

This is the restriction of $L \otimes \operatorname{Sym}(L^\vee)$ after sheafification. The tautological section comes from the module map

$$\varphi : A[t] \longrightarrow Ae \otimes_A A[t], \qquad 1 \longmapsto e \otimes t.$$

The pullback of the local section e is

$$e \longmapsto p^* e = e \otimes 1,$$

so $T = t\, p^* e$. For a closed point $a \in \mathbf{L}|_U$ with $x = p(a)$, we have

$$\begin{aligned} \mathbf{L}_x &= \mathbf{L} \times_X \operatorname{Spec} k(x) = \operatorname{Spec}(A[t] \otimes_A \mathbb{C}) = \mathbb{A}^1_{\mathbb{C}} \\ &= \operatorname{Spec} \operatorname{Sym}_{\mathbb{C}}\big((Ae \otimes_A \mathbb{C})^\vee\big). \end{aligned}$$

From the functorial point of view,

$$\begin{aligned} \mathbf{L}_x(\mathbb{C}) &= \operatorname{Hom}_{\mathbb{C}\text{-schemes}}(\operatorname{Spec} \mathbb{C}, \mathbf{L}_x) \\ &= \operatorname{Hom}_{\mathbb{C}\text{-algebras}}\big(\operatorname{Sym}_{\mathbb{C}}((Ae \otimes_A \mathbb{C})^\vee), \mathbb{C}\big) \\ &\cong \operatorname{Hom}_{\mathbb{C}}\big((Ae \otimes_A \mathbb{C})^\vee, \mathbb{C}\big) \\ &\cong Ae \otimes_A \mathbb{C} = \mathbb{C}e. \end{aligned}$$

Thus we can write $a = \lambda e$, with $\lambda \in \mathbb{C}$, and

$$T(a) = (t\, p^* e)(a) = t(a)(e \otimes 1) = \lambda e = a,$$

under the identification $(p^*L)_a = L_x$. This is why it is called the "tautological" section: at a point of the geometric line bundle it returns that vector itself.

Remark 4.1.10. From the construction,

$$p_* \mathcal{O}_{\mathbf{L}} = \mathrm{Sym}(L^\vee) = \mathcal{O}_X \oplus L^{-1} \oplus L^{-2} \oplus \cdots .$$

Locally this is $A[t] = A \oplus At \oplus At^2 \oplus \cdots$. Passing to the quotient defining Y, the relation $t^m = f$ gives

$$A[t]/(t^m - f) = A \oplus At \oplus \cdots \oplus At^{m-1}.$$

These descriptions glue to the canonical isomorphism of $\mathcal{O}_X$-modules

$$\pi_* \mathcal{O}_Y = \mathcal{O}_X \oplus L^{-1} \oplus L^{-2} \oplus \cdots \oplus L^{1-m}.$$

The multiplication is determined by s. Moreover, $D' \sim_{\mathrm{lin}} \pi^* L$ because D' is the divisor of the section s' of $\pi^* L$.

Remark 4.1.11. This cyclic covering construction is all that is needed for the classical Kodaira vanishing theorem in the next section.

Proposition 4.1.12 (An snc divisor behaves well)**.** *In the above situation, suppose X and D are smooth, and $D + \sum_{i=2}^r D_i$ is a reduced snc divisor on X. Put $D_i' = \pi^* D_i$ for $i \geq 2$. Then*

$$mD' + \sum_{i=2}^{r} D_i'$$

has simple normal crossing support on Y; each D_i' is smooth and reduced.

Proof. We use the tangent description in Remark 4.1.7. Let $q \in Y$, and put $p = \pi(q)$. If $p \notin D$, the covering is *étale* near q, so the snc condition is preserved.

If $p \in D$, choose regular parameters $f, f_2, \ldots, f_n$ at p, where D has equation $f = 0$ and the other components through p have equations among the f_i. Then

$$\mathcal{O}_{Y,q} = \mathcal{O}_{X,p}[t]/(t^m - f).$$

To justify that this ring is already local, note that it is finite over the local ring $\mathcal{O}_{X,p}$, so every maximal ideal contracts to $\mathfrak{m}_p$. Its quotient by $\mathfrak{m}_p$ is $\mathbb{C}[t]/(t^m)$, which has only the maximal ideal (t). Thus its unique maximal ideal is

$$\mathfrak{m}_q = (t, f, f_2, \ldots, f_n) = (t, f_2, \ldots, f_n).$$

These n elements form a regular system of parameters, since Y is smooth of dimension n. The relevant divisors are $t = 0$ and some of the $f_i = 0$, so their reduced union is snc. □

Bloch–Gieseker and Kawamata coverings.

Lemma 4.1.13. *On an arbitrary quasi-projective variety X, every line bundle L can be written*[70]

$$L = A \otimes B^{\vee},$$

where A and B are very ample line bundles.

Proof. Choose an immersion $X \hookrightarrow Z \hookrightarrow \mathbb{P}^N$, with X open in the projective variety Z, and let $H = \mathcal{O}_Z(1)|_X$. This is very ample and ample. Indeed, every coherent sheaf F on X extends to a coherent sheaf F' on Z, and Serre's theorem makes $F' \otimes \mathcal{O}_Z(n)$ globally generated for $n \gg 0$. Its restriction $F \otimes H^n$ is therefore globally generated.

Choose n_1 such that $L \otimes H^{n_1}$ is globally generated. Its tensor product with the very ample line bundle H is very ample. Hence, for a suitable $n > 0$, we can take

$$A = L \otimes H^n, \qquad B = H^n.$$

□

Theorem 4.1.14 (BG coverings)**.** *Let X be a quasi-projective variety of dimension n, let L be a line bundle on X, and fix a positive integer m. There exist a variety Y, a finite flat surjective morphism $f : Y \to X$, and a line bundle N on Y such that $f^*L \cong N^{\otimes m}$.*[71]

If X is non-singular, Y can be chosen non-singular. In this case, given an snc divisor $D = \sum D_i$ on X, we can arrange that each f^*D_i is smooth and f^*D is reduced and snc. Finally, if $\dim X \geq 2$, each f^*D_i can also be chosen irreducible.

Proof. **Step 1:** We first make some reductions. If $\dim X = 0$, then $X = \operatorname{Spec} \mathbb{C}$ and L is trivial, so the identity suffices. Assume $n \geq 1$. Suppose the result is known for the pullback of the hyperplane bundle by a morphism $X \to \mathbb{P}^r$ with finite fibers. By the preceding lemma, write $L = A \otimes B^{\vee}$ for two very ample line bundles. First take a covering $f_1 : Y_1 \to X$ with $f_1^*A = N_A^{\otimes m}$. Apply the special case again to f_1^*B: its map to projective space still has finite fibers. This gives $f_2 : Y \to Y_1$ with $f_2^* f_1^* B = N_B^{\otimes m}$. Then, for $f = f_1 \circ f_2$,

$$\begin{aligned} f^*L &= f_2^*(f_1^*A) \otimes f_2^*(f_1^*B^{\vee}) \\ &\cong f_2^* N_A^{\otimes m} \otimes N_B^{-m} = \left(f_2^* N_A \otimes N_B^{\vee}\right)^{\otimes m}. \end{aligned}$$

[70] For Serre's global-generation theorem, extension of coherent sheaves, and tensoring with a globally generated line bundle, see [Har77, II, 5.17], [Har77, II, Exercise 5.15], [Har77, II, Exercise 7.5(d)].

[71] For general translates, generic smoothness, and smooth fibres over the ground field, see [Laz04a, 3.3.10], [Har77, III, 10.7], [Har77, III, 10.0.3]. For the smooth action map and the regular-parameter statements used in the transversality argument, see [Har77, III, 10.8], [Mat86, 15.1], [Mat86, 14.2].

The extra smoothness and snc requirements can be retained at both stages.

Step 2: Thus assume $L = \varphi^*\mathcal{O}_{\mathbb{P}^r}(1)$ for a finite-to-one morphism $\varphi : X \to \mathbb{P}^r$. Fix the Fermat covering

$$\nu : \mathbb{P}^r \longrightarrow \mathbb{P}^r, \qquad [x_0 : \cdots : x_r] \longmapsto [x_0^m : \cdots : x_r^m].$$

It satisfies $\nu^*\mathcal{O}(1) = \mathcal{O}(m)$. On standard affine charts its algebra is free with basis the monomials whose exponents are between 0 and $m-1$, so ν is finite flat of degree m^r.

For $g \in G = \mathrm{GL}(r+1, \mathbb{C})$, let $\nu_g = g \circ \nu$ and set

$$Y_g = X \times_{\mathbb{P}^r} \mathbb{P}^r, \qquad \begin{array}{ccc} Y_g & \xrightarrow{\bar\varphi} & \mathbb{P}^r \\ f_g \downarrow & & \downarrow \nu_g \\ X & \xrightarrow{\varphi} & \mathbb{P}^r. \end{array}$$

Here the fiber product uses φ and ν_g. The morphism f_g is finite flat and surjective by base change, and

$$f_g^* L = \bar\varphi^* \nu_g^* \mathcal{O}(1) = \bar\varphi^* \mathcal{O}(m) = \big(\bar\varphi^* \mathcal{O}(1)\big)^{\otimes m}.$$

We show that a general g gives the remaining properties. The space Y_g is the inverse image of the graph of g^{-1} under

$$\varphi \times \nu : X \times \mathbb{P}^r \longrightarrow \mathbb{P}^r \times \mathbb{P}^r.$$

The image has dimension $n + r > r$, since φ has finite fibers. The generic irreducibility result [Laz04a, Example 3.3.10] therefore shows that Y_g is irreducible for general g.

For reducedness, choose g so that none of the translated coordinate hyperplanes contains $\varphi(X)$. Over a dense open subset of X, the covering is then *étale*, since the Fermat covering is *étale* off these hyperplanes. This gives a reduced generic fiber. Finite flatness over the integral variety X then gives reducedness everywhere by the following lemma. □

Lemma 4.1.15. *Let $f : V \to W$ be a finite flat morphism of schemes of finite type over $\mathbb{C}$, with W integral and V generically reduced. Then V is reduced.*

Proof. Locally write $V = \operatorname{Spec} B$ and $W = \operatorname{Spec} A$, where A is a domain and B is finite flat over A. Flatness implies that every irreducible component of V dominates W, and that

$$B \longrightarrow B \otimes_A \operatorname{Frac}(A)$$

is injective. The generic fiber is a zero-dimensional Noetherian scheme; its local rings are the reduced local rings at the generic points of V. Hence it is reduced. A nilpotent element of B maps to zero in this generic fiber and is therefore zero. Equivalently, after some localization B_δ is reduced, and $\delta^a b = 0$ implies $b = 0$ by flatness. □

Continuation of the proof of Theorem 4.1.14. The same argument, applied to a reduced irreducible Cartier divisor $E \subseteq X$, makes $f_g^* E$ reduced and irreducible if $\dim X \geq 2$, so that $\dim E \geq 1$.

Step 3: Suppose X is non-singular. Consider

$$\begin{array}{ccccc} W = X \times_{\mathbb{P}^r} (G \times \mathbb{P}^r) & \longrightarrow & G \times \mathbb{P}^r & \xrightarrow{\mathrm{pr}_1} & G \\ n \downarrow & & \downarrow m & & \\ X & \xrightarrow{\varphi} & \mathbb{P}^r, & & \end{array}$$

where $m(g, y) = g \cdot \nu(y)$ and $b : W \to G$ denotes the top composite. Its fiber over g is Y_g. The transitive group action makes m smooth, and hence n is smooth by base change. Since X is smooth, so is W. Generic smoothness applied to b makes Y_g smooth for general g. Applying the same argument to a smooth divisor $E \subseteq X$ makes $f_g^* E$ smooth as well.

Step 4: Let $D = \sum D_i$ be snc. The subvarieties

$$F_i = n^{-1}(D_i) = D_i \times_{\mathbb{P}^r} (G \times \mathbb{P}^r) \subseteq W$$

are smooth and meet transversally. Indeed, local equations of the D_i form part of a regular system of parameters on X. Their pullbacks under the smooth map n remain part of a regular system of parameters on W: a smooth local map is flat with regular fibers, and one adds parameters along the fiber to those from the base.

The next lemma shows that, for general g, the fibers $(F_i)_g = f_g^* D_i$ are smooth and meet transversally in Y_g. Thus $f_g^* D$ is snc. All the conditions imposed on g hold on finitely many dense open subsets, so they can be imposed simultaneously. □

Lemma 4.1.16. *Let $b : W \to G$ be a morphism of non-singular varieties, and let $V_i \subseteq W$ be a finite collection of smooth subvarieties meeting transversally. For general $g \in G$, the fibers $(V_i)_g$ are smooth and meet transversally, or not at all, in W_g.*

Proof. Apply generic smoothness to W and to all nonempty intersections $V_I = \bigcap_{i \in I} V_i$. Remove the closures of the images of components that do not dominate G. There are only finitely many such strata, so one dense open subset of G works for all of them. Over it, each nonempty $(V_I)_g$ is smooth, with codimension in W_g equal to the codimension of V_I in W. These codimensions are the sums of those of the V_i, since the intersections in W are transverse. The same is therefore true in W_g, which gives transversality there. □

By combining cyclic and BG coverings, we can pass to a covering on which each component of an snc divisor becomes divisible by a prescribed positive integer.

Proposition 4.1.17 (Kawamata coverings)**.** *Let X be a non-singular quasi-projective variety, let $D = \sum_{i=1}^{r} D_i$ be snc, and fix positive integers $m_1, \dots, m_r$. There exist a smooth variety Y and a finite flat covering $h : Y \to X$ such that*

$$h^* D_i = m_i D_i',$$

where each D_i' is smooth and $D' = \sum D_i'$ is snc.

Proof. By induction, it suffices to construct a cover with

$$h^* D_1 = m_1 D_1', \qquad h^* D_i = D_i' \quad (2 \le i \le r),$$

and $\sum D_i'$ snc. First take a BG covering $f : X_1 \to X$ such that

$$D_1^+ := f^* D_1 \sim_{\text{lin}} m_1 A_1^+,$$

for some divisor A_1^+, while f^*D is snc and D_1^+ is smooth. Then take the m_1-fold cyclic covering $\pi : Y \to X_1$ branched along D_1^+. We have

$$\pi^* D_1^+ = m_1 D_1', \qquad D_1' \cong D_1^+.$$

Set $h = f \circ \pi$. For $i \ge 2$, put $D_i' = h^* D_i$. Proposition 4.1.12 says that $\sum D_i'$ is snc. The cyclic cover is integral: for $m_1 > 1$, its equation is Eisenstein at the generic point of any component of the nonempty reduced divisor D_1^+; its finite flat algebra embeds into the resulting function field. Thus Y is a smooth variety.

Replace X by Y and proceed with D_2', and so on. At every stage the previous root divisors have smooth reduced pullbacks. A smooth divisor may be disconnected, but its components are disjoint; the same construction applies to it. The multiplicities already extracted are therefore retained. □

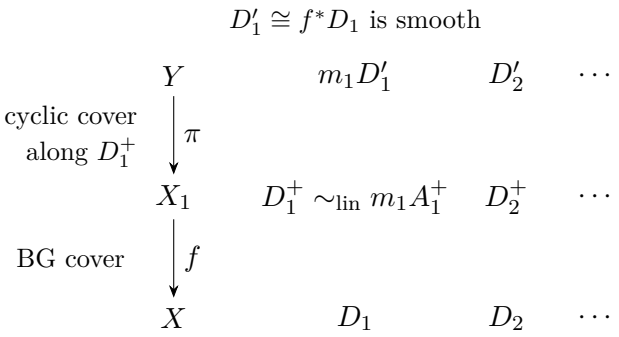


Figure 4.2: Inductive construction. Then replace X by Y and proceed with D_2'.

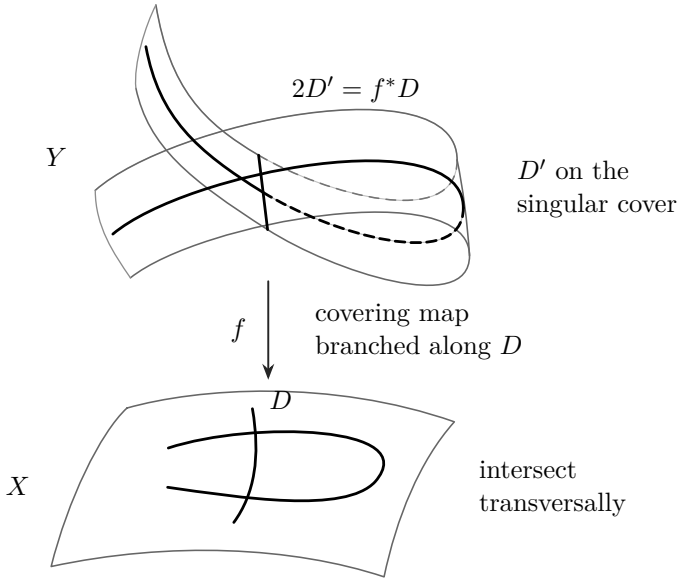


Figure 4.3: A schematic double cover branched along two crossing components. The local equation $t^2 = xy$ is singular at the origin. This differs from taking roots successively along smooth divisors as in Figure 4.2.

Injectivity.

We now use these covering constructions to establish vanishing theorems. Starting from a line bundle L on X, the general strategy is to pull it back to a covering $f : Y \to X$ on which the problem simplifies. The next lemma shows that vanishing on Y is enough.

Lemma 4.1.18. *Let $f : Y \to X$ be a finite surjective morphism of complex algebraic varieties, with X normal, and let E be a vector bundle on X. Then the natural map*[72]

$$H^i(X, E) \longrightarrow H^i(Y, f^*E)$$

is injective. In particular, $H^i(Y, f^*E) = 0$ implies $H^i(X, E) = 0$.

Proof. Finiteness and the projection formula give

$$H^i(Y, f^*E) \cong H^i(X, f_*f^*E) \cong H^i(X, E \otimes f_*\mathcal{O}_Y).$$

On the other hand, regard $\mathbb{C}(Y)$ as a finite-dimensional $\mathbb{C}(X)$-vector space. For $b \in \mathbb{C}(Y)$, multiplication

$$\mathbb{C}(Y) \longrightarrow \mathbb{C}(Y), \qquad x \longmapsto bx$$

[72]For comparison of cohomology under the finite morphism, see [Har77, III, Exercise 8.2].

is a linear transformation. Its matrix trace defines $\mathrm{Tr}(b) \in \mathbb{C}(X)$. Locally, f is $\operatorname{Spec} B \to \operatorname{Spec} A$ with B finite, hence integral, over the normal domain A. For $b \in B$, its trace is integral over A and lies in $\mathrm{Frac}(A)$, so it belongs to A. These local maps glue to

$$\mathrm{Tr}_{Y/X} : f_*\mathcal{O}_Y \longrightarrow \mathcal{O}_X.$$

Write $d = [\mathbb{C}(Y) : \mathbb{C}(X)]$. The composite of the natural inclusion with the trace is multiplication by d. Thus the normalized trace

$$\frac{1}{d}\,\mathrm{Tr}_{Y/X} : f_*\mathcal{O}_Y \longrightarrow \mathcal{O}_X$$

splits the natural inclusion $\mathcal{O}_X \hookrightarrow f_*\mathcal{O}_Y$. Locally this splits the exact sequence

$$0 \longrightarrow A \longrightarrow B \longrightarrow \mathrm{coker} \longrightarrow 0.$$

Tensoring the splitting with E gives a splitting of $E \hookrightarrow E \otimes f_*\mathcal{O}_Y$. Hence $H^i(X, E)$ embeds as a direct summand of $H^i(X, E \otimes f_*\mathcal{O}_Y) \cong H^i(Y, f^*E)$, as required. □

4.2 Kodaira and Nakano Vanishing Theorems

In this section we prove the classical vanishing theorems of Kodaira and Nakano. We first connect singular cohomology with the cohomology of the constant sheaf. On a locally contractible paracompact space, such as a smooth manifold, the comparison theorem gives

$$H^i_{\mathrm{sing}}(X, G) \cong H^i(X, \underline{G}),$$

where G is an abelian group (in particular, the additive group of a commutative ring). For $G = \mathbb{C}$, the de Rham theorem further identifies these groups with the cohomology of the complex of smooth differential forms. Here and below, topological statements concern the analytification of X.

We also need the Lefschetz hyperplane theorem; see [Laz04a, 3.1.17], [Voi03, 1.23], or [GH78, p. 156].

Theorem 4.2.1 (Lefschetz hyperplane theorem)**.** *Let X be a non-singular complex projective variety of dimension n, and let D be an effective ample divisor on X. The restriction*[73]

$$r_i : H^i_{\mathrm{sing}}(X, \mathbb{Z}) \longrightarrow H^i_{\mathrm{sing}}(D, \mathbb{Z})$$

is an isomorphism for $i \leq n-2$ and is injective for $i = n-1$.

After changing coefficients to $\mathbb{C}$ and using the comparison above, we can use Hodge decomposition and the Dolbeault theorem; see [GH78] or [Voi02].

[73] For the comparison between singular and sheaf cohomology, see [Voi02, 4.47], [GH78, p. 44].

Theorem 4.2.2 (Hodge decomposition)**.** *For a compact Kähler manifold X, the cohomology of the constant complex sheaf satisfies*

$$H^i(X,\mathbb{C}) = \bigoplus_{p+q=i} H^{p,q}(X), \qquad H^{p,q}(X) = \overline{H^{q,p}(X)}.$$

Here $H^{p,q}(X) \subseteq H^{p+q}(X,\mathbb{C})$ consists of classes represented by closed forms of type (p,q). Moreover, the Kähler identity $\Delta_d = 2\Delta_{\bar\partial}$ gives

$$H^{p,q}(X) \cong H^{p,q}_{\bar\partial}(X).$$

The Fubini–Study metric makes every smooth complex projective variety a compact Kähler manifold.

Theorem 4.2.3 (Dolbeault theorem)**.** *For a complex manifold X, we have*

$$H^q(X,\Omega^p_X) \cong H^{p,q}_{\bar\partial}(X),$$

where Ω^p_X is the sheaf of holomorphic p-forms.

For a projective algebraic variety, GAGA identifies these coherent analytic cohomology groups with their algebraic counterparts. We use this identification without changing notation.

4.2A Kodaira Vanishing Theorem

Lemma 4.2.4. *Let X be a smooth complex projective variety of dimension n, and let $D \subseteq X$ be a smooth effective ample divisor. The restriction maps*

$$r_{p,q} : H^q(X,\Omega^p_X) \longrightarrow H^q(D,\Omega^p_D)$$

are bijective for $p+q \le n-2$ and injective for $p+q = n-1$.

Proof. Combine the preceding three theorems. Restriction preserves the Hodge decomposition, so the isomorphism or injection on total cohomology restricts to an isomorphism or injection on each (p,q) summand. □

We shall now turn to the central topic.

Theorem 4.2.5 (Kodaira vanishing)**.** *Let X be a smooth complex projective variety of dimension n, and let A be an ample divisor on X. Then*

$$H^i(X,\mathcal{O}_X(K_X + A)) = 0 \qquad (i > 0).$$

Equivalently, by Serre duality,

$$H^j(X,\mathcal{O}_X(-A)) = 0 \qquad (j < n).$$

Proof. The case $n = 0$ is immediate. Since A is ample, Bertini's theorem gives a smooth divisor $D \in |mA|$ for $m \gg 0$. Let $\pi : Y \to X$ be the cyclic covering in Proposition 4.1.8, and put $A' = \pi^* A$. The tautological section gives a smooth divisor $D' \in |A'|$ with $\pi^* D = mD'$. It is ample because π is finite. By Lemma 4.1.18,

$$H^j(X, \mathcal{O}_X(-A)) \hookrightarrow H^j(Y, \pi^* \mathcal{O}_X(-A)) = H^j(Y, \mathcal{O}_Y(-D')).$$

Thus it suffices to prove vanishing for the negative of a smooth effective ample divisor.

Now let $D' \subseteq Y$ be such a divisor. Lemma 4.2.4 with $p = 0$ says that

$$H^j(Y, \mathcal{O}_Y) \longrightarrow H^j(D', \mathcal{O}_{D'})$$

is bijective for $j \leq n-2$ and injective for $j = n-1$. The exact sequence

$$0 \longrightarrow \mathcal{O}_Y(-D') \longrightarrow \mathcal{O}_Y \longrightarrow \mathcal{O}_{D'} \longrightarrow 0$$

therefore gives $H^j(Y, \mathcal{O}_Y(-D')) = 0$ for $j \leq n-1$: the map in degree $j-1$ is surjective, and the map in degree j is injective. For $j = 0$, the negative-degree group is zero. This proves the theorem. □

4.2B Differential Forms with Logarithmic Poles

The preceding proof uses only a small part of Lemma 4.2.4. The remaining instances lead to Nakano's generalization of Kodaira vanishing. Before giving the proof, we investigate differential forms with logarithmic poles; see [EV92].

For an snc divisor $D = \sum_{i=1}^{r} D_i$ on a non-singular n-fold X, let $\tau : U = X \setminus D \hookrightarrow X$ and write

$$\Omega^p_X(*D) = \varinjlim_{\nu} \Omega^p_X(\nu D) = \tau_* \Omega^p_U.$$

These forms may have poles of arbitrary order along D. Of course, $(\Omega^\bullet_X(*D), d)$ is a cochain complex.

Definition 4.2.6. The sheaf $\Omega^p_X(\log D)$ of differential forms with logarithmic poles along D is the subsheaf of $\Omega^p_X(*D)$ whose sections α satisfy the condition that both α and $d\alpha$ have at most simple poles along D.

Proposition 4.2.7 (Properties of differential log-forms)**.** *(a)* $(\Omega^\bullet_X(\log D), d)$ *is a subcomplex of* $(\Omega^\bullet_X(*D), d)$.

(b) $\Omega^p_X(\log D) = \bigwedge^p \Omega^1_X(\log D)$.

(c) $\Omega^p_X(\log D)$ *is locally free. More precisely, at a closed point* x *through which exactly* $D_1, \ldots, D_s$ *pass, choose parameters* $f_1, \ldots, f_n$ *with* $D_i = (f_i = 0)$ *for* $i \leq s$. *Set*

$$\delta_i = \begin{cases} df_i/f_i & i \leq s, \\ df_i & i > s, \end{cases} \qquad \delta_I = \delta_{i_1} \wedge \cdots \wedge \delta_{i_p}$$

for $I = \{i_1 < \cdots < i_p\}$. Then the δ_I, for $|I| = p$, form a free system of local generators.

In particular, $\Omega^0_X(\log D) = \mathcal{O}_X$.

Proof. (a) follows from $d^2 = 0$, and (b) follows from the generators in (c).

For (c), each δ_i is closed, since

$$d(df_i/f_i) = d(f_i^{-1}) \wedge df_i + f_i^{-1} d(df_i) = 0.$$

Hence the forms δ_I and the differentials of their regular linear combinations have at most simple poles. If Ω denotes their $\mathcal{O}_X$-span, then $\Omega \subseteq \Omega^p_X(\log D)$. The generators are independent because they are nonzero scalar multiples of the usual basis df_I over the function field.

First consider a single component, with equation $f_1 = 0$. Write

$$\alpha = \alpha_1 + \alpha_2 \wedge \frac{df_1}{f_1},$$

where α_1 and α_2 involve only wedge products of $df_2, \ldots, df_n$. If α and $d\alpha$ have at most simple poles, then

$$f_1\alpha = f_1\alpha_1 + \alpha_2 \wedge df_1, \qquad f_1 d\alpha = f_1 d\alpha_1 + d\alpha_2 \wedge df_1$$

are regular. Comparing the terms with and without df_1, we see that α_2 and $f_1\alpha_1$ are regular. Also,

$$d(f_1\alpha_1) = df_1 \wedge \alpha_1 + f_1 d\alpha - d\alpha_2 \wedge df_1.$$

Thus $df_1 \wedge \alpha_1$ is regular, and, since α_1 does not involve df_1, α_1 is regular. This proves the assertion for one component.

For several components, expand $\alpha = \sum h_I \delta_I$ with rational coefficients. At the generic point of each D_i, the other local equations are units, so the one-component argument shows that every h_I is regular there. Away from D they are already regular. A regular local ring is normal, and a rational function regular at all its height-one primes is regular. Thus every h_I is regular, proving $\Omega = \Omega^p_X(\log D)$. □

Each D_i may also denote a smooth divisor with several disjoint connected components, since locally it is still cut out by one parameter. Thus components carrying the same coefficient can be grouped together when their union is smooth.

Using the notation above, the residue map for one-forms is

$$\mathrm{Res}_D : \Omega^1_X(\log D) \longrightarrow \bigoplus_{i=1}^{r} \mathcal{O}_{D_i}, \qquad \sum_{i=1}^{n} a_i \delta_i \longmapsto (a_i|_{D_i})_{1 \le i \le s}$$

near a point on exactly $D_1, \dots, D_s$. The other summands have zero stalk there. In this display Res_D denotes the usual total residue map. For $p \geq 1$, there is also

$$\mathrm{res}_{D_1} : \Omega_X^p(\log D) \longrightarrow \Omega_{D_1}^{p-1}\big(\log((D - D_1)|_{D_1})\big).$$

Locally write, with the logarithmic factor placed on the left,

$$\varphi = \varphi_1 + \frac{df_1}{f_1} \wedge \varphi_2, \qquad \varphi_2 = \sum_{1 \in I} a_I \delta_{I \setminus \{1\}},$$

where φ_1 is spanned by the δ_I with $1 \notin I$. Then

$$\mathrm{res}_{D_1}(\varphi) = \varphi_2|_{D_1}.$$

The order of the wedge factors fixes the sign. The maps res_{D_i} are defined in the same way.

Finally, ordinary restriction gives

$$\rho_{D_1} : \Omega_X^p\big(\log(D - D_1)\big) \longrightarrow \Omega_{D_1}^p\big(\log((D - D_1)|_{D_1})\big).$$

The sheaf on the left has generators $f_1\delta_I$ when $1 \in I$, and δ_I when $1 \notin I$. Thus

$$\rho_{D_1}\left(\sum_{1 \in I} f_1 a_I \delta_I + \sum_{1 \notin I} a_I \delta_I\right) = \sum_{1 \notin I} (a_I|_{D_1})(\delta_I|_{D_1}).$$

Sheaves on D_i are regarded as sheaves on X by pushforward under the closed immersion.

Proposition 4.2.8. *We have the following exact sequences:*

(a) $0 \longrightarrow \Omega_X^1 \longrightarrow \Omega_X^1(\log D) \xrightarrow{\mathrm{Res}_D} \bigoplus_{i=1}^r \mathcal{O}_{D_i} \longrightarrow 0.$

(b) For $p \geq 1$,

$$0 \longrightarrow \Omega_X^p\big(\log(D - D_1)\big) \longrightarrow \Omega_X^p(\log D) \xrightarrow{\mathrm{res}_{D_1}} \Omega_{D_1}^{p-1}\big(\log((D - D_1)|_{D_1})\big) \longrightarrow 0.$$

(c) For $p \geq 0$,

$$0 \longrightarrow \Omega_X^p(\log D)(-D_1) \longrightarrow \Omega_X^p\big(\log(D - D_1)\big) \xrightarrow{\rho_{D_1}} \Omega_{D_1}^p\big(\log((D - D_1)|_{D_1})\big) \longrightarrow 0.$$

Proof. All assertions can be checked on the displayed local bases. In (a), vanishing residue means that the coefficient of df_i/f_i is divisible by f_i, so the form is regular. In (b), the coefficients of terms containing df_1/f_1 must be divisible by f_1, which removes the pole along D_1. Both residue maps are locally surjective.

For (c), restriction kills the terms $f_1\delta_I$ with $1 \in I$ because they contain df_1. For $1 \notin I$, restriction vanishes precisely when the coefficient is divisible by f_1. Thus the kernel has basis $f_1\delta_I$ for all I, which is exactly $\Omega_X^p(\log D)(-D_1)$. Restriction is locally surjective as well. □

Corollary 4.2.9. *Assume D is a smooth divisor.*

(a) We have exact sequences

$$0 \longrightarrow \Omega^p_X \longrightarrow \Omega^p_X(\log D) \xrightarrow{\mathrm{res}_D} \Omega^{p-1}_D \longrightarrow 0,$$

$$0 \longrightarrow \Omega^p_X(\log D)(-D) \longrightarrow \Omega^p_X \xrightarrow{\rho_D} \Omega^p_D \longrightarrow 0.$$

In the first sequence we set $\Omega^{-1}_D = 0$ if $p = 0$.

(b) Let $\pi : Y \to X$ be the cyclic covering of Proposition 4.1.8, and $D' \subseteq Y$ the divisor mapping isomorphically to D, with $\pi^* D = mD'$. Then pullback gives

$$\pi^* \Omega^p_X(\log D) \cong \Omega^p_Y(\log D').$$

Proof. (a) follows from Proposition 4.2.8. For (b), locally $t^m = f$, so

$$\pi^*(df/f) = \frac{d(t^m)}{t^m} = m\frac{dt}{t}.$$

Since m is invertible, the logarithmic bases correspond. Away from the branch divisor the map is *étale*, giving the assertion there too. □

4.2C Nakano Vanishing Theorem

Theorem 4.2.10 (Nakano vanishing)**.** *Let X be a smooth complex projective variety of dimension n, and let A be ample. Then*[74]

$$H^q(X, \Omega^p_X \otimes \mathcal{O}_X(A)) = 0 \qquad (p + q > n).$$

Equivalently, by Serre duality,

$$H^s(X, \Omega^r_X \otimes \mathcal{O}_X(-A)) = 0 \qquad (r + s < n),$$

where $s = n - q$ and $r = n - p$.

For $p = n$ this is Kodaira vanishing. The strategy is again to use a cyclic covering to represent the pulled-back ample divisor by a smooth effective divisor.

Proof. We prove the second statement by induction on n. For $n = 0$ there is nothing to prove; for $n = 1$, only $(r, s) = (0, 0)$ occurs, and $H^0(X, \mathcal{O}_X(-A)) = 0$ since $\deg(-A) < 0$.

Choose a smooth divisor $D \in |mA|$ by Bertini. We first claim that it suffices to prove

$$H^s(X, \Omega^r_X(\log D) \otimes \mathcal{O}_X(-A)) = 0 \qquad (r + s < n). \tag{4.1}$$

[74] For positivity of degree on a curve and the vanishing of sections of negative degree, see [Har77, IV, 3.3], [Har77, IV, 1.2].

Indeed, the first sequence in Corollary 4.2.9, twisted by $-A$, gives

$$0 \longrightarrow \Omega_X^r(-A) \longrightarrow \Omega_X^r(\log D)(-A) \longrightarrow \Omega_D^{r-1}(-A|_D) \longrightarrow 0.$$

By induction on D, the term $H^{s-1}(D, \Omega_D^{r-1}(-A|_D))$ vanishes whenever $r+s<n$. Hence the map from $H^s(X, \Omega_X^r(-A))$ to the group in (4.1) is injective. If $r=0$ the sheaf with exterior degree -1 is zero; if $s=0$ the negative cohomology group is zero.

For (4.1), take the m-fold cyclic covering $\pi : Y \to X$ branched along D. Lemma 4.1.18 reduces us to

$$H^s\big(Y, \pi^*(\Omega_X^r(\log D) \otimes \mathcal{O}_X(-A))\big) = 0.$$

Corollary 4.2.9(b) identifies $\pi^*\Omega_X^r(\log D)$ with $\Omega_Y^r(\log D')$, and the tautological section gives $D' \sim_{\text{lin}} \pi^*A$. Thus the group is

$$H^s(Y, \Omega_Y^r(\log D')(-D')).$$

The divisor D' is smooth and ample. By the second sequence in Corollary 4.2.9,

$$0 \longrightarrow \Omega_Y^r(\log D')(-D') \longrightarrow \Omega_Y^r \longrightarrow \Omega_{D'}^r \longrightarrow 0.$$

Lemma 4.2.4 makes the restriction in degree $s-1$ surjective and the restriction in degree s injective, since $r+s<n$. Therefore the desired group vanishes. □

4.3 Vanishing for Big and Nef Line Bundles

This section is devoted to the basic vanishing theorem, due to Kawamata and Viehweg, for big and nef divisors.

4.3A Statement and Proof of the Theorem

We can weaken the ampleness hypothesis in Kodaira vanishing to nefness and bigness. This result was independently discovered by Kawamata [Kaw82] and Viehweg [Vie82].

Theorem 4.3.1 (Kodaira vanishing for big and nef divisors)**.** *Let X be a smooth complex projective variety of dimension n, and let D be a nef and big integral divisor on X. Then*[75]

$$H^i(X, \mathcal{O}_X(K_X + D)) = 0 \qquad (i > 0).$$

Equivalently, by Serre duality,

$$H^j(X, \mathcal{O}_X(-D)) = 0 \qquad (j < n).$$

[75]The form of Kodaira's lemma used in the proof is Corollary 3.2.7.

Remark 4.3.2. Recall from Theorem 3.2.15 that a nef divisor D is big if and only if its top self-intersection is strictly positive, i.e. $(D^n) > 0$. However, we do not use this characterization in the proof.

Remark 4.3.3. The papers [Kaw82, Theorem 2] and [Vie82, Proposition 2.3] also contain the fundamental vanishing theorem for $\mathbb{Q}$-divisors, which we discuss in the next section.

Example 4.3.4. The analogue of Nakano vanishing can fail for big and nef divisors. Let $X = \mathrm{Bl}_p \mathbb{P}^3$, and let $H \cong \mathbb{P}^2$ be the pullback of a general hyperplane not passing through p. Then H is nef and big, but[76]

$$H^1(X, \Omega^1_X \otimes \mathcal{O}_X(-H)) \neq 0.$$

Indeed, the blow-up formula [Voi02, 7.31] gives an isomorphism of complex vector spaces

$$H^k(X, \mathbb{C}) \cong H^k(\mathbb{P}^3, \mathbb{C}) \oplus H^{k-2}(\{p\}, \mathbb{C}) \oplus H^{k-4}(\{p\}, \mathbb{C}).$$

For $k = 2$ this gives $h^2(X) = 2$. Hodge decomposition also gives

$$H^2(X, \mathbb{C}) = H^{0,2}(X) \oplus H^1(X, \Omega^1_X) \oplus H^{2,0}(X).$$

The Hodge numbers $h^{p,0}$ are birational invariants of smooth projective varieties. Since projective space has only (p,p)-type cohomology, $h^{2,0}(X) = h^{0,2}(X) = 0$. Consequently,

$$\dim H^1(X, \Omega^1_X) = 2, \qquad \dim H^1(H, \Omega^1_H) = 1.$$

The second sequence of Corollary 4.2.9 gives

$$\cdots \longrightarrow H^1(X, \Omega^1_X(\log H)(-H)) \longrightarrow H^1(X, \Omega^1_X) \xrightarrow{\varphi} H^1(H, \Omega^1_H) \longrightarrow \cdots.$$

Since $\dim \ker \varphi \geq 2 - 1 = 1$, the left-hand group is nonzero. The first sequence, twisted by $-H$, gives

$$\begin{aligned} \cdots \longrightarrow H^1(X, \Omega^1_X(-H)) &\longrightarrow H^1(X, \Omega^1_X(\log H)(-H)) \\ &\longrightarrow H^1(H, \mathcal{O}_H(-H)) \longrightarrow \cdots. \end{aligned}$$

The last group is $H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(-1)) = 0$. In fact, $H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(-1)) = 0$ too, so the first map in this display is an isomorphism. This proves the claimed nonvanishing.

The point is that, although the $h^{p,0}$ are birational invariants, the Hodge numbers $h^{p,q}$ for $p, q \geq 1$ need not be. Thus we cannot expect Nakano-type statements under these weaker positivity hypotheses. The term "plurigenus" instead refers to $h^0(X, mK_X)$.

[76] For the Hodge numbers of projective space and birational invariance of holomorphic forms, see [Voi02, 7.14], [Har77, II, Exercise 8.8], [Har77, III, Exercise 7.3].

We turn to the proof of Theorem 4.3.1. Following Kawamata's original article, we reduce it to a statement that weakens positivity in the presence of a strong geometric hypothesis. This is a logarithmic version of Kodaira vanishing.

Lemma 4.3.5 (Norimatsu's lemma)**.** *Let X be a smooth projective variety, A an ample divisor, and E an snc divisor. Then*

$$H^i(X, \mathcal{O}_X(K_X + A + E)) = 0 \qquad (i > 0).$$

Equivalently,

$$H^j(X, \mathcal{O}_X(-A - E)) = 0 \qquad (j < \dim X).$$

Proof. Write $E = \sum_{i=1}^{r} E_i$. We induct on $n = \dim X$ and, for fixed n, on r. Dimension zero is immediate, and $r = 0$ is Kodaira vanishing. Consider

$$0 \longrightarrow \mathcal{O}_X\left(-A - \sum_{i=1}^{r} E_i\right) \longrightarrow \mathcal{O}_X\left(-A - \sum_{i=1}^{r-1} E_i\right) \longrightarrow \mathcal{O}_{E_r}\left(-A|_{E_r} - \sum_{i=1}^{r-1} E_i|_{E_r}\right) \longrightarrow 0.$$

The restriction $A|_{E_r}$ is ample, and the sum of the nonempty $E_i|_{E_r}$ is snc on E_r. The induction on dimension gives vanishing of its cohomology in degrees $j - 1 < n - 1$. The induction on r gives vanishing of the middle term in degrees $j < n$. The long exact sequence therefore gives the desired vanishing of the left term for $j < n$. Negative cohomological degrees are understood to be zero. □

Proof of Theorem 4.3.1. Since D is big, Kodaira's lemma gives, for a suitable $m \gg 0$,

$$mD \sim_{\text{lin}} H + N, \tag{4.2}$$

where H is ample and N is effective. We proceed in two steps.

Step 1: We first reduce to the case where N has snc support. Take a log resolution $\mu : X^* \to X$ of N by smooth blow-ups. The divisor $D^* = \mu^* D$ remains nef and big. Corollary 4.1.6 gives

$$\mu_* \mathcal{O}_{X^*}(K_{X^*}) = \mathcal{O}_X(K_X), \qquad R^i \mu_* \mathcal{O}_{X^*}(K_{X^*}) = 0 \quad (i > 0). \tag{4.3}$$

The pullback $\mu^* H$ is not necessarily ample, so we modify the decomposition. Write

$$\mu^* N = \sum_j a_j F_j, \qquad a_j \geq 0,$$

including all exceptional divisors among the F_j, with zero coefficients where necessary. The reduced union of these divisors is snc. By Corollary 4.1.6(ii), for $p \gg 0$

there are $b_j \geq 0$, positive on the exceptional divisors and zero on the others, such that

$$\mu^*(pH) - \sum_j b_j F_j$$

is ample. Hence

$$\mu^*(pmD) \sim_{\text{lin}} \left(\mu^*(pH) - \sum_j b_j F_j\right) + \sum_j (pa_j + b_j)F_j.$$

This is the promised decomposition $m'D^* \sim_{\text{lin}} H^* + N^*$, with H^* ample and N^* effective with snc support.

Suppose the vanishing is known on X^*. Both assertions in (4.3), the projection formula, and the Leray spectral sequence give

$$\begin{aligned} H^i(X^*, \mathcal{O}_{X^*}(K_{X^*} + \mu^* D)) &\cong H^i(X, \mu_* \mathcal{O}_{X^*}(K_{X^*} + \mu^* D)) \\ &\cong H^i(X, \mathcal{O}_X(K_X + D)). \end{aligned}$$

Thus vanishing descends to X. We may therefore assume that N in (4.2) has snc support.

Step 2: Write $N = \sum_{i=1}^r e_i E_i$, with $e_i > 0$ integral. If $N = 0$, then $mD \sim_{\text{lin}} H$ makes D ample, and Kodaira vanishing applies. Otherwise put

$$e^* = e_1 \cdots e_r, \qquad e_i^* = e^*/e_i.$$

Use the Kawamata covering with integers $me_1^*, \ldots, me_r^*$ to obtain $h : Y \to X$, with Y smooth, such that

$$h^* E_i = me_i^* E_i', \qquad E' = \sum_i E_i' \text{ is snc.}$$

Let $D' = h^*D$ and $H' = h^*H$. Then

$$mD' \sim_{\text{lin}} H' + me^* E'. \tag{4.4}$$

The divisor D' is nef, and H' is ample since h is finite. Consequently,

$$me^*(D' - E') \sim_{\text{lin}} H' + m(e^* - 1)D'$$

is ample, being a sum of an ample and a nef divisor. Thus $A' = D' - E'$ is ample, and Norimatsu's lemma gives

$$H^j(Y, \mathcal{O}_Y(-D')) = H^j(Y, \mathcal{O}_Y(-A' - E')) = 0 \qquad (j < n).$$

Finally, Lemma 4.1.18 gives

$$H^j(X, \mathcal{O}_X(-D)) \hookrightarrow H^j(Y, \mathcal{O}_Y(-D')) = 0,$$

as required. □

Example 4.3.6 (An extension of Theorem 4.3.1)**.** Let D be a nef divisor on a smooth projective n-fold X. Suppose

$$(D^{n-k} \cdot H^k) > 0$$

for some integer $0 \leq k \leq n$ and an ample divisor H. Then

$$H^i(X, \mathcal{O}_X(K_X + D)) = 0 \qquad (i > k).$$

This statement is also due to Kawamata and Viehweg and contains Theorem 4.3.1 as the case $k = 0$.

Proof. For $k = 0$, the nef divisor D is big by its positive top intersection, so Theorem 4.3.1 applies. For $k = n$, the result is Grothendieck vanishing. It remains to consider $1 \leq k \leq n - 1$.

After replacing H by a sufficiently high multiple, choose a smooth very ample representative, still denoted H. Induct on n. On H, the nef divisor $D|_H$ satisfies the condition with $k' = k - 1$, since

$$\left((D|_H)^{(n-1)-(k-1)} \cdot (H|_H)^{k-1}\right)_H = (D^{n-k} \cdot H^k)_X > 0.$$

Adjunction gives

$$0 \longrightarrow \mathcal{O}_X(K_X + D) \longrightarrow \mathcal{O}_X(K_X + D + H) \longrightarrow \mathcal{O}_H(K_H + D|_H) \longrightarrow 0.$$

The middle term has no higher cohomology by Kodaira vanishing, since $D + H$ is ample. For $i > k$, the induction hypothesis gives

$$H^{i-1}(H, \mathcal{O}_H(K_H + D|_H)) = 0$$

because $i - 1 > k - 1$. The long exact sequence proves the assertion. The dimension-one cases were already covered by $k = 0$ and $k = n$. □

We give two theorems without proof.

Theorem 4.3.7 (Kollár's injectivity theorem)**.** *Let X be smooth and projective, and L a semi-ample divisor. Given $k \geq 1$, fix $D \in |kL|$. The homomorphisms given by multiplication by its defining section,*

$$H^i(X, \mathcal{O}_X(K_X + mL)) \xrightarrow{\cdot D} H^i(X, \mathcal{O}_X(K_X + (m + k)L)),$$

are injective for all $i \geq 0$ and $m > 0$.

This implies Kodaira vanishing: if L is ample, take $m = 1$ and $k \gg 0$ with $|kL| \neq \varnothing$. The target is $H^i(X, \mathcal{O}_X(K_X + (k + 1)L))$, which vanishes for $i > 0$ by Serre vanishing.

The second theorem shows that the direct images of the canonical bundle have exceptionally good cohomological properties.

Theorem 4.3.8 (Kollár vanishing theorem)**.** *Let $f : X \to Y$ be a surjective morphism of projective varieties, with X smooth. Then:*

(i) $R^j f_* \mathcal{O}_X(K_X)$ *is torsion-free for every* $j \geq 0$.

(ii) $R^j f_* \mathcal{O}_X(K_X) = 0$ *for* $j > \dim X - \dim Y$.

(iii) For every ample divisor A on Y and every $j \geq 0$,

$$H^i(Y, R^j f_* \mathcal{O}_X(K_X) \otimes \mathcal{O}_Y(A)) = 0 \qquad (i > 0).$$

We used this result in the proof of Fujita's vanishing theorem.

4.3B Some Applications

Theorem 4.3.9 (Grauert–Riemenschneider vanishing)**.** *Let $f : X \to Y$ be a generically finite, surjective projective morphism of varieties, with X smooth. Then*

$$R^i f_* \mathcal{O}_X(K_X) = 0 \qquad (i > 0).$$

Proof. See [Laz04a, 4.3.9] for a proof. We will also obtain this as a special case of generalized Kawamata–Viehweg vanishing at the end of §4.4. □

4.4 Kawamata–Viehweg Vanishing Theorem

We now come to the basic vanishing theorem for $\mathbb{Q}$-divisors. First we need some operations on divisors.

Definition 4.4.1 (Rounds)**.** Let $D = \sum_i a_i D_i$ be a $\mathbb{Q}$-divisor, with distinct prime divisors D_i. Its round-up and round-down are the integral divisors

$$\lceil D \rceil = \sum_i \lceil a_i \rceil D_i, \qquad \lfloor D \rfloor = \sum_i \lfloor a_i \rfloor D_i,$$

where $\lceil x \rceil$ is the least integer at least x, and $\lfloor x \rfloor$ the greatest integer at most x. The fractional part is

$$\{D\} = D - \lfloor D \rfloor.$$

The same definitions apply coefficientwise to $\mathbb{R}$-divisors.

Remark 4.4.2. Rounding is defined for actual divisors, at the cycle level, rather than for their linear or numerical equivalence classes. On a smooth variety, Cartier and Weil divisors are identified, so rounding a $\mathbb{Q}$-Cartier divisor in this sense does make sense and gives an integral Cartier divisor. On a singular variety, however, its round-up or round-down need not be Cartier; additional hypotheses may be needed for pullbacks and other operations.

Warning 4.4.3 (Warning on rounds)**.** Pullback does not commute with rounding, and numerical equivalence does not determine the rounding. For example, in $X = \mathbb{A}^2$ let $A = V(y - x^2)$, let Y be the x-axis, and let P be the origin. Then Y is tangent to A at P. For $D = \frac{1}{2}A$,

$$\lfloor D \rfloor|_Y = 0, \qquad \lfloor D|_Y \rfloor = \lfloor \tfrac{1}{2} \cdot 2P \rfloor = P.$$

Likewise, if $L, C \subseteq \mathbb{P}^2$ are a line and an irreducible conic, then $L \equiv_{\text{num}} \frac{1}{2}C$, but

$$\lfloor \tfrac{1}{2}C \rfloor = 0, \qquad \lfloor L \rfloor = L.$$

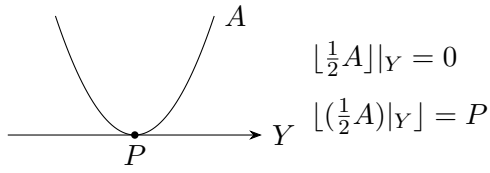


Figure 4.4: Rounding does not commute with restriction.

4.4A Statement and Proof of the Theorem

Theorem 4.4.4 (Kawamata–Viehweg vanishing)**.** *Let X be a non-singular projective variety of dimension n, and N an integral divisor on X. Assume*

$$N \equiv_{\text{num}} B + \Delta,$$

where B is a nef and big $\mathbb{Q}$-divisor and $\Delta = \sum_i a_i \Delta_i$ has snc support, with $0 \leq a_i < 1$. Then

$$H^i(X, \mathcal{O}_X(K_X + N)) = 0 \qquad (i > 0).$$

Equivalently, by Serre duality,

$$H^j(X, \mathcal{O}_X(-N)) = 0 \qquad (j < n).$$

Remark 4.4.5. The hypothesis on Δ says that it is effective, its support is snc, and $\lfloor \Delta \rfloor = 0$. In standard terminology a boundary has coefficients in $[0, 1]$; here we require all coefficients to be strictly less than 1. This strict inequality is part of the vanishing hypothesis.

Proof of Theorem 4.4.4. We allow each Δ_i to have several disjoint smooth connected components, as in the discussion following Proposition 4.2.7. This is useful because the pullback of a point on a curve may be disconnected. We induct on the number of nonzero fractional terms.

If $\Delta = 0$, then N is nef and big because these properties are numerical, and Theorem 4.3.1 applies. Otherwise write $a_1 = c/d$, with integers $0 < c < d$. By

Lemma 4.1.18, it suffices to find a finite surjective morphism $f : X' \to X$, with X' non-singular, for which

$$H^j(X', f^*\mathcal{O}_X(-N)) = 0 \qquad (j < n). \tag{4.5}$$

Take a BG covering such that $\Delta'_i = f^*\Delta_i$ are smooth, their sum is snc, and

$$\Delta'_1 \sim_{\text{lin}} dA' \tag{4.6}$$

for some divisor A' on X'. Set $N' = f^*N$ and $B' = f^*B$.

Next take the d-fold cyclic covering $\pi : X'' \to X'$ branched along Δ'_1. Its root divisor Δ''_1 satisfies

$$d\Delta''_1 = \pi^*\Delta'_1, \qquad \Delta''_1 \sim_{\text{lin}} A'', \qquad A'' = \pi^*A'.$$

The linear equivalence comes from the tautological section. For $i \geq 2$, put $\Delta''_i = \pi^*\Delta'_i$, and set $N'' = \pi^*N'$, $B'' = \pi^*B'$. Then

$$N'' \equiv_{\text{num}} B'' + cA'' + \sum_{i\geq 2} a_i\Delta''_i,$$

so

$$N'' - cA'' \equiv_{\text{num}} B'' + \sum_{i\geq 2} a_i\Delta''_i.$$

The divisor B'' is nef and big, and the remaining fractional part has snc support by Proposition 4.1.12. The induction hypothesis gives

$$H^j(X'', \mathcal{O}_{X''}(-N'' + cA'')) = 0 \qquad (j < n),$$

and, since π is finite,

$$H^j(X', \pi_*\mathcal{O}_{X''}(-N'' + cA'')) = 0 \qquad (j < n). \tag{4.7}$$

By Remark 4.1.10 and the projection formula,

$$\begin{aligned}\pi_*\mathcal{O}_{X''}(-N'' + cA'') &= (\pi_*\mathcal{O}_{X''}) \otimes \mathcal{O}_{X'}(-N' + cA') \\ &\cong \bigoplus_{a=0}^{d-1} \mathcal{O}_{X'}(-N' + (c-a)A').\end{aligned}$$

The summand with $a = c$ is $\mathcal{O}_{X'}(-N')$, since $1 \leq c \leq d-1$. Thus (4.7) implies (4.5), proving the theorem. □

Corollary 4.4.6. *Let X be smooth and projective, and B a nef and big $\mathbb{Q}$-divisor whose fractional part has snc support. Then*

$$H^i(X, \mathcal{O}_X(K_X + \lceil B \rceil)) = 0 \qquad (i > 0).$$

Proof. Since $\lceil B \rceil = -\lfloor -B \rfloor$, we have

$$B - \lceil B \rceil = B + \lfloor -B \rfloor = -\{-B\}.$$

Thus $\lceil B \rceil = B + \{-B\}$, where $\{-B\}$ has the same support as $\{B\}$ and all coefficients less than 1. Apply Theorem 4.4.4. □

Corollary 4.4.7. *Let X be smooth and projective, L an integral divisor, and D a $\mathbb{Q}$-divisor. If $L - D$ is nef and big and D has snc support, then*

$$H^i(X, \mathcal{O}_X(K_X + L - \lfloor D \rfloor)) = 0 \qquad (i > 0).$$

Proof. Use $L - \lfloor D \rfloor = (L - D) + \{D\}$ and Theorem 4.4.4. □

4.4B Generalizations of the Vanishings in Relative Version

Following [Kaw24] and [KMM87], we generalize the vanishing theorem in directions needed for higher-dimensional algebraic geometry. Recall that an $\mathbb{R}$-Cartier divisor is f-nef if its restriction to every fiber is nef, and f-big if its restriction to the generic fiber is big. We first need a lemma.

Lemma 4.4.8. *Let X be smooth, and let D be an $\mathbb{R}$-divisor such that $\{D\}$ has snc support. If $f : Y \to X$ is a proper birational morphism with Y smooth, then*[77]

$$f_* \mathcal{O}_Y(K_Y + \lceil f^* D \rceil) \cong \mathcal{O}_X(K_X + \lceil D \rceil).$$

Proof. Choose compatible canonical divisors. Put $\Delta = \lceil D \rceil - \lfloor D \rfloor$, the reduced snc divisor supported on the nonintegral components of D. The discrepancy formula gives

$$K_Y + f_*^{-1}\Delta = f^*(K_X + \Delta) + \sum_{E_i \text{ exceptional}} a_i E_i.$$

For a reduced snc pair the coefficients a_i are integers at least -1; see the local snc calculation in Corollary 5.2.20. We only need this lower bound, not equality of the infimum with -1.

Adding $\lceil f^* D \rceil$ and rearranging, using that $\lfloor D \rfloor$ is integral Cartier, gives

$$K_Y + \lceil f^* D \rceil = f^*(K_X + \lceil D \rceil) + F,$$

where

$$F = \lceil f^* \{D\} \rceil - f_*^{-1}\Delta + \sum_i a_i E_i.$$

The coefficients on nonexceptional divisors cancel, since $\lceil \{D\} \rceil = \Delta$. For an exceptional divisor E_i, its coefficient is

$$\operatorname{mult}_{E_i} \lceil f^* \{D\} \rceil + a_i.$$

[77] The purity of the exceptional locus over a smooth target is Proposition 2.1.3.

If $a_i \geq 0$, this is nonnegative because $\{D\}$ is effective. If $a_i = -1$, its center lies in $\operatorname{Supp}\Delta$: outside that divisor, the relative canonical divisor of a birational morphism between smooth varieties is effective. Since every component of Δ has a strictly positive coefficient in $\{D\}$, we then have

$$\operatorname{mult}_{E_i}(f^*\{D\}) > 0, \qquad \operatorname{mult}_{E_i}\lceil f^*\{D\}\rceil \geq 1.$$

Thus this coefficient is again nonnegative. Hence F is effective and exceptional. Algebraic Hartogs gives $f_*\mathcal{O}_Y(F) = \mathcal{O}_X$, and the projection formula yields

$$\begin{aligned} f_*\mathcal{O}_Y(K_Y + \lceil f^*D\rceil) &\cong \mathcal{O}_X(K_X + \lceil D\rceil) \otimes f_*\mathcal{O}_Y(F) \\ &\cong \mathcal{O}_X(K_X + \lceil D\rceil). \end{aligned}$$

□

Theorem 4.4.9 (Generalized Kodaira vanishing in the projective case)**.** *Let X be a smooth projective variety, $f : X \to S$ a surjective morphism onto a projective variety, and D an f-ample $\mathbb{Q}$-divisor with $\{D\}$ of snc support. Then*[78]

$$R^i f_*\mathcal{O}_X(K_X + \lceil D\rceil) = 0 \qquad (i > 0).$$

Proof. **Step 1:** We can add the pullback of any integral Cartier divisor A on S. Indeed,

$$\begin{aligned} R^i f_*\mathcal{O}_X(K_X + \lceil D + f^*A\rceil) &= R^i f_*\mathcal{O}_X(K_X + \lceil D\rceil + f^*A) \\ &\cong R^i f_*\mathcal{O}_X(K_X + \lceil D\rceil) \otimes \mathcal{O}_S(A). \end{aligned}$$

Vanishing is therefore unchanged.

Step 2: Clear denominators so that rD is an integral f-ample divisor on X. Choose an ample integral Cartier divisor A on S and replace it by a sufficiently high multiple. Relative ampleness then makes $D + f^*A$ ample on X. Merely increasing the common denominator does not do this; the multiple of A must be large enough.

Replace D by $D + f^*A$. Its fractional part is unchanged. Fix an ample divisor H on S and consider the Leray spectral sequence

$$\begin{aligned} E_2^{p,q} = H^p\big(S, R^q f_*\mathcal{O}_X(K_X + \lceil D\rceil) \otimes \mathcal{O}_S(mH)\big) \quad \Longrightarrow \\ H^{p+q}\big(X, \mathcal{O}_X(K_X + \lceil D\rceil + mf^*H)\big). \end{aligned}$$

For $m \gg 0$, Serre vanishing makes $E_2^{p,q} = 0$ for $p > 0$. There are only finitely many nonzero higher direct images, so the same m works for all q. Thus

$$H^0\big(S, R^i f_*\mathcal{O}_X(K_X + \lceil D\rceil) \otimes \mathcal{O}_S(mH)\big) \cong H^i\big(X, \mathcal{O}_X(K_X + \lceil D\rceil + mf^*H)\big).$$

[78] For the spectral sequence and its degeneration on the q-axis, see Theorem D.6.3 and Proposition D.5.11. For adding a sufficiently positive pullback from the base to a relatively ample divisor, see Proposition 1.5.8.

Since D is ample and f^*H is nef, $D + mf^*H$ is ample. Corollary 4.4.6 shows that the right side vanishes for $i > 0$. For $m \gg 0$, Serre's global generation theorem also makes

$$R^i f_* \mathcal{O}_X(K_X + \lceil D \rceil) \otimes \mathcal{O}_S(mH)$$

globally generated. A globally generated sheaf with no global sections is zero. Twisting back proves the assertion. □

Theorem 4.4.10 (Generalized Kodaira vanishing in the general case)**.** *Let X be smooth, $f : X \to S$ a projective surjective morphism onto a variety, and D an f-ample $\mathbb{Q}$-divisor whose fractional part has snc support. Then*[79]

$$R^i f_* \mathcal{O}_X(K_X + \lceil D \rceil) = 0 \qquad (i > 0).$$

The statement also applies when f is proper and admits the given relatively ample divisor: such a morphism is projective, as in Remark 1.5.2.

Proof. The idea is to turn the problem into the projective case. The assertion is local on S, so assume S affine. Take a projective compactification $\bar{S}$ and, using a projective embedding of X over S, take the closure of X over $\bar{S}$. Resolve outside X to obtain a smooth projective $\bar{X}$ and a projective morphism $\bar{f} : \bar{X} \to \bar{S}$ fitting into

$$\begin{array}{ccc} X & \hookrightarrow & \bar{X} \\ f\downarrow & & \downarrow \bar{f} \\ S & \hookrightarrow & \bar{S}. \end{array}$$

The square is Cartesian: $X = \bar{f}^{-1}(S)$. This property is preserved by the modifications outside X below.

Step 1: We construct a projective birational morphism $\bar{\mu} : \bar{Y} \to \bar{X}$, with $\bar{Y}$ smooth, and a $\mathbb{Q}$-divisor $\bar{C}$ such that:

(1) $\bar{C}$ is nef and big over $\bar{S}$;

(2) $\operatorname{Supp}\{\bar{C}\} \cup \operatorname{Exc}(\bar{\mu})$ is contained in an snc divisor;

(3) on $Y = \bar{\mu}^{-1}(X)$, we have $\bar{C}|_Y = \mu^* D$, where $\mu = \bar{\mu}|_Y$.

Lemma 4.4.8 will then give

$$\mu_* \mathcal{O}_Y(K_Y + \lceil \mu^* D \rceil) \cong \mathcal{O}_X(K_X + \lceil D \rceil). \tag{4.8}$$

Let $\bar{D}$ be the closure of D in $\bar{X}$. Choose m sufficiently divisible and large so that $m\bar{D}$ is integral and mD is f-very ample. Consider evaluation

$$\varphi_m : \bar{f}^* \bar{f}_* \mathcal{O}_{\bar{X}}(m\bar{D}) \longrightarrow \mathcal{O}_{\bar{X}}(m\bar{D}).$$

[79] For the morphism defined by a generated relative linear system and ampleness over an affine base, see [Har77, II,7.12], [GW20, 13.59]. For degeneration of the spectral sequence on the p-axis, see Proposition D.5.11. The resolution of the base ideal is the construction of Example 3.1.34.

It is surjective on X. Its image has the form $\mathcal{I}\mathcal{O}_{\bar{X}}(m\bar{D})$, with $\mathcal{I}|_X = \mathcal{O}_X$. Blow up this ideal and resolve, only outside X. Replacing the compactification and pulling back $\bar{D}$, we obtain an effective Cartier divisor E supported outside X, and an invertible quotient

$$\bar{f}^*\mathcal{E} \twoheadrightarrow \mathcal{O}_{\bar{X}}(m\bar{D} - E),$$

where $\mathcal{E}$ is the coherent sheaf on $\bar{S}$ used in the original evaluation. This gives an $\bar{S}$-morphism

$$\pi : \bar{X} \longrightarrow \mathbb{P}_{\bar{S}}(\mathcal{E}).$$

Let $\bar{P}$ be its reduced image. The relatively ample line bundle $\mathcal{O}_{\bar{P}}(1)$ corresponds to a divisor H, and

$$\pi^*H \sim_{\text{lin}} m\bar{D} - E.$$

On X, this is the map given by the complete relative linear system of mD, so $\pi|_X$ is a closed immersion, in particular finite. One can also see finiteness from positivity: a contracted complete curve $C \subseteq X$ would satisfy

$$(mD \cdot C) = (H \cdot \pi_* C) = 0,$$

contrary to relative ampleness, since π is over S. A positive-dimensional projective fiber would contain such a curve. Thus π is generically finite, and

$$D_0 = \bar{D} - \frac{1}{m}E \sim_{\mathbb{Q}} \frac{1}{m}\pi^*H$$

is $\bar{f}$-nef and $\bar{f}$-big, with $D_0|_X = D$.

Now resolve the support of $\{D_0\}$ together with the exceptional locus. Set $\bar{C} = \bar{\mu}^*D_0$. This gives (1)–(3), since the pullback of the integral part of D_0 is integral, and any new fractional components are among the exceptional divisors.

Step 2: The divisor $\bar{C}$ is only relatively nef and big; Theorem 4.4.9 cannot yet be applied to it. We use the small-coefficient perturbation construction of Lemma 4.4.13 below, over the projective base $\bar{S}$. That construction uses only Chow's lemma, Kodaira's lemma, resolution, and relative ampleness, so this forward reference does not use the vanishing theorem being proved.

It gives a projective birational morphism $\nu : \bar{Z} \to \bar{Y}$, with $\bar{Z}$ smooth, and an effective divisor $\sum \delta_j F_j$ with arbitrarily small positive rational coefficients such that

$$\bar{A} = \nu^*\bar{C} - \sum_j \delta_j F_j$$

is ample over $\bar{S}$ and has fractional part of snc support. Choose the coefficients small enough that

$$\lceil \bar{A} \rceil = \lceil \nu^*\bar{C} \rceil.$$

This is possible coefficient by coefficient: subtract less than the fractional part from a nonintegral coefficient, and less than 1 from an integral coefficient.

Put $g = \bar{\mu} \circ \nu : \bar{Z} \to \bar{X}$ and $h = \bar{f} \circ g : \bar{Z} \to \bar{S}$. Since $\bar{A}$ is h-ample, it is also g-ample. Both bases and $\bar{Z}$ are projective. Applying Theorem 4.4.9 to g and h gives, for

$$\mathcal{G} = \mathcal{O}_{\bar{Z}}(K_{\bar{Z}} + \lceil \bar{A} \rceil),$$
$$R^q g_* \mathcal{G} = 0 \quad (q > 0), \qquad R^i h_* \mathcal{G} = 0 \quad (i > 0).$$

The spectral sequence

$$R^p \bar{f}_* (R^q g_* \mathcal{G}) \Longrightarrow R^{p+q} h_* \mathcal{G}$$

therefore collapses to the p-axis, giving

$$R^i \bar{f}_* (g_* \mathcal{G}) = 0 \qquad (i > 0).$$

On $Z = g^{-1}(X)$, write $g_X = g|_Z$. We have $\nu^* \bar{C}|_Z = g_X^* D$, so Lemma 4.4.8 gives

$$(g_* \mathcal{G})|_X = (g_X)_* \mathcal{O}_Z(K_Z + \lceil g_X^* D \rceil) \cong \mathcal{O}_X(K_X + \lceil D \rceil).$$

Restriction to the open subset $S \subseteq \bar{S}$ commutes with higher direct images. Thus

$$0 = (R^i \bar{f}_* (g_* \mathcal{G}))|_S \cong R^i f_* \mathcal{O}_X(K_X + \lceil D \rceil),$$

which is the desired vanishing. □

The relatively ample statement extends to $\mathbb{R}$-divisors by perturbing coefficients.

Theorem 4.4.11 (Generalized Kodaira vanishing for $\mathbb{R}$-divisors)**.** *Let X be smooth, $f : X \to S$ projective and surjective, and D an f-ample $\mathbb{R}$-divisor whose fractional part has snc support. Then*[80]

$$R^i f_* \mathcal{O}_X(K_X + \lceil D \rceil) = 0 \qquad (i > 0).$$

Proof. By openness of the relative ample cone, we can approximate the finitely many coefficients of D by rational numbers while preserving relative ampleness. Keep the integral coefficients fixed, and keep each nonintegral coefficient strictly between its original floor and ceiling. The resulting $\mathbb{Q}$-divisor has the same round-up and the same fractional support. Theorem 4.4.10 applies. □

The following is the relative Kawamata–Viehweg theorem; see [KMM87, Theorem 1-2-3].

Theorem 4.4.12 (Generalized Kawamata–Viehweg vanishing)**.** *Let X be nonsingular, $f : X \to S$ a proper surjective morphism, and D an f-nef and f-big $\mathbb{Q}$-divisor whose fractional part has snc support. Then*[81]

$$R^i f_* \mathcal{O}_X(K_X + \lceil D \rceil) = 0 \qquad (i > 0).$$

[80] Openness of the relative ample cone is Proposition 1.5.16.

[81] For the spectral-sequence degeneration in this proof, see Proposition D.5.11.

Before proving this, we give the perturbation lemma used above. The general statement is [KMM87, Corollary 0-3-6]. Its construction is independent of the vanishing theorems.

Lemma 4.4.13. *Let $f : X \to S$ be a proper surjective morphism, with X normal, and let D be an f-nef and f-big $\mathbb{Q}$-Cartier divisor. There exist a projective birational morphism $\mu : Y \to X$, with Y smooth and projective over S, and finitely many prime divisors F_j on Y, such that*[82]

$$\operatorname{Supp}(\mu^* D) \cup \bigcup_j F_j \cup \operatorname{Exc}(\mu)$$

is contained in an snc divisor, and

$$\mu^* D - \sum_j \delta_j F_j$$

is $(f \circ \mu)$-ample for suitable positive rational coefficients δ_j that can be made arbitrarily small on the chosen model.

Construction for the bases used here. We give the construction when S is quasi-projective. This covers both the projective base in the proof of Theorem 4.4.10 and every affine neighborhood needed to prove the relative vanishing statements. The form over an arbitrary algebraic base is the relative perturbation lemma cited above; the local construction alone is not a gluing argument for one global model over such a base.

Step 1: By Chow's lemma, choose a variety X' projective over S and a proper birational morphism $\varphi : X' \to X$. It is projective: its graph is a closed immersion into $X' \times_S X$, which is projective over X, since X is separated over S. Normalize X' and compose with the finite normalization map. We retain the notation X', φ, and $f' : X' \to S$.

Step 2: The divisor $D_0 = \varphi^* D$ is f'-nef and f'-big. Relative Kodaira's lemma gives

$$mD_0 \sim_S A' + N',$$

where A' is f'-ample and integral, and N' is effective. For completeness, choose such an A' first. Kodaira's lemma on the generic fiber makes $f'_* \mathcal{O}_{X'}(mD_0 - A')$ nonzero for a suitable divisible m. Since S is quasi-projective, twisting this coherent sheaf by a sufficiently ample line bundle from S gives a nonzero global section. Its divisor gives N', with the twist from S absorbed by $\sim_S$.

[82] For projectivity of the modifications, the relatively ample exceptional divisor, and composition of relatively ample bundles, see [Har77, II, Exercise 4.8], [Har77, II, 7.13(a)], [GW20, 13.65]. For the birational invariance of bigness, relative Kodaira decomposition, and nef plus relatively ample, compare Proposition 3.2.36 on the generic fibre, Corollary 3.3.7, and Corollary 1.5.20.

Set $E = N'/m$. For any rational $0 < \varepsilon < 1$,

$$D_0 \sim_{S,\mathbb{Q}} (1-\varepsilon)D_0 + \frac{\varepsilon}{m}A' + \varepsilon E.$$

The divisor

$$A_\varepsilon = (1-\varepsilon)D_0 + \frac{\varepsilon}{m}A'$$

is f'-ample, being a sum of a relatively nef and a relatively ample divisor. Thus $D_0 \sim_{S,\mathbb{Q}} A_\varepsilon + \varepsilon E$, with the coefficients of εE arbitrarily small.

Step 3: Take a log resolution $\mu' : Y \to X'$ of the *union* of the supports of D_0 and E, including the exceptional loci in the construction. We can arrange that $\operatorname{Supp} \mu'^* D_0$, $\operatorname{Supp} \mu'^* E$ and $\operatorname{Exc}(\varphi \circ \mu')$ are contained in an snc divisor. Resolving the union is necessary: components of the sum $D_0 + E$ could cancel.

$$\begin{array}{ccccc} Y & \xrightarrow{\mu'} & X' & \xrightarrow{\varphi} & X \\ & & \downarrow f' & & \downarrow f \\ & & S & = & S. \end{array}$$

The resolution can be made by blow-ups, and there is an effective exceptional Cartier divisor F with $-F$ μ'-ample. This follows by combining the relatively ample $\mathcal{O}(1) = \mathcal{O}(-E_i)$ of the blow-ups, as in the induction in Corollary 4.1.6(ii). For a sufficiently small rational $\xi > 0$,

$$\mu'^* A_\varepsilon - \xi F$$

is ample over S: a sufficiently large multiple of the pullback of an f'-ample divisor, minus F, is ample over S. Consequently,

$$\mu'^* D_0 - \left(\mu'^*(\varepsilon E) + \xi F\right) \sim_{S,\mathbb{Q}} \mu'^* A_\varepsilon - \xi F$$

is ample over S.

Step 4: Set $\mu = \varphi \circ \mu'$, so $\mu^* D = \mu'^* D_0$, and write

$$\sum_j \delta_j F_j = \mu'^*(\varepsilon E) + \xi F.$$

The support is contained in the snc divisor constructed in Step 3. Fix the resolution first, choose ε small, and then choose ξ sufficiently small for ampleness. This makes all the positive coefficients δ_j as small as required, and proves the asserted construction in the cases used here. □

Proof of Theorem 4.4.12. Work locally on S, so assume S affine. Lemma 4.4.13 provides $\mu : Y \to X$, with Y smooth and projective over S, and an $(f \circ \mu)$-ample divisor

$$P = \mu^* D - \sum_j \delta_j F_j$$

with snc fractional support. Choose the coefficients small enough that $\lceil P \rceil = \lceil \mu^* D \rceil$. There are only finitely many coefficients to check: at a nonintegral coefficient a, subtract less than $a - \lfloor a \rfloor$; at an integral coefficient, subtract less than 1.

The divisor P is also μ-ample. Applying Theorem 4.4.10 to μ gives

$$R^i \mu_* \mathcal{O}_Y(K_Y + \lceil P \rceil) = R^i \mu_* \mathcal{O}_Y(K_Y + \lceil \mu^* D \rceil) = 0 \qquad (i > 0). \tag{4.9}$$

Lemma 4.4.8 identifies the degree-zero direct image with $\mathcal{O}_X(K_X + \lceil D \rceil)$. The spectral sequence

$$R^p f_* \big(R^q \mu_* \mathcal{O}_Y(K_Y + \lceil P \rceil) \big) \Longrightarrow R^{p+q} (f \circ \mu)_* \mathcal{O}_Y(K_Y + \lceil P \rceil)$$

collapses to the p-axis by (4.9). Applying Theorem 4.4.10 to $f \circ \mu$, we conclude, for $i > 0$,

$$\begin{aligned} 0 &= R^i (f \circ \mu)_* \mathcal{O}_Y(K_Y + \lceil P \rceil) \\ &\cong R^i f_* \big(\mu_* \mathcal{O}_Y(K_Y + \lceil \mu^* D \rceil) \big) \\ &\cong R^i f_* \mathcal{O}_X(K_X + \lceil D \rceil). \end{aligned}$$

These local vanishings prove the assertion on S. □

Theorem 4.4.14 (Generalized Kawamata–Viehweg vanishing for $\mathbb{R}$-divisors)**.** *Let X be non-singular, $f : X \to S$ proper and surjective, and D an f-nef and f-big $\mathbb{R}$-divisor whose fractional part has snc support. Then*[83]

$$R^i f_* \mathcal{O}_X(K_X + \lceil D \rceil) = 0 \qquad (i > 0).$$

Proof. Work over an affine open subset of S. The crucial point is the real-divisor version of the construction in Lemma 4.4.13. After the projective modification, relative Kodaira's lemma for real divisors gives

$$D_0 \sim_{S,\mathbb{R}} A_0 + E,$$

where A_0 is f'-ample and E is effective. This form follows by expressing a big real divisor as a positive combination of nearby rational big divisors and applying the rational relative Kodaira lemma to each of them. We do not approximate a nef divisor and assume the approximations remain nef.

For a small real $0 < \varepsilon < 1$, put

$$A_\varepsilon = (1 - \varepsilon) D_0 + \varepsilon A_0.$$

This is f'-ample, and $D_0 \sim_{S,\mathbb{R}} A_\varepsilon + \varepsilon E$. Resolve the union of supports as above and choose $-F$ relatively ample for μ'. We need

$$\mu'^* D_0 - \big(\mu'^* (\varepsilon E) + \xi F \big) \sim_{S,\mathbb{R}} \mu'^* A_\varepsilon - \xi F$$

[83] For composition of relatively ample line bundles, see [GW20, 13.65].

to be ample over S for small $\xi > 0$.

Write the ample real divisor as a *positive* linear combination

$$A_\varepsilon = \sum_i r_i H_i, \qquad r_i > 0,$$

where the H_i are f'-ample integral divisors. For each i, the divisor $n\mu'^* H_i - F$ is ample over S for all sufficiently large integers n. Choose one such n for every i. Then

$$\sum_i r_i \left(\mu'^* H_i - \frac{1}{n} F \right) = \mu'^* A_\varepsilon - \frac{\sum_i r_i}{n} F$$

is ample over S. Taking $\xi = (\sum_i r_i)/n$ makes ξ arbitrarily small. Thus the desired effective perturbation $\sum_j \delta_j F_j = \mu'^*(\varepsilon E) + \xi F$ exists with arbitrarily small real coefficients.

Choose them so that rounding is unchanged. Apply Theorem 4.4.11 to μ and to $f \circ \mu$, and then use Lemma 4.4.8 and the same spectral sequence as for Theorem 4.4.12. This proves the real-divisor statement. □

We can now finish the proof of Grauert–Riemenschneider vanishing.

Proof of Theorem 4.3.9. Since f is generically finite, its generic fiber has dimension zero. The divisor $D = 0$ is f-nef, and it is f-big because its restriction to that zero-dimensional generic fiber is big. Its fractional part is empty. Theorem 4.4.12 therefore gives

$$R^i f_* \mathcal{O}_X(K_X) = R^i f_* \mathcal{O}_X(K_X + \lceil 0 \rceil) = 0 \qquad (i > 0).$$

□

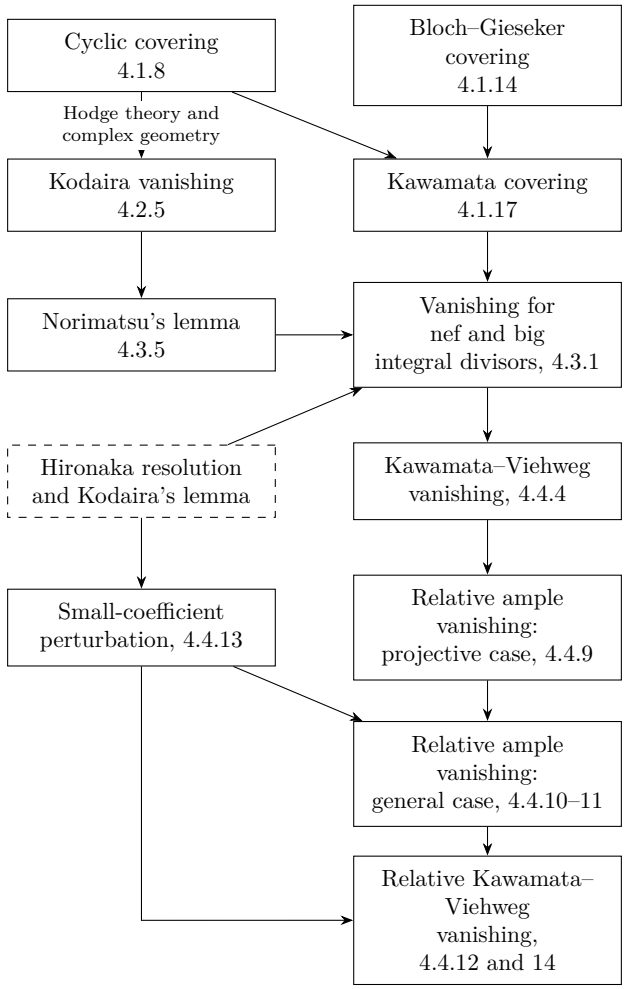


Figure 4.5: Knowledge framework. The perturbation lemma is proved without vanishing and is used in 4.4.10; there is no arrow from relative vanishing back to that lemma. Cyclic and BG coverings also enter the proof of 4.4.4.

Chapter 5
Introduction to the Minimal Model Program

5.1 Introduction to Mori's program

We work over $\mathbb{C}$ in this section. Recall that in Chapter 2, we proved the result for contracting rational curves on a smooth projective surface:

Theorem 5.1.1 (MMP for Surfaces)**.** *Let X be a smooth projective surface and $R \subseteq \overline{\mathrm{NE}}(X)$ an extremal ray such that $(R \cdot K_X) < 0$, i.e. R is generated by a K_X-negative curve. Then the contraction morphism $\mathrm{cont}_R : X \longrightarrow Z$ exists and is one of the following types:*

(1) Z is a smooth surface and X is obtained from Z by blowing up a closed point; $\rho(Z) = \rho(X) - 1$.

(2) Z is a smooth curve and X is a $\mathbb{P}^1$-bundle over Z; $\rho(X) = 2$.

(3) Z is a point, $\rho(X) = 1$ and $-K_X$ is ample. (In fact, $X \cong \mathbb{P}^2$, but this is harder to prove.)

The cases 5.1.1(2) and 5.1.1(3) provide a structure theorem for X, and we call these two contractions Mori fibre spaces. Case 5.1.1(1) introduces a new surface Z. We can apply Theorem 5.1.1 to Z and continue if possible. At each birational step the Picard number drops by one, and so this process terminates. This gives the following:

Theorem 5.1.2. *Let X be a smooth projective surface. There is a sequence of birational contractions $X \to X_1 \to \cdots \to X_n = X'$ such that X' satisfies exactly one of the following conditions:*

(1) $K_{X'}$ is nef;

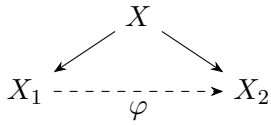
(2) X' is a $\mathbb{P}^1$-bundle over a curve C;

(3) $X' \cong \mathbb{P}^2$.

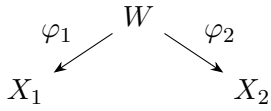
Definition 5.1.3. If $K_{X'}$ is nef as above, then X' is called a minimal model of X.

Remark 5.1.4. When X has non-negative Kodaira dimension, $K_{X'}$ being nef is equivalent to X' being a minimal surface. Without this assumption, $K_{X'}$ being nef still implies that X' is a minimal surface with no (-1)-curves, but the reverse direction fails; see Figure 2.1 for minimal surfaces which are not minimal models in this sense. Therefore, the concepts of minimal surface and minimal model are different. The terminology "minimal model" in the surface theory of [Har77] uses the former convention, whereas Definition 5.1.3 requires nefness of the canonical divisor.

It turns out that, when a minimal model exists, the morphism $X \to X'$ is unique up to a unique isomorphism of the targets compatible with these morphisms. Thus X' is determined by the surface X. Indeed, suppose that there are two minimal models X_1 and X_2 such that K_{X_1} and K_{X_2} are nef. Via the diagram



we obtain a rational map $\varphi : X_1 \dashrightarrow X_2$ and eliminate the indeterminacy to find a smooth common resolution



Choose such a W with minimal Picard number. If φ_1 is not an isomorphism, there is a (-1)-curve E contracted by φ_1. It cannot also be contracted by φ_2: otherwise both morphisms factor through the contraction of E, contradicting the choice of W. Write $C = (\varphi_2)_* E$ on X_2. Then, by

$$K_W = \varphi_2^* K_{X_2} + \sum_i a_i E_i, \qquad a_i > 0,$$

where the E_i are the φ_2-exceptional curves, we have

$$(K_W \cdot E) = (K_{X_2} \cdot C) + \sum_i a_i (E_i \cdot E).$$

The left hand side is -1, and the right hand side is non-negative since K_{X_2} is nef and E is distinct from all the E_i. This is a contradiction. Similarly, φ_2 must be an isomorphism.

Now, run the minimal model program on a given smooth projective surface X:

Step 0: We start with a smooth projective surface X.

Step 1: If K_X is nef, then we go to Step 5. If K_X is not nef, then by the Cone Theorem we can choose a K_X-negative extremal ray $R \subseteq \overline{\mathrm{NE}}(X)$.

Step 2: By Theorem 5.1.1, the contraction morphism $\mathrm{cont}_R : X \longrightarrow Z$ exists. There are two possibilities.

Step 3: If $\dim Z = \dim X$, then go back to Step 0 with Z replacing X.

Step 4: If $\dim Z < 2 = \dim X$, then 5.1.1(2)–(3) gives a structure theorem for X.

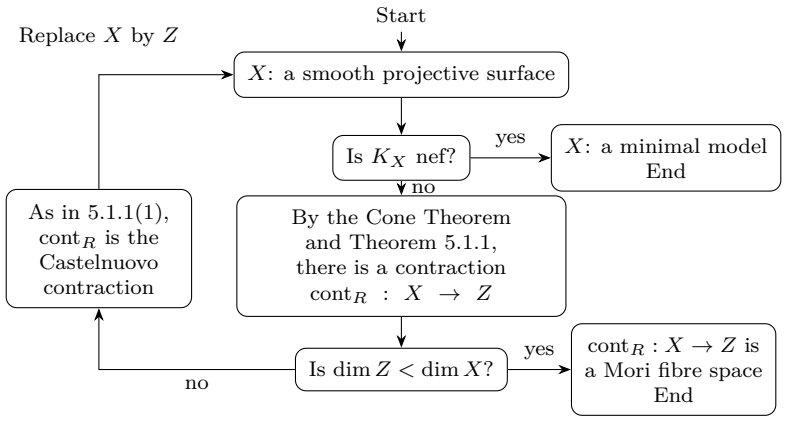


Figure 5.1: Minimal Model Program for Smooth Surfaces.

Step 5: If K_X is nef, then we stop. These surfaces should be investigated by other methods.

The aim of the Minimal Model Program is to provide a similar step by step approach to higher dimensional varieties. The next step is to develop an analogue of Theorem 5.1.1 for higher dimensional varieties. There are already numerous cases for smooth threefolds over $\mathbb{C}$; see [Mor82, §3].

Theorem 5.1.5 ([Mor82])**.** *Let X be a non-singular projective threefold over $\mathbb{C}$ and $\mathrm{cont}_R : X \longrightarrow Y$ the contraction of a K_X-negative extremal ray $R \subseteq \overline{\mathrm{NE}}(X)$. The following is a list of all possibilities for cont_R:*

E: (Exceptional) $\dim Y = 3$, cont_R *is birational, and there are five types of local behaviour near the contracted surface:*

- **E1:** cont_R *is the blow-up of a smooth curve in the smooth threefold Y.*
- **E2:** cont_R *is the blow-up of a smooth point of the smooth threefold Y.*
- **E3:** cont_R *is the blow-up of an ordinary double point of Y. Locally analytically, an ordinary double point is given by $x^2 + y^2 + z^2 + w^2 = 0$.*
- **E4:** cont_R *is the blow-up of a point of Y which is locally analytically given by $x^2 + y^2 + z^2 + w^3 = 0$.*
- **E5:** cont_R *contracts a smooth $\mathbb{P}^2_{\mathbb{C}}$ with normal bundle $\mathcal{O}(-2)$ to a point of multiplicity 4 on Y which is locally analytically the quotient of $\mathbb{C}^3$ by the involution $(x, y, z) \mapsto (-x, -y, -z)$.*

C: (Conic bundle) $\dim Y = 2$, *Y is smooth, and* cont_R *is a fibration whose fibres are plane conics. General fibres are, of course, smooth.*

D: (Del Pezzo fibration) $\dim Y = 1$, *Y is a smooth curve, and general fibres of* cont_R *are Del Pezzo surfaces.*

F: (Fano variety) $\dim Y = 0$, *$-K_X$ is ample, and hence X is a Fano variety with $\rho(X) = 1$.*

[84]

Note that because of the singularities occurring in cases E3, E4 and E5, we cannot apply Theorem 5.1.5 to the resulting variety Y. Thus we need to go back and see if one can prove a version of Theorem 5.1.5 when we allow X to have certain singularities. It turns out that the singularities in E3 and E4 do not cause much trouble. But the singularity in E5 is not a hypersurface singularity, and it is much harder to give a direct geometric analysis of varieties with such singularities.

So it seems that a more conceptual approach is required. As with many questions, there are two opposing points of view:

- Minimalist: We should try to identify the smallest class of singularities which we encounter starting with smooth varieties and applying Theorem 5.1.5 and its generalizations.
- Maximalist: We should try to identify the largest class of singularities where an analogue of Theorem 5.1.1 is possible.

For the beginning of the section we try to follow the maximalist approach. This is most suitable for general discussion. Unfortunately, essentially none of the results needed for the entire program hold without restrictions on the singularities. Thus, when it comes to proofs, we have to get close to the minimalist approach.

In Step 1 we first ask if K_X is nef. For this question to make sense, the intersection numbers of K_X with curves have to be defined. These numbers make sense if K_X is Cartier, or more generally, when K_X is $\mathbb{Q}$-Cartier. We retain this assumption in the following discussion.

Next we have to establish the Cone Theorem for X. For smooth projective varieties this was done in Chapter 2. The main use of smoothness is in the bend-and-break method, to get the lower bound for the dimension of the deformation space; recall Corollary 2.1.17 and the deformation estimates in §2.1. This lower bound frequently fails when X is singular. There are, however, large classes of singularities where the estimate holds, and this leads to the Cone Theorem for many singular varieties. In dimension three a Cone Theorem holds for normal projective varieties with K_X $\mathbb{Q}$-Cartier, whereas the corresponding assertion fails for arbitrary normal varieties in higher dimensions; see [KM98, §2.3] and [Kol92]. This does not by itself give the contraction theorem for arbitrary singularities.

In Step 2 we need to contract the extremal ray $R \subseteq \overline{\mathrm{NE}}(X)$. This is a more delicate result, which holds only under suitable assumptions on the singularities. For the moment we recall a result proved in Chapter 2, which classifies the resulting extremal contractions into three types.

[84] For the threefold contractions described here, see [Mor82, Section 3].

Lemma 5.1.6. *Let X be a normal projective variety and let $R \subseteq \overline{\mathrm{NE}}(X)$ be an extremal ray. Assume that X is $\mathbb{Q}$-factorial and that its contraction to a normal projective variety exists:*

$$f := \mathrm{cont}_R : X \longrightarrow Z.$$

Then one of the following holds:

- *f is birational and $\mathrm{Ex}(f)$ is an irreducible divisor; we say that f is a divisorial contraction;*
- *f is birational and $\mathrm{Ex}(f)$ has codimension at least two; we say that f is a small contraction;*
- *$\dim Z < \dim X$; we say that f is of fibre type.*

The three cases of Lemma 5.1.6 behave very differently from the point of view of our program. Write the target as Y in the following discussion.

Case 1. Fibre type contractions

These correspond to Step 4. If $(R \cdot K_X) < 0$, then $-K_X$ is relatively ample, since the relative cone is generated by R. The general fibre F is normal, and $-K_F = -K_X|_F$ is ample. We call this a Fano fibre space. Thus, at least in principle, we reduce the problem of understanding X to the study of the lower dimensional variety Y and the fibres of f. Moreover, the general fibres are of a very special kind: they are analogues of $\mathbb{P}^1_{\mathbb{C}}$ and of Del Pezzo surfaces.

Case 2. Divisorial contractions

These should correspond to Step 3. For this to be true, we need to establish that K_Y is $\mathbb{Q}$-Cartier, and preferably that Y is $\mathbb{Q}$-factorial. If this holds, then Y can be considered simpler than X, as the Picard number goes down by one.

Case 3. Small contractions

This is a new situation which never happens for surfaces for dimensional reasons. It also does not occur among the K_X-negative extremal contractions of smooth threefolds in Theorem 5.1.5. Small K_X-trivial contractions of smooth threefolds can occur, as the auxiliary construction in Example 5.1.7 will show.

For a small K_X-negative contraction, we claim that Y has singularities for which no positive multiple of K_Y is Cartier. Indeed, assume that mK_Y and mK_X are Cartier for some $m > 0$. Choose compatible canonical divisors. Then mK_X and $f^*(mK_Y)$ agree outside $\mathrm{Ex}(f)$, since f is an isomorphism there. Their difference is a Weil divisor supported in codimension at least two, so it is zero. Hence

$$mK_X = f^*(mK_Y).$$

This is impossible: $(R \cdot K_X) < 0$, whereas $(R \cdot f^*(mK_Y)) = 0$, because the curves in R are contracted to points.

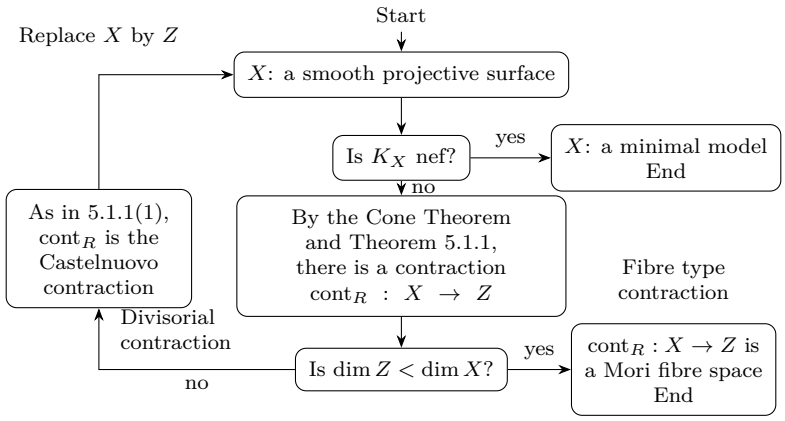


Figure 5.2: Two possibilities correspond to different contractions.

So we have been led out of the class of varieties that we can control. In order to continue at this point, we introduce a new operation called a flip. The flip produces a new variety X^+ and a canonical divisor K_{X^+} which is positive on the new contracted ray R^+. In this sense, it makes a step toward the goal of obtaining a minimal model, where the canonical divisor is nef.

Example 5.1.7 (Example of flips)**.** In this example, for $n \geq 2$, the flip removes the curve $C_n \cong \mathbb{P}^1$ from the singular variety X_n and replaces it with $C_n^+ \cong \mathbb{P}^1$ to obtain the variety X_n^+, which in this case is non-singular.

We start with an auxiliary construction. Consider $Y = V(xy - uv) \subseteq \mathbb{A}^4$, which has an isolated singularity at the origin.

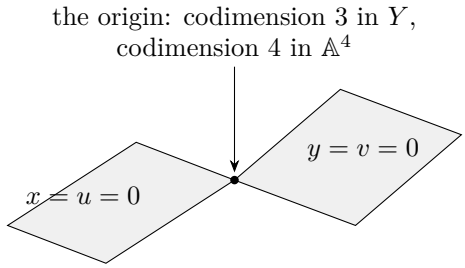


Figure 5.3: Informal diagram of the two planes through the origin.

To investigate $\widetilde{X} = \mathrm{Bl}_0 Y$, we blow up $\mathbb{A}^4$ at the origin:

$$\mathrm{Bl}_0 \mathbb{A}^4 = \left\{ ((x,y,u,v),[X:Y:U:V]) \in \mathbb{A}^4 \times \mathbb{P}^3 : \operatorname{rank} \begin{pmatrix} x & y & u & v \\ X & Y & U & V \end{pmatrix} = 1 \right\}.$$

The projection $\pi : \mathrm{Bl}_0 \mathbb{A}^4 \to \mathbb{A}^4$ is given by the affine coordinates. Thus we have

the equations

$$\begin{cases} xY = yX, & xU = uX, & xV = vX, \\ yU = uY, & yV = vY, & uV = vU. \end{cases}$$

At the origin, the preimage is $\{0\} \times \mathbb{P}^3 \cong \mathbb{P}^3$, with no relation among X, Y, U, V. Now $\mathbb{P}^3$ is covered by four affine pieces:

- On $X \neq 0$, set $X = 1$. The equations of the total transform of Y are

  $$xU = u, \qquad xY = y, \qquad xV = v, \qquad xy - uv = 0.$$

  Substituting into the last equation gives $x^2(Y - UV) = 0$. Here $x = 0$ defines the exceptional divisor E of $\mathrm{Bl}_0 \mathbb{A}^4$, the factor x^2 records its multiplicity two in the total transform, and $Y - UV = 0$ defines the strict transform $\widetilde{X}$. The exceptional set Q of $\widetilde{X}$ is $E \cap \widetilde{X}$.

- On $Y \neq 0$, set $Y = 1$. We have

  $$x = yX, \qquad u = yU, \qquad v = yV, \qquad xy - uv = 0.$$

  Substituting gives $y^2(X - UV) = 0$, where $y = 0$ defines E and $X - UV = 0$ defines $\widetilde{X}$.

Similarly, we obtain the charts for $U \neq 0$ and $V \neq 0$. Therefore the strict transform is

$$\widetilde{X} \subseteq \mathbb{A}^4 \times \mathbb{P}^3, \qquad XY - UV = 0,$$
$$\operatorname{rank} \begin{pmatrix} x & y & u & v \\ X & Y & U & V \end{pmatrix} = 1.$$

Its fibre over the origin is $Q = V(XY - UV) \subseteq \mathbb{P}^3$. This is the quadric surface given by the Segre embedding $\mathbb{P}^1 \times \mathbb{P}^1 \hookrightarrow \mathbb{P}^3$, so Q has two families of lines:

$$\begin{cases} tX = sV, \\ sY = tU, \end{cases} \quad \text{and} \quad \begin{cases} tX = sU, \\ sY = tV, \end{cases} \qquad [s : t] \in \mathbb{P}^1.$$

For $st \neq 0$, writing $c = s/t$, these become the original expressions

$$\begin{cases} X = cV, \\ Y = c^{-1}U, \end{cases} \quad \text{and} \quad \begin{cases} X = cU, \\ Y = c^{-1}V. \end{cases}$$

Indeed, $\det \left(\begin{smallmatrix} X & V \\ U & Y \end{smallmatrix}\right) = 0$; the first family fixes the proportionality of the columns, and the second that of the rows. The homogeneous form also includes the lines for $c = 0, \infty$.

These two families can be blown down to smooth threefolds X and X^+. They can also be obtained by blowing up the ideals (x, v) and (x, u) on Y. In detail:

- Blowing up $(xy - uv = 0)$ along the plane $(x = v = 0)$ gives

$$\begin{aligned} X = \mathrm{Bl}_{(x,v)}\, Y &= \mathrm{Proj}_Y \bigoplus_{d\geq 0} (x,v)^d \\ &\cong \mathrm{Proj}_Y \big(A[S,T]/\ker\varphi\big) \\ &= \big\{\big((x,y,u,v),[S:T]\big) : xT = vS,\ yS = uT\big\} \\ &\subseteq \mathbb{A}^4 \times \mathbb{P}^1_{[S:T]}, \end{aligned}$$

  where $A = \mathbb{C}[x,y,u,v]/(xy-uv)$ and the graded surjection

$$\varphi : A[S,T] \longrightarrow \bigoplus_{d\geq 0} (x,v)^d, \qquad S \longmapsto x, \quad T \longmapsto v$$

  has kernel $(xT - vS, yS - uT)$. Here x, v on the right are in degree one. Equivalently, the defining ideal in $\mathbb{C}[x,y,u,v,S,T]$ consists of the 2×2 minors of

$$\begin{pmatrix} x & u & S \\ v & y & T \end{pmatrix}.$$

  We may omit $xy - uv = 0$ from the incidence equations in $\mathbb{A}^4 \times \mathbb{P}^1$: the other two equations give

$$T(xy-uv) = 0, \qquad S(xy-uv) = 0,$$

  and S, T cannot vanish simultaneously.

- Blowing up $(xy - uv = 0)$ along the plane $(x = u = 0)$ gives

$$\begin{aligned} X^+ = \mathrm{Bl}_{(x,u)}\, Y &= \mathrm{Proj}_Y \bigoplus_{d\geq 0} (x,u)^d \\ &\cong \mathrm{Proj}_Y \big(A[S',T']/\ker\varphi'\big) \\ &= \big\{\big((x,y,u,v),[S':T']\big) : xT' = uS',\ yS' = vT'\big\} \\ &\subseteq \mathbb{A}^4 \times \mathbb{P}^1_{[S':T']}, \end{aligned}$$

  where $\ker\varphi' = (xT' - uS', yS' - vT')$ for the analogous surjection

$$\varphi' : A[S',T'] \longrightarrow \bigoplus_{d\geq 0} (x,u)^d, \qquad S' \longmapsto x, \quad T' \longmapsto u.$$

Now consider the two rulings:

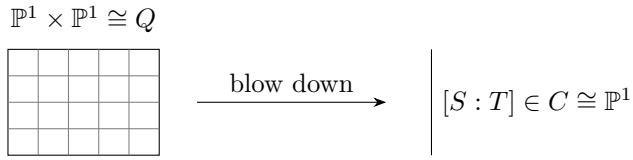
(1) For $X = cV$, $Y = c^{-1}U$, we have $X/V = U/Y = c$ wherever the ratios are defined. Thus

$$\widetilde{X} \longrightarrow X \longrightarrow Y, \qquad Q \longrightarrow C \longrightarrow \{0\}.$$

For $X = V(xT - vS, yS - uT) \subseteq \mathbb{A}^4 \times \mathbb{P}^1$, we have

$$\begin{aligned} &\text{on } \{S \neq 0\}: \quad T/S = v/x = y/u = c^{-1}, \\ &\text{on } \{T \neq 0\}: \quad S/T = x/v = u/y = c. \end{aligned}$$

The ratios on Q are interpreted using $[X : Y : U : V]$ and extend across the vanishing denominators through the incidence equations. They determine the point $[S : T]$ of $C \cong \mathbb{P}^1$.



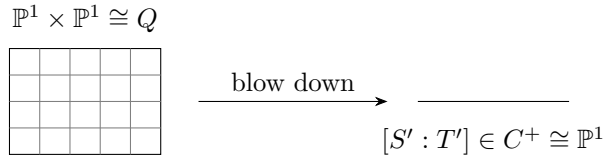
(2) For $X = cU$, $Y = c^{-1}V$, we have $X/U = V/Y = c$. Thus

$$\widetilde{X} \longrightarrow X^+ \longrightarrow Y, \qquad Q \longrightarrow C^+ \longrightarrow \{0\}.$$

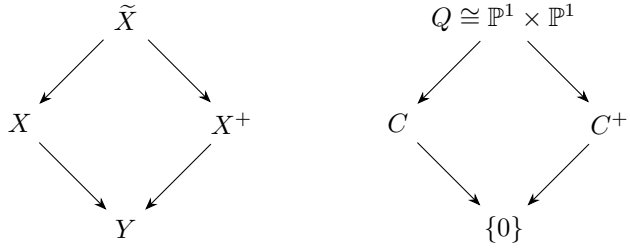
For $X^+ = V(xT' - uS', yS' - vT')$, the corresponding ratios are

$$\begin{aligned} &\text{on } \{S' \neq 0\}: \quad T'/S' = u/x = y/v = c^{-1}, \\ &\text{on } \{T' \neq 0\}: \quad S'/T' = x/u = v/y = c. \end{aligned}$$

They determine a point $[S' : T']$ of $C^+ \cong \mathbb{P}^1$ in the same way.



Therefore we have the following diagram, together with its exceptional loci:



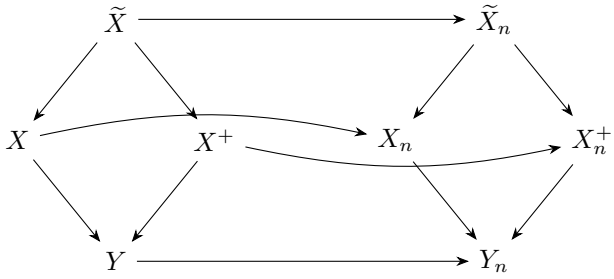
Now consider the action of the cyclic group μ_n, where $n \geq 2$ and $\zeta = \exp(2\pi i/n)$:

$$(x, y, u, v) \longmapsto (\zeta x, y, \zeta u, v).$$

This lifts to $\widetilde{X}, X, X^+$ by the actions

$$[X : Y : U : V] \longmapsto [\zeta X : Y : \zeta U : V],$$
$$[S : T] \longmapsto [\zeta S : T], \qquad [S' : T'] \longmapsto [S' : T'].$$

We denote the quotients by $\widetilde{X}_n, X_n, X_n^+$ and Y_n. The natural morphisms are equivariant, giving the diagram



On Y, the fixed point set is the two-plane $(x = u = 0)$. On the quadric $Q \subseteq \widetilde{X}$, the projective action has two fixed lines:

- $X = U = 0$, on which $[\zeta X : Y : \zeta U : V] = [X : Y : U : V]$;
- $Y = V = 0$, on which $[\zeta X : Y : \zeta U : V] = [\zeta X : \zeta Y : \zeta U : \zeta V]$.

On X, the fixed locus has two components:

- the strict transform of the plane $(x = u = 0) \subseteq Y$, a surface meeting C at one point;
- the image of the line $(Y = V = 0) \subseteq Q$, an isolated fixed point. On the chart $S = 1$, local coordinates are $(x, v' = v/x, u)$, and the action is

$$(x, v', u) \longmapsto (\zeta x, \zeta^{-1} v', \zeta u).$$

On X^+, the blow-up relation is $xT' = uS'$. On $S' = 1$, local coordinates are $(x, u' = u/x, v)$, and the action is

$$(x, u', v) \longmapsto (\zeta x, u', v).$$

The fixed locus here is $x = 0$. On $T' = 1$, the coordinates are $(u, x' = x/u, y)$, with action $(u, x', y) \mapsto (\zeta u, x', y)$, and the fixed locus is $u = 0$. These pieces glue to an irreducible surface containing C^+. Moreover, X_n^+ is smooth: its two charts are

$$\operatorname{Spec} \mathbb{C}[x^n, u', v], \qquad \operatorname{Spec} \mathbb{C}[u^n, x', y].$$

Now we compute the intersections of the canonical classes with the exceptional curves. At the isolated quotient singularity of X_n, the weights are $(1, -1, 1)$. The

action is free away from the origin of this chart, and its character on a top form is ζ. Consequently K_{X_n} has index n there. On the other chart the quotient is smooth, so nK_{X_n} is Cartier everywhere.

On $S = 1$, the curve C is the v'-axis and the volume form

$$\omega_1 = dx \wedge dv' \wedge du$$

is sent to $\zeta\omega_1$. Thus

$$\omega_1^{\otimes n} \longmapsto \zeta^n \omega_1^{\otimes n} = \omega_1^{\otimes n}.$$

It descends to a non-vanishing local section of $\mathcal{O}_{X_n}(nK_{X_n})$ on this quotient chart. On $T = 1$, use coordinates $(y, u' = u/y, v)$, with C the u'-axis. The form

$$\omega_2 = dy \wedge du' \wedge dv$$

is also sent to $\zeta\omega_2$. Its nth tensor power is invariant, but, because this chart has a divisorial fixed locus, it need only descend as a *rational* pluricanonical form.

More precisely, the quotient coordinates are (y, w, v), where $w = (u')^n$, and C_n is the w-axis. A regular frame of $\mathcal{O}_{X_n}(nK_{X_n})$ is $(dy \wedge dw \wedge dv)^{\otimes n}$, whose pullback equals

$$\bigl(n(u')^{n-1}\bigr)^n \omega_2^{\otimes n} = n^n (u')^{n(n-1)} \omega_2^{\otimes n}.$$

Thus, as rational forms on the quotient,

$$\omega_2^{\otimes n} = \frac{1}{n^n} w^{-(n-1)} (dy \wedge dw \wedge dv)^{\otimes n}.$$

On the overlap, $v' = v/x = (u/y)^{-1} = (u')^{-1}$ and $x = u'v$, $u = u'y$. Hence

$$\begin{aligned} \omega_1 &= d(u'v) \wedge d((u')^{-1}) \wedge d(u'y) \\ &= -dv \wedge du' \wedge dy = dy \wedge du' \wedge dv = \omega_2. \end{aligned}$$

Gluing these expressions, we obtain a rational section of $\mathcal{O}_{X_n}(nK_{X_n})$. Its restriction to C_n is regular and non-zero on the first chart and has a pole of order $n-1$ at $w = 0$ on the second chart. Therefore

$$(K_{X_n} \cdot C_n) = -\frac{n-1}{n}.$$

Similarly, for the smooth variety X_n^+, on $S' = 1$ use $(x^n, u' = u/x, v)$ with volume form

$$\omega_1 = d(x^n) \wedge du' \wedge dv.$$

On $T' = 1$, use $(u^n, x' = x/u, y)$ with

$$\omega_2 = d(u^n) \wedge dx' \wedge dy.$$

On the overlap, $x^n = u^n (x')^n$, $u' = (x')^{-1}$ and $v = x'y$, so

$$\omega_1 = -(x')^{n-1} \omega_2.$$

The coordinates u', x' on C^+ are invariant under μ_n, and so also give coordinates on C_n^+. The transition formula therefore shows that

$$(K_{X_n^+} \cdot C_n^+) = n - 1.$$

The quotient morphisms $X_n \to Y_n$ and $X_n^+ \to Y_n$ are projective and are isomorphisms away from these exceptional curves. Their relative cones are each generated by the corresponding curve. The two intersection computations imply that $-K_{X_n}$ is relatively ample and $K_{X_n^+}$ is relatively ample. Thus $X_n^+ \to Y_n$ is the flip of $X_n \to Y_n$ for $n \geq 2$. For $n = 1$, the auxiliary construction has zero canonical degree on both exceptional curves and gives the Atiyah flop.

The operation in the above example can be formalized as follows:

Definition 5.1.8 (Flip)**.** Let $f : X \longrightarrow Y$ be a proper birational morphism of normal varieties such that the exceptional set $\mathrm{Ex}(f)$ has codimension at least two in X. Assume furthermore that K_X is $\mathbb{Q}$-Cartier and $-K_X$ is f-ample. A normal variety X^+ together with a proper birational morphism $f^+ : X^+ \longrightarrow Y$ is called a flip of f if

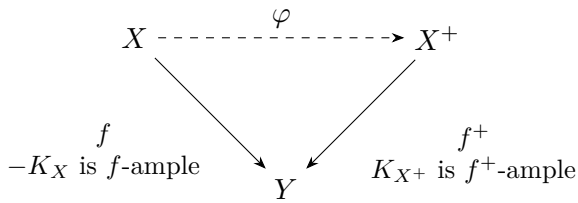
(1) K_{X^+} is $\mathbb{Q}$-Cartier;

(2) K_{X^+} is f^+-ample;

(3) $\mathrm{Ex}(f^+)$ has codimension at least two in X^+.

By a slight abuse of terminology, the rational map $\varphi : X \dashrightarrow X^+$ is also called a flip.

We will see that a flip is unique over Y, and the main question is its existence. A flip gives the following diagram:



If we perform a flip, it is not clear that X^+ is simpler than X. In the example above this happens since X_n is singular but X_n^+ is smooth.

Before we formalize the above discussion, it is necessary to settle on a class of singularities where the Minimal Model Program works. Such a class was already studied in the nineteenth century; see [Die85, VI, 23], in the following setting. Let $Y \subseteq \mathbb{P}^n$, $n \geq 2$, be a normal hypersurface of degree d, possibly singular, and let $f : X \longrightarrow Y$ be a resolution. The adjunction formula gives

$$\omega_Y \cong \mathcal{O}_Y(d - n - 1).$$

One would like to have

$$H^0(X, \mathcal{O}_X(K_X)) \stackrel{?}{=} H^0(Y, \mathcal{O}_Y(K_Y)) = H^0(Y, \mathcal{O}_Y(d-n-1)).$$

The last space has dimension $\binom{d-1}{n}$ (interpreted as zero if $d-1 < n$), by the hypersurface exact sequence and the cohomology of $\mathbb{P}^n$. But the first equality can fail when Y is singular, because its singularities impose "adjunction conditions". Thus it is of interest to understand which singularities impose no adjunction conditions, also for pluricanonical forms. In modern terminology these are canonical singularities:

Definition 5.1.9 (Canonical singularities)**.** Let Y be a normal variety. We say that Y has canonical singularities if

(1) K_Y is $\mathbb{Q}$-Cartier;

(2) for every resolution $f : X \longrightarrow Y$ and every positive integer m divisible by the Cartier index of K_Y, the natural inclusion is an equality:

$$f_*\mathcal{O}_X(mK_X) = \mathcal{O}_Y(mK_Y).$$

Both sides are identified inside the sheaf of rational pluricanonical forms on Y.

The following observation leads to a smaller class. If Y is smooth, $m > 0$, and $\gamma \in H^0(Y, \mathcal{O}_Y(mK_Y))$, then $f^*\gamma$, as a section of $\mathcal{O}_X(mK_X)$, vanishes along every exceptional prime divisor. This is stronger than condition (2) above, which requires only that $f^*\gamma$ have no poles there:

$$H^0(X, \mathcal{O}_X(mK_X)) = H^0(Y, f_*\mathcal{O}_X(mK_X)) \cong H^0(Y, \mathcal{O}_Y(mK_Y)), \qquad f^*\gamma \longleftrightarrow \gamma.$$

This stronger condition gives the next definition, the class used when running the MMP starting from smooth varieties.

Definition 5.1.10 (Terminal singularities)**.** Let Y be a normal variety. We say that Y has terminal singularities if

(1) K_Y is $\mathbb{Q}$-Cartier;

(2) for every resolution $f : X \longrightarrow Y$ and every positive integer m divisible by the Cartier index of K_Y, the natural inclusion is an equality:

$$f_*\mathcal{O}_X(mK_X - E) = \mathcal{O}_Y(mK_Y),$$

where $E \subseteq X$ is the reduced exceptional divisor, i.e. the divisorial part of $\mathrm{Ex}(f)$.

We introduce the concept of higher dimensional minimal models:

Definition 5.1.11 (Minimal model)**.** Let X be a normal proper variety. We say that X is minimal, or is a minimal model, if

(1) X has terminal singularities;

(2) K_X is nef.

If Y is a smooth proper variety birational to X, then X is also called a minimal model of Y.

It should be stressed that, unlike for surfaces (see the discussion after Remark 5.1.4), higher dimensional minimal models are not unique. With these definitions at hand, we can formulate a step by step outline of the Minimal Model Program.

Minimal Model Program

Starting with a smooth projective variety X, we perform a sequence of birational modifications until we arrive at a variety X^* whose global structure is simpler. The following is a more detailed list of the necessary steps.

Step 0 (Initial datum): We start with a projective $\mathbb{Q}$-factorial variety $X = X_0$ over $\mathbb{C}$ with terminal singularities. We set up a recursive procedure which produces intermediate varieties X_i, with the aim of stopping at a final variety X^*. Assume that we have already constructed X_i.

Step 1 (Preparation): If K_{X_i} is nef, then there is nothing to do and we go directly to Step 3(2). If K_{X_i} is not nef, then we use the following two results:

(1) (Cone Theorem)

$$\overline{\mathrm{NE}}(X_i) = \overline{\mathrm{NE}}(X_i)_{K_{X_i}\geq 0} + \sum_j \mathbb{R}_{\geq 0}[C_j].$$

If K_{X_i} is not nef, we can pick a K_{X_i}-negative extremal ray $R_i \subseteq \overline{\mathrm{NE}}(X_i)$.

(2) (Contraction of an extremal ray) Let $\mathrm{cont}_{R_i} : X_i \longrightarrow Y_i$ denote the corresponding extremal contraction.

Step 2 (Birational transformations): By Lemma 5.1.6, there are three possible types of contractions $\mathrm{cont}_{R_i} : X_i \longrightarrow Y_i$. Two of them are used to produce a new variety X_{i+1}:

(1) (Divisorial contraction) If cont_{R_i} is divisorial, set $X_{i+1} = Y_i$. The preservation theorem says that X_{i+1} is again $\mathbb{Q}$-factorial with terminal singularities, so we go back to Step 0 with X_{i+1} and start anew.

(2) (Flipping contraction) If cont_{R_i} is small, set $X_{i+1} = X_i^+$, the flip of cont_{R_i}, provided that it exists. The preservation theorem again gives $\mathbb{Q}$-factorial terminal singularities, so we return to Step 0 with X_{i+1}.

Step 3 (Final outcome): We hope that eventually the procedure stops, giving one of the following two possibilities:

(1) (Mori fibre space) If $\mathrm{cont}_{R_i} : X_i \longrightarrow Y_i$ is a Fano contraction, or fibre type contraction, set $X^* = X_i$. The hope is that the investigation of the lower dimensional variety Y_i and of the fibres provides new methods to study X^*.

(2) (Minimal model) If K_{X_i} is nef, again set $X^* = X_i$. We hope to exploit the semi-positivity of the canonical class in further attempts to understand X^*.

Unlike divisorial contractions, flips do not decrease the Picard number. Thus the surface argument does not establish termination of this program in higher dimensions; termination is a separate issue.

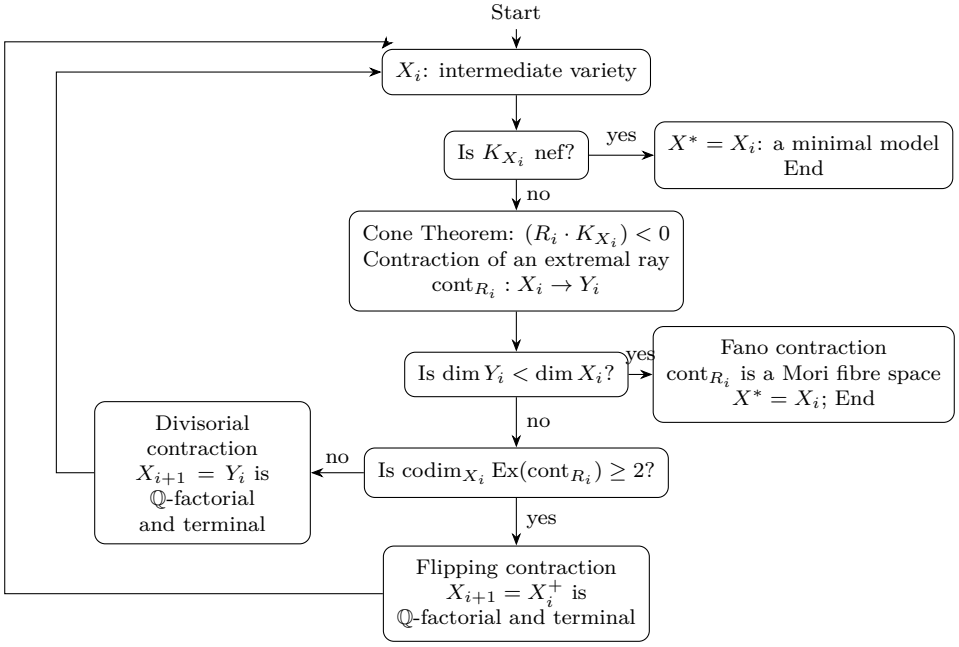


Figure 5.4: Minimal Model Program.

5.2 Singularities in the Minimal Model Program

As we saw in Theorem 5.1.5, contractions of extremal rays lead to singular varieties, as in E3–E5. In order to continue, we have to understand the singularities that occur in the process, and we need a new way to measure how singular a variety is. This new measure is called discrepancy.

Unless otherwise stated, we work over $\mathbb{C}$ and consider varieties of dimension at least two. This dimension convention is needed for the bounds on the infimum over exceptional divisors below. For a normal curve there are no exceptional divisorial valuations over the curve, and the infimum over that empty set would be $+\infty$.

Definition 5.2.1 (Canonical divisor)**.** Let X be a normal variety, and let X_{reg} be its smooth locus. A canonical divisor K_X is a Weil divisor such that

$$\mathcal{O}_X(K_X)|_{X_{\mathrm{reg}}} \cong \omega_{X_{\mathrm{reg}}}.$$

It is well-defined up to linear equivalence.

For a Weil divisor D, the corresponding sheaf $\mathcal{O}_X(D)$ need not be invertible, i.e. a line bundle. It is called a divisorial sheaf and is defined, for an open subset $U \subseteq X$, by

$$H^0(U, \mathcal{O}_X(D)) = \{h \in \mathbb{C}(X)^\times : \operatorname{div}(h)|_U + D|_U \geq 0\} \cup \{0\}.$$

As in the Cartier divisor case in Chapter 1, a non-zero global section s corresponds to a rational function h. Its divisor of zeros is

$$\operatorname{div}(s) = \operatorname{div}(h) + D,$$

which is effective. The same equation defines the divisor of a non-zero rational section, but it need not then be effective. For example, as in Example 1.1.11, the rational section corresponding to $h = 1$ has divisor D.

There is an isomorphism $\mathcal{O}_X(D)_\eta \cong \mathcal{O}_{X,\eta} = \mathbb{C}(X)$ at the generic point, since

$$\mathcal{O}_X(D)_\eta = \varinjlim_{\eta \in U} H^0(U, \mathcal{O}_X(D)),$$

and we can shrink U to avoid the supports of D and $\operatorname{div}(h)$. A morphism $\mathcal{O}_X(D)|_U \to \mathcal{O}_U$ is therefore generically multiplication by a rational function f:

$$H^0(U, \mathcal{O}_X(D)) \longrightarrow H^0(U, \mathcal{O}_X), \qquad h \longmapsto hf.$$

At a prime divisor $P \subseteq U$, write $d_P = \operatorname{coeff}_P D$. Since a generator of $\mathcal{O}_X(D)$ at its generic point has order $-d_P$, this multiplication is regular precisely when

$$v_P(f) - d_P \geq 0.$$

Normality allows us to test this in codimension one. Consequently

$$\mathcal{O}_X(D)^\vee := \mathcal{H}om_{\mathcal{O}_X}(\mathcal{O}_X(D), \mathcal{O}_X) \cong \mathcal{O}_X(-D).$$

Thus $\mathcal{O}_X(D)$ is a reflexive sheaf of rank one: it is isomorphic to its double dual. See also [Sch]. Linearly equivalent divisors have isomorphic divisorial sheaves, though their realizations as subsheaves of $\mathbb{C}(X)$ need not be identical.

5.2A Definition of Discrepancies

Definition 5.2.2 (Discrepancy)**.** Let X be a normal variety such that mK_X is Cartier for some $m > 0$. Suppose that $f : Y \longrightarrow X$ is a not necessarily proper birational morphism from a normal variety Y. Let $E \subseteq Y$ be a prime exceptional divisor. Pick a general closed point $e \in E$ where Y and E are smooth, and choose local parameters $y_1, \ldots, y_n$ with $E = (y_1 = 0)$. Locally near e,

$$f^*(\text{local generator of } \mathcal{O}_X(mK_X)) = y_1^{ma(E,X)}\, u\, (dy_1 \wedge \cdots \wedge dy_n)^{\otimes m},$$

where u is a unit and $ma(E, X) \in \mathbb{Z}$. The rational number $a(E, X)$ is called the discrepancy of E with respect to X.

Proposition 5.2.3. *$a(E, X)$ is independent of the choice of m.*

Proof. Choose positive integers m, m' such that mK_X and $m'K_X$ are Cartier. Then $mm'K_X$ is Cartier. For local generators ω_X, ω'_X of the corresponding sheaves, we have

$$\begin{aligned} f^*\omega_X &= y_1^{ma(E,X)} u\, (dy_1 \wedge \cdots \wedge dy_n)^{\otimes m}, \\ f^*\omega'_X &= y_1^{m'a'(E,X)} u'\, (dy_1 \wedge \cdots \wedge dy_n)^{\otimes m'}. \end{aligned}$$

The powers $\omega_X^{\otimes m'}$ and $(\omega'_X)^{\otimes m}$ are generators of the same line bundle and differ by a unit. Comparing their pullbacks gives

$$mm'a(E, X) = mm'a'(E, X),$$

so $a(E, X) = a'(E, X)$. □

Proposition 5.2.4. *In the above setting, assume that $f : Y \longrightarrow X$ is proper and that mK_Y is also Cartier. Then*

$$mK_Y \sim_{\text{lin}} f^*(mK_X) + \sum_i ma(E_i, X)E_i,$$

where the E_i are the f-exceptional prime divisors.

Proof. On the locus where f is an isomorphism, the canonical sheaves are naturally identified, so the difference is supported on the exceptional divisors. Write

$$mK_Y \sim_{\text{lin}} f^*(mK_X) + \sum_i c_i E_i.$$

At the generic point of E_i, the local ring of the normal variety Y is a discrete valuation ring, with uniformizer y_1. To express the differential form, take parameters at a general closed smooth point of E_i as above. Then

$$f^*(\text{local generator of } \mathcal{O}_X(mK_X)) = y_1^{ma(E_i,X)} u \, (dy_1 \wedge \cdots \wedge dy_n)^{\otimes m}. \tag{5.1}$$

Thus $c_i = ma(E_i, X)$. Equivalently,

$$\mathcal{O}_Y(mK_Y) \cong f^*\mathcal{O}_X(mK_X) \otimes \mathcal{O}_Y\left(\sum_i ma(E_i, X)E_i\right).$$

The sum here is Cartier, although its individual prime components need not be. There is no inclusion in a fixed direction without a sign condition on the coefficients. Choosing compatible canonical divisors gives an equality of Weil divisors, and dividing by m gives

$$K_Y = f^*K_X + \sum_i a(E_i, X)E_i.$$

□

Remark 5.2.5. On a normal variety Y, the one-dimensional local ring $\mathcal{O}_{Y,\eta_E} \subseteq \mathbb{C}(Y)$ at the generic point of a prime divisor E is a discrete valuation ring. It determines a normalized valuation $v(E, Y)$ of $\mathbb{C}(Y) \cong \mathbb{C}(X)$. Such valuations are called algebraic, or divisorial, valuations.

Let $f' : Y' \longrightarrow X$ be another birational morphism and $E' \subseteq Y'$ a prime divisor such that $v(E', Y') = v(E, Y)$. This is equivalent to the induced birational map $Y \dashrightarrow Y'$ identifying open neighbourhoods of the generic points of E and E'. The definition then gives $a(E, X) = a(E', X)$.

Thus the discrepancy depends only on the valuation, not on the particular choice of f and Y. This is why these do not appear in the notation $a(E, X)$.

Definition 5.2.6. Let X be a variety, $f : Y \longrightarrow X$ a not necessarily proper birational morphism from a normal variety, and $E \subseteq Y$ a prime divisor. Such an E is called a divisor over X. The closure of $f(E)$ in X is called its center on X, denoted by $\text{center}_X E$.

As above, the center depends only on the valuation and its center on X. We now generalize discrepancy to pairs (X, Δ) such that $K_X + \Delta$ is $\mathbb{Q}$-Cartier.

Definition 5.2.7 (Generalized discrepancy)**.** Let X be a normal variety and $\Delta = \sum_i a_i D_i$ a sum of distinct prime divisors with rational coefficients. Assume that $K_X + \Delta$ is $\mathbb{Q}$-Cartier. Choose $m > 0$ such that $m(K_X + \Delta)$ is Cartier and $m\Delta$ is integral. Let $f : Y \longrightarrow X$ be a birational morphism from a normal variety, and set

$$f_*^{-1}\Delta = \sum_i a_i f_*^{-1} D_i.$$

For an exceptional prime divisor E_i, choose a general closed point $e_i \in E_i$ and parameters as in Definition 5.2.2, avoiding the strict transform of Δ. Locally near e_i,

$$\begin{aligned} &f^*(\text{local generator of } \mathcal{O}_X(m(K_X + \Delta))) \\ &\quad = y_1^{ma(E_i,X,\Delta)} u \,(\text{local generator of } \mathcal{O}_Y(m(K_Y + f_*^{-1}\Delta))). \end{aligned}$$

This defines $a(E_i, X, \Delta) \in \mathbb{Q}$, with $ma(E_i, X, \Delta) \in \mathbb{Z}$.

Proposition 5.2.8. *In the above situation, on the locus $Y \setminus E$ where f is a local isomorphism, the following line bundles are naturally isomorphic:*

$$\mathcal{O}_Y(m(K_Y + f_*^{-1}\Delta))|_{Y\setminus E} \cong f^*\mathcal{O}_X(m(K_X + \Delta))|_{Y\setminus E}.$$

Here E denotes the non-isomorphism locus; for a proper birational morphism it is $\mathrm{Ex}(f)$*. The sheaf on the left need not be a line bundle on all of Y.*

Proof. This follows from the local isomorphism and the identification of canonical forms there. □

As before, it follows that

$$m(K_Y + f_*^{-1}\Delta) \sim_{\text{lin}} f^*(m(K_X + \Delta)) + \sum_{E_i \text{ exceptional}} ma(E_i, X, \Delta) E_i.$$

We call $a(E, X, \Delta)$ the discrepancy of E with respect to the pair, and write $a(E)$ when no confusion is likely. Twisting by $\mathcal{O}_X(m\Delta)$ allows poles along $m\Delta$; consequently we extend the definition to divisors on X by

$$a(D_i, X, \Delta) = -a_i,$$

and $a(D, X, \Delta) = 0$ for any prime divisor D different from the D_i. As in the case $\Delta = 0$, discrepancy depends on the valuation and not on f.

Remark 5.2.9. Dividing the preceding relation by m gives a $\mathbb{Q}$-linear equivalence of Weil divisors:

$$\begin{aligned} K_Y + f_*^{-1}\Delta &\sim_{\mathbb{Q}} f^*(K_X + \Delta) + \sum_{E_i \text{ exceptional}} a(E_i, X, \Delta) E_i, \\ K_Y &\sim_{\mathbb{Q}} f^*(K_X + \Delta) + \sum_{E_i \text{ arbitrary}} a(E_i, X, \Delta) E_i. \end{aligned}$$

In the second line, $-f_*^{-1}\Delta$ is included in the sum, since

$$a(f_*^{-1}D_i, X, \Delta) = a(D_i, X, \Delta) = -a_i.$$

When the divisors to be intersected are $\mathbb{Q}$-Cartier, these relations imply numerical equivalence on proper curves. In particular, if f is proper, we may use relative numerical equivalence $\equiv_f$ or $\equiv_X$ on curves contracted by f; the base X need not be proper. This interpretation does not make each individual Weil divisor E_i or K_Y $\mathbb{Q}$-Cartier.

Remark 5.2.10. By choosing compatible representatives of the canonical divisors, we obtain the equality of $\mathbb{Q}$-Weil divisors

$$K_Y + f_*^{-1}\Delta = f^*(K_X + \Delta) + \sum_{E_i \text{ exceptional}} a(E_i, X, \Delta)E_i.$$

This was used in the proof of generalized Kawamata–Viehweg vanishing. The equality is meaningful because $K_X + \Delta$ is $\mathbb{Q}$-Cartier; it does not require every term on the left to be $\mathbb{Q}$-Cartier separately.

Lemma 5.2.11. *With notation as above, assume that Δ' is effective and $\mathbb{Q}$-Cartier. Then*

$$a(E, X, \Delta) \geq a(E, X, \Delta + \Delta')$$

for every divisor E over X, with strict inequality if and only if $\operatorname{center}_X E \subseteq \operatorname{Supp} \Delta'$.

Proof. Choose a common multiple $m > 0$ such that $m(K_X + \Delta)$ and $m\Delta'$ are Cartier and all relevant rational coefficients have integral multiples. On a model carrying E, the local generator formulas give the global equalities, with compatible canonical divisors,

$$mK_Y = mf^*(K_X + \Delta) + \sum_i ma(E_i, X, \Delta)E_i,$$
$$mK_Y = mf^*(K_X + \Delta + \Delta') + \sum_i ma(E_i, X, \Delta + \Delta')E_i,$$

where the sums include non-exceptional divisors. Subtracting gives

$$a(E, X, \Delta) - a(E, X, \Delta + \Delta') = \operatorname{ord}_E(f^*\Delta') \geq 0.$$

The order is positive precisely when the local equation of $m\Delta'$ vanishes at the center of E, namely when $\operatorname{center}_X E \subseteq \operatorname{Supp} \Delta'$. □

For us the most important values are the infima of $a(E, X, \Delta)$ as E runs through various sets of divisors. We use several versions:

Definition 5.2.12 (Discrepancy of a pair)**.** The discrepancy of (X, Δ) is

$$\operatorname{discrep}(X, \Delta) = \inf_E \{a(E, X, \Delta) : E \text{ is an exceptional divisor over } X\}.$$

Thus E runs through all prime exceptional divisors on all birational models of X. Equivalently, the infimum is over divisorial valuations whose center on X is non-empty and has codimension at least two. For $\Delta = 0$, write $\operatorname{discrep}(X)$.

The total discrepancy is

$$\operatorname{totaldiscrep}(X, \Delta) = \inf_E \{a(E, X, \Delta) : E \text{ is a divisor over } X\}.$$

This also includes the prime divisors on X, or equivalently all divisorial valuations with a non-empty center on X. For $\Delta = 0$, write $\operatorname{totaldiscrep}(X)$.

5.2B Computation of Discrepancies

First we recall a result of Zariski [Zar39, 19.10], which shows that every divisorial valuation with a center can be reached by a sequence of blow-ups. The proof follows [Art86, 5.2].

Lemma 5.2.13. *Let X be an algebraic variety over an arbitrary field k, not necessarily algebraically closed, and in any positive dimension. Let $(R, \mathfrak{m}_R)$ be a discrete valuation ring of $k(X)$, containing k, such that*[85]

$$\operatorname{trdeg}_k(R/\mathfrak{m}_R) = \dim X - 1.$$

Assume that R has a center on X: there is a morphism $f : Y = \operatorname{Spec} R \longrightarrow X$ which induces the identity on the function field. Let $y \in Y$ be the closed point. Define a sequence as follows:

- *Set $X_0 = X$ and $f_0 = f$.*
- *If $f_i : Y \longrightarrow X_i$ is defined, let $Z_i = \overline{\{x_i\}}$, where $x_i = f_i(y)$. Set $X_{i+1} = \operatorname{Bl}_{Z_i} X_i$, and let $f_{i+1} : Y \longrightarrow X_{i+1}$ be the induced morphism.*

Then f_n induces an isomorphism $\mathcal{O}_{X_n, x_n} \cong R$ for some $n \geq 0$.

Proof. The normalized valuation of the discrete valuation ring is

$$v_R(a) = j \quad \text{if } a = t^j u, \quad j \in \mathbb{Z},$$

where t is a uniformizer and u a unit.

[85] For localization of blow-ups, finite normalization, and domination by valuation rings, see [GW20, 13.27], [Eis95, 13.13], [Har77, I, 6.1–6.2A].

Step 1: Suppose that $f_n : Y \longrightarrow X_n$ is defined and maps y to x_n. The point x_n need not be closed. Dominance gives an inclusion

$$\mathcal{O}_n := \mathcal{O}_{X_n,x_n} \hookrightarrow R,$$

which is local; write $\mathfrak{m}_n = \mathcal{O}_n \cap \mathfrak{m}_R$. Choose generators $z_1, \ldots, z_r$ of $\mathfrak{m}_n$, ordered so that

$$v_R(z_1) \leq v_R(z_2) \leq \cdots \leq v_R(z_r).$$

Set

$$\mathcal{O}'_n = \mathcal{O}_n[z_2/z_1, \ldots, z_r/z_1].$$

This is the coordinate ring of the chart $D_+(z_1)$ of the blow-up of $\operatorname{Spec} \mathcal{O}_n$ at its closed point. To see its relation to X_{n+1}, take the natural flat morphism $g : \operatorname{Spec} \mathcal{O}_n \longrightarrow X_n$. We have the Cartesian diagram

$$\begin{array}{ccc} X_{n+1} \times_{X_n} \operatorname{Spec} \mathcal{O}_n & \longrightarrow & X_{n+1} \\ \downarrow & & \downarrow \\ \operatorname{Spec} \mathcal{O}_n & \xrightarrow[g]{} & X_n \end{array}$$

By base change for relative Proj [EGA II, 3.5.3; GW20, 13.27],

$$\begin{aligned} X_{n+1} \times_{X_n} \operatorname{Spec} \mathcal{O}_n &\cong \operatorname{Proj}_{\mathcal{O}_n} \left(g^* \bigoplus_{d \geq 0} \mathcal{I}_{Z_n}^d \right) \\ &\cong \operatorname{Proj}_{\mathcal{O}_n} \bigoplus_{d \geq 0} \mathfrak{m}_n^d = \operatorname{Bl}_{\mathfrak{m}_n} \operatorname{Spec} \mathcal{O}_n. \end{aligned}$$

Here flatness identifies $g^* \mathcal{I}_{Z_n}^d$ with $\mathfrak{m}_n^d$. The degree-zero part of the localized graded algebra on $D_+(z_1)$ is precisely

$$\left(\left(\bigoplus_{d \geq 0} \mathfrak{m}_n^d \right)_{z_1} \right)_0 = \mathcal{O}_n[z_2/z_1, \ldots, z_r/z_1].$$

Every non-zero ideal of R is principal, so the inverse image of $\mathcal{I}_{Z_n}$ on Y is invertible. The universal property of blowing up gives f_{n+1}. Moreover,

$$v_R(z_i/z_1) = v_R(z_i) - v_R(z_1) \geq 0,$$

so $\mathcal{O}'_n \subseteq R$ and x_{n+1} lies in this chart. Consequently $\mathcal{O}_{n+1}$ is the localization of $\mathcal{O}'_n$ at $\mathcal{O}'_n \cap \mathfrak{m}_R$.

Step 2: We claim that $R = \bigcup_{n \geq 0} \mathcal{O}_n$. We already have $\mathcal{O}_n \subseteq R$. Take $u \in R \backslash \mathcal{O}_n$. Since the blow-ups are birational,

$$k(X) = k(X_n) = \operatorname{Frac} \mathcal{O}_n,$$

so write $u = y_1/y_2$ with $y_i \in \mathcal{O}_n$. The denominator cannot be a unit, since otherwise $u \in \mathcal{O}_n$. Therefore

$$v_R(y_2) > 0, \qquad v_R(y_1) \geq v_R(y_2) > 0,$$

the second inequality following from $u \in R$. Thus both y_i lie in $\mathfrak{m}_n$. Write

$$y_i = \sum_{j=1}^{r} z_j y_{ij}, \qquad y_{ij} \in \mathcal{O}_n,$$

and set

$$y_i' = y_i/z_1 = y_{i1} + (z_2/z_1)y_{i2} + \cdots + (z_r/z_1)y_{ir} \in \mathcal{O}_n' \subseteq \mathcal{O}_{n+1}.$$

Then $u = y_1'/y_2'$ and

$$v_R(y_2') = v_R(y_2) - v_R(z_1) < v_R(y_2).$$

If u is still not in $\mathcal{O}_{n+1}$, both numerator and denominator again belong to its maximal ideal and we repeat. The non-negative integer valuation of the denominator strictly decreases. After finitely many steps it reaches zero, so the denominator is a unit in the corresponding local ring. Hence $u \in \mathcal{O}_{n+s}$ for some s, proving the claim.

Step 3: Choose $t_1, \ldots, t_r \in R$ whose residues form a transcendence basis of $R/\mathfrak{m}_R$ over k. By Step 2, all t_i belong to $\mathcal{O}_n$ for sufficiently large n. Its residue field $k(x_n) = \mathcal{O}_n/\mathfrak{m}_n$ embeds into $R/\mathfrak{m}_R$ and contains these algebraically independent residues. Therefore

$$\operatorname{trdeg}_k k(x_n) = \operatorname{trdeg}_k(R/\mathfrak{m}_R) = \dim X - 1 \quad \text{for } n \gg 0. \tag{5.2}$$

Since $\mathcal{O}_n$ is essentially of finite type over k, the dimension formula gives

$$\dim \mathcal{O}_n = \dim X_n - \dim \overline{\{x_n\}} = \operatorname{trdeg}_k k(X) - \operatorname{trdeg}_k k(x_n) = 1.$$

In particular,

$$\operatorname{trdeg}_{k(x_n)}(R/\mathfrak{m}_R) = \operatorname{trdeg}_k(R/\mathfrak{m}_R) - \operatorname{trdeg}_k k(x_n) = 0.$$

Let R' be the integral closure of $\mathcal{O}_n$ in $k(X)$. Since R is integrally closed,

$$\mathcal{O}_n \subseteq R' \subseteq R \subseteq k(X).$$

The normalization is finite for a ring essentially of finite type over a field, so R' is a finite $\mathcal{O}_n$-module of dimension one. Set $\mathfrak{p} = R' \cap \mathfrak{m}_R$. Then $R'_{\mathfrak{p}}$ is a one-dimensional normal local ring, hence a discrete valuation ring, and it is dominated by R. Two discrete valuation rings with the same fraction field cannot properly dominate one another, so $R'_{\mathfrak{p}} = R$.

Choose finitely many $\mathcal{O}_n$-module generators $v_1, \dots, v_s$ of R'. Step 2 puts all of them in $\mathcal{O}_k$ for sufficiently large k, so $R' \subseteq \mathcal{O}_k \subseteq R$. Every element of $R' \setminus \mathfrak{p}$ is outside the maximal ideal of $\mathcal{O}_k$, hence invertible there. Thus

$$R = R'_{\mathfrak{p}} \subseteq \mathcal{O}_k \subseteq R,$$

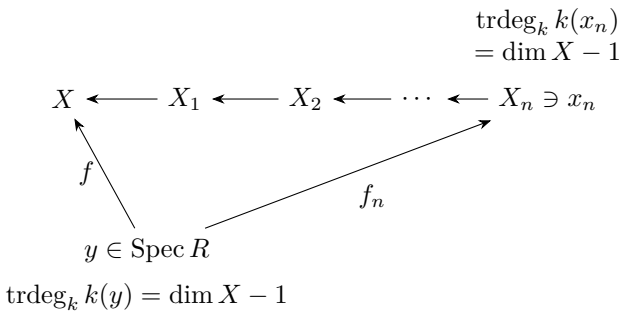
which proves $\mathcal{O}_k = R$. □

We explain the geometrical meaning of the lemma.

Remark 5.2.14. After finitely many blow-ups, we obtain a point x_n with

$$\dim \mathcal{O}_{X_n, x_n} = \operatorname{codim}_{X_n} \overline{\{x_n\}} = 1.$$

Thus $Z_n = \overline{\{x_n\}}$ is the desired prime divisor. The local ring there equals the prescribed valuation ring.



We return to characteristic zero. The next two lemmas make it possible to compute discrepancies in many cases. The first shows what happens under one blow-up, and the second shows how to use this step inductively. Lemma 5.2.13 ensures that all divisorial valuations with centers are covered by this method.

Lemma 5.2.15. *Let X be a normal variety, and $\Delta = \sum_i a_i D_i$ a sum of distinct prime divisors such that $K_X + \Delta$ is $\mathbb{Q}$-Cartier. Let $Z \subseteq X$ be an integral closed subvariety of codimension $k \geq 2$ not contained in $\operatorname{Sing}(X)$. Let $p : \operatorname{Bl}_Z X \longrightarrow X$ be its blow-up, and let E be the unique exceptional component dominating Z. This blow-up is smooth near the generic point of E; if necessary, normalize it when regarding E as a divisor over X. Then*[86]

$$a(E, X, \Delta) = k - 1 - \sum_i a_i \operatorname{mult}_Z D_i.$$

Proof. **Step 1:** If Z is singular, $\operatorname{Bl}_Z X$ need not be normal even when X is smooth. Take

$$U = X_{\mathrm{reg}} \setminus \operatorname{Sing}(Z), \qquad Z_U = Z \cap U.$$

[86] For the exceptional divisor and the canonical divisor of a blow-up along a smooth centre, see [Har77, II, 8.24], [Har77, II, Exercise 8.5(b)].

Then Z_U is a smooth closed subvariety of the smooth variety U, and it contains the generic point of Z. There is a Cartesian diagram

$$\begin{array}{ccc} \mathrm{Bl}_Z X \times_X U & \longrightarrow & \mathrm{Bl}_Z X \\ \downarrow & & \downarrow \\ U & \xrightarrow[j]{} & X \end{array}$$

By base change for Proj,

$$\mathrm{Bl}_Z X \times_X U \cong \mathrm{Proj}_U \bigoplus_{d\geq 0} j^*\mathcal{I}_Z^d \cong \mathrm{Proj}_U \bigoplus_{d\geq 0} \mathcal{I}_{Z_U}^d = \mathrm{Bl}_{Z_U} U.$$

This is smooth. Its exceptional divisor is irreducible, since Z_U is integral. The generic point of E lies here, and

$$\mathcal{O}_{\mathrm{Bl}_Z X, \eta_E} = \mathcal{O}_{\mathrm{Bl}_{Z_U} U, \eta_E}$$

is a discrete valuation ring.

Step 2: We compute on this open set, writing $E_U = E \cap \mathrm{Bl}_{Z_U} U$. The canonical divisor formula for the blow-up of a smooth center gives

$$K_{\mathrm{Bl}_{Z_U} U} - p^* K_U = (k-1)E_U.$$

Also, each $D_i|_U$ is Cartier, and

$$p^*(D_i|_U) = p_*^{-1}(D_i|_U) + \mathrm{mult}_Z D_i \, E_U.$$

Consequently

$$\begin{aligned} a(E_U, U, \Delta|_U)E_U &= (K_{\mathrm{Bl}_{Z_U} U} - p^* K_U) - \left(p^*(\Delta|_U) - p_*^{-1}(\Delta|_U)\right) \\ &= \left(k - 1 - \sum_i a_i \, \mathrm{mult}_Z D_i\right) E_U. \end{aligned}$$

The valuation and discrepancy are unchanged on this neighbourhood of the generic point, so this is also $a(E, X, \Delta)$. □

Lemma 5.2.16. *Let $f : Y \longrightarrow X$ be a proper birational morphism of normal varieties. Let Δ_Y and Δ_X be $\mathbb{Q}$-divisors such that both $K_Y + \Delta_Y$ and $K_X + \Delta_X$ are $\mathbb{Q}$-Cartier, and*

$$K_Y + \Delta_Y \equiv_f f^*(K_X + \Delta_X), \qquad f_*\Delta_Y = \Delta_X.$$

Then, for any divisor F over X,

$$a(F, Y, \Delta_Y) = a(F, X, \Delta_X).$$

Before giving the proof, we need the Negativity Lemma; see [Bir12, 4.15].

Lemma 5.2.17 (Negativity Lemma)**.** *Let $\psi : Y \longrightarrow X$ be a proper birational morphism of normal varieties. Let D be a $\mathbb{Q}$-Cartier divisor on Y such that*

(1) $-D$ is ψ-nef;

(2) $\psi_* D \geq 0$.

Then $D \geq 0$.

Proof. **Step 1:** By Chow's lemma, followed by normalization, there is a proper birational morphism $\mu : Y' \longrightarrow Y$ such that $\psi' = \psi \circ \mu : Y' \longrightarrow X$ is projective. Set $D' = \mu^* D$. For every ψ'-contracted curve C, the projection formula gives

$$(-D' \cdot C)_{Y'} = (-D \cdot \mu_* C)_Y \geq 0.$$

Also,

$$\psi'_* D' = \psi_*(\mu_* \mu^* D) = \psi_* D \geq 0,$$

because $\mu_* \mu^* D = D$: the strict transform pushes forward to D, and the exceptional terms push forward to zero. If $D' \geq 0$, then $D = \mu_* D' \geq 0$. We may therefore assume that ψ is projective.

Step 2: The assertion is local on X, so we may work over an affine open subset of X meeting the image of any proposed negative component. The negative part of D is ψ-exceptional, since $\psi_* D \geq 0$ and the image of ψ-exceptional locus has codimension at least 2. We will construct an effective divisor whose negative is *relatively ample*, and use a contracted curve on a negative component directly.

It is not enough to cut Y by arbitrary general hyperplanes and assert that the restricted divisor still has effective pushforward to the image surface: an exceptional divisor of $Y \to X$ may become non-exceptional after such a cut. Working over an affine base avoids this problem.

Step 3: Choose an effective ψ-very ample Cartier divisor H_Y which contains no exceptional prime divisor. Since X is affine, we can choose an effective principal Cartier divisor H_X whose support contains $\psi(H_Y)$ and the images of all exceptional prime divisors. Taking its defining function to a sufficiently high power ensures

$$E := \psi^* H_X - H_Y \geq 0.$$

Every exceptional prime divisor occurs in E with positive coefficient, and $-E$ is ψ-ample because it is relatively linearly equivalent to H_Y. In particular, for every contracted integral curve C,

$$(\psi^* H_X \cdot C) = 0, \qquad (-E \cdot C) = (H_Y \cdot C) > 0. \tag{5.3}$$

Here E is allowed to have non-exceptional components; only its effectiveness and positivity on every negative component of D are needed.

Step 4: Suppose that D is not effective. Choose the smallest non-negative rational number e for which $D + eE \geq 0$. Explicitly,

$$e = \max_{\operatorname{coeff}_P D<0} \frac{-\operatorname{coeff}_P D}{\operatorname{coeff}_P E} > 0.$$

There is an exceptional prime divisor P with negative coefficient in D and coefficient zero in $D + eE$. Choose a general closed point of P outside $\operatorname{Supp}(D + eE)$. Since P is exceptional, its general fibre over $\psi(P)$ has positive dimension; it is projective. By cutting this fibre with hyperplanes through the chosen point, we obtain a contracted integral curve $C \subseteq P$ which is not contained in $\operatorname{Supp}(D+eE)$.

The effective $\mathbb{Q}$-Cartier divisor $D + eE$ therefore satisfies $(D + eE) \cdot C \geq 0$. On the other hand, $-D$ is ψ-nef and (5.3) gives

$$(D + eE) \cdot C = D \cdot C + e(E \cdot C) < 0,$$

a contradiction. Hence $D \geq 0$. □

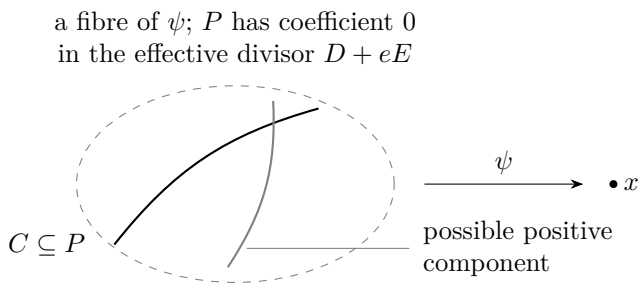


Figure 5.5: A contracted curve C not contained in $\operatorname{Supp}(D + eE)$. Any intersections with its positive components have non-negative degree.

We now prove Lemma 5.2.16.

Proof of Lemma 5.2.16. Our strategy is to use the Negativity Lemma to show that a divisor is zero, and then compare coefficients.

Step 1: Choose compatible canonical divisors, so that $f_*K_Y = K_X$, and set

$$D = (K_Y + \Delta_Y) - f^*(K_X + \Delta_X).$$

This is $\mathbb{Q}$-Cartier. Pushing forward gives

$$\begin{aligned} f_*D &= f_*(K_Y + \Delta_Y) - f_*f^*(K_X + \Delta_X) \\ &= (K_X + \Delta_X) - (K_X + \Delta_X) = 0. \end{aligned}$$

Moreover $D \equiv_f 0$, so both D and $-D$ are f-nef. Applying the Negativity Lemma to D and $-D$ gives $D \geq 0$ and $-D \geq 0$, whence $D = 0$.

Step 2: Given a divisor F over X, take a common birational model $g : W \longrightarrow Y$ on which F appears, and write $h = f \circ g$. Properness of f ensures that the valuation has a center on Y. With compatible canonical divisors,

$$K_W = g^*(K_Y + \Delta_Y) + \sum_i a(E_i, Y, \Delta_Y)E_i,$$
$$K_W = h^*(K_X + \Delta_X) + \sum_i a(E_i, X, \Delta_X)E_i,$$

where the sums range over the prime divisors on W with non-zero coefficients. Since

$$g^*(K_Y + \Delta_Y) - h^*(K_X + \Delta_X) = g^*D = 0,$$

comparison of the coefficient of F proves the assertion. □

Remark 5.2.18. If we assume only the relative numerical equivalence in Lemma 5.2.16, subtraction gives merely

$$\sum_i \big(a(E_i, Y, \Delta_Y) - a(E_i, X, \Delta_X)\big)E_i \equiv_h 0$$

on a common model. Numerical triviality by itself does not identify the coefficients. The exceptional support and the Negativity Lemma are needed to deduce the divisor equality. If the stronger equality $K_Y + \Delta_Y = f^*(K_X + \Delta_X)$ is assumed from the outset, then the comparison in Step 2 is immediate and no Negativity Lemma is needed.

Example 5.2.19. For a proper birational morphism $f : Y \longrightarrow X$ of normal varieties, choose compatible canonical divisors and write

$$K_Y = f^*(K_X + \Delta_X) + \sum_{E_i \text{ arbitrary}} a(E_i, X, \Delta_X)E_i.$$

Set

$$\begin{aligned}\Delta_Y &= \sum_{E_i \text{ arbitrary}} -a(E_i, X, \Delta_X)E_i \\ &= \sum_{D_i \subseteq \operatorname{Supp} \Delta_X} a_i f_*^{-1} D_i - \sum_{E_j \text{ exceptional}} a(E_j, X, \Delta_X)E_j.\end{aligned}$$

Then $K_Y + \Delta_Y = f^*(K_X + \Delta_X)$ is $\mathbb{Q}$-Cartier, and $f_*\Delta_Y = \Delta_X$. This is the direct, crepant-pullback case of Lemma 5.2.16.

The following is the first computation of discrepancies of pairs.

Corollary 5.2.20. *Let X be a normal variety of dimension at least two, and $\Delta = \sum_i a_i D_i$ a $\mathbb{Q}$-divisor such that $K_X + \Delta$ is $\mathbb{Q}$-Cartier. Then:*

(1) Either $\operatorname{discrep}(X, \Delta) = -\infty$*, or*

$$-1 \leq \operatorname{totaldiscrep}(X, \Delta) \leq \operatorname{discrep}(X, \Delta) \leq 1.$$

(2) If X *is smooth, then* $\operatorname{discrep}(X, 0) = 1$*.*

(3) If X *is smooth,* $\sum_i D_i$ *is an snc divisor, and* $a_i \leq 1$ *for every* i*, then*

$$\operatorname{discrep}(X, \Delta) = \min \left\{ \min_{i \neq j,\ D_i \cap D_j \neq \varnothing} (1 - a_i - a_j), \quad \min_i (1 - a_i), \quad 1 \right\}.$$

An infimum over an empty index set is omitted from this minimum.

Proof. **For (1)**

Choose an integral codimension-two subvariety Z whose generic point lies in the smooth locus of X and outside $\operatorname{Supp} \Delta$. We only need Z smooth on a neighbourhood of that generic point. The exceptional divisor E_0 dominating Z on the blow-up has

$$a(E_0, X, \Delta) = 2 - 1 = 1$$

by Lemma 5.2.15. Thus $\operatorname{discrep}(X, \Delta) \leq 1$. The inclusion of the set of exceptional divisors into the set of all divisors over X gives

$$\operatorname{totaldiscrep}(X, \Delta) \leq \operatorname{discrep}(X, \Delta) \leq 1.$$

Suppose now that $\operatorname{totaldiscrep}(X, \Delta) < -1$. There is a divisor E over X with $a(E, X, \Delta) = -1 - c$, where $c > 0$. Take a normal birational model $f : Y \longrightarrow X$ carrying E and define

$$K_Y + \Delta_Y = f^*(K_X + \Delta)$$

as in Example 5.2.19. The coefficient of E in Δ_Y is $1 + c$.

Choose a codimension-two subvariety $Z_0 \subseteq E$ whose generic point lies where Y and E are smooth and no other component of Δ_Y is present. Near that point, E is a smooth divisor and $\operatorname{mult}_{Z_0} E = 1$. We may shrink to make this a blow-up of smooth varieties along a smooth center. For $g_1 : Y_1 \longrightarrow Y$ with exceptional divisor E_1, Lemma 5.2.15 gives

$$a(E_1, Y, \Delta_Y) = 1 - (1 + c) = -c.$$

By crepant pullback, $a(E_1, X, \Delta) = -c$ as well.

Let Z_1 be the intersection of E_1 and the strict transform of E. Put

$$\Delta_{Y_1} = g_{1*}^{-1} \Delta_Y + cE_1, \qquad K_{Y_1} + \Delta_{Y_1} = g_1^*(K_Y + \Delta_Y).$$

Blowing up Z_1 gives an exceptional divisor E_2 with

$$a(E_2, Y_1, \Delta_{Y_1}) = 1 - c - (1 + c) = -2c.$$

Again this equals $a(E_2, X, \Delta)$. Repeating along the intersection of the strict transform of E with the newest exceptional divisor gives

$$a(E_j, X, \Delta) = -jc \longrightarrow -\infty.$$

Therefore $\operatorname{discrep}(X, \Delta) = -\infty$ whenever $\operatorname{totaldiscrep}(X, \Delta) < -1$.

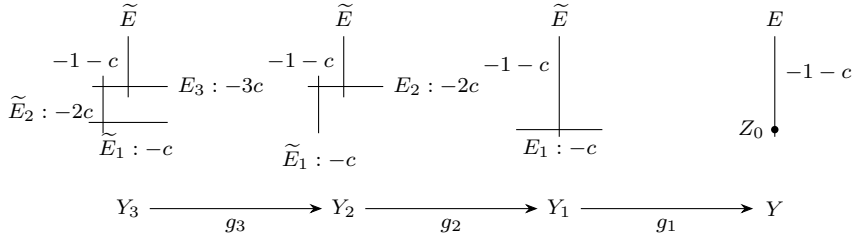


Figure 5.6: Discrepancies of the exceptional divisors under successive blow-ups.

For (2)

This is the special case $\Delta = 0$ of (3); the argument below also shows directly that every exceptional discrepancy over a smooth variety is at least one.

For (3)

Set

$$r(X, \Delta) = \min \left\{ \min_{i \neq j,\ D_i \cap D_j \neq \varnothing} (1 - a_i - a_j), \quad \min_i (1 - a_i), \quad 1 \right\}.$$

First we construct blow-ups realizing the terms in this minimum:

(1) Blow up an irreducible component Z of $D_i \cap D_j$. At its generic point no other boundary component is present, and

$$a(E, X, \Delta) = 2 - 1 - a_i - a_j = 1 - a_i - a_j.$$

(2) Blow up a general codimension-two subvariety $Z \subseteq D_i$, not contained in another D_j. Then

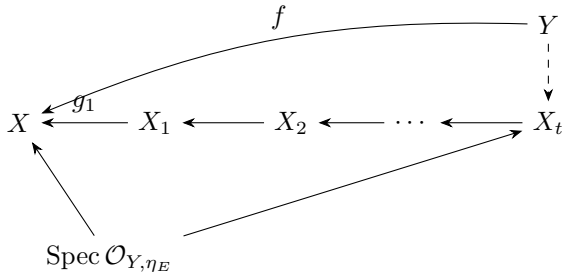
$$a(E, X, \Delta) = 2 - 1 - a_i = 1 - a_i.$$

(3) Blow up a codimension-two center generically outside $\operatorname{Supp} \Delta$. Then $a(E, X, \Delta) = 1$.

These centers can be taken smooth near their generic points; discrepancy is determined there. Hence $\operatorname{discrep}(X, \Delta) \leq r(X, \Delta)$.

For the converse, let E be an exceptional divisor over X. By Lemma 5.2.13, its valuation is obtained after a finite succession of blow-ups of its centers. Near

the generic point of each center, we may shrink so that the center is smooth and has normal crossings with the current snc boundary. Indeed, the boundary components containing the center define a smooth stratum, and the center is generically smooth inside that stratum; the remaining components can be avoided. Shrinking cannot decrease the displayed minimum r, since it only removes terms.



Induct on the number t of blow-ups. Let the first center have codimension $k \geq 2$ and be contained in exactly $D_1, \dots, D_b$. Since the support is snc, $D_1 \cap \cdots \cap D_b$ has codimension b and contains the center, so $b \leq k$. For the first exceptional divisor E_1,

$$e := a(E_1, X, \Delta) = k - 1 - \sum_{l=1}^{b} a_l.$$

There are three cases:

(1) If $b = 0$, then $e = k - 1 \geq 1 \geq r(X, \Delta)$.

(2) If $b = 1$, then $e = k - 1 - a_1 \geq 1 - a_1 \geq r(X, \Delta)$.

(3) If $b \geq 2$, then

$$\begin{aligned} e &= (k - b - 1) + \sum_{l=1}^{b} (1 - a_l) \\ &\geq -1 + (1 - a_1) + (1 - a_2) = 1 - a_1 - a_2 \geq r(X, \Delta), \end{aligned}$$

since every $a_l \leq 1$.

Thus the case $t = 1$ is proved.

Define

$$\Delta_1 = g_{1*}^{-1}\Delta - eE_1, \qquad K_{X_1} + \Delta_1 = g_1^*(K_X + \Delta).$$

The support is snc on the chosen neighbourhood. Since $e \geq r(X, \Delta) \geq -1$, the new coefficient $-e$ is at most one. In computing $r(X_1, \Delta_1)$, the terms involving only strict transforms already occurred in $r(X, \Delta)$. The new terms are $1 + e$ and, for components meeting E_1, $1 + e - a_i$. Each is at least e, because $a_i \leq 1$. Therefore

$$r(X_1, \Delta_1) \geq \min\{r(X, \Delta), e\} = r(X, \Delta).$$

If E is not E_1, the induction hypothesis on the remaining blow-ups and Lemma 5.2.16 give

$$a(E, X, \Delta) = a(E, X_1, \Delta_1) \geq r(X_1, \Delta_1) \geq r(X, \Delta).$$

This completes the proof. □

Under suitable conditions, discrepancy can be computed from the exceptional divisors on a fixed resolution:

Corollary 5.2.21. *Let (X, Δ) be a pair with $K_X + \Delta$ $\mathbb{Q}$-Cartier, and let $f : Y \longrightarrow X$ be a resolution, with exceptional prime divisors E_i.*

(1) In the case $\Delta = 0$, assume K_X is $\mathbb{Q}$-Cartier and

$$0 \leq \min_i a(E_i, X) \leq 1.$$

Then $\operatorname{discrep}(X) = \min_i a(E_i, X)$.

(2) Write $\Delta = \sum_j a_j D_j$, where $a_j \leq 1$. There is a log resolution for which $\sum_j f_*^{-1} D_j$ is smooth, i.e. its components are smooth and pairwise disjoint. For any such resolution, if $a(E_i, X, \Delta) \geq -1$ for every i, then

$$\operatorname{discrep}(X, \Delta) = \min \left\{ \min_i a(E_i, X, \Delta), \quad \min_j (1 - a_j), \quad 1 \right\}.$$

Empty minima are omitted; the assumption in (1) in particular requires at least one exceptional divisor.

Proof. **For (1)**

Define Δ_Y by $K_Y + \Delta_Y = f^* K_X$. Then

$$\Delta_Y = -\sum_i a(E_i, X) E_i \leq 0.$$

Monotonicity and Corollary 5.2.20(2) imply

$$\operatorname{discrep}(Y, \Delta_Y) \geq \operatorname{discrep}(Y, 0) = 1.$$

Every exceptional valuation over X is either one of the E_i or exceptional over Y, so

$$\begin{aligned} \operatorname{discrep}(X) &= \min\{\operatorname{discrep}(Y, \Delta_Y), \ \min_i a(E_i, X)\} \\ &= \min_i a(E_i, X). \end{aligned}$$

For (2)

Start with a log resolution $g : Z \longrightarrow X$, so that the union of the strict transforms of the D_j and the exceptional divisors is snc. Whenever two strict

transforms meet, blow up their whole intersection (a disjoint union of smooth codimension-two subvarieties). Their new strict transforms are disjoint, the total boundary remains snc, and pairs which were disjoint stay disjoint. Thus the number of intersecting pairs of strict transforms decreases. Repeating finitely many times gives a resolution $f : Y \longrightarrow X$ with pairwise disjoint strict transforms of the D_j.

Set $b_i = -a(E_i, X, \Delta) \leq 1$ and

$$\Delta_Y = \sum_j a_j f_*^{-1} D_j + \sum_i b_i E_i, \qquad K_Y + \Delta_Y = f^*(K_X + \Delta).$$

By Corollary 5.2.20, $\operatorname{discrep}(Y, \Delta_Y)$ is the minimum of the applicable numbers

$$1 - b_i - a_j, \quad 1 - b_i - b_{i'}, \quad 1 - a_j, \quad 1 - b_i, \quad 1,$$

where the first two types require the corresponding divisors to meet. There is no term $1 - a_j - a_{j'}$, since the strict transforms are disjoint. Moreover,

$$\begin{aligned} \operatorname{discrep}(X, \Delta) &= \min\{\operatorname{discrep}(Y, \Delta_Y),\ \min_i a(E_i, X, \Delta)\} \\ &= \min\{\operatorname{discrep}(Y, \Delta_Y),\ \min_i(-b_i)\}. \end{aligned}$$

Since all $b_i, a_j \leq 1$,

$$-b_i \leq 1 - b_i - b_{i'}, \qquad -b_i \leq 1 - b_i - a_j, \qquad -b_i \leq 1 - b_i.$$

All terms involving an exceptional coefficient can therefore be removed from the minimum, leaving

$$\operatorname{discrep}(X, \Delta) = \min\{\min_i(-b_i),\ \min_j(1 - a_j),\ 1\}.$$

□

5.2C Singularities for the Pair

Next we define the six classes of singularities which are most important for the Minimal Model Program.

Definition 5.2.22 (Singularities for pairs)**.** Let X be a normal variety and $\Delta = \sum_i a_i D_i$ a sum of distinct prime divisors with $a_i \in \mathbb{Q}$. Assume that $K_X + \Delta$ is $\mathbb{Q}$-Cartier. We say that (X, Δ) is

$$\begin{array}{rll} \text{terminal} & \text{if} & \operatorname{discrep}(X, \Delta) > 0, \\ \text{canonical} & \text{if} & \operatorname{discrep}(X, \Delta) \geq 0, \\ \text{klt} & \text{if} & \operatorname{discrep}(X, \Delta) > -1 \text{ and } \lfloor \Delta \rfloor \leq 0, \\ \text{plt} & \text{if} & \operatorname{discrep}(X, \Delta) > -1, \\ \text{lc} & \text{if} & \operatorname{discrep}(X, \Delta) \geq -1. \end{array}$$

Here klt stands for "Kawamata log terminal", plt for "purely log terminal", and lc for "log canonical".

If (X, Δ) is lc, then $a_i \leq 1$ for every i. Otherwise, at a general smooth point of a component with $a_i > 1$, the blow-up construction in Corollary 5.2.20(1) gives exceptional divisors E_j such that

$$a(E_j, X, \Delta) = j(1 - a_i).$$

Here one first blows up a codimension-two center in D_i, then successively the intersection of its strict transform with the newest exceptional divisor:

$$\begin{array}{ccccccc} X_3 & \xrightarrow{g_3} & X_2 & \xrightarrow{g_2} & X_1 & \xrightarrow{g_1} & X \\ E_3 & & E_2 & & E_1 & & D_i \\ 3(1-a_i) & & 2(1-a_i) & & 1-a_i & & -a_i \end{array}$$

Since $1 - a_i < 0$, eventually $j(1 - a_i) < -1$, contradicting log canonicity.

Definition 5.2.23 (Divisorial log terminal)**.** Let X be a normal variety and $\Delta = \sum_i a_i D_i$ a sum of distinct prime divisors, with rational coefficients $0 \leq a_i \leq 1$. Assume that $K_X + \Delta$ is $\mathbb{Q}$-Cartier. We say that (X, Δ) is dlt, or divisorial log terminal, if and only if there is a closed subset $Z \subseteq X$ such that

(1) $X \setminus Z$ is smooth and $\Delta|_{X \setminus Z}$ has snc support;

(2) if E is a divisor over X with $\operatorname{center}_X E \subseteq Z$, then $a(E, X, \Delta) > -1$.

Each of these notions has an important place in the theory of minimal models:

1. Terminal: The Natural Foundation of the MMP

Core Definition: We require $\operatorname{discrep}(X, \Delta) > 0$. For $\Delta = 0$, terminal singularities give the standard class in which to run the MMP starting from smooth varieties.

Geometric Motivation: Suppose we start with a smooth projective variety X. A divisorial contraction or a flip may produce a singular variety X'. Every exceptional divisor over a smooth variety has positive discrepancy. The preservation theorems for the MMP ensure that a $\mathbb{Q}$-factorial terminal starting variety again gives $\mathbb{Q}$-factorial terminal varieties after these operations. Thus terminal singularities allow us to continue the program after smoothness is lost. This preservation property is the point needed here; it is not a proof of a set-theoretic minimality assertion about all possible classes of singularities.

2. Canonical: General Type and Canonical Models

Core Definition: We require $\operatorname{discrep}(X, \Delta) \geq 0$. In particular, canonical singularities occur on the canonical models of smooth varieties of general type.

Geometric Motivation: For a smooth variety of general type, we consider the canonical ring

$$R(X) = \bigoplus_{m \geq 0} H^0(X, \mathcal{O}_X(mK_X)).$$

If this ring is finitely generated, we can define $X_{\text{can}} = \operatorname{Proj} R(X)$. For a normal variety X with canonical singularities, pluricanonical forms acquire no poles along exceptional divisors when pulled back to a resolution $f : Y \longrightarrow X$. For positive multiples of the index, this gives

$$H^0(X, \mathcal{O}_X(mK_X)) \cong H^0(Y, \mathcal{O}_Y(mK_Y)).$$

This invariance is essential for canonical models. Moduli compactifications require a larger class: for example, KSBA stable varieties may have semi-log-canonical singularities and need not even be normal. They are not restricted to canonical singularities.

3. klt (Kawamata Log Terminal): The Domain of Vanishing Theorems

Core Definition: We require $\operatorname{discrep}(X, \Delta) > -1$ and $\lfloor \Delta \rfloor \leq 0$.

Geometric Motivation: Kawamata–Viehweg vanishing and the Basepoint-Free Theorem naturally apply to klt pairs, together with their respective positivity hypotheses. On a log resolution, the relevant coefficients lie strictly below one, which is the boundary condition underlying the vanishing argument. At discrepancy -1, the standard klt form of these theorems need not apply; this does not mean that every vanishing theorem fails for lc pairs.

Why introduce a larger class for induction? Adjunction to a divisor takes the form

$$(K_X + S + B)|_S = K_S + B_S.$$

If the pair $(X, S + B)$ is required to be klt, the coefficient of S cannot be one. Thus klt pairs do not directly include the coefficient-one boundary needed in this adjunction setup. This motivates plt pairs; it does not rule out inductive arguments involving klt varieties by other methods or by perturbing the boundary.

4. plt (Purely Log Terminal): Adjunction and Induction

Core Definition: We allow coefficients equal to one in the boundary, but require $a(E, X, \Delta) > -1$ for every exceptional divisor E.

Geometric Motivation: If $(X, S + B)$ is plt, with $B \geq 0$ and S a prime divisor of coefficient one, then S is normal. Adjunction produces a boundary B_S, the different, such that

$$(K_X + S + B)|_S = K_S + B_S,$$

and (S, B_S) is klt. In the smooth transverse case $B_S = B|_S$, but in general it includes contributions from the singularities. Together with suitable positivity hypotheses, vanishing and restriction sequences can then relate sections on X and S. The basic sequence is

$$0 \longrightarrow \mathcal{O}_X(-S) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_S \longrightarrow 0;$$

when S is not Cartier, $\mathcal{O}_X(-S)$ here is its divisorial ideal sheaf. Thus the plt condition supplies a useful setup for induction on dimension.

5. lc (Log Canonical) and dlt: The Log Canonical Boundary

Core Definition: Log canonicity requires $\operatorname{discrep}(X, \Delta) \geq -1$. Discrepancies are still defined below this range, but the infimum then equals $-\infty$.

Geometric Motivation: As proved above, once a divisor with discrepancy less than -1 appears, further blow-ups yield discrepancies tending to $-\infty$. The lc condition excludes this behaviour. It nevertheless permits complicated singularities, and the standard klt vanishing and contraction arguments cannot simply be applied without further work.

The dlt class retains lc behaviour while requiring discrepancies over the bad locus Z in Definition 5.2.23 to be strictly greater than -1. Exceptional divisors of discrepancy -1 are still allowed over the snc boundary strata of $X \setminus Z$. For example, blowing up the intersection of two coefficient-one boundary components on a smooth surface gives discrepancy -1, although the original snc pair is dlt. Thus dlt does not prohibit all exceptional divisors of discrepancy -1. It gives a controlled setting for boundary strata and adjunction.

Remark 5.2.24 (Discrepancy in the $\mathbb{R}$-Cartier case)**.** For a pair with $K_X + \Delta$ $\mathbb{Q}$-Cartier, we defined discrepancy using the vanishing order of a pulled-back local generator after clearing denominators. This line-bundle description applies directly to rational coefficients. In algebraic geometry, we can also treat $K_X + \Delta$ $\mathbb{R}$-Cartier by using divisor equalities.

Choose compatible canonical divisors and write

$$K_Y + f_*^{-1}\Delta = f^*(K_X + \Delta) + \sum_{E_i \text{ exceptional}} a(E_i, X, \Delta)E_i.$$

Equivalently, including all prime divisors on Y in the sum,

$$\sum_i a(E_i, X, \Delta)E_i = K_Y - f^*(K_X + \Delta).$$

This defines the real discrepancy $a(E, X, \Delta)$, independently of the model. We also set

$$\begin{aligned} \operatorname{discrep}(X, \Delta) &= \inf\{a(E, X, \Delta) : E \text{ exceptional over } X\}, \\ \operatorname{totaldiscrep}(X, \Delta) &= \inf\{a(E, X, \Delta) : E \text{ a divisor over } X\}. \end{aligned}$$

The previous arguments which use divisor equalities, orders and inequalities remain valid for real coefficients. This includes Lemmas 5.2.11 and 5.2.15–5.2.17 and Corollaries 5.2.20–5.2.21. In the Negativity Lemma, the same proof uses the minimum real number e, rather than a rational one. The definitions of terminal, canonical, klt, plt, lc and dlt extend accordingly.

The following is a Bertini-type comparison between special and general members of a linear series.

Corollary 5.2.25. *Let X be a proper normal variety, $\Delta \geq 0$ an effective $\mathbb{R}$-divisor with $K_X + \Delta$ $\mathbb{R}$-Cartier, and $|L|$ a non-empty linear series of effective Cartier divisors. Let $L_0 \in |L|$ and let $L_g \in |L|$ be a general member. Then, for every $c \geq 0$,*[87]

$$\operatorname{discrep}(X, \Delta + cL_0) \leq \operatorname{discrep}(X, \Delta + cL_g).$$

Reduction to the rational Bertini theorem. For rational coefficients, we use the Bertini discrepancy comparison of [KM98, Corollary 2.33], with properness and effectivity explicitly imposed here. The proof below explains the passage to real coefficients and the discrepancy bookkeeping on a resolution. The rational Bertini comparison is an external input, rather than a consequence of birational invariance of discrep.

Step 1: Choose a resolution $f : Y \longrightarrow X$ such that

(1) $f^*|L| = B + |F|$, with $B \geq 0$ and $|F|$ free;

(2) the strict transforms of the components of $\operatorname{Supp} \Delta$ are smooth and pairwise disjoint;

(3) $\operatorname{Supp} B \cup f_*^{-1} \operatorname{Supp} \Delta \cup \operatorname{Ex}(f)$ is snc.

For $L_\lambda \in |L|$, write $f^* L_\lambda = B + F_\lambda$ and

$$K_Y + \Delta_Y = f^*(K_X + \Delta), \qquad \Gamma = \Delta_Y + cB.$$

Then

$$f^*(K_X + \Delta + cL_\lambda) = K_Y + \Gamma + cF_\lambda.$$

For general F_g, the support upstairs is snc. The conditions on a general member depend on the fixed supports, not on their coefficients.

Step 2: Let E_i be the f-exceptional prime divisors, and set

$$C(\lambda) = \max_i \operatorname{coeff}_{E_i}(\Gamma + cF_\lambda).$$

If there is no exceptional divisor, omit the term $-C(\lambda)$ below. By distinguishing divisors already on Y from those exceptional over Y, the correct identity is

$$\operatorname{discrep}(X, \Delta + cL_\lambda) = \min\{\operatorname{discrep}(Y, \Gamma + cF_\lambda),\ -C(\lambda)\}. \tag{5.4}$$

Individual discrepancies are preserved by the crepant equality in Step 1, but the sets defining the two infima differ. Thus the comparison to prove on this model is

$$\begin{aligned} &\min\{\operatorname{discrep}(Y, \Gamma + cF_0), -C(0)\} \\ &\qquad \leq \min\{\operatorname{discrep}(Y, \Gamma + cF_g), -C(g)\}. \end{aligned} \tag{5.5}$$

[87] For resolving the base ideal of a linear system, see Example 3.1.34.

One cannot in general discard the terms $-C(0)$ and $-C(g)$. If $|F| = \{0\}$, all members are the same divisor and the conclusion is immediate.

Step 3: Suppose now that the coefficients are real. If $(X, \Delta + cL_0)$ is not lc, its discrepancy is $-\infty$ by Corollary 5.2.20, and there is nothing to prove. We may assume this special pair is lc.

Fix the finite support of Δ. Choose local Cartier presentations of $K_X + \Delta$ on a finite open cover, and a common log resolution of this support, the base ideal of $|L|$, and L_0. Choose L_g in the Bertini open set for this data; it can be chosen independently of the coefficients. Refine the resolution for the two fixed members L_0, L_g, separating all strict transforms as in Corollary 5.2.21.

On this fixed resolution, the coefficients of the crepant boundary and the exceptional discrepancies are affine functions of the coefficients of Δ and of c. The local Cartier presentations give rational linear constraints on these parameters. Effectivity, $c \geq 0$, and log canonicity of the special pair give finitely many rational linear inequalities: the strict-transform coefficients must be at most one, and the exceptional discrepancies at least -1. Hence the given real parameters lie in a rational polyhedron, and can be approximated inside it by rational parameters (Δ_j, c_j). These satisfy

$$\Delta_j \geq 0, \quad c_j \geq 0, \quad K_X + \Delta_j \text{ is } \mathbb{Q}\text{-Cartier},$$

and $(X, \Delta_j + c_j L_0)$ is lc.

Step 4: Apply the rational Bertini comparison to each approximation:

$$\operatorname{discrep}(X, \Delta_j + c_j L_0) \leq \operatorname{discrep}(X, \Delta_j + c_j L_g).$$

The left side is at least -1, so every general pair on the right is lc as well. Their coefficient inequalities are closed and pass to the limit, proving that $(X, \Delta + cL_g)$ is lc. On the lc region, Corollary 5.2.21 computes both discrepancies as minima of finitely many affine functions on the chosen resolution. They are therefore continuous there. Passing to the limit proves the required inequality for real coefficients. □

Example 5.2.26 (The assumption of properness is crucial)**.** Take $X = \mathbb{A}^2$ and $L = \mathbb{C}\langle 1, x\rangle \subseteq H^0(X, \mathcal{O}_X) = \mathbb{C}[x, y]$. The parameter space is $\mathbb{P}(L) \cong \mathbb{P}^1$, with $[a : b]$ corresponding to the section $a + bx$. The special member L_0, with parameter $[1 : 0]$, is the zero divisor. A general member L_g is the line $a + bx = 0$, with $b \neq 0$. Put $\Delta = 0$ and $0 < c \leq 1$. Then

$$\operatorname{discrep}(X, cL_0) = \operatorname{discrep}(X, 0) = 1.$$

For a point $p \in L_g$, its blow-up has exceptional discrepancy

$$a(E, X, cL_g) = 2 - 1 - c \operatorname{mult}_p L_g = 1 - c.$$

In fact $\operatorname{discrep}(X, cL_g) = 1 - c$ by the snc formula. Thus the special discrepancy is strictly greater than the general one. Here a non-zero section on a non-proper variety can be nowhere vanishing, even though general sections of the same system have zeros.

Example 5.2.27 (The assumption of effectivity is crucial)**.** Many references reserve "pair" for an effective boundary, often with coefficients at most one. In our convention, negative coefficients are also allowed, sometimes under the name "sub-pair". The preceding comparison need not hold for such boundaries.

Take $X = \mathbb{P}^2$, $\Delta = -dH$ with $d > 0$ and H a line, and $|L| = |\mathcal{O}_{\mathbb{P}^2}(1)|$. Let $L_0 = H$, and let L_g be another line meeting H transversely. For $0 \leq c \leq 1$, Corollary 5.2.20 gives

$$\begin{aligned}\operatorname{discrep}(X, \Delta + cL_g) &= \min\{1 + d - c, 1 - c, 1 + d, 1\} = 1 - c,\\ \operatorname{discrep}(X, \Delta + cL_0) &= \min\{1 - c + d, 1\}.\end{aligned}$$

In particular, for $c = 1$,

$$\operatorname{discrep}(X, \Delta + L_0) = \min\{d, 1\} > 0 = \operatorname{discrep}(X, \Delta + L_g).$$

This is the reverse of the inequality in Corollary 5.2.25.

Definition 5.2.28 (Alternative definition for canonical pairs)**.** Let X be a normal variety and Δ an effective $\mathbb{R}$-divisor. We say that (X, Δ) has canonical singularities if

(1) $K_X + \Delta$ is $\mathbb{R}$-Cartier;

(2) for every resolution $f : Y \longrightarrow X$, writing

$$K_Y + C = f^*(K_X + \Delta),$$

the divisor $-C + f_*^{-1}\Delta$ is effective.

Indeed,

$$\begin{aligned}K_Y + f_*^{-1}\Delta &= f^*(K_X + \Delta) + \sum_{E_i \text{ exceptional}} a(E_i, X, \Delta)E_i\\ &= K_Y + C + \sum_i a(E_i, X, \Delta)E_i.\end{aligned}$$

Thus $-C + f_*^{-1}\Delta = \sum_i a(E_i, X, \Delta)E_i$, and the condition on every resolution is exactly $\operatorname{discrep}(X, \Delta) \geq 0$.

Proposition 5.2.29. *When $\Delta = 0$ and K_X is $\mathbb{Q}$-Cartier, this definition agrees with Definition 5.1.9.*

Proof. Take $m > 0$ divisible by the Cartier index of K_X. We show that effectivity of $-C$ on every resolution is equivalent to the natural equality

$$f_*\mathcal{O}_Y(mK_Y) = \mathcal{O}_X(mK_X).$$

Write

$$mK_Y = f^*(mK_X) + \sum_i ma_iE_i = f^*(mK_X) - mC,$$

where E_i are exceptional. If $-C \geq 0$, the projection formula gives

$$f_*\mathcal{O}_Y(mK_Y) \cong \mathcal{O}_X(mK_X) \otimes f_*\mathcal{O}_Y(-mC).$$

We have $\mathcal{O}_X = f_*\mathcal{O}_Y \subseteq f_*\mathcal{O}_Y(-mC)$. Conversely, a section of the latter is a rational function regular outside the image of the exceptional divisors, a set of codimension at least two. Normality and algebraic Hartogs show that it is regular on X. Thus $f_*\mathcal{O}_Y(-mC) = \mathcal{O}_X$.

Conversely, under this natural equality, the function 1 on every open $U \subseteq X$ is a section of $\mathcal{O}_Y(-mC)$ on $f^{-1}(U)$. The divisor condition for this section gives $\operatorname{div}(1) - mC = -mC \geq 0$. Hence $-C$ is effective. □

Definition 5.2.30 (Alternative definition for terminal pairs)**.** Let X be a normal variety and Δ an effective $\mathbb{R}$-divisor. We say that (X, Δ) has terminal singularities if

(1) $K_X + \Delta$ is $\mathbb{R}$-Cartier;

(2) for every resolution $f : Y \longrightarrow X$, writing $K_Y + C = f^*(K_X + \Delta)$, the divisor $-C + f_*^{-1}\Delta$ is effective;

(3) its support is the divisorial part of $\operatorname{Ex}(f)$.

As above,

$$-C + f_*^{-1}\Delta = \sum_{E_i \text{ exceptional}} a(E_i, X, \Delta)E_i.$$

Conditions (2)–(3) say that every exceptional discrepancy is strictly positive. To pass from these individual inequalities to $\operatorname{discrep}(X, \Delta) > 0$, use Corollary 5.2.21 on a fixed log resolution with disjoint strict transforms. Blowing up a general codimension-two center in a boundary component first shows that its coefficient is strictly less than one. The finite minimum in that corollary is therefore positive. This also justifies the equivalence for real coefficients.

Proposition 5.2.31. *When $\Delta = 0$ and K_X is $\mathbb{Q}$-Cartier, this definition agrees with Definition 5.1.10.*

Proof. Choose $m > 0$ divisible by the Cartier index. On a resolution write

$$mK_Y = f^*(mK_X) + \sum_i ma_i E_i = f^*(mK_X) - mC, \qquad E = \sum_i E_i.$$

If every $a_i > 0$, then ma_i is a positive integer and hence $ma_i \geq 1$. The projection formula gives

$$f_*\mathcal{O}_Y(mK_Y - E) \cong \mathcal{O}_X(mK_X) \otimes f_*\mathcal{O}_Y\left(\sum_i (ma_i - 1)E_i\right).$$

The sum is effective and exceptional, so the same Hartogs argument as above shows that the last pushforward is $\mathcal{O}_X$.

Conversely, the natural equality of these sheaves implies that the function 1 is a section of $\mathcal{O}_Y(\sum_i (ma_i - 1)E_i)$ over inverse images of open sets. Hence every $ma_i - 1 \geq 0$, so

$$a_i \geq \frac{1}{m} > 0.$$

Thus $-C$ is effective with support equal to the divisorial exceptional locus, as required. □

Definition 5.2.32 (Alternative definition for klt pairs)**.** Let X be a normal variety and Δ an effective $\mathbb{R}$-divisor. We say that (X, Δ) is klt if

(1) $K_X + \Delta$ is $\mathbb{R}$-Cartier;

(2) every coefficient of Δ is less than one;

(3) there exists a log resolution $f : Y \longrightarrow X$ such that, writing

$$K_Y + C = f^*(K_X + \Delta), \qquad C = \sum_j c_j C_j,$$

every coefficient $c_j < 1$.

The equivalence with Definition 5.2.22 for $\Delta \geq 0$ follows from the snc computation and Corollary 5.2.21. When negative coefficients are allowed, the corresponding terminology is sometimes "sub-klt".

Definition 5.2.33 (Alternative definition for plt pairs)**.** Let X be a normal variety and Δ an effective $\mathbb{R}$-divisor. We say that (X, Δ) is plt if

(1) $K_X + \Delta$ is $\mathbb{R}$-Cartier;

(2) every coefficient of Δ is at most one;

(3) for every log resolution $f : Y \longrightarrow X$, writing

$$K_Y + C = f^*(K_X + \Delta), \qquad C = \sum_j c_j C_j,$$

we have $c_j < 1$ for every f-exceptional component C_j.

This agrees with Definition 5.2.22 by the finite-resolution formula and $-c_j = a(C_j, X, \Delta)$. The quantifier "for every" is essential.

Definition 5.2.34 (Alternative definition for lc pairs)**.** Let X be a normal variety and Δ an effective $\mathbb{R}$-divisor. We say that (X, Δ) is lc if

(1) $K_X + \Delta$ is $\mathbb{R}$-Cartier;

(2) every coefficient of Δ is at most one;

(3) there exists a log resolution $f : Y \longrightarrow X$ such that, writing

$$K_Y + C = f^*(K_X + \Delta), \qquad C = \sum_j c_j C_j,$$

every coefficient $c_j \leq 1$.

Again the snc computation gives the equivalence for $\Delta \geq 0$. With arbitrary negative coefficients, one sometimes uses the term "sub-lc".

Definition 5.2.35 (Alternative definition for dlt pairs)**.** Let X be a normal variety and Δ an effective $\mathbb{R}$-divisor. We say that (X, Δ) is dlt if

(1) $K_X + \Delta$ is $\mathbb{R}$-Cartier;

(2) every coefficient of Δ is at most one;

(3) there exists a log resolution $f : Y \longrightarrow X$ such that, writing

$$K_Y + C = f^*(K_X + \Delta), \qquad C = \sum_j c_j C_j,$$

we have $c_j < 1$ for every f-exceptional component C_j.

The equivalence with Definition 5.2.23 is due to Szabó; see [Sza94]. In contrast to the plt definition above, only the existence of one such log resolution is required.

Example 5.2.36. Take $X = \mathbb{A}^n$, $n \geq 2$, with coordinate hyperplanes $D_1, \ldots, D_n$, and let $\Delta = \sum_i a_i D_i$ be effective. Then (X, Δ) is

- klt if and only if $0 \leq a_i < 1$ for all i;
- plt if and only if $0 \leq a_i \leq 1$ for all i, with $a_i = 1$ for at most one i;
- lc if and only if $0 \leq a_i \leq 1$ for all i.

Here lc and dlt coincide: the identity is already a log resolution with no exceptional divisors.

Example 5.2.37. Take $X = \mathbb{A}^2$ with coordinates x, y and prime divisor $D = \operatorname{div}(x^2 - y^3)$. We determine when (X, dD) is klt or lc, for $0 \leq d \leq 1$.

Construct a log resolution as in [Har77, V, 3.9.1]. First blow up $p_0 = (0, 0)$:

$$f_1 : Y_1 \longrightarrow X.$$

The exceptional divisor is $E_1 \cong \mathbb{P}^1$. The strict transform $D_1 = f_{1*}^{-1} D$ is smooth and meets E_1 at one point p_1 with intersection multiplicity two.

Blow up p_1 to obtain $f_2 : Y_2 \longrightarrow Y_1$, with exceptional divisor $E_2 \cong \mathbb{P}^1$. The smooth curves E_2, $D_2 = f_{2*}^{-1} D_1$ and $E_1' = f_{2*}^{-1} E_1$ meet at one point p_2 with three distinct tangent directions. Thus their union has multiplicity three at p_2, while each pair has intersection multiplicity one.

Blow up p_2 to obtain $f_3 : Y \longrightarrow Y_2$, with exceptional divisor $E_3 \cong \mathbb{P}^1$. The union of E_3, $D_3 = f_{3*}^{-1} D_2$, $E_1'' = f_{3*}^{-1} E_1'$ and $E_2' = f_{3*}^{-1} E_2$ is snc. The composition $f = f_1 \circ f_2 \circ f_3$ is a log resolution. We have

$$K_Y + dD_3 = f^*(K_X + dD) + \sum_i a(E_i, X, dD) E_i,$$

where

$$K_Y = f^* K_X + E_1'' + 2E_2' + 4E_3, \qquad f^* D = D_3 + 2E_1'' + 3E_2' + 6E_3.$$

Therefore

$$\sum_i a(E_i, X, dD) E_i = (1 - 2d) E_1'' + (2 - 3d) E_2' + (4 - 6d) E_3.$$

Testing these coefficients and the coefficient d of D_3 gives

$$(X, dD) \text{ is klt } \iff 0 \leq d < \frac{5}{6}, \qquad (X, dD) \text{ is lc } \iff 0 \leq d \leq \frac{5}{6}.$$

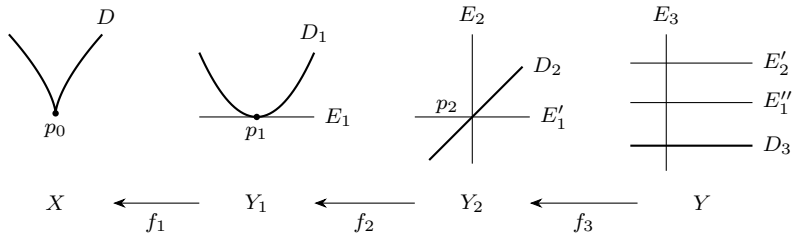


Figure 5.7: Log resolution of (X, dD).

Lemma 5.2.11 and Corollary 5.2.21 imply the following continuity properties.

Corollary 5.2.38. *Let (X, Δ) be a pair and Δ' an effective $\mathbb{R}$-Cartier divisor. Then:*

(1) If $(X, \Delta + \Delta')$ is terminal (resp. canonical, klt, plt, lc), then (X, Δ) has the same property. The same holds for dlt when $\Delta \geq 0$.

(2) If (X, Δ) is terminal (resp. klt), then $(X, \Delta + \varepsilon\Delta')$ is terminal (resp. klt) for $0 \leq \varepsilon \ll 1$.

(3) If (X, Δ) is plt and Δ, Δ' have no common irreducible components, then $(X, \Delta + \varepsilon\Delta')$ is plt for $0 \leq \varepsilon \ll 1$.

(4) If $\Delta \geq 0$, (X, Δ) is dlt, and

$$\operatorname{Supp} \Delta' \subseteq \operatorname{Supp}(\Delta - \lfloor \Delta \rfloor),$$

then $(X, \Delta + \varepsilon\Delta')$ is dlt for $0 \leq \varepsilon \ll 1$.

(5) If (X, Δ) is terminal, then $(X, \Delta+\Delta')$ is canonical if and only if $(X, \Delta+c\Delta')$ is terminal for every $c < 1$.

(6) If (X, Δ) is klt (resp. plt), then $(X, \Delta + \Delta')$ is lc if and only if $(X, \Delta + c\Delta')$ is klt (resp. plt) for every $c < 1$.

For $c < 0$ in (5)–(6), we use the convention of Definition 5.2.22 allowing negative boundary coefficients. If "pair" is reserved for effective boundaries, the equivalences can instead be stated for $0 \leq c < 1$.

Proof. We do not assume Δ effective except in the dlt assertions.

For (1)

For every divisor E over X, monotonicity gives

$$a(E, X, \Delta + \Delta') \leq a(E, X, \Delta).$$

This proves the exceptional-discrepancy inequalities, and decreasing the boundary also preserves the coefficient condition for klt. For dlt with $\Delta \geq 0$, keep the closed set Z from Definition 5.2.23. The support of Δ on $X \setminus Z$ is a union of components of the original snc boundary, and the required discrepancies over Z stay greater than -1.

For (2)

Choose a log resolution $f : Y \longrightarrow X$ of $\operatorname{Supp} \Delta \cup \operatorname{Supp} \Delta'$ on which all the strict transforms of its components are pairwise disjoint. Write

$$\Delta = \sum_j d_j D_j, \qquad \Delta' = \sum_j e_j D_j,$$

using the same index set and allowing zero coefficients. For the finitely many exceptional divisors E_i, set

$$q_i = a(E_i, X, \Delta), \qquad b_i = \operatorname{mult}_{E_i} f^*\Delta' \geq 0.$$

Then

$$a(E_i, X, \Delta + \varepsilon\Delta') = q_i - \varepsilon b_i, \qquad \operatorname{coeff}_{D_j}(\Delta + \varepsilon\Delta') = d_j + \varepsilon e_j.$$

Let $t = 0$ in the terminal case and $t = -1$ in the klt case. The initial assumptions give $q_i > t$ and $d_j < 1$. Choose $\varepsilon_0 > 0$ so that

$$\varepsilon_0 \leq \min\left\{1,\ \min_{b_i>0} \frac{q_i - t}{b_i},\ \min_{e_j>0} \frac{1 - d_j}{e_j}\right\}.$$

Empty minima are omitted. For $0 \leq \varepsilon < \varepsilon_0$, every exceptional discrepancy on this resolution is greater than t, and every boundary coefficient is less than one. Corollary 5.2.21 and the klt coefficient condition prove the assertion. In particular, the klt numerator is $q_i + 1$, rather than q_i.

For (3)

Use the same common resolution. Here $q_i > -1$ and $d_j \leq 1$. The absence of common components means that $e_j > 0$ implies $d_j = 0$. Thus the exceptional inequalities remain strict and no coefficient exceeds one for sufficiently small ε. Corollary 5.2.21 gives discrepancy greater than -1, so the perturbed pair is plt.

For (4)

Choose a log resolution as in the alternative dlt definition, so that its finitely many exceptional discrepancies are greater than -1. Since $\operatorname{Supp}\Delta' \subseteq \operatorname{Supp}\Delta$, this also resolves the support of Δ'. The further condition in (4) says that only components whose coefficients are strictly below one are increased. A sufficiently small common $\varepsilon_0 > 0$ therefore preserves all coefficient bounds and all exceptional inequalities on this chosen resolution. The support is unchanged and remains snc upstairs. Definition 5.2.35 shows that the perturbed pair is dlt. The fixed finite resolution is what supplies one value of ε_0 valid for all valuations.

For (5)

For every divisor E and $0 \leq c \leq 1$, discrepancy is affine in c:

$$a(E, X, \Delta + c\Delta') = (1 - c)a(E, X, \Delta) + c\, a(E, X, \Delta + \Delta').$$

If the endpoint pair is canonical and the initial discrepancy is $\delta > 0$, then every exceptional discrepancy at $c < 1$ is at least $(1 - c)\delta > 0$. Hence the intermediate pair is terminal. Conversely, if every intermediate pair is terminal, taking $c \to 1$ in the inequality for each fixed E gives $a(E, X, \Delta + \Delta') \geq 0$ for all exceptional E. This is canonicity.

For (6)

Use the same interpolation. If the endpoint is lc and the initial exceptional discrepancy is $\delta > -1$, then for $c < 1$ every exceptional discrepancy is at least

$$(1-c)\delta - c = -1 + (1-c)(\delta+1) > -1.$$

For the klt assertion, each initial boundary coefficient is less than one and each endpoint coefficient is at most one, so its affine interpolation is strictly less than one. For plt, both endpoint coefficient bounds are at most one. This proves the forward direction. Conversely, pass to the limit for each exceptional divisor to get discrepancy at least -1 at the endpoint. In both (5) and (6), negative c cause no further difficulty: monotonicity compares the pair with the initial pair at $c = 0$. □

Proposition 5.2.39. *Let (X, Δ) be a klt pair with $K_X + \Delta$ $\mathbb{R}$-Cartier.*[88]

(1) There is a log resolution $f : X' \longrightarrow X$ such that, writing

$$f^*(K_X + \Delta) = K_{X'} + A_{X'} - B_{X'},$$

where $A_{X'}, B_{X'}$ are effective with no common components, the support of $A_{X'}$ is smooth.

(2) If $a(E, X, \Delta) < 1 + \operatorname{totaldiscrep}(X, \Delta)$, then $\operatorname{center}_{X'} E$ is a divisor. In particular, only finitely many exceptional divisors over X have discrepancy strictly less than $1 + \operatorname{totaldiscrep}(X, \Delta)$.

Proof. **For (1)**

Start with a log resolution $f : Y \longrightarrow X$ and write

$$f^*(K_X + \Delta) = K_Y + A_Y - B_Y, \qquad A_Y = \sum_{i \in I} (1 - \alpha_i) A_i,$$

where A_Y, B_Y are effective with no common components and $0 < \alpha_i < 1$. These inequalities follow from klt, since

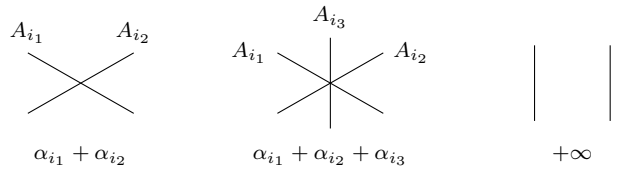
$$f^*(K_X + \Delta) = K_Y + f_*^{-1}\Delta - \sum_{E_i \text{ exceptional}} a(E_i, X, \Delta) E_i.$$

Let $S(Y) = \sum_{i \in I} \mathbb{Z}_{\geq 0} \alpha_i \subseteq \mathbb{R}_{\geq 0}$. For $J \subseteq I$, set

$$w(Y, J) = \begin{cases} \sum_{i \in J} \alpha_i, & \text{if } \bigcap_{i \in J} A_i \neq \varnothing \text{ and } \#J \geq 2, \\ +\infty, & \text{otherwise.} \end{cases}$$

This records intersections among the components of the positive part:

[88] For the canonical divisor calculation in this blow-up, see [Har77, II, Exercise 8.5].



The middle picture denotes a triple stratum in dimension at least three.

Set

$$M(Y) = \min_{J \subseteq I} w(Y, J), \qquad r(Y) = \#\{J \subseteq I : w(Y, J) = M(Y)\}.$$

Either $M(Y) \in (0, \dim X) \cap S(Y)$ or $M(Y) = +\infty$. The latter means that the components of A_Y are pairwise disjoint, i.e. its support is smooth.

Suppose that $M(Y) < +\infty$. Choose J with $w(Y, J) = M(Y)$ and blow up

$$Z = \bigcap_{i \in J} A_i.$$

This is a disjoint union of smooth subvarieties of codimension $c = \#J \geq 2$. Write $p : Y' \longrightarrow Y$ for the blow-up and E for its exceptional divisor; the following calculation applies to each component of E. We have

$$K_{Y'} = p^* K_Y + (c-1)E, \qquad p^* A_Y = \sum_{i \in I} (1 - \alpha_i) p_*^{-1} A_i + \sum_{i \in J} (1 - \alpha_i) E.$$

No component of B_Y contains a component of Z: the union of the supports is snc, and another component cannot contain a whole stratum of these A_i. Thus $p^* B_Y$ has no exceptional term along E. The coefficient of E in the crepant boundary on Y' is consequently

$$1 - c + \sum_{i \in J} (1 - \alpha_i) = 1 - \sum_{i \in J} \alpha_i = 1 - M(Y).$$

The new positive part is

$$A_{Y'} = \sum_{i \in I} (1 - \alpha_i) p_*^{-1} A_i + \begin{cases} (1 - M(Y))E, & M(Y) < 1, \\ 0, & M(Y) \geq 1. \end{cases}$$

If a new positive component occurs, its weight is $M(Y)$, a sum of old weights. All old components persist, so $S(Y') = S(Y)$. Any intersection involving a new positive component and another positive component has weight strictly greater than $M(Y)$. Old intersections can disappear but cannot be created by the blow-up, and the chosen intersection indexed by J disappears. Hence

$$M(Y') > M(Y), \qquad \text{or} \qquad M(Y') = M(Y) \text{ and } r(Y') < r(Y).$$

The semigroup generated by finitely many positive real numbers has only finitely many elements in a bounded interval. Thus $M(Y)$ has only finitely many possible finite values. At each fixed value of M, the non-negative integer r strictly

decreases. The process must terminate with $M = +\infty$, proving (1). This does not require r to have a fixed bound throughout the whole process.

For (2)

Crepant pullback and monotonicity give

$$a(E, X, \Delta) = a(E, X', A_{X'} - B_{X'}) \geq a(E, X', A_{X'}). \tag{5.6}$$

If E is exceptional over X', the smooth support of $A_{X'} = \sum_i (1 - \alpha_i) A_i$ gives

$$a(E, X', A_{X'}) \geq \operatorname{discrep}(X', A_{X'}) = \min\{\min_i \alpha_i, 1\}. \tag{5.7}$$

When $A_{X'} \neq 0$,

$$\operatorname{totaldiscrep}(X, \Delta) \leq \min_i a(A_i, X, \Delta) = \min_i (\alpha_i - 1),$$

so $\min_i \alpha_i \geq 1 + \operatorname{totaldiscrep}(X, \Delta)$. Also $\operatorname{totaldiscrep}(X, \Delta) \leq 0$, since a prime divisor on X outside $\operatorname{Supp} \Delta$ has discrepancy zero. This handles the extra term 1 and the case $A_{X'} = 0$. Thus every E exceptional over X' satisfies

$$a(E, X, \Delta) \geq 1 + \operatorname{totaldiscrep}(X, \Delta).$$

If its discrepancy is *strictly less* than this bound, its center on X' must therefore be a divisor. Only finitely many prime divisors on X' are exceptional over X, proving (2). □

Remark 5.2.40. For any pair with X normal and $K_X + \Delta$ $\mathbb{R}$-Cartier,

$$\operatorname{totaldiscrep}(X, \Delta) \leq 0.$$

Indeed, there is a prime divisor D outside the finite support of Δ, and $a(D, X, \Delta) = -\operatorname{coeff}_D \Delta = 0$.

Proposition 5.2.41. *Let X be a normal variety and $\Delta = \sum_i a_i D_i$, with distinct prime divisors D_i and $0 \leq a_i \leq 1$. Assume that $K_X + \Delta$ is $\mathbb{R}$-Cartier. Then (X, Δ) is dlt if and only if there is a closed subset $Z \subseteq X$ such that*

(1) $X \setminus Z$ is smooth and $\Delta|_{X \setminus Z}$ has snc support;

(2) there is a log resolution $f : Y \longrightarrow X$, also resolving the ideal of Z, such that $f^{-1}(Z)$ is a divisor with snc support together with the transformed boundary, and

$$a(E, X, \Delta) > -1$$

for every prime divisor $E \subseteq f^{-1}(Z)$.

In particular, the inverse image in (2) has pure codimension one when non-empty.

Proof. If the pair is dlt, choose a proper closed bad locus Z as in Definition 5.2.23, and take a log resolution which also principalizes its ideal. Such a Z may be chosen proper because the pair is snc at the generic points of X and of its boundary components. Every component of $f^{-1}(Z)$ has center contained in Z, so the discrepancy inequalities follow from the definition.

Conversely, assume that f exists as in (2). Write

$$K_Y + \Delta_Y = f^*(K_X + \Delta)$$

and choose an effective divisor Δ' supported exactly on $f^{-1}(Z)$. Its components have coefficient less than one in Δ_Y. Every other component of Δ_Y has coefficient at most one: those which meet $f^{-1}(X \setminus Z)$ are governed by the lc snc pair there. For sufficiently small $\varepsilon > 0$, the divisor $\Delta_Y + \varepsilon\Delta'$ therefore has snc support and all coefficients at most one. It is lc by Corollary 5.2.20.

For any divisor E with $\operatorname{center}_X E \subseteq Z$, properness gives $\operatorname{center}_Y E \subseteq f^{-1}(Z)$. Thus

$$a(E, X, \Delta) = a(E, Y, \Delta_Y) > a(E, Y, \Delta_Y + \varepsilon\Delta') \geq -1.$$

This proves Definition 5.2.23(2). □

Proposition 5.2.42. *For an effective $\mathbb{R}$-divisor Δ, the pair (X, Δ) is klt if and only if it is dlt and $\lfloor \Delta \rfloor = 0$.*

Proof. For the forward direction, all divisorial discrepancies of a klt pair are greater than -1, and all boundary coefficients are less than one. In Definition 5.2.23 we may take $Z = X$, so the pair is dlt.

Conversely, suppose that the pair is dlt and $\lfloor \Delta \rfloor = 0$, and let Z be its bad locus. If $\operatorname{center}_X E \subseteq Z$, then $a(E, X, \Delta) > -1$ by definition. Otherwise restrict to $U = X \setminus Z$. The snc formula gives

$$\operatorname{discrep}(U, \Delta|_U) = \min\left\{ \min_{i \neq j,\ D_i|_U \cap D_j|_U \neq \varnothing} (1 - a_i - a_j),\ \min_i (1 - a_i),\ 1 \right\} > -1,$$

since every $a_i < 1$. This equals the corresponding bound for valuations whose centers meet U. All exceptional discrepancies are thus greater than -1; the finite-resolution formula supplies a uniform strict lower bound. Together with $\lfloor \Delta \rfloor = 0$, this proves klt. □

Recall that for a Weil divisor D, its rank-one reflexive sheaf $\mathcal{O}_X(D)$ is coherent and

$$H^0(X, \mathcal{O}_X(D)) = \{h \in \mathbb{C}(X)^\times : \operatorname{div}(h) + D \geq 0\} \cup \{0\}.$$

We can define its complete linear series by

$$\begin{aligned} |D| &= \{D_h : D_h \text{ is effective and } D_h \sim_{\text{lin}} D\} \\ &= \{\operatorname{div}(h) + D : 0 \neq h \in H^0(X, \mathcal{O}_X(D))\}, \end{aligned}$$

and its set-theoretic base locus by

$$\mathrm{Bs}(|D|) = \bigcap_{D_h \in |D|} \mathrm{Supp}\, D_h.$$

For a finite-dimensional subspace $V \subseteq H^0(X, \mathcal{O}_X(D))$, there is a map

$$\Phi : \mathbb{P}(V) \longrightarrow |D|, \qquad [h] \longmapsto \mathrm{div}(h) + D.$$

Here h is the rational function representing the section. In the earlier section notation, the same divisor is simply $\mathrm{div}(s)$; one must not add D twice. The map is well-defined, but on a non-proper variety it need not be injective, since non-constant units can give the same divisor. Its image defines the corresponding linear subseries. If X is proper and integral over $\mathbb{C}$, its global units are constant and this map is injective.

Recall that $D_1 \sim_{\mathbb{R}} D_2$ (resp. $D_1 \sim_{\mathbb{Q}} D_2$) means that $D_1 - D_2$ is a real (resp. rational) linear combination of principal Weil divisors.

A dlt pair can be approximated by klt pairs as follows:

Proposition 5.2.43. *Assume that (X, Δ) is dlt and X is quasi-projective with an ample Cartier divisor H. Let Δ_1 be an effective $\mathbb{R}$-divisor, not necessarily $\mathbb{R}$-Cartier, such that $\Delta - \Delta_1 \geq 0$. Then there are a real number $c > 0$ and an effective $\mathbb{R}$-divisor $D \sim_{\mathbb{R}} \Delta_1 + cH$ such that*

$$(X, \Delta - \varepsilon\Delta_1 + \varepsilon D)$$

is dlt for all $0 < \varepsilon \ll 1$. If $\mathrm{Supp}\, \Delta_1 = \mathrm{Supp}\, \Delta$, these pairs are klt for all sufficiently small $\varepsilon > 0$.

Before proving this result, we need a rational version.

Lemma 5.2.44. *Assume that (X, Δ) is dlt and X is quasi-projective with an ample Cartier divisor H. Let Θ be an effective $\mathbb{Q}$-divisor, not necessarily $\mathbb{Q}$-Cartier, such that $\Delta - \Theta \geq 0$. Then there are a rational number $c > 0$ and an effective $\mathbb{Q}$-divisor $B \sim_{\mathbb{Q}} \Theta + cH$ such that*[89]

$$(X, \Delta - \varepsilon\Theta + \varepsilon B)$$

is dlt for all $0 < \varepsilon \ll 1$. If $\mathrm{Supp}\, \Theta = \mathrm{Supp}\, \Delta$, these pairs are klt for all sufficiently small $\varepsilon > 0$.

Proof. We have $\Delta = (\Delta - \Theta) + \Theta$ with both summands effective.

Let $Z \subseteq X$ be a closed subset witnessing that (X, Δ) is dlt, and put $U = X \setminus Z$. Choose positive integers m, m' such that $m\Theta$ is integral and the coherent sheaf

$$\mathcal{O}_X(m\Theta + m'H)$$

[89] For Bertini applied to the base-point-free system on the smooth open set, see [Har77, III, 10.9].

is generated by global sections. This uses ampleness of H and does not require $m\Theta$ to be Cartier. Since X is quasi-projective, the whole space of global sections need not be finite-dimensional. Nevertheless, quasi-compactness and coherence let us select finitely many global sections which generate the sheaf: on each of finitely many affine opens, finitely many of the given global generators suffice. Let V be their finite-dimensional span.

On the smooth open set U, the sheaf is a line bundle and the induced system is basepoint-free. Apply Bertini both on U and on all strata of $\operatorname{Supp}(\Delta|_U)$. For a general member, its effective divisor

$$A \sim_{\mathrm{lin}} m\Theta + m'H$$

is smooth on U and meets the boundary strata transversely, so $A|_U + \Delta|_U$ has snc support. We also choose A to contain no component of Δ. This is possible because the sheaf is generated at the generic point of each of those components. Applying Bertini on X_{reg} shows that A is reduced in codimension one, hence is a reduced Weil divisor.

Set $B = A/m$ and $c = m'/m$. Then B is effective and

$$B \sim_{\mathbb{Q}} \Theta + cH.$$

In particular, $B - \Theta$ is $\mathbb{Q}$-Cartier, although B and Θ need not be individually $\mathbb{Q}$-Cartier. Therefore

$$K_X + \Delta - \varepsilon\Theta + \varepsilon B = K_X + \Delta + \varepsilon(B - \Theta)$$

is $\mathbb{R}$-Cartier for every real ε.

Choose a log resolution $f : Y \longrightarrow X$ of $(X, \Delta + A)$ which is an isomorphism over the snc open set U. Every exceptional prime divisor E on this resolution has center in Z, so $a(E, X, \Delta) > -1$. We have

$$a(E, X, \Delta - \varepsilon\Theta + \varepsilon B) = a(E, X, \Delta) - \varepsilon \operatorname{ord}_E f^*(B - \Theta).$$

There are only finitely many such E. Hence one common $\varepsilon_0 > 0$ keeps all these numbers greater than -1. Decrease ε_0 further so that $\varepsilon_0 \le \min\{1, m\}$. Then $\Delta - \varepsilon\Theta$ is effective with coefficients at most one, and the new components of εB have coefficient $\varepsilon/m < 1$. Since A contains no component of Δ, the entire boundary has coefficients at most one. Its total transform is supported in the fixed snc divisor on Y. The alternative dlt definition now proves the first assertion.

For the last assertion, write $\Delta = \sum_i a_i D_i$ and $\Theta = \sum_i b_i D_i$, with the same support and $b_i > 0$. We may further require

$$\varepsilon_0 \le \min\left\{ m,\ \min_i \frac{a_i}{b_i} \right\},$$

omitting the minimum if the support is empty. For $0 < \varepsilon < \varepsilon_0$,

$$0 \le a_i - \varepsilon b_i < 1,$$

and the new components from B have coefficients less than one. Therefore

$$\lfloor \Delta - \varepsilon\Theta + \varepsilon B \rfloor = 0.$$

By Proposition 5.2.42, these dlt pairs are klt. □

We now prove Proposition 5.2.43.

Proof of Proposition 5.2.43. Our strategy is to use Lemma 5.2.44 and rational approximation. Write

$$\Delta_1 = \sum_i r_i \Gamma_i, \qquad r_i > 0.$$

Choose rational numbers $0 < q_i < r_i$, and set $\Theta = \sum_i q_i \Gamma_i$. If $\Delta_1 = 0$, take $\Theta = 0$. Then Θ is effective with $\operatorname{Supp}\Theta = \operatorname{Supp}\Delta_1$, and

$$\Delta - \Theta = (\Delta - \Delta_1) + (\Delta_1 - \Theta) \geq 0.$$

Apply Lemma 5.2.44 to get $c \in \mathbb{Q}_{>0}$ and an effective $\mathbb{Q}$-divisor B with

$$B \sim_{\mathbb{Q}} \Theta + cH, \tag{5.8}$$

such that $(X, \Delta - \varepsilon\Theta + \varepsilon B)$ is dlt for $0 < \varepsilon \ll 1$.

Set $D = B + (\Delta_1 - \Theta)$, which is effective. By (5.8),

$$D - (\Delta_1 + cH) = B + (\Delta_1 - \Theta) - \Delta_1 - cH = B - \Theta - cH$$

is a rational linear combination of principal divisors, hence also a real one. Thus $D \sim_{\mathbb{R}} \Delta_1 + cH$. Finally,

$$\begin{aligned} \Delta - \varepsilon\Delta_1 + \varepsilon D &= \Delta - \varepsilon\Delta_1 + \varepsilon(B + \Delta_1 - \Theta) \\ &= \Delta - \varepsilon\Theta + \varepsilon B. \end{aligned}$$

The first assertion follows. If $\operatorname{Supp}\Delta_1 = \operatorname{Supp}\Delta$, then $\operatorname{Supp}\Theta = \operatorname{Supp}\Delta$, so the last assertion of Lemma 5.2.44 gives klt as well. □

Appendix A
Relative Linear Equivalence

All varieties are assumed to be normal to establish the theory of Weil divisors. In this appendix we work over an algebraically closed field of characteristic zero, as in [Kaw24]. In the divisor formulas below, assume that f is dominant, so that pullbacks of Cartier divisors are defined.

In [Kaw24, p. 8], two definitions of relative linear equivalence are given:

Definition A.0.1 (Version I)**.** Given a dominant morphism $f : X \longrightarrow S$ between two normal algebraic varieties, two Weil divisors are *relatively linearly equivalent over* S, which we denote by $D \sim_S D'$, if there exists an open covering $\{S_i\}$ of S such that

$$D|_{f^{-1}S_i} \sim D'|_{f^{-1}S_i}$$

after restriction over each S_i.

Definition A.0.2 (Version II)**.** Under the same conditions, we say that D and D' are relatively linearly equivalent over S if there exists a Cartier divisor B on S such that

$$D \sim D' + f^*B.$$

In general, these two definitions are not the same, and Version I is weaker. But if f is proper and surjective with connected geometric fibres, then, in the characteristic-zero setting above, they are equivalent.

Remark A.0.3. We can instead assume that f is proper and dominant, and that its generic fibre satisfies

$$H^0(X_\eta, \mathcal{O}_{X_\eta}) = k(S), \qquad \eta = \eta_S.$$

For example, a geometrically integral generic fibre is sufficient. A merely connected generic fibre is not sufficient: the finite morphism $\mathbb{P}^1 \to \mathbb{P}^1$, $t \mapsto t^2$, has a connected generic fibre but does not satisfy $f_*\mathcal{O}_X = \mathcal{O}_S$. The displayed generic-fibre condition also works in positive characteristic.

Since Version II is the stronger version, it trivially implies Version I. Now we discuss how Version I implies Version II under the suitable condition in Remark A.0.3.

Since $\{S_i\}$ covers S, we have

$$D|_{f^{-1}S_i} - D'|_{f^{-1}S_i} = \operatorname{div}(g_i), \qquad g_i \in k(X)^*.$$

In particular, $D - D'$ is Cartier. On overlaps,

$$\varphi_{ij} := g_i/g_j \in \mathcal{O}_X(f^{-1}(S_i \cap S_j))^\times.$$

To show that $\mathcal{O}_X(D - D') \cong f^*\mathcal{L}$ for some line bundle on S, we construct it as follows:

- We want units $v_{ij} \in \mathcal{O}_S(S_i \cap S_j)^\times$ such that $f^* v_{ij} = \varphi_{ij}$.
- Since f is proper and dominant, S is normal, and the generic-fibre condition holds, [SP25, Tag 0AY8] implies that $f_* \mathcal{O}_X = \mathcal{O}_S$.
- For any open subset $U \subseteq S$, the natural map

$$\mathcal{O}_S(U) \longrightarrow \mathcal{O}_X(f^{-1}U) = f_* \mathcal{O}_X(U)$$

  is therefore an isomorphism.
- Taking $U = S_i \cap S_j$, we obtain the required units v_{ij} via this isomorphism.
- These units satisfy the cocycle condition. Glue local frames ℓ_i by $\ell_j = v_{ij} \ell_i$ to form a line bundle $\mathcal{L}$ on S. The frames g_i^{-1} of $\mathcal{O}_X(D - D')$ satisfy the same relations after pullback, so

$$f^* \mathcal{L} \cong \mathcal{O}_X(D - D').$$

  Since S is integral, $\mathcal{L} \cong \mathcal{O}_S(B)$ for a Cartier divisor B on S. Thus $D - D' \sim f^* B$.

Appendix B
Local Properties of Algebraic Schemes

B.1 Bertini's Theorem

Theorem B.1.1 (Bertini)**.** *Let X be a nonsingular closed subvariety of $\mathbb{P}^n_k$, where k is algebraically closed. Then there exists a hyperplane $H \subseteq \mathbb{P}^n_k$, not containing X, such that the scheme $H \cap X$ is regular at every point. Furthermore, the set of hyperplanes with this property forms an open dense subset of the complete linear series $|H|$, considered as a projective space. We say that the general hyperplane section satisfies the theorem.*[90]

Proof. We fix a closed point $x \in X$ and consider the set of bad hyperplanes

$$B_x = \{H \mid H \supseteq X, \text{ or } H \not\supseteq X \text{ and } x \in H \cap X \text{ is not regular}\}.$$

A hyperplane is cut out by a nonzero section $f \in V = H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(1))$. Fix $f_0 \in V$ such that $x \notin H_0 = V(f_0)$. Then we can define a map of k-vector spaces

$$\varphi_x : V \longrightarrow \mathcal{O}_{X,x}/\mathfrak{m}_x^2, \qquad f \longmapsto f/f_0 \pmod{\mathfrak{m}_x^2}.$$

The quotient is isomorphic, as a k-vector space, to $k \oplus \mathfrak{m}_x/\mathfrak{m}_x^2$. The function $h = f/f_0$ is regular near x, and $H \cap X$ is locally defined by (h). Thus $x \in H \cap X$ if and only if $h \in \mathfrak{m}_x$. If $H \not\supseteq X$, then $h \neq 0$ in the regular local domain $\mathcal{O}_{X,x}$. Writing $r = \dim X$, we have $\dim \mathcal{O}_{X,x}/(h) = r - 1$ whenever $x \in H \cap X$. In this case, if $h \in \mathfrak{m}_x^2$, then

$$\dim_k \frac{\mathfrak{m}_x/(h)}{(\mathfrak{m}_x/(h))^2} = \dim_k \frac{\mathfrak{m}_x}{\mathfrak{m}_x^2} = r,$$

so the quotient is not regular. If $h \notin \mathfrak{m}_x^2$, its embedding dimension is $r - 1$, so it is regular. Hyperplanes containing X also have $\varphi_x(f) = 0$. Consequently, $B_x = \mathbb{P}_{\text{sub}}(\ker \varphi_x)$. Here $\varphi_x(f) = 0$ means that the first jet vanishes; it does not imply that H contains all of X.

Since x is closed and k is algebraically closed, the maximal ideal is generated by restrictions of affine linear coordinates centred at x. Hence φ_x is surjective. Therefore

$$\dim_k \mathcal{O}_{X,x}/\mathfrak{m}_x^2 = r + 1, \qquad \dim_k \ker \varphi_x = (n+1) - (r+1) = n - r.$$

If $n = r$, the kernels are zero and there are no bad hyperplanes, so we are done. Assume $n > r$. The bad hyperplanes at each x form a projective space of dimension $n - r - 1$.

[90] For the closed-image property of the proper projection, see [Har77, II, Exercise 4.4].

Now consider the incidence subset $B \subseteq X \times |H|$ consisting of the bad pairs (x, H). The first-jet equations show that B is closed. Locally on X, the maps φ_x are represented by matrices of constant rank $r+1$; their kernels form a vector bundle of rank $n-r$. Thus

$$p_1 : B \longrightarrow X$$

is its associated projective bundle. In particular, B is irreducible and $\dim B = (n-r-1)+r = n-1$.

For the second projection $p_2 : B \longrightarrow |H|$, we have $\dim p_2(B) \leq n-1 < n = \dim |H|$. Since X is projective, the projection $X \times |H| \to |H|$ is proper; hence $p_2(B)$ is closed. Its complement is an open dense subset. Any hyperplane in this complement does not contain X and has a regular intersection with X at every closed point, hence at every point. This is what we desired. □

B.2 Differential Forms in Algebraic Geometry

We fix our ground field k to be algebraically closed. Recall that, for a variety X and a closed point $x \in X$, the Zariski cotangent space is $T^\vee_{X,x} = \mathfrak{m}_x/\mathfrak{m}_x^2$, a $k(x) = k$-vector space, and the Zariski tangent space is

$$T_{X,x} = \mathrm{Hom}_{k(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2, k(x)).$$

When we say that X is smooth or nonsingular over k at x, we mean that $\dim_k \mathfrak{m}_x/\mathfrak{m}_x^2 = \dim X$. This is an open condition. Moreover, every variety over k has a nonempty open subset on which it is smooth.

Example B.2.1 (Differential forms on $\mathbb{A}^n$)**.** For $X = \mathbb{A}^n$, the differential of a regular function f is a section

$$df = \sum_i \frac{\partial f}{\partial x_i}\, dx_i$$

of the sheaf of differential forms. At a closed point x, its value is the cotangent vector

$$df(x) = \sum_i \frac{\partial f}{\partial x_i}(x)\, dx_i|_x \in T^\vee_{\mathbb{A}^n,x}.$$

The operation d satisfies:

(1) $\lambda_1 df_1 + \lambda_2 df_2 = d(\lambda_1 f_1 + \lambda_2 f_2)$ for $\lambda_i \in k$;

(2) $d\alpha = 0$ for a constant function $\alpha \in k$;

(3) $d(f_1 f_2) = f_1 df_2 + f_2 df_1$.

General differential forms are sums $\sum_i g_i\, dx_i$; they need not be of the form df for a single function.

Definition B.2.2 (Module of Kähler differentials)**.** Let $\varphi : A \to B$ be a ring homomorphism. Define the B-module $\Omega_{B/A}$ as the quotient of the free B-module generated by $\{d\beta \mid \beta \in B\}$, with the relations

(1) $d\beta_1 + d\beta_2 = d(\beta_1 + \beta_2)$;

(2) $d\alpha = 0$ for $\alpha \in \varphi(A)$;

(3) $d(\beta_1\beta_2) = \beta_1 d\beta_2 + \beta_2 d\beta_1$.

There is a canonical map $d : B \to \Omega_{B/A}$, $\beta \mapsto d\beta$.

Example B.2.3. If $A = \mathbb{C}$ and $B = \mathbb{C}[x_1, \ldots, x_n] = \Gamma(\mathbb{A}^n_{\mathbb{C}}, \mathcal{O})$, then

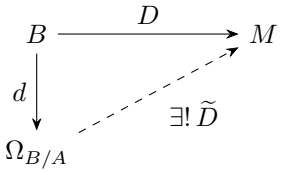
$$\Omega_{B/A} = \bigoplus_{i=1}^{n} B\, dx_i.$$

These are the regular differential forms on affine space.

Proposition B.2.4 (Universal property of $\Omega_{B/A}$)**.** *Let $\varphi : A \to B$ be a ring homomorphism. For a B-module M, an A-derivation $D : B \to M$ is an A-linear map satisfying*

$$D(b_1b_2) = b_1D(b_2) + b_2D(b_1).$$

The set of such maps is a B-module, denoted by $\mathrm{Der}_A(B, M)$. The canonical map $d : B \to \Omega_{B/A}$ is an A-derivation. For every $D \in \mathrm{Der}_A(B, M)$ there is a unique B-linear map $\widetilde{D} : \Omega_{B/A} \to M$ making the following diagram commute:



Thus $\Omega_{B/A}$ represents the functor $\mathrm{Der}_A(B, -)$:

$$\mathrm{Der}_A(B, M) \cong \mathrm{Hom}_B(\Omega_{B/A}, M).$$

Indeed, $\widetilde{D}(d\beta) = D(\beta)$ is well-defined by the defining relations. Notice that D and d are generally not B-linear; it is $\widetilde{D}$ that is B-linear.

Example B.2.5. Take $A = \mathbb{C}$ and $B = \mathbb{C}[x] = M$, corresponding to the affine line.

- $\mathrm{Der}_{\mathbb{C}}(\mathbb{C}[x], \mathbb{C}[x]) \cong \mathbb{C}[x]\langle\partial_x\rangle$ is free, with generator $\partial_x : f \mapsto f'$.
- $\Omega_{\mathbb{C}[x]/\mathbb{C}} \cong \mathbb{C}[x]\langle dx\rangle$, and

$$d : \mathbb{C}[x] \longrightarrow \Omega_{\mathbb{C}[x]/\mathbb{C}}, \qquad f \longmapsto f'(x)\, dx.$$

- $\mathrm{Hom}_{\mathbb{C}[x]}(\mathbb{C}[x]\langle dx\rangle, \mathbb{C}[x]) \cong \mathbb{C}[x]\langle \partial_x\rangle$. Here ∂_x is dual to dx, as in differential geometry.

Example B.2.6. (1) If $M = B$, then

$$\mathrm{Der}_A(B,B) \cong \mathrm{Hom}_B(\Omega_{B/A}, B) = \Omega^{\vee}_{B/A}.$$

Thus derivations with values in B form the dual module of differentials.

(2) If $A \to B \to 0$ is exact, then every $\beta \in B$ is the image of an element of A. Hence $d\beta = 0$ and $\Omega_{B/A} = 0$: there are no relative differentials.

(3) Take $A = \mathbb{C}$ and $B = \mathbb{C}[x_1, \ldots, x_n]$. Then

$$\mathrm{Der}_A(B,B) \cong \bigoplus_{j=1}^{n} B\langle \partial_{x_j}\rangle, \qquad \Omega_{B/A} \cong \bigoplus_{j=1}^{n} B\langle dx_j\rangle,$$

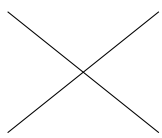
and the canonical map sends f to $df = \sum_j (\partial f/\partial x_j) dx_j$. To check this formula, apply the product rule repeatedly to a monomial:

$$d(x_1^{a_1} \cdots x_n^{a_n}) = \sum_j a_j x_1^{a_1} \cdots x_j^{a_j - 1} \cdots x_n^{a_n}\, dx_j,$$

and then use linearity. The partial derivations show that the dx_j are independent.

(4) Take $A = \mathbb{C}$ and $B = \mathbb{C}[x,y]/(xy)$, corresponding to two crossing lines, singular at the origin. Then

$$\Omega_{B/\mathbb{C}} = \frac{B\langle dx, dy\rangle}{\langle x\, dy + y\, dx\rangle}.$$

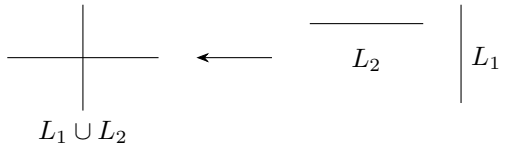


Since $d(xy) = 0$, the nonzero element $w := x\, dy = -y\, dx$ satisfies $xw = yw = 0$. Thus $B \cdot w = \mathbb{C}\langle w\rangle$ is a one-dimensional vector space supported at the origin. Furthermore,

$$\Omega_{B/\mathbb{C}}/B \cdot w \cong \mathbb{C}[x]\langle dx\rangle \oplus \mathbb{C}[y]\langle dy\rangle = \Omega_{B_2/\mathbb{C}} \oplus \Omega_{B_1/\mathbb{C}}.$$

We have the short exact sequence

$$0 \longrightarrow B \cdot w \longrightarrow \Omega_{B/\mathbb{C}} \longrightarrow \Omega_{B_2/\mathbb{C}} \oplus \Omega_{B_1/\mathbb{C}} \longrightarrow 0.$$



Taking the B-dual kills the torsion $B \cdot w$. More explicitly,

$$\mathrm{Der}_{\mathbb{C}}(B, B) = (x)\partial_x \oplus (y)\partial_y.$$

This is a module of derivations, whereas the tangent space at the origin is

$$T_{X,0} = \mathrm{Hom}_{\mathbb{C}}(\Omega_{B/\mathbb{C}} \otimes_B \mathbb{C}, \mathbb{C}) \cong \mathbb{C}^2.$$

One must distinguish this fibre dual from the dual module tensored with the residue field. Differentials retain the torsion information that the module dual loses.

Example B.2.7. Let $\mathbb{C} \hookrightarrow K$ be a finitely generated field extension.[91]

- $\{df_1, \ldots, df_n\}$ is a K-basis of $\Omega_{K/\mathbb{C}}$ if and only if $\{f_1, \ldots, f_n\}$ is a transcendence basis of $K/\mathbb{C}$. The dual basis belongs to $\mathrm{Der}_{\mathbb{C}}(K, K)$. Here the algebraic extension of $\mathbb{C}(f_1, \ldots, f_n)$ is separable, since we are in characteristic zero.
- A polynomial relation $P(f_1, \ldots, f_s) = 0$ gives the differential relation

$$\sum_i \frac{\partial P}{\partial x_i}(f_1, \ldots, f_s)\, df_i = 0.$$

There is no one-to-one correspondence between polynomial relations and their differentiated relations.

- If $K = k(X)$ is the function field of a variety over $\mathbb{C}$, then

$$\dim_K \mathrm{Der}_{\mathbb{C}}(K, K) = \mathrm{trdeg}_{\mathbb{C}}\, K = \dim X.$$

Indeed,

$$\dim_K \mathrm{Hom}_K(\Omega_{K/\mathbb{C}}, K) = \dim_K \Omega_{\mathcal{O}_{X,\eta}/\mathbb{C}} = \dim_K (\Omega_{X/\mathbb{C}})_\eta,$$

using compatibility of differentials with localization [Har77, II, 8.2A].

Proposition B.2.8 (Canonical construction)**.** *Let $\varphi : A \to B$ be a ring homomorphism, and consider multiplication*

$$\delta : B \otimes_A B \twoheadrightarrow B, \qquad f \otimes g \longmapsto fg.$$

Set $I = \ker \delta$. Then $B \cong (B \otimes_A B)/I$, and

$$I/I^2 \cong \Omega_{B/A}$$

as B-modules. The B-module structure comes from $I/I^2 = I \otimes_{B \otimes_A B} B$.

[91] For the field-extension properties of differentials used here, see [Har77, II, 8.6A].

Proof. For $\beta \in B$ and $u \in I$, define

$$\beta \cdot \bar{u} = \overline{(\beta \otimes 1)u} = \overline{(1 \otimes \beta)u}.$$

The equality holds because $\beta \otimes 1 - 1 \otimes \beta \in I$. Now define

$$\varphi : \Omega_{B/A} \longrightarrow I/I^2, \qquad d\beta \longmapsto \overline{1 \otimes \beta - \beta \otimes 1}.$$

Additivity and the relation $d\alpha = 0$ are respected. Moreover,

$$\begin{aligned} 1 \otimes \beta_1\beta_2 - \beta_1\beta_2 \otimes 1 = {} & (1 \otimes \beta_2)(1 \otimes \beta_1 - \beta_1 \otimes 1) \\ & + (\beta_1 \otimes 1)(1 \otimes \beta_2 - \beta_2 \otimes 1), \end{aligned}$$

which modulo I^2 is the image of $\beta_2 d\beta_1 + \beta_1 d\beta_2$. Hence φ is well-defined and B-linear.

To construct its inverse, use the square-zero ring $B \oplus \Omega_{B/A}$ with

$$(b, \omega)(c, \eta) = (bc, b\eta + c\omega).$$

The homomorphism

$$B \otimes_A B \longrightarrow B \oplus \Omega_{B/A}, \qquad b \otimes c \longmapsto (bc, b\,dc)$$

sends I into $0 \oplus \Omega_{B/A}$, whose square is zero; hence it kills I^2. For instance, for $b, c \in B$,

$$(1 \otimes b - b \otimes 1)(1 \otimes c - c \otimes 1) = 1 \otimes bc - c \otimes b - b \otimes c + bc \otimes 1$$

maps to $(0, d(bc) - c\,db - b\,dc) = 0$. It therefore induces a map on $(B \otimes_A B)/I^2$, which restricts to

$$\psi : I/I^2 \longrightarrow \Omega_{B/A}, \qquad \overline{\sum_i b_i \otimes c_i} \longmapsto \sum_i b_i\,dc_i, \quad \sum_i b_i c_i = 0.$$

In particular, $\psi(\overline{1 \otimes \beta - \beta \otimes 1}) = d\beta$. These elements generate I/I^2, and the $d\beta$ generate $\Omega_{B/A}$. Thus ψ and φ are inverse. □

Definition B.2.9. For a point x of a variety X, distinguish the stalk

$$(\Omega_{X/k})_x \cong \Omega_{\mathcal{O}_{X,x}/k}$$

from the fibre

$$\Omega_{X/k}(x) := (\Omega_{X/k})_x \otimes_{\mathcal{O}_{X,x}} k(x).$$

The first is an $\mathcal{O}_{X,x}$-module and the second a $k(x)$-vector space. On an affine open, the stalk is obtained by localizing the corresponding module at the prime of x. We often omit the subscript k.

We also use nonclosed points. For the generic point η,

$$\Omega_X(\eta) = \Omega_{k(X)/k} \otimes_{k(X)} k(X) = \Omega_{k(X)/k} = (\Omega_X)_\eta.$$

Thus the generic rank of Ω_X is $\dim X$.

Theorem B.2.10. *For a closed point $x \in X$, the following are naturally isomorphic k-vector spaces, with the last set given the vector-space structure from the others:*

(1) The Zariski tangent space $T_{X,x} = \mathrm{Hom}_k(\mathfrak{m}_x/\mathfrak{m}_x^2, k(x))$.

(2) The space of point differentials $\mathrm{Der}_k(\mathcal{O}_{X,x}, k(x))$, *namely k-linear maps satisfying*

$$D_x(fg) = f(x)D_x(g) + g(x)D_x(f).$$

(3) $\mathrm{Hom}_{\mathcal{O}_{X,x}}((\Omega_X)_x, k(x)) \cong \mathrm{Hom}_{k(x)}(\Omega_X(x), k(x))$.

(4) $\{f : I \to X \text{ over } k \mid f((\varepsilon)) = x\}$, *where* $I = \operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$.

Here $k(x) = k$, since x is closed and k is algebraically closed.

Proof. "(2) $\Leftrightarrow$ (3)" follows from the universal property:

$$\mathrm{Der}_k(\mathcal{O}_{X,x}, k(x)) \cong \mathrm{Hom}_{\mathcal{O}_{X,x}}(\Omega_{\mathcal{O}_{X,x}/k}, k(x)) = \mathrm{Hom}_{\mathcal{O}_{X,x}}((\Omega_X)_x, k(x)).$$

Since the target is killed by $\mathfrak{m}_x$, this is the dual of $\Omega_X(x)$.

"(1) $\Leftrightarrow$ (2)": a point differential kills $\mathfrak{m}_x^2$, because for $f, g \in \mathfrak{m}_x$,

$$D_x(fg) = f(x)D_x(g) + g(x)D_x(f) = 0.$$

It therefore induces a k-linear map $\bar{D}_x : \mathfrak{m}_x/\mathfrak{m}_x^2 \to k(x)$. Conversely, for a k-linear map $\varphi : \mathfrak{m}_x/\mathfrak{m}_x^2 \to k(x)$, define

$$D_{x,\varphi}(f) = \varphi((f - f(x)) \bmod \mathfrak{m}_x^2).$$

It is easy to verify the product rule. These constructions are inverse. The correspondence with (4) is described below. □

Corollary B.2.11. *For a closed point $x \in X$, there is an isomorphism of $k(x)$-vector spaces*

$$\Omega_X(x) = (\Omega_X)_x \otimes_{\mathcal{O}_{X,x}} k(x) \cong \mathfrak{m}_x/\mathfrak{m}_x^2, \qquad df \otimes 1 \longmapsto (f - f(x)) \bmod \mathfrak{m}_x^2.$$

Thus $\Omega_X(x)$ is the Zariski cotangent space.

Proof. The previous theorem identifies the duals of the two finite-dimensional vector spaces, so taking the $k(x)$-dual gives the claimed isomorphism. Equivalently, the inverse sends the class of $f \in \mathfrak{m}_x$ to $df \otimes 1$. This does not require that $(\Omega_X)_x$ be free. □

Now we explain (4) in Theorem B.2.10. The dual numbers $I = \operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ have one underlying point $p = (\varepsilon)$, together with a first-order infinitesimal direction. We have

$$\mathfrak{m}_p/\mathfrak{m}_p^2 = k\langle\varepsilon\rangle, \qquad T_{I,p} = k\langle\mathcal{L}\rangle, \qquad \mathcal{L}(\varepsilon) = 1.$$

Proposition B.2.12. *Let X be a variety, $x \in X$ a closed point, and $V \in T_{X,x}$ a tangent vector. Then there is a unique k-scheme morphism $f : I \to X$ such that $f(p) = x$ and $df(\mathcal{L}) = V$.*

Proof. Define the local k-algebra homomorphism

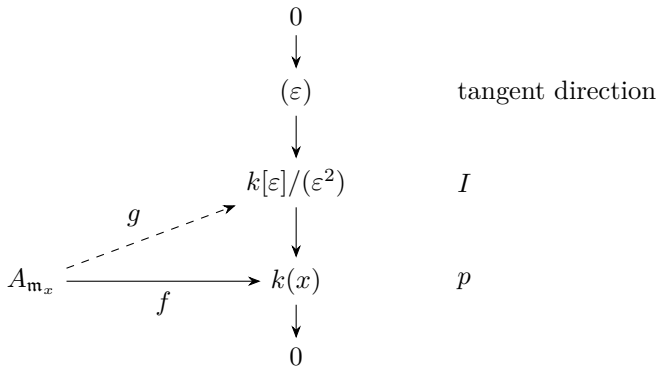
$$f^{\#} : \mathcal{O}_{X,x} \longrightarrow k[\varepsilon]/(\varepsilon^2), \qquad a \longmapsto a(x) + V\big((a - a(x)) \bmod \mathfrak{m}_x^2\big)\varepsilon.$$

The product rule proves multiplicativity, so this gives the required morphism. Conversely, any such local homomorphism induces

$$\bar{f} : \mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow (\varepsilon)/(\varepsilon^2) = k\langle\varepsilon\rangle.$$

Its tangent map sends $\mathcal{L}$ to $\mathcal{L} \circ \bar{f}$. Thus $df(\mathcal{L}) = V$ determines the coefficient of ε in $f^{\#}(a)$, while $f(p) = x$ determines the constant coefficient. This proves uniqueness. □

Returning to "(4) ⇔ (1)", take an affine neighborhood $X = \operatorname{Spec} A$ of $x = \mathfrak{m}_x$. The quotient $A_{\mathfrak{m}_x} \to k(x) = k$ can be placed in



The quotient $k[\varepsilon]/(\varepsilon^2) \to k$ has a k-algebra section, so there is always the constant lift $a \mapsto a(x)$. Other lifts have different coefficients of ε.

Proposition B.2.13. *The set of these lifts g is in bijective correspondence with $\Omega_X(x)^\vee = T_{X,x}$, where x is a closed point of $X = \operatorname{Spec} A$.*

Proof. Remember that $k(x) = k \cong k\langle\varepsilon\rangle$ as vector spaces.

Step 1: If g_1, g_2 are lifts, then $g_1 - g_2$ takes values in (ε). This ideal is a $k[\varepsilon]/(\varepsilon^2)$-module on which ε acts as zero, so the action of $A_{\mathfrak{m}_x}$ factors through evaluation at x. Write

$$g_i(a) = a(x) + h_i(a)\varepsilon.$$

Then $(g_1 - g_2)(a) = (h_1(a) - h_2(a))\varepsilon$ is k-linear. Moreover,

$$g_i(ab) = a(x)b(x) + \big(a(x)h_i(b) + b(x)h_i(a)\big)\varepsilon,$$

so $\theta = h_1 - h_2$ satisfies

$$\theta(ab) = a(x)\theta(b) + b(x)\theta(a).$$

Thus $\theta \in \mathrm{Der}_k(\mathcal{O}_{X,x}, k(x))$.

Step 2: Given any such derivation θ, the formula

$$a \longmapsto f(a) + \theta(a)\varepsilon$$

defines a lift of the evaluation map $f : A_{\mathfrak{m}_x} \to k$. This follows directly from the product rule and $\varepsilon^2 = 0$.

Step 3: Taking the constant lift as origin, all lifts are parametrized by

$$\mathrm{Der}_k(\mathcal{O}_{X,x}, k(x)) \cong \mathrm{Hom}_{\mathcal{O}_{X,x}}((\Omega_X)_x, k(x)) \cong \Omega_X(x)^\vee,$$

as desired. □

Now we give some computations of Ω_X.

Proposition B.2.14. *Let $Y \hookrightarrow X$ be a closed immersion of affine varieties, and let $x \in Y$ be closed. For the exact sequence*

$$0 \longrightarrow I_x \longrightarrow \mathcal{O}_{X,x} \longrightarrow \mathcal{O}_{Y,x} \longrightarrow 0,$$

there is a short exact sequence

$$0 \longrightarrow W_x \longrightarrow \Omega_X(x) \longrightarrow \Omega_Y(x) \longrightarrow 0, \qquad W_x = \sum_{f \in I_x} k(x)(df \otimes 1) \subseteq \Omega_X(x).$$

Before the proof, let us make some analysis. The conormal sequence is right exact:

$$I_x/I_x^2 \longrightarrow (\Omega_X)_x \otimes_{\mathcal{O}_{X,x}} \mathcal{O}_{Y,x} \longrightarrow (\Omega_Y)_x \longrightarrow 0.$$

Tensoring with $k(x)$ preserves right exactness, although $k(x)$ is generally not flat over the local ring. Its left-hand image is W_x. Dualizing over $k(x)$ gives

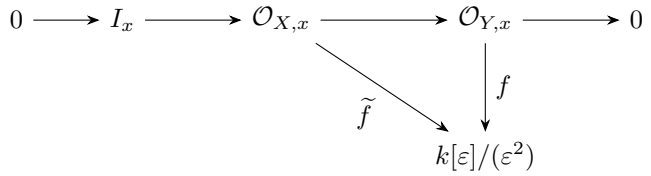
$$0 \longrightarrow T_{Y,x} \longrightarrow T_{X,x} \longrightarrow W_x^\vee \longrightarrow 0.$$

Geometrically, $T_{Y,x}$ is the subspace annihilated by the differentials of the local defining equations of Y.

Proof. The inclusion of tangent spaces is dual to the surjection on cotangent spaces. We also give a set-theoretic description using dual numbers:

$$T_{X,x} = \{\widetilde{f} : \mathcal{O}_{X,x} \to k[\varepsilon]/(\varepsilon^2) \text{ over } k \mid \widetilde{f}^{-1}((\varepsilon)) = \mathfrak{m}_x\},$$
$$T_{Y,x} = \{f : \mathcal{O}_{Y,x} \to k[\varepsilon]/(\varepsilon^2) \text{ over } k \mid f^{-1}((\varepsilon)) = \mathfrak{m}_{Y,x}\}.$$

The inclusion comes from composing with the quotient map:



A tangent vector $\widetilde{f}$ factors through $\mathcal{O}_{Y,x}$ if and only if $I_x \subseteq \ker \widetilde{f}$. Writing its corresponding linear functional as $\bar{f} \in \Omega_X(x)^\vee$, we have

$$\widetilde{f}(\varphi) = \varphi(x) + \langle \bar{f}, d\varphi \otimes 1 \rangle \varepsilon.$$

Here the pairing comes from

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \cong \Omega_{\mathcal{O}_{X,x}/k} \otimes k(x), \qquad \varphi - \varphi(x) \longmapsto d\varphi \otimes 1.$$

Since $I_x \subseteq \mathfrak{m}_x$, the constant term vanishes for $\varphi \in I_x$. Thus the factorization condition is precisely $\langle \bar{f}, d\varphi \otimes 1 \rangle = 0$ for every $\varphi \in I_x$. It says that $\bar{f}$ annihilates W_x, not that the differentials themselves are zero. This proves the description. □

Corollary B.2.15. *Let $X \subseteq \mathbb{A}_k^n$ have ideal $I = (g_1, \ldots, g_m)$, and suppose that $x = 0 \in X$. Then $T_{X,0}$ is the kernel of the Jacobian matrix $(\partial g_i/\partial x_j)(0) : k^n \to k^m$, as in differential geometry.*

Proof. By the previous proposition,

$$0 \longrightarrow T_{X,0} \longrightarrow T_{\mathbb{A}^n,0} \xrightarrow{\varphi} \left(\sum_i k\, dg_i|_0 \right)^\vee \longrightarrow 0.$$

Furthermore,

$$\begin{aligned} \Omega_{\mathbb{A}^n}(0) &= (\Omega_{\mathbb{A}^n})_0 \otimes_{\mathcal{O}_{\mathbb{A}^n,0}} k(0) \\ &\cong \Omega_{k[x_1,\ldots,x_n]/k} \otimes_{k[x_1,\ldots,x_n]} k(0) \\ &= \mathrm{Span}_k\{dx_1|_0, \ldots, dx_n|_0\}. \end{aligned}$$

Thus $T_{\mathbb{A}^n,0}$ has dual basis $\partial_{x_j}|_0$. For $v = \sum_j v_j \partial_{x_j}|_0$, we compute

$$v(dg_i|_0) = v\left(\sum_j \frac{\partial g_i}{\partial x_j}(0) dx_j|_0 \right) = \sum_j \frac{\partial g_i}{\partial x_j}(0) v_j.$$

Hence $v \in \ker \varphi$ if and only if

$$\begin{pmatrix} \frac{\partial g_1}{\partial x_1}(0) & \cdots & \frac{\partial g_1}{\partial x_n}(0) \\ \vdots & & \vdots \\ \frac{\partial g_m}{\partial x_1}(0) & \cdots & \frac{\partial g_m}{\partial x_n}(0) \end{pmatrix} \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = 0.$$

All values are taken in $k(0) = k$. □

Proposition B.2.16. *Let $B = A[x_1, \ldots, x_n]/I = R/I$ be a finitely presented A-algebra, with $I = (f_1, \ldots, f_m)$. Then $\Omega_{B/A}$ is generated over B by $dx_1, \ldots, dx_n$, with relations*

$$\sum_{j=1}^{n} \frac{\partial f_i}{\partial x_j}\, dx_j = 0 \qquad (1 \le i \le m),$$

where coefficients are read in B. If A is a k-algebra, there is also the transitivity sequence

$$\Omega_{A/k} \otimes_A B \longrightarrow \Omega_{B/k} \longrightarrow \Omega_{B/A} \longrightarrow 0.$$

Proof. The second exact sequence for differentials [Eis95, 16.12] gives

$$I/I^2 \xrightarrow{\delta} \Omega_{R/A} \otimes_R B \longrightarrow \Omega_{B/A} \longrightarrow 0, \qquad \bar{f} \longmapsto df \otimes 1.$$

The module I/I^2 is generated by $\bar{f}_1, \ldots, \bar{f}_m$, and $\Omega_{R/A} \otimes_R B = \bigoplus_j B\, dx_j$. On these generators,

$$\delta(\bar{f}_i) = df_i \otimes 1 = \sum_j \frac{\partial f_i}{\partial x_j}\, dx_j.$$

Consequently,

$$\Omega_{B/A} \cong \frac{\bigoplus_{j=1}^{n} B\, dx_j}{\left\langle \sum_j (\partial f_i/\partial x_j) dx_j \right\rangle_{i=1}^{m}},$$

as claimed. □

B.3 Implicit Function Theorem in Algebraic Geometry

Recall that in analytic geometry, if $X \subseteq \mathbb{C}^n$ is a submanifold of dimension $n - r$, then locally there is an analytic change of coordinates such that $X = V(z_1, \ldots, z_r)$. This is the Implicit Function Theorem [GH78, p. 20]. In algebraic geometry, we have:

Theorem B.3.1. *Let X be a nonsingular variety and $i : Y \hookrightarrow X$ a closed subvariety. For a closed point $x \in Y$, the variety Y is nonsingular at x if and only if there is an affine open neighborhood U of x and functions $f_1, \ldots, f_r \in \mathcal{O}_X(U)$, where $r = \operatorname{codim}_X Y$, such that*

(1) $Y \cap U = \operatorname{Spec}(\mathcal{O}_X(U)/(f_1, \ldots, f_r))$*;*

(2) $df_1(x), \ldots, df_r(x)$ *are linearly independent in* $\Omega_X(x)$*.*

In this situation the f_i *form a regular sequence in* $\mathcal{O}_{X,x}$*. Equivalently, after shrinking around* x*,* i *is a regular immersion and the conormal map identifies* I/I^2 *with a rank-*r *subbundle of* $\Omega_X|_Y$*.*

Here "subbundle" means that the quotient is locally free. A regular sequence alone does not ensure independence of its differentials at x.

Corollary B.3.2. *If* Y *is nonsingular, there is a short exact sequence of locally free sheaves on* Y*:*

$$0 \longrightarrow I/I^2 \longrightarrow \Omega_X|_Y \longrightarrow \Omega_Y \longrightarrow 0.$$

Taking duals gives

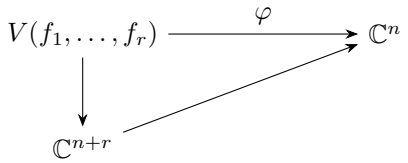
$$0 \longrightarrow T_Y \longrightarrow T_X|_Y \longrightarrow N_X Y \longrightarrow 0,$$

where $N_X Y = \mathcal{H}om_{\mathcal{O}_Y}(I/I^2, \mathcal{O}_Y)$ *is the normal bundle of* Y *in* X*.*

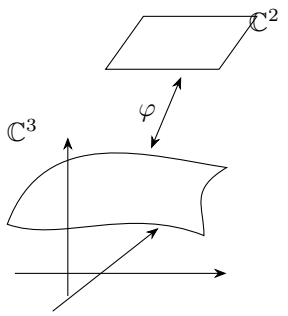
Now return to the analytic Implicit Function Theorem. Let $f_1, \ldots, f_r$ be analytic functions near a common zero $x \in \mathbb{C}^{n+r}$, with

$$\det\left(\frac{\partial f_i}{\partial z_j}(x)\right)_{1 \le i,j \le r} \neq 0.$$

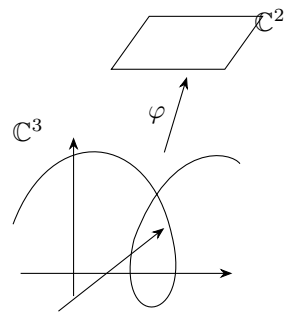
Then the projection to the last n coordinates induces a local isomorphism



$$(z_1, \ldots, z_{n+r}) \longmapsto (z_{r+1}, \ldots, z_{n+r})$$



local isomorphism



not a local isomorphism

In algebraic geometry, a Zariski open neighborhood may be too large for this conclusion. We need to modify two things:

(1) the condition on a map corresponding to a "local isomorphism";

(2) the notion of a sufficiently small neighborhood.

Thus we turn to *étale morphisms*. Let R be a finite type k-algebra.

Definition B.3.3 (Étale morphism)**.** Let $Y = \operatorname{Spec} R$ and

$$X = \operatorname{Spec} R[x_1, \dots, x_n]/(f_1, \dots, f_n) \hookrightarrow Y \times \mathbb{A}^n_k,$$

with $f : X \to Y$ induced by $R \to R[x_1, \dots, x_n]/(f_1, \dots, f_n)$. This morphism is *étale at* x if

$$\det\left(\frac{\partial f_i}{\partial x_j}\right)(x) \neq 0 \quad \text{in } k(x).$$

In general, a morphism of algebraic schemes is étale at x if, on suitable open neighborhoods of x and $f(x)$, it has such a Jacobian presentation, allowing an open subset of the displayed affine scheme:

$$\begin{array}{ccc} U & \xrightarrow{\text{open immersion}} & \operatorname{Spec} R[\mathbf{x}]/(\mathbf{f}) \\ {\scriptstyle f|_U}\downarrow & & \downarrow \\ V & \xrightarrow[\cong]{} & \operatorname{Spec} R \end{array}$$

The Jacobian determinant is required to be nonzero at x. Such morphisms are flat at x [SP25, Tag 02GS].

Remark B.3.4. Set $S = R[x_1, \dots, x_n]/(f_1, \dots, f_n)$. The transitivity sequence is

$$S \otimes_R \Omega_{R/k} \longrightarrow \Omega_{S/k} \longrightarrow \Omega_{S/R} \longrightarrow 0.$$

The S-module $\Omega_{S/R}$ is generated by $dx_1, \dots, dx_n$ with relations

$$\begin{pmatrix} \partial f_1/\partial x_1 & \cdots & \partial f_1/\partial x_n \\ \vdots & & \vdots \\ \partial f_n/\partial x_1 & \cdots & \partial f_n/\partial x_n \end{pmatrix} \begin{pmatrix} dx_1 \\ \vdots \\ dx_n \end{pmatrix} = 0.$$

For this square presentation,

$$\text{étale at } x \quad \Longleftrightarrow \quad \Omega_{X/Y}(x) = 0 \quad \Longleftrightarrow \quad \det(\partial f_i/\partial x_j)(x) \neq 0.$$

By Nakayama's lemma, $\Omega_{X/Y}(x) = 0$ implies that the coherent sheaf $\Omega_{X/Y}$ is zero on a neighborhood of x. In this presentation, f is therefore étale everywhere if and only if $\Omega_{X/Y} = 0$. For arbitrary finite type morphisms one must also require flatness.

Now we temporarily drop the assumption that k is algebraically closed.

Proposition B.3.5. *Let X be a scheme of finite type over k. Then $X \to \operatorname{Spec} k$ is étale if and only if*[92]

$$X = \coprod_{i=1}^{s} \operatorname{Spec} k_i,$$

where every k_i/k is finite separable. In particular, if $\bar{k} = k$, then X is a finite disjoint union of copies of $\operatorname{Spec} k$.

Proof. "⇒": on an affine open $U = \operatorname{Spec} R$, the étale Jacobian presentation has $\Omega_{R/k} = 0$. The finite type algebra criterion [Eis95, 16.16] implies that R is a finite product of finite separable extensions of k. Thus each point of X is open and has this form. Quasi-compactness makes the number of points finite.

"⇐": a finite separable extension is étale over k. Indeed, by the primitive element theorem it has a presentation $k[t]/(h)$ with h' invertible. The finite disjoint union is therefore étale. In particular, X is reduced. □

Proposition B.3.6. *Let $f : X \to Y$ be an étale morphism of algebraic schemes.*

(1) For $y \in Y$, the scheme-theoretic fibre is a finite disjoint union of $\operatorname{Spec} k_i$, *with each $k_i/k(y)$ finite separable.*

(2) For $x \in X$ and $y = f(x)$,

$$\mathfrak{m}_y \mathcal{O}_{X,x} = \mathfrak{m}_x, \qquad k(y) \hookrightarrow k(x) \text{ is finite separable.}$$

Proof. (1) Étaleness is stable under base change, so in the fibre square

$$\begin{array}{ccc} X_y & \longrightarrow & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ \operatorname{Spec} k(y) & \longrightarrow & Y \end{array}$$

the morphism f' is étale. Apply the previous proposition over $k(y)$.

(2) Locally write $Y = \operatorname{Spec} R$, $X = \operatorname{Spec} S$, and let $\mathfrak{p} \subset R$, $\mathfrak{q} \subset S$ represent y, x, with $\mathfrak{q} \cap R = \mathfrak{p}$. The base-change and localization diagram is

$$\begin{array}{ccc} S & \longrightarrow & S_{\mathfrak{q}} \\ \downarrow & & \downarrow \\ S \otimes_R k(\mathfrak{p}) & \longrightarrow & S_{\mathfrak{q}}/\mathfrak{p}S_{\mathfrak{q}}. \end{array}$$

The bottom-right ring is $\mathcal{O}_{X_y,x} = \mathcal{O}_{X,x}/\mathfrak{m}_y\mathcal{O}_{X,x}$. By (1), it is a field, necessarily $k(x)$. Thus $\mathfrak{m}_y\mathcal{O}_{X,x} = \mathfrak{m}_x$, and the residue-field extension is finite separable. These are the local unramifiedness conditions. □

[92] For finite separable algebras and vanishing of their differentials, see [Eis95, 16.16], [Har77, II, 8.6A].

We return to an algebraically closed field k.

Proposition B.3.7. *Let $f : X \to Y$ be an étale morphism of nonempty varieties over k. Then*[93]

(1) f is dominant;

(2) $k(Y) \hookrightarrow k(X)$ is a finite separable extension;

(3) for a closed point $y \in Y$, the fibre is a finite set of reduced points $\operatorname{Spec} k$ *(possibly empty).*

Proof. (3) follows from the previous proposition and algebraic closedness. For (1), an étale morphism is open, so $f(X)$ is a nonempty open subset of the irreducible variety Y. Thus $\overline{f(X)} = Y$. For (2), take $x = \eta_X$ and $y = \eta_Y$ in Proposition B.3.6. □

Example B.3.8. An open immersion is always étale: locally it is the isomorphism

$$\begin{array}{ccc} U & \xrightarrow{\sim} & \operatorname{Spec} R[t]/(t) \\ \downarrow & & \downarrow \\ i(U) & \xrightarrow{\sim} & \operatorname{Spec} R. \end{array}$$

Thus étale morphisms need not be surjective or finite. For example, $\mathbb{A}^1 \setminus \{0\} \hookrightarrow \mathbb{A}^1$ is étale but not finite.

Remark B.3.9. For a finite type morphism of algebraic schemes,

$$f \text{ is étale} \iff f \text{ is flat and } \Omega_{X/Y} = 0 \iff f \text{ is flat and unramified.}$$

Definition B.3.10 (Étale neighborhood)**.** For a point $x \in X$, an étale neighborhood means a pointed morphism

$$\varphi : (U', x') \longrightarrow (U, x),$$

where $U \subseteq X$ is a Zariski open neighborhood, φ is étale, and $\varphi(x') = x$. When equality of residue fields is needed, we additionally require the induced map $k(x) \to k(x')$ to be an isomorphism.

To find a "smaller neighborhood", we consider the formal neighborhood of an algebraic scheme.

[93] The dimension argument for dominance can also be expressed using [SP25, Tag 03WT], [Har77, II, Exercise 3.22(b)].



nodal curve
$y^2 - x^2(x+1) = 0$

crossing lines
$x^2 - y^2 = 0$

Over $\mathbb{C}$, these curves are analytically isomorphic near the origin, but their Zariski local rings are not isomorphic. In fact,

$$\mathcal{O}_{C_1,0} = \mathbb{C}[x,y]_{(x,y)}/(y^2 - x^2(x+1))$$

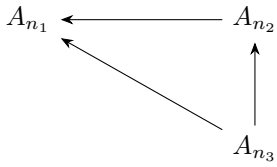
is an integral domain, whereas

$$\mathcal{O}_{C_2,0} = \mathbb{C}[x,y]_{(x,y)}/(x^2 - y^2) \cong \mathbb{C}[u,v]_{(u,v)}/(uv)$$

is not. The last isomorphism is a linear change of coordinates.

But suppose that we adjoin a unit u with $u^2 = 1 + x$. Then $y^2 - x^2(x+1) = (y - ux)(y + ux)$, and replacing x by ux straightens the two branches. Analytically, the branch $\sqrt{1+x}$ exists near 0. In algebraic geometry we can make the same change in the formal completion.

Let A be a finite type k-algebra with maximal ideal $\mathfrak{m}$. Put $A_n = A/\mathfrak{m}^n$. For $n_1 < n_2 < n_3$ there is a commutative diagram



More explicitly, the maps are the quotient maps $A_{n_3} \twoheadrightarrow A_{n_2} \twoheadrightarrow A_{n_1}$. This inverse system has surjective transition maps and satisfies the Mittag–Leffler condition. Its inverse limit is the $\mathfrak{m}$-adic completion

$$\widehat{A} = \varprojlim_n A/\mathfrak{m}^n.$$

Its infinite expansions converge in the $\mathfrak{m}$-adic topology; no analytic convergence is being asserted.

Example B.3.11. For $A = k[x,y]$ and $\mathfrak{m} = (x,y)$, the completion is the formal power-series ring $\widehat{A} = k[[x,y]]$.

Proposition B.3.12. *For a maximal ideal $\mathfrak{m} \subset A$, the following diagram commutes:*

$$\begin{array}{ccc} A & \longrightarrow & A_{\mathfrak{m}} \\ \downarrow & & \downarrow \\ \widehat{A} & \xrightarrow{\sim} & \widehat{A_{\mathfrak{m}}}, \end{array}$$

where the right-hand completion is along $\mathfrak{m}A_{\mathfrak{m}}$. Thus completion at a maximal ideal depends only on the corresponding local ring.

Indeed, $A/\mathfrak{m}^n \to A_{\mathfrak{m}}/\mathfrak{m}^n A_{\mathfrak{m}}$ is an isomorphism for every n.

Example B.3.13. For $C_1 = V(y^2 - x^2(x+1))$ and $C_2 = V(x^2 - y^2)$ over $\mathbb{C}$, we have

$$\begin{array}{ccc} \mathcal{O}_{C_1,0} & \longrightarrow & \widehat{\mathcal{O}}_{C_1,0} \\ \not\cong & & \cong \\ \mathcal{O}_{C_2,0} & \longrightarrow & \widehat{\mathcal{O}}_{C_2,0}. \end{array}$$

Indeed, $u = \sqrt{1+x} \in \mathbb{C}[[x]]$ is a unit, and the substitution $x \mapsto x\sqrt{1+x}$, $y \mapsto y$ is an automorphism of $\mathbb{C}[[x,y]]$. It carries $y^2 - x^2$ to $y^2 - x^2(x+1)$.

Remark B.3.14. The completed local ring records the formal local structure. Over $\mathbb{C}$, for a closed point we have the comparison

$$\mathcal{O}_{X,x} \hookrightarrow \mathcal{O}^{\mathrm{an}}_{X,x} \hookrightarrow \widehat{\mathcal{O}}_{X,x} \cong \widehat{\mathcal{O}^{\mathrm{an}}_{X,x}}.$$

Thus the analytic and algebraic local rings have the same formal completion, although the local rings themselves need not agree.

Definition B.3.15 (Formal neighborhood)**.** Let X be locally noetherian and $x \in X$. Its formal neighborhood at x is

$$\mathrm{Spf}(\widehat{\mathcal{O}}_{X,x}), \qquad \widehat{\mathcal{O}}_{X,x} = \varprojlim_n \mathcal{O}_{X,x}/\mathfrak{m}_x^n,$$

with the maximal-ideal-adic topology. It has one underlying point and the complete topological local ring $\widehat{\mathcal{O}}_{X,x}$ as its ring of sections.

Recall that the completion of a noetherian *local* ring is faithfully flat over that local ring.

Proposition B.3.16. *Let $\mathcal{F} \subseteq \mathcal{G}$ be coherent sheaves on a locally noetherian scheme X, and let $i : \widehat{X} \to X$ be the formal neighborhood at x. If the induced inclusion $i^*\mathcal{F} \to i^*\mathcal{G}$ is an isomorphism, then $\mathcal{F} = \mathcal{G}$ on a Zariski open neighborhood of x.*

Proof. Take $x = \mathfrak{p} \in U = \operatorname{Spec} A$, and write $\mathcal{F}|_U = \widetilde{M} \subseteq \widetilde{N} = \mathcal{G}|_U$. The relevant local diagram is

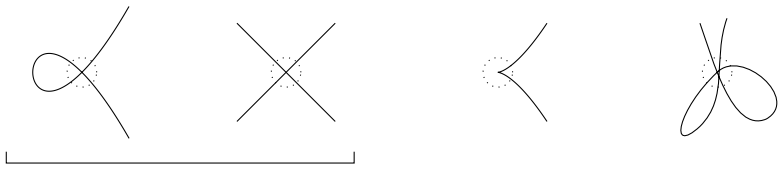
$$\begin{array}{ccccc} A & \longrightarrow & A_{\mathfrak{p}} & \longrightarrow & \widehat{A_{\mathfrak{p}}} \\ & & \downarrow & & \downarrow \\ & & k(x) & \xrightarrow{\sim} & \widehat{A_{\mathfrak{p}}}/\mathfrak{m}_{\widehat{A_{\mathfrak{p}}}}. \end{array}$$

It corresponds to $\operatorname{Spec} k(x) \to \widehat{X} \xrightarrow{i} X$. The given inclusion becomes

$$M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \widehat{A_{\mathfrak{p}}} \xrightarrow{\sim} N_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \widehat{A_{\mathfrak{p}}}.$$

Faithful flatness over $A_{\mathfrak{p}}$ implies $(N/M)_{\mathfrak{p}} = 0$. Since N/M is finite, its support is closed, and it vanishes on a neighborhood of x. Hence $\mathcal{F} = \mathcal{G}$ there. $\square$

We can use $\widehat{\mathcal{O}}_{X,x}$ to distinguish formal singularity types of curves.



same formal singularity

Example B.3.17. Over $\mathbb{C}$, a curve C has an A_n-singularity at x if

$$\widehat{\mathcal{O}}_{C,x} \cong \mathbb{C}[[u,v]]/(v^2 - u^{n+1}) \qquad (n \geq 1).$$

Changing the sign is immaterial over $\mathbb{C}$. The two left-hand curves in the picture have an A_1-singularity, since

$$\mathbb{C}[[u,v]]/(v^2 - u^2) \cong \mathbb{C}[[s,t]]/(st)$$

by an invertible linear change of coordinates $s = v - u$, $t = v + u$.

The formal structure can distinguish curves with the same local topological branch configuration.

Example B.3.18. Let X and Y be unions of four distinct lines through $0 \in \mathbb{C}^2$:

$$X = V\left(\prod_{i=1}^{4}(x + a_i y)\right), \qquad Y = V\left(\prod_{i=1}^{4}(x + b_i y)\right).$$

Assume that their unordered sets of four directions are not equivalent under $\mathrm{PGL}_2(\mathbb{C})$.



X $\qquad\qquad$ Y

Their completed local rings are

$$\widehat{\mathcal{O}}_{X,0} = \mathbb{C}[[x,y]]/\left(\prod_i (x + a_i y)\right), \qquad \widehat{\mathcal{O}}_{Y,0} = \mathbb{C}[[x,y]]/\left(\prod_i (x + b_i y)\right).$$

Suppose they were isomorphic. The isomorphism would induce an isomorphism of the associated graded rings

$$\mathrm{gr}_{\mathfrak{m}}\, \widehat{\mathcal{O}}_{X,0} = \mathbb{C} \oplus \mathfrak{m}/\mathfrak{m}^2 \oplus \mathfrak{m}^2/\mathfrak{m}^3 \oplus \cdots,$$

and similarly for Y. In degree one it gives $\varphi_1 : \mathfrak{m}/\mathfrak{m}^2 \xrightarrow{\sim} \bar{\mathfrak{m}}/\bar{\mathfrak{m}}^2$. This invertible linear change of coordinates must carry one tangent cone to the other, and hence send the four directions to the other four directions by a Möbius transformation. This contradicts the assumption. Equivalently, the cross ratios must agree up to the six values obtained by permuting four points. Different coefficients alone do not guarantee different formal types. For example, direction sets with cross ratios 2 and 3 are not equivalent, since the respective orbits are

$$\{2, 1/2, -1\} \quad \text{and} \quad \{3, 1/3, -2, -1/2, 3/2, 2/3\}.$$

Before the main theorem, we need two lemmas from commutative algebra.

Lemma B.3.19. *Let $f : (A, \mathfrak{m}_A) \to (B, \mathfrak{m}_B)$ be a local homomorphism of noetherian local rings, inducing an isomorphism of residue fields. Then the induced map $\operatorname{gr}_{\mathfrak{m}_A} A \to \operatorname{gr}_{\mathfrak{m}_B} B$ is an isomorphism if and only if $\widehat{f} : \widehat{A} \to \widehat{B}$ is an isomorphism.*

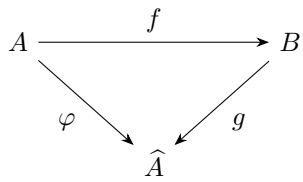
Proof. "⇒": use the exact sequences

$$0 \longrightarrow \mathfrak{m}_A^r/\mathfrak{m}_A^{r+1} \longrightarrow A/\mathfrak{m}_A^{r+1} \longrightarrow A/\mathfrak{m}_A^r \longrightarrow 0$$

and their counterparts for B. Induction and the short five lemma show that $A/\mathfrak{m}_A^r \to B/\mathfrak{m}_B^r$ is an isomorphism for every r. Take inverse limits.

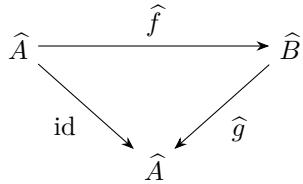
"⇐": completion preserves the quotients by powers of the maximal ideal, and hence the associated graded ring [Eis95, 7.13]. Apply this to the induced isomorphism of completions. □

Lemma B.3.20. *Let $(A, \mathfrak{m}_A)$ and $(B, \mathfrak{m}_B)$ be noetherian local rings, with local homomorphisms $f : A \to B$ and $g : B \to \widehat{A}$ fitting into*



where φ is the completion map. If $\mathfrak{m}_A B = \mathfrak{m}_B$, then $\widehat{f} : \widehat{A} \to \widehat{B}$ is an isomorphism.

Proof. Complete the maps to obtain $\widehat{f} : \widehat{A} \to \widehat{B}$ and $\widehat{g} : \widehat{B} \to \widehat{A}$, with $\widehat{g}\widehat{f} = \mathrm{id}_{\widehat{A}}$. Thus $\widehat{f}$ is injective and $\widehat{g}$ is surjective:



Set $I = \ker \widehat{g}$. The split exact sequence of $\widehat{A}$-modules

$$0 \longrightarrow \widehat{A} \xrightarrow{\widehat{f}} \widehat{B} \longrightarrow I \longrightarrow 0, \qquad b \longmapsto b - \widehat{f}(\widehat{g}(b)),$$

gives $\widehat{B} = \widehat{A} \oplus I$ as modules. The assumption yields

$$\mathfrak{m}_{\widehat{A}}\widehat{B} = \mathfrak{m}_A\widehat{B} = (\mathfrak{m}_A B)\widehat{B} = \mathfrak{m}_B\widehat{B} = \mathfrak{m}_{\widehat{B}}.$$

Since $\widehat{g}$ is a local surjection, under the decomposition we also have $\mathfrak{m}_{\widehat{B}} = \mathfrak{m}_{\widehat{A}} \oplus I$. On the other hand,

$$\mathfrak{m}_{\widehat{A}}\widehat{B} = \mathfrak{m}_{\widehat{A}}(\widehat{A} \oplus I) = \mathfrak{m}_{\widehat{A}} \oplus \mathfrak{m}_{\widehat{A}} I.$$

Hence $I = \mathfrak{m}_{\widehat{A}} I \subseteq \mathfrak{m}_{\widehat{B}} I \subseteq I$. The ideal I is finite over the noetherian ring $\widehat{B}$, so Nakayama's lemma implies $I = 0$. Therefore $\widehat{f}$ and $\widehat{g}$ are inverse isomorphisms. □

Our main theorem says that an étale morphism is the algebraic analogue of a local diffeomorphism.

Theorem B.3.21. *Let $f : X \to Y$ be a finite type morphism between noetherian schemes. Let $x \in X$ and $y = f(x)$, and assume that the induced map $k(y) \to k(x)$ is an isomorphism. Then f is étale at x if and only if the induced map*

$$\widehat{f}_x^* : \widehat{\mathcal{O}}_{Y,y} \longrightarrow \widehat{\mathcal{O}}_{X,x}$$

is an isomorphism.

The proof is given in B.3A below. Over an algebraically closed field, the residue fields at closed points are both k, and we obtain:

Corollary B.3.22. *Let X, Y be schemes of finite type over an algebraically closed field k. A k-morphism $f : X \to Y$ is étale if and only if, for every closed point $x \in X$, the induced map $\widehat{\mathcal{O}}_{Y,f(x)} \to \widehat{\mathcal{O}}_{X,x}$ is an isomorphism.*

Indeed, the étale locus is open, and a nonempty closed subset of a finite type k-scheme contains a closed point.

Corollary B.3.23. *Let $f : X \to Y$ be a morphism of varieties over an algebraically closed field. For a closed point x and $y = f(x)$, consider the induced diagram of tangent cones and tangent spaces:*

$$\begin{array}{ccc} C_xX & \longrightarrow & T_{X,x} \\ \downarrow & & \downarrow{\scriptstyle df_x} \\ C_yY & \longrightarrow & T_{Y,y} \end{array}$$

Then f is étale if and only if the induced maps $C_xX \to C_yY$ are isomorphisms for all closed $x \in X$.

Proof. "⇒": the theorem gives an isomorphism of completed local rings. By Lemma B.3.19, it induces an isomorphism of associated graded rings. Recall that

$$C_x X = \operatorname{Spec}\left(\bigoplus_{n\geq 0} \mathfrak{m}_x^n/\mathfrak{m}_x^{n+1}\right) = \operatorname{Spec} \operatorname{gr}_{\mathfrak{m}_x} \mathcal{O}_{X,x}.$$

Thus the induced tangent-cone map is an isomorphism.

"⇐": first note that the image of a closed point is closed. Locally, if $X = \operatorname{Spec} B$, $Y = \operatorname{Spec} A$, and $x = \mathfrak{m}$, its image corresponds to $\mathfrak{p} = \mathfrak{m} \cap A$, and

$$A/\mathfrak{p} \hookrightarrow B/\mathfrak{m} = k(x).$$

The right-hand field is finite-dimensional over k by Zariski's lemma. Hence $A/\mathfrak{p}$ is a finite-dimensional k-algebra and an integral domain. Multiplication by a nonzero element is injective and therefore surjective, so $A/\mathfrak{p}$ is a field. Thus y is closed, and here $k(y) = k(x) = k$.

An isomorphism of the *induced* cone maps gives an isomorphism of the induced graded ring maps. Lemma B.3.19 gives the isomorphism on completions, and Theorem B.3.21 implies étaleness at x. Apply Corollary B.3.22. Abstract isomorphism of the cones, without reference to the map induced by f, would not be sufficient. □

Remark B.3.24. An isomorphism on completions does not imply that the original local ring map is an isomorphism. For example, take $A = B = \mathbb{C}[x]_{(x)}$ and let $f : A \to B$ send x to $x + x^2$. This map is not an isomorphism (the corresponding function field extension has degree two), but its completion[94]

$$\widehat{f} : \mathbb{C}[[x]] \longrightarrow \mathbb{C}[[x]], \qquad x \longmapsto x + x^2$$

is an isomorphism. Its inverse sends x to the formal power series $h(x)$ with $h(0) = 0$ and $h(x) + h(x)^2 = x$. Equivalently, $h(x + x^2) = x$. Thus completion does not recover the full Zariski local ring map.

A finite étale morphism is the algebraic analogue of a finite covering map in analytic topology.

B.3A Proof of the Main Theorem

Proof of Theorem B.3.21. Set $A = \mathcal{O}_{Y,y}$ and $B = \mathcal{O}_{X,x}$, and identify their residue fields by the given map.

"⇒": if f is étale at x, then $\mathfrak{m}_A B = \mathfrak{m}_B$. We construct a local A-algebra map $g : B \to \widehat{A}$ extending the completion map of A. Choose a local Jacobian presentation for f near x. The residue-field point gives a solution of its equations

[94] For the structure of the completed regular local ring, see [Eis95, 10.16].

modulo $\mathfrak{m}_A$. Suppose a solution has been lifted modulo $\mathfrak{m}_A^r$. Lift its coordinates arbitrarily modulo $\mathfrak{m}_A^{r+1}$. The errors lie in $\mathfrak{m}_A^r/\mathfrak{m}_A^{r+1}$, and the invertible Jacobian matrix gives a correction that kills all errors. This is the linear Taylor formula, since products of two corrections vanish modulo $\mathfrak{m}_A^{r+1}$. Any denominators in the local presentation remain units. Induction produces compatible solutions for every r, and their inverse limit defines g. Lemma B.3.20 now gives $\widehat{A} \xrightarrow{\sim} \widehat{B}$.

"$\Leftarrow$": suppose the induced map $\widehat{A} \to \widehat{B}$ is an isomorphism. Since $\widehat{A}$ is flat over A and $\widehat{B}$ is faithfully flat over B, flatness descends to $A \to B$. The isomorphism also gives

$$\mathfrak{m}_A \widehat{B} = \mathfrak{m}_{\widehat{A}} \widehat{B} = \mathfrak{m}_{\widehat{B}} = \mathfrak{m}_B \widehat{B}.$$

Faithful flatness over B therefore implies $\mathfrak{m}_A B = \mathfrak{m}_B$. Together with the isomorphism of residue fields, this is the local unramifiedness criterion for a map essentially of finite type. Thus $A \to B$ is flat and unramified, so f is étale at x. This also agrees with [SP25, Tag 0257]. □

B.4 Blowing-up Properties and Infinitesimal Lifting

Definition B.4.1 (Regular sequence and depth)**.** Let A be a ring and M an A-module. A sequence $x_1, \dots, x_r \in A$ is regular for M if $M/(x_1, \dots, x_r)M \neq 0$ and

- x_1 is not a zero divisor on M;
- x_2 is not a zero divisor on M/x_1M;
- in general, x_i is not a zero divisor on $M/\sum_{j<i} x_j M$.

For a noetherian local ring $(A, \mathfrak{m})$ and a nonzero finite A-module M, $\operatorname{depth} M$ is the maximal length of an M-regular sequence with all $x_i \in \mathfrak{m}$.

Definition B.4.2 (Cohen–Macaulay ring)**.** A noetherian local ring A is Cohen–Macaulay if $\operatorname{depth} A = \dim A$.

Example B.4.3. The following rings are Cohen–Macaulay:

- $k[[x_1, \dots, x_n]]$;
- $k[x_1, \dots, x_n]_{\mathfrak{m}}$ for any maximal ideal $\mathfrak{m}$;
- the ring $\mathbb{Z}_p$ of p-adic integers;
- $k[\varepsilon]/(\varepsilon^2)$.

Theorem B.4.4. *Let $(A, \mathfrak{m})$ be a noetherian local ring.*

(a) If A is regular, it is Cohen–Macaulay.

(b) If A is Cohen–Macaulay, then so is $A_\mathfrak{p}$ for every $\mathfrak{p} \in \operatorname{Spec} A$.

(c) If A is Cohen–Macaulay and $x_1, \dots, x_r \in \mathfrak{m}$, then these elements form a regular sequence if and only if

$$\dim A/(x_1, \dots, x_r) = \dim A - r.$$

(d) If A is Cohen–Macaulay and $x_1, \dots, x_r \in \mathfrak{m}$ is regular, then $A/(x_1, \dots, x_r)$ is Cohen–Macaulay.

(e) Under the hypotheses of (d), put $I = (x_1, \dots, x_r)$. The natural map

$$(A/I)[t_1, \dots, t_r] \longrightarrow \operatorname{gr}_I A = \bigoplus_{n \geq 0} I^n/I^{n+1}, \qquad t_i \longmapsto x_i \bmod I^2$$

is an isomorphism. In particular, I/I^2 is free over A/I of rank r, and $\operatorname{Sym}^n(I/I^2) \to I^n/I^{n+1}$ is an isomorphism for every $n \geq 1$.

Theorem B.4.5. *Let X be a nonsingular variety over $\mathbb{C}$ and $Y \subsetneq X$ a nonempty nonsingular closed subvariety, with ideal sheaf $\mathcal{I}$. Let $\pi : \widetilde{X} = \operatorname{Bl}_Y X \to X$ be the blowing-up, and let $Y' \subseteq \widetilde{X}$ be defined by*

$$\mathcal{I}' = \pi^{-1}\mathcal{I} \cdot \mathcal{O}_{\widetilde{X}}.$$

Then:

(a) $\widetilde{X}$ is nonsingular;

(b) $Y' \to Y$ is isomorphic to the projective bundle $\mathbb{P}(\mathcal{I}/\mathcal{I}^2)$, with convention $\mathbb{P}(\mathcal{E}) = \operatorname{Proj} \operatorname{Sym} \mathcal{E}$;

(c) under this isomorphism, the normal sheaf is

$$N_{Y'/\widetilde{X}} = \mathcal{H}om_{\mathcal{O}_{Y'}}(\mathcal{I}'/\mathcal{I}'^2, \mathcal{O}_{Y'}) \cong \mathcal{O}_{\mathbb{P}(\mathcal{I}/\mathcal{I}^2)}(-1).$$

Proof. (b) Since $\widetilde{X} = \operatorname{Proj}_X \bigoplus_{n \geq 0} \mathcal{I}^n$, base change to Y gives

$$\begin{aligned} Y' &\cong \operatorname{Proj}_Y \bigoplus_{n \geq 0} (\mathcal{I}^n \otimes_{\mathcal{O}_X} \mathcal{O}_X/\mathcal{I}) \\ &\cong \operatorname{Proj}_Y \bigoplus_{n \geq 0} \mathcal{I}^n/\mathcal{I}^{n+1}. \end{aligned}$$

By Theorem B.3.1, $\mathcal{I}$ is locally generated by a regular sequence of length $r = \operatorname{codim}_X Y$. Theorem B.4.4(e) gives

$$\mathcal{I}^n/\mathcal{I}^{n+1} \cong \operatorname{Sym}^n(\mathcal{I}/\mathcal{I}^2),$$

where $\mathcal{I}/\mathcal{I}^2$ is locally free of rank r. Hence

$$Y' \cong \operatorname{Proj}_Y \operatorname{Sym}(\mathcal{I}/\mathcal{I}^2) = \mathbb{P}(\mathcal{I}/\mathcal{I}^2).$$

(a) The scheme Y' is locally $Y \times \mathbb{P}^{r-1}$, so it is nonsingular. The tautological sheaf on the blow-up satisfies

$$\mathcal{O}_{\widetilde{X}}(1) \cong \mathcal{I}' = \mathcal{O}_{\widetilde{X}}(-Y').$$

This can be checked on each standard blow-up chart: the pulled-back ideal is generated there by the chosen homogeneous generator. Thus Y' is an effective Cartier divisor. At a point $p \in Y'$, its local ring is the quotient of $\mathcal{O}_{\widetilde{X},p}$ by a nonzero divisor. A noetherian local ring whose quotient by such an element is regular is itself regular. Hence $\widetilde{X}$ is regular along Y'. Outside Y', π is an isomorphism to $X \setminus Y$, so $\widetilde{X}$ is nonsingular everywhere.

(c) Since $\mathcal{I}' \cong \mathcal{O}_{\widetilde{X}}(1)$,

$$\mathcal{I}'/\mathcal{I}'^2 \cong \mathcal{I}' \otimes_{\mathcal{O}_{\widetilde{X}}} \mathcal{O}_{Y'} \cong \mathcal{O}_{Y'}(1).$$

Taking the dual on Y' gives $N_{Y'/\widetilde{X}} \cong \mathcal{O}_{Y'}(-1)$. □

Corollary B.4.6 (Blowing up a nonsingular subvariety)**.** *Let X be a nonsingular variety and $Y \subseteq X$ a nonempty nonsingular irreducible closed subvariety of codimension $r \geq 2$. For $\pi : \widetilde{X} = \operatorname{Bl}_Y X \to X$ and $Y' = \pi^{-1}(Y)$,*[95]

(a) $\operatorname{Pic} \widetilde{X} \cong \operatorname{Pic} X \oplus \mathbb{Z}$*;*

(b) $\omega_{\widetilde{X}} \cong \pi^* \omega_X \otimes \mathcal{O}_{\widetilde{X}}((r-1)Y')$.

Proof. (a) Both X and $\widetilde{X}$ are nonsingular, so we can use divisor class groups. Since $\operatorname{codim}_X Y \geq 2$,

$$\operatorname{Pic}(X \setminus Y) \cong \operatorname{Pic} X, \qquad \widetilde{X} \setminus Y' \cong X \setminus Y.$$

Restriction therefore gives a surjection $\operatorname{Pic} \widetilde{X} \to \operatorname{Pic} X$, whose kernel is generated by $[Y']$ [Har77, II, 6.5]. The map $\mathbb{Z} \to \operatorname{Pic} \widetilde{X}$, $n \mapsto \mathcal{O}_{\widetilde{X}}(nY')$, is injective: restricting to a fibre $F \cong \mathbb{P}^{r-1}$ gives $\mathcal{O}_F(-n)$, which is trivial only for $n = 0$. Thus

$$0 \longrightarrow \mathbb{Z} \longrightarrow \operatorname{Pic} \widetilde{X} \longrightarrow \operatorname{Pic} X \longrightarrow 0$$

is exact. The pullback π^* splits it. Note the sign convention

$$\mathcal{O}_{\widetilde{X}}(1) = \mathcal{O}_{\widetilde{X}}(-Y').$$

(b) By (a), write

$$\omega_{\widetilde{X}} \cong \pi^* L \otimes \mathcal{O}_{\widetilde{X}}(qY')$$

[95] For restriction of divisor classes and adjunction, see [Har77, II,6.5(b)], [Har77, II, 6.18].

for some $L \in \operatorname{Pic} X$ and $q \in \mathbb{Z}$. Restricting to $\widetilde{X} \setminus Y' \cong X \setminus Y$ gives $L|_{X\setminus Y} \cong \omega_X|_{X\setminus Y}$. The isomorphism $\operatorname{Pic} X \to \operatorname{Pic}(X \setminus Y)$ therefore gives $L \cong \omega_X$. To determine q, apply adjunction to Y':

$$\begin{aligned}\omega_{Y'} &\cong (\omega_{\widetilde{X}} \otimes \mathcal{O}_{\widetilde{X}}(Y'))|_{Y'} \\ &\cong (\pi^*\omega_X)|_{Y'} \otimes \mathcal{O}_{Y'}(-q-1).\end{aligned}$$

Now choose a closed point $y \in Y$ and the fibre $F = \pi^{-1}(y) \cong \mathbb{P}^{r-1}_{k(y)}$. The smooth projective-bundle map $Y' \to Y$ gives

$$\omega_{Y'}|_F \cong \omega_F \otimes_{k(y)} \det \Omega_Y(y).$$

The last factor is a one-dimensional vector space, so it gives a trivial line bundle on F. Also $(\pi^*\omega_X)|_F$ is trivial. Consequently,

$$\omega_F \cong \mathcal{O}_F(-q-1).$$

But $\omega_{\mathbb{P}^{r-1}} \cong \mathcal{O}_{\mathbb{P}^{r-1}}(-r)$, so $q = r-1$, as required. □

There is also a useful generalization of Proposition B.2.13.

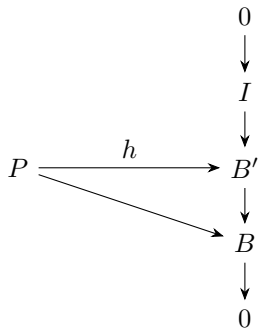
Lemma B.4.7. *Let $B' \twoheadrightarrow B$ be a surjective homomorphism of k-algebras with kernel I satisfying $I^2 = 0$, and let $P \to B$ be a k-algebra homomorphism.*

(a) If $h, h' : P \to B'$ are lifts, then $\theta = h - h'$ is a k-derivation from P to I.

(b) Conversely, if h is a lift and $\theta : P \to I$ is a k-derivation, then $h' = h + \theta$ is another lift.

Since $I^2 = 0$, the ideal I is naturally a B-module and hence a P-module. Thus $\operatorname{Der}_k(P, I) = \operatorname{Hom}_P(\Omega_{P/k}, I)$.

Proof. The lifting diagram is



If h' is another lift, then $h - h'$ takes values in I and is additive. For $a_1, a_2 \in P$,

$$\begin{aligned}\theta(a_1a_2) &= h(a_1)h(a_2) - h'(a_1)h'(a_2)\\ &= h(a_1)\big(h(a_2) - h'(a_2)\big) + \big(h(a_1) - h'(a_1)\big)h'(a_2)\\ &= a_1\theta(a_2) + a_2\theta(a_1),\end{aligned}$$

where the final action is through B. For $c \in k$, $\theta(c) = c - c = 0$, so θ is a k-derivation.

Conversely, $h + \theta$ is k-linear and preserves 1. Since products of two values of θ lie in $I^2 = 0$,

$$\begin{aligned}(h+\theta)(a_1a_2) &= h(a_1)h(a_2) + h(a_1)\theta(a_2) + h(a_2)\theta(a_1)\\ &= (h(a_1) + \theta(a_1))(h(a_2) + \theta(a_2)).\end{aligned}$$

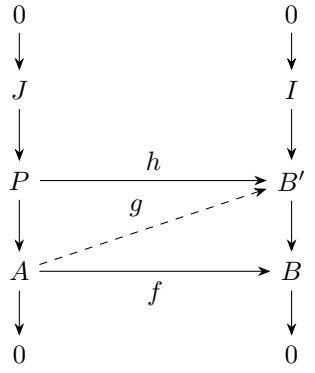
Thus it is a homomorphism, and it has the same reduction in B. □

Proposition B.4.8 (Infinitesimal lifting property)**.** *Let k be algebraically closed, and let A be a finitely generated k-algebra such that* Spec A *is nonsingular. Given a square-zero extension of k-algebras*

$$0 \longrightarrow I \longrightarrow B' \longrightarrow B \longrightarrow 0, \qquad I^2 = 0,$$

and a homomorphism $f : A \to B$, there exists a k-algebra homomorphism $g : A \to B'$ lifting f.

Proof. Write $A = P/J$ with $P = k[x_1, \ldots, x_n]$. Since $B' \to B$ is onto, choose lifts of the images of the variables to obtain $h : P \to B'$ in the diagram



Then $h(J) \subseteq I$, so $h(J^2) \subseteq I^2 = 0$, inducing $\bar{h} : J/J^2 \to I$. This map is A-linear, since

$$\bar{h}((p+J)(j+J^2)) = f(p+J)\bar{h}(j+J^2).$$

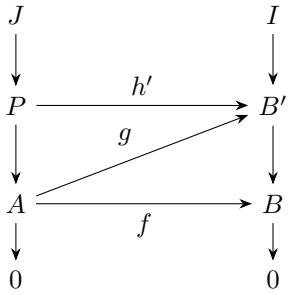
Smoothness gives the exact conormal sequence

$$0 \longrightarrow J/J^2 \longrightarrow \Omega_{P/k} \otimes_P A \longrightarrow \Omega_{A/k} \longrightarrow 0.$$

The module $\Omega_{A/k}$ is finite locally free, hence finite projective [Har77, II, 8.15; SP25, Tag 00NX]. Therefore $\operatorname{Ext}^1_A(\Omega_{A/k}, I) = 0$, and applying $\operatorname{Hom}_A(-, I)$ gives

$$0 \longrightarrow \operatorname{Hom}_A(\Omega_{A/k}, I) \longrightarrow \operatorname{Hom}_P(\Omega_{P/k}, I) \longrightarrow \operatorname{Hom}_A(J/J^2, I) \longrightarrow 0.$$

Choose a preimage of $\bar{h}$, viewed as a derivation $\theta : P \to I$. By Lemma B.4.7, $h' = h - \theta : P \to B'$ is a homomorphism with the same reduction in B. For $j \in J$, $h'(j) = h(j) - \theta(j) = 0$ by the choice of θ. Thus h' factors through $A = P/J$, giving the desired lift g:



□

Appendix C
Chevalley's Constructibility Theorem

Definition C.0.1. For a noetherian scheme X, a constructible subset $S \subseteq X$ is a finite union

$$S = S_1 \cup \cdots \cup S_t,$$

where every S_i is locally closed, that is, the intersection of an open subset and a closed subset.

Theorem C.0.2 (Chevalley)**.** *Let $f : X \to Y$ be a finite type morphism with Y noetherian. Then the image $f(S)$ of every constructible subset $S \subseteq X$ is constructible in Y.*

Proof. First, X is also noetherian. Choose finite affine covers $\{U_{ij}\}$ of X and $\{V_i\}$ of Y, with $f(U_{ij}) \subseteq V_i$. Since

$$f(S) = \bigcup_{i,j} f(S \cap U_{ij}),$$

it is enough to treat affine X and Y.

Write $X = \operatorname{Spec} R$ and $Y = \operatorname{Spec} S$, where $R = S[a_1, \ldots, a_n]$ is a finitely generated S-algebra. The factorization

$$\operatorname{Spec} S[a_1, \ldots, a_n] \longrightarrow \operatorname{Spec} S[a_1, \ldots, a_{n-1}] \longrightarrow \cdots \longrightarrow \operatorname{Spec} S$$

reduces the assertion to one generator. A one-generator algebra is a quotient of $S[x]$, and a closed immersion preserves constructibility. It therefore suffices to prove the assertion for

$$f : \operatorname{Spec} S[x] \longrightarrow \operatorname{Spec} S.$$

Use noetherian induction on closed subsets of $\operatorname{Spec} S$. If the assertion fails, choose a minimal closed subset $V(I)$ over which it fails. Replace S by S/I, and then by its reduction; these operations do not change the underlying topological question. Thus we may assume that S is reduced, and that the assertion holds after restriction to every proper closed subset of $\operatorname{Spec} S$.

The base must also be irreducible. Otherwise, write $\operatorname{Spec} S = Z_1 \cup Z_2$ with both Z_i proper closed. For a constructible $C \subseteq \operatorname{Spec} S[x]$, induction gives constructibility of

$$f(C \cap f^{-1} Z_i) = f(C) \cap Z_i \quad (i = 1, 2).$$

Their union is $f(C)$, contradicting the choice of the base. Hence S is an integral domain. The following lemma supplies the nonempty open subset needed to finish the induction. □

Lemma C.0.3. *Let S be a noetherian integral domain, with generic point $\eta \in \operatorname{Spec} S$. Let $C \subseteq \operatorname{Spec} S[x]$ be irreducible closed, and let $C_0 \subseteq C$ be open. For the projection $f : \operatorname{Spec} S[x] \to \operatorname{Spec} S$, there is a nonempty open subset $U \subseteq \operatorname{Spec} S$ such that either $U \subseteq f(C_0)$ or $U \cap f(C_0) = \varnothing$.*

Completion of the proof of Theorem C.0.2. Apply the lemma to a locally closed irreducible subset C_0. If $U \subseteq f(C_0)$, then

$$f(C_0) = U \ \cup \ \big(f(C_0) \cap (\operatorname{Spec} S \setminus U)\big).$$

The second term is the image of $C_0 \cap f^{-1}(\operatorname{Spec} S \setminus U)$ under the map over the proper closed subset $\operatorname{Spec} S \setminus U$. By the induction hypothesis it is constructible, so $f(C_0)$ is constructible. If $U \cap f(C_0) = \varnothing$, the whole image lies in that proper closed subset, and the same argument applies. Every constructible subset of the noetherian space $\operatorname{Spec} S[x]$ is a finite union of locally closed irreducible subsets. Its image is therefore constructible. This contradicts the assumed failure and proves the theorem. □

Proof of Lemma C.0.3. If C_0 is empty, take $U = \operatorname{Spec} S$. Assume $C_0 \neq \varnothing$, and set $K = \operatorname{Frac} S$. The generic fibre is

$$f^{-1}(\eta) = \operatorname{Spec} S[x] \times_{\operatorname{Spec} S} \operatorname{Spec} K \cong \operatorname{Spec} K[x] = \mathbb{A}^1_K.$$

The intersection $C \cap f^{-1}(\eta)$ is either empty or an irreducible closed subset of this affine line. There are three cases:

Case i) C contains the whole generic fibre;

Case ii) $C \cap f^{-1}(\eta) = \{\zeta\}$, a closed point of the generic fibre;

Case iii) $C \cap f^{-1}(\eta) = \varnothing$.

In Case (i), C contains the generic point of $\operatorname{Spec} S[x]$, so $C = \operatorname{Spec} S[x]$. The nonempty open C_0 contains a distinguished open $D(g)$, with

$$g = a_0 x^n + a_1 x^{n-1} + \cdots + a_n \neq 0, \qquad a_0 \neq 0.$$

Take $U = D(a_0) \subseteq \operatorname{Spec} S$. For $s \in U$, the polynomial $\bar{g} \in k(s)[x]$ is nonzero, so $D(\bar{g}) \subseteq \mathbb{A}^1_{k(s)}$ is nonempty. It is contained in $C_0 \cap f^{-1}(s)$, hence $s \in f(C_0)$. Thus $U \subseteq f(C_0)$.

In Case (ii), write $C = V(\mathfrak{p})$. Since $\mathfrak{p} \cap S = (0)$,

$$\mathfrak{p}K[x] = gK[x]$$

for an irreducible polynomial of positive degree. Clearing denominators, we can choose its generator in $\mathfrak{p}$ and write

$$g = a_0 x^n + a_1 x^{n-1} + \cdots + a_n, \qquad a_i \in S, \quad a_0 \neq 0, \quad n \geq 1.$$

Then $V(g) \supseteq C \supseteq C_0$. Since C_0 is nonempty open in C, it contains the generic point of C, which is the point ζ above η. Thus all three sets meet the generic fibre in exactly $\{\zeta\}$.

Write $C_0 = C \cap W$ with W open in $\operatorname{Spec} S[x]$. The set

$$V(g) \setminus C_0 = (V(g) \setminus C) \cup (V(g) \setminus W)$$

is constructible and disjoint from $f^{-1}(\eta)$. Decompose it into finitely many irreducible locally closed subsets W_i, with generic points w_i. None of the $f(w_i)$ equals η, and

$$f(W_i) \subseteq \overline{\{f(w_i)\}} \subsetneq \operatorname{Spec} S.$$

The union of these finitely many proper closed subsets is proper because S is a domain. Choose $0 \neq \alpha \in S$ with

$$f(V(g) \setminus C_0) \subseteq V(\alpha).$$

Set $U = D(a_0\alpha)$. For $s \in U$,

$$f^{-1}(s) \cap C_0 = f^{-1}(s) \cap V(g) = V(\bar{g}) \subseteq \mathbb{A}^1_{k(s)}.$$

The polynomial $\bar{g}$ still has positive degree n, so it has an irreducible factor and $V(\bar{g})$ is nonempty, even when it has no $k(s)$-rational zero. Thus $s \in f(C_0)$, proving $U \subseteq f(C_0)$.

In Case (iii), let ζ be the generic point of C. Its image is not η, and

$$f(C) \subseteq \overline{\{f(\zeta)\}} \subsetneq \operatorname{Spec} S.$$

The nonempty open complement of this closure is disjoint from $f(C_0)$. This completes the proof. □

Appendix D
Spectral Sequences in Algebraic Geometry

D.1 Bicomplexes

Many spectral sequences arise from bicomplexes, so we begin with them. We work with modules over a ring R, and mostly use cohomological conventions. For the tensor-product examples, R is assumed commutative.

Definition D.1.1 (Graded module)**.** A graded module is a $\mathbb{Z}$-indexed family $M = (M^i)_{i\in\mathbb{Z}}$ of R-modules, often denoted by $M^\bullet$.

Example D.1.2. (i) If $(C^\bullet, d)$ is a cochain complex

$$\cdots \longrightarrow C^{n-1} \xrightarrow{d} C^n \xrightarrow{d} C^{n+1} \longrightarrow \cdots,$$

then $C^\bullet = (C^n)_{n\in\mathbb{Z}}$ is a graded module.

(ii) The cohomology $H^\bullet(C^\bullet) = (H^p(C^\bullet))_{p\in\mathbb{Z}}$ is a graded module.

Definition D.1.3 (Graded maps)**.** For graded modules $M^\bullet, N^\bullet$ and $a \in \mathbb{Z}$, a graded map of degree a is a family

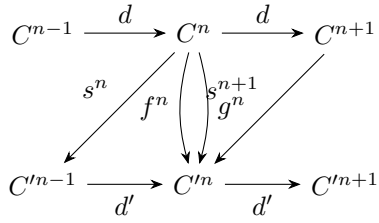
$$f = (f^p : M^p \to N^{p+a})_{p\in\mathbb{Z}}.$$

We write $\deg f = a$.

Example D.1.4. (i) The differential of a cochain complex is a graded map of degree $+1$.

(ii) A cochain map $f : C^\bullet \to C'^\bullet$ is a graded map of degree 0.

(iii) A homotopy $s = (s^n : C^n \to C'^{n-1})$ between cochain maps f, g has degree -1, and $f - g = d's + sd$:



Proposition D.1.5. *If $M^\bullet \xrightarrow{f} N^\bullet \xrightarrow{g} P^\bullet$ have degrees a, b, respectively, then $g \circ f$ has degree $a + b$.*

The usual category of graded modules uses degree-zero morphisms. Maps of other degrees can be regarded as maps into a shifted grading. For a fixed degree a, the group of homogeneous maps is

$$\mathrm{Hom}^a(M^\bullet, N^\bullet) = \prod_p \mathrm{Hom}_R(M^p, N^{p+a}).$$

The collection of all homogeneous maps is obtained by allowing all a; when the degree is part of the data, this is a disjoint union of these groups. It should not be confused with the degree-zero Hom group.

Definition D.1.6 (Exactness of graded modules)**.** A graded submodule $M'^\bullet \subseteq M^\bullet$ consists of submodules $M'^p \subseteq M^p$. Its quotient is $(M^p/M'^p)_p$. Both inclusion and quotient maps have degree zero. For a degree-a map $f : M^\bullet \to N^\bullet$,

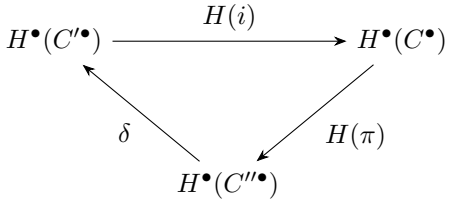
$$\ker f^\bullet = (\ker f^p)_p \subseteq M^\bullet, \qquad (\mathrm{im}\, f)^p = \mathrm{im}\, f^{p-a} \subseteq N^p.$$

Thus exactness of $A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{g} C^\bullet$ means $\mathrm{im}\, f^{p-a} = \ker g^p$ for every p.

A short exact sequence of cochain complexes

$$0 \longrightarrow C'^\bullet \xrightarrow{i} C^\bullet \xrightarrow{\pi} C''^\bullet \longrightarrow 0$$

gives a long exact cohomology sequence, which can be displayed as an exact triangle:



The first two maps have degree zero, while δ has degree $+1$. More generally, an exact triangle $(A, B, C; \alpha, \beta, \gamma)$ with degrees a, b, c gives

$$\cdots \longrightarrow B^{p-b-c} \xrightarrow{\beta} C^{p-c} \xrightarrow{\gamma} A^p \xrightarrow{\alpha} B^{p+a} \longrightarrow \cdots.$$

Definition D.1.7 (Bigraded modules)**.** A bigraded module is a doubly indexed family $M^{\bullet,\bullet} = (M^{p,q})_{(p,q)\in\mathbb{Z}^2}$. We put p on the horizontal axis and q on the vertical axis.

Definition D.1.8 (Bigraded maps)**.** A map $f : M^{\bullet,\bullet} \to N^{\bullet,\bullet}$ of bidegree (a, b) is a family

$$f^{p,q} : M^{p,q} \longrightarrow N^{p+a,q+b}.$$

If $\deg g = (a', b')$, then $\deg(gf) = (a + a', b + b')$. For a fixed bidegree,

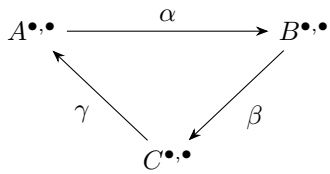
$$\mathrm{Hom}^{a,b}(M, N) = \prod_{p,q} \mathrm{Hom}_R(M^{p,q}, N^{p+a,q+b}).$$

As in the singly graded case, the usual category uses bidegree $(0, 0)$ morphisms; other homogeneous degrees can be recorded separately.

Submodules and quotients are taken in each bidegree. If $\deg f = (a, b)$, then

$$(\ker f)^{p,q} = \ker f^{p,q}, \qquad (\operatorname{im} f)^{p,q} = \operatorname{im} f^{p-a,q-b}.$$

Thus exactness means equality of these submodules at each position. An exact triangle of bigraded modules



with bidegrees $(a, a'), (b, b'), (c, c')$ gives long exact sequences

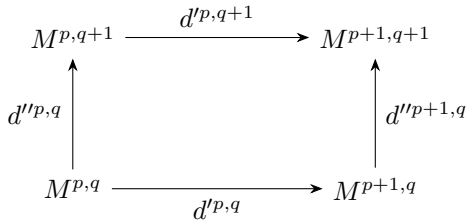
$$\cdots \to C^{p-c,q-c'} \xrightarrow{\gamma} A^{p,q} \xrightarrow{\alpha} B^{p+a,q+a'} \xrightarrow{\beta} C^{p+a+b,q+a'+b'} \to \cdots .$$

Conversely, the family of such sequences determines the exact triangle.

Definition D.1.9 (Double complex)**.** A bicomplex, or double complex, is a triple $(M^{\bullet,\bullet}, d', d'')$, with $\deg d' = (1, 0)$, $\deg d'' = (0, 1)$, $(d')^2 = (d'')^2 = 0$, and

$$d'^{p,q+1} d''^{p,q} + d''^{p+1,q} d'^{p,q} = 0.$$

Thus the following square anticommutes:



Example D.1.10. Suppose that d', d'' are differentials of bidegrees $(1, 0), (0, 1)$ whose squares commute rather than anticommute:

$$\begin{array}{ccc} \bullet & \xrightarrow{d'} & \bullet \\ {\scriptstyle d''}\uparrow & & \uparrow{\scriptstyle d''} \\ \bullet & \xrightarrow{d'} & \bullet. \end{array}$$

This need not be a double complex under our convention. Define $\Delta''^{p,q} = (-1)^p d''^{p,q}$. Then $(\Delta'')^2 = 0$ and

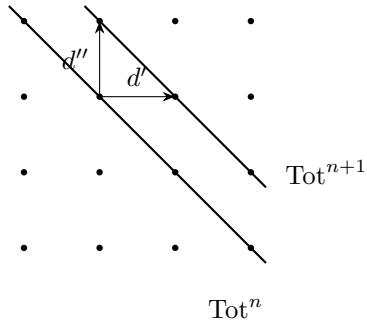
$$d'^{p,q+1}\Delta''^{p,q} + \Delta''^{p+1,q} d'^{p,q} = (-1)^p d'^{p,q+1} d''^{p,q} + (-1)^{p+1} d''^{p+1,q} d'^{p,q} = 0.$$

This is the sign change trick, and (M, d', Δ'') is a double complex.

Definition D.1.11 (Total complex)**.** For a bicomplex, its total complex $\mathrm{Tot}(M)$ has terms

$$\mathrm{Tot}(M)^n = \bigoplus_{p+q=n} M^{p,q},$$

with differential D^n restricting on $M^{p,q}$ to $d'^{p,q} + d''^{p,q}$.



Lemma D.1.12. *If $M^{\bullet,\bullet}$ is a double complex, then $(\mathrm{Tot}(M), D)$ is a complex.*

Proof. The image of $M^{p,q}$ lies in $M^{p+1,q} \oplus M^{p,q+1}$, so D raises total degree by one. Moreover,

$$D^2 = (d')^2 + (d'd'' + d''d') + (d'')^2 = 0,$$

where the middle term vanishes by anticommutativity. □

Example D.1.13. (i) Let $A_\bullet$ and $B_\bullet$ be nonnegative chain complexes:

$$\cdots \to A_p \xrightarrow{\Delta'} A_{p-1} \to \cdots \to A_0 \to 0, \qquad \cdots \to B_q \xrightarrow{\Delta''} B_{q-1} \to \cdots \to B_0 \to 0.$$

Here we use the homological version of the conventions above: differentials lower the relevant index. Define

$$M_{p,q} = A_p \otimes_R B_q, \qquad d'_{p,q} = \Delta'_p \otimes \mathrm{id}, \qquad d''_{p,q} = (-1)^p \mathrm{id} \otimes \Delta''_q.$$

This homological bicomplex is concentrated in the first quadrant.

Definition D.1.14 (First quadrant bicomplex)**.** A cohomological or homological bicomplex is first quadrant if $M^{p,q} = 0$, respectively $M_{p,q} = 0$, whenever $p < 0$ or $q < 0$.

Continuing Example D.1.13(i), its total chain complex is the tensor product:

$$(A_\bullet \otimes_R B_\bullet)_n = \bigoplus_{p+q=n} A_p \otimes_R B_q, \qquad D(a \otimes b) = \Delta' a \otimes b + (-1)^p a \otimes \Delta'' b.$$

(ii) If

$$I_A = (0 \to I^0 \to I^1 \to \cdots), \qquad J_B = (0 \to J^0 \to J^1 \to \cdots)$$

are deleted injective resolutions, the same tensor-product construction with cohomological indices gives the first quadrant double complex $I_A \otimes_R J_B$.

D.2 Filtrations and Exact Couples

Definition D.2.1 (Filtration)**.** A filtration of a module M is a family of submodules $(M^p)_{p\in\mathbb{Z}}$ with

$$\cdots \subseteq M^{p+1} \subseteq M^p \subseteq M^{p-1} \subseteq \cdots .$$

The factors form the graded module $(M^p/M^{p+1})_p$.

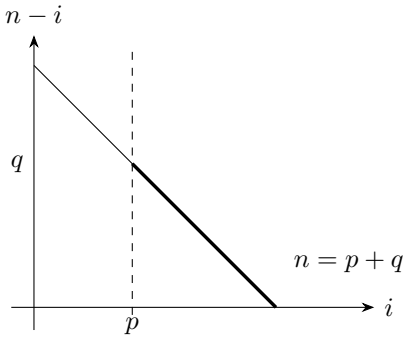
Similarly, in an abelian category, a filtration of a cochain complex is a family of subcomplexes $(F^pC^\bullet)_p$ with $F^{p+1}C^\bullet \subseteq F^pC^\bullet$. The quotients are $F^pC^\bullet/F^{p+1}C^\bullet$. For every n, the corresponding column in the diagram is a filtration:

$$\begin{array}{ccccc}
C^{n-1} & \longrightarrow & C^n & \longrightarrow & C^{n+1} \\
\uparrow & & \uparrow & & \uparrow \\
F^pC^{n-1} & \longrightarrow & F^pC^n & \longrightarrow & F^pC^{n+1} \\
\uparrow & & \uparrow & & \uparrow \\
F^{p+1}C^{n-1} & \longrightarrow & F^{p+1}C^n & \longrightarrow & F^{p+1}C^{n+1}.
\end{array}$$

Filtrations need not reach either 0 or the whole module in any finite degree. Now let (M, d', d'') be a double cochain complex.

Definition D.2.2 (First filtration)**.** The first filtration of $\mathrm{Tot}(M)$ is

$$\begin{aligned}
({}_IF^p \,\mathrm{Tot}(M))^n &= \bigoplus_{i\geq p} M^{i,n-i} \\
&= \cdots \oplus M^{p+2,q-2} \oplus M^{p+1,q-1} \oplus M^{p,q}, \qquad p+q=n.
\end{aligned}$$

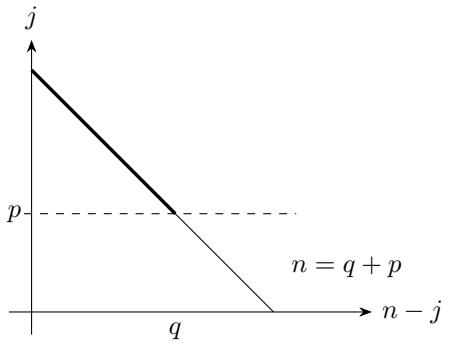


It is a subcomplex because, for $i \geq p$,

$$D(M^{i,n-i}) \subseteq M^{i+1,n-i} \oplus M^{i,n-i+1} \subseteq ({}_IF^p \,\mathrm{Tot}(M))^{n+1}.$$

Definition D.2.3 (Second filtration)**.** The second filtration is

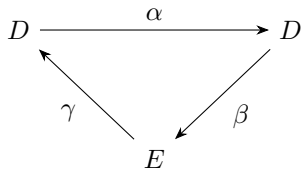
$$\begin{aligned}
({}_{II}F^p \,\mathrm{Tot}(M))^n &= \bigoplus_{j\geq p} M^{n-j,j} \\
&= \cdots \oplus M^{q-2,p+2} \oplus M^{q-1,p+1} \oplus M^{q,p}, \qquad q+p=n.
\end{aligned}$$



Likewise, for $j \geq p$,

$$D(M^{n-j,j}) \subseteq M^{n-j+1,j} \oplus M^{n-j,j+1} \subseteq ({}_{II}F^p \operatorname{Tot}(M))^{n+1}.$$

Definition D.2.4 (Exact couple)**.** An exact couple is a tuple $(D, E; \alpha, \beta, \gamma)$ of bigraded modules and homogeneous maps whose triangle is exact at each vertex:



Proposition D.2.5 (A filtration determines an exact couple)**.** *Every filtration of a cochain complex determines an exact couple with*

$$D^{p,q} = H^{p+q}(F^pC^\bullet), \qquad E^{p,q} = H^{p+q}(F^pC^\bullet/F^{p+1}C^\bullet),$$

and bidegrees $\deg\alpha = (-1,1)$*,* $\deg\beta = (0,0)$*,* $\deg\gamma = (1,0)$*.*

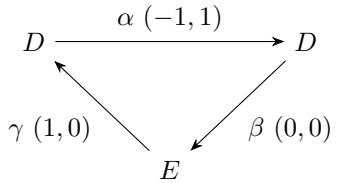
Proof. For each p, the short exact sequence

$$0 \to F^{p+1}C^\bullet \xrightarrow{j^{p+1}} F^pC^\bullet \xrightarrow{\nu^p} F^pC^\bullet/F^{p+1}C^\bullet \to 0$$

gives a long exact cohomology sequence. Write $n = p + q$, and let $\alpha = H(j)$, $\beta = H(\nu)$, and γ be the connecting homomorphism. Since $H^{p+q}(F^{p+1}C^\bullet) = D^{p+1,q-1}$, the sequence becomes

$$\cdots \to D^{p+1,q-1} \xrightarrow{\alpha} D^{p,q} \xrightarrow{\beta} E^{p,q} \xrightarrow{\gamma} D^{p+1,q} \xrightarrow{\alpha} D^{p,q+1} \to \cdots.$$

This proves exactness and gives the displayed bidegrees.



□

Every exact couple determines a derived exact couple.

Definition D.2.6 (Differential bigraded module)**.** A differential bigraded module is a pair (M, d), where d is homogeneous and $d^2 = 0$. If $\deg d = (a, b)$, its cohomology has terms

$$H(M, d)^{p,q} = \ker d^{p,q} / \operatorname{im} d^{p-a,q-b}.$$

A double complex gives (M, d') and (M, d''), but $d' + d''$ does not in general have a single bidegree. It is instead homogeneous for the total grading.

Proposition D.2.7 (Derived couple)**.** *Given an exact couple $(D, E; \alpha, \beta, \gamma)$, the map $d_1 = \beta\gamma$ is a differential on E. There is a derived exact couple $(D_2, E_2; \alpha_2, \beta_2, \gamma_2)$ with $D_2 = \operatorname{im}\alpha$ and $E_2 = H(E, d_1)$.*

Proof. We give the construction with the bidegrees $(-1, 1), (0, 0), (1, 0)$ occurring above.

Step 1: Define

$$d_1^{p,q} = \beta^{p+1,q}\gamma^{p,q} : E^{p,q} \longrightarrow D^{p+1,q} \longrightarrow E^{p+1,q}.$$

The composition is between adjacent long exact sequences in the family, rather than two consecutive arrows of one such sequence. Exactness gives

$$d_1^{p+1,q} d_1^{p,q} = \beta^{p+2,q}\gamma^{p+1,q}\beta^{p+1,q}\gamma^{p,q} = 0.$$

Thus $\deg d_1 = (1, 0)$.

Step 2: Set

$$E_2^{p,q} = H(E, d_1)^{p,q} = \ker d_1^{p,q} / \operatorname{im} d_1^{p-1,q}.$$

Step 3: Set

$$D_2^{p,q} = \operatorname{im}(\alpha^{p+1,q-1} : D^{p+1,q-1} \to D^{p,q}).$$

Define $\alpha_2 = \alpha|_{D_2}$. Thus $\alpha_2(\alpha u) = \alpha^2 u$, and $\deg \alpha_2 = (-1, 1)$.

Step 4: For $y = \alpha v \in D_2^{p,q}$, with $v \in D^{p+1,q-1}$, put

$$\beta_2(y) = [\beta v] = [\beta\alpha^{-1} y].$$

Here $\alpha^{-1}y$ means a chosen preimage. The element βv is a cycle because $\gamma\beta = 0$. If $\alpha v = \alpha v'$, then $v - v' \in \ker\alpha = \operatorname{im}\gamma$, so $v - v' = \gamma w$ and $\beta(v - v') = d_1 w$ is a boundary. Thus β_2 is well-defined and has bidegree $(1, -1)$.

Step 5: If $[z] \in E_2^{p,q}$, then $\beta\gamma z = 0$, so $\gamma z \in \ker\beta = \operatorname{im}\alpha = D_2$. Define

$$\gamma_2 : E_2 \longrightarrow D_2, \qquad [z] \longmapsto \gamma z.$$

Changing z by a boundary $\beta\gamma w$ does not change its image, since $\gamma\beta = 0$. The degree is $(1, 0)$.

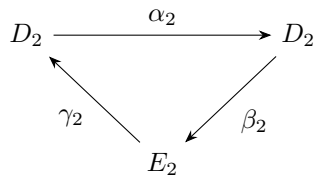
Step 6: We verify exactness. First,

$$\begin{aligned}\beta_2\alpha_2(\alpha u) &= [\beta\alpha u] = 0,\\ \gamma_2\beta_2(\alpha u) &= \gamma\beta u = 0,\\ \alpha_2\gamma_2[z] &= \alpha\gamma z = 0.\end{aligned}$$

For the reverse inclusions:

- If $x \in \ker\alpha_2$, then $x = \gamma y$ because $\ker\alpha = \operatorname{im}\gamma$. Since $x \in D_2 = \ker\beta$, we have $d_1 y = \beta\gamma y = 0$. Thus $x = \gamma_2[y]$.
- If $x = \alpha u \in \ker\beta_2$, then $[\beta u] = 0$, so $\beta u = \beta\gamma w$ for some w. Hence $u - \gamma w \in \ker\beta = D_2$, and $\alpha_2(u - \gamma w) = \alpha u = x$.
- If $[z] \in \ker\gamma_2$, then $\gamma z = 0$, so $z = \beta v$ for some v. Then $\beta_2(\alpha v) = [z]$.

This gives the derived exact triangle



□

Remark D.2.8. If the original bidegrees are $\deg\alpha = \mathbf{a}$, $\deg\beta = \mathbf{b}$, and $\deg\gamma = \mathbf{c}$, the derived degrees are $\mathbf{a}$, $\mathbf{b}-\mathbf{a}$, and $\mathbf{c}$, respectively. Only the degree of β changes.

Definition D.2.9 (Higher derived couple)**.** Put

$$(D_1, E_1; \alpha_1, \beta_1, \gamma_1) = (D, E; \alpha, \beta, \gamma).$$

Inductively, its $(r+1)$st couple is the derived couple of its rth couple.

Corollary D.2.10. *For the exact couple arising from a filtration:*

(i) $\alpha_r, \beta_r, \gamma_r$ *have bidegrees* $(-1, 1)$, $(r-1, 1-r)$, $(1, 0)$, *respectively;*

(ii) $d_r = \beta_r\gamma_r$ *has bidegree* $(r, 1-r)$, *and is induced by* $\beta\alpha^{-r+1}\gamma$ *wherever a representative and the required preimages are chosen;*

(iii) $E_{r+1}^{p,q} = \ker d_r^{p,q} / \operatorname{im} d_r^{p-r,q+r-1}$;

(iv)

$$D_r^{p,q} = \operatorname{im}\bigl(\alpha^{p+1,q-1}\alpha^{p+2,q-2}\cdots\alpha^{p+r-1,q-r+1}\bigr) \subseteq D^{p,q},$$

with the empty composition equal to the identity when $r = 1$.

Definition D.2.11 (Spectral sequence)**.** A cohomological spectral sequence is a sequence $(E_r, d_r)_{r\geq 1}$ of differential bigraded modules, with $\deg d_r = (r, 1-r)$ and specified identifications $E_{r+1} = H(E_r, d_r)$ for every r.

Theorem D.2.12. *Every filtration of a cochain complex yields a spectral sequence as described above.*

D.3 Convergence

Definition D.3.1 (Subquotient)**.** A subquotient of a module M is a module of the form M'/M'', where $M'' \subseteq M' \subseteq M$.

Example D.3.2. (i) Each factor $F^pM/F^{p+1}M$ of a filtration is a subquotient of M.

(ii) $H^n(C^\bullet)$ is a subquotient of C^n, since $\operatorname{im} d^{n-1} \subseteq \ker d^n \subseteq C^n$.

(iii) For a spectral sequence, $E_2 = H(E_1, d_1) = Z_2/B_2$ is a subquotient of E_1. A submodule of E_2 corresponds to a unique submodule S with $B_2 \subseteq S \subseteq Z_2$. Pulling the next cycles and boundaries back to Z_2 gives

$$B_2 \subseteq B_3 \subseteq Z_3 \subseteq Z_2 \subseteq E_1, \qquad E_3 \cong Z_3/B_3.$$

Continuing in this way gives submodules in the fixed ambient module E_1:

$$B_2 \subseteq \cdots \subseteq B_r \subseteq Z_r \subseteq \cdots \subseteq Z_2 \subseteq E_1, \qquad E_r \cong Z_r/B_r.$$

In particular, E_r is a subquotient of E_{r-1}.

Definition D.3.3 (Limit term of a spectral sequence)**.** Using these submodules of E_1, define

$$Z_\infty = \bigcap_r Z_r, \qquad B_\infty = \bigcup_r B_r.$$

The ascending union B_∞ is a submodule, and $B_\infty \subseteq Z_\infty$. The limit term is the bigraded module

$$E_\infty^{p,q} = Z_\infty^{p,q}/B_\infty^{p,q}.$$

Lemma D.3.4. *Let (E_r, d_r) be a spectral sequence.*

(i) The differential d_r is zero if and only if $Z_{r+1} = Z_r$ and $B_{r+1} = B_r$. In that case the prescribed cohomology identification canonically identifies E_{r+1} with E_r.

(ii) If the cycles and boundaries stabilize for all $r \geq s$, then $E_s = E_\infty$.

The same assertions apply in a fixed bidegree when both its incoming and outgoing differentials vanish.

Proof. (i) The cycles in $E_r = Z_r/B_r$ are Z_{r+1}/B_r, and the boundaries are B_{r+1}/B_r. They are all of E_r and zero, respectively, exactly when the stated equalities hold. This is canonical stabilization; abstract isomorphism of two successive modules alone would not suffice.

(ii) Under stabilization, $Z_\infty = Z_s$ and $B_\infty = B_s$, so $E_\infty = Z_s/B_s = E_s$. □

For inclusions $i^p : F^pC^\bullet \hookrightarrow C^\bullet$, the inclusions $F^pC^\bullet \subseteq F^{p-1}C^\bullet$ give

$$\operatorname{im} H^n(i^p) \subseteq \operatorname{im} H^n(i^{p-1}) \subseteq H^n(C^\bullet).$$

Definition D.3.5 (Induced filtration)**.** The induced filtration on cohomology is

$$\Phi^p H^n(C^\bullet) = \operatorname{im}\big(H^n(F^pC^\bullet) \xrightarrow{H^n(i^p)} H^n(C^\bullet)\big).$$

Thus in each column we have inclusions

$$\begin{array}{ccc} H^{n-1}(C^\bullet) & H^n(C^\bullet) & H^{n+1}(C^\bullet) \\ \uparrow & \uparrow & \uparrow \\ \Phi^p H^{n-1} & \Phi^p H^n & \Phi^p H^{n+1} \\ \uparrow & \uparrow & \uparrow \\ \Phi^{p+1} H^{n-1} & \Phi^{p+1} H^n & \Phi^{p+1} H^{n+1}. \end{array}$$

If

$$E_\infty^{p,q} \cong \Phi^p H^{p+q}(C^\bullet)/\Phi^{p+1} H^{p+q}(C^\bullet),$$

the spectral sequence determines the associated graded module of the cohomology, though it does not by itself determine the extensions between these factors.

Definition D.3.6 (Bounded filtration)**.** A filtration $(F^pM^\bullet)$ of a graded module is bounded if for every n there are integers $s = s(n)$ and $t = t(n)$ such that

$$F^sM^n = 0, \qquad F^tM^n = M^n.$$

Thus in each degree there is a finite chain

$$0 = F^sM^n \subseteq F^{s-1}M^n \subseteq \cdots \subseteq F^{t+1}M^n \subseteq F^tM^n = M^n.$$

The bounds may depend on n.

Example D.3.7 (A bounded filtration induces a bounded cohomology filtration)**.** If $(F^pC^\bullet)$ is bounded, the induced filtration on $H^n(C^\bullet)$ is bounded, with the same bounds $s(n), t(n)$.

Proof. If $F^sC^n = 0$, there are no degree-n cycles in $F^sC^\bullet$, so $\Phi^sH^n = 0$. If $F^tC^n = C^n$, every degree-n cycle of $C^\bullet$ belongs to $F^tC^\bullet$, so $H^n(F^tC^\bullet) \to H^n(C^\bullet)$ is surjective. Hence

$$0 = \Phi^sH^n \subseteq \Phi^{s-1}H^n \subseteq \cdots \subseteq \Phi^{t+1}H^n \subseteq \Phi^tH^n = H^n(C^\bullet).$$

We do not need a single pair of bounds for the whole complex. □

Definition D.3.8 (Convergent spectral sequence)**.** In the bounded-filtration setting used here, a spectral sequence converges to a graded module $H^\bullet$, denoted by

$$E_2^{p,q} \underset{p}{\Longrightarrow} H^{p+q},$$

if H^n has a bounded filtration with

$$E_\infty^{p,q} \cong \Phi^pH^n/\Phi^{p+1}H^n, \qquad n = p + q.$$

The index p is the filtration degree. In particular, $E_2^{p,q}$ need not be isomorphic to $E_2^{q,p}$; for instance, cohomology of higher direct images does not in general stay the same when the two indices are exchanged.

Theorem D.3.9. *Let $(F^pC^\bullet)$ be a bounded filtration and (E_r, d_r) the spectral sequence of Theorem D.2.12. Then:*[96]

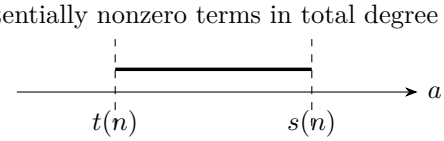
(i) for each fixed (p, q), $E_r^{p,q}$ stabilizes to $E_\infty^{p,q}$ for sufficiently large r;

(ii) $E_2^{p,q} \underset{p}{\Longrightarrow} H^{p+q}(C^\bullet)$, with the induced filtration Φ, which has bounds $s(n), t(n)$ in total degree n.

Proof. (i) Fix total degree n. If $a < t(n)$ or $a \geq s(n)$, the degree-n term of $F^aC^\bullet/F^{a+1}C^\bullet$ is zero. Thus

$$E_1^{a,n-a} = H^n(F^aC^\bullet/F^{a+1}C^\bullet) = 0.$$

Every later $E_r^{a,n-a}$ is a subquotient and is also zero. So there are only finitely many potentially nonzero positions on each diagonal.



[96] For the distinction between enough injectives and enough projectives in categories of sheaves, see [Har77, III, Exercise 6.2]. The exact-couple notation in the proof is that of Proposition D.2.5.

For fixed (p,q) with $n = p+q$, the outgoing differential has target $E_r^{p+r,q-r+1}$ of total degree $n+1$. It vanishes for large r, since $p + r \geq s(n+1)$. The incoming differential has source $E_r^{p-r,q+r-1}$ of total degree $n-1$, and it vanishes for large r, since $p - r < t(n-1)$.

$$E_r^{p-r,q+r-1} = 0 \xrightarrow{d_r} E_r^{p,q} \xrightarrow{d_r} E_r^{p+r,q-r+1} = 0$$

Therefore

$$E_{r+1}^{p,q} = \ker d_r^{p,q} / \operatorname{im} d_r^{p-r,q+r-1} = E_r^{p,q}$$

canonically for large r. Lemma D.3.4 identifies this stable value with $E_\infty^{p,q}$.

(ii) Keep $n = p+q$ fixed. The rth exact couple gives

$$D_r^{p-r+2,q+r-2} \xrightarrow{\alpha_r} D_r^{p-r+1,q+r-1} \xrightarrow{\beta_r} E_r^{p,q} \xrightarrow{\gamma_r} D_r^{p+1,q}.$$

By Corollary D.2.10,

$$D_r^{a,b} = \operatorname{im}\bigl(H^{a+b}(F^{a+r-1}C^\bullet) \longrightarrow H^{a+b}(F^a C^\bullet)\bigr).$$

Choose r large enough that $F^{p-r+1}C^\bullet$ and $F^{p-r+2}C^\bullet$ agree with $C^\bullet$ in degrees $n-1, n, n+1$. This is possible by degreewise boundedness. Their degree-n cohomologies therefore agree with $H^n(C^\bullet)$, and the first two terms above identify as

$$D_r^{p-r+2,q+r-2} = \Phi^{p+1}H^n(C^\bullet), \qquad D_r^{p-r+1,q+r-1} = \Phi^p H^n(C^\bullet).$$

Under these identifications α_r is the inclusion. The last term is

$$D_r^{p+1,q} = \operatorname{im}\bigl(H^{n+1}(F^{p+r}C^\bullet) \longrightarrow H^{n+1}(F^{p+1}C^\bullet)\bigr),$$

which is zero for large r because $F^{p+r}C^{n+1} = 0$. Exactness now yields

$$\Phi^p H^n(C^\bullet)/\Phi^{p+1}H^n(C^\bullet) \cong E_r^{p,q} = E_\infty^{p,q}.$$

This is the asserted convergence. □

D.4 Cartan–Eilenberg Resolutions

If an abelian category $\mathcal{A}$ has enough injectives (respectively, projectives), then the category $\mathrm{Comp}(\mathcal{A})$ of complexes also has enough injectives (respectively, projectives). For example, the category of sheaves of $\mathcal{O}_X$-modules has enough injectives, although generally not enough projectives. When studying coherent sheaves on a

noetherian scheme, we take injective resolutions in this larger category; $\mathrm{Coh}(X)$ itself generally does not have enough injectives.

A Cartan–Eilenberg resolution additionally resolves the cycles, boundaries, and cohomology of a complex. Its rows need not be injective objects of $\mathrm{Comp}(\mathcal{A})$ in the ordinary abelian-category sense. We first discuss that ordinary notion and then make the distinction precise.

Definition D.4.1 (Fundamental exact sequences)**.** For a cochain complex $(C^\bullet, d)$ in an abelian category, the fundamental exact sequences are

$$0 \longrightarrow Z^n \longrightarrow C^n \xrightarrow{d'} B^{n+1} \longrightarrow 0, \qquad 0 \longrightarrow B^n \longrightarrow Z^n \longrightarrow H^n(C^\bullet) \longrightarrow 0.$$

Here d' is the differential with its target restricted to the image, so that d^n factors through $B^{n+1} \hookrightarrow C^{n+1}$.

Definition D.4.2 (Split complex)**.** A complex is split if all its fundamental exact sequences split. This does not require the complex to be exact. A split complex that is also acyclic will be called *split exact*.

Lemma D.4.3. *Every morphism $f : A \to B$ in an abelian category can occur as a differential of a complex.*

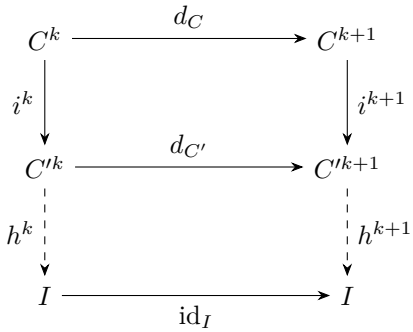
Proof. For any $k \in \mathbb{Z}$, take

$$\Sigma^k(f) = (\cdots \to 0 \to A \xrightarrow{f} B \to 0 \to \cdots),$$

with A in degree k and B in degree $k+1$. □

Lemma D.4.4. *If I is injective in $\mathcal{A}$, then $\Sigma^k(\mathrm{id}_I)$ is injective in* $\mathrm{Comp}(\mathcal{A})$.

Proof. Let $i : C^\bullet \hookrightarrow C'^\bullet$ be a monomorphism of complexes, and let $u : C^\bullet \to \Sigma^k(\mathrm{id}_I)$ be a cochain map. Its components satisfy $u^k = u^{k+1} d_C^k$. Injectivity of I extends u^{k+1} to $h^{k+1} : C'^{k+1} \to I$. Set $h^k = h^{k+1} d_{C'}^k$, and set all other components to zero. Then $h^k i^k = h^{k+1} i^{k+1} d_C^k = u^k$, and $h^k d_{C'}^{k-1} = 0$. Thus h is the required cochain extension. The diagram is



Equivalently, $\mathrm{Hom}_{\mathrm{Comp}}(C^\bullet, \Sigma^k(\mathrm{id}_I)) \cong \mathrm{Hom}_{\mathcal{A}}(C^{k+1}, I)$, an exact contravariant functor. □

Proposition D.4.5. *If $(C^\bullet, d)$ is split, then*

$$C^\bullet \cong \bigoplus_{k\in\mathbb{Z}} \Sigma^{k-1}(\delta^k), \qquad \delta^k : B^k(C^\bullet) \hookrightarrow Z^k(C^\bullet).$$

This sum is locally finite: only two summands contribute to any fixed degree.

Proof. Write $B^k = B^k(C^\bullet)$ and $Z^k = Z^k(C^\bullet)$. Choose splittings

$$0 \longrightarrow Z^k \xrightarrow{i} C^k \xrightarrow{p} B^{k+1} \longrightarrow 0, \qquad q : C^k \to Z^k, \quad j : B^{k+1} \to C^k,$$

with $qi = \mathrm{id}$, $pj = \mathrm{id}$, $pi = 0$, and $iq + jp = \mathrm{id}$. Thus $\theta^k = (q, p) : C^k \xrightarrow{\sim} Z^k \oplus B^{k+1}$. The degree-$n$ term of the displayed sum is $Z^n \oplus B^{n+1}$, and its differential is

$$\Delta^n : Z^n \oplus B^{n+1} \longrightarrow Z^{n+1} \oplus B^{n+2}, \qquad (z, b) \longmapsto (\delta^{n+1} b, 0).$$

Since d^n factors through $p : C^n \twoheadrightarrow B^{n+1}$ and the inclusion into Z^{n+1}, the square

$$\begin{array}{ccc} C^n & \xrightarrow{d} & C^{n+1} \\ {\scriptstyle\theta^n}\downarrow & & \downarrow{\scriptstyle\theta^{n+1}} \\ Z^n \oplus B^{n+1} & \xrightarrow{\Delta} & Z^{n+1} \oplus B^{n+2} \end{array}$$

commutes. Hence θ is an isomorphism of complexes. □

Corollary D.4.6. *If $C^\bullet$ is split exact and every C^n is injective in $\mathcal{A}$, then $C^\bullet$ is injective in* $\mathrm{Comp}(\mathcal{A})$.

Proof. Exactness gives $B^k = Z^k$. Each Z^k is a direct summand of the injective C^k, and hence injective. The proposition gives

$$C^\bullet \cong \prod_{k\in\mathbb{Z}} \Sigma^{k-1}(\mathrm{id}_{Z^k}).$$

This product exists and coincides with the displayed direct sum, since it is finite in each degree. Each factor is injective by Lemma D.4.4, and their product is injective: extensions against a monomorphism can be chosen in each factor. The exactness assumption is essential; a nonzero injective object concentrated in a single degree is split, but is not injective in the category of all complexes. □

Now we prepare for the converse characterization.

Definition D.4.7 (Shifted complex)**.** For $p \in \mathbb{Z}$, define $C^\bullet[p]$ by

$$(C[p])^n = C^{n+p}, \qquad d^n_{C[p]} = (-1)^p d^{n+p}_C.$$

Then $H^n(C^\bullet[p]) \cong H^{n+p}(C^\bullet)$.

Example D.4.8. For the graded cycles $Z^\bullet$ and boundaries $B^\bullet$, given zero differentials, there is an exact sequence of complexes

$$0 \longrightarrow Z^\bullet \longrightarrow C^\bullet \xrightarrow{d} B^\bullet[1] \longrightarrow 0.$$

Definition D.4.9 (Mapping cone)**.** For a cochain map $f : (C^\bullet, d) \to (D^\bullet, d')$, set

$$\mathrm{Cone}(f)^n = C^{n+1} \oplus D^n,$$

with differential

$$\delta^n(x, y) = (-d^{n+1}x, d'^n y - f^{n+1}x), \qquad \delta^n = \begin{pmatrix} -d^{n+1} & 0 \\ -f^{n+1} & d'^n \end{pmatrix}.$$

The cochain-map condition gives $\delta^2 = 0$.

Lemma D.4.10. *There is a short exact sequence of complexes*

$$0 \longrightarrow D^\bullet \xrightarrow{i} \mathrm{Cone}(f)^\bullet \xrightarrow{j} C^\bullet[1] \longrightarrow 0,$$

where $i(y) = (0, y)$ and $j(x, y) = -x$. The connecting homomorphism $H^n(C[1]) = H^{n+1}(C) \to H^{n+1}(D)$ is $H^{n+1}(f)$.

Proof. Degreewise exactness is immediate, and the signs ensure that both maps are cochain maps. A cocycle $x \in C^{n+1}$ lifts under j to $(-x, 0)$. Its differential is $(dx, fx) = (0, fx)$, whose preimage under i is fx. Thus the connecting map, computed as $i^{-1}\delta j^{-1}$ on cocycles, is $H^{n+1}(f)$. □

Lemma D.4.11. *For a complex $(C^\bullet, d)$ in an abelian category:*

(i) $\mathrm{Cone}(\mathrm{id}_C)$ *is acyclic;*

(ii) if a degree--1 map s satisfies $d^n = d^n s^{n+1} d^n$ for every n, then $C^\bullet$ is split;

(iii) $\mathrm{Cone}(\mathrm{id}_C)$ *is split exact; in fact it is contractible.*

Proof. (i) The connecting maps from the previous lemma are identities. The long exact sequence therefore gives $H^n(\mathrm{Cone}(\mathrm{id}_C)) = 0$ for all n.

(ii) Use element notation for the corresponding factorizations in the abelian category. For $b = dc \in B^{n+1}$, put $t(b) = s(b)$. Then $dt(b) = dsdc = dc = b$, splitting $0 \to Z^n \to C^n \to B^{n+1} \to 0$. For the second fundamental sequence, define

$$u : H^n \longrightarrow Z^n, \qquad [z] \longmapsto z - dsz.$$

If $b = dc$ is a boundary, $b - dsb = dc - dsdc = 0$, so this is well-defined. It is a section of $Z^n \twoheadrightarrow H^n$, since $[z - dsz] = [z]$. Thus both fundamental sequences split.

(iii) On the cone, set

$$s^n : C^{n+1} \oplus C^n \longrightarrow C^n \oplus C^{n-1}, \qquad s^n(x, y) = (-y, 0).$$

A direct calculation gives

$$\begin{aligned} \delta s(x, y) &= (dy, y), \\ s\delta(x, y) &= (x - dy, 0), \end{aligned}$$

so $\delta s + s\delta = \mathrm{id}$. In particular, $\delta s \delta = \delta$, and (ii) gives splitting. Together with (i), this is split exactness. □

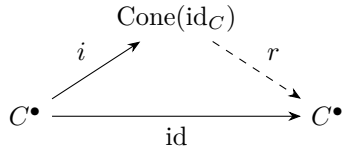
Definition D.4.12 (Quasi-isomorphism)**.** A cochain map $f : (C^\bullet, d) \to (D^\bullet, d')$ is a *quasi-isomorphism* (abbreviated qis) if every induced map $H^n(f) : H^n(C^\bullet) \to H^n(D^\bullet)$ is an isomorphism.

Corollary D.4.13. *A cochain map f is a quasi-isomorphism if and only if* $\mathrm{Cone}(f)$ *is acyclic.*

Proof. This follows from the long exact sequence in Lemma D.4.10. □

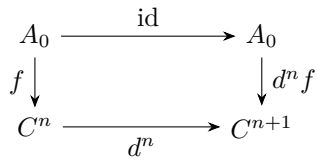
Theorem D.4.14 (Characterizing injective objects)**.** *If $\mathcal{A}$ is an abelian category, a complex $(C^\bullet, d)$ is injective in* $\mathrm{Comp}(\mathcal{A})$ *if and only if it is split exact and every term C^n is injective in $\mathcal{A}$.*

Proof. Sufficiency is Corollary D.4.6. Conversely, if $C^\bullet$ is injective, the inclusion $i : C^\bullet \hookrightarrow \mathrm{Cone}(\mathrm{id}_C)$ admits a retraction:



By Lemma D.4.11, the cone has a contracting homotopy s. Thus rsi is a contracting homotopy on $C^\bullet$, since $d(rsi) + (rsi)d = r(\delta s + s\delta)i = ri = \mathrm{id}$. Hence $C^\bullet$ is split exact.

Now we show that each C^n is injective. Given a monomorphism $g : A_0 \hookrightarrow A$ and a map $f : A_0 \to C^n$, use the disk complexes $\Sigma^n(\mathrm{id}_{A_0})$ and $\Sigma^n(\mathrm{id}_A)$, concentrated in degrees $n, n+1$. The maps f and $d^n f$ define a cochain map $F : \Sigma^n(\mathrm{id}_{A_0}) \to C^\bullet$:



The map $\Sigma^n(g)$ is monic. Injectivity of $C^\bullet$ extends F to a cochain map $H : \Sigma^n(\mathrm{id}_A) \to C^\bullet$. Its degree-$n$ component satisfies $H^n g = f$, proving that C^n is injective. □

Theorem D.4.15. *If an abelian category $\mathcal{A}$ has enough injectives, then so does* $\mathrm{Comp}(\mathcal{A})$.

Proof. For each n, choose an injective object I^n and a monomorphism $j^n : C^n \hookrightarrow I^n$. Define $J^n : C^\bullet \to \Sigma^{n-1}(\mathrm{id}_{I^n})$ by its two possibly nonzero components $j^n d^{n-1}$ and j^n:

$$\begin{array}{ccc} C^{n-1} & \xrightarrow{\quad d^{n-1} \quad} & C^n \\ {\scriptstyle j^n d^{n-1}}\downarrow & & \downarrow{\scriptstyle j^n} \\ I^n & \xrightarrow[\quad \mathrm{id} \quad]{} & I^n \end{array}$$

Put

$$\Sigma^\bullet = \prod_{n \in \mathbb{Z}} \Sigma^{n-1}(\mathrm{id}_{I^n}).$$

This product exists degreewise: only two factors are nonzero in each degree. It also equals the degreewise direct sum. Each factor is injective by Lemma D.4.4, so the product is injective. The maps J^n induce $J : C^\bullet \to \Sigma^\bullet$, which is monic because its degree-n component has the monomorphism j^n as one of its projections. □

Remark D.4.16. These discussions dualize to projective objects. The full subcategories $\mathrm{Comp}^{\geq 0}(\mathcal{A})$ and $\mathrm{Comp}^{\leq 0}(\mathcal{A})$ also have enough injectives when $\mathcal{A}$ does. At a truncation boundary, use the corresponding injective stalk complex in place of a disk extending outside the allowed degrees.

Definition D.4.17 (Cartan–Eilenberg resolution)**.** Let $C^\bullet$ be a complex in an abelian category $\mathcal{A}$. A *Cartan–Eilenberg injective resolution* of $C^\bullet$ is an exact sequence in $\mathrm{Comp}(\mathcal{A})$,[97]

$$0 \longrightarrow C^\bullet \longrightarrow M^{\bullet,0} \longrightarrow M^{\bullet,1} \longrightarrow \cdots,$$

such that, for each p, the following are injective resolutions in $\mathcal{A}$:

(i) $0 \to C^p \to M^{p,0} \to M^{p,1} \to \cdots$;

(ii) $0 \to Z^p(C) \to Z^{p,0} \to Z^{p,1} \to \cdots$;

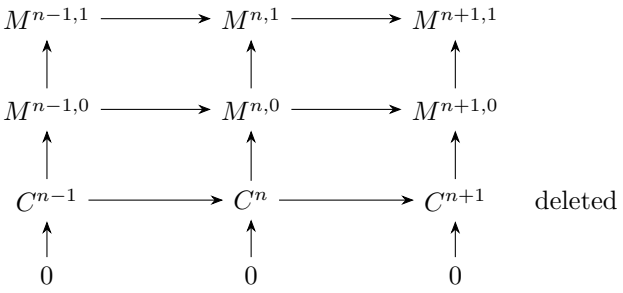
(iii) $0 \to B^p(C) \to B^{p,0} \to B^{p,1} \to \cdots$;

(iv) $0 \to H^p(C) \to H^{p,0} \to H^{p,1} \to \cdots$.

[97] This is distinct from the degreewise embedding into an injective complex in Theorem D.4.15; split rows need not be exact.

Here $Z^{p,q}, B^{p,q}, H^{p,q}$ are the cycles, boundaries, and cohomology of the row $M^{\bullet,q}$ in degree p; every term with a double superscript above is injective in $\mathcal{A}$. There is a dual notion of a Cartan–Eilenberg projective resolution.

A Cartan–Eilenberg resolution is a large commutative diagram. Its columns $M^{p,\bullet}$ are deleted injective resolutions of C^p:



The fundamental exact sequences of each row,

$$0 \to Z^{p,q} \to M^{p,q} \to B^{p+1,q} \to 0, \qquad 0 \to B^{p,q} \to Z^{p,q} \to H^{p,q} \to 0,$$

split because $Z^{p,q}$ and $B^{p,q}$ are injective. Thus the rows are split complexes with injective terms, cycles, boundaries, and cohomology. They need not be exact, and hence need not be injective objects of $\mathrm{Comp}(\mathcal{A})$. A Cartan–Eilenberg resolution is therefore not, in general, an injective resolution in that category. Multiplying the vertical map in column p by $(-1)^p$ gives an anticommuting double complex.

Theorem D.4.18 (Construction of a CE resolution)**.** *If an abelian category $\mathcal{A}$ has enough injectives (respectively, projectives), every complex in* $\mathrm{Comp}(\mathcal{A})$ *has a Cartan–Eilenberg injective (respectively, projective) resolution.*

Proof. Write $B^p = B^p(C)$, $Z^p = Z^p(C)$, and $H^p = H^p(C)$. For each p there are fundamental exact sequences

$$0 \to B^p \to Z^p \to H^p \to 0, \qquad 0 \to Z^p \to C^p \to B^{p+1} \to 0.$$

Choose injective resolutions $B^{p,\bullet}$ and $H^{p,\bullet}$. The horseshoe lemma first constructs an injective resolution $Z^{p,\bullet}$, then one $M^{p,\bullet}$ of C^p, with degreewise split exact sequences of complexes

$$0 \to B^{p,\bullet} \to Z^{p,\bullet} \to H^{p,\bullet} \to 0, \qquad 0 \to Z^{p,\bullet} \to M^{p,\bullet} \to B^{p+1,\bullet} \to 0.$$

Define the horizontal maps by the composites

$$d^{p,q} : M^{p,q} \twoheadrightarrow B^{p+1,q} \hookrightarrow Z^{p+1,q} \hookrightarrow M^{p+1,q}.$$

They square to zero; their kernels, images, and cohomology are precisely $Z^{p,q}, B^{p,q}, H^{p,q}$. They commute with the resolution maps, since all the maps

in the two displayed sequences are cochain maps. We obtain the augmented commutative diagram shown above, whose columns resolve C^p, and all four requirements of Definition D.4.17 hold. If $C^p = 0$ for $p < 0$, all resolutions in those columns may be chosen zero. The dual construction gives the projective case. □

D.5 Cohomology of the Total Complex

Recall that, for a double complex $(M^{\bullet,\bullet}, d', d'')$, the first filtration of $\mathrm{Tot}(M)$ is

$$ {}_{\mathrm{I}}F^p \,\mathrm{Tot}(M)^n = \bigoplus_{i \geq p} M^{i,n-i} = \cdots \oplus M^{p+2,q-2} \oplus M^{p+1,q-1} \oplus M^{p,q}, \qquad p+q=n. $$

These are the terms to the right of the vertical line. The second filtration is

$$ {}_{\mathrm{II}}F^p \,\mathrm{Tot}(M)^n = \bigoplus_{j \geq p} M^{n-j,j} = \cdots \oplus M^{q-2,p+2} \oplus M^{q-1,p+1} \oplus M^{q,p}, \qquad q+p=n, $$

consisting of the terms above the horizontal line.

Lemma D.5.1. *For the transposed double complex (M^t, d'', d'), one has ${}_{\mathrm{II}}F^p \,\mathrm{Tot}(M) = {}_{\mathrm{I}}F^p \,\mathrm{Tot}(M^t)$ under the natural identification of total complexes.*

Proof. Since $(M^t)^{p,q} = M^{q,p}$, we have $\bigoplus_{j \geq p} M^{n-j,j} = \bigoplus_{j \geq p} (M^t)^{j,n-j}$. The total differentials agree. □

Theorem D.5.2. *Let $M^{\bullet,\bullet}$ be a first quadrant double complex, with spectral sequences ${}_{\mathrm{I}}E_r$ and ${}_{\mathrm{II}}E_r$ from these filtrations.*

(i) Both filtrations are bounded.

(ii) For fixed p, q, both sequences stabilize: ${}_{\mathrm{I}}E_\infty^{p,q} = {}_{\mathrm{I}}E_r^{p,q}$ and ${}_{\mathrm{II}}E_\infty^{p,q} = {}_{\mathrm{II}}E_r^{p,q}$ for sufficiently large r.

(iii) Both converge to $H^{p+q}(\mathrm{Tot}(M))$.

Proof. In total degree $n \geq 0$, the first filtration is all of $\mathrm{Tot}(M)^n$ at $p = 0$ and zero at $p = n+1$: if $i \geq n+1$, then $n - i < 0$. Thus $t(n) = 0$ and $s(n) = n+1$ are bounds. Negative total degrees are zero. The same argument applies to the second filtration. Parts (ii) and (iii) follow from Theorem D.3.9, also using $\mathrm{Tot}(M^t) = \mathrm{Tot}(M)$. □

Example D.5.3 (Computation of ${}_{\mathrm{I}}E_2$)**.** Drop the prescript I temporarily and put $n = p + q$. We have[98]

[98] For the filtered-complex construction being used, see Propositions D.2.5 and D.2.7.

$$(F^p)^n = \cdots \oplus M^{p+2,q-2} \oplus M^{p+1,q-1} \oplus M^{p,q},$$
$$(F^{p+1})^n = \cdots \oplus M^{p+2,q-2} \oplus M^{p+1,q-1}.$$

Hence $(F^p/F^{p+1})^n = M^{p,q}$. Its differential is induced by $D = d' + d''$, but the component $d'a \in M^{p+1,q}$ is killed in the quotient. Only d'' survives, so

$$H^n(F^p/F^{p+1}) = \frac{\ker \overline{D}^n}{\operatorname{im} \overline{D}^{n-1}} \simeq \frac{\ker d''^{p,q}}{\operatorname{im} d''^{p,q-1}} = H^q(M^{p,\bullet}) =: H''(M)^{p,q}.$$

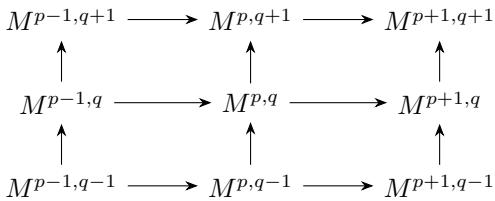
Its elements are $[z]$ with $z \in M^{p,q}$ and $d''z = 0$. For fixed q, the row $H''(M)^{\bullet,q}$ is a complex with differential

$$\overline{d}'^p : H^q(M^{p,\bullet}) \longrightarrow H^q(M^{p+1,\bullet}), \qquad [z] \longmapsto [d'z].$$

Indeed $d''d'z = -d'd''z = 0$, and boundaries map to boundaries. Taking its cohomology gives

$$H^p(H''(M)^{\bullet,q}) = \frac{\ker \overline{d}'^p}{\operatorname{im} \overline{d}'^{p-1}} =: H'^p H''^q(M).$$

An element is represented by $[[z]]$, where $d''z = 0$ and $d'z$ is a vertical boundary; one then also divides by horizontal boundaries in vertical cohomology.



Definition D.5.4 (First iterated cohomology)**.** The first iterated cohomology of $(M^{\bullet,\bullet}, d', d'')$ is the bigraded module with (p, q) term $H'^p H''^q(M)$.

Thus we first take cohomology of the columns, and then of the resulting rows.

Proposition D.5.5. *For a first quadrant double complex,*[99]

$${}_{\mathrm{I}}E_1^{p,q} = H^q(M^{p,\bullet}), \qquad {}_{\mathrm{I}}E_2^{p,q} = H'^p H''^q(M) \implies H^{p+q}(\mathrm{Tot}(M)).$$

Proof. The computation of E_1 is Example D.5.3, and convergence follows from Theorem D.5.2. It remains to identify d_1 with $\overline{d}'$. The exact-couple differential $d_1 = \beta\gamma$ is the connecting map for

$$0 \to F^{p+1}/F^{p+2} \to F^p/F^{p+2} \to F^p/F^{p+1} \to 0.$$

Indeed, if $z + F^{p+1}$ is a quotient cocycle, then $Dz \in F^{p+1}$, and both constructions send its class to $[Dz + F^{p+2}]$. We do not require $Dz = 0$ in F^p.

[99] The differential is the one supplied by the exact couple in Proposition D.2.5.

$$\begin{array}{ccccc} & & (F^p/F^{p+2})^n & \xrightarrow{\pi} & (F^p/F^{p+1})^n \\ & & \overline{D}\downarrow & & \downarrow \\ (F^{p+1}/F^{p+2})^{n+1} & \xrightarrow{i} & (F^p/F^{p+2})^{n+1} & \longrightarrow & (F^p/F^{p+1})^{n+1} \end{array}$$

More concretely, use the ordered decompositions

$$(F^p/F^{p+2})^n = M^{p+1,q-1} \oplus M^{p,q},$$
$$(F^p/F^{p+2})^{n+1} = M^{p+1,q} \oplus M^{p,q+1}.$$

Then $\overline{D}(a,b) = (d''a + d'b, d''b)$. For a vertical cocycle z, choose the lift $(0,z)$ under π. Its differential is $(d'z, 0)$, so

$$d_1^{p,q}[z] = [i^{-1}\overline{D}\pi^{-1}z] = [d'z] = \overline{d}'^p[z] \in H^q(M^{p+1,\bullet}) = E_1^{p+1,q}.$$

Taking cohomology yields the stated E_2-term. □

Example D.5.6 (Computation of ${}_{\mathrm{II}}E_2$)**.** Since ${}_{\mathrm{II}}F^p\,\mathrm{Tot}(M) = {}_{\mathrm{I}}F^p\,\mathrm{Tot}(M^t)$, put $\delta' = d''$ and $\delta'' = d'$. The p-th column of M^t is the p-th row of M, so

$${}_{\mathrm{II}}E_1^{p,q} = H^q(M^{\bullet,p}).$$

For fixed q, these groups form a complex under $\overline{d}''$:

$$\cdots \longrightarrow H^q(M^{\bullet,p-1}) \xrightarrow{\overline{d}''^{p-1}} H^q(M^{\bullet,p}) \xrightarrow{\overline{d}''^{p}} H^q(M^{\bullet,p+1}) \longrightarrow \cdots.$$

Its p-th cohomology is

$$H^p(H'(M)^{q,\bullet}) = \frac{\ker \overline{d}''^p}{\operatorname{im} \overline{d}''^{p-1}} =: H''^pH'^q(M).$$

The diagram of Example D.5.3 is transposed: the original horizontal maps are δ'', and the original vertical maps are δ'.

$$\begin{array}{ccccc} M^{q-1,p+1} & \longrightarrow & M^{q,p+1} & \longrightarrow & M^{q+1,p+1} \\ \uparrow & & \uparrow\ \delta' = d'' & & \uparrow \\ M^{q-1,p} & \xrightarrow{\delta'' = d'} & M^{q,p} & \longrightarrow & M^{q+1,p} \\ \uparrow & & \uparrow & & \uparrow \\ M^{q-1,p-1} & \longrightarrow & M^{q,p-1} & \longrightarrow & M^{q+1,p-1} \end{array}$$

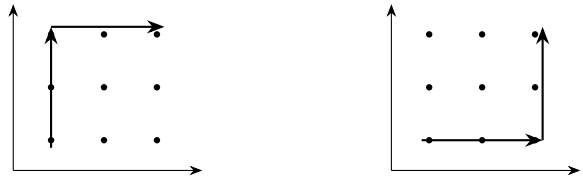
Definition D.5.7 (Second iterated cohomology)**.** The second iterated cohomology of $(M^{\bullet,\bullet}, d', d'')$ is the bigraded module with (p,q) term $H''^pH'^q(M)$.

Proposition D.5.8. *For a first quadrant double complex,*

$${}_{\mathrm{II}}E_1^{p,q} = H^q(M^{\bullet,p}), \qquad {}_{\mathrm{II}}E_2^{p,q} = H''^pH'^q(M) \implies H^{p+q}(\mathrm{Tot}(M)).$$

Proof. Apply Proposition D.5.5 to M^t. □

Remark D.5.9. Informally, the first iterated cohomology takes columns and then rows; the second takes rows and then columns.



columns, then rows rows, then columns

Although both converge to $H^n(\mathrm{Tot}(M))$, their filtrations need not be the same, so ${}_{\mathrm{I}}E_\infty^{p,q}$ and ${}_{\mathrm{II}}E_\infty^{p,q}$ need not be isomorphic.

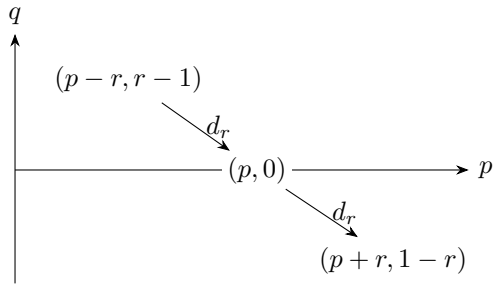
Definition D.5.10 (Collapsed spectral sequence)**.** A spectral sequence collapses on the p-axis if $E_2^{p,q} = 0$ for $q \neq 0$, and on the q-axis if $E_2^{p,q} = 0$ for $p \neq 0$.

Proposition D.5.11. *Suppose a first quadrant spectral sequence converges to* $H^{p+q}(\mathrm{Tot}(M))$.

(i) If it collapses on either axis, then $E_\infty^{p,q} = E_2^{p,q}$.

(ii) In the p-axis case, $H^n(\mathrm{Tot}(M)) \simeq E_2^{n,0}$*; in the q-axis case,* $H^n(\mathrm{Tot}(M)) \simeq E_2^{0,n}$.

Proof. (i) In the p-axis case, all later terms off that axis vanish since they are subquotients of E_2. For $r \geq 2$, the outgoing differential from $E_r^{p,0}$ has target $E_r^{p+r,1-r} = 0$, and the incoming one has source $E_r^{p-r,r-1} = 0$. Thus $E_{r+1}^{p,0} = E_r^{p,0}$ canonically.



The q-axis case is similar.

(ii) The filtration in total degree n is

$$0 = \Phi^{n+1}H^n \subseteq \Phi^n H^n \subseteq \cdots \subseteq \Phi^0 H^n = H^n.$$

In the p-axis case, $E_\infty^{p,n-p} = 0$ for $0 \leq p < n$, so $H^n = \Phi^0 H^n = \cdots = \Phi^n H^n$. Hence $E_2^{n,0} = \Phi^n H^n / \Phi^{n+1} H^n \simeq H^n$. In the q-axis case, the only nonzero quotient is $E_2^{0,n} = \Phi^0 H^n / \Phi^1 H^n$, and $\Phi^1 H^n = 0$. □

D.6 Grothendieck Spectral Sequence

In this section we discuss the spectral sequence introduced by Grothendieck in his Tôhoku paper.

Definition D.6.1 (Right/left acyclic)**.** Let $\mathcal{B}$ be an abelian category with enough injectives (respectively, projectives), and let $F : \mathcal{B} \to \mathrm{Mod}_R$ be additive. An object B is *right F-acyclic* if $R^iF(B) = 0$ for all $i \geq 1$, and *left F-acyclic* if $L_iF(B) = 0$ for all $i \geq 1$. Here right derived functors of an additive functor are defined by applying it to an injective resolution and taking cohomology; without left exactness, R^0F need not equal F. The analogous convention applies on the left.

Example D.6.2. (i) The covariant additive left exact functor $\mathrm{Hom}_R(M, -) : \mathrm{Mod}_R \to \mathrm{Mod}_R$ sends injectives to right acyclic objects: $\mathrm{Ext}^i_R(M, I) = 0$ for $i \geq 1$.

(ii) The covariant additive right exact functor $M \otimes_R - : \mathrm{Mod}_R \to \mathrm{Mod}_R$ has every flat module N as a left acyclic object: $\mathrm{Tor}^R_i(M, N) = 0$ for $i \geq 1$. In particular, projectives are left acyclic. Here R is commutative, as in our module examples.

(iii) For the covariant additive left exact functor $\Gamma : \mathrm{Mod}_{\mathcal{O}_X} \to \mathrm{Mod}_{\Gamma(X,\mathcal{O}_X)}$, every flasque sheaf is right acyclic: $H^i(X, \mathcal{F}) = 0$ for $i \geq 1$. Every injective $\mathcal{O}_X$-module is flasque and hence right acyclic.

(iv) For a morphism of schemes $f : X \to Y$, the functor $f_* : \mathrm{Mod}_{\mathcal{O}_X} \to \mathrm{Mod}_{\mathcal{O}_Y}$ is covariant, additive, and left exact. Every flasque sheaf is right f_*-acyclic, since its restrictions to $f^{-1}(V)$ are flasque. In the noetherian situations considered below, higher direct images of coherent sheaves under proper morphisms are coherent.

Theorem D.6.3 (Grothendieck)**.** *Let $\mathcal{A} \xrightarrow{G} \mathcal{B} \xrightarrow{F} \mathcal{C}$ be covariant additive functors between abelian categories with enough injectives. Assume that F is left exact and that GI is right F-acyclic for every injective I in $\mathcal{A}$. For every $A \in \mathcal{A}$, there is a first quadrant spectral sequence*

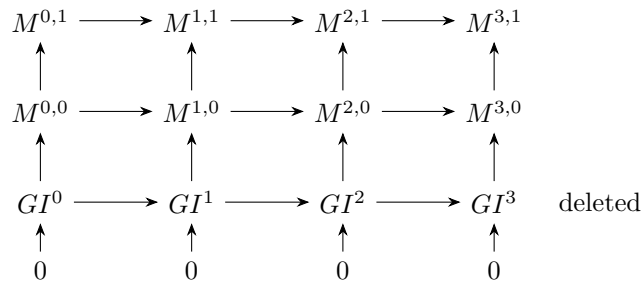
$$E_2^{p,q} = (R^pF)(R^qG)A \implies R^{p+q}(FG)A.$$

The derived functors are understood in the sense just specified; if G is also left exact, then $R^0G = G$ as usual.

Proof. Choose an injective resolution $0 \to A \to I^0 \to I^1 \to \cdots$, and apply G to its deletion I_A to obtain

$$GI_A = (0 \to GI^0 \to GI^1 \to GI^2 \to \cdots).$$

By Theorem D.4.18, this has a first quadrant Cartan–Eilenberg injective resolution $M^{\bullet,\bullet}$ with columns resolving GI^p.



The displayed squares commute. Change the vertical differential in column p by the sign $(-1)^p$ to obtain an anticommuting double complex, and apply F. We compute both spectral sequences of $\mathrm{Tot}(FM)$.

For fixed p, the column $M^{p,\bullet}$ resolves GI^p, so

$$H^q(FM^{p,\bullet}) = (R^qF)(GI^p) = \begin{cases} F(GI^p), & q = 0, \\ 0, & q > 0. \end{cases}$$

We used F left exact in degree zero and GI^p right F-acyclic in positive degrees. Consequently the first iterated cohomology is

$${}_{\mathrm{I}}E_2^{p,q} = \begin{cases} R^p(FG)A, & q = 0, \\ 0, & q > 0, \end{cases}$$

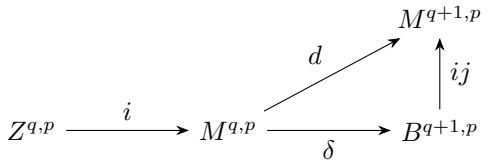
since the surviving row is FG applied to I_A. The first spectral sequence collapses on the p-axis, and Proposition D.5.11 gives

$$H^n(\mathrm{Tot}(FM)) \simeq {}_{\mathrm{I}}E_2^{n,0} = R^n(FG)A.$$

To compute the second iterated cohomology, first take horizontal cohomology. Write $d^{q,p} : M^{q,p} \to M^{q+1,p}$ for the horizontal differential. The fundamental sequences in each row are split exact:

$$0 \to Z^{q,p} \xrightarrow{i} M^{q,p} \xrightarrow{\delta} B^{q+1,p} \to 0, \qquad 0 \to B^{q,p} \xrightarrow{j} Z^{q,p} \to H^{q,p} \to 0.$$

Here $Z^{q,\bullet}$ resolves $Z^q(GI_A)$, and all the doubly indexed objects are injective. The differential factors as $d = ij\delta$, with indices adjusted at the target:



Since an additive functor preserves split exact sequences, both remain exact under F. Thus Fi, Fj are monic and $F\delta$ is epic. Suppressing indices in the following calculation, we have

$$\ker Fd = (Fi)(FZ), \qquad \mathrm{im}\, Fd = (Fi)((Fj)(FB)).$$

The injectivity of Fi therefore gives

$$\frac{\ker Fd}{\operatorname{im} Fd} \simeq \frac{FZ}{(Fj)(FB)} \simeq FH.$$

Restoring indices,

$$H^q(FM^{\bullet,p}) \simeq FH^{q,p} = FH^q(M^{\bullet,p}).$$

Thus F commutes with the horizontal cohomology of these split rows; we are not asserting that it commutes with arbitrary cohomology. By the CE property,

$$0 \to H^q(GI_A) \to H^{q,0} \to H^{q,1} \to \cdots$$

is an injective resolution of $H^q(GI_A) = (R^qG)A$. The vertical sign on horizontal degree q is constant and does not change this cohomology. Hence

$${}_{\mathrm{II}}E_2^{p,q} = H^p\big(F(H^{q,\bullet})\big) = (R^pF)(R^qG)A.$$

By Proposition D.5.8, this converges to $H^{p+q}(\mathrm{Tot}(FM)) \simeq R^{p+q}(FG)A$. Dropping the prescript II proves the theorem. □

Example D.6.4 (Generalized projection formula)**.** Let $f : X \to Y$ be a surjective morphism of proper varieties over $\mathbb{C}$. Let $D_1, \ldots, D_n$ be Cartier divisors on Y, with $n \geq \dim X$. If f is generically finite, then[100]

$$(f^*D_1 \cdots f^*D_n) = \deg(f)(D_1 \cdots D_n).$$

If f is not generically finite, the left-hand side is zero. Equivalently, the displayed formula holds if one sets $\deg(f) = 0$ in this case. An intersection with more divisors than the dimension is understood to be zero.

Proof. For a coherent sheaf $\mathcal{F}$ on X, the sheaves $R^qf_*\mathcal{F}$ are coherent by properness [Har77, III, Corollary 8.8.1]. Apply the Grothendieck spectral sequence to

$$\mathrm{Mod}_{\mathcal{O}_X} \xrightarrow{f_*} \mathrm{Mod}_{\mathcal{O}_Y} \xrightarrow{\Gamma(Y,-)} \mathrm{Vect}_{\mathbb{C}}.$$

These categories have enough injectives, and f_* sends injective sheaves to flasque, hence $\Gamma(Y,-)$-acyclic, sheaves. Since $\Gamma(Y,-)f_* = \Gamma(X,-)$, we obtain the Leray spectral sequence

$$E_2^{p,q} = H^p(Y, R^qf_*\mathcal{F}) \implies H^{p+q}(X, \mathcal{F}).$$

[100]For vanishing over the finite locus and the generic trivialization of the coherent direct image, see [Har77, III, 11.2], [Har77, II, Exercise 5.7(a)]. For the first-quadrant bounds on the filtration, see Theorem D.5.2 and Example D.3.7.

All terms are finite dimensional, and E_2 is supported in a finite rectangle: $0 \le p \le \dim Y$, $0 \le q \le \dim X$. For $H^n = H^n(X, \mathcal{F})$, the induced filtration has quotients $E_\infty^{p,n-p} \simeq \Phi^p H^n / \Phi^{p+1} H^n$. Hence

$$\begin{aligned} \dim_{\mathbb{C}} H^n &= \sum_{p=0}^{n} \bigl(\dim_{\mathbb{C}} \Phi^p H^n - \dim_{\mathbb{C}} \Phi^{p+1} H^n\bigr) \\ &= \sum_{p+q=n} \dim_{\mathbb{C}} E_\infty^{p,q}. \end{aligned}$$

Taking the alternating sum gives

$$\begin{aligned} \chi(X, \mathcal{F}) &= \sum_{p,q} (-1)^{p+q} \dim_{\mathbb{C}} E_\infty^{p,q} \\ &= \sum_{p,q} (-1)^{p+q} \dim_{\mathbb{C}} E_2^{p,q} \\ &= \sum_{q \ge 0} (-1)^q \chi(Y, R^q f_* \mathcal{F}). \end{aligned}$$

The middle equality follows page by page from $E_{r+1} = H(E_r)$ and the invariance of the Euler characteristic of a finite complex under cohomology. Indeed, the two exact sequences

$$0 \to B^n \to Z^n \to H^n \to 0, \qquad 0 \to Z^n \to C^n \to B^{n+1} \to 0$$

give that equality by cancellation of the boundary terms. All sums here are finite.

Put $L = \mathcal{O}_Y(\sum_i m_i D_i)$ and $\mathcal{F} = \mathcal{O}_X(\sum_i m_i f^* D_i) = f^* L$. The projection formula for a line bundle gives

$$R^q f_*(\mathcal{O}_X \otimes f^* L) \simeq R^q f_* \mathcal{O}_X \otimes L.$$

Thus $(f^* D_1 \cdots f^* D_n)$ is the coefficient of $m_1 \cdots m_n$ in

$$\sum_{q \ge 0} (-1)^q \chi\left(Y, R^q f_* \mathcal{O}_X \otimes \mathcal{O}_Y\left(\sum_i m_i D_i\right)\right).$$

If f is not generically finite, then $\dim Y < \dim X \le n$. Every summand has degree at most the dimension of the support of $R^q f_* \mathcal{O}_X$, which is less than n, so that coefficient vanishes.

Otherwise, f is finite of degree d over a dense open subset $U \subset Y$. A finite morphism is affine, so $R^q f_* \mathcal{O}_X|_U = 0$ for $q > 0$. Their supports lie in $Y \setminus U$, of dimension less than n, and again their degree-n coefficients vanish. At the generic point η,

$$(f_* \mathcal{O}_X)_\eta = \mathbb{C}(X), \qquad \dim_{\mathbb{C}(Y)} \mathbb{C}(X) = d.$$

Choose a basis over $\mathbb{C}(Y)$. Since $f_* \mathcal{O}_X$ is torsion-free, this embeds it in $\mathcal{K}_Y^d$. After shrinking U, this identification gives $f_* \mathcal{O}_X|_U \simeq \mathcal{O}_U^d$. Set $\mathcal{G} = f_* \mathcal{O}_X \cap \mathcal{O}_Y^d$ inside $\mathcal{K}_Y^d$. It is coherent, and there are two exact sequences

$$0 \to \mathcal{G} \to f_* \mathcal{O}_X \to \mathcal{G}_1 \to 0, \qquad 0 \to \mathcal{G} \to \mathcal{O}_Y^d \to \mathcal{G}_2 \to 0,$$

where $\operatorname{Supp}\mathcal{G}_1$ and $\operatorname{Supp}\mathcal{G}_2$ are disjoint from U. Tensoring with L and taking Euler characteristics gives

$$\begin{aligned}\chi(Y, f_*\mathcal{O}_X \otimes L) &= \chi(Y, \mathcal{G} \otimes L) + \chi(Y, \mathcal{G}_1 \otimes L),\\ \chi(Y, \mathcal{O}_Y^d \otimes L) &= \chi(Y, \mathcal{G} \otimes L) + \chi(Y, \mathcal{G}_2 \otimes L).\end{aligned}$$

Subtracting, the right-hand difference has no $m_1 \cdots m_n$ coefficient, because both quotient sheaves have support of dimension less than n. The coefficients on the left are therefore equal. As $\chi(Y, \mathcal{O}_Y^d \otimes L) = d\chi(Y, L)$, we conclude

$$(f^*D_1 \cdots f^*D_n) = d(D_1 \cdots D_n).$$

□

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